Index Calculus for the Elliptic-Curve Discrete Logarithm:

Theory, Algorithms, and a Measured Frontier

A treatise from a four-month research programme,

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Abstract

Pollard rho remains the fastest known algorithm for the discrete logarithm on elliptic curves of cryptographic shape. Index calculus could in principle exploit more structure, but it has never beaten rho there. This treatise collects everything a research programme of July to early November 2026 learned about why. Every number is drawn from committed evidence and carries its boundary: oracle stage, complete solve, cold one-target cost, or online cost. No proof-carrying claim covers a result here unless one is named: a measured figure is its producer's reported value, "validated" means only that a validation file names it, and "kernel-checked" means that a Lean kernel accepted it.

Theory. We give, in a draft proof upheld by an integrity audit (the separability lemma in characteristic 2 still awaits an expert check), the Galois group of Semaev's summation cover. It is $(\mathbb{Z}/2)^{m-2}$ for every curve, field and characteristic, with arithmetic monodromy equal to geometric. We also give an exact factorisation law for its fibres over $\mathbb{F}_q$. Two consequences follow:

- fibre statistics are constant on the factor-base locus;
- at fixed factor-base size the mean decomposition yield is fixed.

So levers that rely on curve-dependent monodromy or on raising mean yield at matched base size are closed without experiment. Coverage, oracle cost and solving difficulty remain free.

On fall degrees, we prove for the binary Weil descent of S_3 that:

- the first fall degree (Caminata–Gorla) is 2 for every n whenever every descended equation has degree 2, as on every instance computed, including Semaev's chained systems, where 4 was stated;
- the last fall degree is at most $k+2$, or $(t-1)k+t$ for chains.

We also certify that the last fall degree is at least 5 on named instances at $n = 33$–45 and in the $m = t = 3$ chain, with certificates that an independent program checks. Every F4 run on those systems therefore exceeds degree 4, and Semaev's Assumption 1 fails as stated. For an explicit-membership chain we prove $(d_{\mathrm{ff}}, d_F) = (4, 5)$ exactly at $B = 3$ for every $p > 3$, and $\max(5, B+1) \le d_F \le 3B - 4$ in general. For PKM Kummer towers we prove a constant first fall degree of 4 and show that the last fall degree depends on the presentation, not the ideal. Throughout, the first fall degree is a constant of the presentation and the last is not. A hardness audit refutes the literal Petit–Quisquater Assumption 2 on all 119 census instances, and a left-kernel theorem, kernel-checked in the producer's compile, shows that one target needs no rank heuristic; the census numbers are unverified assertions. A generic collision floor, derived in the generic group model and checked by Monte Carlo, shows that a single-target index calculus whose base logarithms are unknown costs at least 1.5× rho, and 1.875× with two base elements, the constant at which the measured generic incumbent sits.

Algorithms. The programme engineered:

- root-finding and meet-in-the-middle decomposition oracles, about 400–$5 \cdot 10^4\times$ cheaper per decomposition than F4 where they apply;
- inherited and gated F4 engines, and an $\mathbb{F}_p$ F4 at msolve parity;
- compact Frobenius-orbit and known-logarithm-closed factor bases;
- FGHR's D_4 symmetry at $k = 4$, cutting one decomposition 370–481×;
- double-large-prime genus-3 index calculus after the cubic GHS transfer, from a fitted $q^{1.91}$ to $q^{1.37}$ (local exponent 1.356 at $q = 2^{22}$, still above 4/3);
- Diem's plane-quartic index calculus on the same family, at exponent ≈ 1.

Frontier. A priced campaign of 4,848 executions found no index calculus below rho on any prime or prime-degree binary curve at any weight of a pricing band applied to every arm. On Koblitz curves of 37–47 bits the gap to a matched rho is 3.4–11.5× on the validated board (rule v7), and 4.7–18× against the same-binary rho of rule v8. On prime curves the validated rows are 5.5–13.5× at 10–24 bits; the campaign's two-summand pair table is 10.5–102× at 20–40 bits; the closest prime arms, generic collision searches at rho's exponent, are 1.3–1.7× at 36–42 bits ($w = 0$). Only in weak extension-field families does index calculus reach or cross rho:

- genus 3 via GHS: below the strongest rho at $q = 2^{16}$ under the registered unit price, above it at the protocol's sensitivity price, unresolved at the other sizes (no validation file);
- plane quartics: 0.005× a negation rho at $q \approx 2^{21}$ (no validation file);
- Joux–Vitse covers: the sieve route at 0.0065× a matched negation rho in group additions (0.016× in $\mathbb{F}_p$ multiplications, at most 0.12× across the band) at $p = 1511$, on the weak class only, validated as accounting (`VAL-20261103-c31c76`, no `pcc` claim); the plain Nagao arm is above a matched negation rho at $p = 1301$ and 1451 and unresolved at 1601, so its extrapolated crossing near $p \approx 1330$ does not hold;

- $k = 4$ in characteristic two: $5.2\times$ rho at $m = 15$ after corrections; a crossover near $m \approx 17.6$ is an extrapolation.

Every online win was retired by an equal-advice audit against Bernstein–Lange rho. We record the corrections that matching references, pricing native work and replaying records forced. Nothing here concerns a deployed curve or key recovery.

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Introduction

For a subgroup $\langle G \rangle$ of prime order r on an elliptic curve $E/\mathbb{F}_q$, Pollard rho [34, 40] computes discrete logarithms in about $\sqrt{\pi r/(2A)}$ group operations, where A is the number of efficiently computable automorphisms [19, 44]. No algorithm is known to beat it on curves of cryptographic shape.

Index calculus breaks finite-field discrete logarithms in subexponential time. It reached elliptic curves through Semaev's summation polynomials [35] and the extension-field analyses of Gaudry [20] and Diem [10]. It is the method that should exploit the most structure. The question is *where it crosses rho*: on which curves, at which sizes, under which accounting.

This treatise records what one research programme learned about that question between July and early November 2026. The programme spanned three repositories (`crypto`, `crypto-autoresearcher`, `cryptanalysis`), several thousand commits and several hundred preregistered studies. Two companion documents organise the same work differently:

- a programme record [3], chronological and exhaustive, with study catalogues;
- an improvement ledger [2], organised as a ratchet of incumbents.

This document organises it as a treatise:

- the theory that constrains index calculus on elliptic curves;
- the algorithms engineered against those constraints;
- the measured frontier against a fairly priced rho;
- what remains open.

It runs no new experiment. It re-verified one certificate (§6.3), adds no improvement of its own, and cites the committed artefact behind every number; the one exception, the session-reported Kummer-tower last fall degrees of §6.5, is labelled where it is used.

The argument in brief

1. Where the cost lives (Part I). An index calculus pays for factor-base set-up, relation collection, linear algebra and the target's descent. Relation collection costs (trials per relation) × (cost of one decomposition). Chapter 2 writes this out and maps every lever the programme pulled to the term it moves.

2. What is fixed for every curve (Chapter 5). The Galois group of the summation cover is $(\mathbb{Z}/2)^{m-2}$ for every curve, field and characteristic (a draft proof upheld by an integrity audit; the separability lemma in characteristic 2 awaits an expert check). Its fibres over $\mathbb{F}_q$ split by an exact law, and they split completely on the factor-base locus. The mean decomposition yield at fixed base size is a double count. Three consequences follow:

- no curve has anomalous monodromy to exploit;
- no fibre census can find an anomalous relation rate;
- no factor-base geometry can raise mean yield.

The design space that remains is the cost of the oracle, the algebraic difficulty of the decomposition system, the linear algebra, and where the problem lives.

3. How hard is the decomposition system (Chapter 6)? Every subexponential claim for prime-degree binary curves assumes that the solving degree stays near the first fall degree. In three presentations we prove that the first fall degree is a constant of the presentation (for binary Weil descent, whenever every descended equation has degree 2, as on every instance computed), and show that the last fall degree, which equals the solving degree for unsatisfiable systems, is not bounded by it:

- it is proved to differ for an explicit-membership chain ($(4, 5)$ exactly at $B = 3$);
- it is certified to differ, by independently checked closure certificates, for binary Weil descent (≥ 5 against 2 at $n = 33$–45);
- it is reported to differ for prime-field Kummer towers (6 against 4, session-reported; the raw data were not committed).

A proof that the last fall degree grows without bound remains the open problem, as Huang, Kosters and Yeo [23] and Kousidis and Wiemers [29] left it.

4. What hardness rests on (Chapter 7). An index-calculus complexity claim is a chain of six obligations: solver growth, yield, relation rank, geometry, a non-generic interface and end-to-end composition. The literal Petit–Quisquater Assumption 2 fails on all 119 census instances; a kernel-checked left-kernel theorem shows that a single target needs no rank condition; PKM's "$\Delta = 10$ suffices" is contradicted at toy sizes; Nagao's Lemma 4 is refuted in three reading–placement cells and proved in the fourth. These are census and derivation results, unverified assertions except where a Lean kernel checked them, and none moves a row of Tables A–C.

5. What engineering bought (Part III). Root-finding oracles made the decomposition step between about $400\times$ and $5 \cdot 10^4\times$ cheaper per decomposition against F4 where they apply (up to 10^5 in wall time against the PDP baseline). Gröbner engines improved one to two orders in counted work. Factor bases closed under maps of known logarithm cut linear algebra by constant factors. On the cold one-target frontier none of this changed the conclusion: the stages these changes leave alone (relation yield per probe, table set-up) now dominate.

6. Where index calculus wins (Chapter 25 and Chapter 26). Only in weak extension-field families does the measured cost cross or approach a fairly priced rho:

- $E_0/\mathbb{F}_{q^3}$ through a genus-3 GHS cover, by double-large-prime hyperelliptic index calculus or by Diem's plane-quartic index calculus;
- the Joux–Vitse cover of $E(\mathbb{F}_{p^6})$, whose sieve route is $0.0065\times$ a matched negation rho at $p = 1511$ (validated as accounting, weak class only), while its plain Nagao arm is above matched rho at $p = 1301$ and 1451 and unresolved at 1601, so its extrapolated crossing near $p \approx 1330$ does not hold;
- possibly $k = 4$ over $\mathbb{F}_{(2^m)^4}$, at $5.2\times$ rho at $m = 15$ after corrections, with an extrapolated crossover only.

These are known theoretical results. The contribution here is matched, priced and replayed measurement, validated in one case. None of these families is deployed.

7. The rho side (Part IV). Every distance is a ratio to rho, so the denominator gets its own part. A collision floor, derived in the generic group model and checked by Monte Carlo, shows that a single-target index calculus with F base elements of unknown logarithm costs at least 1.5, 1.875 and $2.188\times$ rho for $F = 1, 2, 3$; the measured generic incumbent, near $1.8\times$, sits at the $F = 2$ constant (Proposition 32.1). Chapters 33 and 34 ask how well rho itself was measured and varied.

8. What the measurements say (Part V). At a matched reference, rho is cheaper on every prime and prime-degree binary curve measured. On Koblitz curves of 37–47 bits the gap is 3.4–$11.5\times$ on the validated board (rule v7) and 4.7–$18\times$ against rule v8's same-binary rho. On prime curves the validated rows are 5.5–$13.5\times$ at 10–24 bits; the two-summand pair table is 10.5–$102\times$ at 20–40 bits, its gap widening as about $r^{1/6}$; the generic k-sum table (5.3–$13.1\times$, no validation) narrows that growth to about $r^{0.02}$ against rho's $r^{1/2}$, but its limit is baby-step giant-step; and the closest prime arms, generic residual collectors, are 1.3–$1.7\times$ at 36–42 bits ($w = 0$). Every apparent online win disappeared against Bernstein–Lange rho with equal advice [4]. Matching the reference, pricing native work and replaying records moved most recorded gaps upward. Chapter 41 lists the corrections that moved the leaderboard, and Appendix B the remaining corrections, retractions and withdrawn readings.

Conventions

- *Unit.* Costs are group-addition equivalents per $\sqrt{r}$ (S) unless a family uses its own counted unit, such as $\mathbb{F}_q$ multiplications.
- *Ratios.* "$X\times$" is baseline over candidate when it describes an improvement, and IC over rho when it describes a distance. Each sentence says which.
- *Intervals.* Intervals are 95%, as the source recorded them.
- *Standing.* No proof-carrying (`pcc`) claim covers a result in this treatise unless one is named. A measured figure is the producer's reported value, an unverified assertion, unless a `VALIDATION.json` is named, and "validated" means only that. A theorem is "kernel-checked" only where a Lean kernel accepted it.

- *Change classes.* Every change is an ADVANCE (an exponent or crossover moves under complete accounting), ENGINEERING (a constant moves), RELABELLING or ACCOUNTING. Against rho, the programme produced no ADVANCE on any prime-order curve of cryptographic shape.
- *Paths.* Artefact paths are relative to the `crypto` repository, unless prefixed `crypto-autoresearcher:` or `cryptanalysis:`.

What this treatise does not claim. No speed-up on a deployed curve, no key recovery, no asymptotic sub-rho algorithm for prime fields, and no statement about the security of ECC2K-130, P-256 or secp256k1. The programme's GPU Pollard-rho work on ECC2K-130 is published separately (IACR ePrint 2026/2469) and is not repeated here.

Part I

Foundations

1 Background

Rho and its floor. With A efficiently computable automorphisms acting on $\langle G\rangle$, a rho walk on orbit representatives has expected cost $\sqrt{\pi r/(2A)}$ group additions. $A = 2$ (negation) on every curve; $A = 2n$ on a Koblitz curve $y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$, where the Frobenius $\pi(x, y) = (x^2, y^2)$ acts on $\langle G\rangle$ as a scalar λ [19, 44]. ECC2K-130 is the case $a = 0$, $n = 131$ of this family; its record GPU rho attempt is a separate paper in this repository (`paper/ecc2k130/`).

Index calculus on elliptic curves. Fix a factor base $\mathcal{F} \subset E(\mathbb{F}_q)$. A *relation* is a decomposition $aG + bQ = R = P_1 + \cdots + P_m$ with $P_i \in \mathcal{F}$; once about $|\mathcal{F}|$ relations are found, linear algebra over $\mathbb{Z}/r$ yields $\log_G Q$. The pipeline has five stages (factor base, decomposition oracle, relation yield, rank/linear algebra, verification) and a sixth, the comparison with rho on the same target. The *decomposition oracle* decides whether R splits over $\mathcal{F}$. Semaev [35] reduced it to the roots of the summation polynomial $S_{m+1}(x_1, \ldots, x_m, x_R)$, which vanishes exactly when $\pm P_1 \pm \cdots \pm P_m = R$ for points with those abscissae. Over $\mathbb{F}_{2^n}$ or $\mathbb{F}_{p^k}$ the factor base is chosen as the points whose abscissa lies in a subspace or subfield, and Weil restriction turns $S_{m+1} = 0$ into a polynomial system over the base field; the textbook oracle solves that system with a Gröbner-basis engine (F4/F5, [12, 13]) or a SAT solver [39]. Gaudry [20] and Diem [10] give the asymptotics over extension fields; for prime fields no factor base is known that makes the method competitive, and summation-polynomial systems over $\mathbb{F}_p$ are the subject of the Kummer-tower and rational-map bases of Petit–Kosters–Messeng [33].

Frobenius-invariant bases. Galbraith, Granger, Merz and Petit [18] observe that a factor base with $\pi(\mathcal{F}) = \mathcal{F}$ splits into Frobenius orbits of size dividing n, with $\log \pi^k(P) = \lambda^k \log P$. The unknowns collapse by $\approx n$, the matrix by $\approx n^2$, and ordered decompositions save $m!$—more than rho's $\sqrt{2n}$, but from a base too far behind to overtake it at deployed sizes. Invariant bases come from linearised polynomials attached to the factors of $x^n - 1$ over $\mathbb{F}_2$; the subspace dimension is $\ell = \mathrm{ord}_n(2)$, so the base size is fixed by n (Table 1 of `RESEARCH_KOBLITZ_INDEX_CALCULUS.md`).

Symmetries of the system. The decomposition system is invariant under the symmetric group S_m; rational 2-torsion adds sign changes, and Faugère, Gaudry, Huot and Renault [14] show that in the coordinate Y with $Y(P + T) = -Y(P)$ the invariance group becomes the Weyl group D_m, with a much smaller quotient. We measure this at $m = 4$ in §21.10.

2 Where the cost of index calculus lives

An improvement is only useful if it moves a term that matters. This chapter writes the cost of one index-calculus solve as a sum of terms, and maps each lever of the programme to the term it acts on.

2.1 The pipeline

Fix a factor base $\mathcal{F} \subset E(\mathbb{F}_q)$ of size N. A *relation* is a decomposition $aG + bQ = P_1 + \cdots + P_m$ with every $P_i \in \pm\mathcal{F}$. Once about N independent relations are known, linear algebra over $\mathbb{Z}/r$ yields $\log_G Q$. A heuristic count is enough to place the levers. A uniformly random point decomposes into m base points with probability about $N^m/(m!\,r)$. Collecting N relations therefore needs about $m!\,r/N^{m-1}$ trials. The cold cost of one target is then

$$T \approx \underbrace{T_{\mathrm{setup}}(N)}_{\text{base, tables}} + \underbrace{\frac{m!\,r}{N^{m-1}}\,\frac{1}{\rho_{\mathrm{yield}}}\,C_{\mathrm{dec}}(N,m)}_{\text{relation collection}} + \underbrace{T_{\mathrm{LA}}(N,m)}_{\text{linear algebra}} + \underbrace{T_{\mathrm{target}}}_{\text{descent, check}} . \tag{2.1}$$

Here C_{dec} is the cost of one decomposition attempt (the *oracle*). The factor $\rho_{\mathrm{yield}} \geq 1$ collects everything that makes a trial more productive than the heuristic: free relations, large primes, symmetry, re-randomised targets. T_{LA} is roughly mN^2 for sparse elimination. With rho's cost $\sqrt{\pi r/(2A)}$ on the other side, the classical balance at $m = 2$ with a constant-cost oracle gives $N \approx r^{1/3}$ and $T \approx r^{2/3}$. That is why two-summand prime-field index calculus loses ground to rho as $r^{1/6}$ (Chapter 40). The k-sum oracle, a generic table of signed k-sums, moves the exponent to $k/(2k-1)$, and no generic single-target arm whose base logarithms are unknown can go below the collision floor of Proposition 32.1.

2.2 The levers

Table 2.1 lists, for each term of (2.1), the levers the programme pulled and the sections that report them.

2.3 Which term dominates now

Early in the programme, C_{dec} dominated. A Gröbner solve per trial cost milliseconds to seconds. After root finding (Chapter 9), the oracle is a handful of field operations and table probes, and the cost has moved elsewhere.

- On the Koblitz descent of the leaderboard's Table A, reusable set-up is 98.7–99.6% of cold S, and descent is 0.4–1.3% (`docs/ic/LEADERBOARD.md`, phase table).
- On the $n = 83$ cold pipeline with base K1107, a mean 113.755 s process splits into 47.166 s of set-up and 66.486 s of rank stage. The target stages take 0.099 s (`research/n83_cold_base_confirm_20261021/RESULT.md`).

So the relation-collection and set-up terms dominate. Their cost is set by N and by the yield per probe, not by the oracle. This is the main structural fact behind the ledger. Oracle speed-ups of about 400–$5 \cdot 10^4\times$ per decomposition against F4 (up to 10^5 in wall time against the PDP baseline) moved the end-to-end gap by one order at most.

Term	Levers pulled	Where
C_{dec}	root finding of S_3; pair and meet-in-the-middle tables; k-sum tables; inherited F4, F5, F6-IC; gate-first order; harvest gates; $\mathbb{F}_p$ F4; D_4 invariants; the Hilbert early stop on the weak-class cover	Ch. 9, 10, 25, 40; §22.2, §30.4
ρ_{yield}, N	compact Frobenius orbits; free relations; stride-renewed targets; greedy coverage; $n = 83$ base selection; double large primes; the Joux–Vitse sieve route	Ch. 15, 21, 26; §30.4
T_{setup}	stacked producer; cache-blocked pair table; u64 root batching	Ch. 15, 21
T_{LA}	CADO-style filter and Wiedemann; structured Gaussian elimination; block Wiedemann; first-dependency stop; post-Gröbner linear stage	Ch. 15, 21, Ch. 25, 26, 41
the exponent	genus-three transfer (cubic GHS); $k = 4$ extension fields	Ch. 25, 26
fixed floor	none: the collision floor bounds every generic single-target arm with unknown base logarithms at $\geq 1.5\times$ rho	§32.2

Table 2.1: The terms of (2.1) and the levers that act on them.

3 The theory, by field type

3.1 What a decomposition costs

Every index-calculus algorithm for elliptic curves in the literature spends its time in three places: finding about $|\mathcal{F}|$ relations (each a decomposition of a random combination $aG + bQ$ over $\mathcal{F}$), solving the sparse linear system over $\mathbb{Z}/r$, and, in some variants, a descent of the target. The second and third are well understood. The first is the crux: a uniformly random point decomposes over a base of size N into m summands with probability about $N^m/(m!\,r)$, so relations need about $m!\,r/N^{m-1}$ trials, and each trial costs whatever the decomposition oracle costs. Every result below is a statement about one of three things: how large N can be while membership stays algebraically testable, how cheap the oracle is, and how the two trade against the linear algebra.

3.2 Extension fields $\mathbb{F}_{q^k}$ with $k > 1$ fixed or small

Gaudry [20] takes $\mathcal{F}$ to be the points with abscissa in the base field $\mathbb{F}_q$ and decomposes into $m = k$ summands by Weil restriction of S_{k+1}, a system of k equations in k unknowns over $\mathbb{F}_q$. For fixed k this gives $\tilde{O}(q^{2-2/k})$, asymptotically below rho's $q^{k/2}$ for $k \geq 3$; the constant hidden in the Gröbner solve grows very quickly with k, which is what makes the crossover large in practice; the measured handovers at $k = 4$ are in Chapter 25. Diem [10] extends the analysis to families where k grows, proving subexponential complexity over sequences of fields with suitably balanced q and k. Joux and Vitse [27] combine Weil descent with decomposition-based index calculus into one algorithm, well suited to curves over $\mathbb{F}_{p^6}$, and computed logarithms on a 149-bit group over a degree-6 extension in about 110,000 CPU-hours; they also give an $(n-1)$-decomposition variant. FGHR [14] exploit the action of S_m and of the 2-torsion to replace S_{m+1} by its invariant form, cutting the Gröbner work by large factors; our measurement at $k = 4$ (§21.10) is a direct instance.

3.3 Binary fields $\mathbb{F}_{2^n}$ with n prime

Prime n leaves no subfield, so the factor base is taken to be the points whose abscissa lies in an $\mathbb{F}_2$-vector subspace V of dimension ℓ. Weil restriction then gives a quadratic-to-low-degree Boolean system whose difficulty is governed by its degree of regularity. The whole debate about binary curves is a debate about that degree.

The subexponential claims. Petit and Quisquater [32] conjectured, from experiments and heuristic evidence, that these systems have degrees of regularity only slightly above the degrees of their equations (the *first fall degree assumption*), and deduced a heuristic $2^{c\,n^{2/3}\log n}$, $c < 2$, for a variant of Diem's algorithm, beating generic methods only for $n \gtrsim 2000$. Semaev [36] split summation polynomials into cubic Boolean systems, argued for regularity degree at most 4 under the same kind of assumption, and obtained a heuristic $2^{c\sqrt{n \ln n}}$ with $c \approx 1.69$; its first version concluded that several FIPS binary curves fall faster than Pollard rho, a conclusion its revision says does not hold under the weaker assumption that the regularity degree is $o(\sqrt{n/\ln n})$.

The evidence against. Shantz and Teske [37] tested the Petit–Quisquater heuristics experimentally. Huang, Kosters and Yeo [23] introduced the *last fall degree*, independent of the monomial order, gave computational evidence against the first fall degree assumption in characteristic 2, and constructed summation-polynomial Weil descent systems where it is unlikely to hold. Galbraith and Gebregiyorgis [17] improved the variable choice (binary Edwards curves), found symmetry-breaking factor bases and SAT solvers inconclusive, and concluded that rho remains much faster for prime-degree binary fields of reasonable size. The Galbraith–Gaudry survey [16] reaches the same verdict for characteristic-two curves over prime-degree fields.

Where it stands. No proof or refutation of a subexponential algorithm for prime-degree binary curves exists. The open quantity is the solving degree of Weil-descended S_{m+1} as n and m grow; recent work supplies diagnostics (saturation-exponent bounds, multi-graded regularity, tests for where systems stop looking generic) but no bound for these systems: rigorous last-fall-degree bounds for summation-polynomial descents are named as open by Huang, Kosters and Yeo [23] and by Kousidis and Wiemers [29], and Kosters and Yeo [28] show that the first fall degree of the descent of S_3 is usually 2, through a trace identity. Chapter 6 proves and certifies what the programme could add. Early toy measurements (§16.1) found last fall degrees of 3–4 at small ℓ while the solving degree grows with ℓ and stays within one of the semi-regular prediction; closure algorithms refute a degree or two earlier but rise too. Chapter 6 then proves a first fall degree of 2 for the $m = 2$ descent (under the degree-2 hypothesis met on every instance computed), finds a last fall degree of exactly 5 at $n = 35$, and certifies ≥ 5 on named instances at $n = 33$–45. Neither supports or refutes a subexponential claim.

3.4 Subfield (Koblitz) curves

When the curve is defined over $\mathbb{F}_2$ the Frobenius acts on the group, and Galbraith, Granger, Merz and Petit [18] show that a Frobenius-invariant factor base saves a factor $\approx n$ in relations and $\approx n^2$ in linear algebra—more than rho's $\sqrt{2n}$—while concluding that Koblitz-curve security is not threatened at deployed sizes. Our Koblitz work (§15) builds on this.

3.5 Prime fields

Over $\mathbb{F}_p$ there is no subfield and no subspace, and no factor base is known for which membership is both dense enough and algebraically testable. Semaev's original small-abscissa base leads to systems of very high degree. Petit, Kosters and Messeng [33] construct bases from rational maps (including the Kummer tower μ_{2^t}) that give the summation system extra structure; Amadori, Pintore and Sala [1] build the base from the instance and need only one relation search, improving on both; McGuire and Mueller [30] avoid Gröbner bases altogether by fast evaluation of S_m through iterated resultants, and in a second variant by summing $m - 1$ factor-base points and testing membership of the result—the same mechanism as the two-summand root-finding oracle of §9. All of these remain exponential and slower than rho; a 2024 analysis (Hu, IACR ePrint 2024/1923, an unrefereed preprint) again concludes that no subexponential prime-field method is known.

3.6 Lower bounds

Generic-group lower bounds bound everything that does not use curve structure. With preprocessing, Corrigan-Gibbs and Kogan [8] show $ST^2 = \tilde{\Omega}(\varepsilon N)$ for S bits of advice and online time T; this is the bound against which the retained-set-up (online) results of §15 must be read. A 2026 structured-generic-group bound (Corrigan-Gibbs, Henzinger and Wu, IACR ePrint 2026/384; recorded at abstract level, body not read) states that an algorithm exploiting structure on only a δ fraction of elements needs $\Omega(\min(\sqrt{q}, 1/\delta))$ time—for index calculus, the decomposable set would have to be dense *and* cheaply testable. Whether it covers coordinate access is an open lead (§37.4).

4 Measuring index calculus against rho

The programme's first lesson was that index-calculus numbers from different studies did not compose: one study timed a decomposition, another a relation, a third a whole multi-target batch with shared tables, each on a different host and in a different unit. Four mechanisms, built in this order, fixed that.

4.1 The boundary ledger (10 September)

`docs/ic/BOUNDARY_TARGETS.md` and its machine twin `docs/ic/boundary_targets.json` give each regime (Koblitz, binary, prime) and each pipeline stage (`factor_base`, `decomposition`, `relation_yield`, `rank`, `end_to_end_dlp`, `vs_rho`) a current record, a next target and acceptance gates. A row promotes only when its evidence JSON carries every field the measurement schema requires for that stage—for example, a `decomposition` claim that beats an algebraic frontier must report the first-fall degree, and a `vs_rho` claim needs one target, the identical public point for both methods, verified scalars, five exclusive IC online phases, the automorphism discount, matched resources and a host-isolation receipt. Missing fields fail closed. The primary `vs_rho` question is the online cost of *one* previously unseen target after reusable IC preparation; multi-target and amortised results are secondary and must be declared as such.

4.2 ecbench (from 2 October)

ecbench (`src/cryptanalysis/ecbench/`, `src/bin/ecbench.rs`, documented in `docs/ecbench/README.md`) runs every ECDLP method behind one interface, on the same single-target workloads, in one unit, into sealed records.

Unit.

$$S \;=\; \frac{\text{total group-addition equivalents (GAE)}}{\sqrt{r}}, \qquad S_{\text{floor}} = \sqrt{\pi/(2A)}.$$

An addition or doubling is one GAE; a scalar multiplication is charged as the additions it performs; native work (lookups, row operations, square roots, Artin–Schreier solves, word XORs) is priced at a pinned conversion where the repository has one and is otherwise recorded as *uncharged*, which marks the record's S as a *lower bound*. Plain rho has $S \approx 1.25$.

What a measurement is. One workload is one curve, subgroup, generator and target; nothing survives between executions, so every table a method builds is inside its cost (*cold*). Beside the cold S every record carries the one-target *online window*, split for index calculus into the five exclusive phases `target_query`, `target_PDP`, `target_relation_check`, `target_descent`, `target_recovery_check`. Targets are either planted or *public*—hashed to the subgroup so that nobody knows the logarithm—and the runner verifies $[k]G = Q$ in its own process; `ecbench verify` checks it again from the files. Failed, exhausted and timed-out runs are rows, never dropped.

Identity, sealing and replay. Every object (curve, factor base, method parameters, spec, session, host environment class) is named by a hash of what it is (`ECBS1h...` for sessions, `ECBND1h...` for bounds). Each execution appends one sealed record; `ecbench verify` replays measured records and emits an audit receipt whose SHA-256 is the replay certificate. Operation counts are deterministic, so a replay must match to the last digit; a mismatch is preserved as a failed audit (§4.5).

Isolation. A run earns L0 (recorded under a host capsule), L1 (pinned, affinity read back, no foreign SMT sibling, CPU/wall ≤ 1.05), L2 (quiet: zero steal, foreign busy time $\leq 1\%$, run-queue delay $\leq 0.5\%$, ≤ 50 preemptions/s) or L3 (isolated cores, fixed governor, bare metal). Wall time is admitted only at the spec's required level (default L2) with at least five admitted pairs and an A/A interval; otherwise it is printed as *descriptive*. Operation counts need no isolation. macOS hosts earn L0; a GitHub-hosted runner earned L1 on 72/72 runs and matched a macOS session's operation counts to the last digit. *isolab* prepares lab hosts for L3 and cross-host replay.

Statistics. The comparable statistic is the mean S (the floor is an expectation). Comparisons are paired by (workload, round) with shared seeds, reported as a ratio of totals $\sum S_B / \sum S_A$, never a mean of ratios, with a two-stage cluster bootstrap that resamples workloads, then pairs. An operations A/A is exactly 1 by construction and checks determinism. A claim about scaling fits exponents over at least four sizes.

Bounds and the frontier. `ecbench` also records sealed *bounds*—constant and exponent with intervals, per phase—and a challenge protocol whose verdict says whether a candidate advances, trades, matches or regresses an incumbent and at which level (exponent, constant, primitive). The incumbent for cold one-target Koblitz operations is `rho.signed_frobenius_strong` (bound `ECBND1hcf69c27b4978`, fit version 2): $1.089\times$ the floor $[0.927, 1.261]$, fitted exponent $\alpha = 0.451$ $[0.399, 0.503]$, 144 runs at six sizes (`research/ecbench_frontier_20261013/`).

Definition 4.1 (comparability)**.** A performance point is keyed by the tuple

$$D = \big(\text{curve and subgroup},\ \text{target law},\ \text{cold} \mid \text{online} \mid \text{stage},\ \text{unit and unpriced work},\ \text{set-up amortisation } q,\ \text{memory envelope},\ \text{correctness/replay/isolation class}\big).$$

Two points are ratio-comparable only when every component of D except the method matches; a method is on the frontier of a domain when no admitted point is no worse on every axis and strictly better on one.

This rules out the four collapses we saw most often: online wall as cold cost; word XORs, probes, field operations and GAE as interchangeable; a lower bound as a complete cost; an L0 wall ratio as an admitted speed-up.

4.3 The pricing band

Almost no cost in a record is complete, and the rho baselines are no exception. Rho, BSGS and the kangaroo leave canonicalisations, partition hashes and table inserts and lookups unpriced. An index-calculus arm is complete only on curves that have a pinned timing calibration (`docs/ic/calibration.json`), which converts lookups, row operations, Artin–Schreier solves, Frobenius maps and solver word XORs to GAE at measured ratios of roughly 10^{-3} to 10^{-1}. A "complete" index calculus set against a lower-bound rho is therefore lopsided, and several earlier comparisons in this repository were of that kind.

Since 15 October every cross-method comparison uses a *pricing band* (repository rule, `AGENTS.md` §5a; native implementation `ecbench band`). For a record with charged cost G, calibration-priced native cost C over n_C operations, and U operations behind its unpriced counters,

$$S(w) = \frac{G - C + w\,(U + n_C)}{\sqrt{r}}, \qquad w \in \{0,\ 0.1,\ 1\} \text{ GAE per native operation},$$

applied alike to every arm, references included. Two arms are ranked only if their order is the same at every weight; otherwise the pair is reported as unresolved. An arm that names an unpriced counter without counting it is a lower bound and cannot be ranked. The band brackets the pinned calibrations from below and above, so a ranking that survives it does not depend on the host the calibration came from.

4.4 Claim hygiene and change classes

Every study directory states a protocol before it runs (frozen inputs are committed first), keeps failures, and closes with a decision in one of the four classes of [3]. A controlled-experiment rule requires a negative control from the right null population, a positive control that proves the detector can see the effect, and the same instrument on every arm before any comparative statement. Blockers are recorded as named problems (`problems/`) with exit criteria rather than silently dropped. Measurement code is native (Rust); Python only lays out recorded values.

Generic baselines. The rho, BSGS and Pohlig–Hellman references were engineered alongside index calculus, which is why each round's distance is quoted against the strongest reference at the time. On Koblitz curves the strong signed-Frobenius rho fell from 1,196 to 679.7 instructions per step over four rounds with no established native-time gain (§36.4). On hyperelliptic Jacobians a negation fold and hash mixing each fell short of their registered gains, and full-width branch coefficients showed that the earlier 8-bit reference walk had been slow at large N (`benchmarks/20261018-hec-rho-negation/`, `benchmarks/20261019-hec-rho-branch-width/`). Generic DLP fixes cut Pohlig–Hellman steps 26.6× and BSGS time 1.3–2.0×, and the candidates run at 0.020–0.167 of PARI's `elllog` time, in shared-host wall time the validation sweep left unvalidated (`research/generic_dlp_memory_parallelism_20261017/`).

4.5 Corrections the accounting forced

Building the harness and the ledger exposed errors in earlier claims, including our own. We list them because they change how earlier numbers must be read. Corrections that moved leaderboard rows (word-XOR pricing, cross-ISA drift, lopsided pricing, rule v8, the volcano and quartic withdrawals) are told in full in Chapter 41, and items 5, 6, 8 and 10 below only summarise them; the remaining corrections are in Appendix B.

1. *Results below rho that reversed against a stronger rho* (September; §15.3). A tournament candidate at 0.5347 of rho's instructions lost all five cells once rho used the same canonical form (`research/ic_triple_counted_20260923/`); an orbit-DLP ratio of 0.2403 became 1.923 against a strengthened rho and was withdrawn; a compact-orbit batch lead of 0.309 reversed against a batched-inversion rho.
2. *Online Koblitz ratios relabelled* (7–8 October, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`). Same-point online ratios of 124.85× ($n = 61$), 2,864.1× ($n = 71$), 1,189.7× ($n = 73$) and 8.2× ($n = 83$) had verified answers but no host-isolation receipt; they are *exploratory*. Only the later 64-target pinned panel at $n = 61$ is admitted (§15).
3. *ECC2K-130 extrapolation constants corrected* (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`): the structural model predicts $C = 0.25$ in probes/relation $\approx C\,r/(n^2K^2)$; the fitted C is 0.27 at $n = 61$ but 1.2–1.5 at $n = 71, 73$, an unexplained gap now stated rather than absorbed.
4. *"F4" that was Buchberger* (`research/ecbench_rootfinding_assessment_20261012/REPORT.md`): the NIST benchmark of the root-finding manuscript printed its baseline as F4 but called `groebner_f4::buchberger`; its 3,021–5,802× are ratios to Buchberger. The manuscript's claim that no implementation can use fewer than linear field operations per ideal is too broad once a reusable pair table is allowed.
5. *Unpriced word XORs* (`research/ecbench_rootfind_faster_20261014/`). Inherited F4's counted but uncharged word XORs made its S at Koblitz $n = 13$ look competitive with rho; priced, it is far above (Chapter 41, item 2).
6. *Cross-host counter drift* (`research/ecbench_frontier_20261013/audits/`): a replay-all audit of the root-finder session on x86_64 matched 120/168 records, the inherited-F4 rows differing in solver word-XOR counts. The 26 October addendum states the cause precisely and closes it (Chapter 41, item 4).
7. *A soundness bug* (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md` §2.1): a degree-bounded `f4_fp::solve` could certify solutions of an inconsistent system (over $\mathbb{F}_5$, $\{x^3 + 1, xy - 1, x + y\}$ returned two "solutions"). Fixed by evaluating every candidate on the input.
8. *F6-IC charged cost* (§10): a stage win in word XORs became a 64.6–1,353× loss in charged S once its batched geometry was counted—recorded as an accounting result, not an advance. With inherited F4's own word XORs priced the $m = 13$ loss is about 2.5× (§10).
9. *Lopsided pricing* (§4.3): comparisons of a calibration-priced index calculus with a lower-bound rho. Priced on both sides, the compact-orbit arm at Koblitz $n = 13$ is unresolved inside the band rather than a win for either side (Chapter 41, item 2).
10. *Wall-time figures recomputed* (validation sweep). Recomputing from the committed per-target records gives 2,321× for the $\ell = 12$ binary MITM oracle (quoted 2,187×), 98.1× for its $\ell = 8$ run 3 (quoted 100.7×), and 6.08× and 6.14 s for the quartic solve at $d = 6$ (quoted 6.4× and 5.9 s); all are shared-host wall times left unvalidated, and the counted evidence behind them stands (`research/validation_sweep_20261019/B-benchmarks/VERDICT_TABLE.md`).
11. *An unmatched rho reference* (21–24 October). Matching the Koblitz rule's rho (rule v4, then rule v8) raised the cold distance; the same week withdrew a volcano-crater claim, whose matched reference puts IC at 84.9× rho at $w = 0$ on the third volcano's $j = 0$ crater, and the plane-quartic S values

of the degree-3 reduction that merges distinct classes (Chapter 41, items 3 and 6). Canonical Pic^0 keys later separated those classes, and a re-run found that the earlier toy logarithms came from enumerating the subgroup (§26).

Part II

Theory

5 The summation cover: what is fixed for every curve

Every index calculus on an elliptic curve asks one question many times: does a target R decompose as a sum of $m-1$ factor-base points? Semaev's summation polynomial answers it algebraically, and every design lever of the programme acted on this question. This chapter collects three facts about it that hold for *every* curve. They closed, without an experiment, the levers that rely on curve-dependent monodromy, on fibre censuses, or on raising mean yield at matched base size. They also mark what stays free (§5.6).

The statements and proofs are those of `crypto-autoresearcher:papers/semaev-conservation-specialization/` (15 pages, anonymized draft, verification script `verify.py`, seed 20260807, which replays byte for byte), and of the internal finding `KN-FIND-bdcd60`. An integrity audit (`crypto-autoresearcher:research/sheriff/SHF-20261023-7b9227/`) upheld the mathematics within its scope. It blindly re-derived the group orders $2, 24, 192, 1920, 23040$ and checked 2.1 million admissible fibres with zero violations. It also rated as overclaimed a statement that three *independent* derivations agree: they came from one model family. **Novelty is not adjudicated** (§5.7).

5.1 Setting

Let K be a field, E/K an elliptic curve with inversion $\iota = [-1]$, and $\pi : E \to E/\langle\iota\rangle \cong \mathbb{P}^1$ the quotient (the x-coordinate in odd characteristic). The m-th summation polynomial $S_m \in K[X_1, \dots, X_m]$ is the reduced defining polynomial of the image of $\{(Q_1, \dots, Q_m) : \sum Q_i = O\}$ under π^m. Equivalently, $S_m(\xi_1, \dots, \xi_m) = 0$ if and only if there are points Q_i with $\pi(Q_i) = \xi_i$ summing to O [35]. For $m \geq 3$ it is symmetric, absolutely irreducible, and of degree 2^{m-2} in each variable. It satisfies the resultant recursion $S_m = \mathrm{Res}_X(S_{m-1}(\dots, X), S_3(X_{m-1}, X_m, X))$.

Fix $m \geq 3$, put $n = m-1$, and let $t_1, \dots, t_n$ be independent transcendentals. Write $L = K(t_1, \dots, t_n)$ for the function field of $(\mathbb{P}^1)^n$ and $M = K(E^n)$ for that of E^n. The group $\Gamma = \langle\iota\rangle^n \cong (\mathbb{Z}/2)^n$ acts on E^n coordinatewise; $\Delta \subset \Gamma$ is its diagonal. M/L is Galois with group Γ. With P_i the generic points ($\pi(P_i) = t_i$), the roots of $S_m(t_1, \dots, t_n, T)$ in T are the abscissae of the signed sums $\theta_\varepsilon = \pi(\varepsilon_1 P_1 + \cdots + \varepsilon_n P_n)$, $\varepsilon \in \{\pm 1\}^n$, with $\theta_\varepsilon = \theta_{-\varepsilon}$.

5.2 The Galois group

Theorem 5.1 (Monodromy of the summation cover)**.** *For every field K, every elliptic curve E/K and every $m \geq 3$:*

1. *$S_m(t_1, \dots, t_n, T)$ is separable of degree 2^{m-2} over L, with root set $\{\theta_\varepsilon\}$;*
2. *its splitting field is $N = M^\Delta = K(E^n/\Delta)$;*
3. *N/L is Galois with group $\Gamma/\Delta \cong (\mathbb{Z}/2)^{m-2}$, acting simply transitively on the roots;*
4. *K is algebraically closed in N, so arithmetic and geometric monodromy coincide, $\mathrm{Mon}^{\mathrm{arith}} = \mathrm{Mon}^{\mathrm{geom}} \cong (\mathbb{Z}/2)^{m-2}$.*

None of the four statements depends on E, on K or on its characteristic.

Proof outline. A root-identification lemma shows that the roots are among the θ_ε. A generic-separability lemma shows that the 2^{m-2} classes $\{\pm\varepsilon\}$ give distinct values. Γ permutes the roots by $\sigma \cdot \theta_\varepsilon = \theta_{\sigma\varepsilon}$, and the kernel of this action is exactly Δ. So Γ/Δ acts faithfully and regularly on 2^{m-2} roots. The stabiliser of one root is Δ, which gives $N = M^\Delta$. Finally, E^n/Δ is geometrically integral, so K is algebraically closed in its function field. □

For a symmetric (unordered) factor base the relevant group is the wreath extension $(\mathbb{Z}/2)^{m-2} \rtimes S_{m-1}$, of order $2^{m-2}(m-1)!$ for $m \geq 4$. For $m \geq 4$ it is primitive if and only if $m-1$ is odd (`KN-FIND-bdcd60`; an integrity audit upheld the finding with this scope, `crypto-autoresearcher:research/sheriff/SHF-20261023-7b9227/REPORT.md`). Statement (4) is the operative one: a cover whose arithmetic monodromy exceeds its geometric monodromy can split differently over different base fields, and hence on different curves. Here no such room exists.

5.3 An exact factorisation law over a finite field

Over $K = \mathbb{F}_q$, define for $a \in \mathbb{P}^1(\mathbb{F}_q)$ the *splitting invariant*:
- $c(a) = 0$ if both points above a are rational;
- $c(a) = 1$ if neither is;
- $c(a) = *$ at a branch point.

For odd q and $E : y^2 = f(x)$ this is the quadratic character of $f(a)$.

Theorem 5.2 (Factorisation law)**.** *Let $a = (a_1, \dots, a_n) \in \mathbb{F}_q^n$ lie outside the degenerate locus, which has $O_m(q^{n-1})$ points. Then $S_m(a_1, \dots, a_n, T)$ is squarefree of degree 2^{m-2}, and:*
1. if $c(a_1) = \cdots = c(a_n)$, it splits into 2^{m-2} distinct linear factors, with roots $x(\pm P_1 \pm \cdots \pm P_n)$;
2. otherwise it is a product of 2^{m-3} distinct irreducible quadratics and has no root in $\mathbb{F}_q$.
No irreducible factor of degree ≥ 3 ever occurs.

Proof. On the étale locus, the factorisation type is the cycle type of Frobenius in Γ/Δ (Dedekind's theorem for specialisations of a Galois cover [15]). The Frobenius element is $((-1)^{c(a_i)})_i$. It is trivial in Γ/Δ exactly when every $c(a_i)$ agrees. Otherwise it is a non-trivial element of order 2 of a regular group, and so has cycle type $2^{2^{m-3}}$. □

The density of totally split fibres is $2^{-(m-2)} + O(m^2/(2^{m-1}q))$. The first-order Hasse terms $\pm t/q$ cancel, so the statistic carries no first-order information about the curve.

5.4 Specialisation to the factor-base locus

Theorem 5.3 (Complete splitting on the factor-base locus)**.** *If each $a_i = x(P_i)$ with $P_i \in E(\mathbb{F}_q)$ and a is non-degenerate, then $S_m(a_1, \dots, a_n, T)$ splits into 2^{m-2} distinct linear factors over $\mathbb{F}_q$, with roots $\{x(\varepsilon_1 P_1 + \cdots + \varepsilon_n P_n)\}$.*

This is Theorem 5.2(1) with every $c(a_i) = 0$. It is the statement behind the root-finding oracle of Chapter 9: given $m - 1$ base points, the admissible last coordinates are the signed sums, computable in $O(2^{m-2})$ group operations. It is *not* an algorithm for relation search. Relation search fixes the target and asks for the base points. The theorem says nothing about how many tuples must be tried, nor about the algebraic difficulty of the system posed in that direction. Its value is negative:

Corollary 5.4 (Census obstruction)**.** *Every statistic of the form "factorisation type, cycle type or Frobenius class of the fibre of S_m" is constant on the factor-base locus, for every curve and every m. A census of generic fibres measures a quantity whose value where relation search operates is known in advance and is the same for all curves.*

The programme ran such censuses before this corollary was proved. The census-as-relation-rate probe was closed by ledger decision `DEC-20261017-435b01` (`reject_scoped`, replicated).

5.5 Conservation of decomposition yield

Proposition 5.5 (Conservation)**.** *Let G be a finite abelian group of order N and $\mathcal{D} \subseteq G \setminus \{0\}$ a set of size B. The number of size-m multisets from $\mathcal{D}$, summed over all targets, is $\binom{B+m-1}{m}$. The mean number of decompositions per target is therefore $\binom{B+m-1}{m}/N$, for every $\mathcal{D}$ of size B.*

The proof is a double count, and no novelty is claimed for it. Its consequences shaped the programme:
- at matched base size, no factor-base geometry can raise the mean yield;
- only the distribution of decompositions is free, and coverage (the fraction of targets with at least one decomposition) is at most $\min(1, \mu)$;
- additive structure that repeats sums *lowers* coverage;
- typing the summands, each from its own sub-base, is a fixed penalty: $\prod B_i \leq (B/m)^m < \binom{B+m-1}{m}$.

Gaudry and Diem's $1/m!$ decomposition probability over extension fields *is* this mean. The extension-field advantage lives in the solve, not in the yield.

5.6 Which levers are fixed and which are free

Together the results above divide the design space as follows:

- *Fixed for every curve:*
  - the monodromy group of the summation cover (Theorem 5.1);
  - the fibre statistics on the factor-base locus (Corollary 5.4);
  - the mean yield at fixed base size (Proposition 5.5).
- *Free:*
  - the coverage distribution (bounded by the mean);
  - the cost of deciding membership (the oracle, Part III);
  - the algebraic difficulty of the decomposition system when it is posed as a polynomial system (Chapter 6);
  - the linear algebra;
  - where the problem lives (transfers, Chapter 26).

Every measured improvement in Part III acts on a free item. Chapter 26 shows that transfers by isogeny conserve the counts as well.

5.7 Prior work and the limits of the novelty assessment

Absolute irreducibility of S_m is classical [20, 35]. Corollary 5.4's ingredient, that the roots are the signed sums, is the defining property of S_m. The action of 2-torsion and of sign changes on summation systems is used by Faugère, Gaudry, Huot and Renault [14] and by Galbraith and Gebregiyorgis [17]. Theorem 5.1's group may be folklore: it is the deck group of $E^n \to (E/\pm)^n$ modulo the diagonal. Theorem 5.2 is in essence a 2-descent statement. The draft could not reach ePrint during its literature search, and says so. Corollary 5.4 and the exact dichotomy of Theorem 5.2 are the parts most likely to be new. Two items must be settled before any of this is submitted:

- an expert check of the separability lemma in characteristic 2, where $E[2]$ is non-reduced;
- a full literature search.

6 Fall degrees of point-decomposition systems

Every claim that elliptic-curve index calculus can be subexponential rests on the cost of solving the point-decomposition system. In practice that means its *solving degree*: the largest degree a Gröbner-basis computation reaches. Petit and Quisquater [32] conjectured that the solving degree stays close to the *first fall degree*, which is small and easy to compute. Semaev [36] assumed it is at most 4 for his chained systems. Kosters and Yeo [28], Huang, Kosters and Yeo [23] and Kousidis and Wiemers [29] gave reasons and experiments to doubt this. All three name rigorous bounds on the *last* fall degree, which equals the solving degree for the systems that matter, as the open problem.

This chapter collects what the programme proved and certified about these degrees, in three presentations of the decomposition system:

- binary Weil descent;
- explicit membership over any field;
- prime-field Kummer towers.

The message is the same in all three. The first fall degree is a constant of the presentation, whereas the last fall degree is not. Where the last fall degree rises above 4, the chapter gives a certificate that a separate program checks without trusting the producer.

6.1 Definitions

Let $F = \{f_1, \dots, f_M\} \subset R = k[x_1, \dots, x_N]$, graded by total degree.

- *Closure.* $V_{F,d}$ is the smallest k-subspace of $R_{\le d}$ that contains $F \cap R_{\le d}$ and is closed under $g \mapsto hg$ whenever $\deg(hg) \le d$ [5, 23].
- *Last fall degree.* $d_F = \min\{d : f \in V_{F,\max(d,\deg f)}\ \forall f \in (F)\}$.
- *First fall degree.* We use the Caminata–Gorla convention [5]: $d_{\mathrm{ff}}(F)$ is the least d at which the top-form syzygies modulo the trivial (Koszul and field) syzygies are non-zero.
- *Boolean closure.* Over $\mathbb{F}_2$ with field equations it is convenient to work in $B = \mathbb{F}_2[x]/(x_i^2 + x_i)$. There $W_d(F)$ is the same closure computed on multilinear representatives, and $M_d(F)$ is the Macaulay space spanned by the products mf with $\deg m + \deg f \le d$.

Remark 6.1 (Definitions matter). Three readings of "first fall degree" are in use. They are:

- Petit–Quisquater: a combination $\sum g_i f_i \neq 0$ whose degree drops below $\max \deg(g_i f_i)$;
- Caminata–Gorla: non-trivial syzygies of the top forms;
- operational: the least c with $V_{F,c} \cap R_{\le c-1} \neq V_{F,c-1}$.

They can disagree. Two documented cases follow.

- Nagao's ePrint 2015/984 [31] defines a "fake" first fall degree d'_F computed modulo the field equations, and states $d_F \le d'_F$ as its Lemma 4. Under the equality reading of its condition (1) the lemma fails over $\mathbb{F}_2$:
  - $\{X^2Y,\ XY + X\}$ has $d'_F = 2 < d_F = 3$, and so does $\{X_1X_2 + X_3,\ X_1X_3 + X_2\}$;
  - 94 of 400 random multilinear pairs in three variables violate it.

  The paper's own applications check the unreduced degrees instance by instance, so they are unaffected (`crypto-autoresearcher:papers/nagao-lemma4-note/`, `KN-FIND-936151`; two implementations, 15 tests).
- A finite descent of S_3 over $\mathbb{F}_{2^4}$ ($n = n' = 4$, seed 32201, $x_R = 1$, basis $[9, 3, 2, 12]$) has operational first fall 3. Its closure dimensions are $\dim V_D = 1, 11, 53$ and $\dim(V_D \cap R_{\le D-1}) = 0, 1, 12$ for $D = 1, 2, 3$. Lean finite certificates check the ranks, the semantics on all 256 assignments, and an explicit witness. The literal Petit–Quisquater reading of the same system gives 2, through a field-equation identity.
  - This refutes an internal claim of a uniform degree-2 first fall, since withdrawn.
  - It contradicts no published theorem: Kosters and Yeo say "usually 2" [28].
  - Source: `crypto-autoresearcher:docs/reports/2026-10-22-semaev-first-fall-replay/`.

Every statement below names its convention.

6.2 General closure facts

The following hold for any Boolean system F, with $\tilde{F}$ its lift to R together with the field equations. They are proved in `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/theorem-dossier.md`, §1.

Proposition 6.2 (Bridges). *1.* $\pi(V_{\tilde{F},d}) = W_d(F)$, *where* $\pi : R \to B$ *is the quotient map. In particular* $1 \in V_{\tilde{F},d}$ *if and only if* $1 \in W_d(F)$.
2. If F *has no Boolean zero, then* $d_{\tilde{F}} = d_{\mathrm{ref}}(F) := \min\{d : 1 \in W_d(F)\}$. *This also equals the solving degree for every degree-compatible order (Caminata–Gorla, reduced basis* $\{1\}$*).*
3. If F *has no Boolean zero, then* every *F4 run on* $\tilde{F}$*, under any selection strategy, produces a polynomial of degree* $\geq d_{\mathrm{ref}}(F)$.
4. If F *has Boolean zero set* S *and* $\operatorname{codim} W_d(F) > |S|$, *then* $d_{\tilde{F}} \geq d+1$, *and every F4 run reaches degree* $d+1$.

Lemma 6.3 (Branching). *If both restrictions* $F|_{x=0}$ *and* $F|_{x=1}$ *satisfy* $1 \in W_d$, *then* $1 \in W_{d+1}(F)$.

Proposition 6.4 (Dual certificate). *Let* $\pi : B_{\leq d} \to Z$ *be linear and onto, with kernel* U. *Suppose:*
(i) $\pi(f) = 0$ *for every* $f \in F$ *of degree* $\leq d$;
(ii) $\pi(xu) = 0$ *for every variable* x *and every* $u \in U$ *of degree* $\leq d-1$.
Then $W_d(F) \subseteq U$, *so* $\operatorname{codim} W_d(F) \geq \dim Z$. *If moreover* $\pi(1) \neq 0$, *then* $1 \notin W_d(F)$. *Conversely, the quotient map* $B_{\leq d} \to B_{\leq d}/W_d(F)$ *is such a certificate whenever* $1 \notin W_d(F)$.

Proposition 6.4 is what makes the lower bounds below checkable. A certificate is a matrix. Checking it needs the curve parameters, an independent descent of the system, and three linear-algebra tests: (i), (ii) and $\pi(1) \neq 0$. Nothing about how the certificate was found has to be trusted. With Proposition 6.2(3), one accepted certificate for $1 \notin W_4(F)$ proves that *no* F4 run on that instance stays at degree 4. This is a statement about the system, not about a solver run.

6.3 Binary Weil descent of S_3

Setting.
- The field is $K = \mathbb{F}_{2^n}$ with trace Tr, and $K_0 = \ker \mathrm{Tr}$.
- The curve is $E : y^2 + xy = x^3 + a_2x^2 + a_6$ with $a_6 \neq 0$.
- The target abscissa is $z \in K^\times$.
- $V \subset K$ is an $\mathbb{F}_2$-subspace of dimension k; Semaev's diagonal is $k = \lceil n/2 \rceil$.
- $X_1 = \sum a_j v_j$ and $X_2 = \sum b_j v_j$ are linear in the $N = 2k$ Boolean variables.
- The system F is the Weil descent of $S = S_3(X_1, X_2, z)$: the coordinates $\mathrm{Tr}(\gamma_i S)$. This is Semaev's system (5) for $m = t = 2$.

Fix ϑ with $\mathrm{Tr}\,\vartheta = 1$, and an Artin–Schreier section G (the half-trace for odd n). Put

$$w = \frac{X_1X_2}{z} + G(s^2) + G(a_6/z^2), \qquad s = X_1 + X_2, \qquad \ell_0 = \mathrm{Tr}(s) + \mathrm{Tr}(a_6/z^2).$$

Theorem 6.5 (Exact reformulation). *For every* n, V *and* z:
1. $S = z^2(w^2 + w + \ell_0\vartheta)$;
2. $\mathrm{span}_{\mathbb{F}_2} F = \{\mathrm{Tr}(\delta w) : \delta \in K_0\} + \mathbb{F}_2\ell_0$;
3. in particular, the affine-linear form $\ell_0 = \mathrm{Tr}(S/z^2)$ *always lies in the span of the equations.*

Proof. (1) Squaring is a ring endomorphism, so $S = z^2(u^2 + u + s^2 + a_6/z^2)$ with $u = X_1X_2/z$, and $h = G(s^2 + a_6/z^2)$ satisfies $h^2 + h = s^2 + a_6/z^2 + \mathrm{Tr}(s^2 + a_6/z^2)\vartheta$, with $\mathrm{Tr}(s^2) = \mathrm{Tr}(s)$. (2) Put $\gamma' = \gamma z^2$. Then $\mathrm{Tr}(\gamma S) = \mathrm{Tr}((\gamma'^{1/2} + \gamma')w) + \mathrm{Tr}(\gamma'\vartheta)\ell_0$, and $\gamma' \mapsto (\gamma'^{1/2} + \gamma', \mathrm{Tr}(\gamma'\vartheta))$ is a bijection $K \to K_0 \times \mathbb{F}_2$. (3) Take $\gamma' = 1$. □

Part (3) is Kosters and Yeo's Corollary 4.11 [28], seen through a different coordinate. The reformulation makes the falls explicit.

Corollary 6.6 (First fall degree 2, for every n). *Suppose every descended equation has degree exactly 2; this held on every instance computed. Then:*
- $d_{\mathrm{ff}}(F) = 2$ *in the Caminata–Gorla sense, for every* $n \geq 3$ *and every* V;
- *if moreover* $\ell_0 \neq 0$, *the Petit–Quisquater first fall degree is also 2.*

The same holds for Semaev's chained system (5) for every $t \geq 2$*: the last equation* $S_3(y, x_t, R_X)$ *always contributes the linear form* $\mathrm{Tr}(Y + X_t) + \mathrm{Tr}(a_6/R_X^2)$.

Semaev states the value 4 for the first fall degree of his systems [36, §4.5]. His argument concerns only the cubic equations of the chain. Under the definition he quotes, the trace relation makes the value 2.

Lemma 6.7 (Explicit degree-3 falls)**.** *Let* $V^\perp$ *be the trace-orthogonal of* V*, and* $\omega = \mathrm{Tr}(\mu w)$ *for any* μ *with* $\mathrm{Tr}\,\mu = 1$.

- *For* $\lambda \in V^\perp$ *and* $i \in \{1, 2\}$, $\mathrm{Tr}(\lambda X_i w)$ *lies in* $M_3(F)$ *and has degree* ≤ 2.
- $\mathrm{Tr}(sw) + \mathrm{Tr}(a_6/z^2)\,\omega$ *lies in* $W_3(F)$ *and has degree* ≤ 2.

These are $2(n - k) + 1$ *explicit degree-3 falls, independent of* n *in form. On random subspaces they span the whole degree-*≤ 2 *part of* W_3*, of dimension exactly* $3n$*; this was checked on six instances with* $n \in \{23, 25, 27\}$*. Polynomial subspaces* $\langle 1, t, \ldots, t^{k-1} \rangle$ *have* $2n - 1$ *further falls, which is why they resist longer.*

Theorem 6.8 (Upper bound)**.** *If* F *has no Boolean zero, then* $1 \in M_{k+2}(F)$*. So* $d_{\tilde{F}} \leq k + 2$*, which is* $\lceil n/2 \rceil + 2$ *on the diagonal. For Semaev's chained system (5) with* t *summands the bound is* $d_{\tilde{F}} \leq (t - 1)k + t$*: each auxiliary variable adds 1, not* n.

Proof for $t = 2$. Substitute $c_j = a_j + b_j$. Then $S = X_1^4 + X_1^2 s^2 + z^2 s^2 + z X_1^2 + z X_1 s + a_6$ is at most linear in a and in c, so each $f_i = \sum_j a_j L_{ij}(c) + M_i(c)$. For each $\sigma \in \mathbb{F}_2^k$ the affine system $f_i(a, \sigma)$ in a is inconsistent, so some combination $\sum_i \lambda_i^\sigma f_i(a, \sigma) = 1$. With the indicators $e_\sigma = \prod_j (c_j + \sigma_j + 1)$,

$$1 = \sum_\sigma \sum_i \lambda_i^\sigma e_\sigma f_i \in M_{k+2}(F). \qquad \square$$

Proposition 6.9 (Lower bound $d_{\tilde{F}} \geq 3$)**.** *Under two explicit genericity conditions on* V *and* z*, which a script checks and which held on every random-subspace instance, together with* $\ell_0 \neq 1$*:*

- $W_2(F) = \mathrm{span}\, F + \ell_0 B_{\leq 1}$*, and its degree-*$\leq 1$ *part is* $\mathbb{F}_2 \ell_0$*;*
- *so an unsatisfiable* F *has* $d_{\tilde{F}} \geq 3 = d_{\mathrm{ff}} + 1$.

The bound is attained at $n = 17, 19$.

Computer-verified instances.

- *Implementations.*
  - Implementation A: a C closure over M4RI, `code/lfdclose2.c`.
  - Implementation B: independent Python sharing no code with A; `verify/b_sweep.py` and `verify/indep_verify.py`.
  - Certificate checker: `verify/cert_check.py`, which rebuilds F from the curve parameters by its own Möbius interpolation of S_3 values.
  - Implementation C: a native Rust closure in this repository, `examples/fd_m2_degrees.rs`, sharing no code with A or B. It builds the descent in the repository's Boolean-ring library and runs the full MutantXL closure ($= W_D$) and Boolean F4. In a pilot on a 4-core development host, which is not a measurement and has not been independently checked, it reproduced the recorded $d_{\tilde{F}}$ of eight instances (R11a, R13a, R15a, R19a, P17a, P21a, P21c, U33a) and $1 \notin W_4$ on T33a with $\mathrm{rank}\, W_4 = 48{,}068$, as recorded. Planted solvable controls were never refuted up to $D = 5$. The calibration stage of the frozen protocols below repeats this check before any new row is read.
- *Instances.*
  - Every instance on the diagonal $k = \lceil n/2 \rceil$, other than the two satisfiable ones, was shown unsatisfiable by exhaustive count, and the count was repeated by B.
  - B reproduced all 61 instances with $n \leq 29$: the same solution count, the same dimensions of $W_{d-1} \cap B_{\leq e}$ for every e, and $1 \in W_d$.

Figure 6.1 shows the result: as n grows, the first fall degree stays at 2 while the last fall degree climbs.

Theorem 6.10 (Exact last fall degrees; computer-verified on instances)**.** *On the diagonal* $k = \lceil n/2 \rceil$*:*

- $d_{\tilde{F}} = 3$ *on all six random-subspace instances at* $n = 11, 13$ *and on all ten polynomial-subspace instances at* $n = 17, 19$*;*
- $d_{\tilde{F}} = 4$ *on all 29 random-subspace instances with* $15 \leq n \leq 33$*, and on all 25 polynomial-subspace instances with* $23 \leq n \leq 35$ *or* $n = 41, 43$*;*
- $d_{\tilde{F}} = 4$ *on Semaev's own cell (*$n = 40$*,* $k = 20$*, uniform* z*), matching his reported* $d_{F4} = 4$*;*

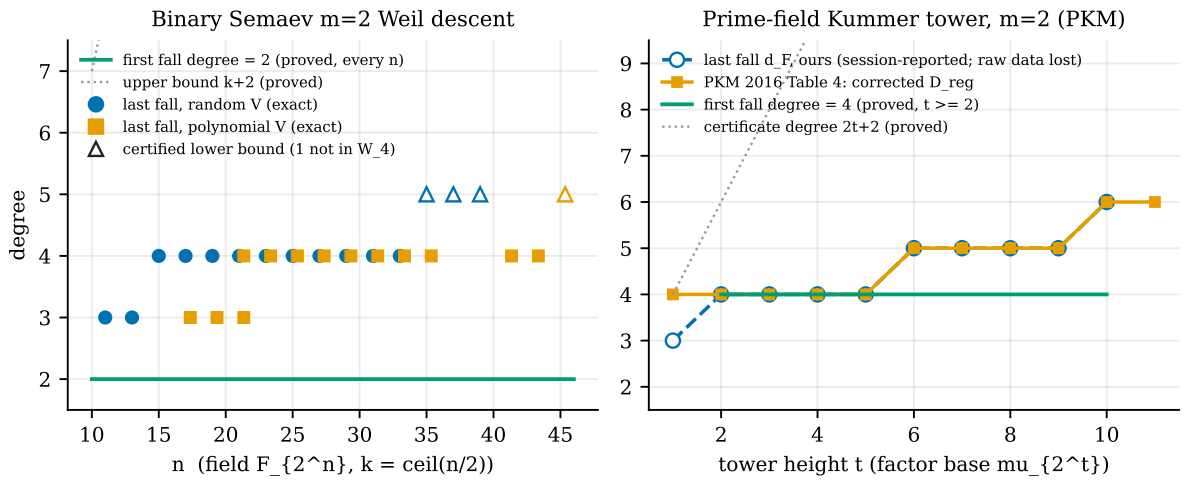


Figure 6.1: *Left:* binary Semaev $m = 2$ descent. The first fall degree is 2 (Corollary 6.6, proved). Filled marks are exact last fall degrees per unsatisfiable instance. Open triangles are certified lower bounds $d_{\tilde{F}} \geq 5$ (Theorem 6.11). The dotted line is the proved upper bound $k + 2$. *Right:* prime-field Kummer tower, $m = 2$ (§6.5). The session-reported last fall degree lies on the corrected degree of regularity of PKM 2016, Table 4. The first fall degree is 4 (proved). Values and sources: `figures/data-new.json`.

- $d_{\tilde{F}} = 5$ *exactly on instance R35a* ($n = 35$, $k = 18$, *random* V): $1 \notin W_4$, *and both restrictions* $a_1 = 0, 1$ *are refuted at* W_4 *(Lemma 6.3).*

Table 6.1 gives every instance behind Theorems 6.10 and 6.11, one row per subspace family and n, in the layout of Kousidis and Wiemers' Table 1.

Theorem 6.11 (Certified $d_{\tilde{F}} \geq 5$)**.** *For each instance below, a certificate (Proposition 6.4) for* $1 \notin W_4(F)$ *was accepted by the independent checker. Hence* $d_{\tilde{F}} \geq 5$*, and every F4 run reaches degree* ≥ 5*.*

instance	n	k	*subspace*	*certified* codim $W_4 \geq$
T33a, T33b	*33*	*17*	*random* $\subset K_0$	*4,888*
R35a, R35b, U35a, U35b	*35*	*18*	*random (U: uniform z)*	*5,287*
Q35a, Q35b (satisfiable)	*35*	*18*	*random*	*5,287*
R37a, R37b	*37*	*19*	*random*	*10,066*
P45a	*45*	*23*	*Semaev's default* $\langle 1, t, \ldots, t^{22} \rangle$	*12,742*
C19a, C19b (chain $m = t = 3$*)*	*19*	*7*	*random*	*9,253, 6,817*

Two further instances, R39a and R39b ($n = 39$)*, and C21a and C21b (the chain at* $n = 21$*) have* $1 \notin W_4$ *by implementation A alone.*

For Q35a and Q35b, which have 6 and 2 Boolean solutions, the bound follows from Proposition 6.2(4), since $5{,}287 \gg |S|$. The P45a closure stalls after three rounds at $\dim W_4 \cap B_{\leq d} = (0, 1, 224, 8950, 166705)$, out of 179,447 monomials. Its quadratic part, $224 = 5n - 1$, is exactly the polynomial-subspace count of Lemma 6.7: at degree 4 the closure finds no new falls.

Corollary 6.12 (Semaev's Assumption 1 fails as stated)**.** *Semaev's Assumption 1 [36, §4.5] asserts* $d_{F4} \leq 4$ *for Boolean systems equivalent to his system (5), for* $q = 2^n$, $k = \lceil n/m \rceil$*, any subspace of dimension* k *and any* $2 \leq t \leq m$*. Theorem 6.11 contradicts it in three cells:*

- $(n, m, t, k) = (35, 2, 2, 18)$*, with random subspaces;*
- $(45, 2, 2, 23)$*, with his own default subspace;*
- $(19, 3, 3, 7)$*, inside the parameter range of his tables, where his polynomial subspace gives 4 and random subspaces do not.*

Implementation A alone also gives $1 \notin W_4$ *at* $(21, 3, 3, 7)$*, with no checked certificate.*

Table 6.2 lists the chain instances. With Semaev's polynomial subspace the closure refutes at degree 4 in every cell; with random subspaces it does not at $n = 19, 21$.

m	V	n	k	inst.	D_{ff}	$d_{\tilde{F}}$	cols at $D=4$	s at $D=4$	codim $W_4 \geq$
2	P	17	9	6	2	3			
		19	10	4	2	3			
		21	11	4	2	3×2, 4×2	9,109	7.3	
		23	12	4	2	4	12,951	7.3	
		25	13	4	2	4	17,902	11.1	
		27	14	4	2	4	24,158	20.3	
		29	15	3	2	4	31,931	40.9	
		31	16	3	2	4	41,449	51.9	
		33	17	3	2	4	52,956	100.1	
		35	18	2	2	4	66,712	160.9	
		41	21	1	2	4	124,314	1076.1	
		43	22	1	2	4	149,986	2001.3	
		45	23	1	2	≥ 5	179,447	1850.7	12,742
	S	40	20	2	2	4	102,091	537.9	
	R	11	6	3	2	3			
		13	7	3	2	3			
		15	8	3	2	4	2,517	1.0	
		17	9	3	2	4	4,048	1.3	
		19	10	3	2	4	6,196	2.1	
		21	11	3	2	4	9,109	3.2	
		23	12	3	2	4	12,951	4.1	
		25	13	3	2	4	17,902	6.6	
		27	14	3	2	4	24,158	15.4	
		29	15	3	2	4	31,931	24.0	
		31	16	3	2	4	41,449	45.2	
		33	17	2	2	4	52,956	91.9	
		35	18	2	2	≥ 5	66,712	121.0	5,287
		37	19	2	2	≥ 5	82,993	312.9	10,066
		39	20	2	2	≥ 5	102,091	445.9	
	U	33	17	2	2	4	52,956	95.4	
		35	18	2	2	≥ 5	66,712	126.4	5,287
	T	29	15	2	2	4	31,931	38.3	
		31	16	2	2	4	41,449	68.3	
		33	17	2	2	≥ 5	52,956	128.8	4,888

Table 6.1: Binary Weil descent of S_3 ($m = 2$, $k = \lceil n/2 \rceil$): the first fall degree $D_{\mathrm{ff}} = 2$ (Corollary 6.6, proved for every n) beside the last fall degree $d_{\tilde{F}}$, the least D with $1 \in W_D$, for each instance. Subspace families: P, Semaev's polynomial subspace $\langle 1, t, \ldots, t^{k-1} \rangle$; S, the same with uniform z (Semaev's own cell); R, random; U, random with uniform z; T, random inside $\ker \mathrm{Tr}$. "≥ 5" means $1 \notin W_4$; R35a is exactly 5 (Theorem 6.10). Columns: instances; columns of the degree-4 closure matrix; median closure wall time at $D = 4$ in seconds, over the instances run to that degree (host not recorded); certified codimension of W_4 where an independent certificate was checked. Source: `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/results/`, laid out by `figures/make_tab_fall_degrees.py` from `figures/data-falldegree-instances.json`.

instance	$m = t$	V	n	k	$d_{\tilde{F}}$	cols at $D = 4$	s at $D = 4$	codim $W_4 \geq$
D17a	3	P	17	6	≤ 4	59,536	294.6	
D17b	3	P	17	6	≤ 4	59,536	297.4	
D19a	3	P	19	7	≤ 4	102,091	1283.7	
D19b	3	P	19	7	≤ 4	102,091	1047.1	
D21a	3	P	21	7	≤ 4	124,314	1270.3	
D21b	3	P	21	7	≤ 4	124,314	1225.4	
C17a	3	R	17	6	≤ 4	59,536	281.3	
C17b	3	R	17	6	≤ 4	59,536	270.9	
C19a	3	R	19	7	≥ 5	102,091	1088.5	9,253
C19b	3	R	19	7	≥ 5	102,091	1184.4	6,817
C21a	3	R	21	7	≥ 5	124,314	1806.2	
C21b	3	R	21	7	≥ 5	124,314	1790.4	

Table 6.2: Chained system, $m = t = 3$, $k = \lceil n/3 \rceil$: closure at $D = 4$ per instance. "≤ 4" means $1 \in W_4$; "≥ 5" means $1 \notin W_4$. Certified codimensions as in Table 6.1; C21a and C21b have no checked certificate: $1 \notin W_4$ by implementation A alone. Source: `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/results/instances_t3.json`.

Kousidis–Wiemers, Table 1 ($m = 2$, F4)						this work, $d_{\tilde{F}}$	
n	n'	D_{ff}	D_{reg}	s	GB	P or S	R, U or T
34	17	2	4	188	1.2		
35	18	2	4	1,237	16.1	4	$\geq 5^{\text{R}}$; $\geq 5^{\text{U}}$
36	18	2	4	1,342	16.4		
37	19	2	5	2,542	29.2		$\geq 5^{\text{R}}$
38	19	2	5	2,815	25.2		
39	20	2	5	4,785	45.6		$\geq 5^{\text{R}}$
40	20	2	5	4,858	46.3	4^{S}	
41	21	2	5	7,930	65.3	4	
42	21	2	5	8,901	66.7		
43	22	2	5	16,816	95.5	4	
44	22	2	5	15,690	96.8		
45	23	2	5	38,352	140.0	≥ 5	
46	23	2	5	31,735	140.7		
47	24	2	5	103,200	207.7		
48	24	2	5	86,636	208.2		

Table 6.3: The $m = 2$ rows of Kousidis and Wiemers' Table 1 [29] (F4, averages over ten repetitions; $n' = \lceil n/2 \rceil$; time in seconds, memory in GB), transcribed from the published table, beside the last fall degree of our instances at the same n (Table 6.1). Superscripts give the subspace family. Their $m = 3, 4$ rows ($D_{\text{ff}} = D_{\text{reg}} = 7$ and 13, $n \leq 21$) concern S_4 and S_5 and have no counterpart here.

What is new, and what is not. That the degree of regularity reaches 5 has been *observed* before:

- Kosters and Yeo saw Magma step degrees of ≥ 5 at $n = 40$ and $n = 45$ [28];
- Kousidis and Wiemers saw F4 at step degree 5 for $n = 37$–48, $m = 2$, and wrote that a lower bound on the degree of regularity "can potentially answer an open question" [29].

Table 6.3 sets their $m = 2$ rows beside ours at the same n. The quantities differ: D_{reg} is the highest step degree of their F4 runs, while $d_{\tilde{F}}$ is the closure degree of the system, and no F4 run refutes below it. So their 5 at $n = 41, 43$ against our 4 on polynomial subspaces is not a contradiction; F4 overshoots the closure there. At $n = 35$ their $D_{\text{reg}} = 4$ agrees with our polynomial subspace and differs from every random subspace we drew, which are all ≥ 5: which subspace a table uses decides the degree it reports. At $n = 45$ both reach 5.

Those are properties of solver runs. Theorem 6.11 is a property of the systems, carried by a certificate any reader can check. With Corollary 6.6 it makes the separation between first and last fall degree rigorous on named instances. What remains open is the same as before: a proof that $d_{\tilde{F}} \to \infty$. Semaev's asymptotics survive if the regularity degree is $o(\sqrt{n/\ln n})$, so only a growing lower bound bears on the subexponential claim.

Re-verification. For this treatise, the T33a certificate was regenerated from the committed producer (M4RI 20200125, 4 min 40 s on 4 vCPUs) and checked with `cert_check.py`:

- the certificate is byte-identical to the recorded SHA-256 `cf9dcae8...22cf`;
- the checker reports "1 is not in $W_4(F)$, and codim $W_4(F) \geq 4888$" (38.8 s);
- the closure dimensions $(0, 0, 64, 2968, 48068)$ match the record.

Source: `research/paper_opportunities_20261029/REPORT.md`, §6.

6.4 Explicit membership over any field: the guarded chain

A second presentation encodes factor-base membership directly. Let S be a set of B abscissae of rational points, and $g(T) = \prod_{s \in S}(T - s)$. The two-$S_3$ chain with an inverse guard is

$$F_S = \{g(x),\ g(y),\ g(z),\ S_3(x, y, u),\ S_3(u, z, r),\ v(x - y) - 1\},$$

with $r = x(R)$. Degrees are ordinary total degrees with explicit inputs. The first fall degree is the homogeneous-top syzygy degree modulo Koszul syzygies. Source: `research/guarded_fall_degree_20261028/`, which holds written proofs, independent audits, a native exact replay and partial Lean.

Theorem 6.13 (Guarded chain)**.** *For unsatisfiable instances:*

1. *Short Weierstrass curves, $p > 3$, $B = 3$: $(d_{\mathrm{ff}}, d_F) = (4, 5)$, exactly, for every p.*
2. *Same presentation, any $B \geq 3$: $d_{\mathrm{ff}} = B + 1$ and $\max(5, B + 1) \leq d_F \leq 3B - 4$.*
3. *Ordinary binary curves $Y^2 + XY = X^3 + a_2X^2 + a_6$, $B = 3$:*
   - *$d_F = 4$ if and only if $g = T^3 + c$, which is possible only for even n;*
   - *$d_F = 5$ otherwise.*

   So for odd n, which covers every SEC 2 binary size, every unsatisfiable instance has $(4, 5)$.
4. *Binary curves, $B \geq 4$: $B + 1 \leq d_F \leq 3B - 4$.*

Proof sketch of (1). Upper bound. The divided difference $Q = (g(x) - g(y))/(x - y)$ enters V_4 through the guard. The pair is then replaced by the missing root $w = s_1 - x - y$. The two quadratics in u are normalised and their resultant extracted in stages inside V_5. Finally the unit $T(w, z)$ is inverted in $K[w, z]/(g(w), g(z)) \cong K^9$.

Lower bound. The space $W = (I_0 \cap R_{\leq 4}) + Kf_1 + Kf_2$ is closed under the degree-4 operations and excludes 1. This uses characteristic > 3, through $-3 \neq 0$. □

Exact degree-5 refutation DAGs were replayed in native arithmetic on public fixtures:

- $p = 101$ and $p = 5$;
- secp256k1, P-256, P-384 and P-521, with $S = \{x(G), x(2G), x(4G)\}$ and $R = 16G$; the DAGs have 413–415 nodes, and OpenSSL rechecks the points.

Twenty polynomial identities used in the proofs are checked in Lean (`FallDegreeIdentities.lean`, `PrimeBitLocal.lean`; axioms `propext` and `Quot.sound` only). The degree-4 lower bound is now machine-checked: `GuardedModel.no_degree_four_refutation` proves that 1 is not derivable at degree four from the six generators, over every field with $6 \neq 0$ and three distinct base abscissae, and `lastFall_lower` states the last-fall consequence under an explicit unit-ideal hypothesis (`research/guarded_fall_degree_20261028/formal/GuardedModel.lean`; six modules, fresh `leanchecker` replays and a transitive axiom audit, merged in #2814). The degree-5 upper construction, the bridge from rational unsatisfiability to the unit ideal, and the first-fall syzygy statement are *not* formalised, so the exact pair $(4, 5)$ is not yet a Lean theorem. A `lastFallDegree` definition and Caminata–Gorla's Theorem 2.8 in `crypto-autoresearcher:formal/CryptoResearch/FallDegree/LastFall.lean` are the bridge a complete formalisation would build on.

Two Boolean upper bounds accompany the guarded chain:

- $d_F \leq 3\ell + 1$ for the fully Boolean binary encoding, with ℓ bits per base coordinate;
- $d_F \leq 3\ell + 2$ for a prime-field bit-restricted lookup encoding.

Both are instances of the standard brute-force refutation in Polynomial Calculus [6]: fix the base bits, and refute the affine residue. For the binary chain at $t = 3$ they are weaker than Theorem 6.8's $2k + 3$.

6.5 Prime-field Kummer towers

Petit, Kosters and Messeng [33] take the factor base $V = \mu_{2^t}$ over $\mathbb{F}_p$ (with $2^t \mid p - 1$). They write membership as a tower $y_0 = x/g$, $y_{j+1} = y_j^2$, $y_{t-1}^2 = 1$, and argue the complexity from the first fall

t (PKM: n_1)	D_{ff} (proved)	d_F (session median)	PKM $D_{\text{reg}}^{\text{corr}}$
1		3	4
2	4	4	4
3	4	4	4
4	4	4	4
5	4	4	4
6	4	5	5
7	4	5	5
8	4	5	5
9	4	5	5
10	4	6	6
11			6

Table 6.4: Prime-field Kummer tower, $m = 2$, factor base μ_{2^t}: the proved first fall degree (Theorem 6.14(3); it needs $t \geq 2$), the session-reported median last fall degree at $p = 2{,}013{,}265{,}921$ (raw data not committed), and the corrected degree of regularity $D_{\text{reg}}^{\text{corr}}$ of Petit, Kosters and Messeng, Table 4 (prime case) [33]. Source: `figures/data-new.json`, from `research/prime_fall_degree_20261027/`.

degree. Source for this section: `research/prime_fall_degree_20261027/DERIVED.md`, written proofs D1–D8.

Theorem 6.14 (Kummer towers, $m = 2$)**.** *Let $R = \mathbb{F}_p[y]/T$ be the tower quotient. Items (3)–(5) concern* inconsistent *systems, those whose reduced S_3 is a unit of R; a satisfiable target has no inverse in R, and the refutation bounds do not apply to it.*

1. *T is a Gröbner basis, and $R \cong \mathbb{F}_p^{V^m}$. So every ideal over R is radical, and an $m = 2$ system is inconsistent if and only if the reduced S_3 is a unit of R.*
2. *$R = \mathbb{F}_p[z]/(z^K - 1)$ with $K = 2^t$. The degree of a monomial z^k is* popcount(k)*, and degree falls are exactly binary carries.*
3. *If $t \geq 2$, $g \neq 0$, $3x_R^2 + a \neq 0$ and the system is inconsistent, then $d_{\text{ff}} = 4$ (operational convention), for every t. The degree-2 part of f is a bilinear form with $\det M = -2g^6(3x_R^2 + a) \neq 0$, so nothing falls at 3; the monomial $q = y_{10}y_{11}$ gives a fall at 4.*
4. *For an inconsistent system, the no-fall certificate has degree $d_F \leq 2 + \text{popdeg}(f^{-1}) \leq 2t + 2$.*
5. *For an inconsistent system in the naive presentation $\{S_3,\ x_i^K - g^K\}$ of the* same *set, $d_F \geq 2^t$. So d_F is not an invariant of the ideal; it depends on the presentation.*
6. *A single summand has $d_F = 2$ for every t. Growth at $m = 2$ must therefore come from the genus-one curve $f(z_1, z_2) = 0$; no specialisation to finitely many points can show it.*

Measured, and prior. Table 6.4 sets the study's session-reported median last fall degree for $t = 1..10$ at $p = 2{,}013{,}265{,}921$ beside the corrected degree of regularity in PKM's own Table 4 for $n_1 = 1..11$ (also Figure 6.1, right). The raw records were never committed (the study's `RESTART.md`), so the d_F column is session-reported, not re-derivable. From $t = 2$ on the two columns coincide.

So the *growth* is prior art. What Theorem 6.14 adds is the proved constant first fall degree beside it, and the presentation dependence. A prediction, $d_F = 4 + \lfloor (t-2)/4 \rfloor$, was registered before any $t \geq 11$ row and is untested.

6.6 Measured solving degrees in other presentations

The binary programme measured refutation and closure degrees on many presentations; the full record is in [3, §6]. Table 6.5 lists the last fall degree by base-subspace dimension ℓ. It is flat in n at fixed (m, ℓ) and rises with ℓ.

Three further measurements bear on the solver rather than the system:

- *Plain refutation tracks the semi-regular degree.* The plain Macaulay refutation degree stays within one of the semi-regular prediction in all 14 exact cells.
- *Closure algorithms refute one or two degrees earlier.* A MutantXL-style closure refutes all 92 unsatisfiable draws at degree 3–4, where XL needs 5–7.
- *Guessing variables restores degree 4.* Fixing one or two variables restores degree 4 at $(m, \ell) = (3, 7)$, at 1/116–1/260 of the projected degree-5 cost.

m	ℓ	n	d_F
3	5	10–15	4
	6	16–24	4
	7		≥ 5 (every draw)
4	3	9–14	4
5	3	11–15	4

Table 6.5: Binary decomposition systems in other presentations: last fall degree d_F by m and base-subspace dimension ℓ, over the measured range of n. At $(m, \ell) = (3, 7)$ every draw is refuted only at degree ≥ 5. Source: [3, §6].

Every observed growth is growth in the size of the factor base: in ℓ, k, t, B, or the digit count s of a prime digit presentation (where $d_{\text{ff}} = d_F = 4 + \lfloor s/2 \rfloor$ at $m = 2$, reproduced by a curve-free null). None was observed in n at a fixed base.

6.7 What this settles and what it does not

- *Settled.* In the three presentations the programme studied, the first fall degree is a proved constant of the presentation: 2 for binary Weil descent, $B + 1$ for explicit membership, 4 for Kummer towers. The last fall degree is not bounded by it.
  - This is proved for the guarded chain at $B = 3$ (exactly 5 against 4).
  - It is certified on named binary instances: ≥ 5 against 2, at $n = 33$–45 and in the $m = t = 3$ chain.
  - It is session-reported for Kummer towers (6 against 4 at $t = 10$; raw data not committed).

  Any complexity argument that substitutes the first fall degree for the solving degree is unsupported for these systems.
- *Not settled.* No presentation has a proved last-fall lower bound that grows with n. That is the open problem of HKY [23] and Kousidis–Wiemers [29], and the only result that would bear on the subexponential claims. The upper bounds $k + 2$ and $(t - 1)k + t$ are linear in n, far above every observed value.
- *Missing before submission.* The certificates are committed only as hashes; they regenerate (above) but need public hosting. The sample is small: two instances per n at $n \geq 33$, one polynomial instance at $n \geq 41$. The written proofs need expert review. The Kummer-tower data must be regenerated (`research/paper_opportunities_20261029/`).
- *Requested data.* The runs that would enlarge the binary sample and add F4 step degrees beside Table 1 of Kousidis and Wiemers [29] are preregistered as two protocols, frozen before any data: `FD-BINARY-CLOSURE-EXT` and `FD-F4-STEP-DEGREE` (`research/fd_paper_data_request_20261029/`). Both run on Implementation C. Each must first pass a calibration stage that reproduces every recorded outcome, an A/A check and a negative control. No run has been made yet. New $d_{\bar{F}} \geq 5$ rows will be marked *closure only* until an independent certificate is checked.

7 What index-calculus hardness rests on

Chapter 6 studied one quantity, the solving degree of the point-decomposition system, and showed that its first fall degree does not bound it. A complexity claim for elliptic-curve index calculus needs more than a degree. It needs a bounded solver cost, a decomposition yield with a known distribution, relations of sufficient rank, a usable geometry, an interface that no generic algorithm reproduces, and an end-to-end composition of all of these. Each of the subexponential claims in the literature discharges some of these obligations by proof and the rest by an assumption. This chapter records, assumption by assumption, what the programme proved, refuted, measured or left open between 2 and 4 November 2026.

The work falls into four kinds:

- a typed obligation chain and a reanalysis of committed evidence (`research/ic_weak_assumptions_20261102/`);
- exact measurements of six weak points on Weil-descent systems (`research/ic_weakpoints_20261102/`);
- exact censuses against five named heuristics (`research/ecdlp_weakness_census_20261104/`) and Lean 4 theorems aimed at the same nodes (`research/ecdlp_weakness_lean_20261104/`, `research/lean_attacks_20261102/`, `research/inside_gap_n4_20261102/`);
- a claim graph of the hardness literature and a triage of its weak points (`docs/ecdlp-hardness-corpus/WEAKNESS_TRIAGE.md`).

Standing. None of this work is an ecbench session, a leaderboard row or a validated result. No validator report or replay certificate exists for any census or measurement below. Every measured number in this chapter is therefore an unverified assertion of its producer under the proof-carrying-claims rule; the census README marks its numbers `[pcc:unverified]`. The Lean results are stated with their exact formal status in §7.11: "kernel-checked" is used only where a Lean kernel accepted the declaration and its source contains no `sorry`, `admit`, `native_decide` or user axiom. For the 4 November project (`ecdlp_weakness_lean_20261104`), whose theorems carry that label in their headers below (Theorems 7.1, 7.2 and 7.3), the kernel that accepted them is the producer's local `lake env lean -o` compile: no proof-gate run is recorded, so there "kernel-checked" means *producer's compile; no gate run recorded*. The corpus records every internal theorem here as `proved_internal_unreviewed`: no external reviewer has read the statements against the literature they address. Nothing in this chapter changes a row of Tables A–C, and none of it is a measured improvement of the index-calculus baseline.

7.1 Six serial obligations

The weak-assumption programme of 2 November (`research/ic_weak_assumptions_20261102/README.md`; literature pass and reanalysis of committed records, no new run) writes an index-calculus security argument as a chain of six obligations. The chain is serial in the sense that matters: failure of any one edge invalidates the complexity claim, whatever the others show. Its machine-readable form, `obligations.json`, is checked by `audit.py`, a fail-closed gate that refuses to promote a route while any field is missing. It covers naive prime-field Semaev, generic pair tables, Kummer (PKM) towers, the proposed arity-three hybrid solver, Wagner filters, the hashed-base tensor route (§24.1) and xedni. The route-by-route reading for prime fields is in §22.7; Table 7.1 lists the obligations with the proxy usually offered for each and the section of this chapter that bears on it.

A stable degree does not certify cost. The reanalysis extracted the committed PKM $m = 4$ random-target F4 rows, two per cell (`research/ic_weak_assumptions_20261102/evidence-summary.json`, key `pkm_m4_random_f4`; `extract_existing.py -check` recomputes it from the JSONL records):

- at $N = 8, 12, 16$ the solving degrees are $6, 7, 7$;
- the maximal column counts are 1,049, 9,452 and 76,837;
- the maximal multiply-add counts are 48,447,128, 85,477,602,933 and 27,048,206,854,510, that is $4.84 \cdot 10^7$, $8.55 \cdot 10^{10}$ and $2.70 \cdot 10^{13}$.

From $N = 8$ to $N = 16$ the columns grow 73.25-fold and the arithmetic 558,304-fold, although the last two degrees are equal. Three cells are not an asymptotic fit. They are a direct counterexample to reading a bounded degree as a bounded cost: the cost of a Gröbner computation at a fixed degree d is governed

Obligation	What it needs	Proxy often used instead	Here
Solution geometry	dimension and radicality of the full system, rational lifts, exceptional charts, target strata	`zero_dim=true` on a random target	§7.2 (W5)
Solver growth	maximal matrix dimensions, nonzeros, arithmetic and memory, with timeouts and out-of-memory exits kept	first fall degree; a stable solving degree	§7.1, §7.3, Chapter 6
Yield and variance	rational decompositions per natural target: distribution, zero mass, multiplicity, solver failures	the mean count B^m/r	§7.7
Relation rank	rank growth, dead columns, full-rank probability	raw row count	§7.6
Non-generic interface	a coordinate predicate whose useful candidates no generic table reproduces at equal cost	"uses coordinates"	§7.9
End-to-end composition	one parameter family charging set-up, failed queries, collection, verification, linear algebra, descent, memory	product of the separate bounds	§7.2 (W6)

Table 7.1: The six obligations of an index-calculus complexity claim, after `research/ic_weak_assumptions_20261102/README.md`. The README's table also states, per obligation, the "strongest next falsifier"; the right column points to the evidence in this chapter.

by the number of monomials of degree $\leq d$ in the variables of the system, and that number grows with N. The same six rows are all `inconsistent=true`: their ideal is the unit ideal. So the `zero_dim` flag on a random target refutes a decomposition and says nothing about the geometry of a successful one. The planted toy rows at $N = 8, 12$ are consistent, with finite bases of 28 and 49 elements, and are the positive control. The README's reading is that the useful theoretical target for PKM is now a lower bound on the last fall or closure degree, or on the matrix width, and not another first-fall plot; the $N = 20$ cell is projected at 240–380 GB and is not to be launched to reconfirm the trend. The solving-degree data of the Kummer towers themselves are in §6.5 and §23.1.

The follow-up. The same day's follow-up (`research/ic_weak_assumptions_20261102/FOLLOWUP.md`) added an arbitrary-precision mirror of the tensor route's D.4 admission guards and a finite-field unit-explanation proof kernel for the arity-three canonical representation, which it describes as validated implementation artefacts. It took no scientific run. Its status line per lane: the tensor route's predicate cost and aggregate complexity remain uncertified; the PKM closure-dual target is an exact nullspace certificate but no growing certificate has been found; the Wagner bucket statistic W has no frozen native producer; for xedni no theorem comparing the selected height with the global minimal discriminant is proved.

7.2 Six weak points measured on Weil-descent systems

The obligations were then measured, as exact counts, on one family where every one of them can be computed: Gaudry–Diem descent for $E : y^2 = x^3 + ax + b$ with random $a, b \in \mathbb{F}_{p^n}$, $n \in \{2, 3\}$, factor base $\{P : x(P) \in \mathbb{F}_p\}$, groups up to 2^{28} for the complete solves (`research/ic_weakpoints_20261102/README.md`; protocol frozen before data, `PROTOCOL.md` with Amendments 1–3a; native Rust tool, standard library only, one 4-vCPU container). The README states that it is not an ecbench session, not a leaderboard row and not admitted by the validation gate, and reports no wall time. Table 7.2 lists the six points in the README's numbering.

The README's reading is that on this family W3, W4 and W5 are cleared, with cheap filters for their failure modes (subfield targets, duplicate rows, spurious solutions), that W1 is a constant and not a growth, and that the factor-base size is a seventh, required lever. Its derived figures, which are models and not measurements:

Weak point	Measured (exact counts)	Producer's reading
W1 solving degree, matrix growth	$n = 3$: solving degree 10 in 88/88 systems, $p = 31$ to 1,073,741,789 (2^5–2^{30}); Macaulay matrix 252×286 at that degree, $c(3) = 1.03$–$1.07 \cdot 10^6$ $\mathbb{F}_p$ multiplications (dense, an upper bound for F4); no first fall in 87/88. $n = 2$: degree 3, 6×10, about 87 multiplications	a p-independent constant $c(n)$; its size, not its growth, is the question
W2 non-generic filter	not addressable on this family (the subfield base is the structure; $\mathbb{F}_p$ is untouched)	open; the prime-field question
W3 yield mean and variance	20 curves, 2,000 or 6,000 targets: mean/$\lambda_{\text{pred}} = 0.88$–$1.19$ ($\lambda = F^3/6N$); $P(0) = e^{-\text{mean}}$ to three decimals; dispersion interval contains 1 on 19/20 curves; the two heavy tails are subfield targets $x_R \in \mathbb{F}_p$	Poisson; drop subfield targets
W4 relation rank	15/15 complete solves return the correct $\log Q$; rank shortfall 0 on 7 curves, 1–3 on 7, 6 on one; cofactor scaling, duplicated rows and dropped signs as controls	not a blocker at this scope
W5 geometry	zero-dimensional 176/176; quotient dimension 64 (Bézout) on 87/88, 4 at $n = 2$; at $p \le 127$, 24 targets: 32 rational solutions, of which 5 genuine and 27 non-split (84%), 2 singular, no positive-dimensional component	benign; about five rejections per relation
W6 end-to-end cost against matched rho	complete toy solves, $\ell = 2^{12}$–2^{25}; IC/rho (negation map, analytic $\sqrt{\pi\ell/4}$): 0.29–3.4 at $w = 0$, $2.8 \cdot 10^4$–$3.2 \cdot 10^5$ at $w = 0.1$, $1.5 \cdot 10^5$–$1.7 \cdot 10^6$ at $w = 1$; rho ahead on 7 curves, 8 unresolved, IC ahead on none	4–5 orders of magnitude remaining at $w \ge 0.1$

Table 7.2: The six weak points on Gaudry–Diem descent over $\mathbb{F}_{p^n}$, $n \in \{2, 3\}$, as reported in `research/ic_weakpoints_20261102/README.md`. Unverified producer figures; not an ecbench session; not admitted by the validation gate. The eight unresolved curves are unresolved only because $w = 0$ makes the Gröbner work free.

- *No large primes.* With $\tau = 3.50$ trials per p, $c(3) = 1.06 \cdot 10^6$ and 3.97 nonzeros per row, the best factor-base fraction gives IC/rho $\to 684$ at $w = 0.1$ and 3,761 at $w = 1$, constant for groups of 2^{60} and more. Without large primes $n = 3$ balances at $p^{3/2}$, rho's exponent, so the gap is a constant proportional to $\sqrt{w\, c(3)}$.
- *Double large primes.* At $|\text{FB}_r| \approx p^r/2$ the fitted exponent of trials to a square system is 1.27 ± 0.04 at $r = 2/3$ (theory 4/3, rho 1.5), from $p \le 2039$; $\log Q$ was correct in 105/105 reached cells. Extrapolated with $c(3)$ and the measured row weights, 2LP reaches IC/rho of 58, 7.7 and about 1 at 2^{96}, 2^{132} and 2^{180} at $w = 0.1$, and 319, 42 and 5.3 at 2^{96}, 2^{132} and 2^{180} at $w = 1$, where it does not cross 1 by 2^{240}. The band therefore ranks IC ahead nowhere up to 2^{240}: above 2^{180} the order flips between $w = 0.1$ and $w = 1$. The README calls this crossover optimistic: the fits are extended by up to 70 bits, and the 2LP slopes at $r = 3/4$ and $5/6$ are probably pre-asymptotic.

For the theory of this chapter the result is that, on the one family where the theorems of Gaudry and Diem apply, the obligations other than W1 and W6 hold at toy size, and the distance that remains is a constant set by the cost of one decomposition. For prime fields, where W2 is the open obligation, the family says nothing.

7.3 Petit–Quisquater Assumption 2

Petit and Quisquater [32] derived their heuristic $2^{c\, n^{2/3} \log n}$ for binary curves (§3) from an assumption that the corpus records as `HEUR-GB-PQ-ASSUMPTION-2`: the degree of regularity, read as the maximal degree a Gröbner solver reaches, of a Weil-descended summation system equals its first fall degree up to $o(1)$. At $m = 2$ the first fall degree of the descended S_3 system is 2 for every n (Kosters and Yeo [28], Corollary 4.11; Corollary 6.6 in our Caminata–Gorla convention), so the literal reading predicts solving degree 2. Kousidis and Wiemers [29] read the assumption as $D_{\text{reg}} = m^2 - m + 1 + o(1)$, which is $3 + o(1)$ at $m = 2$. Degrees are integers, so "$+o(1)$" means eventual equality, and a lower bound that exceeds the predicted value for infinitely many n refutes that reading for the family.

Theorem 7.1 (Subspace criterion; kernel-checked)**.** *Let K be a field, $I \subseteq K[x]$ an ideal with $1 \notin I$, and*

n	instances	iterated degree 3 reaches k	d_{iter}	d_{lin}
8–14	21	21	3	4 (20), > 4 (1)
15	3	0 (`iter3` $= 1$)	4	> 4
16	3	3	3	> 4
17–20	12	0 (`iter3` $= 1$)	> 3	> 4
21–44	72	0 (`iter3` $= 1$)	> 3	> 3
45–47	8	not run	–	> 3
17–24 (fresh)	24	0	4 (24/24)	

Table 7.3: `pq2_rank`: exact degree lower bounds for the $m = 2$ descended S_3 system over $\mathbb{F}_{2^n}$, 119 instances in the main sweep. The last row is two separate degree-4 sweeps at $n = 17$–20 and 21–24 on fresh instances. At every n, `low2` $= 1$; `low3` $= 1$ for every $n \geq 9$. Unverified census figures. Source: `research/ecdlp_weakness_census_20261104/pq2_rank/README.md`.

$\prec$ a degree-compatible monomial order. Suppose I contains k polynomials whose leading monomials are k distinct variables. If a Gröbner basis of I lies in a finite-dimensional subspace S, then $k \leq \dim_K\big(S \cap K[x]_{\leq 1}\big)$. In particular, if S is the span of the Macaulay rows of degree $\leq d$ and $\dim(S \cap K[x]_{\leq 1}) < k$, no Gröbner basis of I lies in S.

The Lean declarations are `SolvingDegreeBound.linear_count_of_gb_in_subspace` and `SolvingDegreeBound.no_gb_in_rows_of_low_lt` (`research/ecdlp_weakness_lean_20261104/lean/SubspaceCriterion.lean`). Its degree-2 special case, `solving_degree_ge_three` in `SolvingDegreeBound.lean`, adds the rank–nullity count $\dim(\operatorname{span} F \cap K[x]_{\leq 1}) + \operatorname{rank}(\operatorname{top}_2 F) \leq |F|$: for generators of degree 2, field equations included, if $k + \operatorname{rank}(\operatorname{top}_2 F) > |F|$ then the solving degree is at least 3. A non-vacuity instance over $\mathbb{F}_2$ is $F = \{x_0^2 + x_0,\ x_1^2 + x_1,\ x_0x_1 + x_0 + x_1\}$, with $k = 2$, rank 3 and $|F| = 3$. A further declaration, `not_tendsto_of_frequently_gap`, proves that an integer gap of at least 1 infinitely often is not $o(1)$.

Write d_{lin} for the least d at which the degree-$\leq d$ Macaulay row space contains k independent polynomials of degree ≤ 1, and d_{iter} for the same quantity computed on the iterated degree-d closure (every multiple of a newly found polynomial that stays within degree d is added until the rank stops growing). By Theorem 7.1, d_{lin} is a lower bound on the Caminata–Gorla one-shot solving degree (Proposition 6.2 relates it to the last fall degree), and d_{iter} is a lower bound on F4's maximal step degree under the containment property that F4's output at maximal step degree D lies in the degree-D closure. That property is argued in the census README and is not formalised. The number k of linear consequences is computed as $N - \operatorname{affdim}(Z)$ from the enumerated solution set Z; the Boolean Nullstellensatz step behind it is kernel-checked only for $|Z| = 1$ (`BooleanNullstellensatz.linear_mem_of_unique_zero`), and the general affine-span case is not formalised.

The census. `pq2_rank` (`research/ecdlp_weakness_census_20261104/pq2_rank/README.md`; Rust, no dependencies, seed `0x5eed20261104`) computes these quantities exactly over $\mathbb{F}_2$, with no Gröbner run, on the curve $y^2 + xy = x^3 + ax^2 + b$ with a random $\lceil n/2 \rceil$-dimensional factor-base subspace and a target with a non-empty solution set. Each instance checks, on 20 random assignments, that the descended equations agree with a direct S_3 evaluation. The main sweep was stopped during $n = 47$, before the binary wrote its own JSON; a script copied the log lines and recomputes nothing. $n = 48$–56 were not reached. Table 7.3 gives the results.

- *Literal Assumption 2 fails on all 119 instances:* $d_{\text{iter}} \geq 3$ everywhere, against the predicted first fall degree 2.
- *The Kousidis–Wiemers value is not reached* at any of the 87 instances with $n = 15$ or $17 \leq n \leq 44$: the iterated degree-3 closure reaches only one of the k linear consequences. On the 24 fresh instances at $n = 17$–24 the iterated degree is exactly 4.
- *Cross-checks.* A pure-Python closure by naive saturation, sharing no code with the Rust one, agrees on five dumped instances at $n = 14$–18, at degrees 3 and 4. A same-seed rerun for $n = 8$–40 reproduces the log line for line; it also gives $|F'|/|V|$ between 0.84 and 1.31 (mean 1.021) over 99 instances, and 0.997–1.004 for $n \geq 30$.

What this does and does not settle. A finite range cannot refute an $o(1)$ statement. It refutes the literal reading on every instance computed, and shows that the asymptotic value 3 is not reached anywhere from $n = 17$ to 44. The bound on F4's step degree rests on the unformalised containment property; the bound on the one-shot solving degree is a kernel-checked criterion applied to an unverified

computation. The measured degree sits above the value the subexponential binary heuristics assume, which removes support from them; it does not support a subexponential algorithm. The corpus node that records the census, `MEAS-PQ2-DLIN-CENSUS`, obstructs `HEUR-GB-PQ-ASSUMPTION-2`, which remains a heuristic in the corpus. The certified binary instances of Theorem 6.11 reach $d_{\tilde{F}} \geq 5$ at $n = 33$–45 on random subspaces of dimension $\lceil n/2 \rceil$; together the two give a lower bound of 4 at $n = 17$–24 by census and of 5 at $n \geq 33$ by certificate. What remains open is the same as in §6.7: a proof that the bound grows with n.

7.4 Semaev's Assumption 1

Semaev [36] states two degree assumptions for his chained systems over $\mathbb{F}_{2^n}$. Assumption 1 (§4.5) asserts $d_{F4} \leq 4$ for any subspace of dimension $k = \lceil n/m \rceil$ and any $2 \leq t \leq m$, supported, as the corpus records, by 2,300 Magma systems with $t = m$, $n \leq 21$, $m \leq 6$. Corollary 6.12 refutes it as stated, in the cells $(35, 2, 2, 18)$ with random subspaces, $(45, 2, 2, 23)$ with Semaev's own default subspace, and $(19, 3, 3, 7)$ inside his tables' range; implementation A alone also gives $1 \notin W_4$ at $(21, 3, 3, 7)$, with no checked certificate. The revision of the paper (ePrint 2015/310, 10 April 2015) replaces it with a growth condition on the regularity degree that the corpus records as $o(\sqrt{n/\ln n})$ (`HEUR-SP-SEMAEV15-WEAK-REGULARITY`), sufficient for the $2^{c\sqrt{n \ln n}}$ asymptotic and not for the FIPS-curve conclusions. Nothing the programme proved bears on that weaker condition: only a growing lower bound would.

His first-fall-degree assumption $d_{F4} \leq d_{\mathrm{ff}}$ (§4.4), applied to his own $t = 2$ cells with his statement that the first fall degree "is always 4", was tested on 200 instances of the ten cells of his Table 2 (`research/semaev_t2_trace_20261102/REPORT.md`; protocol frozen; exact standard-library arithmetic). The first fall degree is 2 on 200/200, each with an explicit trace certificate: with $\lambda = z^{-2}$,

$$\mathrm{Tr}\big(z^{-2} S_3(x_1, x_2, z)\big) = \mathrm{Tr}(x_1) + \mathrm{Tr}(x_2) + \mathrm{Tr}(B z^{-2}),$$

a constant combination of the coordinate equations of degree ≤ 1, the Kosters–Yeo identity in Semaev's coordinates. None of the 200 systems has a solution in $V \times V$ (brute force). The degree at which a Macaulay computation proves this is 2 on 154 instances and 3 on 46, never 4. So the statement "first fall degree always 4" is refuted, while the inference that these rows refute $d_{F4} \leq d_{\mathrm{ff}}$ splits: the sharp assumption fails on 46/200, where 3 exceeds 2, and Semaev's reported $d_{F4} = 4$ exceeds what the systems require. The corpus inference that had read the rows as a refutation everywhere is corrected in Chapter B. The `RT-SP-SEMAEV-A1-N45` target of the triage, an exact F4 run in the $n = 45$ cell, is still listed as open.

7.5 Nagao's Lemma 4: four cells and three repairs

Nagao's ePrint 2015/984 [31], which the corpus records as "polynomial-time binary ECDLP under FFDA" (status conditional), bounds the first fall degree d_F by a "fake" first fall degree d'_F computed modulo the field equations (Lemma 4: $d_F \leq d'_F$). Remark 6.1 gives the equality-reading counterexample over $\mathbb{F}_2$. The lemma's verdict depends on two choices: the reading of Definitions 5 and 6 ("equality": the maximal degree of the products is exactly d; "literal $\geq$") and the placement of the field equations (outside or inside the generating set). Table 7.4 lists the four cells and the three candidate repairs tested after the refutation.

The fourth cell. `NagaoLiteralInside.lemma4_literal_inside` proves, for any field, any family F and any finite family E of "field equations" placed among the generators, that $d_F \leq d'_F$ whenever a fake fall exists. The proof lifts a fake fall: replace the sum by a least-degree representative modulo $\langle E \rangle$ and pay the difference with multiples of E, which is allowed because E is inside. The file's comment records why the argument fails outside, or under the equality reading, where the added multiples can push the maximal degree above d. The reading axis is now complete, refuted in three cells and proved in the fourth. Which cell Nagao meant is a question of the source text: the triage lists a fresh PDF-level transcription of Definitions 5 and 6 as owed.

A1: outside, the fake degree bounds nothing. Modulo the field equations $X^k Y \mapsto XY$, so every W_k has the reduced system of the original witness and $d'_F = 2$. With the field equations outside, $X^{k-1}(XY + X) + X^k Y = X^k$ is a fall at $k + 1$, and no fall exists below it, so $d_F = k + 1$: the native table gives $3, 4, \ldots, 8$ for $k = 2, \ldots, 7$, and 3 for every k with the field equations inside (`research/lean_`

Statement	Result	Formal status
Lemma 4, equality, outside	refuted: $\{X^2Y, XY+X\}$, $d'_F = 2 < d_F = 3$ (Remark 6.1)	two implementations
Lemma 4, equality, inside	refuted	corpus record
Lemma 4, literal $\geq$, outside	refuted ($d_F = 3$, $d'_F = 2$)	corpus record
Lemma 4, literal $\geq$, inside	*theorem*: every fake fall is a true fall at the same degree, so $d_F \leq d'_F$, over any field	kernel-checked (Mathlib)
A1: $d_F \leq f(d'_F)$, outside	refuted: $W_k = \{X^kY, XY+X\}$ has $d'_F = 2$, $d_F = k+1$	$k=3$ kernel-checked; all k by hand proof
A2: $d_F \leq d'_F + 1$, inside	refuted: $\{YZ+1, XZ+X+1\}$, $d'_F = 1$, $d_F = 3$; 10,752 of 521,731 pairs have gap 2	native, two implementations
Inside gap, $n \geq 2$	$d_F - d'_F \leq 3$, tight ($\{XYW+XY+W, XZW+XW+1\}$); unbounded for $n = 1$	key identity kernel-checked in two variables
A3: $d'_F \leq d_F$ when the reduced generators are independent	survives: 0 counterexamples in 25,536 and 322,896 independent triples	census only

Table 7.4: Nagao's Lemma 4 over $\mathbb{F}_2$ by reading and placement, and the candidate repairs. Sources: `docs/ecdlp-hardness-corpus/WEAKNESS_TRIAGE.md` (W4), `research/ecdlp_weakness_lean_20261104/lean/NagaoLiteralInside.lean`, `research/lean_attacks_20261102/REPORT.md`, `research/inside_gap_n4_20261102/REPORT.md`.

`attacks_20261102/results/family.txt`). `IcNegative.FakeDegreeGap.gap_two_witnessW3` kernel-checks $k = 3$ ($d'_F = 2$, $d_F = 4$) in core Lean with `decide +kernel`; the repository proof gate passed it in run 41399938870 (target `papers/ic-negative-results/lean` PASS, 40 theorems, 0 trusted). The all-k statement is the hand proof plus the table.

A2 and the inside gap. The census of all 521,731 pairs of degree ≤ 2 in three variables (`research/lean_attacks_20261102/results/p3gap.txt`) finds gap 2 inside on 10,752 pairs; the witness $\{YZ+1, XZ+X+1\}$ has a fake fall at 1 and, inside, the true fall $Z(XZ+X+1)+X(Z^2+Z) = Z$ at 3. Two implementations agree (a Rust census and a Python brute force over every admissible multiplier tuple); the witness has three variables and is outside the two-variable Lean encoding. The open question this left was answered the same day (`research/inside_gap_n4_20261102/REPORT.md`; protocol frozen): for $n \geq 2$ variables, $d_F(F \cup S_{\text{fe}}) \leq 4$ for every system, because $Y^2(X^2+X)+X^2(Y^2+Y) = XY^2+X^2Y$ is a fall at 4 built from the field equations alone, so the inside gap is at most 3; a gap-3 witness in four variables was found in a sample of 2,000,000 pairs, and for $n = 1$ the family $\{X^k, X^k+1\}$ has gap $k-1$. `InsideGap.lean` states the field-equation fall in two variables in core Lean; the gate passed it in run 41484047855 (44 theorems, 4 new, 0 trusted). The step "zero multipliers on further generators leave the fall intact" is argued, and the four-variable witnesses are native only. The consequence is that inside-placement d_F under the equality reading is degenerate: it never exceeds 4 in two or more variables, through the trivial falls among field equations that Ding–Hodges and Petit–Quisquater exclude from their first fall degree. No estimate can take its degree from that quantity without first removing them.

A3 and Nagao's Proposition 5. The reverse repair, $d'_F \leq d_F$ outside when the reduced generators are linearly independent, has no counterexample among the triples tested: 266 counterexamples among the dependent triples of degree ≤ 2 in two variables, none among the 25,536 independent ones, and none among the 325,500 multilinear triples in three variables (`research/lean_attacks_20261102/REPORT.md`). Nothing here touches Nagao's Proposition 5 or the paper's own applications, which check unreduced degrees instance by instance; only a bound through d'_F alone is ruled out, and only outside.

7.6 Relation rank and the left kernel

The corpus node `HEUR-RELATION-RANK` conditions sparse linear algebra and, through it, marked three externally proved results as resting on a heuristic: Theorem 1 of Gaudry, Thomé, Thériault and Diem [22], Diem's small-genus theorem [9] and the GHS Weil-descent attack [21]. These are 4 of the 17 status inflations of §7.10. Galbraith and Gaudry [16] (§6.2) already note that target-dependent relations are

easier to analyse rigorously than relations reused across targets. The triage wrote the argument out (W2) and the Lean round of 4 November formalised its finite core.

Theorem 7.2 (Left-kernel relations; kernel-checked)**.** *Let K be a field and $M \in K^{n\times B}$ a relation matrix with $a_i + b_i x = \sum_j M_{ij} f_j$ for $i = 1, \ldots, n$, where f_j are the factor-base logarithms and x the unknown logarithm.*

1. *If $n > B$, a nonzero $\lambda \in K^n$ with $\lambda M = 0$ exists, for every M.*
2. *For any such λ, $\sum_i \lambda_i a_i + x \sum_i \lambda_i b_i = 0$; hence $x = -\sum \lambda_i a_i / \sum \lambda_i b_i$ whenever $\sum \lambda_i b_i \neq 0$.*
3. *Let K be finite with q elements and let Λ be any function of the vector of relation logarithms $\ell = (a_i + b_i x)_i$ alone whose value is always a nonzero vector. Exactly q^{2n-1} of the q^{2n} pairs (a, b) give $\Lambda(\ell) \cdot b = 0$: the failure density is exactly $1/q$.*

The declarations are `LeftKernel.exists_left_kernel`, `left_kernel_identity`, `left_kernel_recovers` and `card_bad_pairs` (stated without subtraction as $|\text{bad}| \cdot \text{q} = \text{q}^{2\text{n}}$) in `research/ecdlp_weakness_lean_20261104/lean/LeftKernel.lean`. Two steps are not formalised: that a decomposition of $R_i = a_i P + b_i Q$ gives the row M_i (true by the definition of the factor-base logarithms), and the reduction from uniform (a_i, b_i) to the counting statement, which is the uniform-measure statement itself. The consequence for the corpus: for a single target, with relations drawn from target-dependent points, index calculus needs no rank condition at all, and the four inflation rows are edge errors in the graph, not fragile theorems. Removing the heuristic *strengthens* the closures that rest on it, such as the GHS and two-summand closures. The triage's own toy check (12 seeds at $p = 10007$, every matrix 2–7 short of full rank, left-kernel recovery correct 12/12) was run in scratch space and is labelled there as an illustration, not evidence.

The heuristic stays load-bearing where relations are reused across targets: right-kernel amortisation, multi-target batches, and PKM's surplus of $\Delta = 10$ relations (`HEUR-PF-PKM-RANK-DELTA`). `rank_deficit` (`research/ecdlp_weakness_census_20261104/rank_deficit/README.md`; results in `results/tables.md`) measures that regime exactly: ten prime-order curves, two per $p \in \{1009, 2003, 4001, 8009, 16001\}$, small-$x$ bases with B closest to $\sqrt{r}$ ($B = 32$ to 126), rows $aG = P_1 + P_2$, five samples each.

- With negation pairs merged into $B/2$ unknowns, $\Delta = 10$ gave full rank in all five samples on 2 of the 10 curves; on the other eight the largest deficit at $\Delta = 10$ is 1–3. At $\Delta = 40$ four curves still have a deficit in some sample, at most 2. On four curves no tested $\Delta \leq 40$ gives full rank in all five samples.
- On the unmerged B columns no tested $\Delta \leq 40$ gives deficit 0 in all five samples on any curve.
- Free relations $P_1 + P_2 = P_3$ among factor-base points span only 3 to 15 of the $B/2$ merged unknowns.

At these sizes PKM's "$\Delta = 10$ suffices" is contradicted (`MEAS-RANK-DEFICIT-CENSUS` obstructs `HEUR-PF-PKM-RANK-DELTA`). The census README also states that the unmerged deficit is "at least about $B/4$ at every Δ"; its own table does not show this (at $\Delta = 40$ the minimum unmerged deficit over five samples is 0 or 1 on seven curves), so that sentence is not used here. The measured complete solves agree with the theorem: on the descent family 15/15 solves returned the correct logarithm with rank shortfalls up to 6 (§7.2), and the 842 prime $m = 2$ rows of the reanalysis report rank 842, a correctness control and not a rank law, since each episode stopped before a full matrix was collected (`research/ic_weak_assumptions_20261102/evidence-summary.json`).

7.7 Yield: the mean is a theorem, the shape is a heuristic

The corpus node `HEUR-DECOMPOSITION-PROBABILITY` has the largest impact on the frontier of the attack graph (13). Its mean, B^m/r, follows by counting: `MeanCount.sum_card_reps` kernel-checks that the ordered representation counts over all targets sum to $|F|^m$, the ordered form of Proposition 5.5. Only the shape is heuristic: Poisson zero mass, variance, independence between targets, rationality and visibility to the solver. The triage (W3) splits the node in two. Over $\mathbb{F}_{q^n}$ with n fixed, Diem's rigorous results bypass the heuristic. Over prime fields it is genuinely heuristic. Under the left-kernel formulation the cost needs only the fraction of points that decompose, a set size, exact on small fields.

Prime field, $m = 2$. `pf_yield` (`research/ecdlp_weakness_census_20261104/pf_yield/results/table.md`) enumerates every target on 18 prime-order curves, p from 2003 to 80021, with three base shapes (small x; random; random closed under negation) at $\mu = B^2/r$ near 0.5, 1 and 2: 162 rows. With ordered pairs the decomposable fraction is 0.55–0.78 of $1 - e^{-\mu}$. With unordered pairs $U(R)$, of mean $\mu/2$, $P(U \geq 1)$ matches $1 - e^{-\overline{U}}$: the mean absolute deviation of the ratio is 0.011, 0.007 and 0.011 for the

three shapes. The variance-to-mean ratio of U is not Poisson for the bases index calculus uses: averaged over 54 rows it is 1.455 for small x and 1.447 for the random negation-closed base, against 0.981 (range 0.91–1.01) for the random base that is not closed under negation. The overdispersion follows closure under negation, through $U(R) = U(-R)$, which couples targets in pairs, and not the small-x shape. The heuristic's zero class is right once the pairing convention is stated; its variance is off by a factor of about 1.45.

The descent family and the $1/n!$ proportion. On the Gaudry–Diem family of §7.2 the yield is Poisson: mean/$\lambda_{\text{pred}} = 0.88$–$1.19$ and $P(0) = e^{-\text{mean}}$, the heavy tails being subfield targets. `ext_decomp` (`research/ecdlp_weakness_census_20261104/ext_decomp/results/table_primary.md`) measures the n-fold decomposable fraction over $\mathbb{F}_{q^n}$ with factor base $x \in \mathbb{F}_q$ against the survey value $1/n!$: 0.350–0.413 at $n = 2$ ($q = 101, 211, 401$) and 0.101–0.165 at $n = 3$ ($q = 23, 41, 61$), that is 0.70–0.83 and 0.61–0.99 of $1/n!$ (the README's summary, 0.70–0.99 for both, overstates the lower end at $n = 3$). The $n = 4$ rows have factor bases of 10–18 points and scatter widely. For control curves defined over $\mathbb{F}_q$ the premise fails: the factor base is $E(\mathbb{F}_q)$ or lies in a small subgroup, and the fraction is 0.0003–0.002 at $n = 3$. The $1/n!$ is the mean of Proposition 5.5 and carries the hypothesis that the curve is not defined over the subfield.

A withdrawn closure. The triage first argued by a dimension count that the trace pull-back (base-change $E/\mathbb{F}_p$ to $\mathbb{F}_{p^n}$, decompose over points with x in an $\mathbb{F}_p$-subspace V, pull back by the trace) cannot give a small factor base, and withdrew the argument on 4 November: the count bounds $|F'|$, not $|\mathrm{Tr}(F')|$. `trace_pullback` (`research/ecdlp_weakness_census_20261104/trace_pullback/results/summary_table.md`) computes the trace image of every one-dimensional subspace for ten curves, $p = 103$ to 2003, over $\mathbb{F}_{p^2}$: the smallest image is 0.319–0.389 of $\#E(\mathbb{F}_p)$, attained at $V = \mathbb{F}_p \cdot i$ on every curve, and two-point decomposition over it covers every target. The lead `GAP-PF-TRACE-PULLBACK-FACTOR-BASE` is narrowed, not closed: higher extension degrees and higher-dimensional V are not covered.

7.8 Waiting times: marginals suffice, hazards are not implied

The Jacobson–Koblitz–Silverman–Stein–Teske obstruction to xedni, audited in Lean on 1 November (`research/jksst_xedni_audit_20261101/`; see Chapter B for the correction of the treatise's earlier "closed" wording), derives a linear expected waiting time from a conditional-hazard recurrence $(1 - \varepsilon)s(n) \le s(n + 1)$ on the survival masses $s(n) = P(\tau > n)$, and listed the recurrence for repeated adaptive trials as open. The triage (W7) promoted that free-text assumption to a node, `ASM-XEDNI-HAZARD-RECURRENCE`.

Theorem 7.3 (Marginals suffice; kernel-checked)**.** *Let μ be a probability mass on a finite set, τ a stopping time, and S_t the event that trial t succeeds, with $\tau \le n$ implying some S_t, $1 \le t \le n$. Trials may be correlated or adaptive.*

1. *If $\mu(S_t) \le \varepsilon$ for every $t \le N$, then $s(n) \ge 1 - n\varepsilon$ for all $n \le N$.*
2. *If moreover $N\varepsilon \le 1/2$, then $\mathbb{E}[\min(\tau, N)] \ge N/2$.*
3. *The converse inference fails: on three equally likely outcomes, outcome k succeeding at trial $k + 1$, every trial has marginal success mass at most $1/3$, the recurrence with $\varepsilon = 1/3$ fails at $n = 1$ ($\frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9} > \frac{1}{3} = s(2)$), and still $\mathbb{E}[\min(\tau, 3)] = 2$.*

The declarations are `survival_ge_of_marginals`, `truncated_expectation_ge_half_horizon_of_marginals`, `marginals_U`, `marginals_do_not_give_hazard` and `wait_U` in `research/ecdlp_weakness_lean_20261104/lean/HazardSeparation.lean`. Part 2 is the conclusion of the JKSST audit's `truncated_expectation_ge_half_horizon` without its hazard hypothesis; part 3 refutes the corpus inference `CLAIM-MARGINAL-IMPLIES-HAZARD`. The open bridge for xedni is therefore a per-trial marginal bound for the adaptive trial actually run (`ASM-XEDNI-TRIAL-MARGINAL`: each trial succeeds with marginal probability at most the one-shot bound C_0/p), a weaker and more natural premise than a history-uniform hazard bound. This strengthens the obstruction; it does not reopen xedni, and it leaves the other conditional links of the JKSST chain (heights and discriminants, Lang's conjecture, selected lifts, useful relations, cofactors) as they were.

7.9 Generic bounds and what they cover

The non-generic-interface obligation exists because a generic algorithm cannot beat $\sqrt{r}$ (§32.1). Two Lean results this autumn fix the scope of that statement.

Shoup's bound is relative to random encodings. `TransparentEncoding.dlog_identity_encoding` and `zero_operation_dlog` (`research/ecdlp_weakness_lean_20261104/lean/TransparentEncoding.lean`) prove that with the identity encoding of $\mathbb{Z}/r$ the logarithm is the encoding, read with zero group operations. This refutes the corpus claim `CLAIM-SHOUP-ALL-ALGORITHMS` ("every algorithm needs $\Omega(\sqrt{r})$") and qualifies the bound and "BSGS is optimal" as generic-model statements. It is a scope statement, not an attack: it is the reason every subexponential claim must name an encoding property, and the reason the obligation of Table 7.1 asks for a coordinate predicate that no generic table reproduces. The generic-interface audit of 1 November (`docs/ecdlp-hardness-corpus/GENERIC_INTERFACE_AUDIT.md`; an internal argument, not a reviewed theorem) sorts the programme's oracles: subtract, meet-in-the-middle, k-sum and pair-enumeration oracles are generic table algorithms once their set-up is charged; the root-finding oracles return the same relations as group subtraction, with representation-specific set-up; Frobenius-orbit methods need an augmented interface with a cheap known-scalar automorphism; and F4/F5/SAT, Weil-descent, PKM-tower and small-root lattice oracles are coordinate-algebraic, so Shoup's and the Corrigan-Gibbs–Kogan [8] bounds do not exclude them. Only the last group can discharge the obligation. The weak-assumption reanalysis credits the PKM/F4 and compact Boolean predicates with a genuinely coordinate-algebraic interface; none of them yet has a solver cost that discharges the obligations after it (§7.1, §22.1).

Rho is conductor-blind in distribution. The conductor-gap programme asked whether the depth of $\mathrm{End}(E)$ in an isogeny volcano changes the cost of rho. The theorem that answers it in distribution is `research/conductor_gap_rho_large_sample_20261018/lean/RepresentationHashInvariance.lean`.

Theorem 7.4 (Representation-hash invariance; checked by the producer)**.** *Let $\varphi : G \to H$ be a bijection compatible with binary operations on G and H. For an r-adding walk $w = (h, c)$ on G, with partition $h : G \to \{0, \dots, r-1\}$, multipliers c and step $x \mapsto x \cdot c(h(x))$, let $\varphi_* w = (h \circ \varphi^{-1}, \varphi \circ c)$. Then the trajectory of $\varphi_* w$ from $\varphi(x)$ is φ of the trajectory of w from x; the rho length (first repeat, capped at the budget) is the same at every budget; and $w \mapsto \varphi_* w$ is a bijection of walk spaces.*

No group axiom is used. Read outside Lean, the push-forward of the uniform distribution on walks under a bijection is uniform, so over a uniformly random partition the cost distribution of rho is the same on the crater and on the floor, and on every curve of an isogeny class of prime order (abstract groups of equal prime order are isomorphic). The conductor, the j-invariant or any invariant that is not an invariant of the group can enter only through the one fixed hash an implementation uses; whether ecbench's fixed partition does so is an empirical question, and the measurements are in §34.4. Formal status: 11 theorems in core Lean 4.34.1 without Mathlib or a Lake project; the committed `build.log` of 18 October reports `[propext, Quot.sound]` for the three audited declarations and no `sorryAx`. The file is not among the targets of proof-gate run 41399938870. Nothing about curves, isogenies, endomorphism rings or conductors is formalised, and the file says nothing about index calculus or about the cost of computing the connecting isogeny. It complements `GenericInvariance.lean` of `research/conductor_gap_cm_pairs_20261010/lean`, which covers only algorithms whose control flow depends on equality answers (the gate run above passes that project with 60 theorems); see also §33.7.

7.10 The hardness corpus

`docs/ecdlp-hardness-corpus/` is a typed claim graph of what supports elliptic-curve discrete-log security: definitions, assumptions, reductions, algorithms, model lower bounds, conditional attacks, heuristics, measurements, obstructions and open obligations, each with its model and scope, and every edge typed (`supports`, `conditions`, `obstructs`, `qualifies`, ...). The triage of 3 November (`docs/ecdlp-hardness-corpus/WEAKNESS_TRIAGE.md`) read the assembled version of 1 November, 583 claims, 950 edges and 236 sources, and marks its own findings `[pcc:unverified]`. The generated `stats.tsv` on the current main branch counts 608 nodes, 32 of them refuted and 68 `proved_internal_unreviewed`; the corpus README's "current snapshot" still gives an older 525. Counts below are the triage's.

Check	What it finds	Count
Status inflation	proved, algorithm or measured claims whose support closure reaches an open, heuristic or assumed claim	17
Unwired dependence	a claim's assumption text invokes a heuristic node with no edge to it	11
Live obstruction	an established claim obstructs, qualifies or does not cover a claim that still supports others	49
Route closure on weak support	an `obstructs` edge whose obstruction rests on an assumed, heuristic or open claim	16
Hidden shared assumption	the same assumption text on 3 or more nodes, never a node itself	18 clusters
Source contagion	a primary source with a refuted claim and other live claims	21
Model-scope leap	a generic-model or measured claim supporting a broader one without a model bridge	9

Table 7.5: Structural checks of the hardness corpus (version of 1 November: 583 claims, 950 edges, 236 sources), as counted by `weakness.mjs` and reported in `docs/ecdlp-hardness-corpus/WEAKNESS_TRIAGE.md`. It then ranks 246 attackable claims by impact times breakability and keeps a separate hygiene queue of 100; the triage states that the scores are a heuristic, not evidence.

The triage's findings, W1–W7, are the targets of the censuses and Lean modules of this chapter:

- *W1: the first-fall pillar is obstructed and still load-bearing.* `HEUR-FIRST-FALL-DEGREE` has four incoming `obstructs` edges from established claims and still conditions seven claims, among them the ECC2K-130 2^{86} projection, Nagao's complexity, the PKM algebraic route and binary Weil descent. Its siblings `HEUR-GB-SEMIREG-ASSUMPTION` (six `obstructs` edges) and `HEUR-GB-PQ-ASSUMPTION-2` (four) head the attack queue with it. Because the graph keeps them at "heuristic", the seven conditional estimates read as one discharged condition away from usable, which overstates them (§7.3).
- *W2: relation rank* (§7.6).
- *W3: yield*, two regimes in one node (§7.7).
- *W4: the Lemma 4 refutation rests on a reading choice*, and two internal papers disagreed about it; the wording of `papers/ic-negative-results/` was corrected on 4 November to the four-cell table (§7.5).
- *W5: eleven unwired dependences* understate the impact of the top frontier nodes; refuting the heuristic would not reach them.
- *W7: route closures on weak support.* Xedni is closed twice, both softly; lifted index calculus is closed by three heuristics; the naive prime-field Semaev method is closed by a lower bound conditional on a statistical conjecture validated only to about 25-bit primes; the quasi-subfield routes are closed by results resting on the yield heuristic. Rank-only closures are not reopenings, by W2.

The triage states that W1–W7 change no claim's status by themselves; the Lean and census rounds added nodes and edges. Its recommended order puts corpus and proof work first (re-wire the Weil-descent chain to the left-kernel theorem, add the eleven edges, transcribe Nagao's definitions) and lists the remaining measurements (YYTK's conjecture past 25 bits, Semaev's $n = 45$ cell, the per-trial xedni marginal) as runs that go through the experiment route and the validation gate before any number enters the corpus.

7.11 Formal status

Table 7.6 gathers the formal artefacts of this chapter with what a kernel has checked and what is only stated or argued. The degree-four lower bound of the guarded chain (`GuardedModel`, Theorem 6.13) and the other fall-degree identities are in §6.4; the seven theorems of the Semaev-alternatives audit, which correct an oracle gate to $s < m - 3$, are in §22.1.

For the 4 November project the exact status is as follows. Its README says each module was compiled with `lake env lean -o` against the Mathlib v4.34.1 cache, that a full local `lake build` was not possible for lack of disk, and that CI's `lean-proof-gate` runs the full build and the transitive axiom audit. The triage records that every theorem audited after round 3 (35) depends only on `propext`, `Classical.choice` and `Quot.sound`. A search of the committed sources finds no `sorry`, `admit`, `native_decide` or `axiom`. No proof-gate run identifier is recorded for this project in its directory or in the corpus fragments, which say the gate "runs on the landing PR" or "on main after merge"; the pcc claim records are described as pending CI signing. The corpus marks the nodes `kernel_checked` with

Artefact	Kernel-checked	Not formalised
`ecdlp_weakness_lean_2026 1104` (Lean and Mathlib v4.34.1): `LeftKernel`, `HazardSeparation`, `SolvingDegreeBound`, `SubspaceCriterion`, `MeanCount`, `BooleanNullstellensatz`, `NagaoLiteralInside`, `TransparentEncoding`, `ProtocolCountermodels`	Theorems 7.1, 7.2, 7.3; the fourth Lemma 4 cell; the mean count; the $\lvert Z\rvert = 1$ Nullstellensatz; 45 `#print axioms` lines in `AxiomAudit.lean`	F4 containment; $k = N - \operatorname{affdim} Z$ for $\lvert Z\rvert > 1$; that PQ's "maximal degree" is the Caminata–Gorla solving degree; relation-to-row step
`IcNegative.FakeDegreeGap`	W_3: $d'_F = 2$, $d_F = 4$ (gate run 41399938870)	W_k for all k (hand proof)
`IcNegative.InsideGap`	the field-equation fall at 4 in two variables (gate run 41484047855)	zero multipliers on further generators; four-variable witnesses
`RepresentationHashInvari ance` (core Lean 4.34.1)	Theorem 7.4, 11 theorems; producer's `build.log`	curves, isogenies, conductors; the fixed hash

Table 7.6: Formal status of the results of this chapter.

status `proved_internal_unreviewed`. The theorems are general statements of linear algebra, counting and finite probability; their bearing on the literature is through the bridges named in the right column, which are argued, not proved.

7.12 What the literature's subexponential claims now rest on

Table 7.7 sets the claims of §3 and the two recent prime-field claims against the obligations of Table 7.1, with what is proved, refuted, conditional and open. Statuses in the second column are the corpus's.

Read by kind:

- *Proved here.* Five general statements, all kernel-checked and unreviewed: the subspace criterion, the left-kernel theorem with its exact $1/q$ failure density, the marginal waiting-time bound with its separation, the fourth Lemma 4 cell, and representation-hash invariance of rho. None is a lower bound on an index-calculus algorithm. Two of them *remove* a heuristic: the rank heuristic from the rigorous Weil-descent chain, and the hazard recurrence from the xedni obstruction.
- *Refuted.* Literal readings, all at the level of a stated assumption: Semaev's Assumption 1 and his $t = 2$ first-fall statement; Nagao's Lemma 4 in three cells, and the repairs A1 and A2 (+1); Petit–Quisquater Assumption 2 read literally, on every instance computed; PKM's $\Delta = 10$ at toy sizes; the corpus inferences `CLAIM-MARGINAL-IMPLIES-HAZARD` and `CLAIM-SHOUP-ALL-ALGORITHMS`.
- *Conditional.* Every subexponential claim for binary or prime fields remains conditional: on an asymptotic degree assumption (Petit–Quisquater's $o(1)$, Semaev's revised growth condition), on a first-fall heuristic (Nagao, PKM), or on an external theorem (the tensor route).
- *Open.* A last-fall or solving-degree lower bound that grows with n for any presentation (§6.7); a per-trial marginal bound for adaptive xedni lifts; the trace pull-back beyond one-dimensional V over $\mathbb{F}_{p^2}$; a non-generic prime-field predicate with a simulator-or-separation proof; and whether ecbench's fixed rho hash sees the conductor.

The direction is consistent. Each refutation removes support from a subexponential claim, and none supplies one; each theorem either confines a heuristic to a narrower regime or shows a closure to be stronger than its stated support. On the only family where every obligation was measured, the Gaudry–Diem descent, the distance to matched rho at toy size is 4–5 orders of magnitude at $w \geq 0.1$, and IC is ahead nowhere in the band.

Claim	Corpus status	What this programme established
Diem 2011: subexponential ECDLP over $\mathbb{F}_{q^n}$ for n growing with $\log q$ [10]	proved (external)	Untouched. Its scope, as the corpus records it: for fixed n the bound is exponential, and for $q = 2$, n prime it is $e^{O(n^2)}$, far above rho. For the rigorous Weil-descent chain (GTTD, Diem 2006, GHS) the rank step needs no heuristic for single targets (Theorem 7.2).
Petit–Quisquater 2012: $2^{c\, n^{2/3} \log n}$ for binary curves [32]	heuristic premise	Literal Assumption 2 fails on 119/119 instances; the $m = 2$ Kousidis–Wiemers value 3 is not reached at $n = 17$–44; $d_{\bar{F}} \geq 5$ certified at $n = 33$–45. The $o(1)$ form is not refuted: no growing lower bound.
Semaev 2015: $2^{c\sqrt{n \ln n}}$, $c \approx 1.69$ [36]	conditional	Assumption 1 ($d_{F4} \leq 4$) refuted as stated in four cells; "first fall degree always 4" refuted at $t = 2$ (200/200 have 2). The revised growth condition is open.
Nagao 2015/984: polynomial time under FFDA [31]	conditional	Lemma 4 refuted in three reading–placement cells, proved in the fourth; A1 and the +1 form of A2 refuted; inside gap bounded by 3 because inside $d_F \leq 4$ is degenerate; A3 survives on the triples tested. The complexity claim rests on the first-fall heuristic of W1.
PKM 2016: prime-field algebraic route [33]	conditional	$m = 4$ solving degrees $6, 7, 7$ above the threshold 4 and work $\times$558,304 from $N = 8$ to 16; $\Delta = 10$ contradicted at $p \leq 16001$ in the right-kernel regime; no cheap root-finding algorithm.
Hashed-base Boolean tensor contraction (Akshima et al. v0.2)	conditional	The only inspected route with an explicit end-to-end subexponential bound; conditional on the pinned contraction theorem; predicate cost and aggregate complexity uncertified (§24.1).
Xedni and lifted index calculus	obstruction conditional	JKSST is a chain of conditional links; the waiting-time link needs only per-trial marginals (Theorem 7.3); the marginal bound for the adaptive trial is open.

Table 7.7: The subexponential claims and the obligations they discharge. Corpus statuses from `docs/ecdlp-hardness-corpus/`; the right column summarises §§7.1–7.9. Census figures are unverified; Lean results as in Table 7.6.

8 Obstructions: what moving the problem cannot buy

The programme kept 153 leads against ECDLP in a lead atlas at its 30 October review (`crypto-autoresearcher:knowledge/leads/ecdlp/REVIEW_STATE.json`). Of these:

- 74 are closed with a named obstruction or exhausted at scale;
- 53 are open with a decided next step;
- 17 are blocked;
- 9 are unexamined.

Later rounds, including the split of 4 November, added more (Chapter 37). Most obstructions are textbook results applied to the case at hand: the Shoup–Nechaev generic bound, the Corrigan-Gibbs–Kogan preprocessing bound $ST^2 \gtrsim N$ [8], isogeny invariance of the group order, and the Balasubramanian–Koblitz embedding-degree bound. This chapter records the few that the programme *proved* itself and that constrain index calculus specifically. Each states a lever that cannot work, and why.

8.1 Transport conservation

A homomorphic transfer $\varphi : E \to E'$ with rational kernel κ changes the number of ordered m-decompositions of a target by exactly $|\kappa|^{m-1}$ (Lemma 26.1 in Chapter 26). So an isogeny buys only three things:

- the kernel symmetry, which is trivial on prime-order curves;
- the factor-base geometry available on the destination curve;
- the cost of arithmetic there.

Chains of isogenies compose to one homomorphism. A loop is an endomorphism acting as a known scalar. Multi-curve factor bases glued by isogenies are single-curve bases in disguise. Measured across sizes at fixed cofactor, fibre-aware index calculus loses ground to rho as $\#E^{1/6}$; the full measurements are in Chapter 26.

8.2 The summation cover is the same everywhere

Chapter 5 proved:

- the monodromy of the summation cover is $(\mathbb{Z}/2)^{m-2}$ for every curve;
- its fibres split completely on the factor-base locus;
- mean yield at fixed base size is a double count.

Together these close three families of leads: curve selection by anomalous monodromy, fibre censuses as a relation-rate probe, and factor-base geometries claimed to raise yield.

8.3 Oracle-augmented index calculus: label-valued oracles stay generic

Several leads proposed giving index calculus an extra oracle on the group. Since the split of 4 November the programme separates two classes.

- *Label-valued oracles* are Shoup-simulable with overhead $\Theta(\log N)$ or $O(m)$: the endomorphism image (by an eigenvalue argument; the premise $\mathrm{End}_{\mathbb{F}_p}(E) = \mathbb{Z}$ of the earlier derivation is false), label-valued halving, an incidence check over a label factor base, and a label-valued net. They stay closed at the $\sqrt{r}$ floor (`LEAD-EXO-augmented-oracle-ggm`; derivation tier, not yet ratified).
- *Coordinate-revealing oracles* (x, y, the full point, x-valued halving, field-valued jet and net, the threshold oracle C_t) are not simulable, and the generic group model gives no bound for them (`KN-FIND-194294`, `KN-FIND-982fdf`; `LEAD-EXO-coordinate-revealing-oracles`, open; §37.4). The earlier jet verdict is withdrawn.

Table-based decomposition obeys the preprocessing bound $ST^2 \gtrsim N$. Balanced generic index calculus costs about $r^{2/3}/n^{4/3}$, consistent with the measured growth of the Koblitz gap to rho as $r^{0.165\,[0.117,0.213]}$

(`LEAD-ICB-generic-table-oracle`). These are known instances of generic-group reasoning [8]; they are recorded because each retired or scoped a concrete proposal.

8.4 Lifting sections does not produce relations

Silverman's xedni calculus lifts points to a curve over a global field, in the hope that rank forces relations. On a frozen family of rational elliptic surfaces (with $\deg b = 6$ and monic quadratic x), the programme derived the exact lift probability

$$P_{\text{lift}}(p) = \frac{1 - 2/p + p^{-3}}{2p^3},$$

so the exponent is $\alpha = 3$, far above the $\alpha < 1/2$ that a relation source would need. This was checked against 596,268,874 enumerated slots with no disagreement, and a parameter count reproduces $\alpha = 3$ (`LEAD-ICP-xedni-function-field-sections`, `EV-XEDN-001`). The scope is one family ($\deg b \leq 10$, $p \leq 31$). A dimension-count theorem over all section shapes would be needed to close the route in general.

The classic obstruction to xedni is also conditional. A kernel-checked Lean audit of Jacobson, Koblitz, Silverman, Stein and Teske's analysis (`research/jksst_xedni_audit_20261101/`) does three things:

- it repairs their conditional theorem;
- it refutes several overbroad readings of it;
- it leaves open the premises it rests on: Lang's height conjecture, a uniform height–discriminant comparison, and the algorithm-specific bridge from a one-shot relation probability to the actual attack.

The audit calls the result a strong conditional asymptotic obstruction for classic xedni, not a proof that the route is closed.

8.5 Twist points carry no relations

A curve and its quadratic twist share x-coordinates, and therefore share summation polynomials. Could twist points supply relations for the curve? They cannot. Frobenius acts as $+1$ on $E(\mathbb{F}_q)$ and as -1 on the embedded twist, so the twist part B of any sum satisfies $2B = 0$.

The study `research/twist_obstruction_20261026/` checked this exactly on toy binary curves in the ECC2K-130 model (odd n, twist $a \mapsto a + 1$):

- 0 of 356,530,036 mixed sums are rational;
- 0 violations on 4,954 decomposition targets at $m = 2, 3$;
- the off-parity solution sets equal the predicted ones on 1,154 of 1,154 targets;
- positive controls found all 18,042 and 95 planted solutions.

A heuristic that ignored the obstruction predicted that 74–89% of solutions would contain a twist-lifting coordinate; 0% did at $m = 2$.

A follow-up asked whether a factor-base subspace can *avoid* the twist altogether (`research/lift_dense_factor_base_20261027/`; 11 Lean theorems, standard axioms only).

- *Composite degree $N = dn$, $n \geq 3$: yes.* On suitable $\mathbb{F}_{2^d}$-lines the lift condition collapses to the constant $\mathrm{Tr}(a)$. When $\mathrm{Tr}(a) = 0$ every abscissa lifts, giving 2^n relations per Gaudry–Diem trial. This applies to c2pnb208w1 and c2pnb368w1, but the modelled gain is $+0.50$ bits, a model output, not a measurement.
- *Prime degree (ECC2K-130): this construction does not exist.* A prime-degree field has no proper intermediate subfield, so there are no $\mathbb{F}_{2^d}$-lines to use. This does not rule out other twist-avoiding subspaces. At the small prime degrees tested, all-lift subspaces of dimension 4–5 exist, though fewer than for density-paired random sets (exact all-lift 4-subspace counts 0.921 ± 0.018 of the null at $N = 11$, permutation $p = 2 \cdot 10^{-5}$). This is toy evidence. Whether useful twist-avoiding subspaces exist at cryptographic prime degree is unresolved.

8.6 What the obstructions leave open

None of the obstructions above concerns the cost of *solving* the decomposition system, nor the linear algebra. Chapter 6 shows that the solving cost is genuinely open. The algorithms of Part III attack the remaining free terms.

Part III

Algorithms and measurements

9 Root-finding decomposition

9.1 Origin: a suggestion in an audited manuscript

The textbook decomposition oracle builds Semaev's polynomial system and solves it with a Gröbner-basis engine. On 12 October a 13-page manuscript, *Point Decomposition by Root Finding*, was supplied for assessment. We froze it (`research/ecbench_rootfinding_assessment_20261012/source-paper.pdf`, SHA-256 `743d3d3e...7c01`) and audited it against its source branch before measuring anything. Its central suggestion is simple: for two summands, fixing one factor-base point turns S_3 into a quadratic in the remaining variable whose roots are already known.

Proposition 9.1 (roots of S_3)**.** *Let $E : y^2 = x^3 + ax + b$ over a field K of characteristic $\neq 2, 3$, and let $R, F \in E(K)$ be affine with $x_R \neq x_F$. Then*

$$S_3(x_R, x_F, X) = (x_R - x_F)^2 \, (X - x_+)(X - x_-), \qquad x_\pm = x(R \pm F),$$

so $S_3(x_R, x_F, X) = 0$ iff $X \in \{x(R+F), x(R-F)\}$.

Proof sketch. With $\lambda_\pm = (\pm y_F - y_R)/(x_F - x_R)$ the chord slopes, $x_\pm = \lambda_\pm^2 - x_R - x_F$. Expanding $(X - x_+)(X - x_-)$ and eliminating y_R^2, y_F^2 with the curve equation gives exactly Semaev's $S_3/(x_R - x_F)^2$. The identity is machine-checked over an arbitrary field in Lean 4 with Mathlib (`papers/rootfinding-decomposition/lean/S3Roots.lean`, theorems `SemaevS3.S3_factor` and `SemaevS3.S3_root_iff`; no `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound`). □

Hence the two-summand system has a parametrised variety: $R = \pm F_i \pm F_j$ iff $x(R \mp F_i) \in x(\mathcal{F})$. No polynomial system is solved; the oracle is one subtraction and one table probe per base point.

Prior art. Neither the identity nor the oracle is new. The identity is a direct consequence of how S_3 is defined: it is the polynomial whose roots are the abscissae of $P_1 \pm P_2$. The oracle mechanism—sum $m-1$ factor-base points in the group and test whether the result lies in the base—is the second algorithm of McGuire and Mueller [30], who also evaluate S_m by iterated resultants instead of solving it, building on the instance-derived bases of Amadori, Pintore and Sala [1]. What this section adds is engineering and measurement: a machine-checked proof, a fail-closed verification protocol, binary pair-sum tables for up to four summands, GPU, Kummer-tower and dense-index variants, and complete-cost measurements in one harness against rho.

What the audit kept and corrected. The identity is exact under the stated generic affine condition, and a returned witness can be re-added in the group for soundness. Completeness of the exhaustive enumeration holds only with an explicit sign and factor-base convention (the binary table stores full point keys over a base containing both signs; the manuscript's schematic x-only table needed that convention stated). The manuscript's NIST baseline was Buchberger, not F4 (§4.5), and its statement that no implementation can use fewer than linear field operations per ideal fails for repeated queries against a reusable pair table. Larger-summand enumeration has a growing exponent in $|\mathcal{F}|$, and the manuscript did not benchmark it end to end. All of this is recorded in `research/ecbench_rootfinding_assessment_20261012/REPORT.md`.

9.2 The algorithms

Prime fields (Algorithm 1, `src/cryptanalysis/prime_fast.rs`): $\Theta(N)$ field operations and $2N$ lookups per target. *Binary fields* (`src/cryptanalysis/koblitz_index_calculus.rs`, `DecompositionStrategy::PairTable`, made the default): a precomputed table of pair sums $F_i + F_j$ turns an m-decomposition into 1, N or N^2 lookups for $m = 2, 3, 4$—the meet-in-the-middle half of a birthday search, exact and complete like exhaustive search, at the price of $\Theta(N^2)$ set-up and memory. Both are verified fail closed: every decomposition is re-added in the group, S_3 is checked to vanish, Gröbner-variety membership and agreement on refutation are checked against F4, and planted positive controls must be found.

Algorithm 1 Prime two-summand oracle by direct subtraction

Require: target R, base $\mathcal{F} = \{F_1, \ldots, F_N\}$, hash set H of abscissae (or full points) of $\mathcal{F}$

1: $d_i \leftarrow x_R - x_{F_i}$ for all i; invert all d_i with one field inversion and $3(N-1)$ multiplications (Montgomery's trick)
2: **for** $i = 1..N$ **do**
3: **for** $\varepsilon \in \{+1, -1\}$ **do**
4: $\lambda \leftarrow (y_R + \varepsilon y_{F_i})\, d_i^{-1}$; $x_3 \leftarrow \lambda^2 - x_R - x_{F_i}$ ▷ $x(R - \varepsilon F_i)$
5: **if** $x_3 \in H$ **then** recover the sign of F_j from y_3; emit $R = \varepsilon F_i + \tau F_j$
6: **end if**
7: **end for**
8: **end for**

Step	What it adds; artefact
Prime subtraction, binary pair table	Prop. 9.1; batched inversion; $m \leq 4$ tables. `research/semaev_root_vs_groebner_20261012/`
NIST P-192…P-521	standard parameters, 6- and 64-point bases; dedicated P-256 field 11–12× over generic big integers. `research/semaev_rootfind_nist_20261012/`
End to end vs msolve	complete DLPs with only the oracle swapped. `research/semaev_rootfind_e2e_20261012/`
PDP oracle family	`rf-mitm`, `mitm3`, `mitm4` (2-sum table); `rf-pairs`, `pairs3` (pairs then solve by the group law); `subtract2`; binary rungs $\ell = 8..14$, prime Kummer towers $m = 2, 3, 4$. `research/pdp_root_finder_20261012/`
CUDA kernel	one thread per (target, slice); per-thread Montgomery inversion; SoA base; shared-memory hash for $\lvert\mathcal{F}\rvert \leq 2048$; 64-bit and P-256 fields. Checked in host emulation only (no GPU available); no validation file. `RESEARCH_GPU_ROOTFIND_DECOMPOSITION.md`
ecbench engines	`pdp3-koblitz:m=2|3,engine=rootfind`, `rootfind-subtract`, later `rootfind-window` and `rootfind-negfold`. `research/ecbench_rootfind_engine_20261012/` and successors
Kummer-tower base	PKM's μ_{2^t} base with the root-finding oracles, 20–28-bit primes. `research/pkm_tower_ecbench_20261013/`
Dense root index, $n = 83$	packed five-byte S_3 root table, prefix widths K1271–K1798, 8 GiB of slots. `research/n83_dense_root_20261013/`
Prime tuning	pair-table size, arity, batched table inversions, synchronised lanes. `research/ecbench_rootfinding_\{tuning,arity,batchtable,lanes\}_2026101*/`

Table 9.1: Root-finding generalisations, 12–14 October. Every row is ENGINEERING: no slope fell below its generic floor.

9.3 Generalising the observation

Over three days the identity was carried through every layer of the pipeline. Table 9.1 lists the steps; each is its own study with a protocol registered before it ran.

9.4 Results

Oracle stage. Per decomposition, root finding is 383× ($\mathbb{F}_{2^9}$, $\lvert\mathcal{F}\rvert = 55$, 254 targets) and 1,535× ($\mathbb{F}_{2^{13}}$, $\lvert\mathcal{F}\rvert = 4{,}005$, 256 targets) faster than matrix F4; 3.5–$4.7 \cdot 10^4$× faster than `groebner_f4` on 16–24-bit prime curves ($\lvert\mathcal{F}\rvert = 6$, 40 targets each); and 3,021–5,802× faster than Buchberger on the decomposition ideal of the five NIST curves (Figure 9.1). Every binary verdict agreed with F4; every prime solution was in F4's variety and every prime refutation was F4's unit ideal. In the PDP study, `rf-mitm` refutes a binary target in 3.1 μs at $\ell = 8$ and 482 μs at $\ell = 14$, against the PS-0 baseline's 9.3 ms and 55.5 s (3,007–115,160×), costs $5\lvert\mathcal{F}\rvert - 3$ multiplications and $2\lvert\mathcal{F}\rvert$ lookups a target, and sits on the pair-table floor at 1.000 ± 0.027 bits per rung; at prime $m = 3$, $N = 18$ it uses 197 multiplications where F4 uses $8.99 \cdot 10^{13}$ multiply-adds; 707/707 brute-force agreements (`research/pdp_root_finder_20261012/`).

Complete solves. End to end, the gain is two orders of magnitude smaller and not one-sided: 50–105× against msolve on 16–24-bit prime curves; 122× on the one binary instance ($n = 23$) large enough

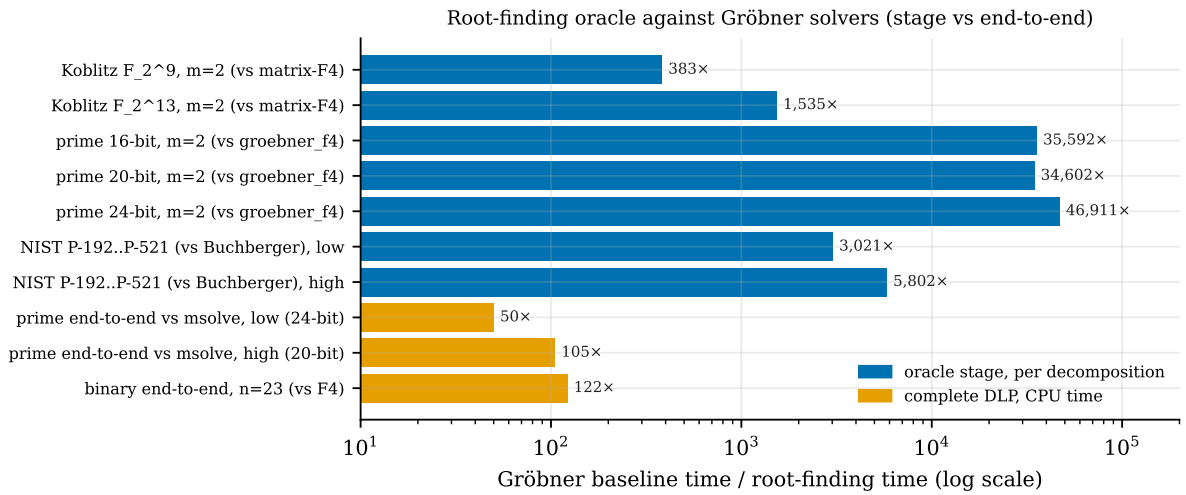


Figure 9.1: Root finding against Gröbner solvers on identical targets. Blue: one decomposition (oracle stage only). Orange: complete discrete-log computations with the oracle swapped and everything else fixed (prime: CPU time against msolve, 20/24-bit totals composed from sampled per-query cost; binary: one cold solve including table build). Sources: `RESEARCH_SEMAEV_ROOTFIND_VS_GROEBNER.md`, `papers/rootfinding-decomposition/README.md`.

to amortise the pair table; and F4 *faster* at $n = 5, 9, 13$, where a solve needs too few decompositions to pay for the table. Inside ecbench, with every group operation charged, the root-finding IC beats inherited F4 in solve wall by about 350× at $n = 13$ (admitted) and about 10^4× at $n = 23$ (descriptive), yet stays 2.9–25,116× the frozen strong rho in S on Koblitz curves $n = 13$–37 (the jump at $n = 31$ is mostly the curve's cofactor, 1,492) and 29–141× rho on prime curves, with the gap widening (`research/pdp_root_finder_ecbench_20261013/`).

The Koblitz loops. Figure 9.2 follows the best complete-count root-finding arm at Koblitz E_0, $n = 13$ through four registered improvement loops (8 public targets × 3 rounds each, L0 operation counts). The first engine had $S = 38.82$ ($m = 3$) and 28.38 ($m = 2$). A windowed third-summand scan cut $m = 3$ by a paired 0.828 $[0.783, 0.863]$; a fold over negation (`rootfind-negfold`) took $m = 2$ to 23.22 and then 22.97. Two further cycles of denser and sparser bases, windowed negfold and plain root finding were dead ends (`research/ecbench_rootfind_vs_rho_20261014/`). With F4's word XORs priced, the paired ratio negfold/F4 is 0.0545 $[0.047, 0.064]$, a complete-against-complete comparison; against the frozen strong signed-Frobenius rho ($S = 14.82$) the best complete arm is still 1.55×. A compact-orbit base with a counted Frobenius pair table reads $S = 15.55$ against F4's then-unpriced 16.75 and about 1.06× the strong rho. All three figures are lower bounds (factor-base construction natives, F4's word XORs and rho's table work unpriced), so neither order is ranked. Amortised over 64 targets the same arm reads $S \approx 11.2$. That is a ratio of lower bounds with no band, and the ledger counts it as a secondary, multi-target diagnostic, not a result below rho (`research/ecbench_rootfind_faster_20261014/README.md`).

Prime tuning in ecbench. On a 20-bit prime curve, the pair table's counted S is 0.018861 $[0.018547, 0.019242]$ of direct subtraction (24 pairs; prime native work unpriced, so a bounded ratio), and 0.057047 $[0.041159, 0.092069]$ on a 17-bit Koblitz curve, fully priced, where the table set-up is about 68% of the arm's cold cost. Doubling the table from 32 to 64 abscissae gives 0.830 $[0.697, 0.962]$; three summands instead of two costs 1.431× $[1.225, 1.716]$; batching the table inversions leaves counted GAE unchanged (1.000; 4,096 versus 64 inversions, set-up wall 243 versus 128 μs, descriptive); and a train-set lane selection did not resolve a hold-out winner (0.979 $[0.919, 1.038]$; 1,088/1,088 verified executions) (`benchmarks/README.md`, rows of 12–14 Oct).

Dense root index at $n = 83$. Widening the prefix base from K1500 to K1798 cut mean probes per relation from 300,021.9 to 169,069.8 (paired difference 130,952, 95% [52,058, 209,991]) and gave a one-point online ratio of 18.528 over same-point rho (IC 63.3 ms, rho 1,173.3 ms), with 339.0 s of reusable preparation and an 8 GiB table outside the window. The host was shared, so the result is exploratory;

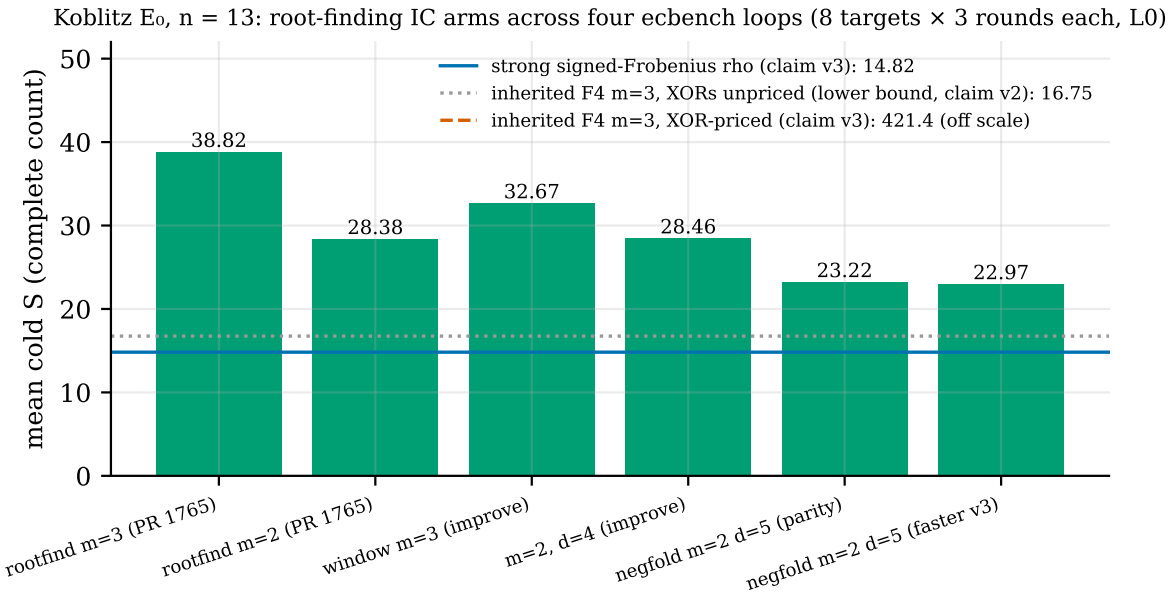


Figure 9.2: Mean cold S of root-finding IC arms at Koblitz E_0, $n = 13$, $d = 5$ unless stated, across the engine, improve, parity and faster loops. All bars are complete counts. Sources: `research/ecbench_rootfind_engine_20261012/`, `research/ecbench_rootfind_improve_20261013/`, `research/ecbench_rootfind_parity_20261013/`, `research/ecbench_rootfind_faster_20261014/`. Each loop had its own seeds and rho arm (rho 14.12–14.82).

two K1500 runs hit the 900 s timeout and are retained. The probe reduction stands; the online ratio does not: it was taken against rho with an empty table, and rho with a precomputed table matched to the index calculus's set-up is faster in every matched $n = 83$ pair (validation sweep, 20 October; §36.4).

Reading. Root finding is the largest constant-factor change of the programme at the oracle stage, and the one with a proof behind it. It does not change the asymptotic count of index calculus: its cost is $\Theta(|\mathcal{F}|^{m-1})$, every measurement is at $m \leq 4$, and it leaves relation yield, factor-base size and linear algebra untouched. Its effect on the cold frontier is to move the decomposition stage from dominant to minor, which exposes the stages that now are.

10 Gröbner-basis engines

Before root finding, the decomposition oracle was a Gröbner solve, and for $m \geq 3$ over binary fields it still is. The repository grew a family of engines (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md`, §1; file names below are under `src/cryptanalysis/`):

Matrix F4 and F5 over $\mathbb{F}_2$
(`koblitz_groebner.rs`, `matrix_f5_f2.rs`): Macaulay matrices of the Weil-restricted Boolean system, reduced by a shared Four-Russians kernel (`gf2_elim.rs`, four 8-bit tables, AVX-512 or AVX2).

Inherited F4
(`inherited_f4.rs`, from 22 September, the default algebraic oracle): a DPLL-style tree over partial assignments in which each child *inherits* its parent's echelon form and re-reduces only what the assignment changed, instead of building a fresh Macaulay matrix per node.

F6-IC
(opt-in): inherited F4 plus a factor-base domain gate (a fully assigned summand whose abscissa is not a base point refutes; partial support pruning refutes earlier), and meet-in-the-middle leaves (with $m-1$ summands fixed the last is looked up; at $m = 3$ and $|\mathcal{F}| \leq 256$ every compatible second point is added and the third looked up).

$\mathbb{F}_p$ F4
(`f4_fp.rs`, `f4_fp_tower.rs`, `sig_fp_tower.rs`): degree-bounded F4 with sparse-by-reducer then dense reduction, a Faugère–Lachartre block for the Kummer-tower oracle, and a signature variant.

Correctness. The audit pinned every engine against every other and against brute force (88+15+9+14+13+12+21+6 passing library tests in the touched modules), and found one soundness bug (§4.5, item 7), two contract defects and several latent holes, all fixed with regression tests.

Engineering results. Figure 10.1 collects the headline ratios and Table 10.1 lists them all; all are ENGINEERING with identical outputs.

Matrix and inherited F4 can return different decompositions, so their splitting trees may differ; only matrix F4 and F5 share a tree by construction. F6-IC's S ratios, once its batched group additions are charged, are an accounting loss, not a frontier move. Those ratios used inherited F4 before its word XORs were priced (#1774); priced, F6-IC is still a loss at K13, while the R13 and $n = 19$ cells remain lower bounds.

The $\mathbb{F}_2$ engine campaigns (about 90 directories, 22 September–17 October; the programme record [3], clusters "boolean", "gf2-elim", "f5-kernel", "f5-support-cut", "f4-f5-shared") gated each micro-change against the strongest reference, and most failed. Kept as defaults: the branchless strip, the bitmap seen-set, batched pair inserts, five-column tables, and direct packing in F5 (default since 17 October). About thirty unpack, pivot, layout and packing variants were refuted, and graded reuse in F5 fell short of the campaign's 2× goal. These are stage speed-ups of the decision engine; none changes an index-calculus distance to rho. Every reduced basis of `f4_fp`, after Gebauer–Möller criteria, packed monomials, interned symbolic preprocessing and a parallel symbolic pass, agrees with Singular or msolve. The Boolean comparison is a difference of task (decide versus compute the basis), not of engine quality.

What did not work. Degree-2 early-abort probes after a 2025–26 proposal: word-XOR ratios 1.00083–1.00378 (no change) and 0.99714–0.99928 (worse). Gate-first geometry: 1.57–1.78× fewer word XORs but lane-batched F4 roots cost 2.6–4.0× a sequential control. Convolution splitting of the summation polynomial: 1.21–2.07× *more* word operations despite lower degree; Frobenius auxiliary variables: 57–701× more (`research/lit2026_w\{4,5,8\}_*_20261011/`). Symmetry-breaking clauses in SAT did not help (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). A registered law for the $m = 3$ refutation degree resolved all 68 units but ended with the verdict "neither": one variant met its generic degree window on 32/32 draws, the other fell below it on 8, and the heaviest degree-10 Macaulay solves were $2.4 \cdot 10^5 \times 2.0 \cdot 10^5$ matrices at 42–51 minutes each (`research/ic_semiregular_law_20261005/`).

Change	Case	Measure	Ratio or value
Matrix / inherited F4	K13	counted matrix work	198.6×
	random-13		222.1×
	$n = 19$	wall, matrix F4/F5 vs inherited F4	timed out at 1,800 s vs about 6 s
F5 / F4	K13	word XORs	0.6465
	random-13		0.6414
	K13, random-13	wall	0.79–0.86
F6-IC / inherited F4	$n = 13$	word XORs	about 0.82
	$n = 19$		0.77–0.78
	—	relation wall	1.7–1.9× faster
	$n = 13$	S, group additions charged	64.6–68.8×
	$n = 19$		1,303–1,353×
	K13	S, word XORs priced (#1774)	1,572.1/634.12 ≈ 2.5×
	R13, $n = 19$		lower bounds
	$n = 31$	first target	both hit 3,600 s cap
Every echelon call to the Four-Russians kernel	oracle's own Macaulay matrices	speed-up	1.75–3.22×
Word-level `pext`/`pdep` rewrite plan	$n = 23$	F4 and F6-IC faster	10–14%
Eager / lazy rewrite of kept rows		speed-up	1.40–1.62×
Fused construction and readback	F4	instructions	−5.28%
	F5		−11.02%
Native hashing	F5		−3.715% [3.714, 3.726]
	standalone criterion		−45.1%
Bounded direct indexing in the F5 packer	F5		−3.947%
Row allocation		requested bytes	42–50% fewer
Quotient-curve model / symmetrised control	$m = 2$, five interleaved cells	F4 word operations cut	1.340–1.642×
Branchless AVX2 strip	one thread	speed-up	1.210×
Bitmap seen-set in Boolean F4	full F4		1.199×
Combined F5 options			1.530×
Fixed support cut	one complete call		3.02–3.07×
Graded reuse in F5			1.78–1.84×
`f4_fp` / its predecessor	square generic systems	speed-up	29.8–54.1×
msolve / ours	single-threaded	ratio	0.96–1.22
Ours / msolve, blocked pair reduction	two largest square systems	time	0.75–0.78×
	four others		1.11–1.49×
msolve faster, a day earlier		time	12–24×
Boolean decomposition systems, $n = 17$, 8 targets	our oracle (decides)	time per target	7–24 ms
	msolve, PolyBoRi (full basis)		40–52 s
	Singular		exceeds 120 s

Table 10.1: Gröbner-engine engineering results, all ENGINEERING with identical outputs. *Change*: candidate / baseline where a slash is shown; *Case*: target set or size (K13, random-13 and R13 as named in the sources; n the field degree, which the sources write m; m the number of summands; "—": no case given); *Measure*: the counted or timed quantity; *Ratio or value*: as recorded. Groups, top to bottom, with sources: inherited versus matrix F4 (`research/f4_f5_f6_cross_family_20261011/`); F5 versus F4; F6-IC versus inherited F4 (priced and lower-bound cells: `research/f4_pricing_jump_20261018/`); kernel routing and rewrite (`research/inherited_f4_eager_rewrite_20261011/`); instruction and allocation work, by paired A/B/A′ instruction counts on a frozen eleven-system suite, elapsed time not claimed; the quotient-curve model (`research/lit2026_w8_summation_models_20261011/`); $\mathbb{F}_2$ campaign results (strip: `research/gf2_branchless_strip_20260929/`; seen-set: `research/boolean_f4_bitmap_20260926/`; combined options: `research/f5_combined_2x_20260928/`; support cut: `research/f5_support_cut19_confirm_20261003/`; graded reuse: `research/boolean_f5_graded_batch_20261002/`); $\mathbb{F}_p$ F4 to msolve parity (`research/f4_fp_msolve_parity_20261011/`; blocked pairs: `research/f4_fp_blocked_pairs_20261016/`; the day-earlier msolve reading: `research/gb_cas_comparison_20261011/`).

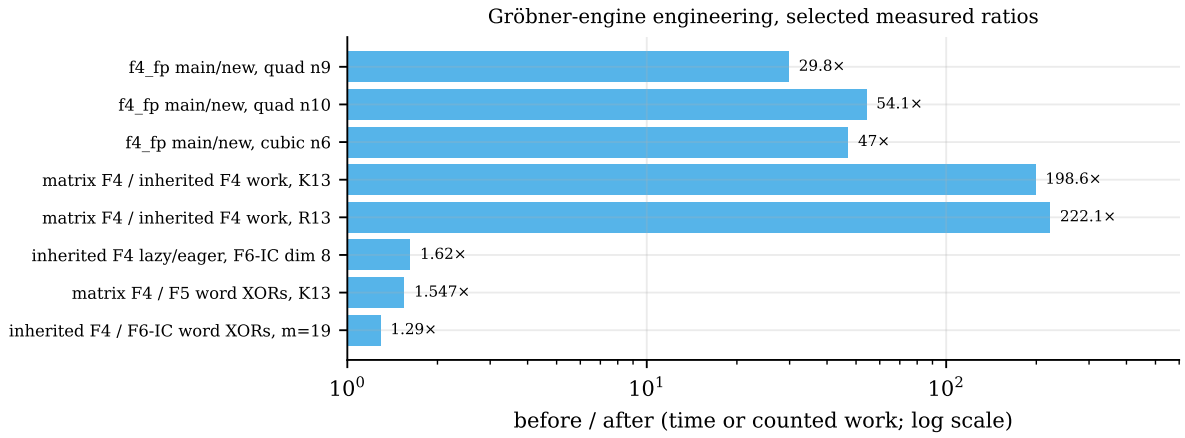


Figure 10.1: Selected engine ratios (before/after or baseline/candidate): `f4_fp` successive versions (single-threaded wall, L0); matrix versus inherited F4 (counted matrix work); lazy versus eager inherited rewrite (median wall); F4 versus F5 and inherited F4 versus F6-IC (word XORs, stage only). Sources in `figures/data-program.json`.

11 Measuring an engine

Chapter 4 describes the instrument as it was built: the unit S, sealed records, replay, isolation levels, the paired statistic and the pricing band of §4.3. Chapter 33 describes how the rho references inside that instrument were calibrated and corrected. This chapter is about the other side of the same instrument: what it takes to measure a decomposition engine, a linear-algebra stage or a whole index-calculus pipeline so that the number survives a replay and a second reader. Between 2 October and 5 November the harness changed its accounting twice, its statistics once, and its notion of a passing audit three times. Each change moved numbers that earlier chapters quote, and each is stated here as what it is: a correction of the instrument, not an improvement of any method.

Three distinctions run through this part of the treatise.

- A *component speed-up* lowers the cost of one stage (a Macaulay build, an echelon, a lookup) with identical outputs. It is measured in the stage's own unit—word XORs, multiply-adds, instructions, stage wall time—and says nothing about the distance to rho until the stage is priced inside a whole solve.
- An *end-to-end change* moves the cold one-target S of a complete pipeline, paired against the same pipeline before the change. It is an improvement under the programme's success criterion when it is verified and measured; it is a frontier move only when it also changes the order against the matched rho across the band of §4.3.
- An *accounting change* moves a number without changing any algorithm: a counter that was not charged is charged, a reference is resized, an interval is recomputed. It is never an improvement, even when it moves the distance to rho in the method's favour.

Every result in Chapters 11–14 carries one of these three labels.

11.1 What the calibration does and does not cover

The calibration study of 2 October (§33.1, Table 33.1) is the harness's only test against a closed-form constant: seven generic methods reach their analytic constants up to $r = 2^{25.35}$ on prime curves and 2^{39} on Koblitz curves. For an engine this has two consequences.

- The counted charge of every generic reference used below is calibrated in its own unit up to those sizes. A ratio of an index-calculus S to a rho S therefore has a calibrated denominator at toy size, and an uncalibrated one above 2^{39}.
- No index-calculus method has a closed-form constant, so no engine in this part is calibrated in the same sense. An engine's charge is checked only by replay (the same counts on a second run, host or instruction set) and by the independent recomputations named `VALIDATION.json` below; neither says that the unit is the right price for the work.

The second point is why the band of §4.3 exists: native work (word XORs, conflicts, lookups, matrix entries) is priced at 0, 0.1 and 1 GAE per operation because no calibration can price it once for all hosts, and two arms are ranked only where the order is the same at all three prices.

11.2 Accounting version 2

PR #2465 (23 October) gave every frozen method identifier a versioned twin and changed what each twin charges (`research/ecbench_accounting_v2_20261023/REPORT.typ`; preregistered, four dated amendments). The study's own first sentence classifies it: an accounting and instrument correction, not an algorithmic improvement and not a speed result. Its effect on the strong Koblitz reference is recorded in §B.3; its effect on engines is recorded here.

- *The verification phase.* Every v2 identifier charges one final check $[d]G = Q$; `ic.large_prime_v2` also charges the membership check $[r]Q$. On matched (workload, round, seed) pairs the v2 total minus the frozen total equals the verification phase exactly on 40 of 40 pairs for each of six generic methods and 16 of 16 for `ic.large_prime`, and every other phase is equal. The preregistered outcome that the phase costs $1.5 \log_2 r$ failed for every record: it is the exact double-and-add count of the planted scalar (19 to 29 GAE here), so its size follows the scalar. The post hoc replacement held on 240 of 240.

- *What it costs.* On these toy curves a fixed 19–29 GAE sits on totals of 235–1,500 GAE, so v2 identifiers cost 1.7–8.8% more S (for example 1.022 $[1.021, 1.024]$ for `bsgs.textbook` and 1.0003 for `ic.large_prime`). The study states that this is a sizing statement for these curves and not for larger r, since one scalar multiplication is $O(\log r)$ against $O(\sqrt{r})$.
- *No frozen count moved.* The same frozen-only specs run on the branch binary and on the main binary agree on every count field of 504 of 504 matched records. Two committed sessions fail a replay identically on both binaries, in their root-finding and inherited-F4 arms only (96 and 16 problems): a pre-existing condition, not caused by the change, that CI's `-replay 0` audit does not see.
- *The distance moves the other way.* The index-calculus arms of the committed claim specs are unchanged; their reference became cheaper. The best arm across the band, `ic-rootfind-m2`, reads 10.6–11.0× the v2 strong reference at $n = 13$ (against 2.0–2.3× the frozen one) and 34–49× at $n = 19$ (against 10–11×). The report states that the corrected reference makes the measured gap larger.

The v2 floor rule (`automorphisms/v2`) also counts six automorphisms on $j = 0$ and four on $j = 1728$; under the first rule a cubic-automorphism rho read 0.720 $[0.591, 0.867]$ of a floor computed with two automorphisms, which is impossible for a correct floor. That correction belongs to the rho side (Chapter 33); for engines its lesson is that a floor is part of the instrument and is versioned with it.

11.3 Statistics version 2: censoring and the wall statistic

PR #2442 replaced the comparison statistic; §33.8 gives the interval widths, the coverage simulation and the censored-rho rows. Three of its changes bear on engines specifically (`research/ecbench_statistics_v2_20261105/README.md`; no new runs, the old and new binaries run on the same committed records).

- *Censored engine runs are rows.* An engine that times out or exhausts its budget now enters the ratio as a censored lower bound. On the yield sweep the two descent-SAT arms over strong rho move from 1.351 (9 of 16 pairs) to 1.429 $[1.241, 1.646]$ with 7 censored, and from 1.892 to 2.486 $[1.442, 4.288]$ with 4 censored; both are lower bounds. The point estimate over verified pairs is unchanged in every row, so the old binary and the new agree where both define a number.
- *Wall time per operation, not per solve.* The v1 headline wall statistic was the median per-pair solve-wall ratio, whose interval was often a single point. The v2 headline is the ratio of nanoseconds per GAE. The two answer different questions and can disagree in direction: on the yield sweep, `sub8-mitm` over strong rho reads 20.26 as a median wall ratio and 0.627 $[0.489, 0.804]$ in ns per GAE, because the arm does 45× the operations at a lower cost per operation. The descent-SAT arms read $5.1 \cdot 10^7$ and $8.8 \cdot 10^6$ ns per GAE: their solver work is unpriced, and the harness now flags them as upper bounds on that statistic.
- *The tail flag needs runs.* A heavy-tail flag (largest run above 25% of an arm's total) fired on 71 of 71 table rows with four runs and on 6 of 6 with two: with n runs the largest share is at least $1/n$, so at $n \leq 4$ the flag carries no information. The same PR made the threshold depend on n.

The engine-side lesson is the first item. A decomposition engine that times out on a hard target is not cheaper on the targets it finished; a comparison that drops it reads the engine as cheaper than it was. Every engine ratio quoted from a session with censored runs before 5 November is a lower bound on the engine's cost, and some (`ic-sub6-mitm`, $6.98 \to 648$) were understated by two orders of magnitude.

11.4 The allocator confound

A runner change (PR #2455) reported its index-calculus online window "about 15% lower". A preregistered study of the same day (`research/ecbench_allocator_confound_20261023/REPORT.md`; ABBA order, three scratch ablations) measured what the change did.

- *Counts are identical.* Ten base-versus-variant session pairs match on every count-bearing field of every record (72 of 72 and 66 of 66 per pair); a tampered copy is flagged by the same script.
- *The small-set arms' wall time collapses.* Eight rho, BSGS and kangaroo arms on a 20-bit prime curve read pooled medians of 5.26–6.77 ms on the merge-base binary and 0.137–0.330 ms on the PR binary, 18.4–38.8× shorter; the index-calculus `ic-mitm` arm (about 1.9 ms) does not move (ratio 1.03).
- *The online window is not distinguished.* On the Koblitz $n = 17$ claim spec the IC online median moves from 127.1 to 119.1 μs (ratio 1.067; 1.135 and 0.983 by order). The "15% lower" is not reproduced; the preregistered verdict is inconclusive.

- *Attribution.* Removing `malloc_trim(0)` alone, or the new pre-solve `/proc` reads alone, leaves the PR binary fast; removing both returns every arm to the merge-base level (0.75–1.05). Either consolidates the allocator before the clock starts.

The study measured the distance to rho on the same claim spec, which no binary changes: the band gives the IC arm 16.85, 17.28 and 21.16 against the strong rho's 4.00, 4.14 and 5.36 at $w = 0, 0.1, 1$, so IC is 4.21, 4.17 and 3.95× the strong rho, and a plain signed-Frobenius rho (1.54–2.48) is cheaper than both at this size because the strong walk's set-up dominates at $r = 2^{16}$. For engines the lesson is that wall time on sub-millisecond solves measured the allocator; the PR's own grader now refuses wall time below 1 ms of solve.

11.5 Sessions that cannot be cited in full

Committed sessions are evidence and are never edited. Two registries record those that cannot pass the audit as committed (`docs/ecbench/`).

- *Incomplete sessions* (`docs/ecbench/INCOMPLETE_SESSIONS.tsv`, since 18 October): four frozen scratch sessions of the $k = 4$ quartic studies, committed without spec and host capsules. Each row pins the SHA-256 of `session.json`, `plan.json` and `records.jsonl`; the CI audit reports them as incomplete evidence, never as audited passes, and fails if a byte changes or a capsule appears.
- *Defective sessions* (`docs/ecbench/DEFECTIVE_SESSIONS.md`, one row, owner-approved 5 November): the interrupted timing smoke of the Curve25519 algebraic ladder. Its `session.json` claims 7 records and a stop at sequence 6, but `records.jsonl` holds sequences 0–3, and the sealed records hash is that of the 4-line file, so the records were lost on the producer's host before sealing. Recovery from Git and from the S3 checkpoints found nothing. The row pins all five capsule hashes and the exact four problem checks (`record_count`, `status_counts`, `last_seq`, `stop`); the audit passes the row only if verify fails with exactly those checks. The session is citable as its four sealed records only (§21.7).

A third class is reported but not fatal: a session whose audit receipt says `records_bound: false` (no whole-file hash binds its records, for example a runner killed before finalisation) is listed as an unbound prefix, not integrity-passed (`.github/scripts/ecbench-audit-sessions.sh`).

11.6 The registry, inventoried

By 1 November the harness exposed 79 top-level methods and 24 nested algorithms behind them: 11 system solvers, 9 three-summand decomposition (PDP3) engines and 4 prime Kummer-tower engines (`research/ecbench_registry_optimization_20261101/README.md`). A native audit (`ecbench optimize inventory` and `audit`) admits a development roster of 114 specs covering 79 of 79 methods and 24 of 24 nested algorithms, with no structural errors or seed leakage.

- *Execution coverage.* 38 fresh L0 sessions (1,095 records, 1,042 verified outcomes) exercised 73 of 73 runnable native methods and every nested engine; 854 of 854 eligible deterministic measured replays reproduced with zero audit problems. Six entries are external CM wrappers without a hash-bound backend and are not executable.
- *A replay that found a defect.* The first pair-index F6 replay exposed an accounting-cache regression; resetting the cache at the independent-run boundary makes the corrected session reproduce 16 of 16.
- *What the count does not include.* A standalone $n = 83$ chained-S_3 lookup (§12.8) and a `chained-x` decomposition oracle registered in `ic bench` are outside the 79+24, because neither is ecbench-selectable; the README lists wiring and pricing them as a coverage gap, not an assumed result.

The inventory changes no distance. Its use is negative: an engine that is in the registry has been run at least once under the same accounting as rho, and an engine that is not has not.

11.7 The end-to-end frontier refresh

The refresh of 31 October and 1 November (`research/ecbench_e2e_frontier_20261031/README.md`, `research/ecbench_e2e_frontier_20261031/REPORT.md`) bound the frontier to one source commit, inventoried every live method and every nested engine, and prepared a 92-shard isolated campaign that did not run (no checkpoint credential; the blocker is recorded). Its answer is a boundary statement, with the domains kept apart as Definition 4.1 requires.

- *Generic prime, random binary and medium Koblitz ladders are rho-led.* No admitted cold one-target index-calculus result in those panels is below its strongest matched reference. The closest markers are 2.28× rho for inherited F4 on one tiny random-binary cell, 2.59× for the prime residual collector at 36 bits (§22.3) and 3.43× on Table A's rule-v7 Koblitz ladder, whose rho reference was 0.57–0.85 of the strong walk per step, so that figure understates the gap by about 1.7–2.0× (§15.6, §17.4); the report states that these are different studies and units, not entries of one table.
- *Engines on the current binary.* The pair-index F6 reduces inherited-F6 work to 0.2792 $[0.246, 0.317]$ and is still 139.30 $[85.48, 227.02]\times$ strong rho end to end; the prime PKM F4, F5, F6 and enumeration engines are 4.62–5.03× negation rho; the best matched $n = 17$ system solver is wide FES/Gray code at 4.28× strong rho. All are one-target L0 operation comparisons with unpriced work.
- *Historical engine counts are bound to their binary.* A fresh replay of the F4/F5/F6 cross-family sessions on the current binary reproduces 72 of 144 deterministic rows, while the optimised-rho session reproduces 720 of 720. The engine claims of Table 10.1 are therefore statements about the binary that produced them.
- *One restricted family is IC-led.* The plane-quartic attack on the GHS-weak family $E_0/\mathbb{F}_{q^3}$ with $\#E_0 = 4\ell$ is below matched rho in its own comparison (§26); the report keeps it out of the generic ranking because it is a different problem class.

The $n = 29$ crossover is Frobenius-folded BSGS. One panel does cross below rho, and its classification is the instructive part (`research/ecbench_generator_orbit_20261022/README.md`; protocol frozen before data). The study measured an orbit-layout pair table and a generator-orbit factor base on seven Koblitz curves ($n = 13$–37). The layout table finds the same relations as the lazy table on 960 of 960 run pairs at 1.02–1.09× lower cost; with the generator-orbit base and stride the net gain over the lazy baseline is 1.26–3.20×, with the order holding across the band—an end-to-end improvement over an index-calculus baseline. At $n = 29$ and one factor-base column the IC arm reads $S = 0.546$, 0.611 and 1.196 at $w = 0, 0.1, 1$ against 1.569 for `rho-frob`, a paired ratio of 0.366 $[0.241, 0.523]$. The directory carries a `VALIDATION.json` with a replay certificate and a passing validator report (a validation-gate pass; no pcc claim, so an unverified assertion under the 4 November rule), and its own scope statement says what the crossing is.

- At one column the pipeline is a Frobenius-folded baby-step giant-step with a known generator-orbit column, not a factor base with unknown logarithms; in the language of §32.2 it has $F = 0$.
- The crossing rests on `rho-frob`'s fixed 133-GAE set-up at $\log_2 r = 15.4$. The gap to rho grows with r: 1.15× at $n = 31$, 1.93× at $n = 23$ and 3.89× at $n = 37$; at $n = 17$, 23, 31 and 37 `rho-frob` is ahead at every weight, at $n = 13$ the pair is unresolved, and at $n = 19$ the band cannot re-price the IC natives (no pinned calibration row).
- In the validation file's like-for-like band, which prices the same native operation types on both arms, rho rises from 1.503 to 1.58 at $w = 0.1$ while the IC arm reads 0.413, 0.478 and 1.063; the crossing is robust to that choice at this size and does not survive growth in r.

The refresh therefore keeps the panel as a fixed-set-up engineering frontier and does not pool it with the uniform-target ladder; the 1 November current-binary snapshot reads the same cell at 0.4405 $[0.363, 0.534]$ of signed-Frobenius rho and calls it a holdout candidate, not a family-level or leaderboard result. It is the clearest example in the programme of a validated number below rho that is not an index-calculus advance.

11.8 The cost of the harness itself

A measured child is a fresh `ecbench exec` process, so the harness's own start-up sits inside every wall time it records. Two engineering studies measured that share without changing any counted unit.

- *Registry and alias parses* (`research/ecbench_child_overhead_20261024/README.md`; Callgrind instructions, frozen protocol). Before #2184 every child parsed the whole 4.7 MB curve registry, about 190 M instructions or 98% of a rho or BSGS child; after it a `rho.negation` child runs 4.26 M instructions (4.64 M on `main`). The index-calculus child had meanwhile grown a parse of the 2.8 MB alias table to match one curve name against a 25-row calibration table: 169.47 M instructions on `main`, 9.86 M after the fix. Median spawn-to-exit wall time per child fell from 37.20 to 2.92 ms after #2184 (0.0786 $[0.0769, 0.0802]$); the alias fix's own wall ratio, 0.9910 $[0.9754, 1.0083]$, is not distinguished.
- *Inversion in the default IC oracle* (`research/ecbench_ic_asm_euclid_20261103/README.md`). A 32-bit extended-Euclid inversion in x86-64 assembly lowered instructions per GAE of the default

`mitm:negation_folded=1` oracle from 537.0 to 476.8 (−11.2%) and of the prime-batch arm from 427.2 to 397.7; the target had been a halving. Every counted field of every record is identical to `main`'s.

Neither changes an S. Both matter for the wall-time column: an index-calculus child that spends most of its instructions finding its curve name, or a rho child that spends 98% parsing a registry, makes an admitted wall-time ratio a ratio of start-up costs. This, with §11.4, is why wall time is admitted only above a minimum solve length and with an A/A interval.

11.9 What a measured engine result must carry

The corrections above add up to a contract that every result in the following chapters is read against. An engine result is citable as a component speed-up when it carries identical outputs on frozen inputs, a same-binary A/A, a preregistered gate, and its own unit. It is citable as an end-to-end change when, in addition, it is a sealed ecbench session (or an equivalent native record) with every phase charged or explicitly unpriced, censored runs kept, and a paired comparison against the pipeline it replaces. It is citable against rho only when the rho arm is the strongest available for the curve (Chapter 33), measured in the same process and unit, and the order is stated per band weight. It is *validated* only where its directory names a `VALIDATION.json` whose validator report passed; everything else in this part is a producer-reported figure, which is how the text marks it.

12 Boolean decision engines

Over a binary field the decomposition oracle receives the Weil restriction of a summation polynomial: a Boolean system in N unknowns whose solutions are the decompositions of one target over the factor base. The index calculus needs a *decision*—a witness or a proof that there is none—not a Gröbner basis, and almost every target has none. Chapter 10 describes the four engine families and Table 10.1 lists their headline engineering ratios. This chapter gives what that chapter compresses: the generic Boolean search campaign that preceded the algebraic engines, the gate methodology of the $\mathbb{F}_2$ micro-studies and the variants it refuted, the inherited-F4 rounds after 5 October, the F6-IC engine's convergence to the meet-in-the-middle oracle, the crossbred and SAT engines, and the $n = 83$ engine campaign. One observation recurs: as an engine is made faster on index-calculus systems it tends to turn into an enumeration with a table, and the table is the generic oracle whose cost Chapter 32 bounds from below.

12.1 Generic Boolean search, 22 September to 3 October

Before the descended Semaev systems were fed to an algebraic engine, a campaign of standalone studies measured complete generic Boolean solving on generated public systems of up to 36 variables: exact specialisation, affine propagation, fixed branch selection (most frequent variable, zero first) and exhaustive search below a node cap, with node and resource limits returning UNKNOWN and never UNSAT. None accepts a curve or a target; every result is a component result on its own fixture distribution. Each study froze its protocol, held out seeds, rotated arm order, and compared a candidate against the *pointwise fastest* of all retained methods rather than against the method it replaced. Most preregistered a universal 2× gate (every size and family group with a paired 95% lower bound above 2), which almost nothing passed. Table 12.1 lists the outcomes.

Two results of this campaign shaped what followed. First, at 24 variables the degree-3 Macaulay kernels were slower than plain search: on the packed residual study's planted holdouts the flat kernel took 253.1 ms per solve, the sparse-bucket kernel 1,050 ms and search-only 53.8 ms, against 12.0 ms for packed residuals (`research/boolean_quadratic_frontier_20260923/CONCLUSION.md`). On small systems a closure that sees further is more expensive than a search that looks less far. Second, every accepted gain was a faster representation of the same search tree: node counts, logical counters and canonical trace digests are identical across arms, so the campaign moved constants and never an exponent. The lead atlas records the same reading for every Boolean engine (`crypto-autoresearcher: knowledge/leads/ecdlp/ic-engine/LEAD-ICE-boolean-f4-f5-f6-engines.yaml`).

12.2 The $\mathbb{F}_2$ micro-studies and their gate

About thirty matrix-F5 and Boolean F4 variants were tested between 26 September and 3 October against one fixture: the $n = 24$, degree-4 complete F5 call of a seven-case suite (`f5_n24_m24_d4`), and the `n20_m30` Boolean F4 workload. The gate was the same in every study, and it is why most variants failed.

- The protocol and an unmodified baseline build were committed before candidate code or timing; the candidate was opt-in and was removed again if it failed.
- One release binary ran both arms. Each selected block had two warm-ups, five reference/reference A/A pairs and five alternating A/B pairs, 22 processes and all seven cases per process; the first isolated exact block per seed was selected before its timing was read.
- Every arm had to match raw and canonical row fingerprints, rank, terms, columns, builder and criterion counts and reduction word operations on every case.
- Ratios are medians of the five paired ratios, with exact 3,125-resample bootstrap intervals; a candidate advanced only with a frozen one-thread complete-call ratio of at least 1.05 and a lower bound above 1.02, and no smaller case below its own A/A minimum.

Table 12.2 lists the refuted variants with the ratio that refuted them; the accepted ones are in Table 10.1.

What passed, and its size. The one complete-call result above 2× is the support cut at column 19: a CI recheck on the same four seeds passed every cell, with one-thread medians of 3.006–3.049× (lower bounds 2.970–3.024) and two-thread medians of 2.415–2.476×, the candidate's counted reduction XORs

Study	Result (paired, against the fastest retained method)	Verdict
Packed direct construction	construction only: 3.48–4.53× (primary), 3.55–4.90× (confirmation)	passes 2× on construction; no solve claim
Construction plus reduction	charging the full RREF, the staged path is 1.38–1.50× faster at $n = 36$ than either streaming candidate; candidates pass 0 of 9	both rejected
Fixed quadratic state	2.40–2.51× on 9 of 9 primary comparisons; confirmation 2.10–2.55×, 7 of 9; 6,144 verified solves	engineering; same search tree
Packed residual coefficients	1.56–1.67× against the fastest of 8 methods, 18 of 18 lower bounds above 1; 5,832 observations on 72 systems	incremental gate passes; 2× rejected
Linear tail ($W \cap L$ kernel)	3.59× [3.57, 3.62] on the large-active mixture; four small cells 0.964–0.975	guarded portfolio rejected
Whole-solve traces	degree-3 hybrid kernel slower than the flat kernel with identical traces; search-only often faster with more nodes; specialisation gain 1.04–1.14×	2× rejected
Row-space transport	full-RREF paired ratios 0.199–2.667 (3 of 18 groups); affine-tail 0.235–0.952 (0 of 18)	rejected
Affine substitution, SIMD Gray	16/18 and 15/18 groups above 2	universal gate rejected
Sibling pivot-trace screen	0 of 64 eligible $n \geq 12$ branch pairs keep a replayable pivot trace	rejected at the structural gate

Table 12.1: Generic Boolean search on generated systems, September–October. All complete-solve timings, correctness pass on every row. Sources, top to bottom: `research/boolean_packed_construction_20260922/`, `research/boolean_construction_reduction_20260922/`, `research/boolean_quadratic_state_20260923/`, `research/boolean_quadratic_frontier_20260923/`, `research/boolean_linear_tail_20260923/`, `research/boolean_solve_trace_20260923/`, `research/boolean_basis_transport_20260923/`, `research/boolean_affine_substitution_20260923/`, `research/boolean_pivot_screen_20261003/` (each `CONCLUSION.md` or `README.md`).

being 28.08–28.10% of the reference's (`research/f5_support_cut19_final_recheck_20261003/RESULT.md`). The README calls it a recheck, not an independent second holdout. Its predecessor, the support split, cut reduction word operations to 33.7% but its ARM64 timings were non-promoting, with one seed's lower bound at 0.966 (`research/f5_support_split_call_20261002/RESULT.md`), and the mainline rank port read 1.450× with an interval of 0.582–1.664 (`research/f5_mainline_port_20261002/RESULT.md`).

Defaults, 17 October to 29 October. Three switches that had passed their incremental gates on x86 CI were left opt-in because their protocols tied a default change to the 2× gate. A decision rule committed before the deciding run made one of them a default: direct F5 packing (1.006–2.173× on every F5 cell); direct unpack stayed opt-in at 0.986× on one four-thread cell, and table reuse at 0.955× on a matrix-F4 cell, and with all three on the largest F5 cases ran 1.11–1.27× faster than the previous defaults (`research/f5_gf2_defaults_20261017/README.md`; L0, round-to-round variation up to ±25%). On GitHub's Arm runners (Neoverse-N2 and a virtual Apple M1) every x86-chosen default stayed: the lazy rewrite with the bit scatter, Arm's path before #2043, takes 1.08–1.69× more CPU on Linux; turning gate-first F6-IC off costs 1.69–2.10× on Linux and 1.62–2.15× on macOS; and no Gray-code table count beats four (`research/arm64_f4_defaults_20261022/README.md`). The follow-up of 29 October tried four more candidates and landed none: a four-row multiply-accumulate pass cut the $\mathbb{F}_p$ kernels from 976 M to 633 M instructions with no wall gain (0.88–1.08 per site), because the loop is bound by reading pivot rows; a column-tiled parallel B' build was 0.87–1.02× the control; and F5 direct unpack "where it wins" passed its rule on one output mode while identical code on the degree-3 cases read 1.087 against 0.933 across runs, a disagreement larger than the effect, so it stayed opt-in (`research/f4_f5_followups_20261029/README.md`).

Reading. The campaign's own standard is the right one: a stage gain is an engine result, the complete call is the unit that matters for the oracle, and the oracle is a minor share of an index-calculus solve whose relation phase is dominated by the number of targets tried. None of these studies, accepted or refuted, changes a distance to rho.

Variant	Complete-call ratio (reference/candidate)	Why it failed
AVX2 gather unpack	0.9652 [0.9630, 0.9874] (1 thread, Intel), 0.9140 (2 threads, AMD)	slower
AVX2 nibble compaction	0.912 (0.906–0.925)	slower
Nibble-subset unpack	0.649 (0.646–0.650)	slower
AVX-512 unpack	new over prior 0.996–1.016	no gain
POPCNT unpack	1.0005 (0.9944–1.0067)	below 1.05
Compact paired unpack	0.9998 (0.9986–1.0045)	below 1.05
Bounds-free unpack	1.004, 0.982 (ARM64 screen)	below 1.05
One-pass unpack	0.992, 0.953	below 1.05
x86-guarded direct unpack	0.9984 (0.9927–1.0226)	below 1.05
Contiguous (flat) rows	0.978, 1.002	below 1.05
Byte-colex packing	1.0229–1.0262; build phase 1.46×	below 1.05
Initial-density row order	1.006, 0.987	below 1.05
Bounded-weight pivots ($K = 8, 16$); stratified pivots	terms 0.964–0.984 of reference (bounded)	structural gate (both)
Always-echelon output	large case 1.50–1.55; `n12_m12_d4` 0.821–0.830	smaller-case regression
Fused row build, 4 threads	build 1.382, complete call 0.865 (0.745–1.095)	4-thread gate
Boolean F4 unsorted products	0.857, 0.876	slower
Boolean F4 degree-bucket queue	0.993–1.008	below 1.05
Boolean F4 six-column tables	primary 1.074; holdout `n20_m20` 0.881	smaller-case regression

Table 12.2: Refuted $\mathbb{F}_2$ engine variants, 26 September–3 October: complete F5 call (`n24_m24_d4`) unless marked F4 (`n20_m30`), frozen seed unless a range is given. Each source is `research/<study>/RESULT.md`: `f5_avx2_gather_unpack_20260929`, `f5_avx2_nibble_unpack_20260929`, `f5_nibble_subset_unpack_20260929`, `f5_avx512_unpack_20260929`, `f5_popcnt_unpack_20260929`, `f5_compact_unrolled_unpack_20260929`, `f5_bounds_free_unpack_20260930`, `f5_one_pass_unpack_20260929`, `f5_x86_guarded_unpack_20260930`, `f5_flat_rows_20260929`, `f5_byte_colex_20260930`, `f5_initial_density_order_20260929`, `f5_bounded_weight_pivots_20260930` and `f5_stratified_weight_pivots_20260930`, `f5_echelon_20260927`, `f5_row_build_20260928`, `boolean_f4_unsorted_products_20260926`, `boolean_f4_degree_queue_20260927`, `boolean_f4_wide_tables_20260926`.

12.3 Inherited F4 after 5 October

Inherited F4 has been the default algebraic oracle since 22 September (Chapter 10). Its later rounds were of four kinds.

Instruction rounds. Two preregistered rounds lowered the machine work behind the counted unit without moving the unit. The overhead round ran 0.847× the baseline's instructions on the confirmatory $m = 19$ inputs (threshold 0.90), with 48 of 48 frozen outputs identical; inherited F4 went from about 120 to about 95 instructions per counted word XOR (`research/inherited_f4_overhead_20261005/RESULT.md`). The remap round read 0.923× (threshold 0.97; about 95 to 89 per word XOR) but was superseded before merge by a more general word-wise rewrite on `main` (`research/inherited_f4_remap_20261007/RESULT.md`). The eager rewrite through shift runs then brought hosts without BMI2 onto the eager path: 1.22–1.70× lower median CPU on six of nine cells against the bit scatter, and the two architectures charge the same units again (`research/inherited_f4_eager_runs_20261016/README.md`). Because an eager rewrite charges every kept row, it charges *more* word XORs than the lazy one while using less CPU: a reminder that word XORs are a stage unit, not a cost.

A batching idea that lost. Replaying one F4 trace on several specialisations at once, after Demin's Groebner.jl tracer, was exact but used 2.2–4.0× more stage units than the sequential root and more CPU (`research/lit2026_w5_lane_batched_f4_20261011/README.md`). The same study found the gate-first order for F6-IC (1.57–1.78× lower stage unit), which became a default after an x86 check on nine cells (1.336–1.733× fewer word XORs, 80 of 80 targets decided identically; `research/f6_gate_first_x86_20261016/README.md`).

Online, against rho. The first public-target panel (one registered point, prepared 62-point base, three fixed process orders) put F5 at a median 208.49× (181.29–223.86) inherited F4's online time and inherited F4 about 315× slower than the rho reference online; both engines spent over 99.9% of the

Curve	dim V	default/off	95% interval	targets up/down	S/frozen strong rho, off→default
$n = 11$, K1	4	1.0958	$[1.0940, 1.0980]$	24/0	$7.17 \to 7.86$
$n = 13$, K0	5	1.0699	$[1.0646, 1.0759]$	24/0	$36.9 \to 39.5$
$n = 17$, K1	6	0.9969	$[0.9955, 0.9985]$	6/18	$424.1 \to 422.8$
$n = 19$, K1	7	0.9593	$[0.9576, 0.9611]$	0/24	$1{,}248 \to 1{,}198$

Table 12.3: Inherited F4 at $m = 3$: total S with the root linear-row harvest on (the default since #1862) over off, 24 paired public targets per curve. At $n = 19$ (K0) the word XORs are unpriced and S is flat; priced at 0.1 or 1 GAE the ratio is 0.9755. Source: `research/f4_harvest_remeasure_20261024/README.md`.

online window in `target_PDP` (`research/ic_solver_online_20261003/RESULT.md`). A capture defect (the wrapper's `time` exited 1 on the sandboxed host) makes the panel diagnostic only. Its useful datum is the split count: F5 used 2,546 splits on the first, unsuccessful target attempt against inherited F4's 68, which is why a matrix-kernel gain cannot be projected onto the online path.

The harvest default. PR #1862 made the root linear-row harvest the default; on the one workload a pricing bisect saw it raised counted word XORs by about 7%. A preregistered re-measure on five curves (`research/f4_harvest_remeasure_20261024/README.md`; five sessions of 144 records, 144 of 144 replays identical, counts only) found that the sign depends on size (Table 12.3). The $m = 2$ negative control reads 1.0000 on all five curves.
The default is a regression at $n = 11$ and 13 and a 4.1% engineering gain at $n = 19$; at $n = 17$ the whole gain comes from a different gate (#1891's root-rank gate). Every priced F4 arm sits 7 to 1,250 times the frozen strong rho, and the harvest moves that by at most 10%. The correction is listed in Chapter B.

Determinism. The cross-host drift of §4.5 (item 6) was first searched for within one host: thread count, a second process with new hash seeds and a replay under a different thread count all gave byte-identical word counts on 48 of 48 rows (`research/inherited_f4_replay_determinism_20261018/README.md`), which excluded the within-host sources and left the instruction-set path, later confirmed as the cause.

The priced distance. With its word XORs priced, inherited F4 is about 22–33× the frozen strong rho at $n = 13$, and the complete pair table is 17–22× cheaper than F4 on the same cells; the atlas records the whole Boolean-engine line as *closed, dominated* by the pair-table oracle on Koblitz and random binary curves up to $n = 31$, $m \leq 4$ (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-engine/LEAD-ICE-boolean-f4-f5-f6-engines.yaml`). A later diagnostic on a `chained-x` oracle read inherited F4 at 24.71, 144.09 and 794.37× matched rho at $m = 2, 3, 4$ on Koblitz $n = 13$; matrix F5 cut matrix F4's work at $m = 3$ to 0.6276 and still cost 16,296.76× rho, and both $m = 4$ matrix arms were censored (`research/ecbench_registry_optimization_20261101/README.md`, `research/ecbench_e2e_frontier_20261031/REPORT.md`). The atlas also records that on the S_3 chain F4 and F6-IC grow faster in n than the meet-in-the-middle floor at $m = 3$ and $m = 4$, with the slope-difference interval above zero.

12.4 F6-IC: from closure to lookup

F6-IC adds to inherited F4 a factor-base domain gate and meet-in-the-middle leaves (Chapter 10). Its record after 5 October is a single argument in three steps.

Step 1: the closure is charged, and loses. On the E_0 ladder (eight public targets per size, 72 verified runs per arm and size, word XORs counted but not charged, geometric additions charged) F6-IC's S was 54.75× (37.81–84.22) the same-target rho at field degree $n = 13$ (the source writes m for the degree) and 2,091× at $n = 19$, against inherited F4's 1.105 and 1.582; at $n = 23$ the two IC arms were 2.72 and 2.09×, and at $n = 31$ both hit the 3,600-second cap on the first target, so the protocol reached no decision (`research/f6_ic_ecbench_ladder_20261005/RESULT.md`). The frozen strong rho of that ladder carried its set-up (§11.2), and the inherited-F4 figures are the unpriced lower bounds that §4.5 (item 5) corrects.

Step 2: the time is in the matrix build, not the gate. Online on the prepared $n = 17$ curve, rho was faster on both frozen public points: rho/IC online 0.0219–0.0221 on T1 and 0.000515–0.000530 on T7 (`research/f6_ic_rho_online_gap_20261006/RESULT.md`). A nested profile of T7 found the Macaulay stage at 81–84% of target PDP for all three arms, of which the matrix construction was 98.5%, and the F6 oracle callback at 1.6–2.3%; eliminating every nanosecond inside the oracle would save about 1.6% of the online window (`research/f6_ic_phase_profile_20261006/RESULT.md`).

Step 3: the few-candidate closure makes the algebra vanish. On 5 November the gate was changed to run the exact one-fixed closure as soon as a summand's assigned bits leave at most 64 base codes, deciding the whole subtree; the change is now on by default (`KIC_F6_FEW_CANDIDATES=64`; `=0` is the same-binary control) (`research/f6_few_candidates_20261105/README.md`; nine Koblitz cells, 5×3 rounds, L0).

- Verdicts and indices are identical in both arms (80 target–engine pairs); F6-IC process CPU falls 2.67–10.29× (median of five; worst round pairing at least 1.66), word XORs 1.84–10.13×, and nodes from 204–1,049 reductions per cell to 5–16. Inherited F4, with no gate, is unchanged (0.87–1.05× CPU).
- The gate's pair lookups and group additions are *identical* with and without the closure. On these cells the algebra was not ruling out a single candidate abscissa: F6-IC with the pair index was the pair-index meet-in-the-middle scan wrapped in Gröbner work.
- The README states the conclusion, and marks it unverified: the change does not beat ecbench's `mitm` oracle, because in the limit it is that oracle.

Two companion items of the same study bound what remains. Re-reducing a node's displaced rows (36–61 median per node, 7.9–24.5 pivots crossed per row) by Four-Russians batching would spend about 32 row operations per pivot on its tables, several times the roughly 1,500 row XORs per node it replaces, so nothing was changed. A parallel split of one target's tree into 2^k subtrees is 1.59–2.88× faster in wall time on refuting cells and 0.94–1.11× on cells where most targets decompose, at 1.12–1.49× (refuting) and 2.67–2.86× (decomposing) the counted work, and stays opt-in because it changes the count.

On the current binary the pair-index F6 is at 0.2792 of inherited F6 and still 139.30× strong rho (§11.7). The F6 line is therefore an engine that became a lookup, and a lookup is priced by §32.2.

12.5 MutantXL closure as an engine

The closure instrument `dreg_closure` (Chapter 14) is also a decision engine: MutantXL-style, it multiplies the fallen lower-degree polynomials by variables and re-closes, and it refuted all 92 unsatisfiable draws of 23 cells at degree 3–4 where plain XL needed 5–7 (§16; `research/dreg_closure_mutantxl_20261016/RESULTS.md`). As an engine its costs are those of a dense degree-4 closure: at $(m, n, \ell) = (3, 22, 8)$ one branch has 124,314 degree-4 columns, a scan of 124–331 s and a peak of 10.6 GB (`research/dreg_closure_guess_20261022/RESULTS.md`). At $(3, 19, 7)$ a hybrid guess that restores degree 4 still costs $5.3 \cdot 10^{11}$ to $1.5 \cdot 10^{12}$ word operations per draw (`research/dreg_hybrid_guess_20261024/RESULTS.md`). The closure's value is as an instrument that measures a degree, which is how Chapter 14 uses it; no study priced it as an oracle against rho.

12.6 Crossbred, exhaustive search and engine suites

A matched engine suite (22 September) froze the contract every algebraic engine is compared under: identical targets in one process, systems fingerprinted, timeouts never read as UNSAT, units named, missing values null (`research/ic_framework_engines_20260922/README.md`). Under it the first end-to-end panel with the strategy as the only variable put enumeration far ahead of every algebraic route: at $n = 9$, $m = 3$, enumerate collected 19,696 relations per second against 51.7 (SAT), 33.2 (Gröbner) and 11.7 (crossbred), and the panel notes that relation collection stays $\Theta(2^n)$ with any oracle polynomial in the factor base (`research/crossbred_20260920/e2e_panel.txt`). A cost-sensitive solver selector fitted on matched queries chose pairs every time; the hindsight oracle and the best constant policy had identical cost, so the portfolio offered zero routing headroom (`research/ic_solver_selection_20260915/README.md`). The current-binary refresh read the best matched $n = 17$ system solver as wide FES/Gray-code enumeration at 4.28× strong rho, with SAT-CDCL, crossbred, exhaustive search, every F4 variant, F5, F6 and every root-finding variant rho-led in their cells (§11.7).

Arm	mean cold S	S/strong rho	class
Strong signed-Frobenius rho	3.998	1.000	reference; native work unpriced
IC, meet in the middle	387.277	96.880	baseline
IC, Frobenius meet in the middle	17.073	4.271	engineering
IC, native-XOR SAT	3.413 (lower bound)	0.854 (diagnostic)	accounting incomplete

Table 12.4: Native-XOR SAT against strong rho, Koblitz $n = 17$, 40 measured runs per arm. The SAT row omits 2,755,966 conflicts in 749 solver calls. Source: `research/ecbench_sat_pdp_20261009/README.md`.

12.7 Native-XOR SAT

SAT engines with native XOR reasoning (CryptoMiniSat-style Gaussian elimination, WDSat [39]) are the literature's fastest point-decomposition deciders. The matched pipeline study (`research/ecbench_sat_pdp_20261009/README.md`) needed three protocols to produce one clean panel, and its history is part of the result.

- The first pilot's Buchberger arm timed out and one SAT replay differed in the low bits of its GAE, exposing a cost-replay defect in the historical `ic.pipeline` method; the failed receipt is kept and the pilot is not used.
- The counted method `ic.pipeline_counted` reconstructs charged cost from counts. Its pilot replayed exactly but hit 56 per-call conflict-budget exits in the measured CNF runs, a missed stop gate; the main panel started in error was stopped and is retained outside every comparison.
- The third protocol dropped only the budget-hitting CNF arm.

Its session (Koblitz $n = 17$, $r = 65{,}587$, eight public targets, five measured rounds; 240 of 240 verified, 200 of 200 measured replays identical) gives Table 12.4.

The SAT row's counted S is below rho, and the README says in so many words why that is not a result: rho's native table work is unpriced too, the SAT row also omits the solver, and a ratio of two lower bounds bounds nothing. The phase split makes the point concrete: SAT charges zero oracle set-up and 31,930 GAE of relations, the same order as the MITM arms' 30,383, while its exclusive online window summed to $4.16 \cdot 10^{10}$ ns against rho's $1.22 \cdot 10^{6}$ ns (L0, descriptive). The atlas adds a derived identity: the CNF-XOR model of the point decomposition is a complete enumerator of V^m modulo S_m, with no pruning above the leaves, so its next step is to price conflicts and fit their growth against $2^{m\ell}/m!$ (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-engine/LEAD-ICE-sat-xor-pdp-oracle.yaml`; lead disposition *open, next step*). The identity is measured directly in §19.5: about one conflict per sorted triple, and no learnable signal. Until conflicts are priced the row is unranked, and statistics version 2 flags such arms as upper bounds on the wall statistic (§11.3). Prime-field SAT and SMT presentations are in §23.6.

12.8 The $n = 83$ engine campaign

Degree 83 was the largest Koblitz target with a working pipeline (§15.5). Its engine history has three layers; the decomposition results after the field-table correction are in §18.2.

October 4–5: F6 components on the $a = 0$ field. About two dozen preregistered micro-studies kept kernel components and failed every ordinary solve (§15.5). The retained components are stage results: a sign-folded pair index stores one representative per pair-sum abscissa and is 1.261–1.618× faster on bases of 258–4,054 points (`research/f6_n83_sign_fold_20261004/RESULT.md`); an ARM64 PMULL implementation of the degree-83 arithmetic made exact queries 6.11–6.87× faster and the index build 1.40–3.05× faster at 258–1,048 points but 0.912× at 4,054 (`research/f6_n83_pmull_xonly_20261005/RESULT.md`). The same days found a correctness barrier: the union-of-charts SAT path held each allowed abscissa in a `u64` and extracted only the first 64-bit word, so abscissae differing above bit 63 were indistinguishable and an UNSAT answer could not prove absence of a decomposition (`research/f6_n83_domain_20261004/RESULT.md`). It was fixed in that path.

The algebraic gates of the same days ended open or negative. On the registered K0 curve a standard dimension-18 subspace gives 261,447 geometric points (130,722 sign-folded columns), and the direct five-summand S_3 chain is exact on a planted control with 339 Boolean variables and 332 cubic equations, smaller than the six-summand chain's 428 and 415, but its root-pruning gate fails (`research/f6_n83_fi`

`vesum_d18_20261005/RESULT.md`). A recursive carry-less (Karatsuba) circuit for the two dense 83×83 products of the middle links passed its encoding gate and the solver timed out (`research/f6_n83_karatsuba_sat_20261005/RESULT.md`). No natural relation came from F6 at $n = 83$.

October 24: every descent engine had been wrong. A summand sweep with every engine on the pinned $n = 83$ fixture (53-bit subgroup) found the same truncation one level lower: `FieldStructure` truncated the field table to one 64-bit word, and every Weil-descent engine—matrix F4, matrix F5, inherited F4, F6-IC, the F6-IC pair index, crossbred, SAT descent in three encodings and MQ-FES—returned "proved UNSAT" on *planted* $n = 83$ relations (`research/n83_msweep_20261024/README.md`). The descent builders now refuse such a field, a regression test pins it, and every earlier $n = 83$ engine refutation is void (Table B.1). On the corrected build:

- chained-S_3 full-width SAT is the only decomposer that runs at every m, and within its 180-second cap it decides only $m = 2$: 17.3 s per public query (4 of 4 refuted) and 2 of 2 planted witnesses recovered. Its derived projection is $2^{75.9}$ s, $2^{75.3}\times$ rho;
- the $m = 4$ root finder solves the complete DLP at $K = 1107$ in 188.5 and 175.2 s against a strong-rho mean of 1.509 s, 116–125× from rho;
- on the degree-61 companion field at $m = 2$, native-XOR SAT is the fastest decider at $D \geq 12$ (0.79–2.41 ms per query) and inherited F4 the fastest Gröbner engine (6.2–33.4 ms); $m \geq 3$ is refused by the 64-variable descent mask.

October 31 to November 1: chained SAT becomes a lookup. Six frozen-protocol follow-ups extended chained SAT to $m = 9$ by putting root-finder tables and a $|B|^d$ meet-in-the-middle lookup in front of it; by the last step no decided target makes a SAT call (Table 18.4, §18.2). The engine became a generic meet in the middle. These are decomposition-stage measurements with no complete DLP and no matched-rho comparison, and the registry counts the lookup outside its 79+24 (§11.6).

12.9 What the Boolean engines settle

Across the four families the record is consistent.

- Every accepted engine change is a component speed-up of the same search: identical verdicts, identical trees or traces, a faster representation. The largest complete-call factor is the F5 support cut (3.0×); most preregistered candidates failed a 1.05 gate.
- Priced inside a whole solve, the algebraic engines are 22–33× (inherited F4, $n = 13$) to 10^2–$10^4\times$ the frozen strong rho, and the gap grows with m (§12.3). Against the v2 reference of §11.2 these distances are roughly 3–5× larger at $n = 13$–19.
- Where an engine became much faster on index-calculus systems—F6-IC with the few-candidate closure, chained SAT with tables and lookups—it did so by becoming the meet-in-the-middle oracle, whose relation cost is bounded below by §32.2.
- Two field-width defects (4 and 24 October) made engines report refutations that were not proofs; the second was found by planted relations and voided every earlier $n = 83$ engine refutation.

The open engine question the atlas leaves is the one the closure measurements of Chapter 14 address: a descended cell where the pair table is infeasible and the algebraic degree stays bounded.

13 Prime and extension-field Gröbner engines

Over $\mathbb{F}_p$ and $\mathbb{F}_{p^k}$ the decomposition system is not Boolean, and the engine is an $\mathbb{F}_p$ F4: `f4_fp` for generic systems, `f4_fp_tower` with a Faugère–Lachartre block for the Kummer-tower systems of §21.2, a signature variant, and for $k = 4$ the weighted Macaulay solves of Chapter 25. Chapter 10 lists `f4_fp`'s headline ratios (29.8–54.1× over its predecessor, msolve parity 0.96–1.22). This chapter gives the arithmetic rounds behind them, the external references, the one study that priced a faster `f4_fp` end to end, the Kummer-tower engine at large primes, the chain-solve levers on the GLS $\mathbb{F}_{p^4}$ line, and F4 trace replay. As in Chapter 12, an engine gain is a component result until a whole solve is priced against rho, and on every line in this chapter rho stays ahead.

13.1 Arithmetic of the $\mathbb{F}_p$ F4

The $\mathbb{F}_p$ rounds used the gate of §12.2: one binary, paired A/B blocks with an A/A floor, exact basis and solution fingerprints, step counts and matrix shapes on 13 cases, holdout seeds, and a preregistered promotion threshold on the complete F4 call of `quad_n8_p65521`.

- *Narrow 32-bit rows* (`research/fp_f4_narrow_20260926/RESULT.md`). The branch-free 32-bit path read 1.113 [1.111, 1.121] and missed its 1.2× gate. Deferring reductions across each row's pivots (mode 2) read 1.641 [1.631, 1.649] on one thread and 1.580 on four, with 2.101× on the overdetermined $n = 12$ case; a default-on confirmation (1.640 one thread, 1.480 four threads) made mode 2 the default for $p \le 65{,}536$.
- *Lazy vector accumulation.* An AVX2 lazy accumulator read 1.059× [1.055, 1.062] over the deferred default and 1.756× over the original 64-bit arithmetic, below its 1.15× promotion gate and with two smaller-case regressions (`research/fp_f4_lazy_avx2_20260928/RESULT.md`); the AVX-512 version read 1.047× and 1.513× on an Intel host and stays hardware-gated and opt-in (`research/fp_f4_lazy_avx512_20260928/RESULT.md`). The two hosts differ, so the two results are not multiplied.
- *Where the time went by 29 October.* On a shared 4-vCPU host, one thread, `f4_fp` was level with msolve 0.6.5 on `quad_n8` (47.8 against 47.4 ms, msolve including start-up) and faster on seven other systems; 44% of instructions on `quad_n9` were in one multiply-accumulate kernel, and at four threads the reduce phase was still 66–84% of the run, with one in-block dependency loop running on a single thread (`research/f4_f5_followups_20261029/README.md`). The two candidates aimed at those findings were exact and did not save time (§12.2).

An independent campaign in `cryptanalysis` reported symbolic preprocessing 2.34× ($fb = 3$) and 1.99× ($fb = 4$) faster on Semaev systems and Boolean F5B-family bases 1.36–1.49× faster, on a shared host under load averages above 40; its README calls this local progress, not a speed record (`cryptanalysis: experiments/groebner-perf-20260924/README.md`).

13.2 External references

Two external comparisons frame what an in-house engine ratio means.

- *msolve.* A day before the parity result of Table 10.1, msolve was 12–24× faster than `f4_fp` on square systems; after the rewrite the ratio is 0.96–1.22 single-threaded, and the follow-up above found parity or better on eight small and medium systems. The parity is on generic square systems, not on the decomposition systems.
- *Magma.* A protocol registered on 13 October, before any Magma output, exports the exact systems behind the repository's Gröbner claims: 26 Kummer-tower systems at $p = 2{,}013{,}265{,}921$, five $k = 4$ systems and eight Boolean root systems of the $n = 17$ search, every one a unit ideal in the repository's runs (`research/magma_f4_comparison_20261013/PROTOCOL.md`). The directory holds the export bundles and the repository's own timings; no Magma output is committed, so the comparison has not been made.

The desk review of 24 October (§B.3) puts these in proportion: on the PKM systems the in-house F4 needed 10^3–10^7 more operations than a brute grid on the same base, so ratios of a root finder or a table

to that F4 (such as those of §9) say nothing about published Gröbner solvers. Solving *degrees* measured with msolve are unaffected.

13.3 A faster `f4_fp`, priced end to end

One study measured what the `f4_fp` rewrite (PR #1717) did to a complete index calculus in which every decomposition is one `f4_fp` solve: Nagao decompositions on a genus-2 cover of $E(\mathbb{F}_{p^6})$, the same seeds and relations in both arms, every logarithm correct (`research/f4_fp_ic_effect_20261018/README.md`). Per decomposition test F4 used 7.7× fewer multiplications and the cover's counted cost 5.0× less (at $p = 53, 101, 251$). End to end:

p, seed	S before/after	S/rho before	S/rho after
53, 1	5.05	4,297	850
53, 2	4.99	1,843	370
101, 1	5.06	981	194
101, 2	5.08	1,897	373

This is the one case in the programme where an $\mathbb{F}_p$ engine gain passed through to the whole solve almost unchanged (5.0× of a per-test 5.0×), because the cover's decomposition is the solve. The S/rho columns are a lower-bound comparison: neither arm emits its unpriced native work, so no band applies. The distance after the change is 194–850×; the cover itself is the subject of §26.

13.4 Kummer-tower engines at large primes

The tower F4 engine held coefficients in `u32`, which left the Kummer tower at 2-adic valuations $v_2(p-1) \geq 40$ unmeasured. A generic-field engine (`research/f4_generic_field_20261015/README.md`; protocol frozen before any measured cell; shared host, L0) ports the same algorithm to any prime field.

- At `Fp32` it reproduces `f4_fp_tower` field for field: 182 rows of every round-8 series with $N \leq 18$, zero differences in degree, width, steps, multiply-adds and every counter; and it is not slower (paired $u32$/generic 1.091 and 1.058).
- Above 2^{31} it is 2.18× ($m = 2$, $N = 16$) and 2.24× ($m = 3$, $N = 12$) faster than the `u32` engine with eager Barrett reduction, and at 45 bits an AVX-512 IFMA backend is 2.57× the 64-bit one; five of five preregistered claims pass.
- The solving degree is independent of v_2 and of the prime's size in every valued cell: Kummer towers at $v_2 = 40, 48, 56$ and at the P-224 prime, and the Dickson torus at P-192 and P-256, give the 31–32-bit degree, width and step count. At $m = 2$, $N = 18$ the multiply-add count is 45,952,850,022 at 45, 52, 61 and 224 bits.

The last point is the engine-side complement of §21.2: the tower's degree is a property of the system shape, and the prime only changes the cost of a multiplication. In ecbench the four PKM tower engines (F4, F5, F6 and geometric enumeration) all run and read 4.62–5.03× negation rho on the current binary (§11.7); the presentations that change the system shape are in §23.1.

13.5 The chain-solve levers on the GLS $\mathbb{F}_{p^4}$ line

Chapter 25 gives the $k = 4$ engineering ladder to 15 October. In parallel a second line of work moved the same problem onto a Klein-four (V_4) invariant line on GLS curves over $\mathbb{F}_{p^4}$ and ran pre-registered experiments E17–E42 (the V_4 line from E22, 18 October), each recorded in the research note `research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md` §9 with its own report directory. On this line the decomposition solve is 97.7–98.5% of S and the linear algebra at most 0.006% (E32, §9.36), so the solve's Macaulay matrix—393×448 at degree 11, 176,064 dense cells per call—is the whole cost. The levers below act on that matrix. Each is priced in the line's own unit,

$$S(g) = \text{counted } \mathbb{F}_p \text{ multiplications} + 97g\,(\text{Macaulay entries} + \text{selection work}), \qquad g \in \{0, 0.1, 1\},$$

which is the band of §4.3 with a native operation priced at $97g$ multiplications, and each is ranked against its predecessor only where the order holds at all three weights on every row.

	Lever	S_A/S_X at 0 / 0.1 / 1 GAE	Rank against A	Standing
E33	B with its selection counted ($1.05 \cdot 10^6$ ops/call)	1.82–1.86 / 0.193–0.197 / 0.149–0.152	unresolved (ahead at 0 only)	validated
	B″ (Markowitz estimate, $\approx 1.27 \cdot 10^4$ ops)	1.538–1.558 / 1.046–1.048 / 0.9465–0.9480	unresolved (behind at 1)	validated
E35	T: B's pivots learned per curve, replayed	1.093–1.252 / 1.020–1.055 / 0.998–1.003	unresolved (6 of 20 rows behind at 1)	producer
E39	F: first-arrival pivots, no selection work	1.457–1.537 / 1.094–1.106 / 1.0115–1.0130	**ahead** on 32 of 32 rows	validated
	S: static order	1.402–1.449 / 1.083–1.091 / 1.0077–1.0086	**ahead**	validated
	H: F with static fallback	1.497–1.542 / 1.096–1.103 / 1.0076–1.0083	**ahead**	validated
E40	V: rows stored from their lead	1.0000 / 1.2518–1.2618 / 1.3629–1.3807	equal at 0, ahead at 0.1, 1	validated
E42b	W: V's tails reserved to the fill (vs V)	1.0000 / 1.022–1.023 / 1.032–1.034	equal at 0, ahead at 0.1, 1	producer

Table 13.1: Chain-solve levers on the GLS/V_4 line, end to end, ranges over the paired rows (20 rows for E33 and E35, 32 for E39–E42b); above 1 the lever is cheaper. E42b is against V, not A. Rho is ahead of every row at every weight. "Validated": the study directory or its `benchmarks/` copy carries a `VALIDATION.json` with a passing independent validator report. Sources: `research/e33_solver_b_pivot_count_20261031/REPORT.typ`, `research/e35_chain_trace_20261102/REPORT.typ`, the research note §9.49 and `research/e39_pivot_rules_confirm3_20261102/`, `research/e40_macaulay_entries_20261102/REPORT.typ`, `research/e42_macaulay_tails_20261103/REPORT.typ`.

The baseline: chain solve and handover ladder. E31 replaced the existing solve (E25's) by its P_2-chain solve with identical decompositions on every call: solver A cut the multiplications per call 2.62–2.69× and solver B (sparsest pivots) 4.99–5.32×. End to end, S_{E25}/S_A is 2.50–2.56× at 0 GAE, 1.39–1.43× at 0.1 and 1.034–1.053× at 1, so A is ranked ahead across the band. B is ahead of E25 at every weight as E31 counted it, but with its selection pass priced (E33, below) it falls behind E25 at 1 GAE and is unresolved (`research/e31_chain_solve_20261024/`; validated, `VAL-20261024-24ef8b`, `benchmarks/20261024-e31-chain-solve/VALIDATION.json`; validation-gate pass, no pcc claim, so an unverified assertion under the 4 November rule). E32 then measured the distance over eleven sizes, $\log_2 n = 22.6$–42.4, 33 paired curves: rho is ahead on every curve at every weight, the best curve is 1,143× from rho at 0 GAE, the fitted slope of the distance is -0.243 $[-0.279, -0.205]$ in $\log_2 n$ (Gaudry's $-1/4$), and the extrapolated crossover order is $2^{85.5}$ $[2^{79.2}, 2^{94.3}]$ at 0 GAE, $2^{91.2}$ at 0.1 and $2^{97.6}$ at 1 (`research/e32_handover_ladder_20261031/`; validated, independent replay of 66 of 66 runs). The crossover is an extrapolation 37–66 bits past the top size, not a measured crossing.

The levers. Table 13.1 lists the solver and storage levers that followed, each against the ranked solver A.

Read across the table, the band does the ranking work it was built for.

- *A gain that exists only at 0 GAE.* Solver B multiplies 1.82–1.86× less than A, but its pivot selection reads the matrix about six times per call. Counted, B and its cheaper twin B′ fall behind A from 0.1 GAE, and the Markowitz estimate B″ falls behind at 1 GAE, where its $1.2 \cdot 10^6$ priced multiplications of selection exceed its $2.3 \cdot 10^5$ saving (E33). The validator report states that E33 claims no improvement across the band.
- *A pivot order that is not a property of the curve.* If B's choices were fixed per curve they could be learned once and replayed, as in Joux–Vitse F4 trace reuse [27]. E35 found that no call's pivot sequence equals its curve's trace and that 997–1,000 of 1,000 sequences per curve are distinct; in

contested columns B's row agrees with the trace in 69.8–88.4% of cases, and another curve's trace agrees as well. The replay keeps 18.8–36.8% of B's saving and is break-even with A at 1 GAE.

- *A gain at every weight.* Taking pivots in arrival order (F) removes the selection pass entirely and still cuts A's per-call cost 1.47–1.55×. E36 and E37 showed F, S and H ahead of A at every weight, but each time a pre-registered kill criterion fired on a control's positive check, so neither ranked them; E38's kill fired at its pilot. E39, the third confirmation, used a control that alters a read pivot row after its last use, no kill fired, and F, S and H are ranked ahead of A and E25 at 0, 0.1 and 1 GAE on all 32 rows, under three alternative pricings as well (validated, `VAL-20261102-E39`; `benchmarks/20261102-e39-pivot-rules/`). At 1 GAE the margin is 0.8–1.3%.
- *Gains that exist only at positive weights.* Storing each row from its lead (V) leaves the arithmetic untouched—41.6% of the 176,064 cells lie left of each row's lead and are never read by the echelon—so V equals A at 0 GAE and is ahead only where a stored cell has a price (E40; validated, `VAL-20261102-E40`). W reserves each row only to the block after its build end and is ahead of V by 2.2–2.3% at 0.1 and 3.2–3.4% at 1 GAE (E42b). The report classifies it as a data-structure constant; its size is set by the per-cell price.
- *A kill by the study's own control.* E42 registered a range of $[900, 960]$ re-allocations per call for its growth control, derived from the wrong counter; G0 made 245.6, the kill fired, and W stayed unranked. E42b re-registered the question, with the range derived from W's own overflow rule ($[200, 300]$; measured 246.3), before any ranking data existed (Table B.4). E42b has no validator report.

Distance. None of these levers is an end-to-end change of the distance in kind. The best rows of the table sit at S_X/rho of $5.15 \cdot 10^3$ (H, 0 GAE) to about $6 \cdot 10^3$ (V and W, 0 GAE), rising to $5.97 \cdot 10^4$–$9.4 \cdot 10^4$ at 1 GAE; the worst rows reach $2.3 \cdot 10^5$ at 0 GAE and $7.0 \cdot 10^6$ at 1 GAE. Measured against the success criterion E31, E39 and E40 are validation-gate improvements of the index-calculus baseline (no pcc claim exists for any of them, so each is an unverified assertion under the 4 November rule; E39 across the whole band, E40 at positive weights), each a constant on one stage; measured against rho they leave the gap where E32's ladder put it, closing as $n^{-1/4}$ with an extrapolated crossover near 2^{85}–2^{98}.

13.6 Trace replay as a component speed-up

Joux and Vitse's observation [27] is that the F4 runs of one decomposition family share their sequence of useful rows, so one run can be recorded and later runs replayed. The programme tried it on four objects.

- *Binary S_3 chain.* Learn/apply trace reuse gave 1.4–2.3× in `cryptanalysis`, and the whole Boolean-engine lead is closed as dominated by the pair table (§12.3).
- *The $k = 4$ weighted solve (E21, 18 October).* Over 400 targets at $p = 53$ and 101 the replay agreed on every decomposition set but fell back on 13–23.5% of targets, because which rows reduce to zero is not stable across targets of one curve; the total cost with fallbacks was 0.95–0.98× of the full solve, a net loss (research note §9.11).
- *The chain solve's pivots (E35).* No per-curve order exists (§13.5).
- *The PKM–Semaev oracle on Curve25519 miniatures (2 November).* `f4_fp_tower` gained `record_trace` and `replay`: a recording run keeps, per F4 step, the S-rows whose residue gave a new pivot and the reducers they reached; a replay forms only those rows and stops at the first step whose leads differ. Only a refutation is a result of a replay—every row lies in the ideal, so reaching 1 is sound whatever the trace was learned on—and a non-refuting replay falls back to plain F4, charged (`research/curve25519_f4_trace_20261102/README.md`).

The fourth object is the one where the idea works, and its numbers show the size of a trace-replay gain (Table 13.2).

Replay costs 0.29–0.45 of plain F4, learning costs one plain run per branch system, and it pays back after about two targets. The preregistered prediction that replay would cost at most 0.25 of plain at $G = 0$ failed (0.29–0.41). The gain is a component gain in the oracle's own unit; the README states that nothing is claimed for wall time, other sizes or against rho. Priced end to end, a $t = 4$, $G = 3$ arm with trace replay costs 0.2837 $[0.2413, 0.3335]$ of the previous Curve25519 winner at $w = 0.1$ and 1.567× more at $w = 0$, so the order flips inside the band, and it is still 27.9, $1.83 \cdot 10^4$ and $1.13 \cdot 10^5$× rho at $w = 0, 0.1, 1$ (§21.7; `research/curve25519_t4_candidate_ecbench_20261104/validator-report.json`, status `pass_with_errata`).

Across the four objects the pattern is the one E21 and E35 state: trace replay pays where the useful-row structure is a property of the system shape (the PKM tower, the binary chain) and fails where it

t	G	plain/target	trace/target	trace/plain	break-even n^*	fallbacks/replays
6	0	$5.55 \cdot 10^7$	$2.25 \cdot 10^7$	0.41	1.68	0/100
6	4	$7.10 \cdot 10^5$	$3.23 \cdot 10^5$	0.45	1.83	0/1,600
7	0	$7.46 \cdot 10^8$	$2.35 \cdot 10^8$	0.32	1.46	1/100
7	4	$9.70 \cdot 10^6$	$4.13 \cdot 10^6$	0.43	1.74	0/1,600
8	0	$6.98 \cdot 10^9$	$2.32 \cdot 10^9$	0.33	1.50	4/100
8	4	$8.64 \cdot 10^7$	$3.85 \cdot 10^7$	0.45	1.86	6/1,554

Table 13.2: F4 trace replay in the PKM–Semaev oracle, Curve25519 $k = 20$ miniature, $m = 2$, 100 targets per cell, in F4 elimination multiply-adds per target (learning excluded; n^* is the number of targets after which learning has paid back). Selected cells of nine. Source: `research/curve25519_f4_trace_20261102/README.md`.

depends on the target (the weighted $k = 4$ solve, sparsest-pivot choices). Where it pays it is a constant between about 1.4× and 3.4× on the oracle, which the relation phase's yield then divides.

14 Degree instruments, linear algebra and planners

The engines of Chapters 12 and 13 solve; the tools of this chapter measure, finish or choose. A degree instrument measures the degree at which a system refutes, which sets the exponent of an algebraic oracle long before any end-to-end run is affordable. The linear algebra finishes the index calculus once relations are collected. Planners and autolabs choose factor bases, oracles and parameters. A GPU line and a hashed-ECDL engine measure how far explicit relations reach in practice. Each kind of tool has its own way of being wrong, and the record below is mostly about those.

14.1 Closure and guess instruments

§16 reports the closure-degree measurements to 24 October: a MutantXL-style closure refutes the chained $m = 3$ descent systems at degree 3–4 where XL needs 5–7, stays at degree 4 up to $\ell = 6$, and rises past 4 at $(m, \ell) = (3, 7)$; guessing one or two variables restores degree 4 there at 1/116–1/260 of the projected degree-5 cost. The $\ell = 8$ guessed-variable series that followed (four guesses suffice, three do not for one set, the $(22, 7)$ null, the invariant guesses, and what they price for ECC2K-130) is reported in §19.1 and Table 19.1. This section describes the instrument and the control lesson behind every row of that table.

The instrument. `dreg_closure` builds the Macaulay closure of one draw's system degree by degree, records the closure and XL ranks, the columns and the word operations at each degree, and scores a draw as refuting at the first degree whose closure contains the constant. `dreg_score` scores the records against a preregistration. Every study of Table 19.1 fixes a cell, draws unsatisfiable and satisfiable systems from a seed, and runs three controls: an identity control that reproduces an earlier committed record field for field, a positive control (a guess set known to refute), and a negative or null control that must *not* refute.

A null control that failed. The specialisation study of 17 October is the counter-example that justifies the controls. Fixing one variable brought most $\ell = 7$ draws back to degree 4, but the null control refuted on 6 of 8 draws where at least 7 of 8 had to fail, so the registered verdict was "uninformative" (`research/dreg_closure_specialise_20261017/RESULTS.md`). Without the null, "one fixed variable restores degree 4" would have read as a structural finding.

14.2 The linearization-excess rule

For two summands a structural number, the linearization excess $e = n - \dim V - \dim V^{(2)}$ of a subspace factor base V, predicts the closure solving degree before any solve, $D = 2 + [e < -3.5] + [e < -8.5]$ over 656 bases at mean absolute error 0.06; the rule, the progression bases it singles out and the $m = 3$ formulation point are stated in full in §20.3 (`cryptanalysis:experiments/pdp-degree-heuristics/README.md`). None of it is an index-calculus result: every row is a decomposition-stage profile on small fields. The atlas lead that would transfer it—rank candidate bases of equal size by e inside the $m = 3, 4$ closure cells of §19.1 and measure degree and charged cost with a random control—is *open, unexamined* (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-engine/LEAD-ICE-linearization-excess-fb-prior.yaml`).

14.3 Presentations as an instrument

The same decomposition can be written as different systems, and a system's solving degree, not the decomposition, is what an engine pays for. Two studies of late October measured that freedom directly.

Tree-glued encodings. A tree-glued, cherry-symmetric encoding T of the prime-field point decomposition—each summand pair encoded by its symmetric functions with a degree-2 remainder tower, pairs glued by S_3, every input of degree at most 4 for every arity—was compared with the

plain chain C and the full symmetric arm F under msolve 0.6.5 on identical Kummer-base targets (`research/ic_tree_gluing_20261103/README.md`; stage diagnostic, no S, contended timings).

- Against C, T is better in every completed cell: at $m = 2$ its degree stays 4 for $t = 3$–7 while C rises to 5 at $t = 6, 7$, with matrices 1.8–3.6× narrower; at $m = 3$, $t = 5$ it finishes at $D = 6$ while C is cut off in degree 7; at $m = 4$, $t = 4$ it finishes at $D = 7$ in 697 s.
- Against F it loses on cost at $m \leq 4$: at $m = 3$, F reaches $D = 5$ against 6 at $t = 4$, and at $m = 4$ the recorded F row ($D = 8$, from a different engine) was 59× faster.

Six further rounds (`research/ic_tree_gluing_20261103/STATUS.md`, all `[pcc:unverified]`, four round-5 statements under claim records) measured a coupling excess $E = D_T - 4$ of 0, 2, 3 at $m = 2, 3, 4$ and at least 5 at $m = 5$ (memory exhausted at 14.5 GiB), about +1.5 per summand and +0.5 per tower level; no encoding, symmetry or branching tried moved either slope. The line is recorded as closed at cryptographic sizes under the measured laws and open asymptotically on whether E grows by at most one per summand, a question its deciding cells cannot answer under 15 GB.

Contraction order on $\mathbb{F}_{p^3}$. The atlas lead on planning the contraction order of a Semaev network (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-engine/LEAD-ICE-gag-contraction-planner.yaml`) was tested at $n = m = 3$ over $\mathbb{F}_{p^3}$, $p \in \{1009, 4099, 32749\}$, 30 planted instances, under a frozen `experiment.protocol/v1` (`crypto-autoresearcher:research/gag_presentation_matched_20261031/README.md`). All arms returned identical verified relations. The monolithic S_4 presentation used 0.110× the chained presentation's F4 matrix cells at every prime (verdict *supported within scope*, descriptive); the tree presentation used 1.095× and is inconclusive, because one charged retry set the $p = 1009$ A/A floor. An independent `review-run` recommends replication, and a filed correction fixes two statements of the report (one instance, not three, has two decompositions) without changing a verdict. Matrix cells are a stage unit; the study prices no pipeline and makes no wall-time or rho statement.

Reading. Both studies say the same thing in different units: a presentation can lower the solving degree or the matrix size of a fixed decomposition by a constant factor, and in the tree line the slope in m and t stays. Chapter 6 gives the degree theory these measurements test.

14.4 The degree-of-regularity forecast

A factor-base sweep decides one decomposition system per probe target and learns the cell's degree of regularity for certain only once the relation phase has run. The forecast `dreg_forecast/v1` (`src/cryptanalysis/dreg_forecast.rs`) estimates it from the solves so far.

- A decided solve observes the solving degree D; an exhausted solve observes only $D > c$ for the highest completed degree c, a right-censored observation. The Kaplan–Meier estimator of $P(D > d)$ with Greenwood's standard error gives the predicted degree (the median) and a 95% band; a weak prior centred half on the semi-regular degree keeps it defined before the first decided solve.
- The semi-regular degree of the system shape is the yardstick: a cell whose band lies below it is extracting structure.
- The cost side projects the remaining relation phase from the observed per-degree work and yield, charges the relation matrix from its own row operations, and reports the rho floor over that work at 1, 10 and 100 multiplications per group addition; advice is given after four real solves and ranks only where the order holds across that band.

The `dreg-triage` skill wraps it as a loop (plan, run, triage, refine), and both the module and the skill state its limits: it is advisory, never stops or shortens a candidate, leaves `certified_degree_of_regularity` null, and is a stage diagnostic, never a complete-DLP result (`.agents/skills/dreg-triage/SKILL.md`). Its estimator is the same censoring logic that statistics version 2 applies to engine costs (§11.3), applied to degrees.

14.5 Linear algebra

The model of Chapter 2 lists the linear-algebra levers (structured Gaussian elimination, a CADO-style filter, Wiedemann's algorithm [43] and its block form [7], a first-dependency stop). The measurements say that those levers are available and are almost never the bottleneck.

Against CADO-NFS. On synthetic $m = 3$ relation systems over a 62-bit prime, in one session on a shared host (`research/la_engines_20261014/README.md`; protocol and five amendments, each before its data):

- the linear algebra is 0.11–1.9% of 27 real `ic logs` precomputes, and the largest real index-calculus matrix in the repository (N73, 600 columns) took 5.1 h to collect and milliseconds to solve;
- the repository's block Wiedemann beats CADO-NFS `bwc` on the identical matrix at every size measured, 1.86× at 131,072 columns;
- CADO-NFS's filter left a 5–8× smaller matrix than the repository's; the port of that filter runs 86.57 → 8.35 s at $D = 262{,}144$ (10.40× [9.99, 11.06]) and is 1.93× [1.74, 1.96] faster than CADO-NFS's own pipeline on the same matrices (7.83 against 14.73 s); below about 4,096 columns it is slower (0.40× at 1,024, both in milliseconds);
- ecbench's default `IncrementalGauss` grows 6–9× per doubling, 19.1 s at 8,192 columns.

These are component results in wall time at L0. The filter port is the default of `ic logs` since.

Inside the pipeline: a null. When the four sparse solvers were made reachable from `ic.pipeline`, a paired study found that every one costs *more* end to end than the baseline: 1.206× in total on an $E(\mathbb{F}_{29^4})$ $k = 4$ ladder and 5.72× falling to 1.030× over 128–4,096 columns on a scaling curve (`research/sparse_la_timing_20261018/README.md`). The cause is the completion rule: the deferred solvers answer only at full column rank, while `structured-gauss` and `incremental-gauss` stop as soon as the target's column is pinned, so the full-rank arms collected 1.75× and 5.5–10.5× more relations. A positive control with the same full-rank rule and dense elimination reproduced the sparse arms' S. In counted row operations the filtered solvers lose at every size; the unfiltered Wiedemann and Lanczos count 10^3–10^4× the row operations of structured Gauss, growing as about columns$^{2.2}$.

Where it matters. The exception is the GHS-weak plane quartic of §26, where double large primes shrink the factor base and put linear algebra and relations both near $q^{1.0}$. There, structured elimination (B1) was a stage gain of 0.81–0.91 of the B0 cost up to $q = 65{,}479$ and a regression (1.53) at $q = 262{,}111$, recorded as a trade and not adopted; double large primes (B2) are the measured improvement, 0.297 $[0.295, 0.299]$ of B0 at $q = 262{,}111$ (`research/quartic_n4_la_20261021/README.md`). On the GLS $\mathbb{F}_{p^4}$ line the linear algebra is at most 0.006% of S (§13.5). The relation phase, not the matrix, sets the cost on every line measured.

14.6 Planners

`ic plan` chooses a factor base, oracle and parameters for a curve from a cost model. Its validation on six curves is in §21. Two later versions added weight rules—`ic-planner/2` a base of $1.41N_0$ at $w = 1$ on prime curves, `ic-planner/3` $0.71c_0$ on Koblitz curves at $w \geq 0.3$—and an exhaustive re-test of 26 October on 14 new curves (3,328 records, 2,428 band verdicts restated in integers and checked by Lean's `decide`) withdrew all three weight rules in `ic-planner/4`: the prime rule failed its $w = 0$ clause, the binary rule failed at $w = 1$, and the Koblitz rule was dearer at every weight (`research/ic_planner_exhaustive_20261026/README.md`; §22.5, Table B.1). The same re-test found that the residual collector is cheaper than the plan across the band from 36 bits—as a generic collision search, whose curve-free control costs the same and whose exponent matches rho's (§22.3). A planner, in short, is a model of the engines it plans for, and its weight rules were the part of the model the band refuted.

14.7 Autolab loops

The autolab is the programme's loop for proposing and screening candidates without letting the screen promote them. Its contract, used by the tournament since September and by the native `ic_autolab` loop since 4 November, has six steps: propose, build a portfolio, screen in sample, select one challenger per size, confirm on fresh targets with the incumbent, an A/A copy and the matched rho, and record the delta on a ratchet (`.agents/skills/ic-autolab/SKILL.md`). The tool enforces what earlier loops left to discipline: a round is frozen once, target seeds are disjoint from every exposure, the in-sample screen never promotes, an improvement needs the paired interval below 1 at $w = 0$ and the band order at every weight, and every arm joins the trial history so it is not proposed again. Its first family, the Curve25519 miniatures, ran 35 rounds; the generic incumbent settled at the collision floor derived in §21.7, which is

the loop working as intended: the ratchet found the floor and could not pass it. Earlier cycles (17–19 October) on the native GF(2^{83}) norm predicate are in §16. The loop's rows are development evidence until the validation gate.

14.8 Explicit relations on a GPU

A separate line measured how far a complete prime-field recovery with explicit relations reaches on one GPU: factor-base construction and generic meet-in-the-middle relations on the device, elimination, scalar extraction and certificate replay on the CPU, on public fixtures (`research/explicit_relations_e2e_20261030/RESULTS.md`).

- Nine pinned versions recorded every launch, failures included. Version 8 completed 12 of 12 recoveries through 60-bit fields on an RTX PRO 6000, with cold solves of 770.3 and 845.2 s at 60 bits; version 9 recorded one 61-bit recovery and one memory refusal.
- On an RTX 4090 a full-cache query kernel lowered the 63-bit cold recovery from 1,183.8 and 1,185.8 s (baseline before and after) to 1,053.9 s, 11.05% lower with a 0.164% baseline repeat spread, certificates byte-identical (`research/explicit_relations_fullcache_20261101/RESULTS.md`); linear algebra took 0.11 s of the 1,054.
- Block normalisation by a product tree uses 765 source-level field multiplications against 4,608 for dual scans, with no cold end-to-end gain established (`benchmarks/20261030-explicit-relations-gpu/`).

Every clock is L0 with `wall_valid=false`; there is no matched rho and no fastest-external-tool comparison, and the benchmark copy says it is not an admitted result. The relations are generic meet-in-the-middle, so the method is bounded by the collision floor of §32.2; what the line measures is engineering reach, not a distance.

14.9 Hashed-ECDL engines

The hashed factor base and the conditional tensor-contraction route are in §24.1. Their CPU engine (`crates/hashed-ecdl-cpu/`) recorded two matched panels of 120 executions with byte-identical certificates across arms; with persistent meet-in-the-middle tables and batched inversions the complete known-answer solve fell from 45.99 to 3.90 ms ($p = 65521$, $m = 3$) and from 80.54 to 4.12 ms ($m = 4$), and from 0.316 to 0.283 ms at $p = 211$ (`research/hashed_ecdl_cpu_20261030/README.md`; L0, one EKS VM). The larger-field ladder expired with zero executions. The README states that these establish exact software behaviour on fixed fixtures and no speed-up, matched-rho advantage or large-curve attack.

14.10 Component speed-ups and end-to-end changes

Table 14.1 sorts the main results of this part by the three labels of Chapter 11.

Three conclusions follow, each with its scope.

- *The machinery works.* Engines reproduce each other and brute force, sessions replay to the last digit, defects are found by planted controls and pinned rather than hidden, and the band ranks levers whose sign depends on a price. That is a statement about the instrument, made by the instrument's own audits.
- *Engine gains are real and small against the gap.* The largest end-to-end engine change in this part is a factor of about 5 (the `f4_fp` cover pipeline; producer-reported, lower-bound comparison). The largest validated ones are the generator-orbit base (1.26–3.20×) and E31 (2.50–2.56× at 0 GAE, 1.03–1.05× at 1 GAE), both validation-gate passes without a pcc claim. These stand against distances to rho of 10^2 (Koblitz, binary, prime) to 10^3–10^6 ($\mathbb{F}_{p^4}$). On the binary and prime lines no engine change moved the distance's slope.
- *Where an engine got fast, it became generic.* F6-IC's closure and the $n = 83$ chained SAT both turned into meet-in-the-middle lookups; the GPU and hashed-ECDL lines measure meet-in-the-middle relations; the one validated crossing below rho is a Frobenius-folded BSGS. Chapter 32 explains why each of these is bounded by the square-root floor.

Result	Effect	Against matched rho
Component speed-ups (identical outputs, stage unit)		
F5 support cut 19	complete call 3.0×	not priced
Few-candidate F6-IC closure	CPU 2.67–10.29×	reduces to MITM
Chained SAT with lookups, $n = 83$	m up to 9 decided	no complete DLP
F4 trace replay (PKM–Semaev)	0.29–0.45 of plain F4	27.9–$1.13 \cdot 10^5\times$
CADO filter port	10.40× (filter)	LA $< 2\%$ of solves
Generic-field tower F4	2.18–2.57×	not priced
End-to-end changes over an IC baseline		
Faster `f4_fp` in the cover pipeline	S 4.99–5.08× lower	194–850× (lower bound)
E31 chain solve (validated)	S 1.03–2.56× lower across band	$4.42 \cdot 10^3\times$ at best
E39 arrival-order pivots (validated)	ahead of A at every weight	$5.15 \cdot 10^3\times$ at best
E40 lead-stored rows (validated)	ahead of A at $w > 0$ only	$6.00 \cdot 10^3\times$ at best
Generator-orbit base, Koblitz (validated)	1.26–3.20×	below rho only at $n = 29$, one column (BSGS)
Harvest default, inherited F4	+9.6% to −4.1% by size	7–1,250× (frozen strong)
Accounting and instrument changes		
Accounting v2	references resized	gap larger
Statistics v2	censored runs kept	engine ratios lower bounds
Allocator confound	wall time of small arms	counts identical
64-bit field truncation, $n = 83$	refutations voided	—

Table 14.1: The part's results by class. "Validated": a `VALIDATION.json` with a passing validator report; every other row is producer-reported. Sources in §§11.2–14.8.

15 Koblitz curves

Koblitz curves $E_a : y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$, $a \in \{0, 1\}$, carry the Frobenius endomorphism τ, which acts on the prime-order subgroup as a known scalar λ; ECC2K-130 is the $a = 0$, $n = 131$ member. They were the programme's main family: the compact-orbit producer, the $n = 83$ campaign, leaderboard Table A and the ECC2K-130 cost model all live here. Random and prime-degree binary curves without τ are in §16, and the Weil-descent degree studies that price ECC2K-130's algebraic route are there too, because their systems use random subspaces rather than Frobenius structure. Every study in this section is listed in the programme record [3].

15.1 Frobenius-invariant and compact-orbit bases

The Koblitz implementation (`src/cryptanalysis/koblitz_index_calculus.rs`) builds GGMP's invariant base from the linearised polynomial of a factor of $x^n - 1$, checks $\pi(\mathcal{F}) = \mathcal{F}$ point by point, rewrites each relation into orbit columns with known λ^k weights, and solves the cofactor-corrected system $ha + hbd \equiv \sum \lambda^{k_i} x_{o_i}$. On small fields the orbit collapse is as predicted: $|\mathcal{F}|/\#\text{orbits}$ of 7.86, 10.89, 12.96 at $n = 9, 11, 13$ for relations and 61.7–168.0 for linear algebra, against rho's 4.24–5.10 (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). Because $\ell = \text{ord}_n(2)$ fixes the invariant base's size, the programme moved to bases sized to the instance: divisor bases, SAT-optimised bases, a rank-guided base at $n = 53$ (220 orbits, a complete implicit S_5 formula of 22,887 clauses), and finally the *compact signed-Frobenius orbit* base, which stores one representative per orbit and no pair table (`docs/ic/COMPACT_ORBIT_PRODUCER.md`). In the 13 October ecbench factor-base suite it was the best measured Koblitz construction: $S = 9.114$ at $n = 17$ (272 points, 8 columns), 9.985× matched signed-Frobenius rho (L0 operation smoke, a lower bound), against full orbit 18.7× and subspace bases 47× (`research/ecbench_frontier_20261013/`). The fully priced orbit ladder of 14 October and the generator-orbit base of 22 October are later and cheaper (§17.1, §11.7).

15.2 The oracle sequence

The binary oracle went through SAT (XOR-native CDCL after Trimoska et al. [39]; decides the $n = 19$, $\ell = 6$ corpus but does not scale), matrix F4 (whose refutation range was the real frontier), pairs-and-solve for S_4 (validated to $n = 21$, $\ell = 7$, still $O(2^{2\ell})$), meet-in-the-middle pair tables, inherited F4 and F6-IC (§10), and root-finding engines (§9). The boundary ledger's current Koblitz decomposition record is an $n = 31$, $m = 2$ F4 cell with an exact block-6 M4RI acceleration; at $n = 53$ an exact Frobenius-orbit S_5 compression removes the coordinate ROM but every unrestricted solver arm remains UNKNOWN.

The September toy studies that fed this sequence are catalogued as campaigns. At $n = 13$ and $n = 19$, rotated subspace bases covered 1,591 of 2,003 targets with five spaces against 61 with one, and joint rank recovered 58/64 targets against 29/64, without passing a 10% yield gate at $n = 19$ (`research/notes/ecc2k130/`, clusters "rotated-n13" and "rotated-n19"). Finite-fibre S_3 semantics were proved exact and an O-aware CNF made 252.7× smaller at $n = 13$, but no existing solver interface was admitted as an arm; three SAT engines decided 5 satisfiable and 27 unsatisfiable targets at $n = 13$, with the five negatives DRAT-certified. At $n = 131$ a symbolic full-point DAG replayed a single-edge DIMACS instance but could not be admitted past the 64-bit ceiling, and the exact Frobenius census of the balanced $(m, d) = (10, 13)$ rotated base has 1,048,134 columns, the only shape with as many columns as the field. A frozen test of the Frobenius quotient found that toy fields saturate and so test nothing; an independent SAT reproduction solved $n = 13$ and none of four planted targets at $n = 19$ in 1,200 s (`research/frobenius_quotient_sat_20260930/`). At $n = 13$ the presentation of the curve does not matter: yield spread is explained by cofactor balance (correlation 0.850) and there is no curve effect at $n = 19$ (intraclass correlation ≤ 0.006; `research/koblitz-isogeny/`).

15.3 Against a stronger rho: the September reversals

Three September results that put Koblitz index calculus at or below rho were reversed when rho was strengthened, which is why every later row is quoted against the strongest available rho (§4.5). Table 15.1

Result	Case	Measure	Recorded	Stronger rho
Tournament	$n = 43$, $a = 1$, best	retired instr.	0.5347	
	$n = 43$, $a = 1$, triple table		0.409	
	five cells[e]			1.19–2.88×
Orbit DLP	$n = 53$, 1,024 targets	retired instr.	0.2403	
	four rungs			$0.240 \to 0.317 \to 1.294 \to 1.923$
	sweep, $n = 37$			0.794
	sweep, $n = 41$			1.520
	sweep, $n = 53$			1.923
	sweep, $n = 61$			2.297
	single target		0.8392[a]	44.0× [39.4, 45.4]
Compact orbit	$n = 41$, $L = 1{,}024$	corrected rho	0.309×	
	$n = 37$, $L = 1{,}024$	32-walk rho[b]		2.081
	$n = 53$, $L = 1{,}024$			3.086
	$n = 41$, $L = 1{,}024$	cold CPU[c]		1.314
	$n = 53$, $L = 1$			22.675
	$n = 37$, $L = 1{,}024$	instructions[d]		0.5165×
	$n = 41$, $L = 1{,}024$			0.6221×

Table 15.1: IC/rho of the three September results as first recorded and against a strengthened rho. Tournament (15 September): IC against a worker rho's retired instructions; giving rho the index calculus's own canonical form made rho win all five cells (`research/ic_triple_counted_20260923/`; the tournament README, `research/ic_candidate_tournament_20260915/`, carries no correction note). Orbit DLP: the recorded rho used software carry-less multiplication on x86-64; rho was strengthened in four rungs, then the preregistered sweep read rising with n (`research/notes/index-calculus/RESEARCH_STRONG_RHO_LADDER_20260929.md`); [a]the selected-panel crossover withdrawn by the direct arm's cost against a strong single-target rho (`research/notes/index-calculus/RESEARCH_SINGLE_TARGET_STRONG_RHO_N53_20260930.md`). Compact orbit: the recorded value is a point panel; [b]a 32-walk batched-inversion normal-basis rho (`research/notes/ecc2k130/compact_orbit_strong_rho_20260929/RESULT.md`); [c]compact/rho CPU of the orbit-disjoint cold confirmation, which ended with six eligible no-gos (`research/notes/ecc2k130/disjoint_cold_v2_outcome_20261001/RESULT.md`); [d]batched compact IC (`research/notes/ecc2k130/compact_ir_ledger_20260930/RESULT.md`). [e]Rho given the index calculus's own canonical form.

gives each result as recorded and against the stronger rho.

Batched compact IC does use fewer instructions than rho while its cold CPU stays above rho, a gap traced to memory behaviour. Some sixty notes on S3 batch inversion, swap quotients, Bloom prefilters, window screens and K grids each moved a constant and none crossed rho (cluster "compact-orbit-cold"). The boundary ledger of the same weeks reached the same place from the other side. Re-pricing every row against a matched negation rho raised the vs-rho ratios 2.1–6.5× (`research/ic_rho_reference_20260923/`); the best batch row was 4.86× [4.21, 5.55] at $r = 2^{39}$, and a page $r^{1/6}$ law was falsified at the small end (`research/ic_exponent_20260926/`); single-target online IC beat table-less rho at all six sizes (8.84–17.5×) while cold rho won at all six (1.98–31.6×; `research/ic_single_target_20260930/`). At $n = 37$ in ecbench the cold IC/rho was 3.64× in instructions and 13.6× [10.6, 18.7] in counted S, and the $n = 37$ parameters did not transfer to $n = 41$ or $n = 53$ (`research/ecbench_n37_shared_rank_20261004/`, `research/ecbench_n41_n53_shared_rank_20261005/`). An $m = 4$ exponent audit measured a cost slope of 0.985–1.060 bits per unit of n at $n \leq 19$, against the 0.25 the lead needed (`research/ic_m4_exponent_audit_20260928/`).

Other Koblitz lines. The remaining Koblitz campaigns moved constants inside index calculus and are catalogued in the programme record [3]. Instruction-level rounds on the Gröbner stage left the whole relation log at 964× rho on K_0 over $\mathbb{F}_{2^{13}}$ (`research/groebner_instructions_20260925/`); a September autolab campaign cut cold instructions by 47.7% and left the candidate at 4.98× matched rho's instructions (`research/ic_autolab_next30_20260915/`); at $n = 37$ a full-rank compact-orbit gate passed with IC at 22.6× rho (`research/notes/ecc2k130/full_rank_compact_orbit_ic_20261003/RESULT.md`). Pipeline kernels gave laned canonical keys 1.09–1.31× (`research/canon_laned_keys_20261008/`), u64 root batching 3.26× whole-process wall (rank cell corrected to 2.47×), and aimed gathering 1.037× set-up (`research/ic_aimed_gather_20261023/`). Transporting a base through ECC2K-130's degree-263 and degree-7 isogenies was built and certified, but on toy fields the transported

n	Base	Columns	Points	Memory	Rho/IC online	Status
53	shared-log[a]			9.34 GB		
61	compact orbit	600	73,200	2.73 GB	124.85×	exploratory[b]
71			85,200		2,864.1×	exploratory[c]
73			87,600		1,189.7×	exploratory[d]
83	$a = 1$				8.2×	exploratory[e]
61	orbit, $a = 0$	$K = 500$	61,000		13.94× [9.81, 19.13]	admitted[f]
					12.87× [7.30, 19.08]	admitted[g]

Table 15.2: The one-target online ladder: rho online time over IC online time on the same point, reusable IC preparation excluded; peak memory where recorded (peak RSS at $n = 53$). [a]Guided shared-log precompute with an explicit pair table; earlier, a 1,024-target shared-log batch beat batched rho in all 15 paired blocks (secondary, multi-target). [b]Fresh reproduction. [c]First rung past 64-bit packing (u128 words); median of three repeats of one point. [d]The guided-rank precompute took 5.1 h at 57–70 million probes per relation. [e]IC online 5.6–7.9 s against rho 43–71 s; re-measured below. [f]Admitted 12 October: 64 fresh public one-target workloads per host, identical candidate manifest on two hosts; median online speed-up with 95% interval; pinned x86_64, claim-check 64/64. [g]The same, unpinned arm64.

and descendant bases cost 3.6–4.0× the original (`research/ecc2k130_factor_base_pilot_20260924/`); and τ-adic known-scalar arithmetic is 0.413 of the binary cost at $m = 131$ with no measured rho benefit (`research/notes/ecc2k130/tau_adic_arithmetic_20260929/README.md`). On the rho side, GPU iteration redesigns for the ECC2K-130 walk did not beat the shipping distinguished-point walk (`research/ecc2k130_iteration_redesign_20261012/`, `research/dp32_sigma_table_20261012/`).

15.4 The one-target online ladder

The ledger's primary `vs_rho` question is: with reusable IC preparation already done, how long does one unseen public target take, against rho with the automorphism discount on the same point? The ladder (Figure 15.1) is in Table 15.2.

Re-measured on 12 October against a default strong rho on five repeats of the same target, the $n = 83$, $a = 1$ IC was *slower* online, 1.382× [1.036, 1.882], and 218.0× rho in fresh-process wall time (`research/n83_evaluation_20261012/`); the later dense-root reading of 18.5× is in §15.5. For the admitted $n = 61$ point, set-up of about 28 s is outside the window; with it charged, IC costs a median 83.4× rho.

The exploratory ratios are large and irregular because a one-point online window measures the luck of one relation search against one rho walk; the admitted panel's range over 64 workloads is 1.06–648.6×. The admitted point is a constant-factor online result at $n = 61$ (47-bit subgroup) with preparation retained; it is not a cold result, not a scaling law, and not an asymptotic claim.

Retired as an index-calculus win (14 October). Every rung of this ladder compares index calculus with a retained table against rho with none. Given a distinguished-point table of equal or smaller size, Bernstein–Lange preprocessing rho [4] is 53–317× faster online than the compact-orbit arm on the same 64 targets at $n = 41$, 53 and 61 (§36.3). At $n = 83$ (§15.5) rho with a table matched to the index calculus's set-up time or memory won every pair that was run. The ladder therefore measures what a table buys, and generic rho buys more per byte. The validation sweep later found `docs/ic/BOUNDARY_TARGETS.md` still printing the 13.94× as a win; it is superseded (`research/validation_sweep_20261019/B-benchmarks/VERDICT_TABLE.md`).

15.5 The $n = 83$ campaign

Degree 83 was the programme's largest Koblitz target with a working pipeline: 145 study directories between 4 and 21 October (the programme record [3], clusters beginning "n83" and "f6-n83"). In order:

- *F6 micro-studies, 4–5 October* (24 directories, almost all on the $a = 0$ curve). They kept kernel gains (exact queries 3.00–6.87× faster with ARM64 PMULL, resident memory −52%) and shrank SAT encodings, but every ordinary solve timed out; four-summand coverage was at most $4.662 \cdot 10^{-12}$ of the subgroup and no natural relation was found. On the 81-bit $a = 0$ gate curve, SAT and XCNF

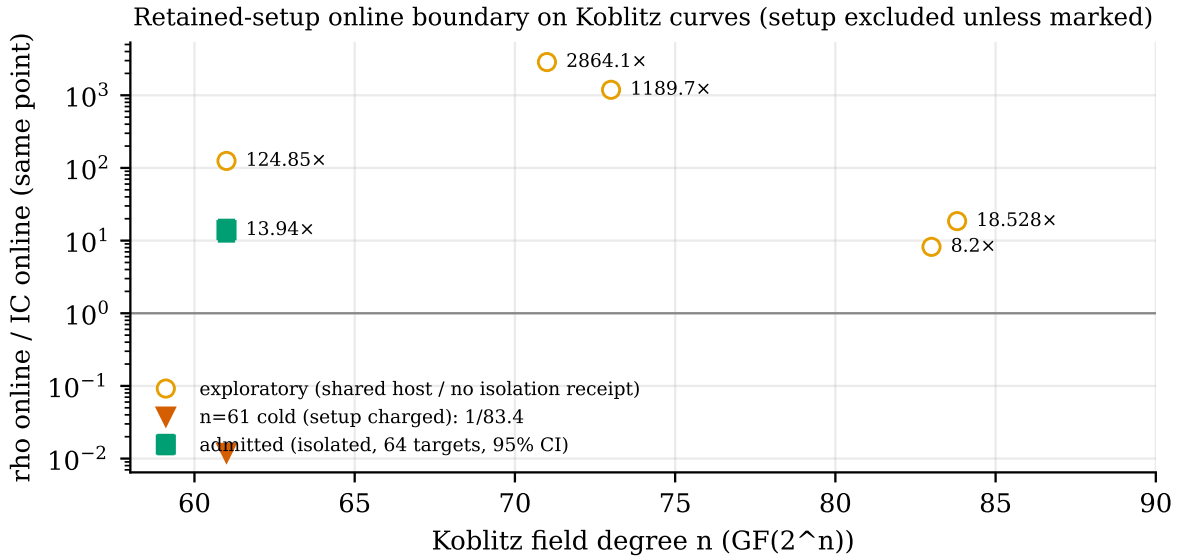


Figure 15.1: Rho online time over IC online time on the identical public point, reusable IC preparation excluded. Open circles: exploratory (verified answers, no qualifying isolation receipt; three repeats of one point or a single point). Filled square: the admitted 64-workload panel with its 95% bootstrap interval. Triangle: the same $n = 61$ candidate with set-up charged (cold). Sources: `docs/ic/boundary_targets.json`, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`, `research/n83_dense_root_20261013/`.

decompositions succeed only with a known witness, and the prebuilt-base question is an open problem record.

- *The $a = 1$ point, 7–12 October.* The archived K600 point read 8.2× online and was re-measured slower (above).
- *Root-index capacity, 13–16 October* (25 directories). Root-index layouts grew the $a = 1$ base from K512 (84,992 points) to K3596 (596,936 points) within 14 GiB. The dense root index cut probes per relation from 300,021.9 to 169,069.8 (K1500 to K1798; paired difference 130,952 [52,058, 209,990]), a within-IC result that stands; its one-point online 18.528× is retired (`research/n83_dense_root_20261013/`).
- *Factor-base selection, 14–16 October* (56 directories). Root4, root11 and root21 layouts, Bloom filters, sparse and canonical variants and K screens from 100 to 20,000 points selected within index calculus; the best means are about 26–29 ms online (K3425, K3450, K3500), and many screens end BOUNDED_UNKNOWN. A three-hash 512 MiB Bloom filter cut a K3300 online mean from 34.77 to 31.32 ms (`research/n83_root11_bloom_results_20261015/`); a K3350 win of 7.18% did not replicate.
- *Tier-1 online readings, 14–16 October* (14 directories): median rho/IC of 54.1, 210.3, 113.0, 104.2, 35.1, 413.3 (one point) and 208.7× (K3072, five fresh points, [38.3, 2,510]; `research/n83_tier1_online_20261016/`), all against a rho with no table and with set-up excluded.
- *The control, 16 October.* Bernstein–Lange rho given a distinguished-point table matched to the index calculus's set-up time or memory was faster online in every pair run: IC/rho 3.74 [2.65, 11.05] (K3300, time-matched), 40.61 (K3300, memory-matched), and 81.5–395.5 (K600) (`research/20261016-n83-precomputed-rho-control/`). Only the K3300 and K600 bases were paired; the K3072 reading was not, so its retirement rests on the same mechanism rather than on its own measurement. The validation sweep and the ledger audit retired all the online readings on 19–20 October.
- *Cold, 16–21 October* (11 directories). The cold one-target process fell from about 150 s (K1024) to 119.8 s (K1120, holdout) and 113.8 s (K1107; `research/n83_cold_base_confirm_20261021/`), against a rho online time of about 2.2–2.6 s on the same hosts. No separate cold rho was run, but plain rho has no set-up, so its online wall is its cold cost: the K1120 holdout reads cold IC over same-point plain-rho wall 55.4 (five points; `research/n83_e2e_holdout_20261016/RESULT.md`), the figure Chapter 40 cites. The hosted single-target benchmark's 133.7× is the other cold figure (`benchmarks/20261014-n71-n83-hosted-single-target/`).
- *A second pipeline in* `cryptanalysis` (K600, $a = 1$; [3]). A Tier 1 ratchet of rank-probe, closure, index-

(a) Rules: cold IC/rho on one unseen public target

Rule	Date	IC/rho, cold	Status
v3		5.23–21×	printed in Table A; unvalidated (pre-gate)
v6		3.78–12.06×	rho reference 0.57–0.85 of strong walk[a]
v7		3.43–11.51×	rho reference 0.57–0.85 of strong walk[a]
v8	19 October	4.67 [4.34, 5.07] to 18.07 [15.98, 20.58]×	matched GFNI rho; `VALIDATION.json` present[b]
v4 recipe	24 October	9.2–42.1×	validation file withheld[c]

(b) Pipeline baselines, 16–17 October: cold speed-up

Baseline	Change	Cheaper cold
v4	head-of-main	
v5	scan-key	1.20–1.24×
v6	scan-fold	1.33–1.40×
v7	build streams	1.07–1.13×
v8	admitted prefetch	1.06×
v4→v8	together	1.90–1.98×

Table 15.3: Leaderboard Table A, Koblitz $n = 41$–61. (a) Each rule's cold IC/rho range over its sizes, against rho on the same point (rules v6 and v7: `benchmarks/20261016-ic-rule-v7/`; v8: `benchmarks/20261019-ic-matched-rho-v8/`; v4 recipe: §36.5). [a]Per step, so the distance was understated. [b]1.30–1.58× the v7 control, fitted slope 0.168 per $\ln r$; its directory carries a `VALIDATION.json` with a replay certificate and a passing validator report (`research/ic_tool_program/rule/v8/VALIDATION.json`); it is not on the board, which still carries v7. [c]A paired retuning improved one size of six; rho is priced in the same process at the canonical batched step (the range is the v4 column, 9.20–42.12; the v3 recipe at the same step reads 9.89–43.46), and the validation file is withheld because that reference may not be the strongest, so it is not an admitted row either. (b) Each baseline's recorded speed-up, and the combined cold factor over v4 (`benchmarks/20261017-ic-r12-admitted-prefetch/`).

build and rank-finder changes brought its latest row to 3.92× a tuned rho end to end, cold, and 0.093× online with set-up excluded (`ca:experiments/koblitz-n83-rankfinder-20261021/README.md`). The study's own limits apply: one run on one host class, a point estimate with no interval, wall time, factor-base construction charged to no arm (so the cold cost is a lower bound), no computable band, and gaps that are not comparable across runs (the same producer read 4.42× on another host). Against Bernstein–Lange rho with equal advice its online time is 22.55× [16.23, 31.44] worse (`ca:experiments/koblitz-n83-bl-rankprobe-20261019/`). It is the closest cold Koblitz figure at $n = 83$ and is still above rho; the two pipelines' numbers are not comparable with each other.

An instrument correction (24 October). On 24 October every Weil-descent engine's $n = 83$ refutation was voided by a 64-bit field-table truncation (§12.8); after the fix the $m = 4$ root finder solves in 175–189 s, 116–125× strong rho (`research/n83_msweep_20261024/`).

The open pull requests at the time of writing include further $n = 83$ work ($m = 4$ queries on a retained 16,627-column base, AWS replays, shared-binary panels); none is on `main` (the programme record [3]).

15.6 Leaderboard Table A: the Koblitz rules

Table A of `docs/ic/LEADERBOARD.md` prices one unseen public target on Koblitz curves at $n = 41$–61 ($\log_2 r = 36.6$–47.2), cold, against rho on the same point. Its history is a sequence of rules, each a pipeline baseline plus a rho reference (`research/ic_tool_program/`). Table 15.3 lists the rules and the pipeline baselines.

Under rules v6 and v7 cold rho was still faster at every size (Table 15.3(a)). Seven further baseline candidates (prefetch gates, filter slots, huge pages, a wider base) did not clear their rounds. Table C prices a 32-target batch on nine Koblitz curves ($\log_2 r = 18$–47.2) against batch rho on the same targets: 3.62–14.9×, a historical diagnostic under the one-target rule. All nine failed the 24 October validation gate, and Table D, the Joux–Vitse historical study, is withheld.

Route	Setting	Cost	Above rho
Decomposition oracle	family bottom	$2^{132.58}$	
	Frobenius-orbit unions, $m = 3$	$2^{124.99}$	$2^{64.18}$ times
Relation sweep	best single relation, $m = 4$		+4.98 bits
	best honest $m = 8$, Frobenius quotient		+13.59 bits
Linearisation	plain		$70.19 - b_{\max}$ bits
	best bilinear XL, $\omega = 2$		71.82 bits
	same, $\omega = 2.81$		78.15 bits
	trilinear		87.16 bits
Weil descent	$m = 3$, free oracle	$2^{68.58}$	
	$m = 4$, degree 4	$2^{68.43}$	+7.62 bits[a]
	$m = 4$, degree 5, $\omega_c = 1$	$2^{71.13}$	+10.32 bits
	$m = 4$, degree 5, $\omega_c = 2$	$2^{106.52}$	+45.71 bits
	$m = 4$, hybrid guess		+8.43 to +58.27 bits

Table 15.4: Priced algebraic routes to ECC2K-130. *Setting*: the variant priced (m summands, ω_c the closure exponent, column touches per Macaulay column, written w in the source and unrelated to the §5a band weight; ω the linear-algebra exponent); *Cost*: the route's cost as recorded; *Above rho*: its distance to rho, as a factor or in bits. The decomposition bottom needs a materialised base of $2^{44.53}$ points at $m = 3$; the $m = 4$ single relation needs $2^{70.79}$ bytes; linearisation gaps charge each Macaulay solve; the descent rows are in unconverted column touches. [a]Now contradicted. Sources: `research/notes/ecc2k130/RESEARCH_ECC2K130_DECOMPOSITION_TARGETS.md`, `research/notes/ecc2k130/RESEARCH_ECC2K130_RELATION_SWEEPS.md`, `research/linearization_reach_20260930/`, `research/xl_bilinear_core_20261006/`, `research/xl_trilinear_t_unknown_20261006/`, `research/notes/ecc2k130/RESEARCH_ECC2K130_DESCENT_DEGREE.md`; the hybrid guess is in §16.1.

The validation sweep found that Table A understates its gap to rho by 1.7–2.0×, and the 24 October validation gate did not check Table A (`research/ic_validation_gate_20261024/`). Rules v8 and v4 measure the same sizes against differently built rho references and are kept apart (Definition 4.1).

15.7 ECC2K-130

ECC2K-130 is the $a = 0$, $n = 131$ member of the same family, with a 130-bit prime subgroup and rho at about $2^{60.81}$ group operations. The compact producer already solves $n = 41, 53, 61, 71, 73$ of the family; about 300 notes under `research/notes/ecc2k130/` price the routes to $n = 131$ itself. No route prices below rho. Table 15.4 lists the priced algebraic routes.

- *Extrapolating the producer.* The fitted probes-per-relation law $C\, r/(n^2 K^2)$ with $C = 1$ gives $1.1 \cdot 10^{18}$ core-years at $K = 600$ and $6.8 \cdot 10^8$ at $K = 10^{12}$ columns (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`, with the unexplained factor-of-5 gap in C of §4.5).
- *Decomposition oracles.* Decompositions exist, but no oracle finds one for less than 2^{131} per family; unions of Frobenius orbits lower the family's bottom but not to rho. Crossbred, RR-solver and WDSat oracles are engineering only: Crossbred is 0.023 of F4's bit operations at $m = 3$ but slower end to end, and the RR solvers' advantage over Semaev at $d = 6$ is worthless against brute force.
- *Linearisation and Weil descent.* Neither reaches rho (Table 15.4). The studies behind the descent rows, and the hybrid guess for the $m = 4$ row, are in §16.1.
- *Covers.* Over $\mathbb{F}_{2^{131}}$ every ordinary curve has GHS magic number 1 or 131, so GHS descends ECC2K-130 to genus 1 (a zero map on the subgroup) or to genus about 2^{130}, and the Hess–Menezes–Teske walk has nowhere to land. One question is open: $\langle G \rangle$ is the group of a simple 130-dimensional abelian variety, and if that variety were isogenous to a Jacobian with an evaluable isogeny, index calculus on the genus-130 curve would cost $2^{37.17}$, $2^{23.64}$ times *less* than rho; no such curve is known, the Torelli codimension at genus 130 is 8,128, CM constructions with prescribed Frobenius stop at genus 3, and the largest published Weil-descent discrete logarithm is at genus 31 over $\mathbb{F}_{32}$ (§26) (`research/notes/ecc2k130/RESEARCH_ECC2K130_HYPERELLIPTIC.md`, with the magic-number correction of `docs/audits/GHS_MAGIC_NUMBER_MMT_CORRECTION_20261013.md`). The $N = 130$ counterfactual is genus 255 at a modelled $2^{55.60}$ against rho's $2^{60.31}$, and measuring one relation trial turns that margin negative (§36.5). The best explicit curve construction reaches genus $2^{28.92}$ (`research/notes/ecc2k130/RESEARCH_ECC2K130_CURVE_CONSTRUCTION.md`). The isogeny class has 262 curves of conductor gap 263 and no measured level effect (§26); rho on the degree-263 leaves

keeps only the negation map, an 11.45× penalty (`research/notes/ecc2k130/isogeny_rho_speed_20260930/`).

- *Rho itself.* The separate GPU rho estimate is about 4.3 RTX PRO 6000 card-years (`paper/ecc2k130/`); on one CPU core the ECC2K-130 client walks 13.2 million steps per second (`research/certicom131_rho_ic_bench_20261022/`). Signed σ steps cut modelled GPU time to 0.961–0.970 (`benchmarks/20261014-ecc2k130-signed-sigma/`).

Index calculus is not a route to ECC2K-130 on this evidence: every route that has been constructed or priced is at least about 5 bits above rho (the $m = 4$ single-relation row, at $2^{70.79}$ bytes), most by tens of bits, and the one hypothetical route below rho needs a genus-130 curve that no known method can construct.

15.8 Factor bases closed under $\mathbb{Z}[\tau]$

The $\mathbb{Z}[\tau]$ small-norm multiplier base of 23 October (`research/ic_improve_20261023/README.md`) is the latest Koblitz change: 1.9–4.4× child CPU at $n = 41$–61 and 6.2–10.5× at $n = 71$, against the programme's own producer, exponent unchanged; it is discussed with the other known-logarithm bases in §36.5.

16 Random and prime-degree binary curves

This section covers binary curves $y^2 + xy = x^3 + ax^2 + b$ over $\mathbb{F}_{2^n}$ with n prime, attacked with random subspace bases that use no Frobenius structure, whether the curve is random or Koblitz: the Weil-descent route and the degree question behind the subexponential claims (§3), the decomposition oracles, the random-binary leaderboard rows, and the 131-bit ECC2-131 target. Many of the descent studies were priced for ECC2K-130 because it was the programme's named target, but their systems are built on prime-degree fields with random subspaces and use no Frobenius structure (the systems at $n = 19, 21$ say so explicitly), so they belong here. The Gröbner and $\mathbb{F}_2$ engines they run on are in §10; every study is listed in the programme record [3].

16.1 The degree question, measured

The subexponential claims for prime-degree binary curves rest on the first-fall-degree heuristic: that Weil-descended summation systems have a solving degree close to their first fall degree (FFD), which stays small. The programme measured both, on toy fields, in about thirty preregistered studies from late September to 24 October.

First fall degree stays low; the solving degree does not. Every Weil-descended S_3 system for $n = 3$–7 has first fall degree 3 (`research/notes/index-calculus/RESEARCH_FFD_MEASUREMENT.md`), and the HKY last fall degree d_F is 3 or 4 at every reachable size—$m = 2$ to $n = 32$, $m = 3$ at $\ell \le 6$, $m = 4$ at $\ell \le 3$, $m = 5, 6$ to $N = 54$ unknowns—while the plain refutation degree rises from 5 to at least 7 and random controls rise with it (`research/fall_degree_bounds_20261008/`). At $m = 2$ the Semaev d_F sits one below the semi-regular degree at $n \le 18$, and the gap does not grow over $n = 10$–18.

The refutation degree tracks ℓ, within one of semi-regular. For chained $m = 3$ systems at fixed surplus the refutation degree reads 5 at $\ell = 2$, 6 at $\ell = 3, 4$, and exactly 7 at $\ell = 5$ on all four draws of the $(10, 5)$ cell, with FFD 3 throughout (`research/dreg_ell_grid_20260925/`, `research/dreg_surplus_control_20260925/`, `research/dreg_degree7_ell5_20260930/`). At $m = 4$ it reads 5, 6 and at least 7 at $\ell = 1, 2, 3$, flat in n at fixed ℓ (`research/dreg_m4_chain_20261006/`). In 10 of 14 exact cells the semi-regular D_{reg} equals the measured degree and in all 14 it is within one. With a dense in-tree engine at $\ell = 6$ ($n = 19$, both Koblitz curves) the x_4 formulation refutes at 11 and the RR formulation at 8 on every draw, 4 above the sharp first-fall bound of 7, rising one degree per rung from $\ell = 3$ (`research/ic_dense_ladder_20261004/`). A registered semi-regular law then fitted none of seven discriminating cells, with x_4 inside $[d_{\mathrm{reg}}, d_{\mathrm{reg}} + 1]$ on all 32 draws (`research/ic_semiregular_law_20261005/`).

Closure algorithms refute earlier, but rise. A MutantXL-style closure refutes all 92 unsatisfiable draws at degree 3–4 where XL needs 5–7, and none of 345 satisfiable draws (`research/dreg_closure_mutantxl_20261016/`). It stays at degree 4 to $N = 42$ at $m = 4$, $\ell \le 4$, but rises past 4 at $m = 3$, $\ell = 7$ on 8 of 8 draws (`research/dreg_closure_scale_20261016/`); the number of unknowns is not the driver, and of two candidate explanations only "$\ell = 7$" survives a run at $(m, \ell) = (4, 6)$, $N = 54$, which refuted at degree 4 (`research/dreg_closure_driver_20261016/`, `research/dreg_closure_m4l6_20261018/`). Guessing one or two variables restores degree 4 at $(3, 7)$ on 10 of 10 draws at 1/116–1/260 of the projected degree-5 cost (`research/dreg_hybrid_guess_20261024/`).

What the measurements price. Applied to ECC2K-130's $m = 4$ row, the closure results contradict the degree-4 floor (+7.62 bits over rho) and leave degree 5 as the optimistic case, +10.32 bits with the closure exponent $\omega_c = 1$ and +45.71 with $\omega_c = 2$ (column touches per Macaulay column, not the §5a band weight); the hybrid gives +8.43 bits if the number of guesses stays flat in ℓ and +58.27 if it grows linearly (§36.5). The $m = 3$, $\ell = 8$ runs since made bound the count at 4 but do not measure it at $m = 4$, so the range stands (§19.1). The $m = 4$ cost slope on this engine is 0.985–1.060 bits per unit of n against the 0.25 an exponent lead would need, and a symmetrised refutation degree rises about one per unit of ℓ (`research/ic_m4_exponent_audit_20260928/`, `research/ic_m4_head_engine_20260929/`, `research/ic_symmetry_lever_slope_20260929/`).

ℓ	Speed-up
8	94–277×
9	2,474×
10	4,244×
11	928×
12	2,321×[a]

Table 16.1: The meet-in-the-middle pair table over Gaudry's pairs-and-solve on $y^2 + xy = x^3 + zx + 1$, by rung ℓ: wall-time speed-up on a shared host, unvalidated; 128 targets per rung, no disagreement. [a]The validation sweep's recomputation from the committed per-target medians; the study quotes 2,187×. Sources: `research/binary_mitm_oracle_l8_20261015/`, `research/binary_mitm_oracle_l9_l10_20261015/`, `research/binary_mitm_oracle_l11_l12_20261016/`.

Levers that did not move the degree. Decomposition trees cut on the curve's own structure have the same sampled cut-out degree as random sets in all 102 cells (`research/ic_tree_split_structured_v2_20261011/`); nonlinear factor-base labels refute nowhere through degree 6 (`research/nonlinear_factor_labels_20261002/`); Hamming-weight ideals answer correctly but resolve only after 7–18 of n coordinates are assigned (`research/hamming_ideal_pdp_20260930/`); the one measured subspace-shape effect on the degree was a trace artefact (`research/notes/index-calculus/RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md`), and frozen shape cells were killed at a gap of zero (`research/shape_search_g1_g6_20261015/`). The offensive first-fall-degree thread found no lever within 9 bits of the 2^N boundary (`research/notes/index-calculus/RESEARCH_DEGREE_REDUCTION.md`). A census over $\mathbb{F}_{2^7}$ found non-subfield quasi-subfield polynomials below the HKPYY Lemma 4.1 bound, which answers a tightness question but moves no cost (`research/notes/index-calculus/RESEARCH_QUASI_SUBFIELD.md`).

The same question in crypto-autoresearcher. The companion repository measured it independently ([3]). msolve's solving degree of the descended $m = 3$ system is 7 at every n from 11 to 23, a lower bound from $n = 17$ on, where the run walls on matrix size (`ar:experiments/EXP-ALBIN-008/analysis.md`); the study reads this as evidence on the bounded side in a narrow range that cannot distinguish bounded from slowly growing. A degree-4 reuse closure refutes all 400 unsatisfiable S_3 descents at $n = 19$ that plain degree-4 Macaulay refutes none of (`ar:experiments/EXP-CERTBIN-060020/`), the same effect as the MutantXL closure above. A crossbred solver's total exponent is 0.891 $[0.869, 0.914]$ against rho's 0.498 over $n = 10$–24 (`ar:experiments/EXP-ICI-001/analysis.md`); on its older range, $n \leq 20$, the fit reproduces the repository's earlier 0.863 as 0.864.

Reading. On toy fields, the evidence says the solving degree of Weil-descended summation systems grows with ℓ and stays within one of the semi-regular prediction, while the first and last fall degrees stay at 3–4. That is the behaviour of a generic system, and it supports no subexponential claim. It does not refute one: the sizes are small, the closure algorithms bend the curve by a degree or two, and the $m = 4$ question is still open. The $(m, \ell) = (4, 6)$ cell has since run and refutes at degree 4 on 2 of 2 unsatisfiable draws (`research/dreg_closure_m4l6_20261018/`). At $\ell = 8$, only $m = 3$ has run, with guessed variables (§36.5). The $m = 4$, $\ell = 8$ cell has not run. The PKM analogue over prime fields (§21) is in the same state.

16.2 Decomposition oracles

The pair-table floor. On the non-Koblitz curve $y^2 + xy = x^3 + zx + 1$ the meet-in-the-middle pair table is the floor every oracle is measured against. Table 16.1 gives its speed-up over Gaudry's pairs-and-solve, with no disagreement on 128 targets per rung. These are wall times on a shared host and unvalidated; the counted evidence—8,068 and 16,092 point additions against 2.1 and 8.4 million quartic solves—carries the result. An earlier pairs-and-root-find oracle was 926× faster than triple enumeration at $\ell = 8$ and decided $\ell = 12$ (11.5 billion triples) in 15 s (`research/notes/index-calculus/RESEARCH_SEMAEV_DECOMPOSITION.md`).

Algebraic oracles against the floor. No algebraic arm beats the pair table: the cheapest is 303× the floor at $n = 13$ and $3.8 \cdot 10^4$× at $n = 31$ for $m = 3$, and the slopes in n are steeper (F4 0.534, F6-IC

0.583, floor 0.207; at $m = 4$ a wide chained solver 0.805 against 0.333; `research/ic_m3_algebraic_vs_mitm_20261014/`).

SAT, SMT and WDSat are enumerators. Every SAT, SMT and WDSat back end spends about one conflict per candidate tuple (CDCL 0.97–1.02 per triple), so the best enumerate-and-root-find oracle beats every solver by 10^2–10^4 (`research/notes/index-calculus/RESEARCH_PDP_SAT_SMT_NAGAO_SURVEY.md`). The SAT pipeline decides all 11 satisfiable $n = 19$, $\ell = 6$ corpus instances in 17.8 s; SMT runs at 0.44–2.63× the native CDCL+XOR solver (`research/pdp_smt_oracle_20261008/`); WDSat solves $n = 5$ (47 against 51,425 conflicts with and without XOR-Gauss) and crashes at $n = 9$ on a static allocation limit, where CryptoMiniSat takes 0.4–2.6 s (`research/wdsat_rr_20260920/`). Run unmodified on the 60 published S_4 instances of Trimoska et al., WDSat is the fastest solver, 2.37 s on the unsatisfiable $n = 19$, $\ell = 6$ cell against 124.6 s for CryptoMiniSat with XOR clauses (`ar:experiments/EXP-ICPERF-66fd51/`; [3]); that is a solver comparison on toy instances, with no relation, linear-algebra or rho column. The SAT route is closed as an exponent lead. Isogenous and random-b neighbours give identical Macaulay ranks and FFD 3 (`research/pdp_isogeny_probe_20261008/`).

The PDP oracle programme. The point-decomposition speed-up programme (`research/pdp_oracle_speedup_20261011/`), run over $\mathbb{F}_{2^{3\ell}}$, measured the oracle in steps before root finding replaced it: a baseline (PS-0; 338–452 ns per pair across five rungs, A/A spread $\leq 8.8\%$), a gate-count model of the group-law form (PS-1; 7,410–14,608 fused ternary-logic issues against 70,695–190,275, 9.5–13.0×), a scalar group-law kernel (PS-2; 29.8–40.8 ns per pair, 0.067–0.123 of baseline, 400 corpus root sets agreeing), and a bit-sliced kernel (50–62× the baseline on two hosts, about equal to the scalar kernel). The root-finding oracle then ran 3,007–115,160× the PS-0 baseline at $\ell = 8$–14, at the generic floor's slope (`benchmarks/20261012-pdp-root-finder/`). A split/hybrid solving-degree study (Koblitz $b = 1$) projected its closest relation phase at 22.8× strong rho, 391× signed-Frobenius rho (`research/pdp_split_dreg_20261011/`); with Frobenius closure the priced S fell to 0.35–0.54 of the standard base but the best arm stayed at 36× strong rho at $n = 11$ and 889× at $n = 19$ (`research/pdp_split_slots_20261012/`). A shape-replay switch cost more on all 9 cells and stays off (`research/pdp_shape_replay_20261015/`).

Norm and circuit searches at $n = 83$. The 17–19 October autolab cycles built a native norm predicate over GF(2^{83}): a cutoff-8 circuit cuts AND gates 60.5% and clauses 32.9%, an affine support limit of 2 cuts internal coordinates from 58,176 to 45,427, and 1,237 of 1,800 positive-dimensional cells were certified empty. Those certificates predate the 24 October field-table correction (§12.8); its list of voided engines does not name the norm predicate, and whether the certificates used the truncated table is not recorded, so the empty cells are unconfirmed. Every hard search returned UNKNOWN with zero verified witnesses, and a primary-source check found the hybrid ANF/CNF approach already established in Bosphorus, SMFF and PolyBoRi (`research/autolab_cycle_20261019_v1/`, `research/autolab_cycle_20261019_v3/`). The companion draft `papers/norm-rootfinding/` reports the same: no measured speed-up.

Toy end-to-end. On toy fields the fastest index calculus, a cached Nagao arm, was 2.33× plain rho, and S_4/S_3 with CryptoMiniSat 14.5–14.7× (`research/toy_ic_e2e_candidates_20260922/`). Frobenius transport preserves zero and lift sets; a constant term c changes zero counts only; neighbouring curves show no consistent F5 advantage; symmetry quotients help three summands (0.26–0.31 of the ordered cost) but not two (1.06–1.32) (`research/toy_binary_elimination_20260924/` and the toy cluster).

16.3 Leaderboard rows

Table B of `docs/ic/LEADERBOARD.md` has five random-binary rows, $\mathbb{F}_{2^{15}}$ to $\mathbb{F}_{2^{27}}$ ($\log_2 r = 14$–24.4), one target, $m = 2$, folded pair table with walked targets: IC is 13.1, 10.6, 18, 23 and 38.2× matched rho. All five passed the validation gate (commit `a8d8e3cd38`; `research/ic_validation_gate_20261024/`), which the ledger paper, written a day earlier, records as no row passing. The cofactor-projected base of LEAD-19 cuts three of these rows by 1.29, 1.58 and 3.36× in cold S, leaving 6.1–14.4× to matched rho (`research/lead19_projected_base/`); it is not yet a leaderboard row. The integer-multiple closure of 23 October cuts S 3–4× against plain seeds at degree 26–28 and remains 6–12× the framework's rho reference (§36.5).

16.4 ECC2-131

The Certicom ECC2-131 target ($\log_2 r = 130.0$, rho $2^{64.83}$ GAE with the negation map) has a cost model, gates and two falsifiers (`docs/ic/CERTICOM131_IC_PROGRAM.md`), all on toy instances. With a free oracle the $m = 3$ floor is $2^{65.2}$–$2^{67.2}$, above rho from linear algebra alone; $m = 4$ is the first usable arity and needs a solving degree of at most 3 (generous pricing) or 1 (plain). On random-b curves at $m = 3$, $n = 5$–11, the measured degree is $D = 4, 5, 6$ at $\ell = 1, 2, 3$, equal to the semi-regular reference with zero spread over 60 draws, and equal to the Koblitz $b = 1$ value on 52 of 52 paired draws; extrapolated to $n = 131$ it is between about 6–7 (if it plateaus) and 39 (semi-regular). One CPU core walks ECC2-131 rho at 8.50 million steps per second with a batched pclmul walk, and 11.46 million bit-sliced (`research/certicom131_rho_ic_bench_20261022/`, `research/certicom131_rho_parity_20261023/`); a CUDA port compiles but has not run on a GPU. The bench's IC ladders on smaller random-binary curves read 11–24× matched rho; it is not an admitted row.

sect113r1. The 113-bit SECG curve is weak by size alone ($2^{56.33}$ generic work); its two degree-5 isogeny routes transfer the problem without an established speed-up, and the GHS and Semaev routes reached no conclusion (`research/sect113r1-weak-curve-20261006/`).

17 Koblitz structure and pipeline constants

Chapter 15 records what the programme built on Koblitz curves $E_a : y^2+xy = x^3+ax^2+1$ over $\mathbb{F}_{2^n}$: the compact-orbit producer, the online ladder, Table A and the priced routes to ECC2K-130. This chapter records why the Frobenius structure buys a constant and nothing more, and what each constant inside the pipeline was measured to be after the leaderboard rules of §15.6. It has nine parts:

- the orbit-fold closure statement and the measurements that bound it (§17.1);
- non-linear Frobenius-stable bases at prime degree (§17.2) and the unrun τ-Horner chain (§17.3);
- the matched rho reference as it applies to Koblitz rows (§17.4);
- the routing between the one-target descent and the pipeline plan (§17.5);
- large primes (§17.6), cofactors and torsion (§17.7), free relations (§17.8) and batch index calculus (§17.9).

Table 17.1 lists the constants in one place.

Standing of the numbers. Under the 4 November rule that every producer may be sabotaging the work, a figure here is either cited with its evidence standing or is a producer-reported, unverified assertion. One study in this chapter carries a `VALIDATION.json`: the large-prime component result of §17.6, whose file, validator report and replay certificate were written by the producer in the same commit and have no independent review. No other study in the chapter has a validation file or a proof-carrying claim, so every other number is a producer-reported measurement or derivation, labelled as such where the source does. Costs are in counted $S = \mathrm{GAE}/\sqrt{r}$ unless stated, against the strongest Koblitz rho the source ran (`rho.signed_frobenius_strong` unless named otherwise), with the §5a band $w = 0, 0.1, 1$ where the source prices it.

17.1 The orbit fold is exactly n

Chapter 15 gives the construction of the invariant and compact-orbit bases. The obstruction that ends the lead is recorded in the lead atlas as a proof, with the ledger hypothesis `H-FROB-6774fb` marked supported (`DEC-20260923-e6cd5d`). It is stated here as the atlas states it; the treatise does not re-prove it.

Proposition 17.1 (Orbit-fold closure; recorded in the ledger, not re-proved here)**.** *Let $E/\mathbb{F}_q$ be attacked over $\mathbb{F}_{q^n}$, let σ be the q-power Frobenius of $\mathbb{F}_{q^n}$ and π the Frobenius endomorphism of E.*

1. *The base $\{P : x(P) \in V\}$ for an $\mathbb{F}_q$-subspace V is π-stable exactly when V is σ-stable.*
2. *The σ-stable subspaces form a lattice fixed by the factorisation of $X^n - 1$ over $\mathbb{F}_q$.*
3. *Folding the base into π-orbits cuts the relations needed by exactly n and linear algebra by n^2, and by no more.*
4. *For prime n with $\mathrm{ord}_n(2) = n - 1$ (as for $n = 131$ and 163) the only stable subspaces over $\mathbb{F}_2$ have dimension 0, 1, $n-1$ or n, so no stable subspace of useful dimension exists.*

Part 4 is the replicated ledger hypothesis `H-BINSTD-b1f016`. Rho receives the matching $\sqrt{2n}$ from the same automorphism group, so the fold changes no gap between index calculus and the matched rho (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-frobenius-orbit-factor-base.yaml`).

The decomposition step gets nothing extra. A pilot in the companion repository (exploratory, "not program evidence" by its own label) enumerated every decomposition of 4,000 targets per cell at $m = 2$ and 1,500 at $m = 3$, on Koblitz curves at $n = 11, 13, 17, 19, 23$, for three bases: the programme's polynomial base V_{poly}, a window of conjugates of a normal element V_{nb}, and a union of random τ-orbits V_{orb} matched in size (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/tau-factor-base-pilot/README.md`). It split the gain into a counting term (orbits per point, $\log_2 g$) and a yield term (distinct projected relations per attempt over the birthday value $\binom{|\mathcal{F}|}{m}/r$).

- V_{orb} has $g = n$ in every cell, so it buys exactly $\log_2 n$ bits of structure. Its yield at matched $|\mathcal{F}|$ is within ± 0.1 bit of the birthday value in every $m = 2$ cell with at least 11 orbits ($+0.01$, -0.09, 0.00 at $n = 23$).

Lever	Measured effect	Distance to matched rho
Frobenius-orbit fold	exactly n on relations, n^2 on linear algebra; $\log_2 n$ bits at matched $\lvert\mathcal{F}\rvert$	IC/rho 2.96 ($n = 13$) to 18.2 ($n = 41$)
Orbit fold, priced as a generic base	$D_{\text{nofold}} = 0.56$ bits $[-0.51, 1.52]$	22.6–33.0 bits (formula)
Trace-spectrum stable base	10.2–12.5× a random subspace per attempt	closed
Hamming ball B_3	0.27–0.43 of random subspaces per attempt; a sparsity effect	not decided
Descent vs pipeline	descent 1.31–2.04× the pipeline	pipeline 9.5–24×
Two large primes, $m = 3$ table	0.604 ($n = 53$), 0.559 ($n = 61$) of the plain $m = 3$ table	30–64×
Two large primes vs `ic-compact`	0.985–1.183 at $w = 0$; null	4.9–28×
Orbit-aware large primes	more verified relations at $n = 17, 29, 31$; none at 23	toy
Torsion anchors	none at $n = 41$; 13.3× dearer at $n = 31$; 7.5× dearer at $n = 37$	$\sim$ 9,700× at $n = 41$
Free relations	0.960× (S3 root pipeline); +5–9% (pair-table pipeline)	unchanged
Batch, 32 targets	Table C 3.62–14.9× batch rho	pre-gate
Many-target crossover (accounting)	$G = \min_k$ IC(k)/batch rho(k) = 3.0–4.1, $n = 41$–61	$G > 1$ at every k (clock units)

Table 17.1: Koblitz constants measured after the leaderboard rules, with the distance each leaves to the matched rho. Every row is producer-reported; the two-large-prime row on the $m = 3$ table carries a producer-written `VALIDATION.json`. Sources are in the section of each row.

- V_{nb} buys about 1 bit, not $\log_2 n$: $g = 1.93$–2.02 at $n = 23$, because a window and its shift share a $(k-1)$-dimensional subspace.
- V_{poly} has a free coincidence of 0.03–0.5 bit that shrinks with n (0.06–0.10 bit at $n = 23$).
- At $m = 3$, few-orbit cells lose yield to duplicated projected rows (decompositions that differ by the kernel identity $\tau^2 - t_1\tau + 2 = 0$): 65% of $m = 3$ decompositions duplicate with one orbit, 0.2% with 22–45 orbits. This is the only place where the τ-structure of a base changes its yield, and it lowers it.

Over the populated cells the V_{orb} gain is 3.81 bits mean ($[2.89, 4.52]$) at $m = 2$ and 3.08 ($[1.89, 4.38]$) at $m = 3$, against $\log_2 n = 3.46$–4.52.

The 28 October recheck. An external toy study (`fb-koblitz-orbit`, $n = 15$ and 31, four resolved cells; its repository is not among the sources read here, so the figures are as the atlas records them) compared the best stable subspace base with the cost-model choice of non-stable base. The stable base wins by $D = 4.73$ bits $[3.44, 6.10]$, all of it the orbit credit ($\log_2 n = 3.91$ and 4.95); priced as a generic base the difference is $D_{\text{nofold}} = 0.56$ bits $[-0.51, 1.52]$, whose interval contains zero. The atlas notes that the protocol was frozen by timestamp only, its stored hash matching no text it holds. The stable arm's formula gap to matched rho is 22.6–33.0 bits.

The constant in counted cost. The fully priced ecbench ladder of 14 October (`research/ecbench_frobenius_orbit_vs_rho_20261014/README.md`; 2,816 records verified, 2,112 replayed identically, every A/A ratio exactly 1.000) prices every native counter of the orbit pipeline and of two signed-Frobenius rho arms. Table 17.2 gives the panel. The paired ratio IC/rho grows with r at slope $\beta = 0.165$ $[0.117, 0.213]$ per $\ln r$ once n is in the fit, against the model value 1/6; the study's class is "engineering, constant-only". Index calculus is below the signed-Frobenius rho on no curve; on two curves ($n = 17$, 29) the interval reaches 1, where rho's own spread is wide. Against the negation-folded subspace base the orbit base costs 0.065 $[0.059, 0.071]$ at $n = 23$, which is the constant the fold buys inside index calculus.

17.2 Non-linear stable bases at prime degree

Part 4 of Proposition 17.1 rules out stable subspaces at $n = 131$, but non-linear stable sets exist: trace-spectrum sets $V_E = \{x : \mathrm{Tr}(x^e) = 0,\ e \in E\}$, Hamming balls in a normal basis, multiplicative subgroups,

a	n	$\log_2 r$	cofactor	S rho-frob	orbit/rho-frob	orbit/negfold
0	13	10.97	4	2.025	2.956 [2.552, 3.242]	0.511
1	17	16.00	2	1.524	1.719 [0.810, 2.980]	0.234
0	19	17.00	4	0.835	2.805 [2.295, 3.236]	0.148
0	23	21.00	4	0.416	3.847 [3.035, 4.819]	0.065
1	29	15.37	12,646	1.665	1.566 [0.992, 2.347]	—
0	31	20.46	1,492	0.394	3.664 [2.907, 4.476]	—
0	37	27.78	596	0.253	5.048 [4.335, 6.086]	—
0	41	39.00	4	0.133	18.16 [15.67, 21.55]	—

Table 17.2: Frobenius-orbit index calculus against the signed-Frobenius rho, every native operation priced; 16 fresh public targets × 3 rounds per curve, L0. Orbit/rho-frob: paired ratio with cluster-bootstrap 95% interval; orbit/negfold: against the negation-folded subspace base where it was run. Source: `research/ecbench_frobenius_orbit_vs_rho_20261014/README.md`.

τ-adic patterns over an orbit union, and $\mathbb{F}_2$-points of low-dimensional abelian varieties (Couveignes–Lercier). If one kept the n-fold orbit gain with a membership test no dearer than a subspace's, it would restore the Koblitz lever at $n = 131$. The lead is open (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-nonlinear-frob-stable-bases.yaml`). Its Hamming-ball and trace-spectrum measurements live in an external repository (`aburan28/ml-cryptanalysis`, pull request 53, open when the atlas was last reviewed) that is not among the sources read for this treatise, so every figure below is as the atlas records it. All are per-attempt CryptoMiniSat conflict ratios at matched $|\mathcal{F}|$ on toy fields, with no orbit credit applied, no complete logarithm and no rho comparison.

- *Trace-spectrum bases: closed.* V_E costs $R = 10.2$ $[8.6, 11.8]$ (curve RC-1, $n = 17$), 11.6 $[8.3, 15.4]$ (Koblitz $n = 19$) and 12.5 $[10.7, 14.4]$ (Koblitz $n = 17$, $a_2 = 1$) times a random subspace, above the lead's 2× band.
- *Hamming ball B_3: not decided.* Against the mean of 8 matched random subspaces ($T = 16$ per workload) $R = 0.27$ $[0.13, 0.46]$ and 0.32 $[0.14, 0.55]$ at $n = 19$ ($a_2 = 0, 1$), 0.43 $[0.20, 0.76]$ (RC-1) and 0.29 $[0.11, 0.54]$ at $n = 17$. All eight point estimates lie at 0.20–0.52, but only three intervals exclude 0.5, so the lead's rule does not decide. An earlier $n = 19$ value of 1.10 came from one noisy subspace denominator. The saving sits in the satisfiable stratum (satisfiable targets about 5× cheaper, unsatisfiable ones the same).
- *Stability against sparsity.* A second run ($T = 32$, weights $w = 2$–4) found the per-attempt saving to be a sparsity effect: balls in a random basis read $R = 0.22$–0.40, and the normal-basis ball against the random-basis ball is $D = +0.20$ bits $[-0.05, +0.45]$, with the null gate failed. The exact yield P_{sys} is higher for the stable ball at $w = 2$ (×1.76 in 8 of 8 units, 48 bases per class); the derived per-relation benefit of -0.66 bits $[-1.10, -0.24]$ is post hoc.
- *Encoding at $n = 131$, size only.* $|B_6| = 2^{32.6}$; a Sinz cardinality counter needs 780 auxiliary variables and 1,684 clauses per block (a totaliser 485 and 1,817), against 98 native XORs for a 33-dimensional subspace. No solver price at $n = 131$ exists.
- *Other stable families.* Cyclotomic-sparse bases are yield-neutral at $n = 23$ (§36); an orbit-union base buys exactly $\log_2 n$ (7.03 bits at $n = 131$); a τ-adic base over an orbit union is a Γ_w-orbit quotient that reaches rho only at $w \geq 13$ (`EV-BINSTD-1e1815`, neutral, preliminary); for Couveignes–Lercier abelian varieties, dimensions 1–4 are excluded and dimensions 5–6 hit an impediment (`EV-FROB-a85eaf`, inconclusive; `EV-BINSTD-2d41c3`, preliminary support at dimension 5).

The lead's own next step is a frozen run with at least 20 normal and 20 random bases per class at $w = 2$, cost per relation as the primary; until then the only non-linear branch that is still open is the Hamming ball, and its measured saving is not yet attributable to stability.

17.3 The τ-Horner chain: an accounting test that has not run

Writing a target as a τ-adic Horner sum $R = \tau^{m-1}P_1 + \cdots + \tau P_{m-1} + P_m$ over one base makes ordered tuples distinct relations. The approved ledger hypothesis `H-FROB-855915` claims $m!$ times the distinct-relation yield of the plain chain at an unchanged descended shape. It has never run (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-tau-horner-chain.yaml`). The discriminating comparison (`IDEA-20261018-f2cd4d`) prices the cost per distinct relation of the plain, symmetry-broken and Horner chains at matched core size, $n \in \{17, 19, 23\}$, $m \in \{2, 3\}$, with a size-matched ordinary-curve

null. Its reading is fixed in advance: Horner cheaper than the symmetry-broken chain is a constant $m!$ advance for collection; equal cost means the Horner $m!$ is the symmetry-breaking $m!$ relabelled, which the success criterion counts as accounting, not an improvement.

17.4 The matched rho reference on Koblitz rows

Chapter 33 records how the strong signed-Frobenius reference was calibrated, engineered and checked, and Chapter 34 the walk variants and the volcano levels. Four Koblitz-specific facts are not there.

The rule-v4 reference is not provably the strongest. Rule v4 (§36.5) prices rho in the same process on the same point at the canonical batched step. Its admissibility check ran the parallel walk `ParallelRho` against the strong fixture on targets T01–T08 at six sizes (`research/ic_tool_program/rule/v4/README.md`): per step the parallel walk was 1.02, 1.20, 1.08, 0.75, 0.92 and 1.08 times the fixture at $n = 47$, 57, 41, 53, 59 and 61, so admissibility failed at four sizes. On a host without GFNI its 16-lane canonicalisation is slower, and at three sizes its processes ran beside other jobs. The validation gate (`research/ic_validation_gate_20261024/README.md`) wrote no `VALIDATION.json` for rule v4 because `strongest_available` could not be set true: on 46 readable pairs the strong fixture needed 0.845 ± 0.08 of `ParallelRho`'s steps. The canonical-step gap does not depend on either walk; it reads 9.89, 9.20, 16.57, 27.17, 24.25 and 42.12 at those sizes, 1.76–1.94× the as-built gap, reproducing the validation sweep's 1.67–1.96.

The Frobenius fold in rho is measured. The volcano-layer study (`research/volcano_koblitz_layers_20261018/RESULT.md`, all sessions audited, L0) ran rho on Koblitz craters and their level-1 neighbours at $n = 19, 23, 37, 41, 61$. Level-1 over crater for the negation walk contains 1 at every size (1.089 $[0.993, 1.195]$ at $n = 19$ to 0.837 $[0.625, 1.073]$ at $n = 61$); the plain walk excludes 1 once in ten intervals ($n = 41$, 1.152 $[1.021, 1.304]$). On the crater the signed-Frobenius solver over the negation solver is 2.153 at $n = 19$ and 0.864 at $n = 23$, where fixed overhead dominates, 0.264 at $n = 37$ and 0.150 $[0.132, 0.170]$ at $n = 41$, against the ideal $1/\sqrt{n} = 0.156$: about a $\sqrt{n}$ gain from $n \geq 41$, the share of the automorphism group that the atlas credits to rho against the fold of Proposition 17.1. The study retracts an earlier "±8%" statement as wrong. At $n = 83$ the same tool built $\Phi_{6473}(X, j_0)$ from 9,864 CRT primes (606,365-bit coefficients, 8,590 s on three threads), reduced it modulo 2 and factored it into 78 Frobenius orbits of size 83, finding all 6,474 level-1 j-invariants (`research/volcano_koblitz_n83_20261018/RESULT.md`).

No counted 131-bit reference. The rho lead records that no 131-bit signed-Frobenius walk exists in the counted harness, so no ECC2K-130 index-calculus lane has a matched counted rho (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-koblitz-rho-reference.yaml`). The throughput client of §15.7 (13.2 million steps per second on one core) is a wall-time walk, and every ECC2K-130 distance in this treatise is a model against $2^{60.81}$.

The step price, cross-checked. The cryptanalysis repository's normal-basis canonicalisation makes signed-Frobenius rho 4.1× cheaper per step, which by itself reverses an apparent index-calculus win there (the atlas cites `cryptanalysis:papers/frobenius-orbits/sec-abstract.tex`, not read for this chapter). The strengthening is accounting: it widens every Koblitz gap and changes no index-calculus count.

17.5 Routing: the one-target descent against the pipeline plan

Table A, as the routing study quotes it, prices the ledger §23 one-target Koblitz pipeline at 4.7–11.8× rho (a range that matches no single rule revision of Table 15.3(a)), but in mixed units: counted index-calculus work against rho in canonical step units, which charge rho its canonicalisation. ecbench gained `ic.koblitz_descent`, the §23 pipeline charged in counted group operations, with logs and native counts identical to `ic price -single-target` on the same seeds, so the descent and the planner's own recipe (`compact-orbit-scan` with `mitm-frobenius-counted:m=2`) can be run in one unit against `rho.signed_frobenius_strong` (`research/ic_koblitz_descent_routing_20261026/README.md`; protocol frozen in `cd3af7078`; 288 of 288 runs verified, 12 exact replays per audit). The six-size retest

Curve	$\log_2 r$	descent/pipeline	pipeline/rho	descent/rho
k47 ($a = 1$, $n = 47$)	36.64	1.313 [1.199, 1.437]	10.8 [8.5, 13.7]	14.2 [11.4, 17.6]
k57	38.00	1.626 [1.544, 1.712]	12.1 [8.5, 17.2]	19.6 [14.4, 26.7]
k41 ($a = 0$, $n = 41$)	39.00	1.471 [1.420, 1.524]	14.1 [11.0, 18.0]	20.7 [16.0, 26.9]
k53	44.26	1.462 [1.410, 1.515]	20.0 [16.6, 24.1]	29.3 [23.9, 35.8]
k59	44.52	1.443 [1.382, 1.507]	22.9 [17.8, 29.5]	33.1 [26.5, 41.3]
k61	47.21	2.039 [1.977, 2.103]	21.4 [17.9, 25.6]	43.6 [36.4, 52.1]

Table 17.3: The §23 descent against the `ic.pipeline` plan, counted S at $w = 0$, 24 pairs per curve, cluster 95% intervals; rho is `rho.signed_frobenius_strong`. The first study (k47, k41, k53) read descent/pipeline 1.421, 1.489, 1.409 and pipeline/rho 9.46, 16.19, 23.99. Source: `research/ic_planner_exhaustive_20261026/README.md`.

(`research/ic_planner_exhaustive_20261026/README.md`; protocol frozen in `0ca110399`; 3,328 records verified, band verdicts restated in integers and checked by Lean's `decide`) is in Table 17.3.

- *Routing.* The descent costs 1.31–2.04× the pipeline at all six sizes, every interval above 1, so the planner keeps `ic.pipeline` for Koblitz curves at $\log_2 r \geq 36$.
- *Where the descent's cost goes.* At k41, relation collection is about 2.05 M GAE and the pair table 0.27 M, while the online descent is 1.0 k–184 k GAE per target; the precomputation is reusable, so a many-target comparison needs batch rho (§34.7).
- *Why Table A read lower.* `rho.signed_frobenius_strong` folds by $2n$ and is charged only group operations, so its counted S is 0.11–0.14; in that unit the Table A descent figures read 13–34× rho (first study), not 4.7–11.8×. The study left re-pricing Table A to the owner, and no row moved.
- *Growth.* Descent/rho rises at every step, 14.2 to 43.6; pipeline/rho rises from 10.8 to 22.9 and reads 21.4 at k61, a literal failure of the preregistered "grows with $\log_2 r$" with overlapping intervals.
- *Band.* `ecbench band` cannot price `ic.koblitz_descent` above $w = 0$: four native categories (field arithmetic outside the group law, hashing, the modular combination, declined AVX-512 lanes) have no count, so the descent is "not ranked" at $w = 0.1$ and 1.
- *Weight rule.* The same retest withdrew the Koblitz planner rule (base $0.71c_0$ at $w \geq 0.3$): held out at 36.6–47.2 bits, the smaller base was dearer at every weight on every curve, 1.06–1.17× at $w = 1$, although it had been cheaper (0.80–0.88×) on 18–26-bit hold-outs. `ic-planner/4` keeps the counted optimum c_0.

17.6 Large primes on Koblitz curves

Large primes appear in this treatise on the genus-3 and extension-field routes (Chapters 25 and 26). On Koblitz curves there are four studies, and the two late-October ones appear to disagree.

Two large primes beat the plain $m = 3$ table. The two-large-prime collector (`mitm-frobenius-counted:m=3,large_primes=2,lp_domain_factor=c`, already on `main`) was run on the classic single-target compact-orbit pair-table pipeline at $n = 53$ and 61, against the plain $m = 3$ table over the same envelope and against the best plain table in the bracket (`research/icb_large_prime_n53_n61_20261029/README.md`, `research/icb_large_prime_n53_n61_20261029/report.html`). Every run verified; the A/A arm and the collector switched off (`large_primes=0`) reproduce the plain arm counter for counter; `ecbench verify -replay-all` replayed every session (612 replays). The benchmarks entry carries a `VALIDATION.json` whose rho reference is the session's own `rho.signed_frobenius_strong` arm, marked `strongest_available` and `order_holds_across_band` (`benchmarks/20261029-icb-large-prime/VALIDATION.json`); it, the validator report and the replay certificate were written by the producer in one commit, so the row is producer-validated, not independently reviewed, and it carries no pcc claim (an unverified assertion under the 4 November rule). The validation file calls it a component result, not a leaderboard row. Table 17.4 gives the best arms.

The order survives the band for the best arms; others win at $w = 0$ and lose at $w = 1$ (`lp2-k20-c10` at $n = 53$: 0.61/0.78/1.28), because their native work exceeds the group operations they save. One large prime is worse than the plain table at equal envelope (1.73 [1.64, 1.82] and 1.47 [1.42, 1.52] at $n = 53$). The optimal large-prime base stays at 40–60 columns while the plain optimum moves from 440 to 880. Memory is not in S: at $n = 53$ both best arms use about 315 MiB; at $n = 61$ the two-LP arm uses

n ($\log_2 r$)	best plain $m = 3$	best two-LP	paired ratio	ratio $w = 0/0.1/1$	arm/rho $w = 0/0.1/1$
53 (44.3)	440 cols, $S = 6.38$	`lp2-k40-c10`, 3.86	0.604 [0.599, 0.610]	0.604/0.635/0.730	30/31/35
61 (47.2)	880 cols, $S = 7.62$	`lp2-k60-c10`, 4.26	0.559 [0.535, 0.578]	0.559/0.623/0.831	47/52/64

Tuned double large prime against `ic-compact`, holdout (96 targets per curve)

n ($\log_2 r$)	selected cell	$w = 0$	$w = 0.1$	$w = 1$
37 (27.8)	$K = 1$, $c = 10$	1.001 [0.945, 1.061]	0.793 [0.751, 0.838]	0.570 [0.536, 0.603]
43 (32.1)	$K = 1$, $c = 20$	0.985 [0.928, 1.051]	0.937 [0.885, 0.995]	0.866 [0.819, 0.919]
41 (39.0)	$K = 1$, $c = 160$	1.183 [1.155, 1.213]	1.112 [1.084, 1.143]	0.955 [0.922, 0.989]
47 (36.6)	$K = 3$, $c = 20$	1.095 [1.060, 1.132]	1.067 [1.032, 1.102]	1.015 [0.980, 1.052]

Table 17.4: Two large primes on Koblitz curves. Top: against the plain $m = 3$ Frobenius pair table, 16 workloads at $n = 53$, 3 at $n = 61$ (`research/icb_large_prime_n53_n61_20261029/README.md`). Bottom: the tuned double-large-prime arm over the frozen `ic-compact` baseline, paired ratio of sums with two-stage cluster bootstrap; every verdict "unresolved inside the band" (`research/ic_koblitz_dlp_tuning_20261023/REPORT.md`).

604 MiB against 1,149 MiB. The $n = 61$ interval is over three workloads, and the column optimum was found on the same curves as the confirmation.

Tuned double large primes do not beat the compact baseline. The preregistered tuning study of 23–28 October (`research/ic_koblitz_dlp_tuning_20261023/REPORT.md`; protocol frozen in `327afb04f`, Amendments 1–3 before the data they affect) tuned the same oracle family's (K, c) on a train panel and compared it on 96 held-out targets per curve with the frozen baseline `ic-compact`. Its class is *null*: no curve beats the baseline at every weight, and at $n = 41$ and 47 the arm is worse at $w = 0$ and 0.1 (Table 17.4). The $w > 0$ gain at $n = 37$ and 43 comes from the arm leaving fewer unpriced native operations, not from a lower count at $w = 0$. The positive control shows the instrument sees a real double-large-prime effect: at $n = 41$ with $K = 10$ columns, double-LP over the plain $m = 3$ oracle is 0.0316 $[0.0267, 0.0374]$ at $w = 0$, 31.6× cheaper. The only train-side win ($n = 43$, 0.893) shrank to 0.985 on holdout, a winner's-curse loss of about 9 points. Both arms stay 4.9–28× the matched rho at every weight.

Why the two studies disagree. The two studies do not state a reconciliation. Their own protocols and the routing study supply one, and it is an inference from unpaired sessions, not a measurement:

- *Different baselines.* The large-prime study's comparator is the plain $m = 3$ oracle `mitm-frobenius-counted:m=3` (`research/icb_large_prime_n53_n61_20261029/PROTOCOL.md`). The tuning study's comparator `ic-compact` is `compact-orbit-scan:columns=Kb` with the $m = 2$ oracle `mitm-frobenius-counted:m=2` (`research/ic_koblitz_dlp_tuning_20261023/PROTOCOL.md`). The double-LP collector is an $m = 3$ oracle in both.
- *The $m = 3$ plain table is the weaker arm.* On the same curves the $m = 2$ pipeline is cheaper than the best two-LP arm. At $n = 53$ (`icv1-f2m53-tm56619371-dac20a85` in both studies) the pipeline reads $S = 2.728$, 24.0× rho, in the routing study and 20.0× in the retest, against 3.856 and 30× for `lp2-k40-c10`; at $n = 61$ (`icv1-f2m61-t158598901-ab42b6c5`) the pipeline reads 21.4× rho against 47× for `lp2-k60-c10`. The plain $m = 3$ table reads 49× and 85× there.
- *So both results hold.* The double large prime roughly halves the cost of the $m = 3$ table, which is the same direction and mechanism as the tuning study's positive control (31.6× at $K = 10$), and leaves it at or above the $m = 2$ compact pipeline, which is what the tuning study found by direct pairing at $n = 37$–47.

The large-prime study itself says its result does not touch Table A, whose aimed $m = 3$ collection with an $m = 2$ descent reads 7.03× ($n = 53$) and 11.5× ($n = 61$) of its own reference; the collector has not been combined with that recipe, and stage 0 of the experiment that would do so (`EXP-BINSTD-e066d8`) has not run (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-large-prime-variants.yaml`). The tuning study's own next step is to draw the large-prime domain from a structured set (small $\mathbb{Z}[\tau]$-norm multipliers) rather than raw orbits, since its selected cells sit at the $K = 1$ edge where the domain does all the work.

Orbit-aware large primes. Quotienting the large-prime vertices by signed Frobenius orbits, with the $\pm\lambda^k$ phase kept, gave more independently verified relations than the plain double large prime from the

same relation stream at $n = 17$, 29 and 31 on every seed, a small and not clearly nonzero gain at $n = 19$, and none at $n = 23$ (`research/orbit_aware_large_prime_20261011/RESULTS.md`; measured 26 October). The gain tracks the number of orbit collisions among the fixed large primes, which a random model predicts at $n = 17$, 19, 29 and which is far above the model at $n = 31$ for a reason not found. All 17,865 emitted relations passed both replays. Online CPU does not separate the arms, so CPU per relation falls only through yield; at $n = 17$, for example, 9.2 independent relations per cell against 6.2 (10.8 against 6.4 emitted), at 0.181 against 0.275 ms of CPU per independent relation. The scale is toy (r between 2^{15} and 2^{21}).

The 131-bit Certicom curves. Symmetry breaking and double large primes at $m = 2$ on ECC2-131 and ECCp-131 are in §20.2: the binary symmetric oracle steps 28.7× faster than the per-point baseline and the distance to rho stays about 2^{22} steps by the $m = 2$ exponent.

17.7 Cofactors, torsion anchors and quotients

The cofactor-projected base of LEAD-19 and its Table B cuts are in §16; the prime-field cofactor-clearing A/B and its `invalid_controls` verdict are in §22.4. Two further Koblitz results are a measured reversal and a closed quotient.

Torsion anchors: a generalisation withdrawn twice. The $m = 2$ cycle-stop study (`research/ic_m2_cycle_leads_20261030/REPORT.md`, exploratory, no frozen protocol) stated that torsion-randomised anchors make large-cofactor curves workable, solving $a = 0$, $n = 37$ (cofactor 596) in 26 and 31 s where the ladder stopped at 99 s, and that they cost 1.5× ($n = 23$) to 16× ($n = 31$) on cofactor-4 curves. Two frozen follow-ups of 1 November corrected both readings:

- $n = 41$, *cofactor 4* (`research/ic_m2_torsion_n41_20261101/REPORT.md`; protocol frozen in `baa2fcfa8`; eight paired targets, 16 of 16 runs verified). The paired geometric mean of trials, torsion on over off, is 0.98 $[0.64, 1.43]$; the per-pair ratios run from 0.30 to 2.60. No difference is detected. A post-hoc replicate at $n = 31$ (32 pairs) gives 13.3× more trials with torsion on, $[10.3, 17.5]$. The $n = 31$ figure is real; the generalisation to cofactor-4 curves is not.
- $n = 37$, *cofactor 596* (`research/ic_m2_torsion_n37_20261101/REPORT.md`; protocol frozen in `7893863417`; eight pairs). Torsion anchors make the solve about 7.5× dearer: off over on is 0.134 $[0.082, 0.205]$ in trials, with all eight pairs below 1. The relation rate is the same (about 17,000 trials per relation in both arms); without torsion the relation graph stays in components of 3–5 columns and a cycle closes after about 126 relations, with torsion components grow to about 117 columns and the first equation needs about 885. The ladder's failure at $n = 37$ came from its 90% coverage stop, not from the cofactor.

So torsion anchors helped nowhere they were measured. The family's distance is set by the method, not by anchors: against the same-process strong rho the $m = 2$ cycle variant is about 9,700× slower at $n = 41$, and 7,700× in mean wall at $n = 37$ without torsion. Both follow-ups are exploratory, on one shared four-vCPU container, with no replay certificate.

Three summands on the same curve. Three companion studies on the $a = 0$, $n = 37$ curve (cofactor 596) ask whether more summands help the cycle stop; all are exploratory, local, with no replay certificate or validator report. With the same stop, relations with three summands cost 26–43× more decomposer calls than two-summand relations at every base size tested (42.9 $[31.5, 68.2]$ at $d = 12$, $K = 2{,}011$), and all 18 pairs have $m = 3$ dearer; the frozen prediction held (`research/ic_m3_cycle_n37_20261101/REPORT.md`). Feeding the stop a mixture of two-summand and "2+1" three-summand relations, at eight mixture fractions q over six seeds, is monotone in q and never below $q = 0$, again as predicted (`research/ic_mix_cycle_n37_20261103/REPORT.md`). A direct S_4 solve at $m = 3$ cannot run at $n = 37$: a linear oracle is determined only for $\ell \leq 1$, and heuristic XL and an F4/F6 extrapolation put it far out of reach; that report is analysis only, with no new run (`research/ic_s4_direct_n37_20261101/REPORT.md`). None of the three moves the gap to rho.

Cofactor and 2-torsion quotients: closed. The atlas records a proof (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-cofactor-torsion-quotients.yaml`). On ECC2K-130 $\gcd(4, \ell) = 1$, so every full-translation homomorphic cofactor quotient reduces to the order-ℓ projection rho already uses; exactly sum-compatible filters on $E(\mathbb{F}_{2^{131}})$ carry 2 bits (the cofactor), so Wagner's

n	K	collisions	verified nontrivial	homogeneous rank	needed
61	600	10,188	588	49	599
61	1,200	12,732	4,800	746	1,199
61	1,500	10,924	6,000	1,197	1,499
83	600	9,672	72	6	599

Table 17.5: Free relations from pair-table self-collisions: the rank they reach against the $K-1$ the rank stage needs. Source: `RESEARCH_FREE_RELATIONS_CENSUS_20261012.md`.

k-tree has no quotient to work in; and the corrected ι-stable classification ($V_d = b^{1/4}\mathbb{F}_{2^d}$) empties the cell (subspace base, 2-torsion quotient) on every prime-degree standardised curve. The production 2-torsion census added no relations. A September study measured the same bound from the other side: the invariants used by translation-invariant bases on $y^2 + xy = x^3 + 1$ are pullbacks by $1+\tau$, $1-\tau$ and [2], held on all 10,238 points at $n = 7, 11, 13$, and each kernel has at most 4 points, so they change a decomposition by at most 2 bits per summand (`research/cheap_closures_20261007/README.md`).

17.8 Free relations

Pair-table self-collisions give relations with right-hand side zero: four base points that sum to O, verified as exact group identities (`RESEARCH_FREE_RELATIONS_CENSUS_20261012.md`). The relations exist and are highly correlated (Table 17.5): the 588 nontrivial rows at $n = 61$, $K = 600$ touch 168 of 600 columns, and the 72 at $n = 83$ span six dimensions.

- *Rank law.* Rank grows about as K^4 with a small constant, so a free rank stage needs $K \approx 1{,}500$–$1{,}900$ at $n = 61$ and $K \approx 2{,}600$ at $n = 83$, where the census scan and the K^2 linear algebra already exceed the guided rank stage (13.1 s at $n = 61$, 4.8 min on 12 threads at $n = 83$), before 36 GB of memory at $n = 83$.
- *In the S3-root pipeline.* On by default for $n \le 63$, the harvest costs $0.960\times$ $[0.946, 0.976]$ of the tuned incumbent pooled over $n = 41$, 53, 61, in S3 root solves per complete solve; wall time agrees, $0.941\times$ (`research/ic_improve_20261015/02-freerel-production/README.md`).
- *In the pair-table pipeline.* Under rule v4 the harvest is correct (6,187 relations, none rejected, in 510 runs) but yields 2–11 relations per solve against 58–327 needed, cuts trials 0.5–9%, and makes the build 1.4–$1.55\times$ slower: the cold cost rises by 5.3–8.8% at every size (1.053–1.088 over 72 pairs per size). The 0.960 does not transfer, because that pipeline costs 59 s per solve at $n = 61$ against 3.7 s here (`research/ic_tool_program/rule/v4/README.md`).

17.9 Batch index calculus and restart portfolios

The many-target accounting correction and its batch-rho reference are in §34.7; Table C is in §15.6. The lead records two statements that bound the route (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-multi-target-and-portfolio.yaml`):

- *The generic multi-instance bound covers set-up.* L logarithms cost at least about $\sqrt{Lr/|\mathrm{Aut}|}$ for a generic-oracle index calculus including its set-up, and Kuhn–Struik batch rho attains it; the L^{-1} amortisation applies only to the online part, after a set-up that obeys the same bound. This is the multi-target form of the Corrigan-Gibbs–Kogan bound [8] of §3.
- *A restart portfolio cannot beat pure rho* when the index-calculus cost concentrates at a floor ratio of at least 1. The ledger hypothesis `H-BINSTD-32867d` is marked supported as a scoped observational hold, strength preliminary (`EV-BINSTD-2d34cb`); its instrument was a deterministic closure, a controlled null rather than a solver.

The four-target union column of the September batch thread is a rare-event linearisation, not a batch speed-up. Table C's 32-target rows stay 3.62–$14.9\times$ batch rho, with the reference taken from a different session (S20) at the canonical step, and all nine failed the 24 October gate.

The many-target crossover. A 30 October accounting study asked directly whether amortising relation collection over k targets ever makes compact-orbit index calculus cheaper than Kuhn–Struik batch rho (`research/ic_many_target_crossover_20261030/README.md`; nothing was run, every number is extracted from recorded runs). With $G = \min_k \mathrm{IC}(k)/\text{batch rho}(k)$ as the distance remaining,

it never does: in clock-priced `ic price` units (same process, both arms) $G = 3.0$–4.1 at $n = 41$–61 for the best swept recipe and 3.7–6.8 for the recipe tuned for $k = 32$, flat to rising above 2^{36}. On the §5a count band, with IC at its emitted additions only (a lower bound), G stays above 1 at $n = 41$, 43, 53 and 61 (3.1–4.8 at $w = 0$, 1.6–2.4 at $w = 1$), so the order holds across the band there; at $n \leq 45$ and $n = 47$ unemitted descent and construction work leaves the sizes unresolved by counts. At $n = 83$, in wall and CPU seconds only, $G = 4.8$–5.1 for K3300. The study classes itself as accounting and audit, not an improvement of the IC baseline; it has no validation file.

18 The $n = 83$ campaign after 21 October

§15.5 covers degree 83 to 21 October, and §12.8 the 24 October field-table correction, after which every Weil-descent engine's earlier $n = 83$ refutation is void. Forty-one study directories were added from 24 October to 1 November. None of them admits a new IC/rho ratio, and none is a leaderboard or `benchmarks/` row. They fall into eight groups (Table 18.1), which this chapter takes in four parts:

- factor-base selection, index capacity and controls on the pipeline curve (§18.1);
- decomposition at $n = 83$ after the correction (§18.2);
- certified degree bounds for the descended systems (§18.3);
- the genus-3 image transfer (§18.4).

Two curves, three families. Degree 83 means two different targets in these studies, and their numbers must not be mixed.

- *The pipeline curve* `EC1N83Ckb1a1haacb69ee9f37`, $y^2+xy = x^3+x^2+1$ ($a = 1$), has a 53-bit subgroup of order 8,569,786,107,849,059 and a cofactor of about $1.1 \cdot 10^9$. Every K-series base, every online and cold time and the matched precomputed-rho control are on this curve.
- *The primary curve* $y^2 + xy = x^3 + 1$ ($a = 0$, modulus $z^{83} + z^{45} + z^2 + z + 1$, trace $-6{,}151{,}469{,}093{,}347$) has the 81-bit subgroup $r = 2{,}417{,}851{,}639{,}230{,}796{,}216{,}685{,}689$ and cofactor 4. The genus-3 transfer and the sixteen factor-base designs target it.
- *The degree sweeps* use full-curve affine bases on three families at the same size: the Koblitz curve, a general binary curve and an 83-bit prime curve, with no subgroup filter.

18.1 Factor-base selection on the pipeline curve

The nearby grid. The ordered K1120 producer dump fixes every complete signed-Frobenius orbit prefix from K1102 to K1120; all 19 candidate manifests were reconstructed from it, and K1104, K1107 and K1120 match their published dumps byte for byte (`research/n83_nearby_full_grid_20261024/README.md`). The same-point Tier 1 screen of all 19 (84 children: 60 IC and 24 rho, each IC block bracketed by K1107 and a rho control) failed its frozen 4 GiB disk gate on the first two hosted attempts, with 2.26 and 2.93 GiB free and zero measured children (`research/n83_nearby_grid_tier1_20261024/README.md`); an AWS Spot variant was blocked by the provider's Spot instance quota (`Client.MaxSpotInstanceCountExceeded`, status `BLOCKED_PROVIDER_SPOT_QUOTA`; `research/n83_nearby_grid_aws_tier1_20261024/admission_20261024.json`). A later hosted screen run (GitHub run 40150483792) supplied the confirmation's candidate set.

The 16-point confirmation. Sixteen fresh public points, sealed before the screen result was inspected, were run on seven candidates in two eight-point Tier 1 strata (AMD EPYC 7763 and Intel Xeon Platinum 8573C runners), each IC run paired with a same-point rho (`research/n83_nearby_grid_confirm16_20261024/RESULT.md`). The independent verifier admitted 112 of 112 rows, and a local replay of the raw artifacts reproduced the hosted aggregate byte for byte. Table 18.2 gives the panel. K1115 has the lowest geometric-mean online time (47.114 ms) and K1111 the lowest mean cold process (159.263 s), but the frozen sign-flip rule with Bonferroni $\alpha = 0.05/21$ returned BOUNDED_UNKNOWN for both metrics (K1115 against K1111 online, $p = 0.959$; K1111 against K1116 cold, $p = 0.773$). No base was selected. The rho/IC column is an online reading with set-up excluded and a table-less rho, the comparison §15 retired; on the same panel the cold IC process takes 159–166 s, while the same-point table-less rho online times at point 0 are 1,455–3,715 ms.

A local five-base screen. On one new public point, five finalists each ran two fresh cold processes in forward and reverse order (`research/n83_local_five_base_screen_20261024/RESULT.md`). Median cold times were 66.2 s (K992/root11, 164,672 points), 62.7 s (K1024), 65.8 s (K1107), 119.7 s (K2304) and 383.6 s (K3300/wide3), with pair spreads of 4.7–19.1%; the two same-point rho controls took 2.015 and 0.989 s with different seeds. Every hash, base, rank, relation and scalar replayed, and no winner was selected because the K1024–K1107 gap (3.1 s) is smaller than the within-arm spread.

Group	Dirs	What it established	Status
Nearby grid K1102–K1120	4	19 exact prefix bases; 16-point confirmation admitted 112 of 112 IC/rho rows; no base selected	Tier 1 cloud; BOUNDED_UNKNOWN selection
Current-source K-series	10	K2304–K5500 bases reconstruct exactly; one K2304 run completes; K3596 and K4405 capped	local controls
Index capacity layouts	7	four-byte, split and fingerprint root indices; K5500 in 25.61–28.61 GiB of allocations	allocations calculated, not measured
Cross-family bridge	2	16 sealed points; never dispatched (Spot quota)	no timing
Precomputed-rho control	1	rho faster online in all three matched arms	Table 40.2
Decomposition engines	7	chained-S_3 SAT decides $m = 2$–9 on a 167-point base with tables and a meet	stage diagnostic
Degree certificates	7	first fall 2; exact top-form degrees 28 and 111–609; no regularity computed	producer certificates
Genus-3 image	3	executable transfer; Prym coprime to r; degree-2 fibres carry no information	structural, negative

Table 18.1: The 41 $n = 83$ study directories of 24 October–1 November, by group. "Tier 1" is the benchmark-contract host tier the $n = 83$ protocols name (`research/n83_cold_neighborhood_20261016/PROTOCOL.md`); the tier file they cite, `docs/IC_BENCHMARK_TIERS.md`, is on neither repository's branch read for this treatise. Sources: the `research/n83_*` directories cited in this chapter.

K	base points	online, ms (geo. mean)	cold, s (mean)	median rho/IC online
1103	183,098	142.358	162.320	16.632
1107	183,762	72.916	160.806	28.414
1109	184,094	99.303	165.435	14.045
1110	184,260	77.205	165.329	28.296
1111	184,426	48.226	159.263	45.384
1115	185,090	47.114	165.754	27.967
1116	185,256	58.977	159.395	56.961

Table 18.2: The nearby-grid confirmation on the pipeline curve: 16 fresh public points, two Tier 1 strata, 112 admitted IC/rho rows. Base points are $166K$ subgroup-usable points in K folded columns. The rho/IC column is online only, against a rho with no table, set-up excluded. Source: `research/n83_nearby_grid_confirm16_20261024/RESULT.md`.

The K-series, re-identified. The large bases of 14–21 October were reconstructed from current-source parent dumps: the first 2,304, 3,300, 3,450 and 3,500 orbits of the K3596 dump (596,936 points) and the first 4,300 and 4,405 of the K5500 dump (913,000 points) reproduce the historical K2304 (382,464 points), K3300 wide3 (547,800), K3450 wide3 (572,700), K3500 root4 (581,000), K4300 (713,800) and K4405 (731,230) bases exactly, by ordered-dump and point-set SHA-256 and the producer's BLAKE3 base hash (`research/n83_k2304_current_source_prefix_20261025/`, `research/n83_k3300_current_source_prefix_20261025/`, `research/n83_crossfamily_current_20261025/README.md`, `research/n83_k4300_k4405_current_source_prefix_20261025/`). Full-process controls on an earlier public point then gave (Table 18.3) one completed large base and three that did not finish on the hosts available.

Index capacity. Seven directories made the root index fit larger bases. Each is an opt-in layout with an exact-key replay rule (a candidate slot reconstructs and compares the full 83-bit root before a hit is accepted), and each stops at calculated allocations for K5500:

- *Exact four-byte slots* (31-bit state index and a root selector in one `u32`): principal allocations of 32,062,766,258 bytes, 29.86 GiB, for K5500 (`research/n83_k5500_fourbyte_20261025/README.md`).
- *ID31 four-byte slots* at the K5500/K4549 split: 12 GiB of tables instead of 15, and state, table and Bloom arrays of 27.86 GiB against 30.86 GiB for five-byte ID31 (`research/n83_fourbyte_id31_20261027/README.md`).
- *A relative-ID split* (primary 2^{31} and extension 2^{30} slots, 1,255,377,750 eleven-byte states): 25.61 GiB of principal allocations, 4.25 GiB below monolithic ID31 (`research/n83_k5500_split4_20261025/README.md`).

Base	Host and cap	Wall, s	Peak RSS	Outcome
K2304 root11	local, 1,200 s, 12 GiB	438.5	5.93 GB	completed; independent replay passed; target stage 420.2 ms
K1107 root4	local macOS ARM64, 1,200 s	438.9	9.94 GB	completed; replay passed; online 46.3 ms
K3596 root4	busy 48 GiB macOS, 1,200 s	792.5; 1,201.0	0.80; 9.58 GB	SIGTERM after CPU starvation; time cap
K4405 wide3	local, 1,200 s	873.2	11.68 GB	stopped by hand under host swap
K4405 wide3	Linux Pod, 64 GiB cgroup	—	—	not started; approved GPU pair out of capacity

Table 18.3: Full-process controls on the pipeline curve, one earlier public point each (fixture zero), outside the sealed bridge points; correctness and resource controls, not Tier 1 pairs. K1107 root4 uses a fixed 8 GiB root table for 101,708,823 distinct roots. Sources: `research/n83_k2304_current_source_control_20261026/README.md`, `research/n83_root4_small_control_20261025/README.md`, `research/n83_k3596_current_source_full_20261025/README.md`, `research/n83_k4405_current_source_full_control_20261026/README.md`, `research/n83_k4405_linux_control_20261026/README.md`.

- *Fingerprint sidecars.* A two-bit sidecar brings the ID31 layout to 28.61 GiB, still 2.25 GiB below five-byte ID31, with the packing and capacity arithmetic proved in Lean 4.34.1 (`research/n83_fourbyte_fp2_20261027/README.md`). A one-byte sidecar kept every digest identical and built the index faster in ABBA order on a shared ARM64 host: geometric-mean baseline over filtered ratios of 1.831 (K128), 1.347 (K512) and 1.505 (K1024), with a K2048 baseline that drifted 42% between bookends (`research/n83_k5500_fp8_20261025/README.md`).

Two whole-process ABBA controls on the K1107 base read the sidecars: the one-byte sidecar gave baseline over fingerprint 1.677 for the full process and 2.157 for one-target online, with a 15.3% baseline drift between bookends (`research/n83_fp8_e2e_control_20261025/README.md`); the two-bit sidecar gave whole-process ratios of 1.017 and 1.294 in the two orders on a host whose load average ran from 22 to 276 (`research/n83_fourbyte_pair_20261027/README.md`). Both are local diagnostics that the Tier 1 gate would not admit.

The bridge that did not run. The cross-family bridge sealed 16 new public points for a same-point Tier 1 comparison of the small-base and wide-base selections, with its nomination rule frozen before either panel reported (`research/n83_crossfamily_bridge_20261025/README.md`). Its current-source panel (K3450 wide3 and K3500 root4, two `g5g.4xlarge` Spot instances, a $4.20 cap) moved from `us-east-1d`, which failed its placement-score gate, to `us-west-2c`; the README still records that the panel has not dispatched or selected a winner (`research/n83_crossfamily_current_20261025/README.md`, `research/n83_crossfamily_current_20261025/AWS_WEST2_ADDENDUM.md`).

The control that did run. The matched precomputed-rho reproduction of 28 October on K1107/root11 found rho faster online in all three matched arms (Table 40.2), every interval above 1, over 16 targets with every scalar replayed (`research/n83_k1107_bl_reproduction_20261028/RESULT.md`). Its plain table-less rho, a positive control only, reads rho/IC 10.7 $[6.33, 88.8]$ online, the same kind of reading as the rho/IC column of Table 18.2.

18.2 Decomposition at $n = 83$ after the correction

After the 24 October field-table correction (§12.8, which also describes the engines) chained-S_3 full-width SAT was the only decomposer that ran at $n = 83$, and it decided only $m = 2$. Six preregistered follow-ups of 31 October–1 November extended it on the pinned pipeline-curve fixture, with the factor base the Frobenius union of span$(1, z)$, 167 points (`research/n83_chain_sat_meet_20261101/PROTOCOL.md`). Each cell has three public targets (not expected to decompose) and two or five planted ones. Table 18.4 gives the sequence. Every run had zero soundness flags: no planted target was refuted and every witness re-adds in the group.

Study	Change	Outcome
order (31 Oct)	all-summand ordering; root-finder tables	$m = 2$: conflicts halved (median public 13,461 → 7,195; 12.8 → 6.2 s); $m = 3$: tables refute every public target with 0 conflicts and find both planted; $m = 4$: public refuted, no planted witness in 120 s; $m \geq 5$: capped
meet (1 Nov)	a meet-in-the-middle lookup for the middle intermediate	$m = 5$–8: public 3/3 refuted with 0 conflicts (median 0.80–76.7 s), planted 0/2; $m = 9$: capped; builds 0.68 s ($m = 5, 6$), 26.1–27.2 s ($m = 7, 8$), 958.8 s ($m = 9$)
rebuild (1 Nov)	rebuild the planted witness from the lookup hit	$m = 5$–8: planted 5/5 with 0 solver calls (median 0.91–81.5 s)
indexed (1 Nov)	indexed lookup, rebuild by peeling	$m = 8$: 78.4 → 39.7 s; $m = 9$: from cap to 3/3 and 5/5 (54.1 s), depth-5 set of 372,406,144 fingerprints in 999 s
parallel (1 Nov)	4-thread back enumeration	$m = 8$: 41.7 → 11.4 s; $m = 9$: 57.0 → 13.7 s; verdicts identical
parbuild (1 Nov)	4-thread depth-5 build	1,006.8 and 993.4 s → 287.7 and 283.3 s, 3.50×

Table 18.4: Chained-S_3 SAT on the pipeline curve, $m = 2$–9, 167-point base; decomposition stage only, L0 four-vCPU hosts. Sources: `research/n83_chain_sat_order_20261031/README.md`, `research/n83_chain_sat_meet_20261101/README.md`, `research/n83_chain_sat_rebuild_20261101/README.md`, `research/n83_chain_sat_indexed_20261101/README.md`, `research/n83_chain_sat_parallel_20261101/README.md`, `research/n83_chain_sat_parbuild_20261101/README.md`.

The sequence turns SAT into a meet-in-the-middle with a SAT residue, and then removes the residue: once the lookup decides the middle intermediate, the public targets are refuted with zero conflicts because the lookup domain is empty on every one of them and non-empty (10–250 entries) on every planted one. The meet study states the cost as about $|B|^{m-d}$ per query after a $|B|^d$ build. That is the generic floor, and the improvements in Table 18.4 are constants of the enumeration (ordering, tables, peeling, threads). The matched rho on this curve is 1.51 s mean online. With 167 points and a 53-bit subgroup, a decomposition over this base is rare, so these are stage measurements: no relation collection, no complete logarithm and no rho ratio.

18.3 Certified degree bounds

The descended $n = 83$ systems were characterised exactly, not solved. Seven directories, all producer certificates with fresh-process replay and independent scientific review pending, give the following.

Solvers do not finish. The balanced degree sweep (Koblitz, general binary and prime; $m = 2$–9; candidate domain $2^{\lfloor 83/m \rfloor}$; a planted and a natural target per cell, two computations each) ran 96 Singular attempts with 40 s and 3 GiB per solver: 53 hit the time limit, 43 the memory limit, and no top-form or affine basis degree completed (`research/n83_ic_degree_sweep_20261027/README.md`). A fixed-base rerun at $\ell = 27$, on full-curve bases that enumerate all $2^{27} = 134{,}217{,}728$ candidate x-values per curve (`research/n83_fixed_factor_base_20261028/README.md`), completed 4 affine bases (all $m = 2$, Koblitz and binary), with 63 time and 29 memory exits (`research/n83_fixed_degree_sweep_20261028/README.md`).

First fall is 2 on binary curves. On every retained input the Koblitz and general binary systems have certified first fall degree 2 at every $m = 2$–9, planted and natural (32 exact witnesses; all 48 replays passed), consistent with Corollary 6.6. On the prime curve the balanced presentation has the proof-assisted lower bounds 24, 17, 14, 12, 10, 9, 9, 8 for $m = 2$–9 (`research/n83_ic_degree_sweep_20261027/README.md`). The fixed $\ell = 27$ catalogue certifies 48 first falls and 32 degree-two closures: at $m = 2$ the binary systems have first and last fall 2 (a restricted global certificate; in each planted case V_2 has 53 independent linear constraints on 54 variables), at $m = 3$–9 the last fall is unresolved with lower bound 2, and the prime systems have first fall at least 5 (`research/n83_fixed_fall_catalog_20261028/R`

m	certified bounds, Koblitz and binary	exact top-form degree
2	$28 \le d_{\mathrm{reg}} \le 55$	28 (prime: 42)
3	$82 \le d_{\mathrm{reg}} \le 165$	111
4	$109 \le d_{\mathrm{reg}} \le 275$	194
5	$136 \le d_{\mathrm{reg}} \le 385$	277
6	$163 \le d_{\mathrm{reg}} \le 495$	360
7	$190 \le d_{\mathrm{reg}} \le 605$	443
8	$217 \le d_{\mathrm{reg}} \le 715$	526
9	$244 \le d_{\mathrm{reg}} \le 825$	609

Table 18.5: Top-form degrees of the fixed $\ell = 27$ descended systems at $n = 83$, both targets. Bounds: certificates of 28 October; exact values: the overlap proof of 29 October, which closes every interval at $111 + 83(m - 3)$ for $m \ge 3$. Neither is an affine solving degree. Sources: `research/n83_top_quotient_20261028/README.md`, `research/n83_candidate_pdp_20261028/README.md`.

EADME.md). A later degree-four witness census raised the certified last-fall lower bound of the twelve retained $m = 3$ Koblitz and binary systems from 3 to 4, with 249 strict degree-four coordinate witnesses each (`research/n83_candidate_pdp_20261028/README.md`).

Top-form degrees are large and exact. The top-form graded quotient was bounded for all 48 fixed-base systems (`research/n83_top_quotient_20261028/README.md`) and the binary chain bounds were then closed exactly by a written overlap proof with native checks (`research/n83_candidate_pdp_20261028/README.md`). Table 18.5 gives both. For the prime systems at $m \ge 3$ the top quotient is positive-dimensional, with no finite first vanishing degree in the saved presentation; the source says this is not a claim of infinite regularity or about the affine ideal. At $m = 2$ the binary systems have FFD = LFD = 2 and top-form degree 28, which separates those quantities explicitly for these presentations.

Encodings and candidate bases. For the prime candidate interval a radix census over all 3,010 partitions of 27 (35 exact Hilbert records and 2,975 certified lower bounds) found the unique minimum exact first vanishing degree, 42, in the all-binary presentation, which excludes improvements within that presentation family (`research/n83_candidate_pdp_20261028/README.md`; the original radix protocol's unequal-arm accounting mismatch is recorded in a correction file). The same directory certifies $7 \le \mathrm{FFD} \le 17$, conditionally, for 42 prime $m = 3$–9 systems. A separate design study lists sixteen factor-base constructions for the primary $a = 0$ curve with a preregistered protocol and proof obligations, and no measurement (`research/n83_factor_base_designs_20261031/README.md`).

Reading. The binary systems fall at degree 2 and their top-form degree grows by 83 per summand. These are certified properties of specific presentations, not attack costs. They say nothing for or against a low affine solving degree, but they show that the descended $n = 83$ systems are far outside the reach of the solvers that were run, which matches the 96 and 92 unfinished attempts above.

18.4 The genus-3 image transfer

A same-field degree-3 cover $\phi : C \to E$ with C of genus 3 descends to $\mathbb{F}_2$: $E : y^2 + xy = x^3 + 1$ and $C : v^2 + u^2 v = u^7 + u$, with equal point counts over $\mathbb{F}_2$, $\mathbb{F}_4$ and $\mathbb{F}_8$ (`research/n83_jacobian_projector_20261031/RESULTS.md`). Three studies of 31 October–1 November asked whether pulling the $n = 83$ instance into $\mathrm{Jac}(C)$ gives anything rho does not have.

- *The transfer executes* (`research/n83_genus3_image_solver_20261031/README.md`). The exact $n = 83$ pullbacks of P and $Q = [37]P$ are reduced Mumford classes of exact prime order r; typed certificates prove $\phi_*\phi^* = [3]$, so the pullback is injective on the subgroup. Rho and BSGS in the Jacobian image solved toy instances at $n = 9, 13, 19, 23$ without the planted scalar, all 400 destination attempts verified ($n = 7$ failed and is retained). The source and destination operation counts have no certified common unit, so no advantage is promoted; the full-Jacobian genus-3 index calculus does not apply to an image of order $r \approx q$, and its ambient $q^{4/3}$ ledger loses to source rho by about 70.17 exponent bits.
- *The Prym shares nothing with the target* (`research/n83_jacobian_projector_20261031/RESULTS.md`). With the closed-point norm, $e = \phi^*\phi_*$ satisfies $e^2 = [3]e$, and on r-torsion $p =$

$[3^{-1} \bmod r]e$ is a checked idempotent. At $n = 83$ the Prym has Frobenius polynomial $X^4 + q^2 = (X^2 - 2^{42}X + q)(X^2 + 2^{42}X + q)$, with elliptic factors of traces $\pm 2^{42}$, neither equal to E's trace, and $\gcd(r, q + 1 \mp 2^{42}) = \gcd(r, q^2 + 1) = 1$. There is no r-primary route through the two-dimensional complement. The identities are checked in Lean.

- *Degree-2 fibres are tautological* (`research/n83_image_closed_degree2_20261101/README.md`). On every odd extension tested a generic rational fibre has closed points of degrees $1 + 2$ whose elliptic labels are exactly P and $[2]P$, so the projected identity is $[3]P = P + [2]P$ and carries no logarithmic information. The frozen information gate fails, so no collector or timing run is admitted; ordinary ambient genus-3 collection and fixed bounded-degree branching also fail the comparison with source rho. A growing-degree or adaptive target-local router is left open.

This is the binary counterpart of the cover results of §26: the map is real and certified, and the destination carries the same r-part and nothing that makes it cheaper.

19 Weil descent and degree, continued

§16.1 measures the degree of Weil-descended summation systems to 24 October: first and last fall degrees stay at 3–4, the refutation degree tracks ℓ within one of semi-regular, closure algorithms refute a degree or two earlier but rise past 4 at $m = 3$, $\ell = 7$, and guessing one or two variables restores degree 4 there. §36 adds the first $\ell = 8$ runs: four guessed variables bring all four unsatisfiable $(22, 8)$ draws back to degree 4, one three-variable set fails on three of four, and the hybrid beats degree 5 at the priced ECC2K-130 cell only if at most six guesses suffice. This chapter records what followed:

- which guessed coordinate matters at $\ell = 8$, not only how many (§19.1);
- the first measured degree-5 closure cell (§19.2);
- the per-attempt criterion and the companion repository's closure and SAT pilots (§19.3);
- a direct check of Semaev's 2015 first-fall statement on his own cells (§19.4);
- the ceiling on learned SAT guidance (§19.5);
- the finding that F6-IC with a pair index is the meet-in-the-middle scan (§19.6).

Every result here is exact algebra or a counted stage diagnostic on toy fields, run on unisolated 4-core, 16 GB containers (the RunPod degree-5 cell excepted). None has a validation file. One carries a proof-carrying claim, recorded as unverified. Closure outcomes do not depend on the host; timings are indicative only.

A desk evaluation bounds what any algebraic-oracle lever in this chapter could do to today's ladder rows. Of 28 candidate point-decomposition strategies checked against the lead atlas, 17 collide with existing leads, 7 overlap partly and 4 have no match, rated least promising (`research/pdp_strategy_evaluation_20261031/README.md`). The ladder rows (random binary curves, cofactor $h = 2$–8) are solved with a pair table and a walk, not with summation polynomials: on arm B at the published rung the walk is 61–82% of S and the matrix at most 0.10% (measured, §3 of the source). On the committed algebraic-oracle rows the oracle's share of S is exactly zero when native operations are unpriced, so an algebraic-oracle lever has a ceiling of 1.00× on those rows (derived).

19.1 Guessing at $\ell = 8$: which coordinate, not how many

All studies in this section use the frozen $m = 3$, $(n, \ell) = (22, 8)$ draws of 22 October ($N = 46$ unknowns, 44 equations, surplus -2; unsatisfiable draws 0, 3, 5, 6) or fresh draws from the same generator, the MutantXL-style closure `dreg_closure` or its guess variants, and a registered reading of each branch as refuting at degree 4 (the constant 1 appears) or not (degree 4 saturates without it); the instrument and its three controls are described in §14.1. Table 19.1 lists them.

What the series shows. At equal guess count, guessing a coordinate of the swap invariant $s = x_1 + x_2$ refutes at degree 4 on more draws than guessing a raw summand coordinate: preregistered on fresh draws, $p = 0.031$ one-sided, and 6 of 6 discordant draws over both studies ($p = 0.016$, secondary). S3 suffices on 7 of 12 draws in all and X3 on 1 of 12. This contradicts the summand-dimension reading proposed after the three-guess study: S3 leaves both summands at dimension 8 and suffices more often than X3, which lowers x_1 to dimension 7. Putting a second invariant guess on the same link (D3) is never better than the raw guess on x_3; putting it on the second link's invariant $e_1 + x_3$ (L3) is as good as the raw guess on these 12 draws. The D3 failures all sit at branches 0 and 4, where both top coordinates of s vanish. Four guesses $\{7, 15, 23, 24\}$ still suffice on every draw measured (68 processes, peak 10.6 GB, 124,314 degree-4 columns per branch), and S3 fails on 5 of 12. The guess study's registered verdict favours the rule $k = m(\ell - 7) + 1 = 4$ over $k = m(\ell - 6) = 6$; $k = 4$ is an upper bound, the smallest count the container could fit under its column cap.

What it does not show. Every row is one cell, $m = 3$, with one guess set per arm. The unspecialised $\ell = 8$ system is not measured at degree 4 (179,447 columns); "does not refute" means a closure degree of at least 5, and degree 5 is not measured here. The null study's 12 of 12 unspecialised $\ell = 7$ failures over $N = 40$, 42, 43 and surplus -2, 0, $+1$ remove the one confound the earlier $\ell = 7$ registration kept (a non-positive surplus), but cannot distinguish "3 of 4" from "4 of 4" against the three-guess $\ell = 8$ result; a scorer defect found in its review (it scored only the files it was given) was fixed and the verdict regenerated unchanged. The second-link study's own reading is that one invariant guess per link extends

Study	Guess set (all set to 0)	Outcome	Verdict
guess (22 Oct)	$x_7, x_{15}, x_{23}, x_{24}$: one top coordinate per summand and one intermediate	64 of 64 branches refute at 4; suffices on 4 of 4 draws	four suffice (an upper bound)
guess3 (23 Oct)	x_7, x_{15}, x_{23}	draws 0, 3, 5 fail at branch 0 (closure rank 129,221, 128,213, 129,893 of 136,698); draw 6 suffices	three do not suffice for this set
null (28 Oct)	none, at $(22, 7)$, $N = 43$	4 of 4 draws do not refute at 4	12 of 12 unspecialised $\ell = 7$ draws now fail
invariant (29 Oct)	S3: $a_7 + b_7$, c_7, $e_1[0]$; X3: a_7, c_7, $e_1[0]$	S3 suffices on 2 of 4, X3 on 1 of 4; S3 never worse	mixed
invariant, more (29 Oct)	S3 against X3 on 8 fresh draws	S3 5 of 8, X3 0 of 8; 5 of 5 discordant draws favour S3, $p = 0.031$	S3 better (preregistered)
pair (29–30 Oct)	D3: $a_7 + b_7$, $a_6 + b_6$, $e_1[0]$ against S3	D3 2 of 12, S3 7 of 12; 5 discordant, all S3; two-sided $p = 0.0625$	no detected difference
second link (5 Nov)	L3: $a_7 + b_7$, $(e_1 + x_3)_{20}$, $e_1[0]$ against S3	L3 6 of 12, S3 7 of 12; one discordant draw; $p = 1.0$	no detected difference; unverified claim `PCC-20261106-607061`

Table 19.1: Guessed-variable closures at $m = 3$, $(n, \ell) = (22, 8)$ (the null row at $(22, 7)$), degree 4, unsatisfiable draws only; a, b, c are the summands' coordinates and $s = a + b$ the swap invariant of the first link. "Suffices" means every branch of the draw refutes at degree 4. Sources: `research/dreg_closure_guess_20261022/RESULTS.md`, `research/dreg_closure_guess3_20261023/RESULTS.md`, `research/dreg_closure_null22_7_20261028/RESULTS.md`, `research/dreg_invariant_guess_20261029/RESULTS.md`, `research/dreg_invariant_guess_more_20261029/RESULTS.md`, `research/dreg_invariant_pair_guess_20261029/RESULTS.md`, `research/dreg_second_link_guess_20261105/RESULTS.md`.

along the chain, so $m = 4$ is its natural test; $m = 4$ at $\ell = 8$ has not run. The preregistered high-arity discriminator that would run it (`research/dreg_high_arity_20261101/README.md`) went through four protocol versions, each superseded before data by a source or manifest change on `main`; v4 is frozen and its launch is blocked.

The price, unchanged. The guess-then-degree-4 oracle costs $2^k\binom{N-k}{\le 4}$ per target against $\binom{N}{\le 5}$ for degree 5. At the priced ECC2K-130 cell ($m = 4$, $\ell = 33$, $N = 2{\cdot}131{+}4{\cdot}33 = 394$, closure exponent $\omega_c = 1$) the ratio $\binom{394}{\le 5}/(2^k\binom{394-k}{\le 4})$ is 78.2 at $k = 0$, 5.09 at $k = 4$, 1.30 at $k = 6$ and 0.66 at $k = 7$, recomputed in the three-guess study. Both registered growth rules give $k \approx 105$–108 there. The measurements—$k \approx 1$ at $\ell = 7$ on 5 of 8 draws, $k \le 4$ at $\ell = 8$ on 4 of 4—do not exclude a bounded k, and the invariant guess lowers k on some draws. Even a bounded k gives at best the degree-4 floor, $2^{+7.62}$ over rho in unconverted units (+8.43 bits under the flat-k rule), and the free-oracle floor of §32.2 still binds every row. Nothing in the series measures k at $m = 4$, so the +8.43 to +58.27-bit range of §36 stands.

Other degree levers of the same weeks. A preregistered invariant arm that adds the summand-swap invariant p as a variable makes every equation quadratic but does not refute at degree 3 at $(22, 6)$, where the plain system refutes at 4; its degree-4 matrix would have 5.1–5.5× the plain system's columns (523,686–637,393 against 102,091), and at $(21, 7)$ degree 4 is unmeasured (`research/dreg_pair_invariant_20261028/RESULTS.md`). The 17 October specialisation study, whose failed null control made it uninformative, is the control lesson of §14.1.

19.2 A measured degree-5 closure

The descent rows of Table 15.4 price the $m = 4$ ECC2K-130 route at degree 4 (+7.62 bits, contradicted) and degree 5 (+10.32 bits with the closure exponent $\omega_c = 1$, +45.71 with $\omega_c = 2$). Until November no degree-5 closure had been computed. The leads study of 24 October (`research/koblitz_ic_leads_build_20261024/README.md`; Amendment 1 post hoc, Amendments 2–7 frozen before the runs they govern) built four leads in Rust and, in Amendment 7 (4 November), measured cell C1.

- *C1 refutes at degree 5.* On $m = 3$, $(n, \ell) = (19, 7)$, draws 0 and 3, the incremental engine `pe_closure` (partial, low-first, batch 64) found 1 in the degree-5 closure, so each draw's closure degree is 5. Draw 0: 607,530 of 760,099 columns filled at refutation (0.80), 31,632 s, 52.6 GB peak resident; draw 3: 606,973 columns, 31,888 s, 52.5 GB. Host: 32 vCPU AMD EPYC 9965 with 128 GB (RunPod).
- *Memory.* The dense engine aborted on the same draws at 424 and 423 GB resident, more than twice its 217 GB projection and about 8× the incremental engine's measured peak, which was below the derived 73 GB bound.
- *Scope.* One cell, $m = 3$ not $m = 4$, two of four draws, no satisfiable control (the owner chose draws 0 and 3). The study reads it as support for the degree-5 rows of `research/notes/ecc2k130/RESEARCH_ECC2K130_DESCENT_DEGREE.md` at $\ell = 7$, and it changes no cost row. C1 cost about $53 of an $80 cap over six pods, of which only the last measured.

The other leads of the same study change no cost either. Partial enlargement (L2) does not reduce memory in general: in the registered order the basis fills 0.66–0.997 of the column space before 1 appears, and other queue orders and batch sizes keep every run at 0.61–1.00. MXL's variable restriction is the one exception, on $m = 4$, $(11, 5)$: all four draws refute at fill 0.38–0.45 with 1.8–2.5× fewer products, none reaching the 0.25 payoff; elsewhere it does not help. The useful by-product is the incremental engine itself, which holds one row per column plus one chunk: 1.3–2.8 GB resident at degree 4 against 1.9–4.2 GB for the full order.

19.3 The per-attempt criterion and the closure pilots

The companion repository's degree pilots of 9–14 October are labelled by their authors exploratory, "not program evidence", with no run record or ledger write. They are worth recording for the criterion they state and for three measurements that §16.1 does not have.

The criterion. If a failed decomposition attempt costs $2^{\alpha\ell}$, relation collection can beat rho only if $\alpha < m/2 - 1$ (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/m3-closure-pilot/README.md`). A harvest of 411 measured rows and 1,554 per-target values from the repository found that SAT and enumeration solvers spend about $2^{m\ell}$ per failed attempt (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/ctrial-harvest/README.md`): WDSat's unsatisfiable conflict count equals $2^{3\ell}/3!$ to within 0.03–1% for $\ell = 5$–11, slope 2.9985 bits per unit ℓ [2.9979, 2.9992]. At $m = 3$ the budgets need -1.7 to -3.1 bits per unit ℓ against a measured $+3.36$; at $m = 4$, -0.12 to -0.61 against $+2.32$ for meet-in-the-middle. Extrapolated to $n = 131$ at the budget ℓ, WDSat with symmetry breaking lies at least 91 bits outside every budget line, a Riemann–Roch $L(4O)$ oracle measured at $n = 131$ at least 55, and the $t = 4$ chained meet-in-the-middle at least 52; $m = 5$, 6 had no usable cells. Only a bounded-degree algebraic refutation could meet the criterion.

The $m = 4$ SAT exponent. Run on the published Trimoska et al. [39] object and configuration, the pilot reproduces $\alpha = 2.981$ [2.973, 2.997] at $m = 3$; on chained S_3 at $m = 4$ with Gaussian elimination it measures $\alpha = 4.328$ [4.315, 4.340] (4.310 at fixed $n = 17$), against the $\alpha < 1$ the $m = 4$ budget needs (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/m4-sat-cost-pilot/README.md`). The pilot also found that WDSat's stock `-b` mode returns "SAT" with undefined bits on unsatisfiable $m = 4$ instances.

Closure degree stays at 4 on toy chains, and is local.

- $m = 3$. The degree-4 mutant closure W_4 refutes 60 of 60 unsatisfiable chained-S_3 instances at $n = 9$ and 40 of 40 at $n = 11$, with no false refutation of a satisfiable one, where plain Macaulay M_4 and M_5 refute none (plain needs degree 6 at $n = 9$; semi-regular predicts 7–11). A support-matched null is refuted and a dense null is not, so the low-degree refutation follows the sparse, block-bilinear support of the descended system, not Semaev's coefficients.
- $m = 4$, *5.* W_4 refutes 25 of 25 at $m = 4$, $n = 7$; 10 of 10 at $n = 9$; 2 of 2 at $n = 11$; 10 of 10 at $m = 5$, $n = 7$; and W_3 refutes 30 of 30 at $m = 5$, $n = 9$, $k = 2$, while plain M_4 refutes nothing (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/m45-closure-pilot/README.md`).
- *Locality.* Restricting monomials to consecutive blocks (locality 1) keeps the refutation on 124 of 135 unsatisfiable instances with 39–69% fewer columns at $m = 3, 4$; every miss refutes at locality 2, and none needed locality 3. At $m = 5$, $n = 9$, $k = 3$ locality 1 refutes none of 10 and locality 2 all 5 tried

(`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/block-local-closure-pilot/README.md`).

These are the toy-scale version of the MutantXL result of §16.1, and they meet the same wall: the closure stays at degree 4 while the columns stay within a host's memory, and the $\ell = 7$–8 cells of §19.1 and the degree-5 cell above are where it leaves degree 4.

19.4 Semaev 2015, §4.5, on its own cells

Chapter 6 proves that the binary $m = 2$ descent and Semaev's chained system have first fall degree 2 for every n (Corollary 6.6) and notes that Semaev states 4 in his §4.5 [36]. A frozen check of 2 November ran his own $t = 2$ cells of Table 2, $S_3(x_1, x_2, z)$ with $x_1, x_2 \in V$ and random z and B, 20 instances per cell and field polynomial, 200 in all (`research/semaev_t2_trace_20261102/REPORT.md`; exact $\mathbb{F}_{2^n}$ and Boolean-ring arithmetic in standard-library Python; source arXiv:1504.01175v1).

- *Refuted: "the first fall degree ... is always 4".* On all 200 instances the first fall degree is 2, each with an explicit trace certificate: with $\lambda = z^{-2}$, $\mathrm{Tr}(z^{-2}S_3) = \mathrm{Tr}(x_1) + \mathrm{Tr}(x_2) + \mathrm{Tr}(Bz^{-2})$, a constant combination of the coordinate equations of degree at most 1.
- *Split: the corpus inference that these rows refute $d_{F4} \le d_{\mathrm{ff}}$.* The least degree at which the Boolean Macaulay span contains 1 is 2 on 154 instances and 3 on 46, never 4 (per cell: $(15, 4)$ 18/2 and 19/1; $(15, 5)$ 20/0 twice; $(16, 4)$ 18/2 and 20/0; $(19, 3)$ 6/14 and 4/16; $(21, 3)$ 15/5 and 14/6). So the sharp first-fall assumption fails on 46 of 200 instances, not everywhere, and Semaev's reported $d_{F4} = 4$ is Magma's step degree, not what the systems require.
- *Why.* The quadratic parts depend only on $x_1x_2 \in V \cdot V$, of dimension $2k - 1 < n$, so most $t = 2$ systems are inconsistent at degree 2 or 3. None of the 200 has a solution in $V \times V$, which matches Semaev's success probabilities of 0.0006–0.0155 for these rows.

It complements the Semaev 2015 audits of Chapter 35, which tested eq. (11) and the typed-image ratio but not the first-fall statement: on these cells the sharp assumption fails on 23% of instances, and the stated first fall degree is wrong under Semaev's own definition.

19.5 The ceiling on learned SAT guidance

The SAT engines and encodings are described in §12.7. The SAT survey dismissed ML-guided SAT without measuring it. A preregistered study of 27 October measured the most any learned component could gain inside a SAT point-decomposition oracle before anyone builds one (`research/pdp_ml_sat_ceiling_20261027/README.md`; protocol frozen in `8a85584fd`; symmetrised S_4, $m = 3$, native CDCL+XOR, the Trimoska-corpus curves, uniform targets; cost in conflicts). It tested triage on target features or early solver telemetry, a fixed conflict cutoff, and learned static branching. All 4,200 labels agree with exhaustive sorted-triple search.

- *The search is an enumerator.* A refutation costs 0.963 ($n = 15$, $\ell = 5$; 3,000 targets) and 0.998 ($n = 17$, $\ell = 6$; 1,200 targets) conflicts per sorted factor-base triple, every refutation within 4.1% of its family median (5,781 and 45,649 conflicts).
- *The prize is real.* A perfect unsatisfiability predictor, charged its telemetry conflicts, would raise relations per conflict by 5.2–9.0×.
- *The signal is not.* Test AUC is within noise of 0.5 for every feature group on both families and both splits; the permuted-label control reads 0.45–0.54 and the positive control (labels flipped with probability 0.25) 0.72–0.79, so the learner works. The one interval that excludes 0.5 ($n = 15$ telemetry at 512 conflicts, $0.556\ [0.518, 0.599]$) includes a "decided within 512 conflicts" flag and gains 0.99×. Train-fitted gains up to 4.7× fall to about 1× held out.

The study closed by its preregistered rule. A related instrument finding is recorded in the atlas: five "unverified SAT answers" of an earlier binary solver comparison were quadratic-twist decompositions (`EV-ICPERF-390707`; `crypto-autoresearcher:knowledge/leads/ecdlp/ic-prime/LEAD-ICP-icperf-binary-pdp-solver.yaml`).

19.6 F6-IC with a pair index is the meet-in-the-middle scan

With the pair index, F6-IC is the pair-index meet-in-the-middle scan. The few-candidate closure of 5 November, now on by default, runs the exact one-fixed closure once a summand's assigned bits leave

at most 64 base codes; it cuts F6-IC CPU 2.67–10.29× while leaving the gate's pair lookups and group additions identical, so on these nine Koblitz cells the algebra was ruling out no candidate x-coordinate (`research/f6_few_candidates_20261105/README.md`). The full account, with its controls and the two companion items, is in §12.4.

20 ECC2K-130, ECC2-131, factor-base selection, composite degree and GHS

The named binary targets are in §15.7 (ECC2K-130) and §16 (ECC2-131), with the priced routes of Table 15.4. This chapter adds what those sections do not have:

- two independent budget tables and the isogeny-class census from the `cryptanalysis` repository, the Torelli toy ladder, and the state of the modular carrier search (§20.1);
- the ECC2-131 symmetry-breaking and stage measurements (§20.2);
- factor-base selection by a structural law and by a learned cost model, with the exact standing of the one validated claim (§20.3);
- quasi-subfield polynomials at $n = 131$ (§20.4);
- composite-degree curves (§20.5);
- binary GHS and Weil-restriction screens outside the late-October cubic catalogue (§20.6).

The `cryptanalysis` sources are read at the head of the branch `claude/cryptanalysis-repo-setup-p59kek` (22 October), not a `main` branch. None of the studies here carries a validation file in a repository read for this treatise.

20.1 ECC2K-130 beyond the priced routes

What one decomposition may cost. Two `cryptanalysis` studies turned the ECC2K-130 cost model around: fix rho at $2^{60.81}$ iterations on the $\langle -1, \tau \rangle$ classes, count one group addition as one rho iteration, and ask how much one decomposition may cost if the whole attack is to tie rho. The subspace version (`cryptanalysis:experiments/pdp-scaling/README.md`) finds no budget at $m = 3$ (linear algebra alone exceeds rho), $2^{12.2}$ iterations at $m = 4$, $\ell = 29$ (about 0.5 ms), $2^{32.8}$ at $m = 5$, $\ell = 28$ (about 12 minutes) and $2^{36.8}$ at $m = 6$, $\ell = 24$. Against these, its meet-in-the-middle decomposition costs 2^{58}, 2^{84} and 2^{72} group additions per point, gaps of +46, +51 and +35 bits, and the fitted algebraic costs (SAT, F4) are near 2^{180} at $m = 4$; SAT grows about 4 ($m = 3$) and 7 ($m = 4$) bits per extra dimension of the base. The Frobenius-quotient version (`cryptanalysis:experiments/frobenius-quotient-m4/README.md`) uses the one quotient that remains at $n = 131$, the orbit quotient of the non-linear weight base $F_w = \{P : 1 \le \mathrm{HW}(x(P)) \le w\}$ in a normal basis (§17.2), with columns $B/131$ credited in full. Table 20.1 gives its best weight per arity.

The quotient raises the budget from $2^{12.2}$ to $2^{31.1}$ at $m = 4$, which is the factor 131 on relations and 131^2 on linear algebra that rho already collects through $\sqrt{2 \cdot 131}$. It does not remove the obstruction. A decomposition that uses only the group law needs $|L_1||L_2| \ge \#E \approx 2^{131}$, at least $2^{65.5}$ per attempt; a method that uses the x-coordinate algebra must reach the required exponent column, which is below $m/2$ for every m, against m (exhaustive, WDSat, FES), $m - 1$ (hybrid) and $\lceil m/2 \rceil$ (MITM, floored at $2^{65.5}$). At $m = 4$ the required exponent is 0.98: one decomposition would have to cost less than listing the base once. The same study measured $m = 4$ on the weight base at toy size with CryptoMiniSat (chained S_3, weight as a cardinality bound): every completed instance at $n = 9$–15 returned a lifted, re-added decomposition with no spurious model and no oracle disagreement, and the descriptive slope of median CPU on $\log_2 B$ over four uncensored cells is 1.34.

The isogeny class, by level. The class of $E_0 : y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{131}}$ has four endomorphism levels, conductors 1, 263, p and $263p$ with $p = 146{,}505{,}763{,}881{,}528{,}721$ (`cryptanalysis:experiments/ecc2k130-isogeny-class/README.md`):

- level 1 (E_0): native and effective rho $2^{60.81}$;
- level 263, 262 floor curves: native rho $2^{64.33}$, effective $2^{60.81}$ through a verified 263-isogeny to E_0 costing at most $2^{15.6}$ $\mathbb{F}_q$-multiplications (the 262 j-invariants independently rederived by Vélu from $E_0[263]$, 262 of 262 matching);
- level p, $p + 1 \approx 2^{57.0}$ curves, and level $263p$, $262(p + 1) \approx 2^{65.1}$ curves: native $2^{64.33}$, effective in $[2^{60.81}, 2^{64.33}]$.

No curve in the class is easier than E_0. The p-level interval would close only with a characteristic-2 higher-dimensional (Kani) isogeny pipeline, and constructing any such curve costs about 2^{70} $\mathbb{F}_q$-multiplications,

m	w	$\log_2 B$	$\log_2$ budget/attempt	required exponent in B	MITM exponent	MITM gap, bits
2	—		none: linear algebra alone exceeds rho at every w			
3	6	31.6	0.5	0.01	2	+65.0
4	6	31.6	31.1	0.98	2	+34.4
5	5	27.2	40.6	1.49	3	+44.0
6	4	22.5	45.3	2.01	3	+25.3
8	3	17.5	50.3	2.87	4	+23.7
12	2	12.1	52.9	4.38	6	+25.6

Table 20.1: ECC2K-130 on the Frobenius-orbit quotient of the weight base: the most one decomposition may cost if the attack is to tie rho ($2^{60.81}$), and the meet-in-the-middle floor against it. Arithmetic only (rows $m = 7, 9, 10, 11$ omitted). Source: `cryptanalysis:experiments/frobenius-quotient-m4/README.md`.

more than solving natively on it. The treatise's 262-curve figure and the 11.45× negation-only penalty of §15.7 are the level-263 row.

A Jacobian carrier: the toy ladder. §15.7 leaves one hypothetical route open, a genus-130 Jacobian isogenous to the simple 130-dimensional abelian variety that carries $\langle G \rangle$, and quotes the Torelli codimension 8,128 against it. A frozen toy test of that rarity heuristic (`research/ic_next10_20261014/10-torelli-toy/README.md`) enumerated curves over $\mathbb{F}_2$ whose Frobenius polynomial is divisible by the trace-zero part A_n of $\mathrm{Res}_{\mathbb{F}_{2^n}/\mathbb{F}_2}(E_a)$:

- $n = 5$*: negative, with low power.* No genus-5 Jacobian (hyperelliptic and trigonal completely, canonical curves by a uniform sample of 2^{21} nets), and no plane quintic or hyperelliptic curve of genus 6–8, carries A_5; but the heuristic predicts only about 0.42 classes per sign at genus 5.
- $n = 7$*: the literal reopen criterion is met, by structure* $n = 131$ *lacks.* 1,536 hyperelliptic models of genus 7 (mass 0.5) per sign have Jacobians isogenous to the full Weil restriction, against a prediction of 2^{-14}–2^{-18}. Here $\mathrm{ord}_7(2) = 3$, A_7 splits into two 3-dimensional factors, and genus $7 = 2^3 - 1$ is what GHS gives with magic number 4 (the attribution to GHS is the author's interpretation).
- *Calibration.* The heuristic is right to within $2^{-3.5}$–$2^{+1.2}$ for the A_3 control at excess genus at most 3, and under-predicts every structured class that occurs, by up to 2^{19} at $n = 7$.

So the ECC2K-130 rarity argument rests on the absence of such structure at $n = 131$ (no small magic numbers, a simple A), not on the number 8,128. The atlas lists $n = 11$ and 13 as the next rungs (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-ecc2k130-jacobian-carrier.yaml`).

The modular carrier search. A separate line looks for a low-genus or modular carrier of the relevant CM factor through an explicit modular bridge (`research/koblitz_modular_ic_20261018/README.md`). It has run 26 rounds of exact constructions with Lean identities and no index-calculus rank. Round 22 computed the integral equivariant recovery on norm-zero targets (eight Smith factors 1, kernel rank 24); round 26 shows by exact diamond norms that the invariant quotient does not carry the intended CM factor under the retained bridge, and records the next Riemann–Roch space (degree 128, dimension 68 under the genus-61 bridge) whose basis is still required.

The ranking of 23 October. A leads survey of that date ranked six degree and closure leads and three rho-accounting items by planning cost against best-case effect, both judgements and not measurements (`research/ecc2k130_leads_survey_20261023/report.typ`). Its first lead, the matched null at $(22, 7)$, ran on 28 October (§19.1). It quotes the unconverted distances +7.62 bits ($m = 4$, degree 4), +10.32 (degree 5) and a free-oracle floor of $2^{56.46}$, and no lead moves a priced row below rho.

20.2 ECC2-131

ECC2-131 is a random binary curve over $\mathbb{F}_{2^{131}}$ ($z^{131} + z^{13} + z^2 + z + 1$, cofactor 2, $a \notin \mathbb{F}_2$, so no Frobenius saving), and 131 is prime, so there is no Weil descent to a subfield. §16 gives its program, gates and degree falsifiers. Two measurements were added.

Family	Curve	n	base M/s	sym M/s	sym/base	IC/rho step rate
binary	ECC2-131	131	0.18	5.24	28.7×	1.20
binary	calibration	127	0.36	7.59	20.8×	1.74
binary	calibration	61	0.39	4.52	11.6×	1.04
prime	ECCp-131	131	0.17	0.09	0.6×	0.02
prime	calibration	61	0.37	0.19	0.5×	0.03

Table 20.2: Decomposition-oracle step rates at $m = 2$, single thread: the per-point baseline, the symmetric e_1 oracle, and the symmetric step rate over the rho adding-walk step rate on the same host. A step-rate ratio, not a cost ratio. Source: `research/certicom131_symlp_20261025/REPORT.typ`.

Symmetry breaking and double large primes at $m = 2$. With the base $V_\ell = \{x : \deg x < \ell\}$ and a large-prime domain $V_{\ell'}$, the binary S_3 linearises in $e_1 = x_1 + x_2$ to $t^2 + t = e_1^2 + b/x_R^2$, and because $V_{\ell'}$ is closed under addition the symmetric oracle can sweep e_1 (each unordered pair once, one batched inversion) instead of lifting each x_1 with its own inversion (`research/certicom131_symlp_20261025/REPORT.typ`; single thread, seed 20261025; IC step and rho step in the same field code and process). Table 20.2 gives the step rates. For ECCp-131 the interval $[0, B)$ is not closed under addition and each step needs two square roots, so the symmetric oracle is slower than its baseline. Direct relations (both abscissae in V_ℓ) are about $5 \cdot 10^{-7}$ of decompositions on the binary calibration curves; the two-large-prime graph is what makes the bulk usable, and the planted pipeline recovers a logarithm through it on both real 131-bit groups.

The study's derived cost model (every row computed, not measured) puts an $m = 2$ solve at about $2^{2n/3}$ decomposition steps at the balancing dimension $d = (n + 1)/3$, which agrees with the merged algebraic ladder's measured trial slope 0.647 $[0.639, 0.655]$, against rho's $\sqrt{\pi r/4}$: $\log_2$ steps 87.3 against 64.8 for ECC2-131 and 65.3 for ECCp-131, a distance of about 2^{22} steps before the linear-algebra floor, which the program gates place above rho for $m \leq 3$. Both accelerators are constants on the $2n/3$ exponent; they would matter at $m \geq 4$, which the study does not reach.

Stage rates. On one isolated sibling pair of an Intel Xeon 6975P-C host, five samples per cell (`research/certicom_131_stage_bench_20261023/REPORT.md`):

- rho state steps per second: 207,822 (ECC2K-130), 207,602 (ECC2-131), 87,055 (ECCp-131);
- full-field S_3 quadratic-root queries per second: 48,080, 48,186 and 9,522.

The two binary curves share the field polynomial and differ by 0.11% in point updates; ECCp-131's generic rho step is 0.419× and its root query 0.198× ECC2K-130's. These are generic-binary CPU stage ratios, not solve costs; the optimised clients of §15.7 and §16 walk 8.5–13.2 million steps per second.

20.3 Factor-base selection

Named subspace properties (greedy coverage, cosets, geometric progressions, normal against polynomial bases) are exhausted at scale (§16, §36). Two further results choose the base by the solving degree rather than by the yield.

A structural law for $m = 2$. Over 656 factor bases (7 families, 9 fields) on $y^2 + xy = x^3 + 1$, the closure solving degree of the two-summand system is

$$D = 2 + [e < -3.5] + [e < -8.5], \qquad e = n - \dim V - \dim V^{(2)},$$

with $V^{(2)}$ the span of pairwise products, at mean absolute error 0.06 degrees, 0.08 with each field size held out (`cryptanalysis:experiments/pdp-degree-heuristics/README.md`). For prime n, $\dim V^{(2)} \geq 2\ell - 1$, and below $n - 1$ only geometric progressions $c \cdot \mathrm{span}\{1, g, \ldots, g^{\ell-1}\}$ attain it; the literature's $\mathrm{span}\{1, z, \ldots, z^{\ell-1}\}$ is one. Where the rule puts a random base a degree higher, the progressions are 15–40 times cheaper per query and per relation on average, up to 100; elsewhere 1.5–2 times; the yield is the same. Two further findings matter for pricing: the homogeneous degree of regularity of these systems is exactly $\ell + 1$, because the two-summand top parts are bilinear across the variable blocks, while the solving degree is 2–4, so the semi-regular prediction misses by 0.5–1.2 degrees on average; and at $m = 3$ the formulation decides whether the base matters (in the symmetric formulation the progression is 15–40× cheaper per query at $\ell = 3$ at equal degree, and 500–1,500× where

the random base needs one more degree). Every row is a decomposition-stage profile on small fields, not an index-calculus result.

A learned cost model. The lead (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-learned-cost-model-factor-base-selection.yaml`, last reviewed 6 November) fits a gated log-linear (ridge) model of per-relation cost on structural features of V (predicted exact yield, solving degree, trace and product-space statistics) and compares its pick from a pool with the hand rule "lowest expected solving degree first". The label producer is the structural-law experiment above. As the atlas records it:

- *Stage.* 1.82 bits per relation $[1.36, 2.26]$ better than the hand rule at $\ell \le 4$ (13 cells), 1.13 $[0.45, 1.62]$ at $\ell = 5$.
- *Across curves, frozen, no refit.* $+1.84$ $[0.95, 2.68]$ on $a_2 = 1$ Koblitz curves; $+0.92$ $[0.54, 1.34]$ on ordinary curves, not confirmed on 16 fresh ordinary curves at $\ell = 4$ ($+0.43$ $[0.15, 0.74]$ against a 1.08 threshold), where the hand rule already sits near the pool best.
- *End to end.* 0.93 bits $[0.36, 1.48]$ less counted cost per verified logarithm over 9 cells ($n \le 41$). At $\ell = 4$ on ordinary curves, $n = 19$, the complete-logarithm check is a null with the sign reversed: -0.45 $[-0.98, +0.04]$ bits at $w = 0$.
- *Native ecbench on the isolated EKS route*, 5 November: K1 ($a_2 = 1$) Koblitz curves, $\ell = 3$, `ic.pipeline` with explicit binary-subspace bases, 4 targets $\times$ 5 rounds per size, 192 of 192 records verified. $\log_2(S_{\text{rule}}/S_{\text{ridge}})$ in counted complete-logarithm cost is $+1.83/+0.80/+0.80$ bits at $n = 17$ and $+0.96/+2.47/+2.47$ at $n = 19$ for $w = 0/0.1/1$, every interval's lower end above 0. The ridge pick stays behind `rho.signed_frobenius_strong` by 1.8/14.8/17.7 bits ($n = 17$) and 3.3/15.1/17.9 ($n = 19$). The producer's first "about 9 bits, identical across the band" was a pinned-calibration point and was corrected.

Standing, exactly. The atlas lead on crypto-autoresearcher `main` (commit `8d61579904`, 6 November) records the EKS result as "independently validated 2026-11-06 with VALIDATION.json, 160 of 160 replays reproduced on a second CPU". The experiment, its protocol, its validator report and that `VALIDATION.json` live in the external repository `aburan28/ml-cryptanalysis` (`experiments/fb-cost-model-curves-eks`), which this session could not open; the same atlas entry's evidence note says the validation is "in a follow-up PR". Nothing about the study is on `main` of `crypto` or `crypto-autoresearcher` except the atlas text. So for this treatise the validated claim is an unverified assertion recorded by the atlas: if it holds, it is the family's only validated, measured improvement over a binary index-calculus baseline, about 1–2.5 bits at toy size, $n = 17, 19$, with its scaling in n not yet measured (K1 $n = 23$ was frozen on 6 November). It moves no row toward rho by more than the 1.8–17.9 bits it leaves.

20.4 Quasi-subfield polynomials at $n = 131$

§16 records the $\mathbb{F}_{2^7}$ census that showed the HKPYY Lemma 4.1 bound [24] tight. The cost side is closed by theorem (`crypto-autoresearcher:analysis/qsp-ecc2k130/README.md`; the atlas records it as a theorem from Euler and Petit's work):

- every linearised quasi-subfield polynomial has $\beta \ge 3/4$, so under the HKPYY cost model the exponent is at least 0.93, at least 2^{122} at $n = 131$ against rho's $2^{60.81}$ ($2^{60.9}$ in the source);
- with $\mathbb{F}_2$ coefficients only $X^2 + X$ ($n' = 1$) and $L_{\Phi_{131}}$ ($n' = 130$, $\beta = 0.99994$) exist at $n = 131$;
- multiplicative quasi-subfield polynomials give $|V| = 264$ or $p + 1$, with $\beta \ge 1.02$;
- the cost model needs $\beta \approx 0.0958$ to reach rho's cost.

At toy size the yield is no better than random: at $n = 31$, $k = 15$, the QSP base $t^{15} + t^7 + t^3 + t + 1$ ($\beta = 0.964$) yields 0.214 (0.190–0.240) against 0.204–0.230 for random subspaces, and the route's kill criterion fired (`research/ic_next10_20261014/08-quasi-subfield/README.md`).

20.5 Composite degree

Composite n is the one binary setting where a cost model puts index calculus below rho on time alone, and only for an unmeasured per-decomposition cost; under time×memory the same model is worse on every row. Chapter 8 records the lift-dense twisted-line base on c2pnb208w1 and c2pnb368w1 (+0.50 bits, modelled), and §26 the GHS reachability of the Oakley curves.

c2pnb: Gaudry–Diem $n = 4$, as a cost model. The five ANSI X9.62 example curves c2pnb176v1, 208w1, 272w1, 304w1 and 368w1 have coefficients in $\mathbb{F}_{2^{16}}$ and live over $\mathbb{F}_{Q^4}$, $Q = 2^{4k}$, $k = 11, 13, 17, 19, 23$. A zero-experiment derivation (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/c1-gaudry-n4-c2pnb/README.md`) prices Gaudry–Diem $n = 4$ [10, 20] with a base of points with $x \in \mathbb{F}_Q$ reduced modulo negation and π_{16} ($N = Q/(2k)$ orbits) and double large primes [22], against matched rho of 78.10, 93.98, 125.78, 141.71 and 173.57 bits. Under three metrics:

- *time only:* below rho on all five rows for every per-decomposition cost $c_{\text{trial}} \in [2^{15.5}, 2^{32.5}]$, by +1.46, +5.80, +14.19, +18.34 and +26.60 bits at $c_{\text{trial}} = 2^{30}$;
- *time × memory against van Oorschot–Wiener:* worse on every row, by 35–56 bits at $c_{\text{trial}} = 2^{30}$;
- *mesh area-time:* better only at 304w1 (+1.28) and 368w1 (+0.97), and only for $c_{\text{trial}} \le 2^{27}$.

The verdict turns on c_{trial}, never measured at toy size, the per-worker FGLM memory (D^2 with $D = 4096$, brief-supplied) and the linear-algebra constant; a toy ladder over $\mathbb{F}_{2^4}$ lifted to $\mathbb{F}_{2^{4k}}$, $k \in \{5, 7, 9, 11, 13\}$, about $2 \cdot 10^4$ decompositions, would pin all three and has not run. The atlas adds that the $\mathbb{F}_{2^{16}}$-rational base is structurally empty in G and that a naive trace-zero extrapolation is a regime-invalid trap (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-composite-degree-c2pnb.yaml`).

Oakley groups 3 and 4: $n = 5$. The twisted line exists on both Oakley EC2N curves: a u with $T(u) = T(c/u) = 0$ was verified in $\mathbb{F}_{2^N}$ ($N = 155$ and 185, $n = 5$), and all 200 sampled elements of $u\mathbb{F}_q$ lift against 104 and 99 of 200 for the standard line, so as a Gaudry–Diem $n = 5$ base each decomposition trial would have $2^5 = 32$ times the relations of the standard base; on c2pnb176v1, 272w1 and 304w1 ($\mathrm{Tr}(a) = 1$) none of 200 lift (`research/oakley_lift_dense_20261027/REPORT.typ`). This is a per-trial yield statement. Three approved Oakley contracts have no runs; the atlas prices the GHS genus-16 transfer at +1.1 bits (group 3) and +10.9 bits (group 4) of advantage after the 11 October magic-number correction, by formula (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-oakley-quintic-tower.yaml`).

20.6 Binary GHS and Weil-restriction screens

The late-October cubic GHS catalogue over $\mathbb{F}_4$, $\mathbb{F}_8$ and $\mathbb{F}_{16}$ belongs to the transfer record (§26). Four binary screens outside it bear on index calculus.

Decomposition coverage at $n = 2$ is 3/8, not Poisson. For generic $E/\mathbb{F}_{q^n}$ and the base $x \in \mathbb{F}_q$, the share of points that are a sum of n base points was computed exactly over the whole group (`research/weil_descent_angles_20261018/REPORT.md`). In odd characteristic at $n = 2$–5 it matches a random x-set of the same size (paired residual $|z| \le 0.94$). Binary $n = 2$ is the exception: coverage tends to 3/8, not the Poisson $1 - e^{-1/2} = 0.3935$ ($z = -4.72$), and the cause is proved. Every $x \in \mathbb{F}_q$ has $\mathrm{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_2}(x) = 0$, so the base lies in one coset of $2E$; the descended relation map is $\mathbb{F}_2$-affine with image in the hyperplane $\mathrm{Tr}(w/x_3^2) = 0$; so a point decomposes only if $\mathrm{Tr}(x_3) = \mathrm{Tr}(a)$, half the group, and inside that half the multiplicities are exactly $\mathrm{Bin}(2, 1/2)$. Because attack targets lie in the odd-order subgroup inside $2E$, the relevant coverage is that of $2E$, exactly 3/4 against 0.3935; the study corrected its first reading ("neutral to slightly negative", measured over the whole group) on this point. The identity holds on all 21 curves.

IPsec group 3. An archived search over the isogeny class of RFC 2409 Oakley group 3 ($\mathbb{F}_2[z]/(z^{155} + z^{62} + 1)$, $N = 12r$) screened 24,492 Frobenius orbits and found no low-genus candidate in the implemented tests; the whole class was not enumerated and no resistance claim follows (`research/ghs_ipsec_isogeny_class_20261023/README.md`).

Generalised binary GHS, explicitly. Full nonzero-c Hess screens at absolute degrees 6, 8, 10 and 12 (63, 255, 1,023 and 4,095 values) gave explicit maps and attack controls (`research/ghs_generalized_20261024/REPORT.md`). At degree 12, 147 profiles have genus at most 2, including 15 of genus 1 over $\mathbb{F}_{2^4}$, and the Weil bound ($\#\mathrm{Jac} \le 25$ and ≤ 81) rules out a homomorphism from the order-199 subgroup; the other 232 profiles are index-two genus-2 profiles that a subfield change maps to one canonical form, presenting P as $T - P$ for the 2-torsion point T. Every exact instance (subgroups of order 19, 67, 43 and 199) verified end to end, and all belong to known GHS [21] and Hess families.

A degree-73 descent at $n = 37$. A separable degree-73 isogeny of $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{37}}$ (74 rational cyclic kernels; a squarefree degree-36 kernel polynomial dividing the degree-2,664 division polynomial) was tested as a presentation change for the Boolean stage (`research/koblitz_isogeny_descent_37_results_20260925/RESULTS.md`). All 128 systems completed and every planted decomposition verified on both curves, but the preregistered criterion (at least a 10% F5 matrix XOR reduction in every cell, with no completion-degree increase) failed: the neighbour cut XORs by at least 10% in 4 of 8 cells, regressed up to 2.00×, and raised the completion degree in 7 of 64 paired systems. The committed raw archive was found not to be gzip and was restored from the original workflow artifact; the result is unchanged.

21 Prime fields

Over $\mathbb{F}_p$ there is no subfield to descend to and no Frobenius to fold, so index calculus must find structure in the factor base itself (§1). The programme tried known-answer ladders, root-finding and pair-table oracles, the Petit–Kosters–Messeng Kummer tower, structured bases on the P-256 field, isogeny-class walks on the NIST curves, lattice reformulations, and miniatures of Curve25519. Every prime-field arm is behind rho with negation. The closest are generic collision searches at rho's exponent: the residual collectors at 1.54–1.64× matched rho at $w = 0$ on 36–37-bit curves with a validator report (1.3–1.7× to 42 bits as the producer reports them; Chapter 22), and the Curve25519 walk tables at 1.6–1.9× (unverified; §21.7). Every other arm is further away, and where it was measured over several sizes (Table B, the Kummer tower, the Curve25519 algebraic track) its gap grows. The section ends with Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$, the one odd-characteristic extension field the programme measured end to end. Every study is listed in the programme record [3].

21.1 Known-answer ladders and leaderboard rows

Over prime fields the ledger tracks staged known-answer ladders (`docs/ic/boundary_targets.json`, regime `prime`). On $j = 0$ curves the staged driver recovers a 20-bit logarithm ($p = 1{,}048{,}609$) in 56.8–64.9 s of IC wall against 150–153 ms of rho on the identical public point, with split timers and three seeds; running the relation search on 14 threads made that stage 3–4.4× faster in wall time with counted work unchanged (`research/j0_parallel_relations_20261016/`). On the generic $a = -3$ shape (P-256 and CryptoPro-B class), a Semaev-S_4 three-decomposition family with Cantor–Zassenhaus root finding yields one relation per trial at $p = 65{,}519$ with a 120-point base; at fixed base size, median trials per relation grow from 2 to 28 to 379–506 at 16, 20 and 24 bits, about 16× per four bits. The prime `vs_rho` row is *not achieved* at any measured size.

Table B of `docs/ic/LEADERBOARD.md` has eight prime rows, $\log_2 r = 9.7$–23.4, one target, $m = 2$, folded pair table with walked targets: IC is 6.34, 5.47, 6.43, 6.71, 5.82, 8.3, 10.5 and 13.5× matched rho, the ratio rising from about 18 bits. All eight passed the validation gate on 24 October (`research/ic_validation_gate_20261024/`). In ecbench the root-finding pair table is 16–20× matched rho in counted GAE, and 7.46× [5.82, 9.63] with a lane collector, all lower bounds (`research/ecbench_rootfinding_lanes_20261013/`); a planner that picks the base, stop rule and oracle beat every one-change alternative on six curves at $w = 0$, the root-find-and-subtract variant by 171–439× (`research/ic_planner_validation_20261020/`). The planner's three weight rules, which scaled the base or stop for $w > 0$, failed out of sample on 26 October and were withdrawn (rule set `ic-planner/4`; `research/ic_planner_exhaustive_20261026/`). The frontier campaign's two-summand prime arm goes from 10.5× rho at 20 bits to 102× at 40 bits (§36.1).

Root finding against Buchberger on NIST shapes. Per query, root finding of Semaev's S_3 was 3,021× (P-256 shape) to 5,802× (P-224) cheaper than the comparison arm at a six-point base (`research/semaev_rootfind_nist_20261012/`); the arm was `groebner_f4::buchberger`, not F4, so the ratio is against Buchberger (§4.5). It is an oracle ratio with no complete solve. A host emulation of a GPU decomposition kernel passed six configurations on 40–61-bit P-256-class curves (`research/gpu_rootfind_decomposition_20261012/`); no device ran it. Three frozen SMT formulations of a 20-bit decomposition all returned UNKNOWN at 120 s, where the native reference needed 146 probes (`research/prime_field_smt_20261005/`).

Earlier, in crypto-autoresearcher. The first end-to-end prime-field index calculus of the laboratory predates this repository's: an imported Autolab round of 1 June 2026 solved logarithms with an $m = 3$ Semaev index calculus over a factor base of the smallest valid abscissae, at a fitted cost exponent of 0.889 against rho's 0.5, IC/rho growing from 21.5 to 205.7 over 10–18 bits (`ar:experiments/EXP-ALPF-032/analysis.md`). Structured bases there weakened rather than supported a relation excess (`ar:ledger/evidence/EV-PFDR-f6b76c.yaml`), and on one 13-bit curve an apparent early fall of the degree fired identically when the Semaev form was replaced by a random polynomial but not when the factor-base constraints were, so it came from the factor base, not from the summation polynomial (`ar:experiments/EXP-PFDR-4bfc6f/analysis.md`). A separate repository asked the underlying

question directly—is there any coordinate-defined subset of $E(\mathbb{F}_p)$ that interacts with addition better than a random subset?—and found none on toy curves (`ecdlp-prime-field-smallness`, `README.md` at `4f20be9b36`).

21.2 The Kummer tower

Petit, Kosters and Messeng's base $V = \mu_{2^t}$ (the roots of $x^{2^t} - 1$) gives prime-field summation systems extra structure for an algebraic solver. Its claim rests on the solving degree of those systems staying bounded as $N = m\ell$ grows; the programme measured that degree for four weeks.

The solving degree. The pilot found F4's solving degree at 4–5 for $m = 2$ through $N = 18$ and read the preregistered "grows like N" as failing at toy size (`research/pkm_tower_pilot_20260924/`). Round 2 showed it is not bounded: at $m = 2$ it is 5 to $N = 18$ and 6 at $N = 20, 22$, at $m = 3$ it reaches 7 at $N = 15$, and the null family behaves the same (`research/pkm_tower_round2_20260925/`). At $m = 4$, $N = 16$ finished at $D = 7$, as at $N = 12$ (`research/pkm_tower_round3_20260925/`); at $m = 3$ the line reads $6, 6, 7, 7$ over $N = 9$–18 (`research/pkm_tower_round7_20261006/`). Signature engines agreed with F4 but needed one or two degrees more and 1.7–28× the memory, so F4 stayed (rounds 4–5); a full-echelon exit cut multiply-adds 39.8% at $m = 4$, $N = 12$ without moving D (round 6). The torus and split Dickson presentations give the same D in every cell, and the Kummer D does not depend on the 2-adic valuation (`research/pkm_dickson_torus_20261015/`). At $m = 3$, $N = 21$ two engines finished every degree-7 step without reaching the unit ideal and stopped inside degree 8 on memory—the native `f4_fp_tower` at a 317 GB matrix cap after a 67 GB peak, msolve at 99.6 GiB on 16 threads (`research/pkm_lead1_m3n21_20261016/`, `research/pkm_m3_n21_msolve_20261018/`). That implies $D \geq 8$ at $N = 21$, which both studies read as inconclusive because no degree-8 step completed. The Certicom-131 programme's gate needs $D \leq 3$ at $m \geq 4$; the toy plateau at 7 is above it, at 2^{89}–2^{98} (generous) and 2^{129}–2^{139} (plain) at 131 bits (`docs/ic/CERTICOM131_IC_PROGRAM.md`). The exponent sum of decompositions on the tower is indistinguishable from random at $k = 1$–6 (`research/ic_exponent_falsifiers_20261013/`).

In ecbench. We ran the tower end to end with the root-finding oracles on five curves of 20–28 bits (896 runs, all verified, all replay audits OK; `research/pkm_tower_ecbench_20261013/`). Index calculus costs 34.0, 41.9, 29.4, 96.6 and 140.6× rho-negation in S (95% intervals in Figure 21.1), growing by 0.26 ± 0.28 bits of S_{IC}/S_ρ per bit of r, consistent with the $p^{1/6}$ the floors predict. The tower base costs the same as a plain base of the same size to within 1% (ratios 0.999–1.012): PKM's structure helps an algebraic solver, not a table. An isogeny-tower variant costs 1.00–1.33× the plain base once its auxiliary-curve search is charged, and every arm is 17–137× rho (`research/pkm_isogeny_tower_ecbench_20261015/`); for a P-192-sized tower that search would cost about 2^{74} GAE, or 11.2 core-years, while a supersingular auxiliary curve gives the needed point in 7 ms with the same solving degree (`research/pkm_isogeny_tower_p192_20261015/`, `research/pkm_supersingular_aux_20261015/`). A 30-bit row of 24 October is 158.9× rho (§36.5). Swapping in an A11 presentation could not be priced: ecbench has no method that builds a tower summation system (`research/a11_pipeline_pricing_20261019/`).

21.3 Structured bases on the P-256 field

Twenty-four rounds between 4 and 6 October (`research/icv1-fp256-t89188191154553853111372247798585809583-f188c491_affine_bitbox_degree_round1_20261004/` and its 23 successors; cluster "icv1-fp256") searched for factor bases on the P-256 field whose summation systems have lower degree. They found degree reductions on split systems: Dickson-torus branching lowered the solving degree of components from 4 to 3, image trees for S_9 and S_{17} reached local degree 2, and an indexed quadratic image cut one stage 498×. At the P-256 modulus the images hit their generic size ceilings (width ratio 1.000000), so the degree reduction transferred and the compression did not (round 14). No round found a relation for a target that was not planted: planted length-17 relations were recovered, hash-public targets were not, and a log-transport round found zero relations among 14.4 million images (round 21). Projected end to end, the best schedule is 16.2–19.4 bits above rho in the same unit, after a symmetric projection of $2^{144.625}$ was withdrawn for exceeding its domain (round 20); the final geometric bases are

392–404× rho, and a known-log constant of 0.964× rho in round 24 is a lower bound that the study itself says is not an attack.

21.4 Small-abscissa bases and isogeny classes

Two-point small-x bases have an exact pair yield of $2B^2/(n-1)$ with no pair-sum collisions: on P-192 a base of 8,192 points needs 2^{178} trials for enough relations against rho's 2^{96}, and on all 33 SECG curves the route loses to rho by at least $2^{40.9}$ in trials (`research/prime_small_x_p192_20261014/`, `research/secg_small_x_20261014/`). At $p = 2^{61} - 1$, 1,228,800,000 target trials over 1,024 curves of a horizontal isogeny class found zero relations, while a 20-bit positive control found 33 and recovered the scalar (`research/prime61_factor_base_20261016/`, `research/prime61_disk_extend_20261018/`). Native walkers enumerated 1,064,000 distinct curves isogenous to P-256, 20,000-curve classes of P-224 and P-192, and verified isogenies of degree up to 10,453 (P-192) and 1,471 (P-224); no preregistered structural screen fired on any walked curve, so no factor-base test was ever selected (`research/p256_isogeny_million_20261006/`, `research/p192_p224_large_degree_isogenies_20261009/`). By Lemma 26.1 a homomorphic transfer could change only bounded constants in any case. The class of a 61-bit curve has a certified order of 358,094,304 with adjacency still open (`research/p61_terminal_closure_20261023/`).

21.5 Lattices

Every lattice intrinsic to $(E, P, [k]P)$ has rank 2 (4 over $\mathbb{Z}[\tau]$), and relation lattices have a derived ceiling of $p^{5/6}$ (`research/notes/ecdlp-general/RESEARCH_LATTICE_ECDLP_ROUTES.md`). A Coppersmith reformulation of the $m = 4$ oracle via S_5 reaches $\delta^* = 1/80$ against about $1/5$ needed and is closed (`research/lit2026_w10_s5_coppersmith_20261011/`); on S_3, a frozen falsifier recovered no planted decomposition in any of 96 cells with $\beta \geq 0.20$ at 24–56 bits, so the $\beta = 1/4$ ceiling stands at multiplicity ≤ 3 (`research/coppersmith_s3_theta2_20261023/`).

21.6 ECCp-131

ECCp-131 ($\log_2 r = 130.19$, rho $2^{64.92}$ GAE) has exact parameters wired into the BSGS and IC command paths; a three-point small-x probe found zero natural relations and rank zero at a 128-point base, which blocks any full-parameter IC solution and is an open problem record (`research/eccp131_bsgs_ic_20261020/`, `problems/eccp131-full-parameter-index-calculus.md`). Structured prime bases for the Certicom programme were a null result (N/E 0.969–1.000 against a random control of 0.974). One CPU core walks ECCp-131 rho at 1.23 million steps per second with a 32-bit Montgomery batched walk, and 15.06 million with an AVX-512 eight-lane walk (`research/certicom131_rho_ic_bench_20261022/`, `research/certicom131_rho_parity_20261023/`). The later ECCp-131 work is in §24.3.

21.7 Curve25519 miniatures

Between 2 and 5 November thirty-one studies measured index calculus on miniatures of Curve25519. The real Curve25519 was not attacked; nothing ran at 255 bits. Each target is a miniature built by Curve25519's own selection rule. The prime p is the largest below 2^k with $p \equiv 5 \pmod 8$. The coefficient A is the smallest $A \equiv 2 \pmod 4$ with $\#E = 8r$, $\#E' = 4r'$ (r, r' prime) and a cyclic 2-part. The base point is the smallest abscissa u of order r. At $k = 255$ this rule gives $A = 486662$, $u = 9$; the 255-bit search itself was not rerun, as it needs SEA point counting. Table 21.1 lists the four main curves; later studies added $k = 16, 18, 22$ and four sibling primes per size (`research/curve25519_ic_baseline_20261102/`). Costs are in the section's unit $S = (\text{counted GAE} + w \cdot \text{unpriced native operations})/\sqrt{r}$ at $w = 0, 0.1, 1$. Against rho with negation (`rho-neg`), every arm is ranked only where its order holds across the band.

Generic track. The cofactor costs what it should: an unprojected small-x base needs 7.66–8.38× the trials (8 predicted, 1.00 on a cofactor-1 control). Projection removes the penalty. With the penalty gone, the baseline gap still grows with k (Table 21.1). The first ratchet round found nothing: a retuned base size was null (0.996 $[0.944, 1.051]$ and 0.977 $[0.941, 1.015]$); the closed cofactor fiber and a T_0-closed base were dominated; and the Kummer tower is blocked, because $p \equiv 5 \pmod 8$ forces $t \leq 2$

k	p	A	r ($\#E = 8r$)	baseline `proj-a` / rho-neg ($w = 0$)
20	1,048,573	926	131,203	7.8×
24	16,777,213	318	2,097,349	10.6×
28	268,435,157	5,330	33,557,809	18.1×
32	4,294,967,197	27,842	536,884,639	24.7×

Table 21.1: Curve25519 miniatures and the generic baseline. The baseline is a projected-abscissa factor base with negation-folded meet-in-the-middle (4,032 records, all verified, replay audit passed). The gap is 7.8–7.9×, 18.1–18.4× and 24.3–24.7× across $w \in \{0, 0.1, 1\}$ at $k = 20, 28, 32$. Source: `research/curve25519_ic_baseline_20261102/README.md`.

(`research/curve25519_ic_ratchet_20261102/`). The T_0 symmetry halves the work of a blind base (1.944–2.016×), but projection already beats it by 3.33–4.00× (`research/curve25519_t0_symmetry_20261102/`). Three-term sums improved on the baseline at $k = 20$ (0.758 $[0.716, 0.802]$) and $k = 24$ (0.596 $[0.575, 0.619]$). The gain did not hold at $k = 28$, where its order across the w band is not fixed (`research/curve25519_m3_generic_20261102/`).

The autolab and the collision floor. A 35-round automated ratchet (`research/curve25519_autolab_20261104/`) moved the generic incumbent from four-term sums to coefficient-box sums and then to a walk table over two seed abscissae (`wwalk`). Rounds 22–35 sit at the baby-step giant-step boundary. They count as index calculus only by the owner's ruling of 4 November. The fixed-table distances of 1.43–1.55× rho were single table draws (correction R4-I1). Averaged over 16 walk seeds they are 1.82× ($k = 32$) and 1.86× ($k = 28$). With a fresh table per run, the incumbents reach 1.94×, 1.81× and 1.82× matched rho at $k = 24, 28, 32$. Over five family curves per size the mean is 1.57, 1.78, 1.80 and 1.91× at $k = 20$–32, with per-curve spread about ±0.2. Every autolab result is a statement claim with no proof attached, so its standing is *unverified* (R4-I1 `PCC-20261104-c97d81`, re-baseline `PCC-20261104-d91a51`, multi-curve `PCC-20261105-37a78d`). A 5 November correction of the fast probe records that the legacy affine-equivalent tally is not a complete native-operation cost ledger; no comparative experiment was run for it (`research/curve25519_autolab_parity_20261105/DELIVERY.md`; unverified). A derivation in the generic group model, checked by Monte Carlo and explicitly not a formal theorem (`research/curve25519_rho_parity_bound_20261104/`), explains the plateau. With F unknown base logarithms, a collision-based index calculus costs at least $\Gamma(F + \frac{3}{2})/(\Gamma(F+1)\Gamma(\frac{3}{2}))$ times rho. That is 1.5× for $F = 1$ and 1.875× for $F = 2$, where Monte Carlo gives 1.889×. `wwalk` has $F = 2$, so its 1.8–1.9× is this floor, not an implementation constant. Parity needs $F = 0$, which is rho under another name.

Algebraic track. Semaev summation decided by F4 (`research/curve25519_algebraic_ecbench_20261102/`) ran end to end at $k = 20$ at 18.2×, $9.55 \cdot 10^5$× and $5.85 \cdot 10^6$× rho for $w = 0, 0.1, 1$. A timing smoke of the frozen algebraic ladder showed that ladder would take days. Its interrupted session is citable only for its four sealed records and is pinned as defective (`docs/ecbench/DEFECTIVE_SESSIONS.md`). The protocol was amended before data to measure F4 cost per call instead (`research/curve25519_algebraic_ladder_20261102/`). The later gains came from an isogeny-tower base (the `pkm-iso` arms, a 2^t-point base built by a chain of 2-isogenies, which lifts the $t \leq 2$ limit of the μ_{2^t} tower of the generic-track paragraph), selecting its frame, seed, depth t and guess count G, and from F4 trace replay, which is a port of the Joux–Vitse technique, not a new method.

Three studies have independent validator reports (pass with errata), and none has a `pcc` claim record:

- The best seed at $t = 5$, $G = 4$ costs 0.0578 $[0.0466, 0.0716]$ of the seed-0 incumbent at $w \geq 0.1$ (`research/curve25519_t10_winner_ecbench_20261104/`).
- A $t = 4$, $G = 3$ arm with trace replay costs 0.2837 $[0.2413, 0.3335]$ of that winner at $w = 0.1$, but 1.567× more at $w = 0$ (`research/curve25519_t4_candidate_ecbench_20261104/`). It is still 27.9×, $1.83 \cdot 10^4$× $[1.55 \cdot 10^4, 2.15 \cdot 10^4]$ and $1.13 \cdot 10^5$× behind rho at $w = 0, 0.1, 1$.
- The atlas (`research/curve25519_ic_atlas_20261104/`) is the source of the comparison with an exhaustive grid and of the scaling paragraph below.

Against an exhaustive grid on the same base, the F4 arm is 42× too expensive, and even the pair table on that base trails the generic baseline by 2.39–3.32×. Several presentations and variants gave no gain:

- Edwards coordinates: the same solving degree in 8 of 8 pairs.
- The T_0 w-form: equal degree and F4 cost, and 1.2–3.0× worse per relation.

- Three-term Semaev sums: always worse than two-term sums on this track.
- Cofactor lifts: about 8× the F4 work for 5 hits against 0, so no per-relation gain.

Sources: `research/curve25519_edwards_semaev_20261102/`, `research/curve25519_t0_algebraic_20261102/`, `research/curve25519_cofactor_algebraic_20261102/`, `research/curve25519_leads_wave4_20261104/`.

Scaling. Fitted to modelled F4 work (atlas; validator report, no `pcc` claim), the algebraic cost grows as $r^{1.03}$ against rho's $r^{1/2}$, so the gap grows as $r^{0.53}$; the optimal t is 4 at every k (`research/curve25519_ic_atlas_20261104/`). The 5 November scale-ups ($k = 24, 28, 32$; all standing unverified, no validator report) agree:

- F4 cost per target is the same at every k (1.000–1.019).
- At $k = 24$ the $t = 10$ arm finished 2 of 6 solves. At $k = 28$ it finished none, and the $k = 28$ gap is only an F4-cost estimate.
- At $k = 32$ the engine refuses $p \geq 2^{31}$.
- The cofactor-lift and three-term conclusions of $k = 20$ carry over.

Sources: `research/curve25519_scale_guess_trace_20261105/`, `research/curve25519_algebraic_ecbench_scale_20261105/`, `research/curve25519_scale_lifts_20261105/`, `research/curve25519_scale_m3_t0_20261105/`.

What this settles. On every Curve25519 miniature, track, size and weight measured, rho with negation is ahead of every index-calculus arm. The generic gap grows with k. Collision-based variants sit at a derived floor near 1.9×, and the algebraic gap grows as about $r^{1/2}$. No study makes or implies a statement about the 255-bit curve, and none weakens its security. No Curve25519 row is on the leaderboard or in `benchmarks/`. Most runs were local, on a shared host, without a second-host replay. Three leads remain open:

- the (w, D) presentation, which is untested;
- digit-set bases, at 0.438–0.479× per relation on one arm only;
- an odd-Kummer design, which has no data yet.

21.8 Audits of standard curves

Several studies examined the NIST, SECG and CryptoPro curves themselves. None found an ECDLP weakness. The later audits are in §24.4. The NIST prime-curve seeds were regenerated as 1997–98 `drand48` output, with the skip rule and base points still unexplained (`research/nist_seed_provenance_20261016/`, `research/nist_basepoint_provenance_20261017/`); a claimed P-256 $n - 1$ outlier was not established after controls, and the published b sits at the median of every statistic over a million seed samples (`research/nist_cyclotomic_controls_20261011/`, `research/p256_trapdoor_round2_20261012/`). The $j = 0$ SECG curves have sextic twists of weak order (rho $2^{65.5}$ for secp256k1's), a fact about twists, not about the curves (`research/safecurves_trapdoor_slots_20261017/`). Special prime shapes change neither the distribution of $\#E$ nor the embedding degree, so only small-embedding-degree pairing curves open an SNFS-type route (`research/snfs_prime_families_20261023/`, `research/solinas_distribution_20261023/`). A P-192 "BREAKS" verdict applies to a singular nodal cubic, an implementation weakness (`research/p192-weak-curve-20261006/`), and the P-224 directory is an invalid-curve attack on an unvalidated ECDH oracle (`research/p224_attack/`). The CryptoPro-B and CP6-782 work is constructive: GLV-2 scalar multiplication 1.25–1.50× faster via isogeny-chain endomorphisms (`research/cryptopro_b_chain_sweep_20261006/`, `research/endo_native_bench_20261016/`). Cheon's attack with auxiliary inputs is cheaper than matched negation rho at most sizes from 36 bits in ecbench's counted unit (dearer at 32 and 40 bits; no exponent separation; not validated; §32.4), and only for the DLP with auxiliary inputs; an earlier study's "beats rho from 24 bits up" (`research/torsion_auxiliary_inputs_20260918/`) was not priced across the band, and the P-224-shape Cheon/rho ratios of a different study were withdrawn (§24.4).

21.9 Literature claims that did not survive

The lit2026 programme (`research/lit2026_speedup_program/`) ran each finding of a 2025–26 search as a measured change against an interleaved baseline binary. Workstream W12 implemented the zero-minor guess-and-determine attack (Mahalanobis, arXiv 2607.09814v2): it solved 48/48 planted targets

on 16–26-bit curves, but its search phases scale with slope 0.905 $[0.769, 1.021]$ in $\log_2 r$ and its cost over rho grows from 260× to 6,715×—exhaustive in r, as the source's own table says. A second flagged ePrint (Hu, 2025/015) did not specify an implementable algorithm and is recorded as blocked. On the generic side, W11 cut BSGS from 16.75 to 3.71 multiplication equivalents per visited point with paired steps (wall 1.15–1.67×), which moves rho's and BSGS's constants, not index calculus.

21.10 Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$

Over $\mathbb{F}_{p^k}$ with a base of points whose abscissa lies in $\mathbb{F}_p$, a relation needs a k-decomposition solved by a Macaulay system. We made the `gaudry_quartic` solve generic in the number of unknowns and faster (median whole solve 1.96/4.19/6.08× at $d = 4, 5, 6$, 37.3 → 6.1 s at $d = 6$, as recomputed by the validation sweep from the committed records on a shared host; the S_5 solve from about 30 minutes to 164–203 s; every count and solution identical). Then we built FGHR's D_4 variant (`research/gaudry_quartic_d4_20261013/`): on a curve with rational 2-torsion T, the coordinate $Y = (x - x_0 - c)/(x - x_0 + c)$ satisfies $Y(P + T) = -Y(P)$, so S_5 in Y is invariant under the Weyl group D_4 (order 192). In the invariants $\sigma_k = e_k(Y^2)$, $\pi = \prod Y_i$, the system has Bézout number 512 instead of 4,096, Macaulay degree 14 instead of 29, and a 13,412 × 11,628 matrix instead of 41,780 × 40,920.

At $p = 269$, the cost C_4 of one decomposition falls from $1.21–1.23 \cdot 10^{12}$ to $2.54–2.55 \cdot 10^{10}$ $\mathbb{F}_p$ multiplications, 47.5–48.4×; a solve takes 4.2–5.6 s instead of 164–203 s. Decompositions agree with an independent meet-in-the-middle enumeration on 19 of 19 solved targets; one target in 20 is degenerate (quotient dimension 517) and is discarded, adding about 5% to the relation phase. The Y-line base decomposes about 1/8 as often (pooled ratio 0.133 over 8,000 targets; prediction $1/8 \pm 30\%$ held on one of two curves), but each column covers $\{\pm P, \pm(P + T)\}$, halving the unknowns.

Inserted into the repository's crossover model (`research/notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md` §11.16), the measured C_4 and rate move the $k = 4$ crossover with rho from $2^{149.4}$–$2^{152.7}$ to $2^{135.1}$–$2^{135.8}$ (cofactor $h = 2$, forced by the 2-torsion), and the asymptotic ratio past it from 0.34–0.63 to 0.12–0.22 of rho. This was the first place in the programme where a modelled crossover moved, and the GLV-invariant line later moved it further (Chapter 31); it moves in an *extrapolated model* ($n^{1/4}$ relations, $n^{1/2}$ linear algebra and rho) fed by a measured constant, so it is classed as ENGINEERING, not an advance. The validation sweep notes that this crossover was not re-derived against a rho with the negation map; against the negation walk, a later note puts the plain prime-order $k = 4$ handover at $2^{156.1}$ and a ψ-folded one at $2^{146.0}$ (`research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`). The measured characteristic-two $k = 4$ ladder is in §36.3. Extension fields of degree four are not the fields deployed curves use.

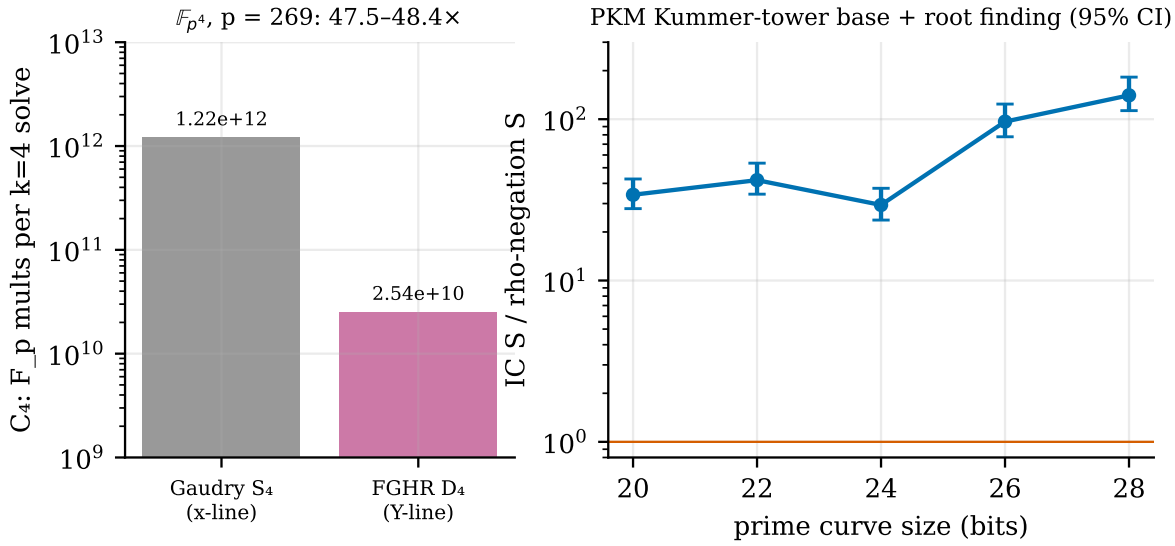


Figure 21.1: Left: cost C_4 of one $k = 4$ decomposition over $\mathbb{F}_{p^4}$, $p = 269$, Gaudry's S_4 system on the x-line against FGHR's D_4 system on the Y-line (`research/gaudry_quartic_d4_20261013/`). Right: index calculus with the PKM tower base and root finding over rho with negation, eight one-target workloads × three rounds per size, paired 95% cluster-bootstrap intervals (`research/pkm_tower_ecbench_20261013/`).

22 Prime fields: the generic track

Chapter 21 gave the prime-field programme as it stood in mid-October: the known-answer ladders, Table B, the Kummer tower, the structured P-256 bases, the isogeny-class walks and the lattice reformulations. This chapter and the next two record what happened after the owner's decisions of 29 October split prime-field index calculus into two counted tracks (`research/prime_ic_program_20261029/DECISIONS.md`, decision R1; the split itself is stated in Chapter 40, Table 40.3). The *generic* track holds every oracle built from tables, pair tables, root finding, subtraction or collision search. Its improvements count and ratchet its own baseline, but they carry the label "bounded by $\sqrt{r}$; cannot beat rho". The *algebraic* track, the subject of Chapter 23, holds relations from a solver that receives the factor-base constraint inside its system.

The chapter has one result that matters for the distance to rho and one correction. The residual collectors of §22.3 are the prime-field arms closest to rho that the programme measured: 1.54–1.64× matched rho at $w = 0$ on 36–37-bit curves with a validator report, and 1.3–1.7× on 36–42-bit curves as the producer reports them, with no growth of that distance over the measured sizes. The Curve25519 walk tables of §21.7 are at about the same distance (1.6–1.9×, unverified). They are generic collision searches with rho's exponent, not index calculus with a better one. Their large gains at $w > 0$ were a pricing artefact, withdrawn when the owner chose option B on 1 November (§22.3.1).

Standing of the numbers. Under the rule of 4 November every producer is assumed to be possibly sabotaging the work, and a result is cited by its proof-carrying claim and standing or labelled an unverified assertion. None of the generic-track studies in this chapter has a `pcc` claim record. Four evidence directories have a `VALIDATION.json` with a replay certificate and a validator report: Table B (Chapter 21), the 20-bit $N = 45$ baseline (Chapter 40), the first $m = 4$ cells of §22.1, and the packed membership filter of §22.5. The residual collector study of 1 November has a replay certificate, a validator report and a `VALIDATION.json`, but is not an admitted row (§22.3). Every other figure below is a producer-reported value from a committed record, re-read from the source for this chapter but not independently replayed, and is so labelled where it is used. Costs are in $S = (\text{counted GAE} + w \cdot \text{unpriced native operations})/\sqrt{r}$ at the band weights $w = 0, 0.1, 1$ of §4.3, and two arms are ranked only where their order holds across the band.

The programme of 29 October. The decisions file that created the tracks also set the programme's execution plan (`research/prime_ic_program_20261029/DECISIONS.md`). Its entries are, in brief:

- `D-EKS`: compute for the published algebraic reference arm, a derived estimate of about 49 CPU-hours (range 15–150) on EKS Spot, approved;
- `D2`: the Table B incumbent may be ranked against that reference, which amends R1 so that arms of the two tracks may be ranked on the same instances, in the same unit and across the band, while each track ratchets only its own baseline;
- `D3`: a pinned, vendored msolve counter patch is allowed, its hash bound into the candidate identity;
- `D-PREFETCH`: the hash-table layout measurement of §22.5, approved;
- `D-FD7` and `D-FD8`: two amendments of the prime fall-degree study, the two-cap bracket for $t \geq 12$ with 56/112 GiB jobs and the $m = 4$ extension, adopted;
- `D-MAC`: the fall-degree study's raw runs, which lived only in one worktree on the owner's laptop, are to be recovered rather than declared lost; until then its $t \leq 10$ reproduction stays at median level;
- `D-ROUTE`: a command route for non-ecbench jobs on the lab, approved for an administrator to implement;
- `D-REPLAY`: a second-host replay of the 20-bit holdout session, then a claims record, approved.

The same file records why root finding is generic: at $m = 2$, root-find-and-subtract fixes $x_1 = c$ for each $c \in V$, solves the remaining quadratic $S_3(c, X, x_R)$ by its roots $x(R \mp P_c)$ and tests each root against a hash index of V. It returns the solution set of a Gröbner solve of $\langle S_3, F_V(x_1), F_V(x_2)\rangle$, but its cost never depends on F_V, and it runs identically on a random V.

Table 22.1 lists the prime-field arms of this chapter and of Chapter 21 that have a distance to matched rho, from the closest to the farthest.

Arm	Bits	IC/rho ($w = 0$)	Standing
Residual collector `res1eq` (§22.3)	36–42	1.32–1.72	validator report at 36–37 bits (1.54–1.64); producer at 36–42 bits
Residual collector `res1dp`	36–42	1.92–2.27	as above
Residual collector `res2`	36	2.6	improvement-log row, no validator
Residual collector `res4`	20.6–32	3.78–4.94	producer
$k = 5$ sum table (§22.2)	20–48	5.4–11.0	unqualified
Table B pair table (Chapter 21)	10–24	5.47–13.5	validated
20-bit pair table, $N = 45$ (Chapter 40)	20	7.75	validated
Cofactor-cleared base, $h = 32$ (§22.4)	census	9.80	`invalid_controls`
$k = 4$ sum table (Chapter 40)	20–40	5.3–13.1	no validation
Kummer tower, P-224 shape (§23.1)	20–28	18–78	producer
Kummer tower (Chapter 21)	20–30	29–159	producer
Packed pair table with filter (§22.5)	32	31.47	validated with caveats

Table 22.1: Prime-field index calculus against matched negation rho, greater than 1 meaning rho is cheaper, as each source states it. Ranges are over sizes or curves; each row's band behaviour is in its section. The residual rows are at rho's exponent; every other row's gap grows with size where more than one size was measured.

22.1 What a generic oracle can buy

The arity gate. Let $s = \log_2 p/m$, so that a factor base of the size index calculus needs has $|V| \asymp 2^s$. Suppose a target-independent setup costs 2as and one natural decomposition query costs 2^{es}. Collecting $\Theta(2^s)$ relations costs $2^{as} + 2^{(e+1)s}$ up to polynomial factors, so beating rho's $p^{1/2}$ needs both

$$a < \frac{m}{2} \qquad \text{and} \qquad e < \frac{m}{2} - 1.$$

At $m = 2$ the second condition is $e < 0$, impossible for a nontrivial query, so exact pair enumeration is optimal at that arity up to polynomial factors; the first arity that can discriminate is $m = 3$, with $a < 1.5$ and $e < 0.5$. The derivation is the arity and solver-model correction of 31 October and 1 November (`DEC-20261031-b51282`, `DEC-20261101-26f1fa`; `research/prime_ic_lead_review_20261031/PDP_ARITY_CORRECTION.md`), which found that every accepted theorem and completed measurement of the prime factor-base degree study was a two-summand result. The same pair of gates is the "complete-cost" row of the six-obligation chain of §22.7.

A wrong gate and its replacement. An audit of 4 November of the programme memo *Alternatives to Semaev summation polynomials for prime-field ECDLP index calculus* (`research/semaev_alternatives_audit_20261104/`) found that the memo's oracle gate, "$e < 1$ at $m = 4$", is not implied by its own cost model. A generic 2+2 meet-in-the-middle oracle passes it, with measured $s = 1.93$ on $T = B^s$ and $e = 0.4$ at $B = p^{1/5}$, while costing $p^{2/3}$, more than rho. The audit's replacement is $s < m - 3$, that is $e < (m - 3)/4$. Seven Lean 4 theorems in `research/semaev_alternatives_audit_20261104/lean/` (`Routes.lean`, `Roots.lean`) cover arities 2 and 3, the $m = 4, 5$ gate, the meet-in-the-middle route and a

two-roots bound; the audit's own ledger counts only these theorems as "ruled out". Its measured numbers are a single local run with no replay certificate, and its README says to treat them as unverified until re-run under the validation gate. The memo's "Idea 10" measured one squarefree collision for all $B \le 2^{16}$; Ideas 2 and 3 are $m = 2$ objects and cannot change the exponent.

The $m = 4$ planted solve. The gate correction has a measured counterpart. A direct 2+2 meet-in-the-middle at $m = 4$ over a random base of size $B = \mathrm{round}(p^{1/5})$ (`research/ic_m4_mitm_scaling_20261031/`; stage diagnostic, producer-reported) recovered every planted four-term decomposition with a fitted exponent of 0.412 in $\log_2 p$ ($R^2 = 0.9986$, eight sizes to 52 bits), 0.404 on a three-rung ladder to 60-bit primes and 0.396–0.450 on an arbitrary-precision ladder to 70 bits. The planted solve is not a relation collector. A uniform target decomposes with probability about B^4/p, so one relation costs about $p/B^2 = p^{3/5}$: the 24–40-bit fit of the uniform arm is 0.469, but the local slope from 40 to 44 bits is $\log_2(1.08 \cdot 10^9/2.09 \cdot 10^7)/4 = 1.41$, and the arm was stopped there. The README's own reading is that the $m = 4$ solve "is not an end-to-end sub-1/2 relation collector", which is the audit's point: an oracle exponent below 1/2 in the planted regime says nothing about the cost of a relation.

The first $m = 4$ cost cells. The blind-spot programme of 18 October priced arity $m = 4$ end to end on 16–28-bit primes (`benchmarks/20261018-blindspots-prime-m4-cells/`; validated, with a negation-map rho added to the same binary and the index-calculus rows reproduced exactly in it). Full index calculus costs 8.2–29.5× the negation-map rho at $w = 0.1$ and 6.5–35.6× across the band, dearer than both rhos at every cell and weight, 16 of 16 solves per cell; the arm's own F4 grows as $|\mathcal{F}|^{11.03}$. The verdict is REGRESSES at the primitive level.

Exponents on the bounds frontier. The sealed bounds frontier (`docs/bounds/FRONTIER.md`, toy tier, 17.7–24.0 bits, 96 runs) gives the folded pair-table `ic.pipeline` a fitted exponent of 0.841 $[0.646, 0.953]$ and the subtraction oracle 1.005 $[0.880, 1.077]$, against 0.504 $[0.316, 0.585]$ for interleaved baby-step giant-step. These are fit-version-2 bounds with wide intervals at four sizes; the campaign exponent 0.620 $[0.557, 0.663]$ of Table 36.2 is the better-determined estimate of the same pair table.

Arity sweep. The crypto-autoresearcher index-calculus sweep of 27 October (`research/ic_arity_sweep_20261027/`; 234 rows, every logarithm checked) ran $m = 2, \ldots, 9$ on generated prime-order curves at 12–18 bits with exhaustive and meet-in-the-middle engines. Its unit is the tool's own, S_3 root solves against rho walk additions, which the tool says are different units. The best exhaustive arity is $m = 2$ at 10.6, 16.2, 20.4 and 60.5× rho at 12, 14, 16 and 18 bits; the best meet-in-the-middle arity is $m = 5$ or $m = 3$ at 2.8, 3.3, 5.8 and 4.3×. Odd arities have the lower fitted exponents (0.63 at $m = 3$, 0.60 at $m = 9$) because the default table split leaves one more level on the search side at even m. None of this is in the counted ecbench unit, and none of it is below rho.

22.2 k-sum tables beyond $k = 4$

Chapter 40 gave the $k = 4$ sum table of 28 October: a negation-folded table of every signed 4-sum of a smaller base, S 83.7 → 11.6 at 40 bits, 13.1× `rho-neg` there. Its generic law is $2^{k-1}\binom{A}{k} + c_k A p/(2^k\binom{A}{k})$, with optimum exponent $k/(2k-1)$, falling toward 1/2 (baby-step giant-step) as k grows and never below it. Three studies followed (Table 22.2). All three are registered before data, every run verified and replay-audited, and none has a second-host replay, a `VALIDATION.json` or a `claims/*.json` record; the two later studies call themselves *unqualified*. The benchmark index lists them only as "not admitted to the results table until it has an independent-host replay and its own `VALIDATION.json`".

$k = 5$. A table of every signed 5-sum, with indices packed 12 bits each into one word and the insert order unchanged so that group-operation counts match, lowers S below the $k = 4$ table at 36 and 40 bits at every band weight, the registered success test (`research/prime_ksum_k5_20261028/`; 1,920 runs, all eight test sessions reproduce every replay; $k = 4$ reproduces the earlier per-run counts on 288 of 288 runs). At 40 bits S is 8.51 against 11.56. The fitted exponent over 36–48 bits is 0.547 $[0.543, 0.554]$ against 0.565 $[0.560, 0.571]$, finite-size fits near the generic optima $5/9 \approx 0.556$ and $4/7 \approx 0.571$. The table is still 7.6× baby-step giant-step at 40 bits and 10.6× at 48, so the method sits above its own generic limit at every size.

Study	Bits	Ratio to its incumbent ($w = 0$)	To `rho-neg`	Exponent
$k = 5$ vs $k = 4$	40	0.74 (0.71 at $w = 1$)	9.7×	0.547 [0.543, 0.554]
	48	0.67	11.0×	($k = 4$: 0.565)
$k = 5$ vs mitm incumbent	20	0.701 [0.661, 0.744]	5.4×	
(same instances)	24	0.453 [0.433, 0.475]	6.3×	
	28	0.331 [0.320, 0.344]	7.2×	
	32	0.218 [0.211, 0.226]	7.3×	
$k = 5$ vs re-tuned incumbent	20	0.700 [0.682, 0.718]	6.0×	0.496 [0.487, 0.510]
(16 fresh curves)	24	0.446 [0.432, 0.461]	5.8×	(incumbent:
	28	0.324 [0.318, 0.331]	7.3×	0.635 [0.627, 0.645])
	32	0.217 [0.212, 0.222]	7.3×	

Table 22.2: Prime-field k-sum tables after 28 October (generic track; all unqualified, producer-reported). Ratio: paired ratio of total cost, cluster-bootstrap 95% intervals. "To rho": mean S over matched negation rho at $w = 0$. Exponent: fitted cost exponent in r at $w = 0$; the pooled study's rho-neg fits 0.457 and `bsgs-neg` 0.501 on the same sessions. Sources: `research/prime_ksum_k5_20261028/`, `research/prime_ksum_vs_incumbent_20261029/`, `research/prime_ksum_pooled_20261029/`.

Against the incumbent. The tracks rule requires a same-instance ranking before a generic change replaces the incumbent. On the incumbent's own four holdout curves, 48 targets, method hashes and measurement seed (`research/prime_ksum_vs_incumbent_20261029/`; 3,456 runs) the $k = 5$ table costs less than the negation-folded pair table of `research/prime_base_oracle_tuning_20261029/` at every size and every weight, and the registered primary test holds in all three of its cells (24, 28, 32 bits). The generic baseline on those curves falls by 1.4× at 20 bits and 4.6× at 32. The widening gap is what the two optima predict, 5/9 against about 2/3; one curve per size gives no exponent. The study compared against the recorded incumbent, whose 32-bit base was null by construction, so a second study re-tuned the incumbent on separate train curves by the same rule as the k-sum families and tested both on 16 fresh curves (`research/prime_ksum_pooled_20261029/`; 13,824 runs). The ranking survived: $k = 5$ is below the tuned incumbent on all 16 curves and in the pooled test at all three weights. Its toy-tier exponent of 0.496 is a finite-size fit over 20–32 bits; the generic law does not allow a limit below 1/2. Tuned for $w = 0$, the incumbent is 8–13% cheaper, and $k = 5$ is still 2.2–4.6× cheaper than it. A train-frozen rule chose $k = 5$ at every size; at 20 bits $k = 4$ is cheaper at $w = 0$ and 0.1 and the two are unresolved at $w = 1$.

The precomputation regime. The $k = 4$ table is built before any target arrives, which invites two accounting questions (`research/prime_precompute_regime_20261028/`; class: control and accounting, not an improvement). At equal memory against preprocessing rho (Bernstein and Lange [4]), a known-log variant `kl-ksum4` answers each target with one probe and no matrix. At 40 bits the equal-size preprocessing rho's online S is below `kl-ksum4`'s by 142, 157 and 229× at $w = 0, 0.1, 1$, and its precomputation costs 333–339× more. Preprocessing rho sits on $E \cdot T^2 \approx N$ (6.96 at $w = 0$); the known-log table sits about $N/E \approx 1.4 \cdot 10^5$ above that line, as the Corrigan-Gibbs–Kogan bound $ST^2 \gtrsim N$ [8] allows. Table 22.3 compares the registered predictions with the 40-bit measurement. The amortised crossover between the two falls at 14.3k, 13.1k and 9.5k targets at 40 bits ($w = 0, 0.1, 1$), against a registered prediction of 15k. Against preprocessing rho at equal precomputation there is no positive crossover at 36 and 40 bits. These are multi-target and preprocessing numbers; no single-target cost in the study falls below rho's.

Known-log targets. A target-blind known-log two-sum method of 21 October (`research/ecbench_known_log_targets_20261021/`) recovered public targets without receiving their logarithms. On 16-, 20- and 24-bit fixtures its cold counted cost was below the abscissa pipeline's on both the primary and holdout panels at every band weight; at 24 bits the pipeline/known-log ratio is 15.6–17.0 on the primary panel. Negation rho remained cheaper at every size (S 1.16 against 2.70 at 24 bits), the 12-bit comparison is unresolved inside the band, wall measurements are L0 and the receipts are same-host replays. Table 22.4 gives the primary panel.

Quantity (40 bits, $w = 0$)	Predicted	Measured
`kl-ksum4` set-up S	5.3	5.01
`kl-ksum4` online S	0.10	0.117
equal-size preprocessing rho, set-up S	1,594	1,668
equal-size preprocessing rho, online S	0.0006	0.0008
equal-precomputation preprocessing rho, online S	0.19	0.33
`kl`/equal-size crossover (targets)	15k	14.3k

Table 22.3: Registered predictions against measurement for the known-log $k = 4$ table and preprocessing rho at 40 bits (class: control and accounting). Producer-reported. Source: `research/prime_precompute_regime_20261028/`.

Curve ($\log_2 r$)	Abscissa S	Known-log S	Rho S	Abscissa / known-log, $w = 0, 0.1, 1$
`icv1-fp12-tm53` (11.807)	27.78	25.06	2.92	1.109, 1.049, 0.755
`icv1-fp16-t295` (15.697)	11.10	7.57	1.76	1.467, 1.447, 1.348
`icv1-fp20-t727` (19.101)	12.09	2.96	1.36	4.081, 4.295, 5.336
`icv1-fp24-t1059` (23.974)	42.23	2.70	1.16	15.65, 15.96, 16.98

Table 22.4: Target-blind known-log two-sum recovery against the abscissa pipeline (32 unsigned pairs, folded two-sum meet-in-the-middle) and negation rho, cold, eight targets per curve and five rounds; S at $w = 0$. Producer-reported, same-host replays, no validation. Source: `research/ecbench_known_log_targets_20261021/`.

22.3 Residual collectors

A partial decomposition $R = P_u \pm P_v$ with $P_v \in \mathcal{F}$ leaves a residual $x(R \mp P_v)$ that misses $\mathcal{F}$. The residual collector (`ic_framework::residual`, `targets=residual`) keeps every residual and turns residual collisions into relations; it is the $|D| = r$ limit of the large-prime variation. The programme ran it on prime curves from 26 October to 2 November, under the atlas lead `LEAD-ICP-residual-partial-relations` and hypothesis `H-PFDR-cb9606`. Table 22.5 gives the arms nearest rho. A residual collision is a birthday collision on $\mathbb{Z}/r$ with a base attached; the collector's cost exponent matches rho's, and on a curve-free carrier it costs the same as on the curve. That is the content of `KeyedInvariance.lean` (`research/ic_planner_exhaustive_20261026/lean/`): an algorithm that sees elements only through a keyed labelling cannot use the curve.

The first measurement. On six registered prime curves of 20.6–32 bits, against the `ic plan` recipe and `rho.negation` (`research/ic_prime_residual_lp_20261026/`; 24 of 24 runs verified per arm and curve), the collector with a four-point base was 2.3–8.4× cheaper than the plan in counted S (`res4`/plan 0.119–0.426 at $w = 0$, every interval below 1). As published, the same arm cost 1.33–3.89× the plan at $w = 1$, so the order reversed inside the band at every size and the two were not ranked. The least-squares exponents over six sizes are 0.470 for `res4` and 0.480 for rho, against 0.634 for the plan; at any fixed base the collector grows as about $r^{1/2}$. On the curve-free carrier ($\mathbb{Z}/r$ with a random-bijection key, same seeds) all nine intervals contain 1. The collector buys its group-operation savings with a table probe and an insert on every step, so the native work it adds is real work; what option B later removed from the $w > 0$ figures was a byte count, not those probes (§22.3.1). The k-sum study derived that large primes do not help the *two-summand pair table*: with two summands a large-prime table of M entries has the same law $M + cp/\sqrt{M}$ (`research/prime_ksum_table_20261028/`). The residual collector escapes that law because it gives up the pair table and becomes a collision search.

The planner re-test. The study extrapolated a crossing of the plan near 35 bits. The planner re-test of the same day (`research/ic_planner_exhaustive_20261026/`; 14 new sessions, 3,328 records, all verified; 2,428 band verdicts restated in integers and checked by Lean's `decide`) measured it. At 36.0 and 36.25 bits `res2` and `res4` are cheaper than the plan at every band weight with intervals below 1 (`res4`/plan 0.888 $[0.817, 0.965]$ and 0.865 $[0.802, 0.933]$ at $w = 1$, the risky registered prediction R3). Over all ten sizes the $w = 1$ ratio falls steadily, from 4.92 at 15.7 bits to 0.865 at 36.25. The curve-free null held (15 of 16 intervals contain 1) and the fitted slope over ten sizes is 0.460 for `res4` against 0.619 for the plan. At 36 bits the collector moved the framework's cheapest one-target arm from 41–47× rho

Arm	Curve	$w = 0$	$w = 1$	Source and standing
`res4` (base 4)	20.6–32 bits	3.78–4.94	3.6–4.8*	residual_lp; producer
`res2`	p36 (36.0 bits)	2.6	2.6*	planner_exhaustive; log row 2.59×
`res2`	p37 (36.25 bits)	2.7	2.7*	planner_exhaustive; producer
`res4f` (fused)	p36 / p37	2.39 / 2.51	—	residual_fused; producer
`res1eq`	p36	1.54	2.66	residual_dp_eq; validator report
`res1eq`	p37	1.64	2.83	residual_dp_eq; validator report
`res1dp`	p36 / p37	2.13 / 2.03	2.22 / 2.07	residual_dp_eq; validator report
`res1eq`	p36 / p40 / p42	1.32 / 1.72 / 1.39	2.27 / 2.99 / 2.42	residual_scale; producer
`res1dp`	p36 / p40 / p42	2.15 / 2.27 / 1.92	2.25 / 2.28 / 1.93	residual_scale; producer
`res1eqc` (chained)	p36	1.591	2.73	res1eq_chain; producer

Table 22.5: Residual collectors: distance to matched negation rho (cost over rho, greater than 1 means rho is cheaper). * With the table-byte counter excluded, as option B prices the band (§22.3.1); the as-published $w = 1$ values were 41–58× (`res4`) and 23–24× (`res2`). The `eq` and `dp` rows are nearly unchanged by option B. Sources under `research/`: `ic_prime_residual_lp_20261026`, `ic_planner_exhaustive_20261026`, `ic_residual_fused_20261031`, `ic_residual_dp_eq_20261101`, `ic_residual_scale_20261101`, `ic_res1eq_chain_20261101`; validator report `benchmarks/20261101-ic-residual-dp-eq/validator-report.json`.

in counted S to 2.6–2.7×. The leaderboard improvement log carries this as "Residual collector res2 vs rho.negation, fp36: 2.59×" (`docs/ic/LEADERBOARD.md`), the $w = 0$ value; the study's own decision notes that no validator has run on it, so it is an unvalidated row.

Fused steps, distinguished points and the equivariant walk. The fused residual step of 31 October (`research/ic_residual_fused_20261031/`; four sessions of 512 records, all verified, A/A 1.0000) cut group additions per step from 2.005 to 1.004 and gave fused/baseline 0.56–0.92 at $w = 0$ on all 12 (curve, base) cells. The fused walk needs 1.1–1.3× more steps per harvested relation, so priced native work ate the saving: the order is mixed at $w = 0.1$ and reversed at $w = 1$, and the preregistered rule C1 was not met. Distinguished points and an equivariant walk followed on 1 November (`research/ic_residual_dp_eq_20261101/`; two sessions of 704 records, all verified, A/A 1.0000; it supersedes the $w > 0$ figures of `research/ic_residual_dp_20261101/`, whose per-step distinguished-point test was unpriced). The best arms are `res1eq` at 1.54, 1.75 and 2.66× matched rho on p36 ($w = 0, 0.1, 1$) and 1.64, 1.86, 2.83 on p37, and `res1dp` at 2.13, 2.14, 2.22 and 2.03, 2.04, 2.07. Between `eq` and `dp` the order is unresolved inside the band: `eq` is lower at $w = 0$, `dp` at $w = 1$.

Scale. On p36, p40 and p42 (`research/ic_residual_scale_20261101/`; three sessions of 288 records, 863 verified, one baseline run exhausted at 10^8 trials) the distance of `res1eq` to matched rho is 1.32, 1.72 and 1.39 at $w = 0$ and 2.27, 2.99, 2.42 at $w = 1$; of `res1dp` 2.15, 2.27, 1.92 and 2.25, 2.28, 1.93. There is no growth of the distance with r over this range, as a method with rho's exponent predicts. Chained restarts, which restart each walk at the previous start plus one random jump instead of two scalar multiplications, were null: 0.994 $[0.953, 1.037]$ at $w = 0$ (`research/ic_res1eq_chain_20261101/`).

Batches and many targets. A residual batch of L targets against pooled multi-target rho at p36 (`research/ic_residual_batch_ecbench_20261102/`; 320 of 320 verified, local L0, L2 replay owed) holds the frozen parity rule at $L = 64$, 1.0077 $[0.9811, 1.0350]$, and fails it at $L = 16$, 1.0465 $[0.9886, 1.1078]$. Per target both batch methods are 3.3–3.4× ($L = 16$) and 6.6–6.7× ($L = 64$) cheaper than single-target rho; the matched reference for a batch is multi-target rho, and against it the collector is at parity, not ahead. An exploratory shared-table collection on a 20-bit curve (`research/ic_largeprime_multitarget_20261102/`; not preregistered) gave a per-target cost at $T = 64$ targets 4.4× below $T = 1$. That is multi-target rho reproduced. The same note sets aside triple and quadruple large primes as a generic k-sum search already measured in §22.2.

22.3.1 Correction: the byte counter

The independent validator of the 1 November study (`benchmarks/20261101-ic-residual-dp-eq/validator-report.json`, finding F1) found that the band as built priced `residual_table_bytes_uncharged`, a byte count of the residual table, as native operations at $w > 0$. The audit of the same day (`research/ic_byte_counter_pricing_20261101/`) confirmed it: the counter is emitted in `ic_framework/residual.rs`, every `*_uncharged` counter is appended to the unpriced list, and `priced_gae(w)` charges each listed unit w GAE, so each byte cost w GAE. The analogous `native_table_bytes_uncharged` is classed as shape, not work. The counter was 83–99% of the residual arms' priced S at $w = 1$ and 40–98% at $w = 0.1$, and it is the whole reason their $w > 0$ distances to rho read 17–58× instead of 2.6–5×. On the first residual study, `res4`/rho at $w = 1$ falls from 41.2–57.6 as published to 3.6–4.8 with the counter excluded; on the planner re-test, `res2`/rho at p36 falls from 23.3 to 2.6.

The owner chose option B on 1 November: the band prices operations, not bytes. `band.rs` now lists the byte counters as `NON_OPERATION_COUNTERS` and does not price them; memory is reported on its own axis. No session record was edited; the four affected studies carry a dated note and the re-run comparisons under `analysis/exbytes/`. What changed:

- *The large $w > 0$ gains are withdrawn.* They were the byte price. On the planner re-test, 36 of 136 weighted comparisons go from "unresolved inside the band" to resolved, 12 stay unresolved and 88 keep their direction; the $w > 0$ distances are flat in w.
- *The rule C1 of the scale study is met in 2 of 8 cells, not 6 of 8.* The equivariant arm fails all four cells at $w = 1$, where its interval reaches or crosses 1 (p40 `res1eq` 1.181 $[0.859, 1.625]$).
- *An unreconciled tally.* The 1 November errata give the 12-cell C1 tally as 9 of 12; the study's own `analysis/summary.json` gives 8 of 12 as built. Nobody has reconciled the ninth cell.
- *What stands.* $w = 0$ results, the leaderboard's 2.59×, every verification, audit and A/A result, and the absence of growth of the distance with r.

A second byte counter, `factor_base_vec_bytes_floor_uncharged`, is priced the same way in the Koblitz sessions, at 0.4–12% of S at $w = 1$; no Koblitz verdict changes direction. The 1 November evidence directory has a replay certificate, the validator report and a `VALIDATION.json`, but is not an admitted row: its raw records live in the artifact store, which the CI gate cannot read (`benchmarks/20261101-ic-residual-dp-eq/README.md`).

What the collectors are. The residual collectors are the nearest the programme came to rho on a prime field: 1.3–1.7× at $w = 0$ and 2.2–3.0× at $w = 1$ for `res1eq`, and about 2× at every weight for `res1dp`. They are measured constant improvements of the generic track over the pair-table incumbent, at rho's exponent. By the curve-free null and `KeyedInvariance.lean` they are birthday searches on $\mathbb{Z}/r$, and the Shoup bound applies to them; the remaining factor is a constant. Whether the parity bound of §21.7 (at least 1.5× rho with one unknown base logarithm) applies to them is open: the cells with a validator report (1.54 and 1.64) are above it, but the producer-reported 1.32 (p36) and 1.39 (p42) of the scale study are below it, and no source states the collectors' number of unknown base logarithms or reconciles the two.

22.4 Cofactor-cleared bases

On prime curves with cofactor h, a small-abscissa base wastes a fraction $1 - 1/h$ of its trials on points outside the prime subgroup. A preregistered A/B test of 3 November on the census V2 prime curves with $h = 32$ (`research/ic_cofactor_clear_ab_20261103/`; 448 matched pairs on 112 workloads per comparison, cloud VM graded L1, counted metrics only) compared the raw base with `prime-projected-abscissa`, which clears the cofactor. The pass line was met: the S gain of clearing is $G_0 = 6.180$ $[5.915, 6.456]$, $G_{0.1} = 6.692$ and $G_1 = 9.087$, and the $h = 1$ negative control gives 0.989. The positive control, rho-plain over rho-neg with a pre-registered lower limit above 1.2, failed on the V2 session at 1.227 $[1.149, 1.311]$. By the frozen decision rule the verdict is `invalid_controls`, and the producer declined to reinterpret it. Descriptively, trials per relation fall from 5,433 to 162 (33.6× against $h = 32$), and the cleared base is 9.80× $[9.33, 10.30]$ matched rho at $w = 0$ against 60.6× for the raw base. The study is therefore not cited as a win. The same lever on random binary curves gave 1.29, 1.58 and 3.36× at $h = 2, 2, 6$ (atlas `LEAD-ICB-cofactor-projected-ladder-base`), and the Curve25519 miniatures of §21.7 use the projected base as their baseline.

22.5 Engineering record

Base size. The 20-bit baseline at $N = 45$ (Chapter 40) was tested against $N = 64$ on three new curves, 576 matched pairs, on an AMD L2 host (`research/p20_large_n_20261029/`; 4,704 records verified, 4,032 measured replays identical). The ratio S_{64}/S_{45} is 1.001 $[0.979, 1.022]$ at $w = 0$, 0.967 at $w = 0.1$ and 0.834 at $w = 1$: unresolved inside the band, so $N = 45$ stays. It is not a relabelling, since the total does not fall at $w = 0$ and rise at $w = 1$. The run also replicated the 32-to-45 step on new curves: $N = 45$ costs 0.842, 0.822 and 0.751 of $N = 32$, within a few percent of the original 0.818, 0.798, 0.726.

Where the cost goes. A phase profile on the deployed lab image (`research/ic_profile_20261029/`; 20-bit curve, $N = 45$, admitted at L2, 144 of 144 executions) puts 71.8% of priced GAE in relation collection, 27.8% in oracle setup and nothing in the factor base, with $S_{\mathrm{IC}} = 9.93$ against $S_\rho = 1.08$. Amortising setup over P targets gives at most a 28% gain ($9.93 \to 7.17$ as $P \to \infty$) and leaves the pipeline at 6.7× matched rho; the relation phase alone is 4.7× rho. To reach parity the relation phase would have to fall by about 7×, which no setup lever touches. `target_guard_probes` is uncharged in that record, so 9.93 is a lower bound.

Table build and membership filter. Three builders of the same negation-folded pair table (scalar hash insert, one batched inversion per row, and the batched arithmetic inserted into cache-sized packed regions) count the same additions and recover the same logarithms on a 32-bit curve, 456 of 456 executions identical, so S is unchanged by construction; the wall comparison is L1 and descriptive (`research/ic_oracle_setup_blocked_20261020/`; leaderboard component section). A one-bit membership filter on the packed table, frozen in the backlog item `XP-20261029-packed-filter`, is the only validated component of the generic track (`benchmarks/20261029-ic-packed-filter/VALIDATION.json`; validated with caveats, single session, not replicated; EKS L2 on an AMD EPYC 9R45). At 32 bits it lowers whole-solve wall per charged GAE to 0.9601 $[0.9569, 0.9633]$, −3.99%, at identical charged GAE. In S it decides nothing: 1.0, 1.0505 and 1.3586 at $w = 0, 0.1, 1$, so the filtered table is costlier at $w > 0$. At 24 bits it reads a regression (1.0032) and at 28 bits it is undecidable. The distance remaining to rho-neg is 31.47–31.63×. The validator re-priced the bit operations with its own code rather than re-running `ecbench work`. The filter is a component speed-up outside tables A–C of the leaderboard and moves no ranked row.

A fixed-window generator. In crypto-autoresearcher, a fixed-window generator of the $aP + bQ$ targets cut complete-solve group operations from 200,856 to 50,490 summed over 54 toy solves per arm, with a paired full-solve reference/candidate ratio of 1.161 $[1.127, 1.198]$ and an A/A of 1.044 (`crypto-autoresearcher:ledger/evidence/EV-ICPERF-92c130.yaml`; nine public curves at 12–16 bits, one machine and one session). Five of the 54 pairs are slower, and the record states that the result does not replicate across machines, runtimes or sessions; the historical generator stays the default.

Curve generation. Every `prime_search` workload had been built by counting points with a Legendre symbol at every abscissa, about p symbols per candidate: 11 s per 24-bit and 218 s per 26-bit curve, paid again by every plan, run, verify and CI replay. Stepping one point across the Hasse interval, about $4\sqrt{p}$ affine additions, answers the only question the search asks, whether $\#E$ is prime; it draws no randomness, so it returns the same curves. Planning time fell by 5× at 16 bits, 239× at 24 and 1,592× at 26 (`research/prime_order_search_20261017/`). This is tooling; it changes no counted cost.

Line layout. A cache-line layout tournament of the pair table (`research/ic_line_layout_tournament_20261031/`) has not produced a measurement. The two preregistered EKS sessions of version 1 passed their contract and build jobs and terminated in the experiment job with remote-result exit code 23 (`TERMINAL-OBSERVATION-20261101.md`); the version 5 qualification job of 6 November stopped at its qualification under stopping rule 1, with no scientific session and no seed consumed. A version 6 redesign was approved on 6 November (`DEC-20261106-662e33`) conditional on a pre-freeze check, and nothing has been dispatched under it. The outcome is inconclusive, and the failure is tracked as `problems/ic-line-layout-eks-failure-cause.md`.

Planner weight rules. Chapter 21 reports the planner's validation on six curves. The re-test of 26 October (`research/ic_planner_exhaustive_20261026/`) held out three weight rules and withdrew all

three in `ic-planner/4`: the prime rule W1 ($1.41 \cdot N_0$ at $w = 1$) held its operative half on 4 of 4 curves but failed its $w = 0$ control half, because at 15.7 and 17.7 bits $1.41N_0$ is cheaper even at $w = 0$; the binary rule W2 failed at $w = 1$; the Koblitz rule W3 was dearer at every weight on every held-out curve at 36.6–47.2 bits (1.06–1.17× at $w = 1$). `ic plan -native-weight` now keeps the counted optimum at every weight.

22.6 Structured bases in measurement

The conservation results of Chapter 5 (Proposition 5.5) and Chapter 26 (Lemma 26.1) say that a factor base chosen by a coordinate rule cannot change the number of relations in expectation. Three prime-field measurements of October tested the finite-size version.

A four-family comparison. Multiplicative-coset, low-x, low-digit-weight and random bases were compared on 48 admitted prime-order curves over 24 fields at 16, 20 and 24 bits, holding the target curve fixed and matching the number of ± point classes (`research/factor_base_evaluation_20261015/`; 26,497 point-relation identities and 402 matrix ranks checked in Sage). In independent rows per CPU second at 24 bits, low-x gave 1,352, coset 1,167, random 1,144 and low digit weight 859. Coset over random is 1.020× with a descriptive 95% interval of 0.959–1.083, so no coset advantage is established. A quadratic-algebra norm solver produced 90 rows in 229.902 CPU seconds on fresh 24-bit inputs, against 115 rows in 0.268 seconds for the exact join; the norm solver is a Sage prototype and the join native C.

Pair-sum coincidences. A structured base might win if its pair-sum coincidence excess κ, the number of four-term relations $P + Q = R + S$ among base points over that of matched random symmetric bases, grew with the base size N. A registered break-even ladder of 29 October (`research/prime_coincidence_harvest_20261029/`) measured window, rank-2 and rank-3 label-progression bases on prime-order curves of 16 to 32 bits at two base-size exponents. The slopes $d \ln \kappa / d \ln N$ have 95% intervals inside $[-0.006, +0.008]$ for all six combinations, 18–41× below the break-even exponents 1/3, 1/5 and 1/7, and every rung's κ lies inside $[0.92, 1.08]$. The $\mathbb{Z}/N$ positive control recovers slopes 0.699 and 0.649 (exact 0.70 and 0.65). A growing harvest would need $\kappa > 1.27\, cN^{1/(2h-1)}$, that is 1,874$c$, 101$c$ and 29c at 32 bits for $h = 2, 3, 4$, against a measured κ of at most 1.07. The ladder is registered as `exhausted_at_scale` at 32 bits; it is not a proof.

Hashed and windowed bases. Two of the skeptic-reviewed leads of §23.6 add the second moment: hashed bases (affine, dense, sparse of weight at least 5) match a size-matched random null to 1.000±0.0003 in paired ratio at $p \approx 2^{14}$–2^{22}, and a shared negation-closed x-window has a yield deficit of 0.713, 0.833 and 0.916 at $m = 4$, an instance of the symmetric-base mechanism already recorded in the knowledge base. Base choice moves constants, in the direction the conservation results predict, and the structured P-256 bases of Chapter 21 are the same finding at a larger modulus.

22.7 The obligation chain

The weak-assumption programme of 2 November (`research/ic_weak_assumptions_20261102/`; reanalysis of committed evidence, no new run) restates an index-calculus security argument as a serial chain of six obligations, each of which a complexity claim must discharge:

- solution geometry: dimension and radicality of the full system, rational lifts and exceptional charts;
- solver growth: matrix dimensions, nonzeros, arithmetic and memory, with timeouts and out-of-memory exits retained;
- yield and variance: rational decompositions per natural target, including zero mass, multiplicity and solver failures;
- relation rank: rank growth, dead columns and full-rank probability, not raw row counts;
- a non-generic interface: a coordinate predicate whose useful candidate set no generic table reproduces at the same cost;
- end-to-end composition: one parameter family charging setup, failed queries, collection, verification, linear algebra and memory, with the fixed-arity gates $a < m/2$ and $e < m/2 - 1$ of §22.1.

The machine-readable chain (`obligations.json`, checked by `audit.py`) covers naive Semaev, generic pair tables, the Kummer tower, the proposed arity-three hybrid solver, Wagner filters, the hashed-base

tensor route of §24.1 and xedni. Three of its reanalyses bear on the generic track. The prime $m = 2$ yield sweep tracks the size-driven $2B^2/r$ model at toy scale, with observed/model hit-rate ratios 0.764–1.108 in nine cells; exact double counting fixes a mean decomposition count, not the probability of at least one, and the adaptively stopped collector cannot estimate the target-wise variance. The same records hold 842 rows at rank 842, a correctness control and not a rank theorem, since each episode stops early and meet-in-the-middle and subtraction expose the same relation set. Its observation about degree alone is in §23.1.

What the generic track settles. On every prime curve measured the generic track is behind rho with negation, and the best arm, the equivariant residual collector, is behind it by a constant at rho's exponent. The k-sum tables improve the incumbent by 1.4–4.6× at 20–32 bits and stay 5.4–11.0× rho, above baby-step giant-step. The pair-table constants (base size, filter, builders, profile) are measured and small. The obstruction is a theorem, not a measurement: any decomposition oracle built from enumeration or table lookup is generic, so Shoup's $T \gtrsim \sqrt{N}$ and, with preprocessing, $ST^2 \gtrsim N$ bound it (atlas `LEAD-ICP-generic-table-decomposition`, closed_obstructed). Every improvement here is a success of the generic track under the iterative criterion and a distance remaining of at least 1.3×, and none of it is evidence for the algebraic track.

23 Prime fields: algebraic presentations

The algebraic track holds relations produced by a solver that receives the factor-base constraint inside its system. Its prime-field baseline was empty when the track was created on 29 October: the owner accepted the small-abscissa base with membership polynomial $F_V = \prod_{c \in V}(x - c)$ as an algebraically defined base, which makes the naive Semaev system with msolve the counted algebraic reference, but noted that $\deg F_V = |V|$ gives a Gröbner solver no low-degree structure (`research/prime_ic_program_20261029/DECISIONS.md`, D1). The reference arm was approved for EKS Spot (decision D-EKS), and the decisions file lists its implementation and execution as not yet done. The end-to-end prime-field costs nearest to that track in the treatise are the Kummer-tower rows of Chapter 21 (28–159× rho), which Chapter 40 places on the algebraic track although their oracles are root finding and tables, so that the tower base is the only algebraic ingredient; the P-224-shaped rows and the derived F4 cost of §23.1; and the F4 rows of the Curve25519 miniatures (§21.7).

This chapter collects seven lines of work, none of which produced a priced pipeline: the Kummer-tower follow-ups (§23.1), the partitioned summation polynomials of Amadori, Pintore and Sala (§23.2), the digit presentation (§23.3), split-graph and secant presentations (§23.4), the divisor-norm direct index calculus (§23.5), the SAT and CDCL(T) lane (§23.6) and the small-root reach theorems (§23.7). Each is a degree study, a correctness implementation or a theorem. The first fall degree and its relatives are defined in Chapter 6, and that chapter's §6.5 gives the Kummer-tower degree theorem (Theorem 6.14). None of the studies below has a verified `pcc` claim; where a claim id exists it is named with its standing. The fixed-arity gate of §22.1, $a < m/2$ and $e < m/2 - 1$, applies to every route here.

23.1 Kummer-tower follow-ups

The symmetric tower. Petit, Kosters and Messeng's [33] base $V = g\mu_{2^t}$ makes membership of distinct summands $x_i = gw_i$ equivalent to $Q(X) = \prod(X - w_i)$ dividing $X^{2^t} - 1$, that is $X^{2^t} \equiv 1 \pmod{Q}$. The symmetric tower of 26 October (`research/pkm_symmetric_tower_20261026/`; stage diagnostic, no S) takes the coefficients e_k of Q and of the remainders $R_j \equiv X^{2^j} \bmod Q$ as unknowns, with equations $R_j = R_{j-1}^2 \bmod Q$, $R_{t-1}^2 \equiv 1$ and S_{m+1} rewritten in the e_k. It sees unordered sets of distinct points, the same relation with $m!$ fewer solutions. Table 23.1 gives the cells. At $m = 3$ the symmetric tower is one degree lower and its widest matrix about 10× narrower, about 50× faster per system at $N = 12$ (14.8 s against 0.3 s), but its degree still climbs: $4, 5, 5, 6$ at $N = 6, 9, 12, 15$. The follow-up of the same day (`research/pkm_symmetric_lead1_20261026/`) added $N = 18$ at $m = 3$, where $D = 6$ again, so the sequence over $N = 6$–18 reads $4, 5, 5, 6, 6$: a staircase that steps up every second cell. The $N = 18$ cell cannot tell "D bounded at 6" from "D rises by one every 6 in N"; the discriminating cell is $N = 21$, which needs more than the 13 GB that host had. From $N = 15$ to 18 the widest matrix grows 3.1× and the time 39×. At $m = 4$, S_5 was written in $e_1, \ldots, e_4$ by interpolation and passed its planted check. At $N = 12$ the symmetric system runs in 8–9 ms against 66–70 s for the plain chain, with 5.9× narrower matrices; its $D = 8$ at $N = 12$ and 16 is the degree of $S_5(e)$, its input, so D cannot rank the arms at $m = 4$, and $N = 20$ aborted under a 6 GB cap.

What the tower instrument does not check. The 29 October audit (`research/prime_one_shot_20261029/`) read the source of `examples/pkm_symmetric_tower.rs` and found that it draws a, b without checking the discriminant or the group order, includes every coset abscissa in the base without checking that it lifts to $E(\mathbb{F}_p)$, and takes x_R as a uniform field element rather than the abscissa of a subgroup point. Its planted checks evaluate assignments built from S_3 roots, not signed group decompositions. The symmetric rows are therefore algebraic instrument diagnostics, not a validated natural prime-order benchmark, and the symmetric arm excludes repeated abscissae that the plain arm permits, so the comparison is not clean evidence for coefficient sharing alone. The same audit recovered a lower bound the earlier summaries had dropped: at $m = 4$, $N = 20$ both raw msolve logs complete three productive degree-8 rounds (adding 1,241, 2,507 and 3,578 basis elements) before exit 139, so $D \geq 8$ there for that solver and presentation, whatever the final verdict.

The audit sorted the tower's failures and nulls into kinds that need different treatment, and its sorting applies beyond the tower:

m	N	plain D	symmetric D	widest matrix, plain / sym	time per system, plain / sym
2	12	5	4	2,451 / 627	0.36 s / 0.09 s
2	16	5	5	9,119 / 6,281	34 s / 24 s
2	18	5	5	14,922 / 12,214	280 s / 176 s
3	9	6	5	3,285 / 248	0.6 s / 0.02 s
3	12	6	5	12,314 / 1,224	14.8 s / 0.3 s
3	15	≥ 6	6	$\geq$ 97,456 / 10,532	$>$ 900 s / 43 s
3	18	—	6	— / 33,153	— / 1,941 s
4	12	7	8	21,619 / 3,663	66–70 s / 8–9 ms
4	16	—	8	— / 13,926	— / 11.8 s

Table 23.1: Plain and symmetric Kummer towers, F4 (`f4_fp`, grevlex) on $p = 786433$, random targets; maximum over two targets. Stage diagnostics with no priced pipeline. The recorded tower-engine D of the plain tower at $m = 3$, $N = 15$ is 7 (§21.2). Sources: `research/pkm_symmetric_tower_20261026/`, `research/pkm_symmetric_lead1_20261026/`.

- *Busy hosts.* The original PKM solver study ran at load 22–25 on four cores; a single-threaded $m = 3$ job received about 19% of one core, so the first degree-7 steps at $N = 18$ and 21 took about 2,024 and 2,041 s wall against 384 and 396 s CPU. The wall-censored inputs justify a controlled retry, not a reading of the exponent. The same study planned a planted-solution control but retained only sixteen refutations with `-planted 0`, so successful root extraction was not validated there.
- *Resource censoring.* At $m = 3$, $N = 21$ the native engine stopped before productive degree-8 reduction because a dense block would need 79,330,205,871 entries, about 317 GB at four bytes, which would not fit the planned 192 GB host either. That is a matrix-memory obstruction, not a contention artefact, and it is why §21.2 reads $D \geq 8$ as inconclusive.
- *Missing raw runs.* The last-fall study's raw runs, table and draft results were never pushed and live only in the owner's laptop worktree (`research/prime_fall_degree_20261027/RESTART.md`), so its quoted medians are session-reported and not independently recoverable. A lost file is not a negative experiment; decision `D-MAC` asks for recovery.
- *Insensitive instruments.* An ecbench Lattès and isogeny study used a degree-s top product polynomial, $s = 16, 24, 32$, that set the solving degree by itself; its null at $m = 2$, $p \leq 1031$ is not a theorem about implicit tower presentations, and rerunning the same design would add little.

Degree alone is not a cost certificate. The weak-assumption reanalysis (`research/ic_weak_assumptions_20261102/`) extracted the committed PKM $m = 4$ random-target rows at $N = 8, 12, 16$. Their F4 solving degrees are $6, 7, 7$, while the maximum column count grows 1,049 $\to$ 9,452 $\to$ 76,837 and multiply-adds $4.84 \cdot 10^7 \to 8.55 \cdot 10^{10} \to 2.70 \cdot 10^{13}$: from $N = 8$ to 16 the columns grow 73.25-fold and the work 558,304-fold, with the last two degrees equal. Three cells are not an asymptotic fit, but they are a direct counterexample to reading a stable degree as a bounded cost. All six selected $m = 4$ natural rows are also inconsistent (their ideal is the unit ideal), so a zero-dimensional flag on a random target does not establish the geometry of a successful decomposition.

P-224's 2-adic shape. A study of 22 October priced the tower at P-224-shaped Solinas primes $p = 2^n - 2^k + 1$ of 20–61 bits and at the P-224 prime itself (`research/pkm_p224_kummer_shape_20261022/`; frozen before any measured run). On the solver axis, D equals the recorded D in all 89 valued tower cells at $m = 2, 3, 4$, the width and step count of every system match, and the shape-matched null gives the same D in all 65 paired cells, although k/n ran from 0.42 to 0.70 and the odd part of $p - 1$ from 6 to 128 bits. The one cell past the recorded frontier, $m = 2$, $N = 24$, was killed at 6.0 GB, and a capped rerun gives only $D \geq 6$. On price, four Solinas curves of 20–28 bits cost 18–78× rho with negation at $w = 0$ and 19.9–134× at $w = 1$, every one of 1,152 runs verified and 864 replays reproduced; at equal t the Solinas curves cost 2–32% more absolute GAE than the earlier Proth curves. The derived end-to-end F4 cost at the one cell matching a curve (20 bits, $m = 2$, $N = 20$) is at least $2.4 \cdot 10^9$ in S, about $1.9 \cdot 10^9\times$ matched rho at $w = 0$. The verdict is inconclusive on the solver axis and negative on price.

Radix gradings. The experiment `EXP-PKM-RADIX-001` (`research/pkm_radix_grading_20261104/REPORT.md`; frozen protocol, two disclosed amendments; every number is a counted metric marked `[pcc:unverified]` in the report) regraded the tower with radix 4 and radix 8 and measured refutation

Cell	radix 2	radix 4	radix 8	D (2/4/8)	widest matrix (2/4/8)
$m = 2$, $t = 5$ ($N = 10$)	$5.43 \cdot 10^6$	$2.20 \cdot 10^6$	$1.64 \cdot 10^6$	4 / 6 / 10	337 / 252 / 260
$m = 2$, $t = 10$ ($N = 20$)	$4.82 \cdot 10^{11}$	$2.59 \cdot 10^{11}$	$8.35 \cdot 10^{10}$	6 / 8 / 13	33,446 / 28,225 / 17,101
$m = 3$, $t = 4$ ($N = 12$)	$1.06 \cdot 10^{10}$	$9.59 \cdot 10^9$	$5.98 \cdot 10^9$	6 / 9 / 12	5,019 / 6,509 / 4,063
$m = 3$, $t = 5$ ($N = 15$)	$3.19 \cdot 10^{12}$	$1.10 \cdot 10^{12}$	$7.47 \cdot 10^{11}$	7 / 9 / 12	31,397 / 28,055 / 20,284

Table 23.2: Refutation multiply-adds of the Kummer tower in three gradings, one value per cell (the two random draws agree to a few parts in 10^7); D is measured in each grading's own degree. Counted metrics, `[pcc:unverified]`. Source: `research/pkm_radix_grading_20261104/REPORT.md`.

multiply-adds. Its frozen verdicts are `gain_constant` at $m = 2$ radix 8 ($r_k = 0.173$ at $t = 10$), $m = 3$ radix 4 (0.344 at $t = 5$) and $m = 3$ radix 8 (0.234), and `no_material_effect` at $m = 2$ radix 4 (0.538). No cell is `gain_growing`. The shape-matched null gives the same verdicts with ratios equal to three digits, so the gain belongs to the support of S_3 on the tower, not to the curve; $p = 786433$ replicates the $t = 10$ bands. The author's registered prediction of no material effect anywhere was falsified in three of four cells. Table 23.2 gives the end cells.

The radices raise the degree in their own grading (at $m = 2$, $t = 10$: 6, 8, 13) while lowering the work, which is the point of the previous paragraph seen from the other side.

Isogeny kernels against μ_k. A component study of 2 November tested a user's toy observation that an isogeny-kernel base, $V = x(\langle G \rangle) \subset E'(\mathbb{F}_p)$ presented as a chain of Vélu 2-isogeny x-maps, behaves differently from Petit's μ_k chain of squarings at $m = 2$ (`research/isogeny_vs_mu_factor_base_20261102/`; toy primes $p \leq 40961$, frozen protocols, 6,144 and 11,532 systems, no gate failures, local host without an isolation tier, counted units only). At equal $|V|$ the isogeny base matches a random base in yield in 15 of 15 cells (median difference -0.020 to $+0.029$). Its extra Gröbner cost comes from the presentation, not the set: writing μ_k in coordinates whose steps look like Vélu steps costs the same as the isogeny chain to within 3%. A search over coordinates cut the lex-order gap to μ_k from 6.4–11.7× to 1.9–2.6×, but found no grevlex improvement (0 of 16 cells), and none of 11,811 valid Vélu representations reaches the density of the μ_k chain. There is no rho row; the distance is stated against μ_k.

The $t = 14$ cell. The decisive cell of the prime fall-degree study (Kummer, $m = 2$, $p = 2013265921$, $t = 14$) is half done (`research/prime_fall_degree_t14_20261101/`). At cap 5 neither random draw refutes and neither is decomposable, so $d_F \geq 6$. The cap-6 cells need a 112 GiB, 86,400 s native job; no route in the lab carried such a job on 1 November, and the cells are owed. The owner approved a command route for non-ecbench jobs (`D-ROUTE`) that an administrator has to implement.

Open. Lead 1 of the tower, whether D stays bounded as N grows, is open in every presentation tried: plain, torus, Dickson, symmetric, radix-graded and P-224-shaped. In each family D either grows with N or is pinned by the input degree, and the shape-matched null matches. The atlas lead `LEAD-ICP-pkm-tower-algebraic-oracle` is `open_blocked` on memory: the cells that would discriminate ($m = 3$, $N = 21$ in the symmetric tower; $t = 14$ at cap 6) need hosts the programme did not have. Chapter 6 gives the proved part.

23.2 Partitioned summation polynomials

Amadori, Pintore and Sala [1] partition the factor base and attach labels to the summands so that one relation comes from a smaller system. The programme implemented the paper's single-relation route and five extensions between 26 October and 5 November (`research/aps_extensions_20261026/`, `research/aps_followup_20261028/`, `research/aps_independent_replay_20261029/`, `research/aps_measurement_20261029/`, `research/aps_export_20261104/`, `research/aps_leads_20261105/`). The native module builds labelled points $R = uG + vQ$ without giving the solver the planted scalar, imposes the partition membership constraints, solves with native F4 and substitution, decodes and sign-checks the summands by curve addition, and recovers $d = -A/B$ from accumulated relations $AG + BQ = O$. The extensions are fixed coefficient labels, unordered divisor and two-pair encodings, automorphism orbits, covering partitions, and expanded, split and hybrid solving; the follow-up added an early stop on the first informative relation. In detail:

- *Informative labels.* For odd arity, points $uG + Q$ have an odd signed coefficient sum, which cannot vanish modulo $n > m$; for even arity, one point from each partition takes weights $1, 2, 4, \ldots$ with $n > \sum$weights. Regrouping or orbit expansion does not inherit the guarantee, so the recovery check stays mandatory.
- *Unordered selections.* A monic cubic $H(T) = T^3 - e_1T^2 + e_2T - e_3$ dividing $f(T) = \prod_{v \in V}(T - v)$ selects three distinct base coordinates exactly when the three remainder coefficients vanish, with the symmetric S_3 rewrite $(e_2 - a)^2 - 4e_1(e_3 + b) = 0$; the study notes that the remainder equations can be much harder to solve and claims no degree reduction.
- *Automorphism orbits.* On a $j = 0$ fixture the map $\varphi(x, y) = (\beta x, y)$ propagates labels through orbits; the intrinsic relation $R + \varphi(R) + \varphi^2(R) = O$ has $A = B = 0$ and is rejected.
- *Covering partitions.* A bit-based three-colour separating family covers every distinct triple with at most $2\lceil \log_2 s \rceil (\lceil \log_2 s \rceil - 1)$ colourings, an $O(\log^2 s)$ construction; coverage does not imply a lower total cost.
- *Solver presentations.* Expanded S_4, the paper's joined-S_3 form and first-coordinate hybrid guessing share one checked relation path.

The field is bounded by 65,521 by construction, and the study states that success of its substitution backend on the fixtures is no evidence that enumeration disappears at large sizes.

What exists is correctness on tiny public fixtures. The independent replay reads the correctness JSON for three fixed fixtures and twelve arms and re-checks group order, every labelled point, partition membership, signed sums, A, B and $dG = Q$ in separate affine arithmetic. The follow-up's release suite reports 5,002 tests passed, 0 failed and 122 ignored, with seven core-only Lean lemmas. The measurement instrument `ic.aps_v1` charges setup and five online phases on one monotonic cold timeline, but fields are capped at 65,521, abscissae at 32 and attempts at 128; its `total_gae` prices only the independent affine checks, so it is a lower bound and `native_work_accounting_complete` stays false. No benchmark has launched, and the paper's 11–22-bit ladder was not reproduced. The obstruction record `problems/aps-single-relation-complete-cost.md` lists what a complete cost needs. The four leads of 5 November (a nonhorizontal line chart $U = t^2$, $V = c/t$ with $H_{U,V}(X) = X^3 - U(X+V)^2 + aX + b$; F4 trace reuse across attempts; incremental factor-base frontiers; unequal partition profiles) are hypotheses with no implementation, filed as `PCC-20261105-aca921` and `PCC-20261105-ac1e93`, both of unverified standing. The APS route is open and unmeasured.

23.3 The digit presentation

Writing each unknown abscissa in s bounded binary digits, $\ell_k = \sum_{i<s} 2^i a_{k,i}$ with $a_{k,i}^2 = a_{k,i}$, gives a prime-field analogue of Weil descent: $n = ms$ Boolean unknowns, and the system $\tilde{S} = S_{m+1}(\ell_1, \ldots, \ell_m, x_R)$ in the function algebra $\mathbb{F}_p[a]/(a^2 - a)$. The closed form of its first fall degree, $d_{\text{ff}} = m2^{m-1} + \lfloor (s - 2^{m-1})/2 \rfloor + 1$, is in Chapter 35, and its $m = 2$ case $4 + \lfloor s/2 \rfloor$ in Chapter 6. A native $\mathbb{F}_p$ ladder with an independent prime, curve and target for every draw (`research/prime_digit_presentation_20261029/`; registered before the counted runs, one amendment) measured the *last* fall degree:

- $d_{\text{last}} = 4 + \lfloor s/2 \rfloor$ on 106 of 106 real draws at $m = 2$ (60 unsatisfiable at $s = 2$–7, 46 planted at $s = 2$–8), equal to the first fall degree in every draw; at $m = 3$, the first fall degree is 13 on 8 independent instances at $s = 4, 5$, and $d_{\text{last}} = 13$ on the 4 at $s = 4$;
- a support-matched null gives $s + 3$ on 44 of 44 draws, and the real system falls strictly earlier from $s = 3$; a block-factored null of random forms on the support of ℓ_k^e reproduces the real first and last fall degrees on 44 of 44 draws, so the early fall is block structure and carries nothing curve-specific;
- the plain presentation $\mathbb{F}_p[x_1, x_2]/(x_i^p - x_i)$ in the same engine has $d_{\text{last}} = p + 2$ on 32 of 32 draws: linear in p, where the digit degree is linear in $\log p$;
- at d_{last} the degree-filtered Macaulay space spans 100, 98, 96, 83, 81 and 60% of the 2^n monomials at $s = 2$–7, and 60% at $s = 8$.

Table 23.3 gives the unsatisfiable ladder. A low degree does not make a small system at this scale, and the lane is exhausted at the measured scale ($m = 2$, $s \leq 8$; $m = 3$, $s = 4, 5$). It would reopen if $s = 9$ broke the closed form (predicted $d_{\text{last}} = 8$, about 2^{18} columns) or if the adaptive staircase arm, not run, fell below the block null. The scale limit is the engine's `u32` accumulators ($p < 2^{12}$), not mathematics; for $s \geq 8$ no unsatisfiable target remains on the 2^{16}-point grid, so $s = 8$ has only two planted draws ($d_{\text{last}} = 8$, 39,203 columns, about 11 minutes a draw). Validation draws preceded registration, and the registered binary's hash is not reproduced by a rebuild, although the rebuilt source reproduces two of panel A's cells line for line. No new theorem is claimed: `lean/NullTable.lean` checks

s	draws	distinct primes	$d_{\text{last}} = d_{\text{ff}}$	support-matched null	columns at d_{last}	columns/2^n
2	12	5	5	5	16	1.00
3	12	10	5	6	63	0.98
4	12	11	6	7	247	0.96
5	12	11	6	8	848	0.83
6	8	8	7	9	3,302	0.81
7	4	4	7	10	9,908	0.60

Table 23.3: Digit-presented Semaev system at $m = 2$, unsatisfiable real draws, each with its own prime, curve and target; none censored. "Columns" is $\sum_{j \le d_{\text{last}}} \binom{n}{j}$, derived from the measured degree. Source: `research/prime_digit_presentation_20261029/`.

the registered table's arithmetic, and a faithful proof of the closed form is blocked on Wilson's rank theorem for inclusion matrices and two further dependencies absent from core Lean. A frozen but unrun protocol for a p^3 second-digit carry experiment is in `research/p3_second_digit_carry_20261101/`.

23.4 Split-graph and secant presentations

Take a factor base of the rational intersections of E with the graph of a low-degree ordinate function $y = h(x)$ whose $h^2 - f$ splits completely, and write a two-point decomposition through the secant through the two summands. The split-graph theorem of 31 October (`research/prime_factor_base_degree_20261031/SECANT_GRAPH.md`) proves a homogeneous top-ideal degree $d+1$ for this presentation against $2d+1$ for the same base in the dense two-summand system, and a Lean continuation proves the polynomial-ring reduction with the actual highest-degree forms, including the exact $\mathbb{F}_{17}$ fixture at degree $5 \to 3$ for every target and all four sign branches (159 declarations in fourteen modules, zero failures, no native trust). The measured part (`research/prime_factor_base_degree_20261031/parallel_20261031/RESULTS.md`) ran stock F4 on 14,220 systems, of which 14,194 verified and 26 hit root-extraction caps (all at $\mathbb{F}_{127}$, degree 5). The handled F4 degree of the secant presentation is 3 against 6 for the dense presentation on all 18 $\mathbb{F}_{17}$ targets, and $d + 1$ against $2d + 2$ on $\mathbb{F}_{83}$, $\mathbb{F}_{101}$ and $\mathbb{F}_{127}$ at graph degree $d = 3$–5, with the matched non-graph secant control in between at $2d$.

The degree gain does not become a cost gain. Against the matched secant control the candidate improves degree, width and pair and row counts. Against its own dense presentation it lowers the degree but increases the branch-summed work: on the $\mathbb{F}_{17}$ toy, 976 reduced pairs and 2,102 reduced rows against 344 and 736 for the dense system. The stock engine exposes no field-operation total, so cost per relation is null. Equal-size bases have the same all-target ordered yield $K^2 - K$. A Boolean-offset successor of 1 November (`research/prime_factor_base_degree_20261031/pencil_search_20261101/`) removes the complete-splitting requirement at canonical degree 7 against a same-base dense 9, and selector recodings prove $7 \to 6$ (XOR) and $7 \to 5$ (one-hot); affine degree, yield and work are unmeasured. The atlas lead `LEAD-ICP-split-graph-secant-presentation` is `open_next_step`: a degree theorem with no cost advantage.

23.5 Divisor-norm decompositions and a direct index calculus

An m-point relation can be written as a norm: a function in $L((m+1)O)$ of the form $A(X) - B(X)y$ whose norm $A^2 - B^2 f$ is divisible by $X - x_R$ and whose other roots must lie in the factor base. Membership in a Kummer base $g\mu_d$ can then be imposed by quotient remainders or by Graeffe coefficient towers. The "one research bet" study of 29 October chose growing-arity divisor-coefficient decomposition with shared quotient and remainder membership towers, and stated that it could not deliver an original sub-exponential algorithm because the essential complexity lemma is unproved (`research/prime_one_shot_20261029/`). Its continuation (`research/prime_one_shot_20261029/continuation/`) built the pieces:

- a native coefficient solver for arity three that receives only the curve, target and base parameters, searches the two free norm coefficients, enforces membership through quotient remainders and extracts every rational selected-chart solution; on $E : y^2 = x^3 + 2x + 2$ over $\mathbb{F}_{17}$ its complete model sets agree with a separate signed-point oracle in all 90 cases (42 SAT, 48 UNSAT, 400 models);

- a mixed-target direct index calculus that recovers the logarithm of every one of the curve's 18 nonzero targets from $R = aP + bQ$, with a left-null vector of the factor-base coefficient matrix alone and no supplied logarithms, retaining 230 attempts with all 217 nonidentity solver calls charged (59,024 coefficient pairs);
- written proofs, re-derived within the session but not formally certified, of rationality, a coverage bound $\rho_g \geq 1/(192m16^m)$ for a constant fraction of admissible bases under stated finite conditions ($1/(320m16^m)$ for the conservative selected chart), and recovery without full base rank.

Each current solver call visits $p(p-1)$ coefficient pairs, so even the coefficient loop is $\Theta(2^{2n})$ in $n = \log_2 p$: the solver is exponential. The supporting reduction would need $2^{4\sqrt{n}+O(\log n)}$ oracle calls and dense linear algebra of $2^{6\sqrt{n}+O(\log n)}$ bit operations, so a proved uniform oracle bound $C(n) = \exp(o(n))$ would give a sub-exponential algorithm in that field family. No such bound is supplied. The obstruction record is `problems/prime-divisor-uniform-solver.md`, the atlas lead `LEAD-ICP-shared-coefficient-membership` is `open_next_step`, and no exponent was measured. The same continuation found the external tensor-contraction preprint of §24.1 and checked that the literal shared-remainder system translates to $O(mb^2)$ Boolean edges, too many for the contraction bound to give $\exp(o(b))$.

23.6 SAT, SMT and CDCL(T)

Chapter 21 reports three frozen SMT formulations of a 20-bit decomposition that returned UNKNOWN at 120 s. The lead review of 31 October (`research/prime_ic_lead_review_20261031/`) asked where SAT-shaped methods belong. It ranked the hashed-base tensor route of §24.1 first, as the only inspected route with an explicit claimed sub-exponential bound, and noted that this is not evidence for a CDCL solver: the source keeps an $O(mb)$-edge tensor network because an ordinary circuit-to-CNF reduction loses the size bound. For the narrower practical-solver question it drafted a fixed-arity-three CDCL(T)/$\mathbb{F}_p$ screen on a one-hot, disjoint-window, chained-S_3 predicate, with strict gates $e < 0.5$ and $a < 1.5$, deliberately not frozen until its generator, theory propagator and preflight cost exist. The shared-coefficient discriminator (`research/prime_shared_coefficient_sat_20261031/`) passed its E0 gate (18 of 18 exact model-set matches on $\mathbb{F}_{17}$; 6 SAT and 12 UNSAT targets) and, on 1 November, its Boolean E1 gate: 36 of 36 target/backend model-set comparisons and 42 of 42 independently checked UNSAT proofs. No timing or scaling was collected; the elimination campaign of §24.1 records the screen as "representation gate passed; performance implementation-blocked".

Twelve leads. On 31 October twelve prime-field leads, L01–L12, were evaluated and then reviewed by skeptic agents, whose digest is authoritative on standing (`research/ic_prime_eth_leads_20261031/SYNTHESIS.md`; `research/ic_prime_eth_leads_20261031/leads/SKEPTIC_DIGEST.md`). Table 23.4 gives the standing of each lead after review; the full account of what the leads measured, the refutations and the synthesis's feasibility limits is in §38.5. The synthesis's verdict is that nothing moved the exponent question, and that the SAT and propagation lane for prime fields "should be closed for exponent purposes"; it leaves open the Kummer-tower cells at $m \geq 4$ (§23.1) and the contraction exponent of the tensor route (§24.1).

The review corrected the lead document in eight places; the corrected sentences are in §38.6, and none changes an exponent argument.

23.7 Small-root reach

A windowed decomposition asks for a root of a block $f = S_3(x_1, x_2, x_3) \bmod p$ with $|x_i| < p^{\delta_i}$, the reach being $\varepsilon = \sum \delta_i$; at $(m, w) = (4, 2)$ index calculus needs $\varepsilon = 8/9$. Chapter 21 records the S_5 Coppersmith reformulation ($\delta^* = 1/80$) and the S_3 falsifier at $\beta = 1/4$. Four studies of 26–29 October turned this into theorems.

Correction. The atlas lead quoted a margin of -0.048 at $(4, 2)$. It came from `tools/smallroot_reach.py`, which compared $\prod X_i^{s_i}$ with p^{s_p}, where s_p is the exponent of p in $\det L$. The Howgrave-Graham criterion compares with $p^{t \dim L - s_p}$, and for nested shift sets $\varepsilon_{\mathrm{HG}} = \varepsilon_{\mathrm{tool}}/n$ in the limit (`CORR-20261024-b2a799`; `crypto-autoresearcher: analysis/smallroot-reach/CORRECTION.md`). The Jochemsz–May extended reach of S_3 falls

Lead	Headline after skeptic review	Standing
L01 char-2 gadget	slope about 2.58 bits of conflicts per window bit, against a bar of 0.5; CPU 3× worse at $\ell = 7$	constant only
L03 sparse-parity bases	paired ratio to a size-matched null 1.000 ± 0.0003, $b \le 22$	constant only
L04 backdoor census	offset $2\log_2(\text{cap}) - 1$ at $m = 3$: enumeration moved into the propagator	contested; cap-bounded statement only
L05 policy pilot	reference slope 2.17 [1.93, 2.38]; renaming is null by construction	refuted as stated (the claim "closed"); policy question untested
L06 contraction width	MMD+ lower bound 15–51 against enumeration 2–10.7 at $b \le 16$	refuted as closed; narrow claim stands
L07 base second moment	hashed bases 1.000 ± 0.005 of null; shared negation-closed window 0.713/0.833/0.916 at $m = 4$	contested; constant only
L08 Koblitz transfer	true margin about 1.4×, not 3.42×; binary only	refuted as stated
L09 BTN census	margin −16.1 to −16.3 bits, no encoding admitted by D.1 ($\lambda \ge 2^{85}$ needed)	tool stands, yardstick downgraded
L10 Lean filings	seven gaps between the lead text and Lean, all confirmed	refuted as fileable
L11 multiplier encodings	pc/school 0.78 [0.67, 0.92] at $b = 9$ only	stands, weak
L12 clause retention	inc/fresh 0.025 [0.014, 0.040], setup uncharged	refuted; retained setup

Table 23.4: The 31 October prime-field leads after skeptic review (L02 was never run). Producer-reported, from the synthesis; no lead has a validation record. Source: `research/ic_prime_eth_leads_20261031/SYNTHESIS.md`.

from 0.8408 to 0.2803, and the best margin at $m = 4$ from -0.0481 to -0.6086. The note is a computation by one agent; its pre-registered audit `EXP-ECDLP-64fa3d` has not run.

Any lattice, equal bounds. The optimal-reach study (`research/smallroot_optimal_reach_20261026/`) bounded every Coppersmith lattice built from $(p, f)^t$, the provably optimal shifts of Ryan included, without building one: with equal per-variable bounds the reach is at most $(n!\,\mathrm{vol}\,N)^{-1/n}$ for Newton polytope N, which is 0.3293 for S_3 in three unknowns ($\mathrm{vol}\,N = 14/3$) and 0.3536 in two. The bound is attained in 11 of 11 exact cases. Against the demand of 0.8889 the margin is -0.56, proved.

Any bounds, any monomial set. The follow-up (`research/smallroot_followups_20261026/`) proved that any such lattice on any monomial set M, convex or not, satisfying the Howgrave-Graham condition needs $h_N(\delta) = \max_{v \in N} \langle \delta, v \rangle < 1$. For S_3 in three unknowns this gives $\varepsilon < 3/4$ and in two $\varepsilon < 1/2$, against 8/9. An exact check found no violation in 9,600 cases. A joint lattice for the three-equation $(4, 2)$ system needs $\nu_s < s$ for some s; the computed $\nu_s/s \ge 2.08$ in the demand region for $s \le 3$ is a computation, not a proof. A third study (`research/prime_eform_reach_20261029/`; derived proof, exact counts) proved Theorem K: for a convex M and the extent W_i of f in coordinate i, every lattice gain satisfies $\sum_i \delta_i W_i \le 1$, so $\varepsilon \le 1/\min_i W_i$, which is 1/2 for S_3, the value the numerical search had only approached. In the symmetric e-variables with full extents the same argument gives $\delta_B \le 2/(m(m+1)2^{m-1})$, that is 1/6, 1/24 and 1/80 at $m = 2, 3, 4$, so the Jochemsz–May asymptotes are optimal there. The closure is conditional: at the exceptional specialisation $z = 0$ the extents are not full, and a review counterexample reaches $7/41 = 0.1707 > 1/6$ for S_3. Finally, any x-variable lattice on a convex set reaches at most $\delta = 1/21.8$ for S_4 and $1/70.8$ for S_5, against 1/4 and 1/5 needed, 5.5 and 14.2× short (`research/prime_s5_small_root_leads_20261028/`, §7c); this rests on the written proof above, which has no Lean formalisation.

The small-root family is closed for one block, for every bound shape and every monomial set by proof, and for convex sets at the e-form asymptotes. It is open for multi-target ideals, the exceptional specialisations and joint lattices beyond $s = 3$.

24 Prime fields: the tensor route and named targets

This chapter has two parts. The first is a review of an external claim: the preprint of Akshima and coauthors, *A Subexponential Time Algorithm for Elliptic Curve Discrete Logarithm*, which claims subexponential prime-field ECDLP conditional on a Boolean tensor-contraction theorem in a second external manuscript. The programme read both, ported the prototype and built instruments. *Nothing was measured*: there is no ecbench count, no wall time and no cost against rho for this route, and the programme did not verify the theorem it rests on. The second part collects the named targets not already in Chapter 21: EcGFp5 and EcMasFp5 over $\mathbb{F}_{p^5}$, a short remainder on $\mathbb{F}_{p^4}$, ECCp-131, and the audits of standard prime curves. Xedni calculus and the JKSST audit are in Chapter 8.

24.1 Boolean tensor contraction: an external claim

The claim. Version 0.2 of the Akshima et al. preprint (pinned at `bochum-crypto/subexponential-ecdl` commit `1ee44e56`, PDF SHA-256 `82e27362...`) claims a classical expected-time algorithm

$$2^{O(b/(\log b)^{1/4})}, \qquad b = \lceil \log_2 p \rceil,$$

for prime-field ECDLP with known odd prime group order, and a fixed-cofactor extension (`research/prime_one_shot_20261029/continuation/tensor-review/README.md`). Its Theorem 4 is conditional on its Theorem 2, which invokes Appendix D.1 of *The Exponential Time Hypothesis is False* (pinned at `openai/math` commit `7ef5c0cc`, PDF SHA-256 `5b1612dc...`; the entry-cost audit of 4 November found the publisher path's later commit `42907c9e` serves the same PDF). That appendix states a contraction bound of $(s + L_{\text{in}})^{O(1)} 2^{O(s/\sqrt{\log s})}$ deterministic bit work, in polynomial space, on an s-edge bipartite Boolean tensor network whose vertex predicates are charged explicitly. The source triage records that repository and author labels establish provenance only, not endorsement, peer review or correctness.

The mechanism. The preprint encodes integer multiplication with $O(b)$ Boolean edges and maximum vertex degree $O(b^{3/4})$, through two stages of exact Fourier transforms over a finite ring; a straightforward bit circuit's extra $\log b$ factor would destroy the saving. Independent affine hashes of injectively compressed points define m separate public factor bases, each membership costing $O(b)$ edges, and a complete m-point sum costs $O(mb)$ edges. The contraction theorem then gives query work $2^{O(mb/\sqrt{\log b})}$. Fresh mixed-target labels and a matrix-only dependency recover the target. With base listing, failed queries, seed restarts, root extraction, prime selection and linear algebra charged, the total is $2^{O(b/m + mb/\sqrt{\log b})}$, and $m \approx (\log b)^{1/4}$ gives the displayed exponent. The hashed bases are generic in the sense of §22.7, so the route's non-generic content is entirely in the contraction theorem.

What the programme checked. Two in-session readings of the decisive contraction proof, D.1 and the full D.4 recurrence, checked packet shrink, degree invariants, interpolation counts, description and evaluation costs, precision absorption, coefficient-ring valuations, Cauchy charts, Hermite recovery and packet factorisation, and found no concrete gap. The parent session read the preprint's multiplication construction, the point-addition encodings, the hash coverage and the complete-cost analysis, and found no transfer gap. The review itself says these are scoped proof readings, not certification of either manuscript or of any implementation. Implementation checks are all small:

- the native tensor crate passes 34 release correctness tests, including forced recursive and direct controls and a tiny complete relation pipeline;
- an independent GMP reconstruction unit passes 72 fixed polynomial cases, 308 Cauchy charts, 8,832 node-error and 1,152 final-error checks and four negative controls, with byte-identical replay; it is not an extension-ring multiplication or contraction test;
- a Rust port of the Go `ecdltensor` prototype passes all 19 Go tests with byte-identical output, and both demos recover the planted scalar (`research/ecdl_tensor_ic_rust_port_20261029/`); its counters are exponential and it does not implement the contraction algorithm.

What was not measured. The speed study (`research/ecdl_tensor_ic_speed_20261030/`) built an `ic.tensor_contraction_v1` arm in ecbench and states that the measurement has not run; it predicts S growing about like r, since one relation costs about $|F_1||F_2| \approx 2r$ additions in the prototype. The counts-only protocol v2 for the tensor and hashed-native panels (`research/hashed_ecdl_counts_20261031/`) superseded v1 before any data, and no session has run under either. The finite-size review (`research/hashed_ecdl_finite_size_20261030/`) states that the preprint's literal constants force a finite-size fallback on every practical graph, which does not refute an asymptotic theorem; it proposes search strategies below the recursion's onset without measuring any. Its ECCp-131 fixture uses a certified hash modulus $s = 3102808443644027376164572183694178153449$.

Census and elimination. Two of the skeptic-reviewed leads, the L09 admission census and the L06 width bound, bear on this route; their numbers are in §38.5. The elimination campaign of 1 November (`research/prime_ic_elimination_20261101/`) re-fetched both PDFs at their pinned commits, matched the hashes, and kept the route at rank 1 of its frozen register: the native finite fallback is rejected; symbolic guards first pass at $\lambda = 2^{480}$; semantic cuts expose 154–354 raw fanout edges but only 7–15 exact signal classes; natural ranks beat support-matched sparse nulls, while all four matched cap-1 typed topology nulls have exactly the same rank. Its state for the route is "legacy census infeasible; minimal typed global-wiring mechanism rejected; theorem/full null open". A rank-census design (`research/tensor_boolean_breakthrough_20261031/`) proposes to test whether natural boundary tensors have lower flattening rank than rewired nulls before any solver search; its E0 core is implemented and nothing affected has been measured. The source-inspection audit of implicit tensor entries (`research/hashed_ecdl_entry_cost_20261104/`) is claim `PCC-20261104-1df881`, and the structured-predicate integration report is `PCC-20261104-458dda`; both are of unverified standing.

What would test it. The rank-census design of 31 October stages the test as six gated experiments plus one SAT/BDD triangulation, none authorised to spend benchmark budget until its predecessor's gate passes:

- E0, canonical IR equivalence and mutation: every compared network regenerated, hashed and replayed with its semantics unchanged;
- E1, a boundary-tensor flattening-rank census against graph- and predicate-preserving rewired nulls, with a held-out target and null gap;
- E1S, SAT and BDD triangulation on the same pairs, to see whether exact Boolean solvers exploit the E1 signature;
- E2, local Fourier and partition-rank certificates that predict at least 20% lower held-out exact work;
- E3, certified rewrites and orders with at least a 2× held-out gain and a lower growth slope;
- E4, a structure-specific contraction engine beating the strongest exact baseline by 2× in operations and memory;
- E5, an end-to-end relation oracle improving on four increasing sizes and an unseen target, still with no asymptotic claim.

The finite-size review proposes five ways to use the hashed relation language below the recursion's onset: making inverse-hash selectors the free inputs, solving for one witness instead of counting completions, width-aware encoding and contraction plans, instance-certified recursion with small packet parameters, and nested bases with partial relations. It notes that enumerating every selector tuple remains generic decomposition search. None of the seven experiments and none of the five routes has a result.

Reading. The route is the programme's best-supported *claimed* sub-exponential algorithm for prime-field ECDLP, and it is external prior art. Its correctness rests on an external result that the programme read but did not verify, its constants make it infeasible at finite size as written, and it has no measurement. It does not move a row of any table in this treatise, and the iterative success criterion does not apply to it until a counted cost exists. The norm and divisor candidate of §23.5 does not inherit its compactness: its literal translation has $O(mb^2)$ edges.

24.2 EcGFp5 and EcMasFp5 over $\mathbb{F}_{p^5}$

EcGFp5 and EcMasFp5 are curves over $\mathbb{F}_{p^5}$ with $p = 2^{64} - 2^{32} + 1$ and $\mathbb{F}_{p^5} = \mathbb{F}_p[z]/(z^5 - 3)$. EcGFp5 is a double-odd curve with one rational 2-torsion point. Chapter 35 records that the September lane on EcGFp5 symmetrisation measured no $m = 5$ ideal degree (`DEC-20260923-d86ff2`). That statement is

p'	W0 CPU-s	BW CPU-s	speed-up	$\log_2 c_{\mathrm{cyc}}$ W0 / BW	G1 band (BW)
4111	31,004	16,912	1.83×	46.30 / 45.43	[43.43, 47.43]
262231	37,036	18,823	1.97×	46.56 / 45.58	[43.58, 47.58]
33554371	38,778	22,516	1.72×	46.63 / 45.84	[43.84, 47.84]

Table 24.1: One complete $m = 5$ solve of the EcGFp5-shaped quotient system at three toy primes, $D = 2^{16}$ in every cell: msolve with rational parametrisation (W0) against the block-Wiedemann finish (BW); one 4-vCPU host, $c_{\mathrm{cyc}} = \text{CPU-s} \times 2.8 \cdot 10^9$. The CLOSE line is 2^{46} and the PROMOTE line 2^{38}. Sources: `research/ecgfp5_quotient_solve_20261101/`, `research/ecgfp5_quotient_bw_finish_20261102/`.

superseded: the 1 November solve below measured $D = 2^{16}$ on EcGFp5-shaped curves over toy primes. It is still true that nothing was measured at the 64-bit p. Chapter 25 gives the Joux–Vitse $(n-1)$-decomposition at $n = 5$ over $E(\mathbb{F}_{p^5})$ for toy p.

Desk evaluation. The four atlas leads were evaluated on 31 October (`research/ecgfp5_lead_evaluation_20261031/`; desk evaluation, every cost a model from cited constants). The design note's "$\geq 2^{142}$" multiplies the large-prime system count $p^{2-2/5} = 2^{102.4}$ by a per-system cost without the $5! = 120$ inverse decomposition probability, and is not a lower bound for the Gaudry–Diem family. The cheapest member is the basic algorithm with a factor base tuned to the per-system cost c, whose total scales as $c^{1/3}$, about $2^{115.5+\log_2(c)/3}$ 64-bit word operations: 2^{126}–2^{136} for the repository's per-system bands. In raw time every variant is 32–45 bits below rho (rho $\approx 2^{159.3}$ steps, 2^{169} word operations), but the tuned base has 2^{57}–2^{63} columns and needs 2^{64}–2^{71} bytes against 2^{35}–2^{45} for rho with distinguished points; capped at a petabyte the method is about 2^{27} above rho. In a memory-aware model rho remains the better attack. The verdicts are: symmetrised Gaudry REVISE (measure c, not D); Joux–Vitse $(n-1)$ for the full logarithm [27] CLOSE ($(n-1)!\,\tilde{c}\,q^2/2 = 2^{131.6}\tilde{c}$ with $\tilde{c} \approx 2^{33.5}$–$2^{34.4}$ cycles lands within ± 2.5 bits of rho, at about 2^{71} bytes, and loses to rho for any base of at most $2^{62.5}$); the odd-characteristic cover CLOSE (genus 65 for EcGFp5 and 97 for EcMasFp5, index calculus at least 2^{206} on the best constructed cover; the atlas keeps this lead `open_blocked` until the source-freeze block is lifted); oracle-assisted static Diffie–Hellman REVISE (about 2^{63} oracle queries, and no deployed service found that exposes the oracle).

The quotient solve. The decisive experiment (`research/ecgfp5_quotient_solve_20261101/`; `benchmarks/20261101-ecgfp5-quotient-solve/`; pre-registered, amendments A1–A5) solved the $\mathbb{F}_p$-rational $(\mathbb{Z}/2)^4 \rtimes S_5$ quotient of the five-summand system on EcGFp5-shaped curves at $p' = 4111$, 262231 and 33554371 with msolve 0.6.5, on one 4-vCPU host. It found $D = 2^{16} = 65{,}536$ at all three primes, exactly the affine Bézout bound $8^5 \cdot 2$, confirming $D = 2^{(m-1)^2}$ with no reduction below Bézout; the planted decomposition was recovered at every prime. One complete solve costs $c_{\mathrm{cyc}} = 2^{46.30}$, $2^{46.56}$ and $2^{46.63}$ cycles. The evaluator's frozen gate G1 reads CLOSE when $c_{\mathrm{cyc}} \geq 2^{46}$, so all three cells close, by margins of 0.30, 0.56 and 0.63 bits. The F4 work is p-independent (the same final 5578×71114 matrix at every prime), F4 alone is $2^{44.72}$–$2^{45.29}$, and about two-thirds of c is msolve's rational parametrisation. Weighted orders and signature-based bases did not lower c. At $m = 3$ all five arms gave the predicted degrees in 30 solves, and msolve was found to crash for every prime from 65537 to 253229, a solver defect, so 65551 was replaced before any $m = 5$ run.

The block-Wiedemann finish. The finish was then replaced by a block-Wiedemann solve [7] on the BNS22 chain matrix of the reduced grevlex basis (`research/ecgfp5_quotient_bw_finish_20261102/`; `benchmarks/20261102-ecgfp5-bw-finish/`; frozen before the first cell, no amendments). The eliminant equals msolve's coefficient for coefficient, the stability check passes on 4,920 chains, and the planted point is recovered at every prime. Whole-solve CPU falls 1.83, 1.97 and 1.72× at the three primes, clearing the frozen 1.5× bar, so the finish is the new per-system baseline; the finish alone is 3.15–4.12× cheaper and the F4 work is unchanged. The new c_{cyc} is $2^{45.43}$, $2^{45.58}$ and $2^{45.84}$, below the CLOSE line by 0.57, 0.42 and 0.16 bits, but the frozen $[c/4, 4c]$ band straddles the line at every prime, so the verdict is unchanged (Table 24.1). The distance to the PROMOTE line of 2^{38} is 7.43, 7.58 and 7.84 bits. This is an engineering improvement of a per-system stage cost at toy primes, not an end-to-end ECDLP cost. An earlier HNF-based finish was a memory-bound null at the development prime $p = 331$: the determinant step was killed at 13.33 GiB on a 15.7 GiB host (`research/ecgfp5_quotient_hnf_finish_20261102/`).

Representations. A desk audit of 2 November (`research/ecgfp5_representation_audit_20261102/`) closed every degree-2 coordinate change and every larger symmetry group: EcGFp5 has one rational 2-torsion point and no rational 4-torsion, so the only usable group gives $D = 2^{16}$ at $m = 5$. It found one open lever, Pornin's $u = x/y$: since $u(P + N) = -u(P)$ and $u(-P) = -u(P)$, the quotient is a polynomial-ring form in $s_k = e_k(u_i^2)$ and $\pi = \prod u_i$, with measured $D = 2^{(m-1)^2}$ at $m = 3$ and $m = 4$ (16 and 512) and the planted point recovered. That corrected a ledger statement: `CORR-20261102-7e3a91` scopes the earlier claim that the ring form is "unavailable" for double-odd curves to coordinates with $\varphi(-P) = \varphi(P)$. Any gain from u is expected to be a constant factor. The atlas lead `LEAD-ICP-gfp5-symmetrized-gaudry` is `closed_dominated`, with the block-Wiedemann finish as a measured per-system improvement still short of the PROMOTE bar.

$\mathbb{F}_{p^4}$. Two items extend §21.10. The D_4 index calculus ran end to end in ecbench on $E(\mathbb{F}_{29^4})$, $\#E = 708{,}208 = 16 \cdot 44{,}263$, with every phase charged: $S = 7.17 \cdot 10^8$ against 1.065 for negation rho, audit OK on three of three records (`research/gaudry_quartic_d4_ecbench_20261013/`). A later leaderboard engineering row cut the plain-S_5 cost C_4 from $1.209 \cdot 10^{12}$ to $8.69 \cdot 10^{10}$ $\mathbb{F}_p$ multiplications (13.90×) with sparsest pivots, chain eigensteps and shape replay (`benchmarks/20261019-s5-plain-c4-options/`), where shape replay alone had been a 2–5% lever (`research/gaudry_quartic_replay_20261013/`).

24.3 ECCp-131

Chapter 21 gives the ECCp-131 parameters, the zero-relation probe and the rho rates. Four further studies are nulls or constructions.

- *A full-width auxiliary tower.* The Kummer tower cannot have $t \geq 2$ on the ECCp-131 field, since $v_2(p - 1) = 1$. A rational eight-step 2-isogeny tower was built instead on the auxiliary curve $E' : y^2 = x^3 + x + 301$ over the exact field (`research/eccp131_full_width_20261023/REPORT.md`; of 301 auxiliary orders computed, two were divisible by 256, at $b = 278$ and $b = 301$, and the curve with $b = 301$ was accepted): 256 distinct coset abscissae, 8 of 8 Vélu maps checked, 115 abscissae lifting to the published curve, 26,450 signed pair points, 8 of 8 planted controls, and zero natural setup or target rows, rank zero.
- *A multiplicative-coset screen.* Twelve $g\mu_5$ cosets and, in a follow-up, sixteen $g\mu_{10}$ cosets (μ_{10} is the maximal smooth subgroup, $p - 1 = 2 \cdot 5 \cdot r$ with r prime) against equal-size small-abscissa bases found zero natural rows and rank zero for both seeds, with every planted control recovered (`research/eccp131_structured_ic_20261022/REPORT.md`).
- *A representation filter.* A Wagner-style coordinate-bit filter trained on true four-sum relations against a rotated null, with 3,450 configurations in each of six base/size cells (`research/eccp131_representation_filter_20261102/REPORT.md`; `benchmarks/20261102-eccp131-representation-filter/`), qualified in no held-out cell: every recorded lower confidence ratio is below 1, and the two superficially positive small-x winners at $B = 64$ and $B = 256$ are post-addition filters that need all $\Theta(B^2)$ pair sums. The verdict is `no_qualifying_filter_in_frozen_family`, with 0 rows and rank 0 at arity 2 and 3.
- *A CUDA port.* The ECCp-131 and ECC2-131 rho walks were written as per-thread functions that compile as device code (`research/certicom131_cuda_port_20261023/`). The kernels compile and assemble for sm_80–sm_120 but have never executed on a GPU; there is no device parity result and no GPU rate.

The full-parameter index calculus remains the open problem `problems/eccp131-full-parameter-index-calculus.md`.

24.4 Audits of standard prime curves

Chapter 21 lists the audits of the NIST, SECG and CryptoPro curves that found no ECDLP weakness. The later ones concern curve provenance and class-group statistics, and none bears on ECDLP hardness.

- *The NIST seed skip rule.* Of the 21 seed candidates the 1997–98 runs accepted under every known criterion, 10 were published. None of 21 new preregistered binary features explains the 11 skips, after about 60 earlier features were ruled out (`research/nist_seed_skip_filter_20261017/`). A later search found an independently checked two-test empirical separator for all 21 eligible curves, with a root-convention dependence and a calibration that blocks reading it as the historical explanation

(`research/nist_skip_criterion_20261025/`). The same question for the binary B- and K-curves is a measured census (`research/nist_binary_skipped_20261020/`), and a factorisation continuation fully factored eight of the nine distinct integers left by an external DiSSECT run, leaving one 444-bit composite cofactor of a P-256 extension-twist order (`research/nist_skipped_curve_dissection_20261021/`).

- *P-224's class group.* The 2-rank of $\mathrm{Cl}(D_K)$ is exactly 6 for P-224 (4 for P-256), and the class number is estimated at $3.5185 \cdot 10^{32} = 2^{108.1167}$, with a GRH-conditional interval $[2^{108.098}, 2^{108.135}]$ under an assumed constant; the exact value is not known (`research/p224_class_number_20261022/`). A 2-rank of 6 is not unusual: P-224 sits at the 91.5th mid-rank percentile of random 224-bit primes, upper tail 12.3% $[10.3, 14.5]$ (`research/p224_two_rank_null_20261022/`).
- *Cheon's attack on P-224's shape.* The DLP with auxiliary inputs was scaled on P-224-shaped curves (`research/cheon_p224_shape_20261022/REPORT.md`). The pre-registered rho-reliability gate H3 failed, with 5 of 84 sealed rho runs outside $S \in [0.3, 4]$, so every Cheon/rho ratio and crossing in that report is withdrawn; the Cheon-only scaling stands. It is in any case not plain ECDLP.
- *Special prime shapes.* Curves built by the CM method over 192-bit Proth, cyclotomic $\Phi_d(m)$ and smooth $p \pm 1$ primes all have embedding degree $k > 4000$, like the random control; only pairing-friendly constructions give k of 10–24 (`research/special_prime_curves_20261023/`).
- *Rho on NIST curves.* Stepping to the better of $P + M_j$ and $P - M_j$ in the `gpu/nist` small-batch kernel stays disabled after an audit; the estimates are historical diagnostics, not measured solve costs (`research/nist_best2_small_batch_20261027/`).

24.5 What the three chapters settle

On every prime curve measured, rho with negation is cheaper than every index-calculus arm, generic or algebraic. The generic track's best arms are the residual collectors at 1.3–1.7× rho ($w = 0$; 1.54–1.64 in the cells with a validator report), a constant at rho's exponent with no growth over 36–42 bits; the k-sum tables are 5.4–11.0× rho and above baby-step giant-step. The algebraic track's priced prime rows are the Kummer-tower rows and the Curve25519 F4 arms; the best of the latter, with a validator report, is still $1.83 \cdot 10^4\times$ rho at $w = 0.1$, and the cost scales as $r^{1.03}$ (§21.7). Its degree studies show growth or input-pinned degrees in every presentation, with matrices that grow when degrees do not. The one claimed sub-exponential route is external, conditional on an unverified theorem, infeasible at finite size as written and unmeasured. On EcGFp5 the measured per-system cost moved by 1.72–1.97× and remains about 7.4–7.8 bits from the promote line.

The chapters state these corrections and withdrawals:

- the residual collectors' $w > 0$ gains were the price of a byte counter, withdrawn under option B (1 November); the scale study's C1 tally falls from 6 of 8 to 2 of 8, and a 9-versus-8 tally of 12 cells is unreconciled (§22.3.1);
- the memo gate "$e < 1$ at $m = 4$" is replaced by $s < m - 3$ (§22.1);
- the cofactor-cleared prime base is `invalid_controls`, not a win (§22.4);
- $N = 64$ at 20 bits is unresolved, and the 24- and 28-bit base selections are relabellings (Chapter 40);
- the small-root margin of -0.048 is superseded by about -0.6 (`CORR-20261024-b2a799`, §23.7);
- the September statement that no $m = 5$ EcGFp5 degree was measured is superseded by $D = 2^{16}$ at toy primes, and the ring-form statement is scoped by `CORR-20261102-7e3a91` (§24.2);
- the symmetric-tower rows are instrument diagnostics, not natural prime-order benchmarks (§23.1);
- the Cheon/rho ratios on P-224's shape are withdrawn (§24.4).

The open items are the discriminating Kummer cells ($m = 3$, $N = 21$ symmetric; $t = 14$ at cap 6), the algebraic reference arm, a counted cost for the tensor route, the $\mathbb{F}_p$ CDCL(T) propagator, the APS complete-cost panel, and a proved uniform bound for a divisor-norm oracle.

25 Extension fields: $k = 4$ and the D_4 symmetry

Over $\mathbb{F}_{q^k}$ with a factor base of points whose abscissa lies in $\mathbb{F}_q$, Gaudry's index calculus [20] decomposes each target into k summands by Weil restriction of S_{k+1}. It costs $\tilde{O}(q^{2-2/k})$, below rho's $q^{k/2}$ for $k \geq 3$, but with a Gröbner constant that grows very fast in k. Faugère, Gaudry, Huot and Renault [14] show that on curves with rational 2-torsion the system has the larger symmetry group D_k, the Weyl group of type D, in a well-chosen coordinate. At $k = 4$ this chapter reports how far that symmetry moves the measured gap to rho. Chapter 21 (§21.10) gives the odd-characteristic engineering in detail; Chapter 31 gives the later $k = 4$ programme and supersedes the odd-characteristic distances below.

25.1 The D_4 system

On E with rational 2-torsion point T, the coordinate $Y = (x - x_0 - c)/(x - x_0 + c)$ satisfies $Y(P+T) = -Y(P)$. In Y, S_5 is invariant under D_4, of order 192. Written in the invariants $\sigma_k = e_k(Y_i^2)$ and $\pi = \prod Y_i$, the system shrinks (Table 25.1).

Table 25.2 gives the engineering ladder, each step against the previous incumbent. At step 1 all 19 solved targets agree with an independent meet-in-the-middle enumeration; at step 3 all 648 of 648 solves are exact. At step 4, double large primes cut the linear algebra to 0.827× but regress the whole solve, to 1.025×.

The D_n symmetry is FGHR's [14]. Invariant rings in characteristic two, including even-power substitutions, were studied by Galbraith and Gebregiyorgis [17]. Whether the Frobenius-reduced weighting is in that work has not been checked.

25.2 Distance to rho

Odd characteristic. Over $\mathbb{F}_{p^4}$ the best line (GLS twists with full rational 2-torsion) measured $1.5{\cdot}10^3\times$ its strongest matched rho at $n = 2^{42.4}$ (E32, validated), falling as $n^{-0.24}$, with a fitted crossover at $2^{85.5}$ $[2^{79.2}, 2^{94.3}]$ at $w = 0$; that is a fit, not a measured crossing (Chapter 31). The earlier single-point models of §21.10 are superseded.

Characteristic two. Over $\mathbb{F}_{(2^m)^4}$ the gap closes with size. IC over the negation-map rho model is:

m	6	8	10	12	14
$\log_2 n$	24	32	40	48	56
IC / rho model	5,080	648	80.6	49.7	6.06
IC / measured rho	4,427	919	98.9	47.2	4.27

These are first targets; $m = 14$ is a single target. The fitted slope is -0.248 in $\log_2 n$, which is Gaudry's $n^{-1/4}$. The extrapolated crossover was $n \approx 2^{68}$, where rho takes about 2^{34} steps, so it could be tested directly.

The red-team verdict. An adversarial review tested that crossover (`research/gaudry_k4_char2_redteam_20261026/REPORT.md`). Its verdict: **not survived (unsupported), not refuted**. Table 25.3 lists its estimates. The only clean point is at $m = 15$, with two targets per arm, so no interval is possible with $n = 2$; the $m = 17$ ratio came from a killed run and is not evidence. The rho reference survives: these are generic ordinary curves with Aut $= \{\pm 1\}$. Pricing partly survives. A prime-field cross-check agrees with the fitted exponents. The corrected crossover is at or above the edge of what was measured.

Linear algebra is 0.4% of the cost at $m = 15$. Basic Gaudry linear algebra grows as q^2, like rho, so the $n^{-1/4}$ trend lasts only while the relation phase dominates. Beyond that point double large primes are required.

The red team froze a plan to settle the question: $m = 14$–17, eight fresh targets per arm, a strong rho with batched inversion and charged canonicalisation, counted and hardware units, about 30 h of wall time. It has not run. It is the cheapest experiment that could put the $k = 4$ characteristic-two family

System	Bézout number	Macaulay degree	Matrix
before invariants	4,096	29	$41{,}780 \times 40{,}920$
D_4 invariants	512	14	$13{,}412 \times 11{,}628$
Frobenius-reduced, char. 2		11	384×448

Table 25.1: Size of the S_5 decomposition system at $k = 4$: before and after rewriting it in the D_4 invariants σ_k, π, and in the Frobenius-reduced form with weights $(2, 2, 1, 1)$ in characteristic two (ladder step 5 of Table 25.2; the source gives no Bézout number for it). Matrix: rows × columns of the Macaulay matrix. Sources: `research/gaudry_quartic_d4_20261013/`, `research/gaudry_quartic_d4_char2_20261015/`.

	Step	Scope	Result
1	D_4 invariants	$p = 269$	$C_4 = 2.54 \cdot 10^{10}$ instead of $1.21 \cdot 10^{12}$, a factor of 47.5–48.4
2	The Y-line on a complete ladder	$p = 101$–1033 (21 curves)	0.0548 $[0.048, 0.061]$ of plain S_4, at the same exponent
3	Sparsest pivots, the weighted grading $\sigma_4 = \pi^2$, and shape replay		C_4 falls to $6.89 \cdot 10^7$, by 369.8× $[367.9, 371.6]$; ecbench S falls 398–406×
4	First-dependency stop		0.979 $[0.973, 0.986]$ of the previous step
5	A Frobenius-reduced form in characteristic two	$m = 8$	one decomposition 481× $[479, 483]$ below the plain form

Table 25.2: The $k = 4$ engineering ladder, each step against the previous incumbent. C_4: $\mathbb{F}_p$ multiplications per decomposition. Result: the step's cost as the source states it, either a factor below the previous incumbent or a fraction of it. Sources: `research/gaudry_quartic_d4_20261013/` (step 1), `research/gaudry_quartic_d4_cost_20261014/` (step 3), `research/gaudry_quartic_d4_char2_20261015/` (step 5).

below rho, as a measured size crossover; other weak families already have measured below-rho rows, the validated Joux–Vitse sieve row among them (Chapter 26, §30.4). It would be a crossover in a weak family: $\mathbb{F}_{(2^m)^4}$ is not a field deployed curves use.

25.3 Other extension-field routes

- *Joux–Vitse $(n-1)$-decomposition.* The $(n-1)$-decomposition [27] at $n = 5$ over $E(\mathbb{F}_{p^5})$ completed verified logarithms at $4.5 \cdot 10^7$–$1.8 \cdot 10^8$× rho, with a crossover extrapolated to groups near 2^{291} (`research/joux_vitse_n1_20261015/`).
- *EcGFp5 and EcMasFp5.* These curves over $\mathbb{F}_{p^5}$, and a short remainder on $\mathbb{F}_{p^4}$, are recorded with the named targets in §24.2.
- *Joux–Vitse cover.* The cover of $E(\mathbb{F}_{p^6})$ is summarised in Chapter 26, with the full account in §30.4.

Part	Quantity	Estimate	Range or note
Data	IC/rho at $m = 15$, recorded	6.30	two targets per arm
	IC/rho at $m = 15$, corrected	5.2	2.4–15.0 over unit-price variants
	IC/rho at $m = 17$	0.72	killed run; not evidence
Rho reference (survives)	floor	0.886	about 3.5% generous to rho
Pricing (partly survives)	ratio moved by the unit price	0.64–0.89	plausible range 0.31–3.1
	ratio moved by unpriced rho work	about 0.93	
Fitted exponents	IC relations	1.07	$[1.01, 1.13]$
	IC linear algebra	2.03	
	rho	2.00	$[1.71, 2.29]$
Corrected crossover	fall of IC/rho per unit of m	about $2^{-0.93}$	
	crossover m^*	≈ 17.6	range 16.3–19.2

Table 25.3: The red team's estimates for $k = 4$ over $\mathbb{F}_{(2^m)^4}$, grouped by the part of the verdict they bear on. Estimate: the point value; Range: the interval or range the review gives, with what it runs over. The corrected IC/rho applies the corrections that favour index calculus. Exponents are of cost in $q = 2^m$. Source: `research/gaudry_k4_char2_redteam_20261026/REPORT.md`.

26 Covers, isogenies and other transfers

A recurring hope was that some other representation of the same instance—an isogenous curve, a curve at another level of the isogeny volcano, a Jacobian, a Weil descent—would make index calculus cheaper. These routes produced the programme's clearest negatives, and one lemma that explains most of them.

Lemma 26.1 (transport conservation, `docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md` §1)**.** *Let $\varphi : E \to E'$ be an $\mathbb{F}_q$-rational isogeny (or any homomorphism of rational points), $\kappa = \ker\varphi \cap E(\mathbb{F}_q)$, $R \in E(\mathbb{F}_q)$, $\mathcal{F}' \subset \varphi(E(\mathbb{F}_q))$ a factor base on the codomain and $\mathcal{F} = \varphi^{-1}(\mathcal{F}') \cap E(\mathbb{F}_q)$. Then the number of ordered m-decompositions of R over $\mathcal{F}$ is $|\kappa|^{m-1}$ times the number of ordered m-decompositions of $\varphi(R)$ over $\mathcal{F}'$.*

Proof. Images of decompositions of R are decompositions of $\varphi(R)$. Conversely, for $\sum Q_i = \varphi(R)$ choose rational preimages P_i of the Q_i; then $\sum P_i - R \in \kappa$, and of the $|\kappa|^m$ choices exactly $|\kappa|^{m-1}$ sum to R (translate P_1 by the kernel element). □

So a homomorphic transfer changes nothing but three curve-specific inputs: the rational kernel symmetry $|\kappa|^{m-1}$ (bounded by the rational torsion; trivial on prime-order curves), the factor-base geometry available on the destination, and the cost of arithmetic there. Sequences of isogenies compose to one homomorphism; a loop is an endomorphism and acts on the subgroup as a known scalar, useful only as a known-log multiplier. Multi-curve factor bases glued by isogenies are single-curve bases in disguise.

Conductor gaps in the ECC2K-130 class. Jao, Miller and Venkatesan [26] prove random self-reducibility among curves with the same endomorphism ring, leaving open whether curves whose conductors differ by a large prime could differ in hardness. The ECC2K-130 class has conductor gap 263; we computed its structure (`docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md`) and ran several rounds of experiments. Native counted-S index calculus on two fixed CM pairs gave floor/crater ratios 0.975 (99% $[0.603, 1.451]$) at $\ell = 8209$ and 1.127 $[0.922, 1.342]$ at $\ell = 131101$: no level effect (192 verified records, 96 exact replays; `research/ecbench_conductor_gap_20261012/`). Two-torsion symmetrisation is a quotient isogeny: at a matched base of abscissae it cut F4 row-reduction work 4.1–4.3× ($|H| = 2$) and 17.6–19.6× ($|H| = 4$) over 1,128 systems—but a correction the next day showed that at a matched number of *columns* the saving disappears; it was the cost of a redundant orbit base. A combined Weil descent used 234–529× the F4 operations of the blind system (`research/isogeny_conductor_gap_round3_20261011/`). The later rounds and the full correction trail are in §27.3.

Fibre-aware index calculus. Isogeny fibres are torsors whose arithmetic is controlled by one cohomology class (`RESEARCH_ISOGENY_FIBERS_ECDLP.md`); relation generation is isogeny-equivariant at zero lifting cost, and a fibre-closed pair table was the best lever of the conductor-gap rounds (12.8× rho in its best cell). Measured against matched rho across sizes at fixed cofactor, the envelope IC/rho ratio grows from 8.4–8.8 to 35.1–37.0 ($h = 4$) and from 13.4–20.1 to 60.4–63.1 ($h = 72$), as $\#E^{1/6}$: the gap widens, it does not close (816 audited records; `research/fiber_ic_scaling_20261011/`).

Other structure. Symmetrised u-frame bases saved 0.84× at $n = 17$ and 0.76–0.77× at $n = 23$, but cost 4.01× at $n = 31$ (`research/koblitz_symmetrised_e2e_20260927/`). GHS and hyperelliptic covers: ECC2K-130 does map to hyperelliptic curves, but the genus that would pay is missing. In odd characteristic the cubic GHS transfer from $E_0(\mathbb{F}_{q^3})$ to a genus-3 Jacobian now runs end to end with double-large-prime index calculus, at about $q^{1.4}$ against rho's $q^{3/2}$, within 1.15× of a measured negation-map rho at $q = 2^{20}$ (§36.3). The isogeny-loop GLV/GLS work produced a measured 1.09–1.15× faster four-dimensional scalar multiplication (`research/isogeny_loop_glv_20261012/`) and a $\sqrt{\text{élu}}$ kernel cost exponent 0.822 $[0.805, 0.838]$; these speed up curve arithmetic, not the discrete logarithm.

Covers, 21–24 October. The Joux–Vitse genus-3 cover of $E(\mathbb{F}_{p^6})$ was, on 24 October, within 1.6–2.0× of a pooled rho constant at $p = 1009$ (a lower bound, superseded in November by a matched negation rho, §30.4); measuring the per-trial cost on a genus-255 binary Jacobian turned the $N = 130$ GHS counterfactual's modelled margin negative (-1.83 bits); and two censuses bound what covers exist

at toy sizes (genus floor 13 for odd-order $E/\mathbb{F}_{q^4}$; a 2^{-l} genus-3 fraction over $\mathbb{F}_{2^{3l}}$). All are in §36.5. None touches a prime field or a prime-degree binary field, where the GHS genus is out of reach.

The published scale of high-genus index calculus. The largest Weil-descent hyperelliptic discrete logarithms in the literature are at genus 31. Jacobson, Menezes and Stein [25] applied GHS to elliptic curves over $\mathbb{F}_{2^{62}}$, $\mathbb{F}_{2^{93}}$, $\mathbb{F}_{2^{124}}$ and $\mathbb{F}_{2^{155}}$ and obtained genus-31 curves C_{62}, C_{93}, C_{124} and C_{155} over $\mathbb{F}_{2^2}$ to $\mathbb{F}_{2^5}$, each with $\#J = 2r$, r prime. They solved C_{62}, C_{93} and C_{124} with Enge–Gaudry (relation collection for C_{124} took 379 CPU-days on a mixed cluster) and only estimated C_{155}, at 26,290 days on one 1 GHz Pentium III. Velichka [42] solved C_{155} ($\#J \approx 2^{155}$) with Vollmer's algorithm and sieving: 136,533 relations, about 1,735 CPU-days in total, 21 days wall-clock, of which 14 days were linear algebra. No genus above 31 appears in that thesis; the journal version [41] was not available to us and is unchecked. Our genus-255 per-trial measurement (§36.5) is a cost sample, not a solved instance, at about eight times the largest published genus. The cover ECC2K-130 would need, of genus about 2^{130} (§15.7), lies about 2^{125} times beyond it.

Genus three against the strongest rho. Rounds three and four of the genus-3 work ran without a rho arm; a 23 October study supplied one (`research/ghs3_vs_rho_20261023/`; it has a frozen protocol and tables but no README, and no review is recorded in it). On round 4's instances, $q = 2^{14}$ to 2^{20}, three curves per size, in round 4's unit and binary, the reference was an ideal negation rho priced with Karatsuba products and 32 walks per inversion (R_{ideal}, the primary reference, the most favourable to rho) and a measured negation walk (R_{meas}). The frozen rule declares "IC below the strong rho" at a size only if every curve is below both references at every weight of the band. At $q = 2^{16}$ it is: IC/R_{ideal} is 0.70, 0.71, 0.70 at $w = 0$ and 0.64 and 0.38 (geometric means) at $w = 0.1$ and 1. At the other six sizes the decision is *unresolved*, with geometric means 0.97–1.52 (1.27 at 2^{20}). Three things qualify the result. The registered unit prices one subgroup multiplication as one $\mathbb{F}_q$ multiplication ($c_\ell = 1$); the protocol's own sensitivity, $c_\ell = 9$, puts every instance at 2.7–12.4× rho. The ratio carries $\sqrt{c}$ of the cofactor, because IC depends on q alone and rho on $\ell = \#E_0/c$: the three curves at 2^{16} all have cofactor 4, and a post-hoc fit gives exponent 0.474 in c (theory 0.5) and -0.081 per bit of q at fixed c (theory -0.167). And the pooled trend over all 21 instances is flat (0.034 $[-0.047, 0.105]$ per bit), so no crossover size is determined. This is the programme's only preregistered decision of index calculus below the strongest rho. It is a measurement of a known pipeline in a structured, weak class ($E_0/\mathbb{F}_{q^3}$ with $4 \mid \#E_0$, never prime order), where theory predicts $q^{4/3}$ against $q^{3/2}$; it depends on the pricing of one operation; and it is not yet reviewed. It updates the ledger paper's "1.15× at 2^{20}" and bears on that paper's prediction of a crossover at $q \le 2^{22}$ once rho is measured: below rho at one size, unresolved at the others, with no measurable pooled trend.

The same family through a plane quartic. The $m = 2$ type of the cubic GHS cover gives a smooth plane quartic over $\mathbb{F}_q$ directly. On a plane quartic, Diem's index calculus is $\tilde{O}(q)$ in theory, against $q^{4/3}$ for the hyperelliptic genus-3 route above. Before 18 October the repository's plane-quartic "pure IC" toys could not be priced: their logarithms came from enumerating the subgroup, and the degree-3 class keys merged distinct classes. A canonical Pic^0 arithmetic now gives every class a unique key (checked against brute-force class counts on 21 curves and full Cayley tables at $q = 5$). A generic addition costs 3,523 multiplications and 44 inversions, flat in q up to $2^{31} - 1$ (`research/quartic_pic_canonical_20261018/`).

On that arithmetic, a pipeline with relations on lines, a reduced factor base with one large prime, Lanczos modulo ℓ and canonical descent solved and verified 51 of 51 units. It ran on the GHS-weak family $E_0/\mathbb{F}_{q^3}$ with $4 \mid \#E_0$, for $q = 223$ to 262,111 (`research/quartic_n4_20261019/`; Table 26.1). Its reference is a matched *plain* rho, without the negation map, on the same subgroup, and the order holds across the pricing band for every $q \ge 4051$. This was a preregistered decision of index calculus below a matched *plain* rho. Its successor, N4 v2 (`research/quartic_n4v2_20261020/`, 20–25 October), took the same family and instances from $q = 2^8$ to 2^{22} (63 instances, 501 index-calculus units and 2,442 rho runs, all verified) and added three changes: double large primes (D), structured Gaussian elimination with Lanczos (L), and a negation-map rho. DL's exponent is the $\tilde{O}(q)$ of Diem's analysis [9, 11]. At $q = 4{,}194{,}271$ one DL solve took 12.3 CPU-seconds against 616 for negation rho.

The qualifications are substantial. The host was L0 (load average 22–60); the intervals resample only three units per size at $k \ge 17$; the conversion constants move D's high-range exponent from 1.009 to 1.048; the frozen B0 records do not replay at any of 11 replayed sizes; the negation rho fails the calibration rule at three sizes; and there is no validation file. The program's closing note calls the result

Study	Arm	Over B0	IC/rho	IC exponent	Rho exponent	Crossover
N4			0.122 $[0.10, 0.16]$	1.083 $[1.040, 1.129]$	1.485 $[1.437, 1.525]$	wall time: only above $q \approx 3 \times 10^4$
N4 v2	D	0.081				
	DL	0.042 $[0.042, 0.042]$	0.005 $[0.004, 0.008]$	0.983 $[0.975, 0.993]$; 0.988 on $k = 17..22$	1.543 $[1.494, 1.583]$	$q^* = 555$ $[389, 752]$ (interpolated); $q^* \approx 1{,}889$ in CPU time
	DL1	0.022				

Table 26.1: Plane-quartic index calculus on the GHS-weak family $E_0/\mathbb{F}_{q^3}$, $4 \mid \#E_0$. N4 against a matched plain rho at $q = 262{,}111$, exponents fitted on $q \geq 4051$; N4 v2 against the frozen pipeline B0 and against negation rho at $q = 2{,}097{,}091$ (the top size). IC/rho in $\mathbb{F}_q$-multiplication equivalents. Exponents: fitted cost exponent in q, over the full range unless stated. Crossover: in counted units unless stated (the N4 crossover is in wall time). D: double large primes; DL: D with structured Gaussian elimination and Lanczos; DL1 is post hoc. Sources: `research/quartic_n4_20261019/`, `research/quartic_n4v2_20261020/`.

Primes	Walks	Quantity	Value
$p = 7$ to 31	exhaustive	pass fraction neighbour	0.53–0.61 (random curves: 0.1857) 0.74–0.98, at 1.5–3.6 steps
$p = 2^{10}$ to 2^{20}	sampled	walk-cost exponent	-0.14 $[-0.80, 0.95]$
$p \equiv 3 \pmod 4$	successful	walk-cost exponent	1.13 $[1.00, 1.26]$

Table 26.2: The 2-isogeny walk to curves on which Smith's construction succeeds. Pass fraction: share of descended curves passing Smith's full test. Neighbour: share of failing curves with a passing curve in their 2-isogeny component, and the mean distance to it. Walk-cost exponent: in p; the first is the frozen fit, which labels the walk cost $o(p)$; the second is post hoc, over successful walks only. Source: `research/smith_elliptic_walk_20261018/`.

"our measurements of a known result" (`research/plane_quartic_program_20261018/CLOSING.md`). It covers the GHS-weak family only; nothing here is about a generic curve, a prime-order curve or a standard curve.

Smith's isogeny and the walk to it. The other genus-3 type of the census is hyperelliptic. Smith's $(2, 2, 2)$-isogeny takes it to a plane quartic when his Propositions 2 and 4 pass. The walk study looked for curves on which Smith's construction succeeds by walking 2-isogenies on the elliptic side (`research/smith_elliptic_walk_20261018/`; Table 26.2). The preregistered expectation of about 0.186 for the pass fraction is falsified, upward. A labelled post hoc analysis attributes the frozen fit's $o(p)$ label to censored walks at the small primes, and reads the $p \equiv 3 \pmod 4$ exponent as $\tilde{\Theta}(p)$. At $p \equiv 1 \pmod 4$ most walks stay inside small trapped sets.

A second study, also open and not yet merged, maps divisor classes through Smith's isogeny and runs the whole route at toy sizes. The program's closing note lists what stays open (`research/plane_quartic_program_20261018/CLOSING.md`).

Joux–Vitse: two arms of one cover. The Joux–Vitse cover of $E(\mathbb{F}_{p^6})$ (full rational 2-torsion, $\#E = 4\ell$, a weak class) has been measured with two solvers, and their comparisons with rho do not agree. The full account, with the limitations and the history, is in §30.4. The *sieve arm* at $p = 1511$ ($\ell = 2^{61.4}$) costs 0.00647 $[0.00545, 0.00799]$ of a matched negation rho in group additions and 0.0156 $[0.0131, 0.0192]$ in textbook $\mathbb{F}_p$ multiplications: rho is 155× and 64× dearer, and the order holds across the band. An independent validator passed it (`VAL-20261103-c31c76`; `research/jv_sieve_negrho_20261102/`). It has no `pcc` claim record, so under the proof-carrying-claims rule it is a validation-gate result whose statement is an unverified assertion. It is the programme's one validated measurement of index calculus below a matched rho; the validator classes it as accounting, not an improvement, and it holds on the weak class only. The *plain Nagao arm* was 1.59–1.98× a pooled rho constant at $p = 1009$ on 24 October, a lower bound (`research/jv_cover_p_scaling_20261024/`). Against a matched negation rho it is above rho at $p = 1301$ and 1451 and unresolved at 1601 in group additions, and above at all three primes in $\mathbb{F}_p$ multiplications (`research/jv_cover_negrho_rescore_20261101/`). A factor-base line search

takes it below rho in additions but not in multiplications; the modelled crossings move to $p^* \approx 1185$ (additions) and 1799 (multiplications), without a validation file. The roughly thirtyfold gap between the two arms is unexplained.

Cover censuses. The minimal cubic GHS genus over an isogeny class follows $\#E(K) \bmod 4$: 13 for odd order, 5 for 2 mod 4, 3 for 0 mod 4, and genus 3 is the only value below rho's $q^{3/2}$ (`research/ghs_cubic_cover_census_20261012/`); at $n = 5$ the floors are 33, 33, 25 and 5 by 2-torsion type (`research/ghs_cover_census_n5_20261013/`). No deployed prime-field or prime-degree binary curve has a usable subfield; only the RFC 2409 Oakley groups 3 and 4 are reachable, at genus 15–16, where exponent models disagree in sign and the measured genus-16 relation phase alone is 5.2 bits slower than rho (`research/cover_route_prime_binary_20261008/`, `research/ghs_oakley_extended_20261023/`). All 15,791 cover maps in the registry were assessed: 165 binary maps have genus-3 arithmetic and the 15,626 odd-characteristic genus-2 covers lack a Jacobian implementation (`research/cover_transfer_assessment_20261013/`). An inventory of about 90 cover and decomposition studies found every route on prime, Koblitz and prime-degree binary curves closed, negative or constant-factor (`research/cover_decomposition_gap_search_20261018/`). Smith's $(2, 2, 2)$-isogeny route was reproduced natively (full success 0.18638 against a theoretical 0.18574 over 10^6 curves) and its sign bias shrinks like $1/\sqrt{p}$ (`research/smith_prop24_native_20261014/`, `research/smith_sign_scaling_20261016/`).

Volcano levels and generators. A 403,168-run census found the level effect equivalent in 33 of 35 comparisons and different in none (`research/volcano_ecdlp_census_20261017/`), and a large-sample rho study found the floor/crater ratio 0.998 $[0.986, 1.010]$ (`research/conductor_gap_rho_large_sample_20261018/`). The one real level effect is the automorphism group: $\sqrt{3}$ on the $j = 0$ crater, confirmed at 1.86–1.92× (`research/drastic_prime_conductor_gap_round3_20261012/`). At $\ell = 2$, floors give 5.8% fewer relations than craters, and 99% of the contrast is explained by the base's cofactor classes (`research/isogeny-volcano-cost-v2_20261017/`, `research/generator_effect_ic_20261019/`).

Other transfer lines. Index calculus on genus-2 to -4 Jacobians over $\mathbb{F}_p$ (the ADH–Gaudry attack, measured against rho on the same Jacobian) read below a distinguished-point rho in early rounds, but the reference was later found slow and the ratios are lower bounds, unranked under the band (`research/notes/index-calculus/RESEARCH_HYPERELLIPTIC_IC_RHO.md`); later rounds are in review. A weak-curve atlas of 922 exact models and 36 published cases ranks covering-model and pairing signals but leaves the selected-subgroup advantage unmeasured (`research/weak_curve_leaderboard_20261012/`); three complete isogeny volcano graphs (24,823 vertices) were built and their graph-wide IC surveys halted at checkpoints (`research/full_volcano_graphwide_ic_20261017/`). Weil restriction of ECC2K-130 to extension fields costs at least 2^{131}, $2^{70.19}$ times rho (cluster "transfer-boundaries").

Named obstructions. Blocked requirements are kept open as problem records (`problems/README.md`), among them the complete-cost calibration of root-finding IC (`problems/rootfinding-complete-cost-calibration.md`), with exit criteria rather than abandoned.

27 Isogenies, volcanoes and conductor gaps

Chapter 26 states the transport-conservation lemma (Lemma 26.1) and records the volcano census, the large-sample floor/crater ratio, the ECC2K-130 conductor-gap rounds and fibre-aware index calculus. Chapter 8 draws the consequence for factor bases: an isogeny buys only the kernel symmetry, the destination's factor-base geometry and its arithmetic. This chapter collects what the programme measured or proved on the isogeny side beyond those summaries: the closure that makes a prime-field isogeny walk to a weak curve impossible, the index-calculus readings across volcano levels, the conductor-gap constructions and their correction trail, and the factor bases and seeds that use isogenies. The rho side of the same studies (one instance on every level, whole classes, the ecbench volcano sessions) is in the matched-reference chapters (§34.4); it is not repeated here. Unless a validation gate is named, every number in this chapter is a producer-reported figure without a pcc standing.

27.1 The Tate closure on prime fields

The statement. Two ordinary elliptic curves over $\mathbb{F}_p$ are $\mathbb{F}_p$-isogenous if and only if they have the same number of rational points (Tate's isogeny theorem). The three weak families an isogeny walk might hope to reach are all functions of $(\#E(\mathbb{F}_p), p)$:

- an anomalous curve has $\#E(\mathbb{F}_p) = p$, trace one;
- the embedding degree is $k = \mathrm{ord}_N(p)$ for the prime $N \mid \#E$;
- a GHS or Weil descent needs a proper subfield, and $\mathbb{F}_p$ has none.

So the weak set inside the isogeny class of a generic prime-field curve is empty at every size, not merely far away. The programme filed this as a derivation, strength *strong*, against its own hypothesis `H-IT-001` (`crypto-autoresearcher:ledger/evidence/EV-IT-511f3d.yaml`). The record notes that an earlier evidence item had stated the obstruction as a path length of at least p/ℓ^2; the theorem gives disconnection, which is stronger. The decision `DEC-20260807-d4f2a9` rejected the lead in scope; `DEC-20260808-86ec3c` later restored its status to "weakened" under the programme's no-pausing directive. That second decision is administrative: it adds no evidence (`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-prime-field-isogeny-to-weak.yaml`).

The measured null. Before the closure was filed, experiment `EXP-IT-001` searched for special curves reachable from generic ones and found none: the special-curve share ρ_{special} was 0 at 20 bits (exhaustive), at 24 bits (78% coverage, stopped on resources) and at 28 bits (a 0.16% sample). Its positive control was a Smart certificate on an anomalous curve (§28.3). Both are in `crypto-autoresearcher:ledger/evidence/EV-IT-aefd12.yaml`. The scope is prime fields. Over composite-degree binary fields descent weakness is *not* an isogeny-class invariant, which is what makes Teske's trapdoor and the walks of Chapter 29 possible.

Walks of standard curves. Several campaigns walked the isogeny graphs of the NIST prime curves as tooling, without an ECDLP cost.

- From registered P-256, the native walker produced 65,536 distinct curve identities and 302,653 certified directed isogeny edges across the 12 walkable odd degrees up to 61; every order was proved prime and every model nonsingular with a valid generator. The artefacts were not published to S3, so the gate passed locally only (`research/p256_isogeny_multigraph_64k_20261006/RESULTS.md`).
- On P-192 and P-224, 100 distinct large prime-degree maps were built and replayed independently: 50 consecutive edges of degree 2341 in the P-192 class and 50 of degree 1117 in the P-224 class, with exact kernel, codomain, equation, subgroup-order and scalar-transport checks (`research/p192_p224_hundred_large_prime_isogenies_20261010/RESULTS.md`). The screen behind it covered every prime from 1010 to 10^6 on both sources.

None of these maps can change the ECDLP cost: by Tate and Lemma 26.1 the destination has the same group and the same factor-base options up to arithmetic.

A battery on the NIST b coefficients. A frozen protocol tested whether the published b of five standard curves is unusual under 17 statistics (`research/b_weak_class_battery_20261022/README.md`). The rule was Bonferroni over $m = 5 \times 17 = 85$ tests, flagging only $p \le 5.88 \cdot 10^{-4}$. No curve–statistic

Metric	Vertical (floor/crater, 90%)	Horizontal τ (null 95th pct.), p	Coordinates τ (null 95th pct.), p
rho `rho.negation` S	1.022 [0.983, 1.062]	1.7% (2.6%), 0.28	2.6% (4.6%), 0.70
IC relation yield	0.980 [0.957, 1.004]; inconclusive	2.2% (1.4%), 0.0005	4.7% (2.5%), 0.0005
IC cost S (reported)	1.039 [1.008, 1.071]	2.9% (2.1%), 0.002	5.6% (3.7%), 0.0005

Table 27.1: Volcano-level effects on index calculus (`ic.pipeline`, the 32 smallest abscissae, meet-in-the-middle) at 24 bits, protocol v1.1. τ: dispersion between curves of one level (horizontal) or between isomorphic models of one curve (coordinates). The vertical IC-yield test is inconclusive because its planted +5% floor control was not detected. Source: `research/isogeny-volcano-cost_20261017/README.md`.

pair was flagged; the smallest p was 0.034 (P-256, statistic `smooth_c`), 57 times the threshold. Positive controls passed in 74 of 85 cells and 11 cells were inconclusive. This is a negative-control result about representation, not about the isogeny class.

Designs that did not run to an answer. The class-wide yield study for the ECCp-131 and ECC2K-130 classes (`research/isogeny_class_ic_yield_20261102/`) has a design and run cards; one ECC2K-130 card timed out at 1,872 s with 1,713 of 2,112 rows written, because an untimed solution count grew by a factor of 8 per step of k (`TIMEOUT_NOTE.md`). Its partial output is not analysed and is not cited here.

27.2 Index calculus across volcano levels

Within one volcano every curve has the same $\#E$, so generic cost is level-blind by the generic-group argument, and the measured rho nulls of §34.4 agree. Index calculus is not generic: its factor base is a set of points chosen by a coordinate rule, so it could in principle see the level, the endomorphism ring or the model. Four studies measured that, and one directory must be excluded.

Vertical, horizontal and coordinate effects. The frozen volcano-cost study ran rho, BSGS and index calculus on six 24-bit classes of depth 3 ($\ell \in \{2, 3\}$), 200 curves in four isomorphic models each: 43,200 measured runs, 46,912 with the two control classes, every run verified and 408 of 408 replays exact (`research/isogeny-volcano-cost_20261017/README.md`). Table 27.1 gives the index-calculus rows under the v1.1 statistic. BSGS cost was identical on every curve and model.

Three readings follow, each stated by the source.

- *The coordinate effect is larger than the isogeny effect.* Isomorphic models of one curve differ in relation yield by $\tau \approx 4.7\%$, more than curves of one level do (2.2%). The factor base "the 32 points of smallest abscissa" is a different point set in each model, and the yield follows the set, not the curve. By the protocol's own rule this is a coordinate effect.
- *The v1 statistic was wrong, and the report says so.* Under the frozen v1 statistic the vertical IC-yield test was nominally significant (0.972 [0.951, 0.993], $p = 0.0012$). The same v1 statistic gave BSGS, whose cost is identical on every curve, a bootstrap interval of [0.88, 1.22], because it pools classes with different shares of floor and crater curves. The amended v1.1 statistic forms the ratio inside each class; under it the test is not significant. The study does not claim the v1 reading of a −2.8% level effect.
- *The distance to rho is untouched.* Per curve, index calculus costs $S \approx 27$–96 at $w = 0$ against rho's 0.8–1.6, so it is about 30–60 times rho at 24 bits, and rho is below index calculus on every one of the 800 workload curves at every weight of the band.

An exploratory per-class contrast (six unadjusted contrasts) suggests that on $\ell = 2$ volcanoes floor curves, whose 2-part is cyclic, give about 6.5% fewer relations per trial than crater curves with full rational 2-torsion (0.930 [0.863, 0.994] and 0.937 [0.888, 0.987] on the ramified and split classes), while $\ell = 3$ volcanoes show nothing. The source files this as a hypothesis for a later protocol. Its successor attributed 99% of the $\ell = 2$ contrast to the base's cofactor classes (Chapter 26). The study's crater controls needed the 24 October class-set-up erratum, which is in Table B.5.

Tall volcanoes. A preregistered study on ten volcanoes with $\ell \in \{2, 3, 5, 7\}$ and depths 6 to 24, whose tallest floors hold 8.4 and 9.6 million curves, ran 14,731 `rho.negation` solves on 608 checked curves (`research/volcano_tall_layers_20261018/RESULT.md`). Every preregistered comparison lies inside the ±5% margin: in tier a (eight volcanoes) floor/crater is 1.0070 [0.9741, 1.0394], middle/crater 0.9898 [0.9654, 1.0139], and the crater-to-floor slope +0.87% [−2.03%, +3.75%]; the Kolmogorov–Smirnov test between crater and floor gives $D = 0.0199$, $p = 0.715$. The positive control, plain over negation rho on the same workloads, reads 1.3485 [1.2380, 1.4780], so the design resolves a 35% difference. None of the 40 per-volcano intervals excludes the null; the widest are 1.079 [0.973, 1.188] (a8, $\ell = 7$, depth 6) and 0.961 [0.879, 1.050] (a7). The raw S of negation rho is 0.947 [0.936, 0.958] against the floor $\sqrt{\pi/4} = 0.8862$, constant from $r \approx 2^{42}$ to 2^{51}. Three tier-b sessions were cut short by a worker restart and kept as measured. The study retracts an earlier "±8%" quoted for the Koblitz layers below.

Prime and Koblitz layers. Two earlier studies have wide intervals and say so.

- On two depth-8 $\ell = 2$ volcanoes over $\mathbb{F}_p$ (864 records, `research/volcano_prime_layers_20261018/RESULT.md`), 14 of 16 tested intervals contain the null. The two that exclude it are both `rho.plain` on one volcano (m_5: middle/crater 0.688 [0.562, 0.891], floor/crater 0.671 [0.524, 0.883]), and both involve its crater, a single curve with six targets, so a crater effect cannot be separated from a curve effect. The non-excluding intervals are about ±30% wide.
- On binary Koblitz curves with $n \in \{19, 23, 37, 41, 61\}$, level-1 over crater for `rho.negation` contains 1 at every n, from 0.837 [0.625, 1.073] ($n = 61$) to 1.089 [0.993, 1.195] ($n = 19$); one of ten plain-rho intervals excludes 1 ($n = 41$, 1.152 [1.021, 1.304]). The Frobenius solver over the negation solver on the crater falls from 2.153 ($n = 19$, fixed overhead) to 0.150 [0.132, 0.170] at $n = 41$, against the ideal $1/\sqrt{n} = 0.156$ (`research/volcano_koblitz_layers_20261018/RESULT.md`). The $\sqrt{n}$ gain is the crater's automorphism, which its neighbours lose.
- At $n = 83$ the native tool built the modular polynomial $\Phi_{6473}(X, j_0)$ from 9,864 CRT primes in 8,590 s, reduced and factored it into 78 Frobenius orbits of size 83, and found the expected 6,474 level-1 j-invariants; no ECDLP was measured there (`research/volcano_koblitz_n83_20261018/`).

An excluded directory. `research/synthetic_volcano_ecdlp_20261018/` states in its first line that its data are invented: no elliptic curve, point, isogeny map, relation matrix or logarithm was constructed, and its 130,410 "observations" come from a planning model. It is not evidence for any statement in this treatise and is not cited for one.

27.3 Conductor gaps

Jao, Miller and Venkatesan [26] leave open whether curves of one order whose endomorphism rings differ by a large conductor gap could differ in hardness. Chapter 26 gives the ECC2K-130 rounds and the fibre-scaling law; the corrections chapter gives the sheriff findings on the CM-pairs banner and the round-2 set-up defect (Table B.7, §B.6). This section records the constructions and the index-calculus measurements those summaries omit.

Certifying a gap without walking it. With a class-number-one CM field the gap can be certified from Frobenius alone (`research/conductor_gap_cm_pairs_20261010/README.md` §1). Take $D = -19$, a prime ℓ, and odd t with $q = (t^2 + 19\ell^2)/4$ and $N = q + 1 - t$ both prime. Then $t^2 - 4q = -19\ell^2$, so every ordinary curve of trace t has conductor 1 or ℓ; the curve with $j = -884736$ is the crater and every other curve of order N is a floor curve. Floor curves are found by sampling (a, b) and testing $[N]P = O$, about $19\ell/4$ samples per hit; no isogeny is computed, and for $\ell \geq 263$ nothing in the repository could exhibit one. The construction fails for $D = -7$ and $D \equiv 1 \pmod 8$ (conductor forced even) and for $D = -11$ ($3 \mid qN$ for every odd t); $D \in \{-19, -43, -67, -163\}$ work. Pairs were built at $\ell = 263$ (19-bit q), 8209 (29 bits) and 131101 (37 bits) and checked independently in Sage (Table 27.2).

The study's banner was the one sheriff cases `SHF-20261017-e4d7f6` and `SHF-20261028-0f3f64` withdrew and then corrected: it now reads that no implemented method shows a *reproducible* conductor effect, pending a rho re-test. Its Lean generic-invariance theorem covers only algorithms whose control flow depends on equality answers and a random tape, that is BSGS-type algorithms; the rho and kangaroo that ecbench runs hash point representations and are outside it (§4 of the source, corrected 16 October).

ℓ	q bits	curves	samples per floor hit	predicted
263	19	1 crater, 4 floor	$\approx$ 1.3k	$\approx$ 1.2k
8209	29	1 crater, 3 floor	$\approx$ 96k	$\approx$ 39k
131101	37	1 crater, 1 floor	$\approx$ 970k	$\approx$ 620k

Table 27.2: Certified conductor-gap CM pairs with $D = -19$: crater $j = -884736$, floor curves found by sampling and one scalar multiplication per sample. Observed rates agree with the prediction within the Poisson noise of one to three hits. Every curve was re-checked in Sage: $\#E = N$ by SEA, the j-invariant, the Frobenius discriminant $-19\ell^2$ and an embedding degree above 20. Source: `research/conductor_gap_cm_pairs_20261010/README.md` §§2–3.

Family, cofactor	Arm	S/blind	yield/blind	cost per relation/blind
prime, $h \le 25$	closed	0.50 [0.21, 2.36]	11.7	0.094
	lifts	1.83 [1.00, 3.02]	11.7	1.01
prime, $h \ge 50$	closed	4.7 [2.3, 75]	116	0.017
	lifts	3.8 [1.8, 29]	114	0.86
random binary, $h \in \{4, 6, 8\}$	closed MITM	5.8 [3.9, 7.9]	3.5	0.65
	lifts MITM	1.02	3.5	0.99
Koblitz, $h \in \{2, 4\}$	closed MITM	3.0 [1.9, 4.0]	2.2	0.78
	lifts MITM	1.01	2.2	1.04

Table 27.3: Fibre-aware arms against the blind oracle on the same curve, factor base and targets (medians, with the range for S). S is the cold whole-pipeline cost over $\sqrt{r}$, a lower bound. Source: `research/isogeny_conductor_gap_ic_20261010/README.md` §3.1.

Fibre-aware relation generation across the gap. The index-calculus study of the same date built 38 prime-field pairs with gaps 121 to 1458, five Koblitz pairs and three random binary pairs, and three fibre-aware oracles: *lifts* (probe every cofactor lift of the target), *closed* (a pair table holding P_i+P_j+K for every cofactor-torsion K) and *combined* (one Weil-descended system with the fibre polynomial adjoined) (`research/isogeny_conductor_gap_ic_20261010/README.md`). Table 27.3 gives Part A.

Yield scales with the fibre, up to the cofactor h, as predicted. Only the closed table cuts the relation phase's cost per relation, and it pays in set-up: it wins on prime curves with $h \le 25$ (S falls to 0.21–0.36 of blind on the 25-bit curves) and loses everywhere else, up to 75× blind at $h = 784$. The F4 solving degree does not move for lifts (mean 2.96–3.00). No fibre arm beats rho: the best cell, closed MITM on the 25-bit curves, has $S \ge 21$ against rho's 1.0–1.4. The combined Weil descent is negative, measured on warm-up runs only, because host reboots ended all four attempts at its session. The study is classified as engineering on the relation phase of cofactor curves; the conductor plays no role in any of these ratios. The next round's correction A0 (a floor/crater set-up ratio of 3.10 that was a counting defect, corrected to 1.00) is in Table B.7.

Drastic gaps. Two rounds pushed the gap to $\ell \approx 2^{19}$ to 2^{21}, 4 to 16 times the largest earlier gap.

- Round 1 built six certified pairs with $D = -19$ and $D = -43$ at $\ell \in \{508433, 1016839, 2033609\}$, eight one-target workloads per curve (`research/drastic_prime_conductor_gap_20261012/README.md`). The rho floor/crater ratios range from 0.529 to 1.293; only one 95% interval excludes 1 ($D = -43$, $\ell = 1016839$: 0.529 [0.306, 0.874], permutation $p = 0.038$), and its 99% interval does not. Index calculus was not run at these sizes.
- Round 2 took the $j = 0$ crater of $D = -3$, whose automorphism group has order 6, at $\ell = 1016839$ and 2033609 with 32 workloads per curve (`research/drastic_prime_conductor_gap_round2_20261012/README.md`). The same-method rho ratio sits at 0.947 and 1.060; the best-method ratio (`rho.cubic` on the crater, negation rho on the floor) is 1.692 [1.217, 2.362] and 1.730 [1.247, 2.461] at 99%, within 3% of $\sqrt{3} = 1.7321$. The round-1 lead at $D = -43$ re-measured on 32 fresh targets at 0.947 [0.662, 1.347]: target-sampling noise.

The $\sqrt{3}$ is the automorphism group, not the conductor. Round 3's confirmation and its later set-up correction are in Chapter 26 and §34.1.

CM orbits and a generic recurrence. The 15 October study ran an orbit-folded index calculus and rho on 100 gap pairs (`research/conductor_gap_cm_orbits_20261015/REPORT.md`). All 500 primary

attempts verified (300 IC, 200 rho). Floor/crater native CPU for index calculus is 0.980 [0.866, 1.111] and the query-count ratio 0.878 [0.764, 1.009]; neither establishes a difference. A post-hoc paired rho diagnostic gives CPU 1.174 [0.999, 1.383] and charged work 1.160 [0.986, 1.367]. The replication cohort shows how fragile such readings are: fresh targets give 0.988 [0.845, 1.149] in queries, but isomorphic coordinates give 1.397 [0.882, 2.056] and $D = -43$ gives 1.375 [0.998, 2.011], whose CPU interval narrowly excludes 1 in an unadjusted exploratory subset. All timings are co-scheduled L0, with crater and floor shards in mostly separate queue waves, so the source calls them descriptive. The study also found that its baseline was too weak: the CM action satisfies $\lambda^2 - \lambda + 5 = 0$ modulo the group order, so the generic recurrence $P_{k+2} = P_{k+1} - [5]P_k$ generates the same orbit on either endpoint without a degree-5 endomorphism; per 8,192-point construction the crater CM map costs 5.265 ms, either recurrence 6.64 ms and a direct floor scalar multiplication 41.494 ms. The comparison against repeated full scalar multiplication overstated the advantage the crater's endomorphism offers. A 96-bit rational-family floor was certified separately; ECDLP timings remain at 53 bits.

Follow-ups and archives. The five follow-up experiments of 16 October (interleaved crossover, relation-yield distribution, orbit geometry, conductor-size trend, rho partition controls) were frozen and launched; at the published checkpoint 182 crossover receipts existed (178 recoveries, four wall-censored), which the source calls execution status, not a comparison (`research/conductor_gap_followups_20261016/README.md`). Rounds 5–7 of the ECC2K-130 series, the replication, size-ladder and crater-design directories, and the 60- and 80-bit archival imports have records or PDFs but no README result statement and no admitted row; this chapter cites no number from them.

Verdict. No index-calculus or generic measurement attributes a cost difference to the conductor. Every apparent effect so far was accounting (round 2's A0), sampling (the round-1 lead), the coordinate model (Table 27.1 and §B.6) or the automorphism group of a $j = 0$ or $j = 1728$ crater.

27.4 Isogeny-derived factor bases

By Lemma 26.1, a factor base built from an isogeny is a base on one curve with a particular point set and presentation. Three studies tested such bases against a non-isogeny control.

- *Horizontal isogenies at $p = 2^{61} - 1$.* On a prime-order curve over $\mathbb{F}_{2^{61}-1}$ (cofactor 1, trace -1604388101), all 15 outgoing kernel-distinct rational isogenies of prime degree at most 61 were built, one hop only, and the factor base was mapped through each (`research/prime61_isogeny_20261016/README.md`). Vertical separable isogenies do not exist in this class ($f_\pi = f_E = 1$). All 16 workloads, 12 paired streams each, reached their caps without a relation: a null relation-stage result, not a measured improvement. Coverage of the whole horizontal class is blocked on memory.
- *Isogeny kernels against Petit's μ_k.* At $m = 2$ and toy primes, the isogeny-kernel base matches a random base in yield and costs more only through its presentation (§23.1).
- *The PKM isogeny tower.* Its rows and their distance to rho are in Chapter 21 and the frontier table (Table 36.4); the seed of that tower is the next section.

27.5 Isogeny seeds

The algebraic arm `pkm-iso-t6-g4` builds its base as $V = x(R' + \langle Q \rangle)$ on an auxiliary curve, and the seed selects the auxiliary curve and the coset; the F4 system has the same shape for every seed. A 3 November study selected the seed by a log-free coverage count J on Curve25519-family miniatures (`research/curve25519_isogeny_seed_20261103/README.md`).

- *Census* (exact, 2,000 seeds per cell, $t = 4$–7): the selected seed covers 1.62–3.24 times the mean seed in all 15 cells with $t \geq 5$ and equals the oracle-best seed in all 20 cells. At $k = 20$, $t = 6$, seed 1093 covers 528 points against seed 0's 172 (3.07×); seed 0 sits at the 6th percentile.
- *Engine*: seed 1093 needs $1.75 \cdot 10^8$ F4 multiply-adds per relation against $5.42 \cdot 10^8$, 3.10× fewer, at the same degree (5) and per-target cost (0.991×).
- *End to end* (ecbench session `ECBS1h405f20219036`, 6 targets, 48 of 48 solves verified): $S = 26.448$ against 29.892 at $w = 0$, and $1.073 \cdot 10^6$ against $1.674 \cdot 10^6$ at $w = 0.1$; negation rho is at 1.447, 1.547 and 2.448 across the band and the projected generic baseline at 12.053, 12.931 and 20.831. The paired ratio is 0.885 [0.570, 1.373] at $w = 0$ and 0.641 [0.379, 1.084] at $w = 0.1$ and 1.

The preregistered criterion V2 (a mean ratio at most 0.6) failed, and the interval includes 1, so the session alone does not establish the gain; the source reports that a later run (T9) replicated it on 16 targets. Two details matter for the reading. Seed 1093 has 46 columns against 25, so it needs about 1.84× more relations, which is why 3.1× cheaper relations net about 1.7×; and the seed search, which ecbench does not see, cost $1.35 \cdot 10^7$ group operations, which charged to one target makes the band order unresolved and amortised per curve leaves it ordered. Neither version changes the order against rho: the algebraic arm with seed 1093 is 18× negation rho at $w = 0$ and about $4 \cdot 10^6\times$ at $w = 1$. The corrections chapter records the failed V2 (Table B.6).

27.6 Isogeny computation as tooling

Several studies built isogeny machinery without an ECDLP claim; they are recorded because later studies rest on them.

- *Large prime degrees.* Every algorithm of Benčina, Kutas, Merz, Petit, Stopar and Weitkämper on fixed-degree isogenies between supersingular curves, and all of $\sqrt{\text{élu}}$ (Bernstein, De Feo, Leroux and Smith) beyond the codomain-and-map core that existed before, were implemented end to end and tested against an independent computation, except the quantum speed-ups, for which there is only a query-count model (`research/fixed_degree_isogenies_20261016/README.md`). The $\sqrt{\text{élu}}$ kernel-cost exponent 0.822 $[0.805, 0.838]$ is in Chapter 26.
- *Modular polynomials.* CM modular polynomials were built by explicit CRT (CM order selection, norm-equation primes, certified Hilbert polynomial seeds, paired class actions on surface and floor), with an exact-equality acceptance run; no speed is claimed (`research/cm_volcano_crt_20261016/REPORT.md`). The $n = 83$ computation of Φ_{6473} in §27.2 is the largest use.
- *An isogenous Boolean system.* Over $\mathbb{F}_{2^{37}}$ a separable degree-73 isogeny of a Koblitz curve (74 rational cyclic kernels, a squarefree degree-36 kernel polynomial) gave an isogenous neighbour whose Boolean decomposition systems were compared with the source's: all 128 systems completed with every planted decomposition verified on both curves, F5 matrix work fell by at least 10% in 4 of 8 registered cells and rose by up to 2.00× in another, and the preregistered criterion failed (`research/koblitz_isogeny_descent_37_results_20260925/RESULTS.md`). By Lemma 26.1 the two systems describe the same relations; a presentation can change the solver's work in either direction.
- *Isogeny-loop endomorphisms.* Round 2 of the isogeny-loop GLV work corrected round 1's pricing of the Verschiebung in characteristic two, $V = [2] \circ F^{-1}$ (a doubling composed with square roots, not two squarings), which turns round 1's lead into a provable loss (`research/isogeny_loop_glv_20261011/README.md`; Table B.5); the four-dimensional GLV–GLS speed-up of 1.09–1.15× is in Chapter 26. These speed up curve arithmetic, not the discrete logarithm.

27.7 What the isogeny side settles

- On prime fields the isogeny class contains no weak curve, by Tate; walks of standard curves are tooling.
- Generic cost is level-blind to within a few per cent on tall volcanoes; the only level effect is the crater's automorphism group.
- Index-calculus yield follows the factor base's point set, which depends on the coordinate model more than on the level; no conductor effect has survived its controls.
- Isogeny-built factor bases are single-curve bases with a presentation cost; selecting the seed is a constant-factor lever that leaves the algebraic arm orders of magnitude above rho.

28 Twists, pairings and lifts

Three classical transfers move the discrete logarithm out of the curve group: to the quadratic twist's eigenspace of $E(\mathbb{F}_{q^2})$, to a multiplicative group by a pairing, and to a p-adic or global lift. Chapter 8 proves that twist points carry no relations for the curve and records the lift-dense subspaces, the function-field xedni bound and the oracle-simulation results. This chapter adds the measured twist studies beyond that obstruction, the pairing and embedding-degree closures, the anomalous-curve control, and the p-adic and global-lift lanes with their corrections. Most items are theorem closures of known results; they are recorded because each retired a concrete proposal of the programme. Measured figures are producer-reported unless a pcc claim is named, and no pcc standing was computed for this draft.

28.1 Twists beyond the obstruction

Over $\mathbb{F}_{q^2}$, $E(\mathbb{F}_q)$ and the twist $E'(\mathbb{F}_q)$ are the $+1$ and -1 Frobenius eigenspaces of $E(\mathbb{F}_{q^2})$, and $\#E(\mathbb{F}_{q^2}) = \#E(\mathbb{F}_q) \cdot \#E'(\mathbb{F}_q)$. Applying Frobenius to a decomposition $R = A + B$ of $R \in E(\mathbb{F}_q)$, with A a sum of E-points and B of twist points, gives $2B = O$ (Chapter 8). Three studies measured what that leaves.

Exact yield on toy prime fields. The 20 October study added twist abscissas to an $x \in \mathbb{F}_p$ factor base on $y^2 = x^3 + 2x + 3$ and solved S_3 for every non-2-torsion target, at $p \in \{101, 211, 233, 307\}$ with 24 abscissas of each kind (`research/twist_index_calculus_20261020/REPORT.md`). The union has 1,176 unordered candidate pairs per target against 300, 3.92 times as many, and gives **0** extra width-two hits on every field, against 569–574 hits from E alone. The twist abscissas do produce relations, 569–573 per field, but for targets on the twist. Each curve has the 2-torsion point $(-1, 0)$; 3–7 twist pairs per field sum to it, and every resulting three-factor equality reduces to $R = F + (-1, 0)$, a shorter relation plus a known translation. The source notes that counting raw roots of a higher summation polynomial without lifting and reducing them can overstate useful yield for exactly this reason.

A case where the twist is usable. For $p \equiv 2 \pmod 3$ the supersingular curve $E_0 : y^2 = x^3 + 1$ has a rational 3-isogeny to $Y^2 = X^3 - 27$, which a rational scaling maps to a nontrivial quadratic twist of E_0. The composition carries a prime-order subgroup to the twist; it was checked on every ordered pair of points at $p = 101$ and 233 (10,404 and 54,756 homomorphism checks) and on a transported scalar. Width-two hit counts on transported targets then differ in both directions with the base (for example 92 against 95 at $p = 101$, $b = 24$, and 12 against 15 at $p = 233$, $b = 16$, where more hits cover fewer targets). The source reads this as transport and local count differences, not a speed-up; these curves also have a small embedding degree (§28.2). It is an instance of Lemma 26.1: an explicit rational map moves the target, and only the destination's base geometry and arithmetic can differ.

Seven twist ideas, preregistered. The 27 October study froze one preregistration per idea before data and had each result replicated and attacked by independent agents (`research/twist_ecdlp_20261027/README.md`; sizes up to 40 bits, operation counts primary). No arm that used the twist beat its matched baseline (Table 28.1).

The reference arm G1, a negation-map rho, solved every instance; the Kummer-line walk cost 1.287–5.907× it, with no row below 1. The study's open items are the mixed-solution rate at 2^{20}–2^{24} and $m \geq 4$, factor bases over extension fields, a measured genus-2 Jacobian arm, and pricing every arm across the band.

A uniform S_3 census of twist transfers. A long-running lane (`research/quadratic_twist_evidence_20261021/README.md`) executed an explicit rational dual of a twist transfer at $\mathbb{F}_{13}$, a uniform census (96 arms, 896 source recoveries, 210 policy comparisons, 0 of 210 strict joint gains in rank or degree) and complete size-two and size-three studies at $\mathbb{F}_{37}$: 336 arms at caps 0–12, 4,368 profiles and zero strongest-source joint records, on eighteen repeated original points. Its own status line keeps the original goal, a useful-yield or degree advantage from an actual twist, open; the registry still lacks 1,921 of its study rows. The lane proves conditional algebra and finite transport, not a yield or speed statement, and this treatise cites it only for that. Two November proposals (a follow-up and a chord preflight, the latter `PCC-20261105-7179e4`) have not run.

	Idea	Status	Key number
E0	separation lemma $2B = O$	supported	0 violations in 8,254 + 34,498 classes
E1	mixed-solution census (S_3, x-only)	structure supported; rate inconclusive from 2^{20}	0 mismatches over 91,200 + 508,800 targets; rate ratio 0.9965 (12 bit), 0.9935 (16 bit), 0.85 [0.29, 1.47] (20 bit)
E2	joint $E + E'$ table for twin targets	gain refuted	1.000–1.062× separate runs
E3	$\mathbb{F}_{q^2}$ eigen-decomposition	a loss	53.3–2,812× more work per E-relation
E4	Koblitz $K_0 \leftrightarrow K_1$, τ-orbit base	no gain	identical 773,404 relations at 2.0246× the solves
E5	Kummer-line multiplicative walk	inconclusive	4.6–6.2× the baseline; falsifiers fired
E6	genus-2 gluing of $E \times E'$	construction only	$\#J(C) = \#E \cdot \#E'$ on 580 of 580

Table 28.1: The twist ideas of the 27 October study. The genus-2 arm's "no gain" comes from a cost model and is not measured. E1, E2, E4 and E6 had replication mismatches that were diagnosed as incomplete replication runs and were not re-replicated. Source: `research/twist_ecdlp_20261027/README.md`.

Lift-dense lines, re-checked. The obstruction chapter records that on composite-degree binary fields suitable $\mathbb{F}_{2^d}$-lines lift entirely when $\mathrm{Tr}(a) = 0$. Two 6 November studies checked claims that arrived from another session as unverified assertions, each with a frozen protocol and a fresh instrument:

- *The 2-torsion pairing.* A valid line stable under adding the 2-torsion point exists exactly when $\mathrm{Tr}_{N/d}(b^{1/4}) = 0$, and it is then $b^{1/4}\mathbb{F}_q$ (48 of 90 toy curves, 0 mismatches); on those lines $x(P+T) = \sqrt{b}/x(P)$, so their columns pair and they give about 2×, not 2^n. None of the valid lines of c2pnb176v1, 208w1, 272w1, 304w1 and 368w1 (180, 180, 270, 210 and 330 lines, reproducing the October counts) is stable, so no earlier count changes (`research/lift_dense_torsion_pairing_20261106/README.md`; `PCC-20261106-b5904a`, `PCC-20261106-74c86c`).
- *Odd characteristic.* For lines $x_0 + u\mathbb{F}_q$ on $y^2 = f(x)$ over $\mathbb{F}_{q^n}$, a line is all-lift or all-twist when, for every Frobenius orbit O of the roots of the shifted cubic, $n\, r_O / s_O$ is even. All 1,287 lines meeting this criterion are constant; at $(q, n) = (101, 2)$ and $(101, 3)$ none of 3,520 lines failing it is; one chance-constant line occurred at $(13, 3)$. On descent lines, which exist on subfield curves up to isomorphism, the dead fraction of lifted points is 0.505–0.615 per cell, but the registered rule for *which* half is dead mispredicts 405 of 27,880 abscissae (`research/lift_dense_orbit_parity_20261106/README.md`; `PCC-20261106-3e6fc4`).

These are the only items in this chapter with pcc claim records; the producers label every statement in them unverified. They refine the count of usable lines; they do not price a factor base against rho.

The order-24 twist action on $j = 0$ curves over $\mathbb{F}_{p^4}$, which changes the matched rho on those curves, belongs to the extension-field programme and is in §31.5.

28.2 Pairings and the embedding degree

MOV and Frey–Rück. The Weil or Tate pairing with an independent N-torsion point maps $\langle P \rangle$ injectively into $\mu_N \subset \mathbb{F}_{p^k}^*$ with $k = \mathrm{ord}_N(p)$, where the discrete logarithm falls to finite-field index calculus. The atlas closes the lead as known (`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-mov-small-embedding-degree.yaml`): k is a function of (N, p) and so an isogeny-class invariant (§27.1), and for random prime-field curves it is of size about N (Balasubramanian–Koblitz). The transfer is a check on a curve, not an ECDLP algorithm. The programme's scoping question for pairings, `RQ-PAIR-21313f`, has toy checks specified and zero runs. Curve constructions in this programme check the embedding degree as a matter of course; the conductor-gap pairs of §27.3, for example, all have embedding degree above 20.

Two routes to a second torsion direction. Two proposals tried to obtain MOV without a second independent point; both are closed by theorem.

- *A self-pairing from one point.* On a cyclic group of prime order every bilinear map is a multiple of one fixed map, so a nontrivial self-pairing on the rational N-torsion line needs all of $E[N]$ rational,

that is $N \mid p^k - 1$ with $k = 1$. An endomorphism of an ordinary curve commutes with Frobenius and preserves the rational eigenline, so it is not a distortion map (`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-one-point-self-pairing.yaml`).
- *Trace-zero varieties and (N, N) graph kernels.* By Balasubramanian–Koblitz, $E(\mathbb{F}_{p^k})[N]$ is larger than the rational line only if the embedding degree divides k; otherwise the trace-zero variety $T_k(E)[N]$ is zero, since trace acts as $[k]$ on the line. An $\mathbb{F}_p$-rational (N, N) kernel of the Kani–Robert type in $E \times E'$ exists only when the embedding degree is 1. Both reduce to MOV (`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-second-torsion-direction.yaml`).

What stays open on the pairing side. Two leads remain, neither with a run.
- *The target-field cost of pairing curves.* For BLS12-381 and BN254 the ECDLP is bounded by the DLP in $\mathbb{F}_{p^{12}}$ after MOV; the quoted security levels come from one family of exTNFS estimates. A recomputation with memory charged has 33 experiment contracts and zero runs; its next step is a certified toy tower-field DLP and a same-shape calibration against a published record (`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-pairing-target-field-recost.yaml`). It bears only on pairing curves.
- *Miller-cocycle maps when the pairing is trivial.* With $\gcd(N, p-1) = 1$ the Tate pairing is trivial, but the Miller function gives a canonical non-homomorphic map $U(R) = f(R)^{1/N}$. If $\log U(kP)$ carried structure in k it would be a cheap transfer into $\mathbb{F}_p^*$; equidistribution at the $1/p$ rate would close it. The lead is unexamined (`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-miller-cocycle-nonhom-transfer.yaml`).

28.3 Anomalous curves and Smart's attack

When $\#E(\mathbb{F}_p) = p$, lifting to $E(\mathbb{Q}_p)$ and applying the formal logarithm makes the discrete logarithm linear in $\mathbb{Z}_p$ (Smart, Satoh–Araki, Semaev). It works exactly on trace-one curves, an isogeny-class invariant absent from generic and standard curves. The programme used it as a positive control: in `EXP-IT-001` a Smart certificate at $p = 771853$ ($N = p$, trace 1) cost $C_{\text{special}} = 157 = \lceil 8 \log_2 p \rceil$ against a matched rho of 779, a ratio of 0.202 below the control threshold 0.7 (`crypto-autoresearcher:ledger/evidence/EV-IT-aefd12.yaml`). A toy study re-derived and replicated the known degeneracy: Smart's attack fails exactly on the canonical-lift class modulo p^2, where the lift is not random (`KN-FIND-3c7e19`, cited in `crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-anomalous-padic-lift.yaml`). The atlas lists the lead as `closed_known`, to be revisited only as a positive control or for a curve with $\#E = p$.

28.4 p-adic sections and the CZLIFT lane

The natural generalisation of Smart to non-anomalous curves is a computable section $E(\mathbb{F}_p) \to E(\mathbb{Q}_p)$ whose formal logarithm or p-adic digits linearise the logarithm.

The torsion section is the only homomorphic one. For $p \nmid \#E(\mathbb{F}_p)$ the only homomorphic section is the prime-to-p torsion section; the formal logarithm vanishes on torsion, as $\log_p$ kills the Teichmüller section of $\mathbb{F}_p^*$. The programme proved this at derivation tier, uniformly in $p \geq 5$ (`KN-FIND-7e2b94`): the section costs $O(\log n + k \log p)$ additions and Smart's functional vanishes on it, so the "full break" clause of the hypothesis is false (`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-padic-canonical-section.yaml`). The toy instrument agreed on 1,116 of 1,116 and 576 of 576 pairs (`EV-CZLIFT-ad28f5`). The one failure, a naive section of $\mathbb{F}_p^*$, was rejected with strength *strong* and explained as a size effect when $g^d < p$, not a p-adic one (`EV-CZLIFT-f2a401`; $p < 2000$). Any non-homomorphic section recovers d only at the random $1/p$ rate. One exponent-relevant question stays open: whether the maximal Fourier mass of p-adic digit characters of the torsion section decays faster than the $n^{-1/4}$ break-even (`KN-OPEN-d3a6f0`). The round synthesis `DEC-20261024-23d64d` records that the question `RQ-CZLIFT-b9f765` cannot close yet, with a control and a review outstanding.

Two November follow-ups. Two preregistered studies tested specific lift-derived observables.
- *H110, the p^2-lift defect.* A deterministic Hensel section into $E(\mathbb{Z}/p^2\mathbb{Z})$ is executable, but the scalar polynomial of its first formal-layer defect had the maximum possible degree $r - 2$ (81 to 945) on

all 28 scalar words, against frozen gates of 49 to 100, so 0 of 28 passed; all 32 random-section controls and both coordinate-isomorphic controls were also at maximum degree. All 9,096 targets passed the arithmetic checks, and every solved torsion lift had $\tau = 0$, which is the vanishing above (`research/p2_lift_defect_20261101/README.md`).

- *H109, a division-polynomial derivative oracle.* The invariant derivative of the order-r division polynomial at the target was tested for a constant-size character formula for scalar parity. With 10,674,040 field multiplications and 211,712 inversions charged and all 8,832 coefficient tuples enumerated, exact formulas were found on 0 of 28 rows; the candidate's best-match rate was 0.500–0.611 against 0.500–0.630 for 32 symmetry-matched controls (`research/torsion_division_derivative_20261031/README.md`).

Each rejects only its frozen family; neither establishes a lower bound.

28.5 Global lifts, heights and the CM canonical lift

Silverman's xedni calculus lifts (E, P, Q) to a curve over $\mathbb{Q}$ or a number field and hopes that the lifts satisfy a short Mordell–Weil relation. The obstruction chapter gives the exact function-field lift probability. Three items refine the global case.

The JKSST theorem, audited in Lean. Jacobson, Koblitz, Silverman, Stein and Teske's analysis derives the classic bound under "certain plausible assumptions". A kernel-checked audit of 1 November separates what is proved from what is assumed (`research/jksst_xedni_audit_20261101/README.md`). It repairs the conditional theorem and refutes several overbroad readings:

- rank, radius and minimum separation do not determine a lattice ball count: two rank-two lattices with the same coarse parameters have radius-one counts 5 and 3 (sup-norm 9 and 3);
- no fixed model discriminant bounds the Néron–Tate heights of *all* multiples of a nontorsion point by one constant, which refutes only the universal reading, not the points xedni selects;
- quadraticity, positivity and finite rank alone do not imply the uniform gap of Lang's height conjecture, which remains open.

The algorithm-specific bridges (selected-point heights against a global minimal discriminant, Lang's conjecture, the random-image model, the good-reduction correspondence) remain open; the audit does not claim the opposite unconditional statement.

Measured toy lifts. In the CZLIFT round of 23 October, small rational points grew as predicted but did not reduce uniformly, and a relation search found one relation, the tangent relation at $p = 101$, in 3,200 valid trials with coefficients bounded by 6, and none of 1,600 at rank 3; a planted dependence was recovered (`EV-CZLIFT-ee9b23`). Three hypotheses were weakened and two run sets declared inadmissible (`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-global-lift-small-height.yaml`). The lead stays open with a named next step: an explicit prime-field lift family that either composes every bridge or falsifies one.

The CM canonical lift: a corrected closure. Lane L9 asked whether the complex multiplication of the Serre–Tate canonical lift shrinks lifted heights enough for xedni. Its closure first rested on a norm-form gap: $N(\alpha) \geq |D_E|/4$ composed with $\hat{h}(\alpha P) = N(\alpha)\hat{h}(P)$ would place every CM companion outside the height budget. An independent red-team review found that this was not the load-bearing argument, and correction `CORR-20260810-930c42` (with `DEC-20260810-3110d5`) restated it (`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-cm-canonical-lift-heights.yaml`). The corrected closure covers ordinary prime-field Serre–Tate and global CM models of fixed arity 2 to 9 with an absolute coefficient bound, and twelve enumerated mechanisms. Its argument is a counting one: a target-independent route chooses a $p^{o(1)}$ output set S before Q, so $\Pr[Q \in \mathrm{red}(S)] \leq |S|/N = p^{-1+o(1)}$, an instance-count exponent of 1 against rho's 1/2. For generic $|D|$, writing the canonical lift through the class polynomial costs degree $|D|^{1/2+o(1)}$, a separate rho-scale gate. The corrected scope excludes target-dependent sections, growing arity or coefficient bounds, and the universal norm-form claim, which was false. The conclusion survives; its reason changed.

28.6 What the classical transfers settle

- The twist contributes only 2-torsion to a decomposition, exactly; every measured twist arm is at or above its baseline, and the lift-dense refinements change counts of usable lines, not distances to rho.
- Pairings help only at small embedding degree, which is a property of (N, p); the post-MOV cost of deployed pairing curves is unmeasured here.
- Smart's attack is a positive control. Every non-anomalous p-adic section the programme tried is either the torsion section, which the formal logarithm kills, or behaves like a random map.
- The global-lift lane is held by conditional theorems with named open premises, one of them repaired in Lean and one corrected in scope.

29 Weil descent and GHS covers

The Gaudry–Hess–Smart construction [21] maps the discrete logarithm on $E/\mathbb{F}_{q^n}$ to the Jacobian of a curve over $\mathbb{F}_q$ whose genus is governed by the magic number m. Where the genus is small, index calculus on the Jacobian is cheaper than rho on E. The treatise already records what this means for ECC2K-130 (magic numbers 1 or 131, with the Maurer–Menezes–Teske correction; §15.7), the measured trial cost on a genus-255 Jacobian that turned the $N = 130$ counterfactual's margin negative (§36.5), the cubic GHS route to genus 3 and its plane-quartic branch (Chapter 26, Table 26.1), the censuses at $n = 3$ and $n = 5$, and the Oakley groups (Chapter 26). This chapter adds three bodies of work: the descent screens of the curve registry and the corrections they went through, the exhaustive binary cubic catalog campaign over $\mathbb{F}_4$, $\mathbb{F}_8$ and $\mathbb{F}_{16}$, and the genus floors in odd characteristic. Every attack cost in it is producer-reported, and most are Magma wall times at toy field sizes.

29.1 Descent screens on the registry

The scan. A native screen read the registry of 321 curves (176 prime-field, 136 binary of which 72 random, 60 Koblitz and 4 subfield, and 9 extension-field toys) and enumerated 1,156 transfer routes, pricing each with one frozen cost model (`research/jacobian_transfer_scan_20261010/README.md`). Of the 321 curves, 314 were `boundary`, 3 `unresolved`, 2 `formula_advantage` and 2 `formula_marginal`. Every row is a formula, not an attack:

- every prime-field curve, including the 54 standardised ones, had best ratio -2.1 to -2.0 bits, on the same-field genus-2 cover, whose Jacobian rho has rho's exponent and four times its operation cost;
- every Koblitz, subfield and prime-degree binary curve was a boundary: Koblitz curves degenerate to $m = 1$, random-b prime-degree curves have $m = n$, and the 60 Koblitz curves sit at -3.3 to -3.2 bits on the genus-3 same-field cover;
- the two formula advantages were Oakley Group 4 (magic number 5, genus 15 over $\mathbb{F}_{2^{37}}$, $+10.9$ bits) and c2pnb368w1 ($m = 4$, genus 7 over $\mathbb{F}_{2^{92}}$, $+10.1$ bits); Oakley Group 3 was marginal at $+1.1$ bits.

The magic-number correction. Three days later an audit found that the trapdoor module had defined the magic number as the plain orbit dimension of $\sqrt{b}$ and decided the genus type by orbit length; both are wrong against Maurer, Menezes and Teske, whose magic number is $\dim \operatorname{span}\{(1, \sigma^i(\sqrt{b}))\}$ and whose type I is $(x-1) \nmid \mathrm{Ord}_b$ (`docs/audits/GHS_MAGIC_NUMBER_MMT_CORRECTION_20261013.md`). The scan's magic numbers were already MMT's; only its type rule was wrong. On regeneration no magic number changed, 252 GHS rows changed genus, 16 rows changed class and four curve verdicts changed (Table 29.1). Every $n = 2$, $m = 2$ row moved from genus 1, not carried, to genus 2, boundary. The ECC2K-130 consequences (the smallest nontrivial genus over $\mathbb{F}_2$ is $2^{130} - 1$, not 2^{129}) are in §15.7.

A second commit replaced the screen's bracket for $\mathrm{Tr}(a) = 1$ with even relative degree ("m or $m+1$") by MMT's Lemma 6 and its condition (8): a never changes the magic number, but it decides whether the descent exists (`research/jacobian_transfer_lemma6_20261013/README.md`). Across 2,091 registry curves, three rows changed (c2pnb176w1, c2pnb272w1 and c2pnb304w1, now exact at $m = 8$, genus 128); two of them changed verdict, c2pnb176w1 to `formula_advantage` ($+23.64$ bits) and c2pnb304w1 to `formula_marginal`. The c2pnb rows over $\mathbb{F}_4$ are the kind of formula row whose genus (128) is beyond any implemented Jacobian arithmetic; their advantage is a model output.

The $N = 130$ counterfactual, priced at the corrected genus. Over $\mathbb{F}_{2^{130}}/\mathbb{F}_2$ the old table priced genus 128 ($m = 8$) at $2^{36.41}$ against rho's $2^{60.31}$. Under MMT no b has $m = 8$; the attainable values start $1, 2, 5, 6, 9, 10, 13, 14, \ldots$ (`research/ghs_counterfactual_n130_20261013/README.md`). The least admissible outcome is $m = 9$, type I, genus 255 ($\mathrm{Ord}_b = \Phi_5(x)^2$), and a concrete b with these invariants and condition (8) was constructed. Its modelled cost is $2^{55.60}$ (smoothness bound 27, relations $2^{54.46}$, linear algebra $2^{54.60}$), a margin of -4.71 bits instead of -23.91, checked against the legacy pricer to 0.01 bit. The model crosses rho between genus 288 ($2^{59.91}$) and 296 ($2^{61.34}$). Table 29.2 gives every row.

A second, independent error in the old table took the attainable magic numbers for $l \geq 2$ from the factorisation of $x^n - 1$ over $\mathbb{F}_{2^l}$ instead of $\mathbb{F}_2$; it changes the $l = 2$ and $l = 26$ rows, which are not below rho either way. The source's own interpretation is that the remaining 4.71-bit margin does not absorb

Curve	Best route	Genus	Ratio (bits)	Verdict
Oakley Group 3	GHS to $\mathbb{F}_{2^{31}}$	$15 \to 16$	$+1.15 \to -3.04$	marginal → boundary
Oakley Group 4	GHS to $\mathbb{F}_{2^{37}}$	$15 \to 16$	$+10.94 \to +6.76$	advantage → marginal
c2pnb368w1	isogeny-extension to $\mathbb{F}_{2^{46}}$	$7 \to 8$	$+10.05 \to +7.92$	advantage → marginal
c2pnb272w1	GHS to $\mathbb{F}_{2^2}$, $m = 8$	$127 \to 128$	$+41.32 \to +72.20$	unresolved → advantage

Table 29.1: The four curve verdicts changed by the GHS type correction of 13 October. Ratio: $\log_2(\text{rho}) - \log_2(\text{route})$ under the frozen cost model, positive when the formula prices the route below rho. None is an attack. Source: `docs/audits/GHS_MAGIC_NUMBER_MMT_CORRECTION_20261013.md` §2.

l	n	old (m, g), cost	now (m, type, g), cost	below rho now
1	130	$(8, 128)$, $2^{36.41}$	$(9, \text{I}, 255)$, $2^{55.60}$	yes (model)
2	65	$(7, 64)$, $2^{299.93}$	$(13, \text{I}, 4095)$, $2^{43242.05}$	no
13	10	$(5, 16)$, $2^{68.63}$	$(5, \text{I}, 15)$, $2^{64.52}$	no
26	5	$(4, 8)$, $2^{60.80}$	$(5, \text{I}, 15)$, $2^{88.78}$	no

Table 29.2: The $N = 130$ GHS counterfactual recomputed under the MMT magic number, against rho's $2^{60.31}$. Rows with $l \geq 2$ use a fixed-genus model the source calls pessimistic for large genus, so "no" means "not shown to break". The $l = 5$, 10 and 65 rows, absent from the old table, are not below rho either. Source: `research/ghs_counterfactual_n130_20261013/README.md`.

its estimated 2^7 curve additions per relation trial; measuring that trial moved the margin to -1.83 bits (§36.5).

Other screens. Three further studies bound what the binary screens can claim.

- *Generalised GHS.* Explicit maps for Hess's generalised binary construction at degree 12 gave 147 screened profiles of genus at most two and 232 index-two genus-two profiles, all members of known families; matched negation rho was run on each source, and the study states that the measurements do not support a calibrated GHS advantage over rho (`research/ghs_generalized_20261024/REPORT.md`).
- *A first comparison.* The October 9 GHS round had no higher-genus solver and so no measured GHS cost; its rho against BSGS raw-cost ratios were 2.35 and 4.13 (`research/ghs_curve_comparison_20261009/RESULT.md`).
- *Teske-type hidden descents.* A trapdoor in Teske's sense needs a weakness that is not a class invariant, a transfer along the isogeny and a weak set sparse in the class (`research/teske_variants_field_types_20261018/REPORT.md`). Prime fields have none of the three. Over binary fields the sparse regime is at relative degree $n = 7$ for the type-I genus-7 descent only (margin $+0.5$, $+11.5$ bits at $N = 161$), while the type-II genus-8 descent and Hess's variant are dense; at $n = 15$, 21, 31 the public sub-descents (for example $n = 5$ inside $n = 15$) already beat rho. In odd characteristic the analogue of Teske's field is $\mathbb{F}_{q^5}$ with a genus-5 family, by a count of weak sets of $\approx q^{r-3}$ curves reproduced at $q \leq 13$.

29.2 The binary cubic GHS catalog campaign

From 26 to 28 October about 25 studies classified, exhaustively, which ordinary elliptic curves over $K = \mathbb{F}_{q^3}$ with $q \in \{4, 8, 16\}$ admit a genus-3 or genus-4 Jacobian transfer to $k = \mathbb{F}_q$, constructed explicit maps where they exist, and ran complete attacks against a matched rho on the source subgroup for a few. The treatise cites only the genus-3 subcover study among them (Table 40.4). The campaign is a closed unit: an exact classification with uniformly negative toy attacks.

The census over $\mathbb{F}_{16}$. For $k = \mathbb{F}_{16}$ and $K = \mathbb{F}_{4096}$ the normalised ordinary cohort has 8,190 K-isomorphism classes and 128 group orders (`research/ghs_cubic_catalog_q16_20261026/REPORT.md`). The LMFDB catalogue of abelian varieties records a genus-3 Jacobian in 126 of the 128 cubic Weil-restriction isogeny classes, which contain 8,142 of the sources. Of these, 5,534 have no direct binary cubic genus-3 GHS cover even after arbitrary K-isogeny walks, and 5,302 of those pass a simple-root gate for an isogeny nonzero on the largest prime subgroup. The source calls this a catalog-backed eligibility

Class of source	Orders	Models
all normalised ordinary curves over $\mathbb{F}_{4096}$	128	8,190
catalogued genus-3 Jacobian in the Weil-restriction class	126	8,142
no direct cubic genus-3 GHS cover, after any K-isogeny walk		5,534
and the largest prime subgroup passes the simple-root gate		5,302
irreducible Frobenius polynomial (simple restriction)	120	7,584
outside the Hess genus-3 closure	46	2,691
with a catalogued genus-3 Jacobian	44	2,643
without one	2	48
reducible Frobenius polynomial	8	606
excluded for all of $\ell = 3, 5, 7, 11, 13$ (one-step (ℓ, ℓ, ℓ))	2	72

Table 29.3: The exact partition of the $\mathbb{F}_{16}$ cubic census. The first block is the catalog join, the second the simple-restriction bound with the Hess closure, the third the odd-isogeny screen on the 44 catalogued simple residual classes. Sources under `research/`: `ghs_cubic_catalog_q16_20261026/`, `ghs_q16_simple_restriction_bound_20261027/`, `ghs_q16_odd_isogeny_screen_20261027/`.

bound, not an attack: the Jacobians are already catalogued and Tate's isogeny is not evaluated. Among the catalog-positive models with order $\equiv 4$ or $6 \pmod 8$, outside the Menezes–Teske walk set defined by order $\equiv 0$ or $2 \pmod 8$, are 4,032 class-C models, 3,852 of which pass the gate; this was noticed after the frozen join and recomputed in the final audit.

What the structure allows. Four exact results narrow the search (Table 29.3).

- *Hess's construction, closed.* Within Hess's Artin–Schreier representation the only genus-3 Frobenius-order pairs are $(1, 2)$ and $(2, 2)$, which gives an if-and-only-if criterion: $\delta = \sqrt{b} \in P$ when $\mathrm{Tr}(a) = 0$ and $\delta \in U^*$ when $\mathrm{Tr}(a) = 1$, with $|U^*| = q^2 - 1 = 255$ and $|P| = (q-1)(q+1)(q+2)/2 = 2{,}295$ (`research/ghs_cubic_hess_closure_q16_20261027/REPORT.md`).
- *A genus lower bound for any correspondence.* If the Frobenius polynomial $T^6 - tT^3 + 4096$ of $\mathrm{Res}_{K/k} E$ is irreducible, the restriction is simple and any nonzero homomorphism to a Jacobian needs genus at least 3; irreducibility holds for 120 of the 128 orders (7,584 models). Independently, a prime-order target $r > 625$ needs genus at least 3 by the Weil bound $r \le 25^g$ (`research/ghs_q16_simple_restriction_bound_20261027/REPORT.md`).
- *One-step odd isogenies.* A rational (ℓ, ℓ, ℓ) isogeny needs a monic cubic divisor of the Frobenius polynomial modulo ℓ. Among the 44 simple classes outside the Hess closure with a catalogued genus-3 Jacobian, $\ell = 3$ excludes 18 classes and $\ell = 5$, 7, 11, 13 exclude 21–23; exactly two classes (orders 4076 and 4118, 72 models) fail for all five primes. At $\ell = 3$ there is a closed form: for $q = 4^m$, a trace $t \equiv 0 \pmod 3$ excludes every rational one-step $(3, 3, 3)$ isogeny (`research/ghs_q16_odd_isogeny_screen_20261027/REPORT.md`).
- *Quotients and neighbours.* The four hyperelliptic genus-3 curves in the class of the order-3974 source have no separable degree-2 map to an elliptic curve (`research/ghs_cubic_quotient_obstruction_20261026/REPORT.md`); automorphism and degree-3 neighbour screens at $q = 4$, 8 and 16 exclude further direct covers, including the order-526 and 542 tails at $q = 8$.

A scalar-3 Frobenius gate for a $(3, 3, 3)$ transfer is self-labelled a rediscovery: the repository's prime-characteristic cubic-transfer work had already derived the scalar-3 and Jordan module screen; the binary study extends it to $q = 16$ with an exact count for every $q = 4^m$ (`research/ghs_scalar3_frobenius_gate_20261027/REPORT.md`).

Witnesses. Searches produced explicit curves in the classes that matter.

- Over $\mathbb{F}_4$, 2,048 drawn plane quartics gave 55 smooth ones with the right point counts, 48 of which match an elliptic source over $\mathbb{F}_{64}$, in 12 order classes; 8 of the 12 have no direct genus-3 source. The LMFDB already lists Jacobians, including plane quartics, in all eight classes, so no new eligible class follows (`research/ghs_binary_cubic_isogeny_20261026/REPORT.md`).
- Over $\mathbb{F}_{16}$, three class-C source orders gained plane-quartic witnesses, and a 30,000-draw search on 13 residual orders found four smooth quartics in matching classes (orders 4006, 4054, 4078, 4126); the source-to-quartic maps were not built (`research/ghs_q16_residual_quartic_search_20261027/REPORT.md`).

Study	Instance	Attack / rho
Hess genus 4, four residual sources	orders 4006, 4054, 4078, 4126; $r = 2003$–2063; 512 rho trials each	512, 698, 698, 688×
Genus-4 frontier	order 3974, $r = 1987$: 13 relations, 0.22 s against 0.00102 s	216×
	order 4178, $r = 2089$	198×
	order 3978, $r = 17$	3,300×
Nonsplit, order 4222	plane-quartic line sieve, same Magma backend	about 271×
Nonsplit V_4, order 4054	attack core 0.360 s CPU against 0.0016 s per rho solve	about 225×
Genus-3 subcovers	15 exact equations, orders 3970, 4098, 4162 (Sage)	549–2,039×
	two nonzero-trace orbits, class `3.16.a_a_ah`	1,236–1,430×
Hess c-parameter audit, $q = 8$	$\#E = 548$, $r = 137$; $\#E = 556$, $r = 139$	2,304, 2,784×
$\mathbb{F}_8$ quartic panel	Frobenius-cyclic quartic, rho median 0.000554 s	about 623,000×

Table 29.4: Complete GHS or cover attacks of the binary cubic campaign against a rho on the same source subgroup and target, in Magma or Sage CPU or wall time at $q \leq 16$, on unisolated hosts. Ratios of medians or means as each source states them. Sources under `research/`: `ghs_q16_4126_hess_g4_20261027/`, `ghs_q16_genus4_frontier_20261028/`, `ghs_nonsplit_order4222_20261028/`, `ghs_nonsplit_v4_cubic_20261027/`, `ghs_q16_genus3_subcovers_20261028/`, `ghs_hess_c_parameter_audit_20261026/`, `ghs_q8_quartic_cover_panel_20261026/`.

- The cubic Tate bridge bounds the number of $\mathbb{F}_{4096}$ classes for which a complete attack beats rho between 0 and 2,608 (orders in $[3969, 4225]$): an eligibility count, not a measurement (`research/ghs_cubic_tate_bridge_20261026/REPORT.md`).
- Four genus-4 Hess curves have explicit degree-4 maps to their elliptic complements, with Lean identities (`research/ghs_q16_hess_elliptic_quotient_20261027/REPORT.md`).

The end-to-end attacks. Table 29.4 lists every complete transfer attack of the campaign against a rho on the same source subgroup and target. Every one loses. The range is wider than "216 to 2,784 times": one genus-4 representative reads 198×, the order-3978 representative, with an order-17 subgroup, 3,300×, and the $\mathbb{F}_8$ quartic panel about 623,000×.

Three caveats apply to the whole table. The times are Magma `Cputime` or `Realtime`, or Sage wall time, with hundredth-second granularity against sub-millisecond rho solves, so a ratio carries a large resolution error at the small end. The subgroups are of order about 2^{11}, where the constant terms of a cover pipeline (map construction, factor-base set-up) dominate. And the rho is a plain or Floyd walk in the same calculator, not the strongest reference. None of these favours rho; they make the ratios imprecise, not reversible. On the asymptotic side the genus-3 route is the one with an exponent below rho's ($q^{4/3}$ against $q^{3/2}$), and its measured behaviour at larger q is in Chapter 26; genus 4 has exponent $q^{2-2/4} = q^{3/2}$, the same as rho's on $E(\mathbb{F}_{q^3})$, so only a constant could separate them.

29.3 Odd characteristic: genus floors

Which degrees reach genus 3. For the GHS compositum attached to a degree-2 function x on $E/\mathbb{F}_{q^n}$, the genus is $g = 1 + 2^{m-2}(r-4)$, where m is the $\mathbb{F}_2$-rank of the conjugate branch sets and r the size of their union; it depends on the Frobenius-orbit type of the branch locus, never on q (`research/odd_char_descent_genus3_20261014/REPORT.md`, Theorems 1–2). Genus 3 occurs only for $(m, r) \in \{(2,6), (3,5)\}$, so for $n \geq 4$ with $\mathbb{F}_q(j) = \mathbb{F}_{q^n}$ every x gives genus at least 5. For $n = 3$ the hyperelliptic genus-3 family $y^2 = e\, g(x)(x-\alpha)(x-\alpha^q)$, $\alpha \in \mathbb{F}_{q^3} \setminus \mathbb{F}_q$, exists for every odd q (Momose–Chao; Joux and Vitse [27] call it the only one). The tool checked Joux–Vitse's explicit model on 81 instances with $p \leq 31$: on every one the L-polynomial is that of $\mathrm{Res}_{\mathbb{F}_{p^3}/\mathbb{F}_p} E$. Explicit genus-3 builds, a plane quartic over $\mathbb{F}_q$ and a hyperelliptic model with the transfer executed by Cantor arithmetic, are in `research/ghs_cubic_genus3_build_20261012/` and `research/ghs_cubic_hyperelliptic_genus3_20261013/`; neither solves a logarithm.

The genus floor reprices the registry. The same derivation gives floors of 9, 49 and 41 at $k = 6$, 10 and 14 for any GHS cover of $E/\mathbb{F}_{p^k}$ with $\mathbb{F}_p(j) = \mathbb{F}_{p^k}$, where every `cover_descent` row of the registry had been priced at genus k (`research/cover_descent_genus_floor_20261015/README.md`). Priced

at the floor, 599 of the 846 rows that were `formula_advantage` stay so (576 at $k = 6$, 23 at $k = 14$), 41 become marginal and 206 boundary (all of $k = 10$, most of $k = 14$); curve verdicts move from 848, 50 and 1,201 to 601, 67 and 1,431 (advantage, marginal, boundary). The floor is a lower bound for GHS covers only, and no cover is exhibited, so the rows stay `proposed`. The corrections chapter lists this repricing (Table B.6).

The census at $n = 4$. Chapter 26 and §36.5 quote the floor 13 for odd-order $E/\mathbb{F}_{q^4}$. The census gives the full picture by 2-torsion type, exhaustive at $q = 3, 5$ (`research/ghs_cover_census_n4_20261024/README.md`):

- odd $\#E(K)$: 13 if $j \notin \mathbb{F}_{q^2}$, else 37;
- $E(K)[2] = \mathbb{Z}/2$: 25 on every curve, whether $\#E \equiv 2 \pmod 4$ or $4 \mid \#E$;
- $E(K)[2] = (\mathbb{Z}/2)^2$: 5 on most curves (26 of 28 at $q = 3$, 158 of 210 at $q = 5$), else 7 or 9.

Two preregistered predictions failed (floor 21 for $\mathbb{Z}/2$, observed 25; odd-class minima in $\{13, 25\}$, observed $\{13, 37\}$), both from errors in the producer's proofs found by the first run; after the amendment there were 0 violations over 1,256, 166 and 23 curve models at $q = 5$, 3 and 7. A structure counts as a descent to $\mathbb{F}_q$ only if the four conjugate quotients are distinct; σ^2-stable branch sets, with genus as low as 2, descend E only to $\mathbb{F}_{q^2}$ and are excluded. Genus 5 at $n = 4$ is $q^{8/5}$ in the index-calculus exponent against rho's q^2, but only on curves with full rational 2-torsion.

Decomposition coverage over $\mathbb{F}_{q^n}$. For a generic $E/\mathbb{F}_{q^n}$ and the base $x \in \mathbb{F}_q$, the share of points that are a sum of n base points was computed exactly over the whole group (`research/weil_descent_angles_20261018/REPORT.md`). In odd characteristic at $n = 2$–5 it matches a random x-set of the same size (paired residual $|z| \le 0.94$). The exception is binary $n = 2$, where the coverage tends to 3/8 instead of the Poisson value $1 - e^{-1/2} = 0.3935$ ($z = -4.72$), for a proved reason: every $x \in \mathbb{F}_q$ has trace zero over $\mathbb{F}_2$, so the base lies in one coset of $2E$ and only half the group can decompose. For an attack the relevant figure is coverage of $2E$, which is exactly 3/4, about 1.9× more relations per target than the model; the source classes this as a cost-model correction. End to end at $n = 2$ the decomposition route still costs 2.7–66× rho in every pricing cell. The Kummer and Dickson tower bases on prime fields show no coverage effect either (18 cells, $|z| \le 1.84$). A genus-2 route at $n = 2$ (a Scholten-type curve with $J(C) \sim \operatorname{Res} E$) needs about $p/3$ Jacobian operations against rho's $0.886\,p/\sqrt{h}$; its ratio to rho spans 0.51–2.83 across the band, so it is unresolved at every size, and at 40 bits rho is cheaper in 10 of 12 cells.

Cubic extensions in odd characteristic. The native follow-up of a historical Sage study certified the rational 3-torsion action and ran 58 native stage and control runs; a complete native transfer is missing (`research/cubic_extension_transfer_20261025/`; pcc claims `PCC-20261105-a43948` and `PCC-20261105-a6723f`, a development and a proposal, standing not computed here).

29.4 Trapdoors in the isogeny class

Teske's trapdoor uses the fact that descent weakness is not a class invariant over composite binary fields: a GHS-weak curve over $\mathbb{F}_{2^{161}}$ is hidden behind an isogeny that only its holder knows. §29.1 locates the fields where such a weakness can be sparse. Two atlas leads ask the same question elsewhere.

- *Prime fields.* By §27.1 no prime-field variant via descent exists; what remains is whether any non-generic index-calculus meter depends on the endomorphism ring (`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-teske-secret-isogeny-trapdoor.yaml`). A heavy-tail screen at 40-bit N found five classes homogeneous (`EV-ECTD-8c098d`, which also discloses a fabrication defect in its rho-substitution prose); a vertical screen was invalid because its meter read the same vector on all 32 curves (`EV-ECTD-4c8a3b`); and every trial of the approved 20-arm panel `EXP-ECTD-0cf008` died with an infrastructure error. The isogeny-fibre pullback theorem, by which summation polynomials pull back along isogenies, predicts homogeneity, and the conductor-gap results of §27.3 agree.
- *The Joux–Vitse weak class over $\mathbb{F}_{p^6}$.* Membership in the weak class is not class-invariant (a random full-2-torsion curve lies in a weak class with probability 0.59–0.63 at $p = 11$ to 47, §30.4), so a hidden weak curve is structurally possible. The open question is whether some classes place their weak curve beyond a cheap public walk; the lead is unexamined (`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-teske-trapdoor-fp6-weak-class.yaml`).

A trapdoor of either kind would be a construction for a designer, not an attack on a given curve.

29.5 What descent settles

- On the registry only composite-degree binary curves have descent rows with a formula advantage, and every one is a model at a genus no implemented Jacobian arithmetic reaches; two type and admissibility corrections moved those rows in both directions.
- Over $\mathbb{F}_{4096}$ the classification is exact: which sources have a genus-3 carrier, which isogeny steps are excluded, and which classes remain. Every complete attack measured on it loses to rho, by factors from about 200 to about $6 \cdot 10^5$ at toy sizes.
- In odd characteristic the genus is fixed by the branch-orbit type; it is 3 only at $n = 3$, and the floors at $n = 4$ depend on the 2-torsion, so prime-order-like curves over $\mathbb{F}_{q^4}$ need genus 13 or 37.

30 Jacobians of genus 2 to 5

Moving the problem into a Jacobian helps only when the Jacobian's index calculus has a better exponent than rho on the source and the transfer is cheap. Genus-g double-large-prime index calculus [22] costs $\tilde{O}(q^{2-2/g})$ over $\mathbb{F}_q$, and Diem's index calculus on a smooth plane curve of degree d [9, 11] costs $\tilde{O}(q^{2-2/(d-2)})$, $\tilde{O}(q)$ for a plane quartic. For a source over $\mathbb{F}_{q^n}$ the Jacobian has dimension at least n, and rho costs $q^{n/2}$: at $n = 3$ the genus-3 routes have exponents 4/3 and 1 against 3/2, at $n = 4$ genus 4 and genus 5 have 3/2 and 8/5 against 2. On a prime field the same argument runs the other way. Chapter 26 records the genus-3 hyperelliptic route against the strongest rho, the plane-quartic studies N4 and N4 v2 (Table 26.1), Smith's walk, and a summary of the Joux–Vitse cover; the rho references on Jacobians are in §34.5. This chapter adds the $n = 3$ Tian route and the N4 iteration between those two studies, the genus-4 and genus-5 Weil-restriction Jacobians, the same-field embeddings on prime fields, and the full Joux–Vitse account, with its validated sieve row (§30.4).

30.1 Plane quartics at $n = 3$

Theorem 4: when a Tian kernel exists. Tian's construction reaches a genus-3 Jacobian from $\mathrm{Res}_{\mathbb{F}_{q^3}/\mathbb{F}_q} E$ through an isogeny with a Frobenius-stable kernel. The 18 October census proved when such a kernel exists (`research/weil_restriction_isogeny_census_20261018/REPORT.md`, Theorem 4): for $\ell \neq p$ prime it exists if and only if $\chi_\varphi(T) = T^2 - tT + q^3$ has a root in $\mathbb{F}_\ell$ and every root is a cube in $\mathbb{F}_\ell$, except that a Jordan φ (a double root, φ not scalar) needs $\ell \equiv 1 \pmod 3$ as well. So a nonsplit φ never qualifies, a split $\mathrm{diag}(\lambda, \mu)$ qualifies exactly when λ is a cube, and a scalar φ always does. The census confirms every cell; the split, scalar and nonsplit cases are proved for all ℓ.

The Tian route measured. A lead agent built the tools and ran every measurement under a frozen protocol; no independent validator has replayed the runs (`research/tian_n3_quartic_route_20261024/REPORT.md`).

- *Census.* For every prime-order class at $q = 7, 11, 13, 17$ that Theorem 4 flags, the isogeny class of $\mathrm{Res}\, E$ contains the Jacobian of a smooth plane quartic over $\mathbb{F}_q$: 26 of 26. Unflagged prime-order classes are hit too, at about half the rate, so a quartic in the class is not evidence of a Tian kernel. No isogeny was computed.
- *Diem index calculus on generic quartics.* A complete double-large-prime index calculus solved and verified every planted instance up to $q = 8191$. With native work priced at 0 it costs 180–340 group additions against rho's 5,000–300,000; at $w = 0.1$ and 1 it loses, because its dense linear algebra grows like $q^{1.4}$. The order does not hold across the band, so the ranking is unresolved.

The Diem arm ran on generic quartics, not on the flagged classes, and no map from a source curve to its quartic was built. The benchmark copy `benchmarks/20261024-tian-n3-quartic-ic/` has no validation file.

The step between N4 and N4 v2. The 21 October iteration added double large primes (B2) and structured elimination (B1) to the frozen N4 baseline B0 on the GHS-weak family $E_0/\mathbb{F}_{q^3}$ with $4 \mid \#E_0$ (`research/quartic_n4_la_20261021/README.md`). B2's lines through two base points meet the quartic in two more rational points, which become large-prime edges; the base shrinks from $3.0\, q^{0.6}$ to $2.0\, q^{0.5}$ points, and both phases scale as about $q^{1.0}$. All 57 B2 units verified. The paired ratio B2/B0 is 0.355 $[0.324, 0.392]$ at $q = 4051$, 0.224 $[0.222, 0.227]$ at 65,479 and 0.297 $[0.295, 0.299]$ at 262,111. Against a matched *plain* rho, IC/rho falls to 0.26 $[0.23, 0.29]$ at $q = 4051$ and 0.024 (0.02–0.03) at $q = 1{,}048{,}507$ ($\ell \approx 2.8 \cdot 10^{17}$), resolved below rho across the band for every $q \geq 991$; at $q = 499$ the band leaves it unresolved and at $q = 223$ rho is lower. The fitted exponent on $q \geq 4051$ is 1.045 $[1.037, 1.052]$ against rho's 1.476 $[1.438, 1.508]$. Table 30.1 gives the ladder. B1 is a trade (0.81–0.91 of B0 up to $q = 65{,}479$, 1.53 at 262,111) and was not adopted. B2 became the baseline that N4 v2 then improved with a negation-map rho; the host was a shared VM and there is no validation file (`benchmarks/20261021-quartic-n4-la/`).

q	units	rho runs	IC/rho, $w = 0$	$w = 1$	band
223	5/5	150	3.85 $[2.67, 5.25]$	2.51	rho lower
499	5/5	150	1.40 $[1.08, 1.79]$	0.85	unresolved
991	5/5	150	0.79 $[0.71, 0.88]$	0.52	IC lower
4051	5/5	150	0.26 $[0.23, 0.29]$	0.18	IC lower
65479	5/5	80	0.05 $[0.04, 0.06]$	0.04	IC lower
262111	3/3	18	0.04 $[0.03, 0.05]$	0.02	IC lower
1048507	3/3	12	0.02 $[0.02, 0.03]$	0.02	IC lower

Table 30.1: N4 iteration 2, arm B2 (double large primes on lines through base points), on the GHS-weak family $E_0/\mathbb{F}_{q^3}$, $4 \mid \#E_0$, against a matched *plain* rho, cold single-target cost in multiplication equivalents with 95% bootstrap intervals; selected rows of thirteen. Band: the order across $w = 0$, 0.1, 1. Shared VM, no validation file. Source: `research/quartic_n4_la_20261021/README.md`.

30.2 Weil restriction at $n = 4$ and $n = 5$

Genus-4 Jacobians. For prime-order $E/\mathbb{F}_{q^4}$ the Weil polynomial of $\operatorname{Res} E$ is $T^8 - tT^4 + q^4$, and a curve matches when its counts over $\mathbb{F}_{q^k}$ for $k \le 4$ do. A lead agent's census (`research/genus4_n4_census_20261024/REPORT.md`) found that the class contains a smooth non-hyperelliptic genus-4 Jacobian at every admissible trace at $q = 3$ (exhaustive, reproducing nine classes of an earlier round), at 13 of 14 traces at $q = 5$ (the exception is $t = 49$) and at all 20 at $q = 7$ (both sampled). No congruence on $(q \bmod \ell, t \bmod \ell)$ predicts which classes are reached, and Tian-kernel existence does not either. Every curve found has plane quintic models, the setting of Diem's plane-curve index calculus at $\tilde{O}(q^{4/3})$ against rho's q^2, a cost the study cites and does not verify. Its Theorem 6 is labelled an agent proof, checked against direct linear algebra on 44 cells and not independently reviewed. No map $J(C) \to E$ was computed: this is existence in the isogeny class, not an attack.

The genus-5 cover. A toy reproduction of the quartic-extension genus-5 cover ran 90 complete attacks ($q = 61$–8009, three seeds, full and ι-folded arms): all 90 logarithms recovered and every invariant check passed, and matched rho was faster at every q (`research/humbert_genus5_cover_20261021/README.md`; `benchmarks/20261026-humbert-genus5-cover/`, no validation file). A first sweep contained one wrong logarithm at $q = 1511$ from a linear-algebra bug (unused factor-base columns) and is kept as superseded. The genus-5 exponent 8/5 is below rho's 2, so the loss at toy sizes is a constant, and no crossover was measured or extrapolated.

Reading the two results together. At $n = 3$ the quartic route's index calculus beats rho at $w = 0$ on generic quartics but not across the band; at $n = 4$ and 5 the Jacobians exist in the right classes, but either no map is built (genus 4) or matched rho wins at every measured q (genus 5). Neither touches a prime-order curve over a prime field.

30.3 Same-field embeddings on prime fields

The closure. For a prime-field source (Table 30.2 summarises the measured collectors), a same-field genus-g cover gives $\#J(C)(\mathbb{F}_p) = N \cdot \#A(\mathbb{F}_p)$ with $\#A = \Theta(N^{g-1})$, and genus-g index calculus costs $\tilde{O}(q^{2-2/g}) \ge q$ for $g \ge 2$: a full exponent above rho's $N^{1/2}$. Quadric-pencil and quartic models of E itself are isomorphic models with the same group. The atlas closes the lead by theorem (`DEC-20260902-4c82fd`; `crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-same-field-genus-g-embedding.yaml`), and the registry scan's -2.1 to -2.0 bits on every prime-field curve is its formula form (§29.1). The studies below measured specific constructions against that closure.

Jacobians in ecbench. The 25 October study added genus-2, plane-quartic and GHS-cover families to ecbench (`research/ecbench_jacobians_transfers_20261025/README.md`).

- On two genus-2 curves (17.5 and 20.2 bits), `ic.hyperelliptic` costs 1.473 $[1.158, 1.874]$ of matched negation rho, above rho at every weight (1.47, 37.5 and 216 at $w = 0$, 0.1, 1), while `bsgs.negation` is 0.737 $[0.609, 0.892]$, below rho at every weight.
- On two plane quartics, `bsgs.textbook` is 0.497 $[0.378, 0.654]$ of plain rho.

Collector	Scope	Against matched rho
genus-2 `ic.hyperelliptic`	two curves, 17.5 and 20.2 bits	1.473×; 216× at $w = 1$
Kummer fixed-difference collector	21 models, 1,260 targets, 16–40 bits	8.6–55.9×; 0 of 45,360 wins (validated)
point-driven pencils, peeling	$p = 739$, 1,151, 1,187; 180 targets	0 of 19,440 cells below (validated)
pencils, Markowitz priority	$p = 739$, 60 targets	1.205× peeling; 0 of 36 below (validated)
genus-2 route at $n = 2$	1,260 runs to 40 bits	0.51–2.83; unresolved
GHS cover from $\mathbb{F}_{p^6}$ (ecbench)	two sources, 20.2 and 23.5 bits	6.62×

Table 30.2: Measured Jacobian and cover collectors on prime-field or small-degree sources. The $n = 2$ genus-2 row is from §29.3. Sources: as cited in the text.

- `transfer.ghs_cover` on two sextic-extension sources is 6.62 $[5.13, 8.54]$ of rho on the source, above at every weight.

The source classifies the study as a new capability, not an improvement: at these sizes BSGS beats rho on the Jacobian, which is a statement about small groups, and index calculus does not.

The Kummer collector. A fixed-difference collector on genus-2 Kummer surfaces (the Scholten cohort: 21 models, 1,260 targets, 180 each at 16 to 40 bits) solved and verified every target in both its uncached and cached arms, processing $2.30 \cdot 10^8$ quotient states (`research/kummer_differential_collector_20261022/REPORT.md`). Both arms lose in all 36 whole-cohort price cells and in all 45,360 target-cell comparisons. Full-cold candidate/rho spans 8.96–55.94 (uncached) and 8.62–54.09 (cached); at $S/M = 0.5$, $I/M = 10$ and 0.1 GAE per native operation the ratios are 14.36 and 12.95. The paired replay has a passing validation gate (`benchmarks/20261022-kummer-paired-replay/`), so these are validated measurements of a loss. On the way the study found that the published Eq. (11) of its Kummer recovery formula produced 1,257 incorrect signed divisors and three $D = 0$ cases on the cohort; the corrected generic equations recover them (`research/kummer_recover_20261022/REPORT.md`).

Point-driven pencils. The pencil collector builds Jacobian relations from three-point pencils on the same frozen cohort (`research/pencil_sparse_20261022/REPORT.md`). With three sparse variable orders and a degree-peeling elimination, at $p = 739$, 1,151 and 1,187 (540 arm-target pairs over 180 targets), the native validator recomputed all 19,440 target-level price cells and found 0 below rho. A Markowitz pivot priority on the 60 targets at $p = 739$ cut modular row updates from 1,336,284 to 1,307,655 but raised logical index operations from 9.48 to 17.51 million, so its cold cost over peeling is 1.204855 and 0 of 36 price cells are below either peeling or rho (`research/pencil_markowitz_20261022/REPORT.md`). Both have passing validation files (`benchmarks/20261022-pencil-degree-peeling/`, `research/pencil_markowitz_20261022/VALIDATION.json`). Later pencil studies (a $p = 739$ large-prime gate, early rejection, lifecycle and inventory work) are engineering and report no new ratio.

Other prime-field covers. Three further results bound the remaining constructions.

- *Enge-type covers.* $C_{d,s} : y^2 = f(t^d + s)$ covers E with genus $(3d - 1)/2$ for odd d; Enge's bound with $d = \Theta(\log p)$ gives an exponent of order $\log p \sqrt{\log \log p}$, larger than rho's $\frac{1}{2} \log p$ (`research/enge_primefield_cover_20261024/REPORT.md`).
- *An external genus-2 backend.* Lindner's divisor-arithmetic library, pinned for comparison with the repository's Cantor arithmetic, returns the identity for `add(D,D)` on a valid divisor that is not 2-torsion, while its `double(D)` returns a degree-2 divisor; its `add` cannot be used as a complete group law without an equal-input dispatch (`research/genus2_external_gate_20261021/README.md`).
- *Genus 3 over* $\mathbb{F}_{2^{83}}$. A faithful transfer of the $n = 83$ source into its genus-3 Jacobian image was replayed at toy sizes $n = 9, 13, 19, 23$ (400 of 400 destination attempts verified); only generic square-root solvers apply to an image of order $r \approx q$, and the ambient genus-3 index calculus loses to source rho by about 70.17 exponent bits there (`research/n83_genus3_image_solver_20261031/README.md`).

Study	Reference, unit	$p = 1301$	$p = 1451$	$p = 1601$
crossover	distinguished-point rho	0.793	$0.775^{\downarrow}$	$0.587^{\downarrow}$
	strong rho	$1.349^{\uparrow}$	1.063	0.910
re-scoring	negation rho, additions (U1)	$1.405^{\uparrow}$	$1.160^{\uparrow}$	0.870
	negation rho, mults (U2)	$3.376^{\uparrow}$	$2.787^{\uparrow}$	$2.090^{\uparrow}$
line search	negation rho, additions (U1)	$0.791^{\downarrow}$	$0.659^{\downarrow}$	$0.548^{\downarrow}$
	negation rho, mults (U2)	$1.900^{\uparrow}$	$1.585^{\uparrow}$	$1.318^{\uparrow}$

Table 30.3: The plain Nagao arm of the Joux–Vitse cover against rho on the same instances, cover/rho. ↑: interval above 1 (rho cheaper); ↓: interval below 1; unmarked: not resolved. The re-scoring and line-search rows share one pooled reference of 126 negation-rho runs. No validation file. Sources: `research/jv_cover_crossover_20261101/`, `research/jv_cover_negrho_rescore_20261101/`, `research/jv_cover_fbline_20261102/`.

30.4 Joux–Vitse in November

The Joux–Vitse cover [27] of $E(\mathbb{F}_{p^6})$, for curves with full rational 2-torsion and $\#E = 4\ell$, maps the problem to a genus-3 Jacobian over $\mathbb{F}_{p^2}$. Chapter 26 summarises the two arms of this cover; this section is the full account: the validated sieve row at $p = 1511$, the November studies around it, and the earlier comparisons as they now stand. All of them are on the weak class; none is on a prime-order curve, and all but the validated row and the Hilbert stop below lack a validation file.

The validated row: the sieve against a matched negation rho. A frozen protocol of 2 November re-measured the sieve arm (`jv_cover -exp sieve`) at $p = 1511$ ($\ell = 2^{61.4}$) against a matched negation-map rho: 16 new rho runs pooled with 126 earlier ones under a frozen homogeneity rule (142 runs), with the sieve rows of the 19 October study unchanged and reproduced by a second build (`research/jv_sieve_negrho_20261102/`). The sieve route costs 0.00647 [0.00545, 0.00799] of rho in group additions (U1) and 0.0156 [0.0131, 0.0192] in textbook $\mathbb{F}_p$ multiplications (U2; seeds 17–24, 95% bootstrap intervals). Rho is therefore 155× [125, 184] and 64× [52, 76] more expensive. Charging the cover's unemitted F4, descent and linear-algebra work at its upper bound leaves the ratio at 0.107–0.123 for $w = 0.1$ and 1. The order holds across the band in all four cells (U1 and U2, emitted and upper-bound layers), for both seed sets and both rho references. An independent validator passed the comparison (`VAL-20261103-c31c76`), with these limitations:

- the replay is sampled: 1 of the 16 new rho runs, 1 of the 126 pooled runs and 2 of the 16 cover rows;
- the cover's F4, descent and linear-algebra additions are not emitted, so only an upper-bound proxy prices them;
- memory is not reported for either arm;
- the run was local, not on the isolated lab.

The validator classifies the result as accounting, not an improvement to any method: it is a matched measurement of an existing solver. It is the programme's one validated measurement of index calculus below a matched rho. It has no `pcc` claim record, so under the proof-carrying-claims rule it is a validation-gate result whose statement is an unverified assertion. It holds on the weak class only, says nothing about prime-order curves, and is not extrapolated. A red-team review of the same study (`research/jv_sieve_fbline_20261102/review/red_team/REPORT.md`, `RT-20261103-fd05ef`) adds two qualifications: U1 prices rho at a schoolbook harness about 2.4× above the tuned price, so the like-for-like cell is U2 (64×); and a minimal $\mathbb{F}_{q^3}$ product would make rho about 1.34× cheaper in $\mathbb{F}_p$ multiplications. Neither reverses the order.

The Nagao arm near parity. The 1 November crossover study ran the plain arm at $p = 1301$, 1451 and 1601 against two rho references on the same instances (`research/jv_cover_crossover_20261101/README.md`; Table 30.3): against the distinguished-point rho it reads 0.793 [0.587, 1.119], 0.775 [0.645, 0.970] and 0.587 [0.448, 0.811], below at the two larger primes under the frozen rule; against the strong rho, 1.349 [1.085, 1.699] (above), 1.063 and 0.910 (not resolved). The source calls this a candidate needing validation. A frozen re-scoring against a pooled matched *negation* rho (126 runs) reversed it (`research/jv_cover_negrho_rescore_20261101/README.md`): in group additions the cover is above rho at 1301 (1.405 [1.206, 1.629]) and 1451 (1.160 [1.020, 1.338]) and unresolved at 1601 (0.870 [0.697, 1.183]), and in textbook $\mathbb{F}_p$ multiplications it is above at every prime (3.38, 2.79, 2.09).

Arm	S/rho (U2)	S^+/rho$^+$ (U2)	S/rho (U1)	Distance
P (plain)	0.0244 [0.0206, 0.0292]	0.0624	0.0101	×41
H (Horner line)	0.0154 [0.0135, 0.0180]	0.0344	0.0064	×65
U (table line)	0.0141 [0.0122, 0.0167]	0.0324	0.0059	×71

Table 30.4: The Joux–Vitse sieve route against a matched negation rho on the same instances, pooled over 24 instances at $p = 1009$, 1511 and 1823, with the per-prime reference the frozen homogeneity rule selected. U2: tuned rho in $\mathbb{F}_p$ multiplications; U1: group additions; S^+: with additions and lookups charged on both sides. Distance: how many times cheaper than rho, in U2. Source: `research/jv_sieve_line_vs_rho_20261103/README.md`.

A factor-base line search. Choosing the factor-base line made the plain arm 1.71× [1.54, 1.89] cheaper ($S_{\text{line}}/S_{\text{plain}} = 0.587$ [0.530, 0.651], as the model predicted, with every instance cheaper) (`research/jv_cover_fbline_20261102/README.md`). Against the same negation-rho reference the cover then reads 0.791, 0.659 and 0.548 in group additions, below rho at all three primes under all three estimators, but 1.900, 1.585 and 1.318 in textbook multiplications. In multiplications the order does not hold across the band, and the cover's native count omits its $\mathbb{F}_p$ additions, so a "below" could narrow under complete counts. The modelled crossovers move to $p^* \approx 1185$ in additions and 1799 in multiplications, inside the range a 64-bit build can test. There is no validation file, so this is a candidate improvement of the index-calculus baseline, not a ranked one.

The line search on the sieve route. A frozen controlled study of 2 November wired the same factor-base line search into the sieve route and compared three arms with the plain route P on 48 paired instances at $p \in \{1009, 1511, 1823\}$, the search's own cost charged (`research/jv_sieve_fbline_20261102/README.md`). Pooled S ratios against P are 0.623 [0.562, 0.689] for the Horner search H, 0.560 [0.500, 0.631] for the exhaustive table T and 0.600 [0.535, 0.674] for the table at $n = 32768$ (U); with additions and lookups charged, H and U stay below P (0.550, 0.525). An independent validator passed the paired comparison (`VAL-20261103-594245`), but the validation gate is not admissible, because the protocol names no rho reference, and there is no `VALIDATION.json` or `pcc` claim. The red team (`RT-20261103-fd05ef`) broke the claim for T: an uncounted integer multiply per table lookup, once charged, puts T's band at $w = 0.1$ at 1.164, worse than P; H and U are unresolved against each other. This study is the cover-only half of the H and U arms below.

The sieve route on the same instances. The 3 November study ran three sieve arms (plain P, Horner line search H, table line search U) and a matched negation rho on the same 24 instances at $p \in \{1009, 1511, 1823\}$, removing the product of ratios that earlier comparisons used (`research/jv_sieve_line_vs_rho_20261103/README.md`; Table 30.4). Every cell is below rho, and the order holds across the band in both cost layers, including the upper-bound layer that prices the unemitted F4, descent and linear-algebra work as native operations, where the ratios are still about 10× below rho at $w = 1$.

The ratio falls with p in every arm (U, S/rho U2: 0.031, 0.0115, 0.0081), because rho grows as $\sqrt{\ell} \sim p^3$ faster than the route. The source lists its limits: one rho run at $p = 1823$ was killed by the time rule and excluded, which biases that prime against the cover; the frozen bootstrap understates rho-run uncertainty (a run-level bootstrap widens the pooled U interval to [0.0114, 0.0182], no verdict changes); H against U is not resolved per prime; the run was local; and there is no validation file. It is accounting and distance, not an improvement.

Cutting the sieve's additions. Two changes that leave every line, hit, relation and logarithm bit-identical remove almost all of the line enumeration's additions and 17–19% of the sieve's. Over 48 paired instances the measured S^+ ratio, new over old, is 0.621 [0.614, 0.629] while S moves by 0.990; in the `bitop/v1` model the effect is 0.880 or 0.953, depending on how linear algebra is priced, because S^+ prices an addition as a full multiplication (`research/jv_sieve_adds_20261103/README.md`). The phase split at $p = 1511$ (line enumeration 25% of multiplications, sieve 24%, descent 42%, linear algebra 9%; $S = 0.0060$, $S^+ = 0.0219$) is in `research/jv_cover_next_leads_20261102/LEADS.md`, which also names the sieve, not the Nagao arm, as the cheapest route for the class.

Earlier comparisons, as they now stand.

- The plain Nagao arm with a sparse-FGLM linear stage read 1.59–1.98× a pooled rho constant at $p = 1009$ on 24 October, closing as about p^{-2} (`research/jv_cover_p_scaling_20261024/`; §36.5). That was a lower bound against an unmatched constant; the matched negation rho above supersedes it, and the crossing it extrapolated near $p \approx 1330$ does not hold against that rho.
- The 19 October sieve study (`research/jv_sieve_strong_rho_20261019/`) measured 0.024 $[0.019, 0.031]$ of a strong rho at $p = 1009$ and 0.0042 at $p = 1823$, but its 20 October review withheld every comparison (no numeric acceptance for the controls, uncharged rho hashes, incomplete accounting). Its sieve rows are the ones the validated row re-scored.
- The replication of the 19 October sieve study failed its preregistered acceptance rule A4 (a rho A/A ratio within $[0.85, 1.15]$: 0.792 $[0.572, 1.073]$), so all its decisions are withheld; its descriptive sieve/rho ratios ran from 0.647 at $p = 503$ to 0.00404 at $p = 1823$ (`research/jv_sieve_strong_rho_v2_20261020/README.md`). The registered prospective re-run (`research/jv_sieve_prospective_20261023/`) never ran and was superseded by the matched study above.
- Re-priced in `bitop/v1` with both arms executed, the whole-route row (an isogeny walk from a non-weak challenge curve to the weak class, then the cover) at $p = 1511$ ($\ell = 2^{61.4}$) reads 0.0098 $[0.0069, 0.0144]$ of the strong rho R3 (largest band upper bound 0.075); it has no validator report (`research/ic_acct1_cover_bitop_20261021/README.md`).
- A 9 October whole-route replay from a non-weak challenge curve, through an isogeny walk to a weak curve and its cover, read 0.058 of a pooled, unmatched rho at $p = 1009$ on one seed per prime, with an earlier revision of the solver whose source includes the sieve module; which arm ran is not recorded (`research/jv_cover_e2e_30bit_20261009/RESULT.md`). The cover-decomposition note still states a measured sieve crossover near $p \approx 430$ (`research/notes/index-calculus/RESEARCH_COVER_DECOMPOSITION_LEDGER.md` §11) with no erratum; it should be read through the review. Leaderboard Table D, the historical study, is withheld.
- On the method side, the sparse-FGLM stage made the plain arm 1.28× cheaper end to end (`research/jv_cover_sparse_fglm_20261018/`), and relation collection is 99.99% of S (`research/ic_cover_leads_20261018/`).
- A large-prime variation of the cover decomposition does not pay at $p \leq 12{,}000$: linear algebra is $2.5 \cdot 10^{-3}$ of S there, and one large prime saves at most 0.02% (`research/jv_cover_large_prime_20261018/README.md`).

The Hilbert early stop, and a relabelling. In the plain arm's F4 stage, a Hilbert-function stop certifies that the six quadrics' leading forms are a regular sequence and drops every pending pair that provably reduces to zero (`research/ic_cover_weakclass_20261023/REPORT.md`). Plumbed into the logarithm mode and measured with a new native counter, it cuts counted S by 1.848–1.875× over 25 cells at equal relations, with the emitted native counter at 0.983–0.990 of the baseline: engineering, with a passing validation gate (`benchmarks/20261029-jv-cover-hilbert-stop/`). The form first found, B0, cut counted S 2.53–2.64× while the unpriced F4-driver work rose 8.3–8.5×; the study classifies B0 as relabelling, which is in the corrections chapter (Table B.6). The baseline's distance to rho falls with p: S/rho of 1361, 205, 35.2 and 7.19 at $p = 53$, 101, 251 and 503, slope -2.27 ± 0.13 in $\log p$ over four sizes, against the matched negation rho of the Hilbert-stop study; the pooled-constant series of Chapter 36 reads 620 at $p = 53$ and slope -2.05.

The weak class. Every number above is on curves with full rational 2-torsion and $\#E = 4\ell$. A norm-one conductor theorem characterises the weak isogeny classes; a random full-2-torsion curve lies in one with probability 0.5908 $[0.5754, 0.6059]$ at $p = 11$ and 0.6320 $[0.6169, 0.6468]$ at $p = 47$, over a census to $p \leq 53$ (`research/iso1_weak_classes_20261007/REPORT.md`). Generalising the cover over 2-torsion configurations gives genera $g \in \{1, 3, 7, 9, 13\}$, and Joux–Vitse's is complete among such covers with $g \leq 3$ (`research/jv_weak_generalization_20261008/README.md`). Depth-1 isogeny classes hold no weak curve over 4,871,062 curves, while 86–95% of deeper classes hold one (`research/jv_weak_class_law_20261007/`). Since weakness is not a class invariant, a Teske-type trapdoor in this class is structurally possible; the atlas lead is unexamined.

30.5 What the Jacobian routes settle

- On prime fields every same-field Jacobian route is closed by theorem, and every measured collector (Kummer, pencils, genus-2 index calculus) is above rho, three of them under a passing validation

gate.

- At $n = 3$ the quartic routes are below rho in the weak family under counted costs; on generic quartics the Diem arm is below at $w = 0$ only.
- At $n = 4$ and 5 the Jacobians exist in the right classes; the one complete attack (genus 5) loses at every q measured.
- The Joux–Vitse sieve is between about 16 and 71 times cheaper than a matched negation rho on the weak class at $p \leq 1823$ in $\mathbb{F}_p$ multiplications, depending on arm and on whether additions are charged, and 155× in group additions (64× in $\mathbb{F}_p$ multiplications) at $p = 1511$; only the $p = 1511$ comparison is validated, and it has no `pcc` claim. The plain Nagao arm is above a matched negation rho at $p = 1301$ and 1451 and unresolved at 1601 in group additions, and above at all three in multiplications; with the line search it is below in additions only, unvalidated.

31 Extension fields: the GLV-invariant programme and quotients

Chapter 25 gives the D_4 system at $k = 4$, its engineering ladder, the characteristic-two distance to rho and the red-team verdict on its crossover; §21.10 gives the first D_4 build and the modelled handover it moved. A second line of work ran beside them: one note, `research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md` ("the GLV note" below; §§1–9.57, about 6,000 lines), registered and measured experiments E1 to E42 on factor bases invariant under automorphisms, Frobenius-type maps and 2-torsion translations, first on subfield and GLS curves, then at $k = 4$ over $\mathbb{F}_{p^4}$. The extension chapter quotes one line of it. This chapter gives the programme's results in order, the corrections it made to its own reference, and the quotient and cofactor levers that belong with it. Validation is named per experiment; the rest is producer-reported.

31.1 Folds on subfield and GLS curves: E1–E18

What a fold can be. E1–E7 established one closed form (GLV note §6.8). The fold of a factor base by any finite group of endomorphisms is the order of the subgroup of $(\mathbb{Z}/r\mathbb{Z})^*$ that the group's eigenvalues generate: 12 on the $j = 0$ GLS twist, 4 (not 8) on the $j = 1728$ twist, 3 (not 9) on the $j = 0$ subfield curve. Rho folded by the same group takes the square root of that number fewer steps, verified (E6, E9: steps $\div 1.40$, 1.74 and 2.16 at $A = 4$, 6 and 12 against the expected 1.41, 1.73 and 2.45). CM endomorphisms of degree 2 and 3 and Q-curve endomorphisms keep base points in the base only at chance (E4, E14), and where a cofactor is fixed by the maps, two-summand relations stay inside one orbit, so the fold buys columns and not relations (E5). Every class in E1–E16 is engineering or accounting.

The best subfield arm and its boundary. With three summands on the Frobenius line, FGHR's 2-torsion symmetry on the system (a degree-16 resultant in place of a 64-dimensional quotient, $S \div 27.0$) and the translation fold on the base ($S \div 2.52$ more), the cheapest arm was 267–5,741 times rho (E13). E16 then stated why no lever on these families can do better than a constant: on a subfield curve the large prime lies in the trace-zero part, $r \approx p^2$, so rho costs $\Theta(p)$ and a line base needs $\Theta(p)$ relations; to $p = 2^{16}$ the best arm's ratio to rho grows as $r^{0.18}$ (267–6,616×). E16 also corrected two earlier exponent readings, E11's and E13's.

Moving the levers to $k = 4$. E17 carried the D_3 solve ($\div 38.0$) and the τ_T fold ($\div 2.00$ exactly) to the full group $E(\mathbb{F}_{p^3})$, still 180–377× rho. E18 folded the $k = 4$ linear algebra: on $j = 0$ curves the ψ fold takes the asymptotic ratio to the matched walk from 1.232 to 0.138 ± 0.008 and the extrapolated handover from $2^{156.1}$ to $2^{146.0}$, the figure §21.10 quotes; the τ_T fold cannot pay for its curves' cofactor $h = 4$ and 4.7× poorer yield. E18 also re-read an earlier note's ratio 0.518 as 0.733 against the negation walk (GLV note §9.4).

31.2 The V_4 line and the $k = 4$ baseline: E19–E26

E19: the first end-to-end run. The τ_T/D_4 pipeline recovered logarithms at $k = 4$ with every phase priced, on two curves each at $p = 53$–269 ($n = 2^{20.9}$–$2^{30.3}$, $h = 4$); the solves are 100.0% of S at a median 1.37–$1.41 \cdot 10^9$ $\mathbb{F}_p$ multiplications per call, and the method is 2.6–$8.8 \cdot 10^6$ times the negation rho (GLV note §9.6). From the pipeline's own constants the modelled handover is $2^{116.7}$, reproducing the 2^{119} assembled from separate harnesses. E20 found that double large primes have the sub-rho exponent at $k = 4$ and lose by the constant (1.66–3.60× the full base, crossing it only near 2^{164}–2^{222}), and E21 that trace replay on the weighted solve is a net loss (0.95–0.98× with fallbacks charged), because the rows that vanish are 7% of the matrix and differ from target to target.

E22: the V_4 line. On curves over $\mathbb{F}_{p^4}$ with full rational 2-torsion and rational halves of two 2-torsion points, a line exists that is stable under the whole Klein group V_4 of translations, and S_5 can be written in $V_4^4 \rtimes S_4$ invariants: the quotient falls from 512 to 64 and the Macaulay degree from 26 to 11. One solve

p	seed	$\log_2 n$	base/cols	C median	S/rho	E19/E24
101	1	24.6	36/9	$1.62 \cdot 10^6$	$1.69 \cdot 10^5$	37×
101	2	24.6	52/13	$1.62 \cdot 10^6$	$3.43 \cdot 10^4$	170×
149	1	26.9	64/16	$1.63 \cdot 10^6$	$3.80 \cdot 10^4$	108×
149	2	26.9	60/15	$1.63 \cdot 10^6$	$5.54 \cdot 10^4$	61×
197	1	28.5	76/19	$1.65 \cdot 10^6$	$1.40 \cdot 10^5$	35×
197	2	28.5	84/21	$1.64 \cdot 10^6$	$5.44 \cdot 10^4$	52×
269	1	30.3	132/33	$1.65 \cdot 10^6$	$2.30 \cdot 10^4$	240×
269	2	30.3	148/37	$1.65 \cdot 10^6$	$2.34 \cdot 10^4$	110×
397	1	32.5	208/52	$1.66 \cdot 10^6$	$1.31 \cdot 10^4$	
397	2	32.5	188/47	$1.66 \cdot 10^6$	$1.58 \cdot 10^4$	
521	1	34.1	240/60	$1.66 \cdot 10^6$	$1.88 \cdot 10^4$	
521	2	34.1	232/58	$1.66 \cdot 10^6$	$2.15 \cdot 10^4$	

Table 31.1: E24: the $k = 4$ pipeline on the V_4 line, $h = 4$ on every curve, against the matched negation rho of the time ($w = 0$). C: $\mathbb{F}_p$ multiplications per solve call. E19/E24: the earlier pipeline's S/rho over this one at the same p and seed, on different curves of that p; E19 did not run $p > 269$. The $p = 101$ seed 2 factor is the validator's 170× (printed 171×). Validated: `benchmarks/20261018-k4-v4-line/`. Source: GLV note §9.14 and §9.19.

costs about $1.7 \cdot 10^6$ multiplications, $\approx 820\times$ less than the weighted D_4 solve, but the larger translation orbits and the thinner base cost about 10× in targets per relation (GLV note §9.12). Its failures are all planted targets with two points of one translation orbit, which give a positive-dimensional system; uniform targets meet this with probability $\approx \text{cols}^2/n$.

E24: the baseline moves, validated. End to end on twelve curves at $p = 101$–521 ($n \le 2^{34.1}$), the V_4 pipeline recovered every logarithm; the solves are 99.3–99.5% of S and every row is below E19 at matched p by 35–240× (GLV note §9.14; Table 31.3). Table 31.1 gives the rows. The fitted slope of S/rho in n is -0.25, the theory's $n^{-1/4}$, and the modelled handover moves from E19's $2^{116.7}$ to $2^{90.1}$ ($2^{91.7}$ from the largest p alone). The class is engineering, a constant, and under the success criterion a verified, measured improvement of the baseline. An independent validator replayed every run of E19, E24 and E25 field for field and recomputed the claims (`benchmarks/20261018-k4-v4-line/`, `benchmarks/20261018-k4-gls-v4/`; both gates pass). It found seven textual errors, none changing a result (§9.19): for example the E19-to-E24 factor at $p = 101$ seed 2 is 170×, not 171×; 392 kept rows were re-added, not 381; and unsolved targets reach 0.033% per curve, not 0.012%. It also priced the band: the 35–240× is at $w = 0$; with one GAE (97 multiplications) per Macaulay entry and walk step it is 19–132× at $w = 0.1$ and 6.5–45× at $w = 1$, and E24 stays below E19 on every curve at every weight. Every E24 row is still $1.3 \cdot 10^4$–$1.7 \cdot 10^5$ times rho, and rho is ahead across the band.

E25: the GLS fold. On GLS twists over $\mathbb{F}_{p^4}$ the GLS map ψ ($\psi^2 = -1$) keeps the V_4 line and acts on it as $z \mapsto \pm i/z$, so the base folds by $\langle V_4, \psi \rangle$ with 8 points a column (GLV note §9.16). Measured on the same curve, base and target stream, the fold cuts S by 1.35–2.67× (median 1.86), and against the matched $\langle \pm\psi \rangle$ walk the ψ arm gains 1.26–1.97× over the control at every $p \ge 197$; below that, walk set-up dominates. The modelled handover on these curves moves from $2^{93.2}$ to $2^{88.0}$ (Table 31.2).

It does not move E24's rows: every curve of this construction has $h = 16$, which makes rho 2× cheaper at a given p and absorbs most of the fold at matched n. E26 then showed, on 560 GLS twists, that $h = 16$ is forced: the halves condition for the V_4 line holds exactly on the 250 twists with $4 \mid t$, each with $v_2(\#E) \ge 4$, and fails on all 310 with $t \equiv 2 \pmod 4$ (§9.18). The validator noted that the $\langle \pm\psi \rangle$ walk applies ψ to both coordinates on every step and estimated that comparing abscissae first would save about 6%, so E25's distances lean that much toward index calculus.

E23 and E28: the other lines. On $j = 0$ the order-24 group exists on the $x_0\mathbb{F}_p$ line, but the line carries no point of the large subgroup (E23, by derivation, confirmed on points). On E19's single-τ_T line with GLS twists, ψ keeps the line with $h = 4$ on all 40 curves and halves the columns, but the solve is unchanged at $\approx 1.4 \cdot 10^9$, so the line stays at least $6.99 \cdot 10^5$ times rho in a model (E28, §9.30). The validator refused E28 at the gate because its end-to-end S is a model.

p	seed	$\log_2 n$	cols A/B	S_A/S_B	$S_B/\text{rho}_{\pm\psi}$	improvement
101	1	22.6	10/5	1.94	$6.77 \cdot 10^4$	2.26
101	2	22.6	14/7	2.11	$2.93 \cdot 10^4$	2.05
149	1	24.9	10/5	2.67	$2.65 \cdot 10^5$	3.13
149	2	24.9	14/7	1.77	$7.01 \cdot 10^4$	2.25
197	1	26.5	22/11	1.35	$6.11 \cdot 10^4$	1.26
197	2	26.5	30/15	1.69	$2.62 \cdot 10^4$	1.51
269	1	28.3	30/15	1.74	$4.84 \cdot 10^4$	1.54
269	2	28.3	40/20	2.02	$1.56 \cdot 10^4$	1.44
397	1	30.5	60/30	1.72	$1.13 \cdot 10^4$	1.76
397	2	30.5	48/24	2.32	$1.71 \cdot 10^4$	1.97

Table 31.2: E25: the GLS fold on the V_4 line, $h = 16$ on every curve. Arm A: V_4 fold, against the negation walk; arm B: $\langle V_4, \psi \rangle$ fold, against the $\langle \pm\psi \rangle$ walk, on the same curve, base and target stream. Improvement: $(S_A/\text{rho}_{\text{neg}})/(S_B/\text{rho}_{\pm\psi})$. Distances predate E27's stronger walk. Validated: `benchmarks/20261018-k4-gls-v4/`. Source: GLV note §9.16.

31.3 The solve and its pivots: E31–E42

With the relation phase priced and the base folded, the per-call cost of the degree-11 solve is all of S. Six experiments worked on it.

E31: the chain solve. The quotient of the V_4 system is a 12-generator module over $\mathbb{F}_p[P_2]$, but the solver paid for a dense 64-wide normal form at each of 240 pivots, a dense Hessenberg reduction and a dense 64-dimensional kernel (GLV note §9.23). Working in the module (left-looking normal forms of the 12 chain ends, $\det R(x)$ of a 12×12 polynomial matrix, kernels of $R(\lambda)$) cut one call from 1.62–$1.66 \cdot 10^6$ multiplications to 3.04–$3.33 \cdot 10^5$ with the sparsest-pivot rule (B), with the same decompositions on all 10,000 calls; end to end on E25's curves S fell 4.53–4.76× (B) and 2.50–2.56× (A, without the pivot rule), with every relation, logarithm and walk unchanged. The independent validation `VAL-20261024-24ef8b` passed (`benchmarks/20261024-e31-chain-solve/`) with seven errata and one qualification that changed the ranking (§9.24): B's sparsest-pivot rule counts the nonzeros of every row it builds, which was neither counted nor priced; priced at the band's rate, B falls behind E25 at $w = 0.1$ and 1, so **B beats E25 and A only at** $w = 0$, its rank is unresolved across the band, and A, not B, is the ranked solver. A stays below E25 at every weight (2.50–2.56× at $w = 0$, 1.39–1.43× at 0.1, 1.034–1.053× at 1): at $w = 1$ the 393×448 matrix the chain solve still builds is almost all of the cost.

E33–E39: pivot rules under the band. The next experiments asked which pivot rule pays for its own selection work.

- *E33.* Counting B's selection pass gave $1.05 \cdot 10^6$ native operations per call, about six matrix reads; at $w = 0.1$ B falls to 0.19× A. A counted estimate B″ with $O(1)$ work per row update is ahead of A at $w = 0$ (1.54–1.56×) and 0.1 (1.046–1.048×) and behind at 1: not ranked (§9.37).
- *E35.* Replaying B's pivot order per curve does not work: no two calls of a curve share a sequence, because ties are broken by accidental zeros of the target's coefficients; the replay keeps 18.8–36.8% of B's saving and splits 14 to 6 at $w = 1$: unranked (§9.42). This agrees with E21's finding on a different object.
- *E36–E38.* Taking each bucket's first-arrived row (F) needs no selection work and cut A's per-call cost 1.47–1.55×; F and two variants were ahead of A and E25 at every weight on all 20 rows, then on all 32 rows on a fresh stream. Each time a registered control's kill criterion fired (on an inverse-map control in E36, on an F-specific control's positive check in E37, at the pilot in E38, because every column the normal forms read is contested), so none was ranked (§§9.43–9.47).
- *E39.* On the third attempt, with controls whose positive checks see what each defect corrupts, no kill fired, and F, S and H are ranked ahead of A and E25 at $w = 0$, 0.1 and 1 on all 32 rows: per row 1.46–1.54× fewer solve multiplications for F at $w = 0$ and $S_A/S_F = 1.0115$–1.0130 at $w = 1$. It is validated (31 of 31 runs replayed, 1,037,565 fields identical, 7,286 summary values recomputed with 0 disagreements; `VAL-20261102-E39`, `benchmarks/20261102-e39-pivot-rules/`).

The sequence is an example of the validation gate doing its job: a measured order held in three runs and was ranked only when a preregistered run with admissible controls held.

E40 and E42b: the matrix itself. At $w = 1$ the band is set by the Macaulay entries (176,064 per call), which no pivot rule removes. E40 found that the forward echelon and the normal forms never read the 73,237 cells left of each row's lead (41.6%), and stored each row from its lead: 1.71× fewer cells, with A's arithmetic, pivots and outputs bit for bit (§9.51). Ranked by its registered rule, the stored form V equals A at $w = 0$ and is ahead by 1.25–1.26× at 0.1 and 1.36–1.38× at 1 on all 32 rows; it has a passing gate (`VAL-20261102-E40`, `benchmarks/20261102-e40-macaulay-entries/`). It is the first change at the $w = 1$ end since E31, and an improvement of the baseline at the positive weights only. E42 stopped on a mis-registered growth-control count; its re-registration E42b reserved V's rows to the block after their build end, cutting cells per call from 102,827 to about 92,970 at about 5,500 charged checks, equal to V at $w = 0$ and ahead by 1.022–1.023× and 1.032–1.034× at 0.1 and 1 (§9.57); no validation is recorded for it.

31.4 The reference, corrected: E27, E30 and E32

E27: the matched walk was too expensive. The rho reference of E24–E31 used 19-multiplication $\mathbb{F}_{p^4}$ products and one $\mathbb{F}_p$ inversion per addition. A Karatsuba tower and Montgomery-batched norm inversions cut the affine addition from 97 to 66.3–66.7 multiplications and made the same walks, with the same points, collisions and logarithms, 1.4533–1.4634× cheaper (GLV note §9.29). The index-calculus side moved only 1.0048–1.0057×, because its $\mathbb{F}_{p^4}$ work is about 2% of S. So every row moved about 1.45× *further* from rho at $w = 0$ (1.24–1.33× at $w = 1$), to $6.39 \cdot 10^3$–$4.06 \cdot 10^5$ times rho. The correction is validated (`VAL-20261031-e27a1c`, `benchmarks/20261030-e27-walk-karatsuba/`) and is also recorded with the other reference corrections (§34.1, Table B.5). Replaying E25's walks with the current code also exposed that a later commit had begun charging the $[d]G = Q$ check inside the walk (0.13–0.78% of a walk), raising every matched rho recorded since; E27 prices walks without it, as E25 did.

E30: an accounting audit. Trivial products are 0.31% of S, so the distance to rho moves by -0.31% at most; validated (GLV note §§9.32–9.33).

E32: the handover ladder, validated. Instead of a single-point model, E32 measured the distance over eleven sizes, p = 101–3089, three curves each ($n = 2^{22.6}$–$2^{42.4}$, 33 paired curves, 528 walks), on the line's best configuration (arm B with solver A and Karatsuba) against its strongest matched walk (§9.36). The per-size median distance falls from $2.1 \cdot 10^4$ to $1.5 \cdot 10^3$ at $w = 0$ (best row $1.14 \cdot 10^3$ at $p = 3089$), and is $4.9 \cdot 10^3$ at $w = 0.1$ and $2.0 \cdot 10^4$ at $w = 1$ at the largest size. The crossover is empty in the reachable range: $n^* > 2^{42.4}$, measured. The free-slope fit gives $\beta = -0.243$ with an interval of ± 0.04 containing $-1/4$, and a fitted crossover at $2^{85.5}$ $[2^{79.2}, 2^{94.3}]$ at $w = 0$, $2^{91.2}$ $[2^{84.4}, 2^{100.7}]$ at 0.1 and $2^{97.6}$ $[2^{90.1}, 2^{107.9}]$ at 1. The 15-bit width of the interval is what an extrapolation over 43 bits beyond the top size can say. The ladder has a passing gate (`benchmarks/20261031-e32-handover-ladder/`); its validator corrected four printed statements without changing a verdict.

Two things change the reading of the earlier single-point figures. The modelled handovers for odd characteristic, $2^{135.1}$–$2^{135.8}$ in §21.10 and the 2^{119} of the GLV note's early frames (E19, `d4-s5`), are superseded on this family by a measured distance to $2^{42.4}$ and a fitted interval of 2^{79}–2^{108} across the band. And the family is narrow: GLS twists over $\mathbb{F}_{p^4}$ with $p \equiv 1 \pmod 4$, full rational 2-torsion, the halves condition and $h = 16$. A curve without rational 2-torsion, such as a prime-order curve, has none of the symmetry this line uses.

31.5 The order-24 twist action: E29, E34 and E41

On E18's $j = 0$ curves over $\mathbb{F}_{p^4}$, the Frobenius-derived map $\psi_p = (u^2 x^p, u^3 y^p)$ with $u = b^{-(p-1)/6}$ has eigenvalue of order 24 on all eight curves (p = 241–1069), and with the automorphisms it acts as a cyclic group of order 24 on $\langle G \rangle$ (GLV note §9.31). It changes two things.

- *The base.* On the canonical model the order-24 fold divides the columns by 4.00 and the linear algebra by 13.3–15.2 against E18's cubic fold, and moves the modelled S_5 handover by about 9–10 bits, to 2^{135}–2^{142}. No S_5 solve was run, so this is a model; the validator's report for E29 is a fail at the gate for that reason alone, with every measured part verified.
- *The walk.* An order-24 walk takes 1.70× fewer steps than an order-6 walk (pooled), but E29's class representative cost 191–210 multiplications a step against 109–122, so the walk's cost ratio was 0.962

Step	Change	Effect	Gate
E19	τ_T/D_4 pipeline end to end	2.6–$8.8 \cdot 10^6\times$ rho; handover $2^{116.7}$ (model)	validated
E24	V_4 line, invariant solve	35–240× below E19 ($w = 0$); handover $2^{90.1}$ (model)	validated
E25	GLS fold on the V_4 line ($h = 16$)	$S \div 1.35$–2.67; handover $2^{88.0}$ (model)	validated
E31	chain solve, solver A	$S \div 2.50$–2.56 ($w = 0$), $\div 1.034$–1.053 ($w = 1$)	validated; B unranked
E27	reference: Karatsuba, batched inversions	every row 1.45× further from rho; $6.39 \cdot 10^3$–$4.06 \cdot 10^5\times$	validated
E32	distance ladder to $n = 2^{42.4}$	$1.5 \cdot 10^3\times$ at the top; fitted $n^* = 2^{85.5}$ $[2^{79.2}, 2^{94.3}]$	validated
E39	arrival-order pivots F	1.46–1.54× fewer solve mults ($w = 0$); 1.01× ($w = 1$)	validated
E40	rows stored from their lead	equal at $w = 0$; 1.25× (0.1), 1.36–1.38× (1)	validated

Table 31.3: The $k = 4$ line over $\mathbb{F}_{p^4}$ in the GLV note, as ranked. Effect: against the previous incumbent unless stated; distances are to the matched rho of the time, which E27 strengthened. Every row has rho ahead at every weight of the band; every handover is a model or a fit. Sources: GLV note §§9.6–9.57; validation directories under `benchmarks/`.

pooled at $w = 0$. The validation added that with one native operation per step priced the ratio is 0.996 at $w = 0.1$ and 1.205 at $w = 1$, so neither walk was the stronger reference across the band.

E34 built a cheaper representative, 15.17 multiplications instead of 86–92, so an order-24 step costs 7–10% more than an order-6 step instead of 80% (§9.39). Pooled over the curves, E29's walk over the cheap one is 1.563, 1.254 and 1.126 at $w = 0$, 0.1 and 1 (schoolbook), order 24 ahead on 7 of 8 curves; the cheap order-24 walk became the matched rho on E18's $j = 0$ curves, and E18's ψ rows moved 1.13–1.63× further from rho (`VAL-20261101-146a90`). E41 then asked why the order-24 walk fell short of its ideal factor of 2 (§9.53). The shortfall was sampling noise in E34's 16-seed design: on fresh seeds, 1,024 walks per arm, the order-6 walk sits at 0.969 ± 0.016 and the order-24 walk at 1.006 ± 0.016 of their ideals, and no order-24-specific cause exceeds 0.1% of steps. The corrected reference is resolved at every weight against both order-6 walks; E18's ψ rows move 1.761, 1.418 and 1.275× from rho at $w = 0$, 0.1 and 1 against E29's walk, and the GLS rows of Table 31.3 do not move (`VAL-20261102-918856`). All three are reference corrections, not index-calculus improvements; E29's base fold leaves the $j = 0$ line $\geq 2^{45}$ behind the V_4 line's modelled handover.

31.6 Quotients and cofactors

Quotienting the problem by a symmetry and clearing a cofactor are the other two ways an extension-field structure enters a factor base.

- *Symmetry quotients of the system.* The EcGFp5 quotient by $(\mathbb{Z}/2)^4 \rtimes S_5$ at $m = 5$ and its block-Wiedemann finish are in §24.2; they are per-system baselines over toy primes and are not repeated here.
- *Shared systems for cofactor lifts.* On Curve25519-family miniatures at $k = 20$, solving one shared F4 system for all eight cofactor lifts uses one call per target instead of eight but costs 30.7–628× the multiply-adds of separate lifts, because the maximum solving degree rises from 5 to 8–10 (`research/curve25519_cofactor_shared_20261102/README.md`); the Curve25519 programme is in §21.7.
- *Cofactor-cleared bases.* On prime curves with $h = 32$ the cleared base's A/B test returned `invalid_controls` by its frozen rule and is not cited as a gain (§22.4); the $m = 2$ torsion-anchor reading at $n = 37$ was reversed on its bootstrap recheck (§B.1).
- *A relabelled quotient count.* In the Joux–Vitse cover, the first form of the Hilbert early stop (B0)

cut counted S 2.53–2.64× while unpriced F4-driver work rose 8.3–8.5×; the study classifies it as relabelling, and only its repair is engineering (§30.4, Table B.6).

The common lesson is the one E21 and E35 drew on the V_4 solve: a quotient or a shared system pays only if the work it moves into an unpriced counter is priced, and the band is where that shows.

31.7 What the extension-field programme settles

- A fold's size is the order of the eigenvalue subgroup; on subfield curves no fold can do better than a constant, because $r \approx p^2$.
- At $k = 4$ on GLS twists with full rational 2-torsion, the baseline moved by validated constants: the V_4 line (35–240×), the GLS fold, the chain solve, arrival-order pivots and lead-stored rows. The reference moved the other way (1.45×), and the measured distance to rho is about $1.5 \cdot 10^3$ at $n = 2^{42.4}$, falling as $n^{-0.24}$.
- The crossover is a fit, $2^{85.5}$ $[2^{79.2}, 2^{94.3}]$ at $w = 0$ and later at higher weights, on a family no deployed curve belongs to.
- On $j = 0$ curves the order-24 action is the strongest rho, which moved those rows further from rho.

Part IV

The rho side

32 The generic floor and the index-calculus boundary

Every distance in this treatise is a ratio to Pollard rho. This part turns to the denominator. The present chapter asks what generic arguments say about index calculus itself: which constructions cannot reach rho no matter how they are engineered, where the generic track ends, and which extra inputs (auxiliary powers, advice tables) change the answer and by how much. Chapter 33 then asks how well the programme measured rho, and Chapter 34 how rho itself was varied.

Standing of the evidence. Two results in this part carry proof-carrying claims in `pcc/claims/`, both of standing unverified: the `wwalk` seed-averaged distances of Table 32.1 (`PCC-20261104-c97d81`, `PCC-20261104-d91a51`). Every other measured figure below is the producer's reported value. Where a sealed ecbench bound (`ECBND1h...`), an audit receipt or a `VALIDATION.json` exists, it is named; "validated" is used only for the last. The theory has the tiers of Part II: the collision floor of §32.2 is a derivation checked by Monte Carlo, not a theorem in the repository's proof system.

32.1 The square-root floor and its scope

Shoup's and Nechaev's generic-group bound says that an algorithm which touches a prime-order group of order r only through the group law and equality tests needs $\Omega(\sqrt{r})$ group operations. The programme records it as a closed obstruction for every generic lead: smarter walks, kangaroos, baby-step giant-step variants, grumpy giants, multi-walk and GPU variants all move the constant and never the exponent (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-generic-sqrt-lower-bound.yaml`). An efficiently computable automorphism group of order A divides the constant by $\sqrt{A}$, which gives the floor $S_{\text{floor}} = \sqrt{\pi/(2A)}$ of §4.2.

The bound has a precise scope, and the treatise's other chapters sit on both sides of it.

- *Inside it* are rho, BSGS, the kangaroo and the pair claw, and also every index calculus whose oracle can be simulated from group operations: the table, pair-table and root-finding oracles of the prime generic track (§32.3) and the oracle-augmented proposals of Chapter 8.
- *Outside it* are methods that read the representation: Gröbner and SAT decompositions that receive the factor-base constraint inside the system, Weil descent, and the cover attacks of Part II. Only these can change the exponent, and none has done so on a generic prime-order curve (Chapter 40).

Measured exponents. The sealed frontier (`docs/bounds/FRONTIER.md`, frontier `ECFR1hac90861c5e8a`, procedure 2, 130 bounds) fits GAE $= C\,r^{\alpha}$ per method. No ranked generic arm fits an exponent above 0.53, and none sits below its floor beyond interval noise. On the prime toy tier ($\log_2 r = 15.7$–25.3), `bsgs.negation` fits $\alpha = 0.529$ $[0.486, 0.595]$ (`ECBND1h9d01a5c2ae56`). The rho arms fit *below* 1/2 there: `rho.negation` 0.398 $[0.315, 0.475]$ and `rho.plain` 0.431 $[0.384, 0.488]$. That is a finite-size effect of fixed set-up (§33.2), not a sub-square-root walk; on the Koblitz medium tier ($\log_2 r = 32.1$–47.2) the strong signed-Frobenius rho fits 0.451 $[0.399, 0.503]$, with 1/2 inside the interval. A sub-$\frac{1}{2}$ fit at toy size is therefore never read as an exponent gain.

32.2 The collision floor for factor bases

Chapter 2 explains why balanced two-summand index calculus loses ground to rho as $r^{1/6}$. A sharper question is whether *any* factor-base construction can reach rho's constant while staying generic. The Curve25519 parity study (`research/curve25519_rho_parity_bound_20261104/`) answers it with a counting argument in the generic group model.

The model. A factor-base element obtained by a non-group operation, such as decoding an abscissa, has an unknown logarithm; in the generic model it is a fresh element. The algorithm learns only from equalities between computed elements, and each equality is one linear equation over $\mathbb{Z}/r$ in the F unknown

F	rows needed	model	Monte Carlo	measured (`wwalk`, $F = 2$)
0 (rho)	1	1.000	1.000	
1	2	1.500	1.517	
2	3	1.875	1.889	1.82–1.94× rho-neg
3	4	2.188	2.196	

Table 32.1: The collision floor: cost of a single-target index calculus with F unknown base logarithms over rho, as modelled, as simulated (4,000 trials at r = 131,203, seed 20261104, `research/curve25519_rho_parity_bound_20261104/data/sim.json`), and as measured on the Curve25519-family autolab incumbent `wwalk`, a walk table on a base of two abscissae. The measured column spans $k = 24$–32 and two seed regimes (`research/curve25519_rho_parity_bound_20261104/README.md`, §2); claims `PCC-20261104-c97d81` and `PCC-20261104-d91a51`, standing unverified.

base logarithms and $\log Q$. An equality that involves no base element already yields $\log Q$: that is rho. A solve that uses its base needs $F + 1$ independent equations.

Proposition 32.1 (collision floor; derived in the generic group model, checked by Monte Carlo)**.** *Let n walk points fall uniformly into the $r/2$ negation classes, so that colliding pairs arrive as a Poisson process of mean n^2/r. The k-th collision appears, in expectation, at*

$$\mathbb{E}[n_k] = \sqrt{r}\,\frac{\Gamma(k+\frac{1}{2})}{\Gamma(k)}.$$

Rho needs $k = 1$, which gives $\sqrt{\pi r/4}$, the negation-map constant. A single-target index calculus with F factor-base elements of unknown logarithm needs $k = F + 1$, and so costs at least

$$\frac{\Gamma(F+\frac{3}{2})}{\Gamma(F+1)\,\Gamma(\frac{3}{2})}$$

times rho: 1.500 *for* $F = 1$, 1.875 *for* $F = 2$ *and* 2.188 *for* $F = 3$.

Derivation. The Poisson count of pairs after n draws has mean $t = n^2/r$, so the k-th arrival time T_k is $\Gamma(k, 1)$-distributed and $n_k = \sqrt{rT_k}$. Then $\mathbb{E}\sqrt{T_k} = \Gamma(k+\frac{1}{2})/\Gamma(k)$, and the ratio for $k = F + 1$ against $k = 1$ is the displayed quotient. Since $\Gamma(F+\frac{3}{2})/\Gamma(F+1) \sim \sqrt{F}$, the floor grows as the square root of the number of unknown base logarithms (this asymptotic is ours, not the source's). Table 32.1 sets the model against the study's Monte Carlo and the lab's measured incumbent.

What it explains. The autolab's generic incumbent converged near 1.8× rho, and the floor says why: it is the $F = 2$ constant, not an implementation shortfall. Earlier fixed-table readings of 1.43–1.55× were single draws of the walk table, as the autolab's own correction I1 found. The Monte Carlo sits about 1% above the model, which the study attributes to the discreteness of pair counts.

What parity would need. The floor leaves three exits, and the study closes or prices each.

- $F = 0$. A base whose every logarithm is known, such as multiples of G, needs one collision and reaches rho's constant. It is rho or BSGS relabelled, which the success criterion excludes.
- *Rows without collisions.* An oracle that finds relations among base elements without group operations leaves the generic model, and its native work is then priced across the band. The best measured such oracle, F4 with trace replay on the $t = 4$ isogeny base, is $1.83 \cdot 10^4\times$ rho at $w = 0.1$ and its modelled cost scales as $r^{1.03}$ against rho's $r^{1/2}$.
- *Algebraic oracles, screened.* The follow-up screen (`research/curve25519_rho_parity_bound_20261104/ALGEBRAIC_SCREEN.md`) prices every known family at $k = 20$ against $S(w) = (F+1)(r/C)(1 + wc)/\sqrt{r}$, for coverage C and c native operations per trial, with set-up and linear algebra uncharged. The measured families fail on coverage before native cost counts. The cheapest design is a full-coverage $m = 2$ base of 256 points with one root-find per base point: 0.44× rho at $w = 0$, but 265× at $w = 0.1$ and $1.6 \cdot 10^3\times$ at $w = 1$. Parity would need $c \le 14$ native operations per trial at $w = 0.1$ and $c \le 3$ at $w = 1$; the whole $w = 1$ budget is about 1,000 GAE. All values are model outputs from measured or derived inputs.

Limits. The argument assumes uniform walk points. Real walks carry an overhead factor; measured negation rho sits at 0.98–1.08$\sqrt{r}$ on these curves against the model's 0.886$\sqrt{r}$, so a ratio compares like with like only when both arms carry the same factor. Linear algebra is uncharged, which favours index calculus, and dependent rows would only raise the count. The study states that the bound holds for one target and for many, but amortising base logarithms over many targets is outside its single-target question. Read with Chapter 2, the floor is the reason the leaderboard's success criterion counts each improvement over the index-calculus baseline separately from the distance to rho: a generic factor base can approach the floor, and the floor is above rho.

An open tension. The prime-field residual collectors are rho walks that need $B+1$ collisions (`research/ic_residual_dp_eq_20261101/PROTOCOL.md`), and their measured distance to matched rho is 1.54 and 1.64 in the cells an independent validator recomputed, above the $F = 1$ floor. Two producer-reported cells of the scale study, 1.32 and 1.39, lie below it. No source states the collectors' number of unknown base logarithms or reconciles the two readings (§22.3).

32.3 Where the generic track ends: the BSGS boundary

The owner's decisions of 29 October (`research/prime_ic_program_20261029/DECISIONS.md`, ruling R1) split prime-field index calculus into a generic and an algebraic track (Chapter 40). The ruling's argument for why root finding is generic is worth stating because it is the template for every later classification. At $m = 2$, `rootfind-subtract` fixes $x_1 = c$ for each c in the base, solves the remaining quadratic $S_3(c, X, x_R)$ through its roots $x(R \mp P_c)$, and tests each root against a hash index of the base. It returns the solution set a Gröbner solve of $\langle S_3, F_V(x_1), F_V(x_2)\rangle$ would, but its cost never depends on F_V and it runs identically on a random base. It is therefore a generic algorithm, subject to $\sqrt{r}$.

The k-sum law tends to BSGS. The negation-folded k-sum table (`research/prime_ksum_table_20261028/`) stores $2^{k-1}\binom{A}{k}$ signed k-sums of a base of size A, costs one probe per walk step, and has the generic cost law $2^{k-1}\binom{A}{k} + c_k Ap/(2^k\binom{A}{k})$ with exponent $k/(2k-1)$: 3/5 for $k = 3$, 4/7 for $k = 4$, falling towards 1/2 as k grows and never below it. The limit is baby-step giant-step, and the study says so. Its measured gains are in Chapter 40.

Generic relabellings measured as arms. Three studies measured generic decompositions as index-calculus arms and set each against the same-binary negation rho.

- *Target-blind known-log two-sum* (`research/ecbench_known_log_targets_20261021/`). Its cold counted cost was below the abscissa pipeline's at 16, 20 and 24 bits on both panels and throughout the band (paired ratio at $w = 0$ from 0.902 of the pipeline's at 12 bits to 0.064 at 24 bits), and the 12-bit pair is unresolved. Negation rho stayed cheaper at every size: at 24 bits rho over known-log is 0.4293, 0.3998 and 0.3025 at $w = 0$, 0.1, 1. Wall time is L0 and the receipts are same-host replays.
- *Curve25519-family $m = 3$ decompositions* (`research/curve25519_m3_generic_20261102/`, L1, counted units). On holdout targets the `ksum:m=3` oracle cut the incumbent's counted GAE by 24% at $k = 20$ and 40% at $k = 24$ across the band, and was unresolved at $k = 28$. The distance to rho stays 5.7–6.0× ($k = 20$), 6.2–6.5× ($k = 24$) and 11.2–17.7× ($k = 28$) across the band; the `mitm:m=3` oracle is worse than the incumbent at every N. No `VALIDATION.json` exists. The collision floor of §32.2 is the theory for this family.
- *The pair claw* of a sibling repository is randomised BSGS on signed-Frobenius classes; it is treated with the memory methods in §34.6.

The end-to-end census. The frontier refresh of 31 October (`research/ecbench_e2e_frontier_20261031/`) inventoried 74 live ecbench method identifiers (31 index-calculus, 32 rho or transfer, seven BSGS, two kangaroo, two claw) and eleven nested solver engines, and found that no admitted result on the generic prime, random-binary or medium Koblitz ladders beats matched cold one-target rho. Its two named exceptions are both classified away from the generic boundary:

- at $n = 29$, on one column, the Frobenius-layout pipeline is 0.366 [0.241, 0.523] of signed-Frobenius rho, but at one column the repository classifies that pipeline as Frobenius-folded BSGS, and the crossing rests on rho's fixed set-up at $\log_2 r = 15.4$ and widens with r;
- the GHS-weak plane-quartic family is below matched rho in its own comparison (Chapter 40; L0, no validation file), a frontier for that family only.

Arm	$w = 0$	$w = 0.1$	$w = 1$	fitted α
comb, $r - 1$ (48 pairs)	0.552 [0.455, 0.651]	0.528	0.418	0.450 [0.406, 0.473]
comb, $r + 1$ (48 pairs)	0.279 [0.216, 0.419]	0.259	0.165	0.428 [0.389, 0.460]
naive, $r - 1$ (40 pairs)	7.660 [6.33, 9.93]	6.983	3.938	0.441 [0.388, 0.497]
`rho.negation` ($r - 1$ curves)				0.496 [0.431, 0.611]
`rho.negation` ($r + 1$ curves)				0.513 [0.386, 0.636]

Table 32.2: Cheon's DLP with auxiliary inputs against matched negation rho: pooled paired ratio of sums over 36–56 bits at three band weights, and fitted exponents. The oracle price moves every pooled ratio by at most 0.003. Source: `research/cheon_ecbench_20261022/REPORT.md`; not validated.

A current-binary audit reproduced only 72 of 144 historical F4/F5/F6 rows, so those solver counts stay bound to the binary that produced them. The census's 92 planning shards had not run at the time of writing.

Engineering towards parity on a toy suite. Before ecbench, a September study pursued full-cost parity with signed-Frobenius rho on ten toy cases (`research/rho_parity_20260915/`), with frozen all-cases gates. Five engineering iterations ran; fresh chart-over-rho runtime ratios after the fourth were 1.08–2.21 on a shared host, and the verdict is "rho parity is not established: the frozen all-cases gates are not met" (`research/rho_parity_20260915/RESULTS.md`).

32.4 Auxiliary inputs: Cheon's algorithm

Chapter 21 records, from the September study, that Cheon's attack beats rho from 24 bits up for the discrete logarithm with auxiliary inputs; that reading was not priced across the band and is superseded by the ecbench measurement below, which finds no exponent separation. That problem is the largest generic line of the programme absent from the treatise, and its boundary with the plain ECDLP is exact.

The floors. Given $G, [\alpha]G, \ldots, [\alpha^d]G$, a generic algorithm needs $\Omega(\sqrt{r/d})$ operations, and Cheon's $r \pm 1$ algorithms attain $\sqrt{r/d} + \sqrt{d}$ when $d \mid r - 1$ and $\sqrt{r/d} + d$, from $2d$ powers, when $d \mid r + 1$ (`research/torsion_auxiliary_inputs_20260918/README.md`).

The acquisition obstruction. From $(P, [x]P)$ a generic algorithm forms only $[a + bx]P$. The point $[x^2]P$ lies outside that span, and computing it is the square-DH problem, which has the same $\Omega(\sqrt{r})$ generic bound. Self-supplied powers therefore cost rho-scale work, and Cheon never touches the ordinary ECDLP generically (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-cheon-aux-acquisition.yaml`, decision `DEC-20260802-204`, closed as a theorem). A non-generic way to compute $[x^2]P$ below $\sqrt{r}$ would reopen it.

Cheon as ecbench methods. Three registered methods, `cheon.pminus1_comb`, `cheon.pminus1_naive` and `cheon.pplus1_comb`, are charged in ecbench's counted unit (`research/cheon_ecbench_20261022/REPORT.md`). The leaked powers come from a planted static-DH oracle whose queries ($d - 1$ for $r - 1$, $2d - 1$ for $r + 1$) and server work are counted outside `total_gae`; a method never sees the scalar, and a public target is refused. Bounds of querying arms live in separate *oracle domains* on the frontier, with oracle queries as an extra axis of dominance. Table 32.2 gives the pooled distances to matched negation rho over 36–56 bits (320 runs at L0, 320 of 320 replayed exactly).

The comb arms' order against rho holds across the twelve-cell band of native price and oracle price, so they are ranked below rho in operations; rho stays on every oracle domain's frontier because it makes no oracle queries. *No exponent separation is established*: the Cheon and rho exponent intervals overlap. Per size, the $r - 1$ comb is cheaper than rho with an interval excluding 1 at 36 and 44–56 bits and dearer at 32 and 40. The original sessions mix the 32-bit toy curve with medium-tier curves, so frontier procedure 2 cannot rank them; a re-run on the medium tier alone is frozen as a protocol with no results yet (`research/cheon_ecbench_medium_20261106/PROTOCOL.md`).

Setup	Curve	top exponent q	best d $(r-1)$	bits below rho
EIP-4844 KZG, mainnet	BLS12-381	4,095	3,648	5.74
EIP-4844 KZG, largest	BLS12-381	32,767	30,531	7.27
Zcash Sapling	BLS12-381	4,194,302	4,142,391	10.82
Filecoin	BLS12-381	268,435,454	265,113,024	13.82
Aztec Ignition	BN254	100,800,000	100,663,296	13.12
Perpetual Powers of Tau	BN254	536,870,910	536,259,126	14.32

Table 32.3: Cheon's generic floor at constant 1 for deployed powers-of-tau setups, against negation rho ($2^{127.25}$ on BLS12-381, $2^{126.62}$ on BN254). Floors, not measurements; the practical $r-1$ algorithm ran at about 20× its floor over 24–48 bits (`research/torsion_auxiliary_inputs_20260918/`), and applying that constant at 255 bits is extrapolation. Source: `research/srs_auxiliary_census_20260927/results/srs_census.md`.

A retraction: Cheon/rho on the P-224 shape. A study built prime-order curves of 32–56 bits whose $r \mp 1$ smooth parts have P-224's ratio $\log d / \log r$ (0.12943 and 0.21106, within 0.015) and ran 126 Cheon and 84 rho solves on them (`research/cheon_p224_shape_20261022/REPORT.md`). Its preregistered gate H3 failed: 5 of the 84 sealed rho runs have S outside $[0.3, 4]$, so under its protocol the rho reference is unreliable, and on 23 October **every Cheon/rho ratio, crossing and advance reading of the study was withdrawn**. What stands is the Cheon-only scaling:

- fitted exponents 0.447 $[0.417, 0.477]$ $(r-1)$ and 0.413 $[0.379, 0.447]$ $(r+1)$, against formula values 0.435 and 0.395;
- formula costs at the P-224 shape of $2^{97.50}$ $(r-1)$ and $2^{88.36}$ $(r+1$, about $2^{48.3}$ sequential powers needed) against rho at $2^{111.83}$;
- extrapolations of the free power law to 224 bits of $2^{101.71}$ $[2^{96.3}, 2^{107.1}]$ and $2^{95.59}$ $[2^{89.3}, 2^{101.8}]$, which are models, not measurements.

Which deployed setups release powers. Two censuses separate the question "how far could Cheon go" from "what does any protocol supply".

- *Standard curves* (crypto-autoresearcher run `RUN-AUXIN-fd1edc`, zero Cheon executions). The best exponent $E(r) = \min_d \log(\sqrt{(r \pm 1)/d} + \sqrt{d}) / \log r$ is about 0.254 for P-256, secp256k1, BN254 and BLS12-381 (each at d between about $2^{126.8}$ and 2^{128}), 0.394 for P-224 $(r+1)$, and at most 0.452 for P-521. These values require degree-d supply; they are not vulnerabilities. Joining them with a typed supply roster is blocked by a refused protocol amendment (`DEC-20260923-a4c3d5`).
- *Powers-of-tau setups* (`research/srs_auxiliary_census_20260927/`, Table 32.3). This is a reproduction of the ethresear.ch analysis (thread #6692), not a discovery. It adds the EIP-4844 ceremony, exact setup sizes from each project's specification, and the largest admissible $d \mid r-1$ instead of the largest power of two, which gains about 0.49 bits for Zcash Sapling, Filecoin and Perpetual Powers of Tau.

Open. A toy reproduction with memory charged and CRT composition of supplied residues is approved but has no runs; an earlier attempt over $\mathbb{F}_{10007}$ could not separate the closed forms by a factor of two (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-cheon-toy-reproduction.yaml`).

32.5 Low-Hamming-weight logarithms

A logarithm of bit length m and weight w admits a meet-in-the-middle solve in $\sqrt{\binom{m}{w}}$ with large memory, and the literature (Esser and May) reaches $2^{0.65m}$ with polynomial memory. This is a different problem from the uniform ECDLP, and no index-calculus row depends on it. Three leads are open and none has a result:

- collision walks on weight-w representations, whose first stage expired its claim before finishing, which is not a negative result (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-lhw-polymem-collision-walks.yaml`);
- the residue-guess price of mod-M representation filters in a prime-order group, an unproved multiplier with an approved experiment and no runs (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-lhw-modular-filter-price.yaml`);

- a preprocessing frontier for the low-weight class, unexamined (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-lhw-preprocessing-frontier.yaml`).

The kangaroo's closure (§34.6) names low-entropy secrets as the one case that would reopen it.

32.6 Advice and query bounds that bind index calculus

Chapter 3 states the Corrigan-Gibbs–Kogan bound $S\,T^2 = \tilde{\Omega}(\varepsilon N)$ for S bits of advice and online time T [8], and Chapter 8 records that table-based decomposition obeys $ST^2 \gtrsim N$. Both matter because the index-calculus set-up is advice: a root or pair table built before any target is seen.

Where the online rows sit against the line. A derived calculation (`research/ic_preprocessing_rho_equal_advice_20261011/REPORT.md`; no run on a curve) gave Bernstein–Lange rho [4] the index calculus's own root-table entries as advice, on the folded group of size $r/(2n)$, with online cost $\approx 2\sqrt{N/S}$ walk steps.

- At $n = 73$ the index calculus's online target stage takes 457,561 probes against about 9,500 BL steps ($\approx 48\times$), and sits $2^{13.2}$ above the generic $ST^2 \approx N$ line.
- At $n = 83$ it takes 8,845,441 probes against about 2,600 ($\approx 3{,}400\times$), $2^{25.4}$ above the line.
- The same calculation corrected its parent note: the note's 3,000–6,000× at $n = 73$ used the rank stage's mean probes per relation, and BL precomputation is $\sqrt{NS} \approx 1.2 \cdot 10^{11}$ steps at $n = 73$ (days on one core), not "seconds to minutes". For one target, neither beats plain rho.

The measured equal-advice controls that later retired every online win are in Chapter 41 (Table 41.1) and Table 40.2.

Two open generic leads. Two constant-level questions about the walk itself remain unexamined and are listed here because each would move every reference at once.

- *Equivalence classes beyond* $\mathrm{Aut}(E)$. Rho could walk on classes $\{[h]P : h \in H\}$ for a subgroup $H \le (\mathbb{Z}/r)^\times$ larger than the automorphism group and gain $\sqrt{|H|}$, but only if a class representative is cheap; for multipliers that are not automorphisms every canonicalisation costs a scalar multiplication. The proposed obligation is an $\Omega(\log h)$ per-step canonicalisation bound for $h > 6$ (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-non-automorphism-equivalence.yaml`; hypothesis `H-EWALK-29cb90`, never tested).
- *Cheaper step primitives.* Tag tracing (Cheon, Hong and Kim) and point-halving steps on binary curves would lower native cost per step without changing the counted constant; neither has been measured here (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-cheaper-step-primitives.yaml`).

33 The matched rho reference

The validation gate of §41.2 asks every row for "a matched rho reference: the strongest available, in the same unit and the same binary, priced across the band". This chapter asks how well that reference is known: whether ecbench charges rho what theory says, how large its finite-size bias is, which rho wins among the references once the band is applied to them, which earlier references were weaker than they looked, and how the strong Koblitz reference was engineered and checked. It complements §4.2 and §4.3, which define the instrument and the band, and Chapter 41, which lists the corrections that matching forced on index-calculus rows.

33.1 Calibration against analytic constants

The calibration study of 2 October (`research/ecbench_calibration_20261002/`) tested whether ecbench's accounting reproduces the constants that seven generic methods' analyses predict. Four sessions ran on one Apple host at L0 (operation counts only): 1,008 prime and 864 Koblitz runs in round 1, and two follow-ups of 256 runs each. All 2,384 executions verified, and each of the four audit receipts reports 12 of 12 replays identical. Table 33.1 gives the predictions at each family's largest curve.

Two preregistered predictions failed as written, and the failures are kept. A follow-up on fresh targets traced both to sample size.

- *Rho's spread.* A rho run's cost has a coefficient of variation near 0.5, so a 24-run mean moves about ±20% at 95%. The [0.85, 1.35] band had been set on a point estimate without that spread; the interval contained the theory constant throughout.
- *The kangaroo is deterministic per target.* The three rounds of each target were identical, so the effective sample was eight targets. Its cost tracks $|k - r/2|$, from $S = 0.13$ at distance $0.017r$ to 2.48 at $0.347r$, and the eight targets averaged $0.119r$ against a uniform $0.25r$, about -2.6σ. The target derivation itself passes a Kolmogorov–Smirnov test over 20,000 draws.

The lesson the study draws binds every later protocol: a tolerance on a mean is set from that mean's sampling spread, and a method that is deterministic per target needs targets, not rounds. The calibration establishes the counted charge for these seven methods up to $r = 2^{39}$; it establishes nothing about wall time (all L0), about index calculus (which has no closed-form constant), or above 2^{39}.

33.2 Finite-size bias and the missing prime medium tier

A calibrated charge is not a calibrated constant at every size. Rho's reference implementations pay fixed set-up (jump tables, distinguished-point bookkeeping, walk starts), which is a visible share of a toy solve.

- On the prime toy tier, `rho.negation` measures 1.453 [1.184, 1.772] times its floor with $\alpha = 0.398$ [0.315, 0.475] (`ECBND1h67e22de83dca`), and `rho.plain` 1.920 [1.633, 2.221] (`docs/bounds/FRONTIER.md`).
- On the Koblitz toy tier the strong reference is 16.8× its floor with $\alpha \approx 0.13$, because its set-up is 71% of its total at $r \le 2^{21}$ (`research/ecbench_bounds_challenges_20261006/`); every Koblitz toy method sits 5–10× above the floor.
- At 28–40 bits the same `rho.negation` measures 1.075–1.146× its floor on four fresh prime-order curves (`research/rho_negation_audit_20261020/`, L0, 64 targets per curve), and the strong Koblitz rho 1.089 [0.927, 1.261] on the medium tier (`ECBND1hcf69c27b4978`).

The toy excess therefore shrinks at medium size, as fixed set-up predicts, but this is descriptive: no prime medium-tier rho bound exists in the plain ECDLP domains of the frontier. Medium-tier `rho.negation` bounds (1.114 and 1.225× the floor at 35.6–55.6 bits) exist only inside the Cheon oracle domains, fitted on one tier of a mixed session and so unranked (§32.4). The open step is to seal `rho.negation`, `rho.plain` and `bsgs.negation` at $\log_2 r = 32$–48 and fit the toy-to-medium ratio (`crypto-autoresearcher: knowledge/leads/ecdlp/generic/LEAD-RHO-matched-rho-calibration.yaml`).

Consequence. Every prime-field distance in this treatise is paired against rho on the same curve and in the same session, so the toy excess sits in the denominator. Where both arms carry fixed set-up the paired ratio is fair at that size; it does not transfer to another size, and an exponent fitted over toy sizes

Method	curve	mean S [95%]	theory	S/theory	verdict
`bsgs.textbook`	fp26	1.466 [1.350, 1.572]	1.500	0.977	pass
`bsgs.interleaved`	fp26	1.396 [1.197, 1.587]	1.333	1.047	pass
`bsgs.negation`	fp26	0.966 [0.850, 1.073]	1.000	0.966	pass
`rho.plain`	fp26	1.306 [1.075, 1.561]	1.253	1.042	pass
`rho.negation`	fp26	0.935 [0.681, 1.198]	0.886	1.055	pass
`rho.signed_frobenius`	f2m41	0.134 [0.106, 0.166]	0.138	0.970	pass
`rho.plain`	f2m41	0.993 [0.688, 1.307]	1.253	0.792	fail
`kangaroo.vow`	fp26	1.203 [0.684, 1.761]	2.000	0.601	fail
follow-up, 64 fresh targets × 2 rounds					
`rho.plain`	f2m41	1.240 [1.109, 1.389]	1.253	0.989	pass
`kangaroo.vow`	fp26	1.972 [1.759, 2.178]	2.000	0.986	pass
`rho.negation`	fp26	0.991 [0.884, 1.109]	0.886	1.119	reference

Table 33.1: ecbench calibration: mean S against each method's analytic constant at $r = 2^{25.35}$ (fp26) and $r = 2^{39}$ (f2m41); 8 targets × 3 rounds in round 1. Selected rows; source `research/ecbench_calibration_20261002/README.md`.

Domain	v1 page	v2 page (current)
prime toy	ops leader `bsgs.negation` (1.100 [0.978, 1.225]) named on a point estimate	no ops leader: `bsgs.negation` and `rho.negation` (1.453 [1.184, 1.772]) overlap, a tie set of six bounds; `rho.plain` (1.920) no longer dominated by `rho.negation`
Koblitz toy	leader a three-size tier-subset bound	no ops leader (tie set of three); both `rho.signed_frobenius` bounds have a large tail (max/median 54.2 and 11.7)
Koblitz medium	ops leader strong rho	ops leader strong rho, 1.089 [0.927, 1.261], $\alpha = 0.451$ [0.399, 0.503], 144 runs, now separating on all six degrees; memory leader the pair claw

Table 33.2: The bounds frontier before and after procedure 2. Source: `research/ecbench_frontier_v2_20261023/README.md`, `docs/bounds/FRONTIER.md`.

inherits rho's sub-$\frac{1}{2}$ slope. This is why Chapter 40 quotes distances per size and fits gap exponents only where the source does.

33.3 The sealed frontier, version 1 and version 2

Bounds are sealed records of constant and exponent per method (§4.2). The first frontier page named leaders that its data did not support. The rebuild of 23 October under frontier procedure 2 (`research/ecbench_frontier_v2_20261023/`) is a **correction**: no session ran; each of the 46 published v1 records stayed byte-identical and gained a fit-version-2 twin; only v2 bounds fitted on four or more curves of a whole session are ranked; and a leader is named only when its interval separates from every other method's on shared curves. Table 33.2 gives the change per domain.

The v1 exponent intervals also carried degenerate ends from one-size resamples; the kangaroo's $[0.179, 2.500]$, on which "α agrees with declared" rested, is $[0.401, 0.724]$ in its v2 twin. Prime medium-tier plain-ECDLP bounds are still absent (§33.2).

The first challenge verdicts. A bound moves the frontier only through a verdict against a sealed challenge on frozen workloads. The first two (`research/ecbench_bounds_challenges_20261006/`, epoch 1, 288 of 288 replays identical in each session, L0) were the reference challenges themselves.

- *Prime toy*: `bsgs.negation` against `rho.negation` reads **trade**, 0.8250 [0.725, 0.945] of the operations at 15.79 [13.38, 19.06]× the table. No single curve's interval excludes 1; the pooled 96 pairs do.
- *Koblitz toy*: the lean `rho.signed_frobenius` against the strong reference reads **advances** at the constant level, 0.2991 [0.235, 0.402] of the operations and 0.560 of the table. It prices the strong walk's toy set-up and says nothing above $r = 2^{24}$.

Neither moves an exponent. Both say which reference is "strongest" at toy size depends on what is charged, which the band makes explicit.

33.4 The band orders the references themselves

The pricing band of §4.3 was introduced to compare index calculus with rho fairly. Applied to the rho references themselves, it changes which of them is strongest.

- *Negation against plain, prime toy.* `rho.negation` is better than `rho.plain` at the charged cost, but the order flips at 0.34–0.55 GAE per unpriced canonicalisation, inside the band, so neither dominates (`docs/bounds/FRONTIER.md`). At 28–40 bits the negation audit finds the same: the order holds at $w = 0$ and 0.1; at $w = 1$ the negation walk's S is 1.89–2.03, above plain, and the ranking is unresolved on all four curves (`research/rho_negation_audit_20261020/band.json`). A real canonicalisation is one $\mathbb{F}_p$ negation and one comparison, far below 1 GAE, but it was not priced.
- *Signed Frobenius against plain, Koblitz toy.* Dominance holds at the charged cost and flips at about 0.123 GAE per uncharged operation; the kangaroo's dominance by signed-Frobenius and negation rho flips at 0.290 and 0.417.
- *Extension-field references.* Two references built for the extension-field programme use the full symmetry: `rho.gls` ($A = 4$, GLS twists over $\mathbb{F}_{p^2}$) and `rho.frobenius_orbit` ($A = 6$, $E/\mathbb{F}_p$ on $E(\mathbb{F}_{p^3})$), each on nine curves with 8 targets (`research/ext_endo_rho_20261023/analysis/`). At $w = 0$ `rho.gls` costs $S = 0.472$–0.921 against negation's 0.812–1.165, and at $w = 1$ 2.316–4.197 against 1.613–2.312; every pair on every curve is unresolved inside the band. `rho.frobenius_orbit` costs 0.482–0.692 at $w = 0$ against negation's 0.783–1.404, and is below negation across the band on four of nine curves; on three curves `rho.plain` is below negation across the band.

The rule this implies is already in the gate's wording: a row's "matched" reference is the strongest available *across the band*, and where the references themselves are unresolved the row is ranked against each. A row ranked against one weight's winner alone overstates index calculus at the other weights.

33.5 References that lacked the negation map

The negation-map audit of 20 October (`research/rho_negation_audit_20261020/`) inventoried every rho walk the programme used as a reference. Two lacked the map: `volcano-rho`'s native walk in this repository, and crypto-autoresearcher's `index_calculus/rho.py`. Both now walk on $\{R, -R\}$ by default. This is a **correction of earlier references**: any index-calculus ratio computed against either before 20 October was against a rho $\sqrt{2}$ weaker than the negation walk.

- *The matched reference delivers the saving.* On four fresh prime-order curves of 28–40 bits (512 records, 16 exact replays), pooled plain over negation is 1.371 $[1.257, 1.491]$, with $\sqrt{2}$ inside the interval.
- *Little of it reaches wall time* in ecbench's implementation: the median paired wall ratio is 1.140 (L0, descriptive).
- *`volcano-rho`.* On a 36-bit curve, 256 solves per arm, off over on is about 1.40 pooled over five arms, at 9–16% more nanoseconds per iteration (an extra branch, a field negation and a 16-entry cycle-window scan), so wall time improves by about 1.2×.
- *crypto-autoresearcher.* 640 paired solves at 16–32 bits give plain over negation 1.353 $[1.281, 1.425]$; its adapter now labels the old walk `rho.plain` and refuses it as a matched reference.

An earlier fix in the collaborative walker (`research/generic_dlp_correctness_20261017/`) found that its negation-map 2-cycle test compared an unflipped sum with a normalised point and never fired; dead trails fell from 353 to 16 across three curves once it compared after the flip.

References that were not the strongest. A register item of the repair framework (`research/r12_rho_ns_per_step_20261026/`) listed every committed ecbench spec that pairs an index-calculus arm with a rho arm: 745 specs in 68 directories, of which 37 directories pair index calculus only with a rho other than the strong reference. On Koblitz curves, counted ratios against `rho.negation`, `rho.plain` or the tuned signed-Frobenius walk are therefore not matched to the strongest rho; on prime curves `rho.negation` is the strongest reference. The same note explains an audit's 599 against 240 ns per step: the unbatched tuned rho is 5.48, 9.34 and 9.56× slower per step than the strong rho at $n = 41, 53, 61$ (L0, Apple arm64, A/A within ±7%), and within the strong algorithm the field backend matters by 1.07–1.76× with the hardware. The 599 ns figure being the tuned rho is a hypothesis supported by the size match.

Round	change	instr./step	native time
canon (8 Oct)	longest-zero-run rotation search; scratch reuse		1.40× per walk ($m = 83$), 1.46–1.61× ($n = 41$–61)
Q4 (16 Oct)	changes declared in two addenda	1,196 → 938	1.02× ($n = 53$), not established; 1.21× ($n = 61$)
canon2 (16 Oct)	six changes in the strong-rho source	938 → 781	
step3 (16 Oct)	target-feature walk; batched walk starts	781 → 718	
batchinv (22 Oct)	fixed-width batch inversion	707.2 → 679.7[a]	not established

Table 33.3: Engineering rounds on the strong Koblitz reference; instructions per walk step at $n = 53$. Sources: `research/strong_rho_canon_20261008/`, `research/rho_assurance_20261016/`, `research/rho_canon2_20261016/`, `research/rho_step3_20261016/`, `benchmarks/20261022-strong-rho-batchinv/` (carries a `VALIDATION.json`; its validator passes the instruction reduction and its findings contradict the native speed-up). [a]From the `main` of 22 October, which already read 707.2.

33.6 Engineering the strong reference

Chapter 36 records that the strong signed-Frobenius rho fell from 1,196 to 718 instructions per step in three rounds, and that a fourth round took `main` from 707.2 to 679.7 with no established native gain. Table 33.3 gives the rounds; every round left walks, step counts and recovered logarithms identical, so counted-unit records are unchanged and only instruction or wall comparisons move.

The assurance round's Q3 profile is the reason the rounds were worth doing: one step cost 1,196 instructions, about 57 field multiplications, against a floor near six or seven, and half of it was canonicalisation. A separate optimisation of ecbench's standing method (`research/ecbench_rho_optimized_20261014/`) cut complete-solve Callgrind instructions to 0.802155 of the trajectory-identical legacy path, with charged GAE exactly 1.0000 over 240 pairs and 720 of 720 replays identical; its native wall gate is unknown. The pattern is the same in every round: counted GAE is the comparison unit and does not move; wall-time comparisons against an older rho overstate index calculus by up to the round's factor (Table 41.3).

Prime-field and GPU walks. The batched affine step with Montgomery's simultaneous inversion is the prime-field engineering baseline. On P-256, a plain step costs 179 ns at 1,024 walks per inversion (`research/rho_walk_leads_20261014/`). A branch-free $\mathbb{F}_p$ 64-bit kernel took `prime_fast::rho_parallel` from 98.9 to 68.2 ns per step (`research/fp64_branchfree_20261016/`). An analytical model of counted operations times measured cycles per operation predicted a batched affine rho iteration to a median 1.8–5.5% on five fields at $B \geq 4{,}096$ walks after one fitted residual per field (`benchmarks/20261016-hwcost-rho-model/`). ECCp-131 throughput is in Chapter 21; GPU walks are in §34.9.

Harness overhead in small wall ratios. Until 22 October ecbench parsed its 4.7 MB curve registry on every lookup. Parsing it once cut the timed solve window of every rho, BSGS and kangaroo arm to 0.03–0.07× at 20 bits and 0.64–0.81× at 32 bits, against 0.96× for the index-calculus pipeline (`benchmarks/20261022-ecbench-harness-overhead/`, which carries a `VALIDATION.json`; counted results unchanged). Any small-size wall ratio recorded before then charged rho for about 4 ms of harness work inside a 0.3 ms solve, which flattered index calculus.

33.7 Formal assurance of the reference

The strong reference is the denominator of every Koblitz row, so the assurance round (`research/rho_assurance_20261016/`) checked its correctness formally as well as empirically.

- *Lean 4* (v4.34.1, no Mathlib): 34 theorems with no `sorry`, `admit` or `native_decide`, using only `propext`, `Classical.choice` and `Quot.sound`. They cover the least-rotation canonicaliser (and its bridge to the Rust zero-run mask), Montgomery batch inversion in one- and four-accumulator forms over any field, and the walk algebra: starts, jumps, canonicalisation and escapes preserve $P = aG + bQ$, and the collision formula returns the logarithm whenever $b \neq b'$.

- *TLA+*: a model of the solve over $\mathbb{Z}_R$ meets all 41 configurations' declared verdicts under TLC, including seven mutants that each fail the property meant to catch them.
- *Steps*: mean steps over $\sqrt{\pi r/(2A)}$ are 0.994, 0.922 and 0.943 at $n = 41, 53, 61$ on 400 fixtures each, with spread 0.50–0.51 against an ideal random mapping's 0.52.

The gaps are stated: the curve group is modelled as the scalar field, the GF(2^m) arithmetic and basis maps are unverified, and word overflow is argued, not proved. Four findings changed no answer; one (the documented fruitless-cycle length) was a documentation fix.

Two further Lean files bear on rho's invariance (§34.4). `research/conductor_gap_rho_large_sample_20261018/lean/RepresentationHashInvariance.lean` proves, in 11 theorems with no `sorry`, that transporting an r-adding walk by a bijection compatible with the operation preserves its trajectory and its cost at every budget, and that transport is a bijection of walk spaces; so over a uniformly random partition rho's cost distribution is the same on isomorphic groups, and a non-group invariant can enter only through the one fixed hash an implementation uses. `papers/isogeny-class-rho/lean/ScalarTransport.lean` proves (3 theorems) that a zero- and addition-preserving injective map transports answers. Neither formalises curves or isogenies.

33.8 Statistics version 2 and censored rho

The statistics paragraph of §4.2 describes the cluster bootstrap of the original comparisons. PR #2442 replaced it with a linearised $t(K-1)$ interval over K workloads and, more consequentially, kept timed-out and exhausted runs in the ratio as censored lower bounds instead of dropping them (`research/ecbench_statistics_v2_20261105/`; no new runs).

- *Width.* On uncensored rows the new interval is 13.1× wider at $K = 2$, 1.23× at $K = 12$ and about 1.0–1.06× at $K \geq 16$. Its simulated coverage is 0.957, 0.948 and 0.947 at $K = 2, 4, 8$; the old percentile bootstrap covered 0.858 at $K = 2$ in the study's own simulation.
- *Censoring.* Dropping exhausted runs had understated `ic-sub6-mitm` over strong rho by 92.9× (6.981 → 648.2 [419.6, 1001], a lower bound), and signed-Frobenius rho over `bsgs.negation` on a GHS panel by 60.4× (3.512 → 212.0), where 13 of the 40 workloads had exhausted rho runs.
- *The A/A gate.* Correcting a wall ratio by its A/A bias flipped one gating decision in the sample: `ic-sub6-mitm`'s ns per GAE ratio 1.248 corrects to 1.147 [0.997, 1.321], inside noise.

The rho-side lesson is the second item: an exhausted rho run is a failure of the reference, and a comparison that drops it reads the reference as cheaper than it was. Earlier "× rho" figures from sessions with censored runs on either arm are lower or upper bounds accordingly.

Independence. A same-machine replay under a different Rust toolchain changed the environment class identifier and would have passed the host-class part of an independent validation (`research/ecbench_independence_20261009/`). A separate hardware class that excludes toolchain, kernel and policy fields now rejects such a receipt; the fingerprints remain self-reported, so an external runner's custody record is still required.

34 Walk design, memory and many targets

This chapter collects the variations of rho the programme measured: walks on automorphism classes, the iteration function, rho across isogenous curves and on Jacobians, the memory-heavy generic methods, many-target and advice-table rho, and rho at deployed scale. Each matters to index calculus in the same way: it moves the reference.

34.1 Automorphism classes

A walk on the classes of an efficiently computable automorphism group of order A saves $\sqrt{A}$ in steps, which is the ceiling for any automorphism walk (§32.1); classes beyond $\mathrm{Aut}(E)$ are the open lead of §32.6. The Koblitz signed-Frobenius reference ($A = 2n$) is the subject of Chapter 15. On prime curves the gains are smaller, and their measurement needed a correction.

$j = 0$ and $j = 1728$, with an erratum. Round 2 of the conductor gap had matched its crater rows against negation rho only. A follow-up ran `rho.cubic` ($j = 0$, $A = 6$) and `rho.quartic` ($j = 1728$, $A = 4$) on the same twelve crater curves (`research/prime_rho_automorphism_20261015/`). Its **erratum of 24 October**: the binary left the class set-up (16–58 GAE per solve) out of each record's total. Corrected,

- `rho.cubic`/`rho.negation` is 0.647 $[0.543, 0.770]$ (was 0.611), with the prediction 0.577 inside;
- `rho.quartic`/`rho.negation` is 0.819 $[0.722, 0.929]$ (was 0.791), whose interval no longer contains the prediction 0.707.

Against the automorphism rho every crater IC/rho ratio of round 2 grows by about 1.55× ($j = 0$) and 1.22× ($j = 1728$); that adjustment applies pooled ratios and is not a joint re-run. The class-representative map is unpriced, so the ratios do not bound full cost. The volcano-census withdrawal of Chapter 41 is the same correction found from the index-calculus side.

The crater is where the fold lives. The volcano study (`research/isogeny-volcano-rho/`) measured that the automorphism-folded walk needs $1.89 \pm 0.09\times$ fewer operations on a $j = 0$ crater ($\sqrt{3}$ predicted) and $1.32 \pm 0.06\times$ on a $j = 1728$ crater ($\sqrt{2}$); in wall time that is 1.33× faster on $j = 0$ and 2.3× *slower* on $j = 1728$, where the tool had no tuned walk. Across a whole $j = 0$ class the folded walk cost 0.439 $[0.401, 0.481]$ of the plain walk against $1/\sqrt{6} = 0.408$, and on the Koblitz crater the fold wins 1.7× (τ) and 2.2× (τ and negation) in wall time although each folded step costs 41 squarings more (`research/isogeny-class-rho/REPORT.md`). Every curve the isogenies lead to has lost the fold (§34.4).

GLS and $\mathbb{F}_{p^3}$ references. The extension-field references and their band behaviour are in §33.4. On E25's ten GLS curves, Karatsuba products with Montgomery-batched norm inversions cut the matched $\langle \pm\psi \rangle$ walk's affine addition from 97 to 66.3–66.7 $\mathbb{F}_p$ multiplications, making it 1.453–1.463× cheaper while the index-calculus side moved only 1.0048–1.0057×; every row moved further from rho, to $6.39 \cdot 10^3$–$4.06 \cdot 10^5\times$ at $w = 0$ (`benchmarks/20261030-e27-walk-karatsuba/`, validated: independent replay 11 of 11 runs, with errata in the validator report). It is a correction to the reference, not an improvement.

34.2 The negation map and fruitless cycles

A negation walk creates fruitless 2-cycles $P \to P \pm R_i \to P$, and every negation reference needs an escape. The walk-leads study (`research/rho_walk_leads_20261014/`, Modal CPU, $N \approx 2^{32}$–2^{40}) measured the cycle rate and two representative rules.

- With a fixed or per-point rule the negation walk takes 1.37× $[1.31, 1.43]$ and 1.33× $[1.28, 1.39]$ fewer steps than plain, with 487 fruitless 2-cycles per 10^6 steps against the predicted $1/(2r) = 488$ for $r = 1{,}024$ jump points.
- Changing the rule per epoch gives 0.83× $[0.79, 0.87]$: worse than no negation map.
- Without an escape, every $\pm R$ arm repeats on a useless collision after 821–1,974 steps against the plain walk's 69,210 to a useful one at $N \approx 2^{32}$: 0 or 1 of 200 solved.

The distributed pool of the sibling ecc-challenges repository walks plain rho on all 23 non-ECC2K-130 campaigns. Its negation-class walk took 0.700 $[0.673, 0.728]$ of the old walk's steps with 0 of 1,800 trapped trails and about 6% per-step overhead, on toy curves of 2^{23}–$2^{34.5}$; it is in an open pull request as a post-hoc amendment, default off. These figures are as the lead atlas reports them; that repository was not available to this draft (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-pool-class-quotient.yaml`).

34.3 The iteration function

Chapter 36 records the best-of-$P \pm R$ lead: 1.129× fewer steps under distinguished points, offset by its cost per step. The same study tested three more walk changes, and the in-degree argument behind the first is the useful part.

- *Best of k candidates.* Choosing the best-scoring of k images gives a point of score rank u the in-degree ku^{k-1}; since a random mapping's collision time scales as $1/\sqrt{\mathbb{E}[\lambda^2]}$, the predicted saving is $\sqrt{k^2/(2k-1)}$, 1.155 for $k = 2$ and 1.512 for $k = 4$. A positive control hits it (1.151, 1.511); under distinguished points best of four gives 1.463 $[1.401, 1.527]$ but costs 2.4–2.6× a plain step, a net of about 0.6.
- *Lazy reduction before the distinguished-point test*: no change (0.998 $[0.995, 1.001]$); indexing the table on a lazy x costs 0.733×, about the predicted $1/\sqrt{2}$.
- *Correlated walks.* A shared index sequence makes useful collisions impossible (0.073×, with 176 of 1,000 trials censored); a precomputed pair table $R_i + R_j$ gives 1.027 $[0.982, 1.073]$, since one addition still advances one step.

Mixing. Teske's r-adding walks are near random for r of 16 to 20. In the isogeny-class study the 32-adding walk ran at 1.0097 of $\sqrt{\pi n/2}$ on 4,400 solves against an expected 1.016; a planted 4-adding walk read 1.291 against Teske's 1.137, and served as the study's positive control (`research/isogeny-class-rho-n45/`). On hyperelliptic Jacobians, mixing the divisor hash through a finaliser did not reach its registered gains (§34.5).

A compiled store-all schedule. A theoretical current-plus-archive walk $X_{t+1} = X_t + X_{t+1-d_t}$ with offsets compiled once per group order (`research/tiered_rho_compiled_walk_20260921/`) has, exhaustively over all 65,537 target logarithms at 2^{16}, an exact capped mean of 19,027,631/65,537 additions against 19,114,086/65,537 for the matched unguided schedule: 0.4523% fewer. It makes no wall-time claim.

34.4 Rho across isogeny classes and volcano levels

Chapter 8 proves that transport by an isogeny conserves index-calculus cost, and Chapter 26 records the volcano census's level effects for index calculus. The generic twin is a measured null: rho's cost does not depend on which isogenous curve, stratum or level carries the instance.

One instance on every level. `volcano-rho` carried one instance (p of 40 bits, $n = 272{,}748{,}451{,}513$) by verified Vélu 3-isogenies to all 80 curves of a depth-3 volcano and ran 100 paired-seed solves on each of 29 curves and three isomorphic controls (`research/isogeny-volcano-rho/`, benchmarks row `20261016-isogeny-volcano-rho`; 32-adding walk, no negation). Table 34.1 gives the levels.

A control of 5,000 solves found that redrawing the secret, the generator or both, or taking $k \le 2^{16}$, matches the walk-seed spread (every $|z| < 0.8$, Kolmogorov–Smirnov $p > 0.6$).

Whole classes. The isogeny-class study (`research/isogeny-class-rho/`, protocol frozen before the run) ran 85,425 solves on 200 curves of four classes: 104 curves of a 347,004-curve prime-field class (12 endomorphism-ring strata, reached through ℓ-isogenies up to $\ell = 61$), 45 curves of a $j = 0$ class, 40 curves of a binary class and the Koblitz class over $\mathbb{F}_{2^{41}}$. No curve, stratum, level, isogeny degree or kernel-field degree left the preregistered band, and the spread of means between curves is 1.1–1.2× what the single-solve spread predicts. The 409- and 1,721-isogenies of the Koblitz curve, whose division polynomials have degrees 83,640 and 1,480,920, were found through torsion abscissae, and the curves they reach cost 0.978–1.009 of the reference by stratum. The detector works: a planted negation fold read 0.63–0.77 of

Level	curves	solves	mean iterations ± 95%
0 (crater)	2	200	700,483 ± 47,849
1	6	600	681,766 ± 26,525
2	10	1,000	637,158 ± 20,410
3 (floor)	11	1,100	659,589 ± 20,993
all isogenous	29	2,900	659,263 ± 12,422
isomorphic control	3	300	674,042 ± 40,816

Table 34.1: Plain rho on one instance carried to every level of a 3-volcano (curve A, ramified crater), against $\sqrt{\pi n/2} = 654{,}547$. Per-curve means span 572,846–733,141, the spread of 100-trial means of a quantity with single-solve standard deviation about 341,000; per-step cost is 641–661 ns on every curve. Source: `research/isogeny-volcano-rho/README.md`.

the unfolded mean (expected 0.71). A 46-bit replica (4,400 solves on 38 curves) found pooled 1.0097 against 1.016 and Kruskal–Wallis $p = 0.38$; one generator family flagged there (KW $p = 1.2 \cdot 10^{-4}$) **did not replicate** on a new draw (KW $p = 0.97$), and generator-level heterogeneity is recorded as unresolved (`research/isogeny-class-rho-n45/README.md`; L0, no ecbench records, so it supports no comparison claim).

In ecbench. A preregistered ecbench study on six 24-bit volcanoes found rho's floor-over-crater ratio 1.022 $[0.983, 1.062]$ (90%) and BSGS identical on every curve and model; its v2 found rho 1.004 (`benchmarks/20261017-isogeny-volcano-cost/`, `benchmarks/20261017-isogeny-volcano-cost-v2/`). The large-sample floor/crater ratio of Chapter 26 agrees. The Lean result of §33.7 is the reason no effect was expected: over a random partition, isomorphic groups give rho the same cost distribution.

34.5 Rho on Jacobians

Genus-2 and genus-3 transfer rows (Chapter 26) are ranked against a distinguished-point rho on the same Jacobian, so that reference needed its own calibration. Three rounds on 18 hyperelliptic instances (genus 2–4, N from 1,321 to 16,790,591, 40 seeds each) tested it in counted group operations.

- *Negation fold.* Folded over plain is 0.67–0.89 at $N \le 6 \cdot 10^4$ but a median 0.968 at $N \ge 10^5$; the registered ≤ 0.80 was falsified. Plain S_{walk} was 1.82 and 2.10 at the two largest rows against an ideal 1.2533 (`benchmarks/20261018-hec-rho-negation/`).
- *Hash mixing.* Passing the divisor hash through a finaliser left the plain median at 1.406 and fold over plain at 0.960; both registered gains were falsified (`benchmarks/20261018-hec-rho-hash-mixing/`).
- *Branch width.* Full-width branch coefficients brought the median S_{walk} at $N \ge 10^5$ to 1.390 and the largest row from 2.100 to 1.481 (`benchmarks/20261019-hec-rho-branch-width/`).

The third round is an accounting correction against index calculus: the 8-bit reference walk was slow at large N, so the early genus-2 to -4 ratios that read below rho are lower bounds (Chapter 26).

34.6 Kangaroo, BSGS, the pair claw and store-all schedules

Kangaroo. `kangaroo.vow` is never an operations leader: dominated on the prime toy tier, at 2.203 $[1.803, 2.527]\times$ its floor (`ECBND1h2e9e2024d8f3`) against negation rho's 1.453; unresolved inside the band on the Koblitz toy tier (the order against signed-Frobenius and negation rho flips at about 0.290 and 0.417 GAE); and unranked on the Koblitz medium tier, where 15.558 $[13.742, 17.597]\times$ its floor against signed-Frobenius rho's 1.188 is a three-size bound and a frontier entry needs four (`docs/bounds/FRONTIER.md`). Its calibration artefact of §33.1 came from its determinism per target. It is the right tool only for a known short interval or a low-entropy secret (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-kangaroo-interval.yaml`).

BSGS: fewer operations, more memory. `bsgs.negation` is in the prime toy tie set at 1.100 $[0.978, 1.225]\times$ the floor, the best point estimate, but with $0.502\sqrt{r}$ table entries, which at 256 bits is 2^{127}. The challenge verdict of §33.3 reads it as a trade: 0.825 of rho's operations at 15.8× its table. The frontier names no operations leader on the prime toy domain because the intervals overlap, and the memory rules BSGS out at cryptographic size.

The pair claw is BSGS. The four-point signed-Frobenius pair claw of cryptanalysis #175 was ported into ecbench and run cold against the strong rho on 48 one-target workloads over six Koblitz curves of $2^{32.1}$–$2^{47.2}$ (`research/ecbench_pair_claw_20261003/`; 960 runs verified, 12 of 12 replays identical, L0). Every factor-base logarithm is known, so it is a randomised BSGS on signed-Frobenius classes, of cost $(c + 1/c)/\sqrt{n}$ against rho's $\sqrt{\pi/(4n)}$.

- At its cost-minimising shape ($c = 1$) it costs 2.195 $[1.897, 2.546]$ strong rho, and at the shape the pull request ran at $n = 83$ it costs 32.65 $[25.29, 41.04]$. The ratio does not shrink with r.
- Without the Frobenius fold, `bsgs.negation` costs 6.969 $[6.063, 8.024]$ strong rho on the same workloads.

On the Koblitz medium frontier the claw at the pull request's shape is the memory leader ($0.004\sqrt{r}$ entries, 35.6× the floor), and the strong rho the operations leader; the claw's other sealed shape (2.390× the floor) is dominated by strong rho across the band. This is the $F = 0$ exit of §32.2 in practice.

Grumpy giants. Bernstein and Lange's two grumpy giants and a baby solve more targets per operation than BSGS or a random walk (0.71875 against 0.6753 of targets at $1.5\sqrt{\ell}$). A certified exhaustive ladder for its exact average-case constant ($\mathbb{Z}/\ell$, 2^{10}–2^{20}) has not run; its first stage met its support outcome at $\ell = 101,257$ (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-grumpy-store-all-schedules.yaml`). A certified constant below `bsgs.negation`'s would be a new toy operations leader, and could not beat rho at cryptographic size, because memory stays $\Theta(\sqrt{\ell})$.

34.7 Many targets: batch rho

The one-target rule of §4 exists because many-target figures were the easiest to misread. The Koblitz collection thread had quoted every "vs rho" as the per-target cost of 32 targets solved together, divided by rho solving one, and had counted each rho step and each stored table pair as one addition. The repository's multi-target margin rule makes batch rho at the same k the reference for such a figure, after Kuhn and Struik. A study frozen before it ran measured that reference and priced both units (`research/ic_rho_koblitz_20260923/`).

- *Batch rho.* k targets solved in sequence by the tuned signed-Frobenius walk with one shared distinguished-point table. Per-target cost at $k = 32$ over $k = 1$ is 0.210, 0.239 and 0.190 at $n = 41, 53, 61$, each interval covering the Kuhn–Struik 0.199; all 1,960 targets were recovered and verified.
- *Prices in the thread's unit.* A canonical step costs 2.74–2.92 units, Bailey et al.'s step 1.38–1.41, and a stored pair 4.5–6.1 (6.14 on the declared base).
- *The closest figure.* The thread's 1.17× reads 6.51× against batch rho as counted, 6.38× with rho's step and the table build both priced, 12.6× with Bailey's step on the canonical walk's count (a model), and 43.7× for one target alone.

This is an **accounting correction**: no index-calculus count changed. Its successor is Table C of the Koblitz leaderboard, priced against batch rho on the same 32 targets at 3.62–14.9× and still pre-gate (§15.6).

34.8 Advice tables

Bernstein–Lange precomputation [4] is the incumbent for every online comparison; its use as the equal-advice control that retired the online wins is in Chapter 41, and the September decisions on tables re-selected from the batch are in Chapter 35. The variants since then (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-advice-table-variants.yaml`) add little.

- A table of $T/2$ entries re-selected from the batch's solved walks falls short of a static table of T by -0.064 $[-0.080, -0.049]$ at 2^{24}, contradicting its claim; a re-scoped selection at 0.65–$0.75\,T$ met its target in three of four toy cells (preliminary).
- A basin-size instrument for table coverage failed its comparability gate.
- Carrying a table across isogenous curves is exact and costs at most 1.3% of a solve, but is never faster than solving at the source (`benchmarks/20261017-volcano-ecdlp-census/`).

34.9 Rho at scale

ECC2K-130. The programme's GPU rho on ECC2K-130 is published separately (`paper/ecc2k130/`); Chapter 15 prices the index-calculus routes to it (§15.7). For the reference the summary is this. The expected cost is $2^{60.9}$ iterations of a Frobenius-class rho, and the walk constant was measured at $2^{60.91}$–$2^{60.92}$. The σ-walk runs at 14.64 billion iterations per second on one RTX PRO 6000 in the audited campaign build and 15.88 in the best later build (a paired 1.0239× over 15.51, A/A drift below 0.1%, 300 of 300 reports replayed; `ecc2k130/SIGMA-TABLE.md`), about 230× the 2010 GTX 295. A roofline puts the ALU pipe at 91% of its peak and the carry-less speed of light at 27.4 billion per second on that card (`ecc2k130/ROOFLINE.md`). The paper separates the σ-walk from the table walk, whose 20.08 was measured under a cycle rule later found to leave cycles inconsistently; the exact rule runs at 8.24. A campaign stored 486 million distinguished points and $2^{56.68}$ iterations (5.4% of the expected work) before its fleet was reclaimed. Two iteration-function changes are derived and not rolled out, because each starts a new distinguished-point corpus:

- signed σ steps (`SIGMA_SIGN=2`) need 0.958–0.967 of the iterations at $n = 41$ at a rate of 0.9965, for 0.961–0.970 of the time per solve (`benchmarks/20261014-ecc2k130-signed-sigma/`);
- a table walk with every fruitless cycle caught is projected at 0.81–0.86× the σ-walk's cost per solve, unmeasured on the current kernel (`ecc2k130/WALK-CONSTANT.md`).

An audit of the live walk measured $2^{28.41}$ steps per completed walk at the $\mathrm{HW}(x) \leq 32$ cutoff and added duplicate-point, seed-reset and fruitless-cycle guards (`research/ecc2k130_rho_audit_20260925/`). None of this improves on the generic bound, and the paper says so.

NIST P-256 and P-384 on a GPU. The same levers applied to the `gpu/nist` engine took P-256 from 1.075 to 1.632 and P-384 from 0.210 to 0.548 billion iterations per second on one RTX PRO 6000 (1.52× and 2.61×), through PTX carry chains and a batch width $W = 32$ with by-reference inversion (`benchmarks/20261013-nist-gpu-rho-ladder/`; single-run sweep cells, no isolation receipt). An inversion-free Jacobian seeding ladder cut P-256 walk seeding from 11.53 to 0.66 ms per seed (17.4×), a set-up change with the step unchanged (`benchmarks/20261017-gpu-nist-jac-seed/`). At these rates P-256 remains about $6 \cdot 10^{21}$ card-years away (`paper/ecc2k130/`). No GPU arm is admitted to ecbench, because the matched-workload contract lacks PCIe telemetry and an admission rule (`problems/ecbench-gpu-matched-workload-contract.md`), and a P-256 best-of-two distinguished-point deficit remains unattributed, with its 0.947 cost estimate derived, not measured (`problems/nist-best2-dp-bias-and-gpu-validation.md`).

Reading. At scale the reference is an engineering artefact as much as an algorithm: its constant is measured, its throughput is set by a pipe, and each walk change has a corpus cost. None of it moves the generic exponent. All of it moves the denominator of every index-calculus row, which is why the gate of §41.2 names the reference per row and why a row ranked against a reference that has not adopted a change overstates index calculus by that change's factor.

Part V

The frontier

35 The summer record: July–September 2026

The October campaign measured in the rest of this part did not begin from nothing. Between 19 July and 30 September the crypto-autoresearcher ledger recorded 1,402 coordinator decisions on ECDLP and index calculus:

- July: 74 decisions on 20 experiments, 977 run directories;
- August: 364 decisions on 92 experiments, 321 run directories;
- September: 964 decisions on 449 experiments, 776 run directories.

Each decision names a verdict, the hypotheses and experiments it rules on, and the evidence it cites. The dated index (`research/ecdlp_jul_sep_index_20261029/`: `INDEX.json`, `INDEX.tsv`, and the generator `build_index.py`) lays these out together with the run-data path of each experiment. A decision is counted as ECDLP work if its goal is one of 19 ECDLP goals or, when it names no goal, if a target or evidence identifier carries an ECDLP family token. That scope rule is a choice and is stated in the generator. Seven decisions filed under late-September names carry October dates and are left out of the counts above. Twenty-three decision files did not parse and are not indexed.

Most of this activity was process (Figure 35.1). Of the 1,402 decisions, 561 approve a contract, 267 revise one and 124 expand one. About 157 carry an outcome on a hypothesis (support, reject, weaken, inconclusive, supersede, pause or replicate). This chapter reports those outcomes by theme, with decision identifiers so that each can be traced. All results are at toy or derivation tier unless a scale is given. None moved an exponent on a prime field.

35.1 July: the first baseline and two retracted wins

The ledger's first ECDLP decision, `DEC-20260719-001`, expanded a toy Semaev S_3 Gröbner study (`EV-SEMAEV-001`). It had twelve runs at 8–12 bits with a factor base of 14 points. The degree proxy stayed at most 4 and the wall time stayed flat near 0.16 s. The decision itself noted that three bit sizes and two seeds cannot establish a trend.

Two supports followed on 26 July, and both were overturned the next day:

- *Index calculus beats enumeration* (`H-IC-001`, 748 runs at 8–36 bits) was weakened by `DEC-20260727-008`. The criterion had been revised after the outcome. There were no decompositions at 20 bits or above, so the measured descent time was the cost of a failed attempt. The relation cost was 2287× the group order at 36 bits. The decision records: "no asymptotic exponent moves".
- *A structured factor base induces sparsity* (`H-STR-002`) was weakened by `DEC-20260727-009`. Three headline instances had no run records. The one-variable ablation gave $\alpha = 27,52,197,386$ against $27,52,199,387$ for a random base, so the factor base induced nothing.

In the same week a generic-group classifier (`EXP-GGM-001`) had its support revoked (`DEC-20260727-007`). It was a lookup table, and its nine runs were byte-identical. Fixed-curve preprocessing showed no robust gain over 72 runs (`DEC-20260726-001`, paused). A subresultant degree exponent was measured at $\beta_{\deg} = 0.5985$ against a gate of 0.3 (`DEC-20260728-003`, weakened).

Xedni-style lifting was closed in exact form (`DEC-20260724-021`, reject_scoped): $P_{\text{lift}} = (1 - 2/p + p^{-3})/(2p^3)$, so $\alpha = 3$. This is the result restated in Chapter 8. The exact count closes one family and slot shape only. A 1 November Lean audit found the classic JKSST obstruction conditional and refuted its overbroad readings (Chapter 8; `research/jksst_xedni_audit_20261101/`).

35.2 August: yield, monodromy, oracles and transfers

Yield matches the heuristic from below. Semaev decomposition yield at $m = 2$ reached 0.80, 0.98 and 1.00 of the $B^m/(m!N)$ heuristic at $p = 1009$, 4001 and 9001 (`EV-YIELD-e1adbf`). `DEC-20260804-9da614` weakened super-heuristic yield at $m = 2$–4. Structured bases gave small-x 0.845, arithmetic progression 0.658 and random 0.563, with no super-heuristic yield (`DEC-20260804-2bfc30`). These measurements came before the conservation proposition of Chapter 5, which explains them: at matched base size the mean is fixed. A complete $m = 2$ index calculus solved a discrete logarithm

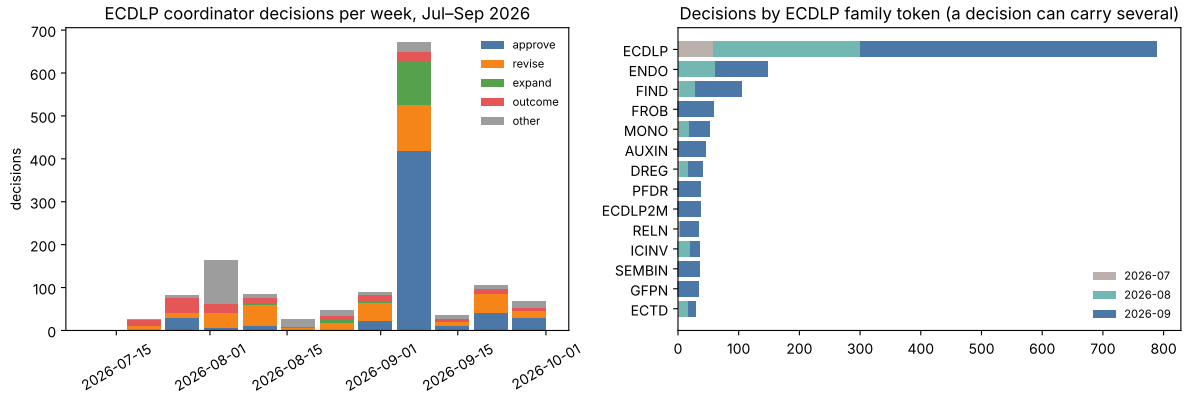


Figure 35.1: ECDLP coordinator decisions, July–September 2026. Left: decisions per week by verdict class. The class *outcome* groups support, reject_scoped, weaken, inconclusive, supersede and pause. Right: decisions per family token, by month; a decision can carry several tokens. The week of 7 September holds 672 decisions, most of them approvals of frozen stage-0 contracts.

at $p = 1009$ (`DEC-20260804-db75bb`). A Fourier-flatness constant for the factor base was fitted as $C(p) \sim p^{0.055}$–$p^{0.079}$ and recorded as an open problem (`H-PSEUDO`). Eight proof routes were closed as non-proving. A frozen re-test on 4 November (`HPSEUDO-SYMNULL-20261104`) explained the recorded offset $R \approx 1.28$–1.34 as an artefact of the null. The factor base is closed under negation, but the null was an asymmetric uniform subset. Against a negation-closed null, R is consistent with 1: the upper 95% bound is 1.09 at 2^{20} and 1.02 at 2^{22}, and the pooled 56 curves give 1.005 ± 0.008. A residual of up to about 9% at 2^{20} is not excluded (`crypto-autoresearcher:docs/reports/hpseudo-symnull-20261104/`).

BKK and meet in the middle: constant factors. A BKK sparsity argument promised a speedup of at least $(m+1)/2$ (`DEC-20260804-1825ea`). It measured 1.99988 against 2.0 at $m = 3$ on a smoke test. `DEC-20260806-26c0e8` then found the multi-target window empty at the planned sizes and the gain about 1.0–1.8×. Meet-in-the-middle structure gave 18×, which rho dominates (`DEC-20260808-030b1f`). A degree-split claw lost its advantage to a matched null, $R = 0.0472$ against 0.0311 (`DEC-20260801-005`). The decision reads: "the instrument, not the hypothesis, has been refuted".

Monodromy. The work that became Chapter 5 was done in August:

- the $m = 3$ derivation (`DEC-20260802-a51c82`);
- the all-m theorem, supported as scoped and "not closed" (`DEC-20260807-64df6a`, `KN-FIND-a8990a`);
- containment in $(\mathbb{Z}/2)^{m-2}$ at $m = 4, 5$ on eight curves, with zero exceptions in an exhaustive census of 2,915,514 triples at $p = 103$ (`DEC-20260830-0f8def`);
- monodromy D_4 for the $m = 3$ symmetric-base cover (`DEC-20260830-1fb2e3`).

The fibre census was ruled blind to relation rate as an instrument (`DEC-20260810-2f86db`, reject_scoped), two months before the census corollary proved why.

Oracles and transfers. Two analysis-only results bounded the oracle leads. Oracles C and D are simulable in the generic group model, and A and B give no sub-birthday advantage (`DEC-20260804-3b4258`, support_scoped). The doubling oracle was then shown equivalent to an x-coordinate oracle (`DEC-20260806-08b9ed`). Three transfer and invariant routes were closed:

- Isogeny transfer: the anomalous, MOV and Weil-descent families are invariants of the isogeny class (`DEC-20260804-2fae6a`).
- Elkies augmentation: with $\ell \leq 23$ it matched a random-bijection null to within 0.004 in all eight cells at 20–28 bits (`DEC-20260804-d0c452`).
- CM: complex multiplication changes the lifted-height obstruction "by exactly nothing" (`DEC-20260810-2c7e3c`).

Within one isogeny class the over-dispersion of ECDLP-relevant statistics traced to the 2-torsion type (`DEC-20260811-a3398a`, `KN-FIND-ddd982`).

Degree of regularity. The boolean chained $m = 3$ system was run at $n = 21$, $D = 5$, and stopped at rank 265,950 of a predicted 268,674 (`EV-DREG-4500cd`). The record calls this "not evidence about d_{reg} in either direction", and `DEC-20260815-6b806c` superseded the unarchived run chain by reference, evidence-neutral. All six $n = 15$ runs then exceeded memory (`DEC-20260828-653923`). The certified fall-degree results of Chapter 6 later replaced this line.

Late August was spent on governance: budget ceilings, pauses and zero-run successors. `DEC-20260820-0c8ffb` paused at the budget ceiling. The "fund every goal" decisions of 28 August lifted the limits.

35.3 September: tables, lifts, audits and the first binary costing

Distinguished-point tables re-selected from the batch. The claim that a table of $T/2$ entries re-selected from solved walks matches a static Bernstein–Lange table of T was contradicted at $a = 1/4$. It reached 0.65–$0.69\,T$ at steady state, with intervals excluding 1, while paying 2.6–$8.9\,T$ of working storage (`DEC-20260906-1f5b71`, weaken: "a tradeoff point, not an improvement"). At $N = 2^{24}$ the feasibility crossing was then bracketed between $a = 1/8$ and $a = 3/16$ (`DEC-20260907-9b1b02`, `DEC-20260907-6d9427`). The result is a ceiling only, with no online arm (`DEC-20260910-34e654`).

The canonical lift carries no logarithm. The p-adic lift statistic D_1 took exactly the values $\{0, 8\}$ on 1,035 pairs at $p = 1009$, and the same in a third independent implementation and on a second instance. It tracked precision $r = 4, 8, 12$ exactly (`DEC-20260903-256c7c`). The digit identity held in 1360/1360 cases, and the Teichmüller contrast showed no inflation: all three ratio intervals contain 1 (`DEC-20260911-74c9dc`, `DEC-20260912-2a626b`). Both results were recorded as "no attack".

Natural statistics are not biased enough. An L^1 Fourier registry over 406 cells and primes 4099–16,777,259 falsified two of its four predictions (`DEC-20260921-4516a0`, weaken):

- interval statistics: $\lambda = 0.0806 \pm 0.0020$, against a prediction of at most 0.05;
- cosets and quadratic residues: 0.50, against a prediction near 0.

Semaev degree and the Semaev 2015 / Nagao audits. The successful-subset loophole in the Semaev degree law was closed: $\beta_{\text{work}} = 0.797$ worst case over 36 cells and 1,152 queries (`DEC-20260929-780751`, reject_scoped, `EV-SDEG-4a1c3d`). The audits of Semaev's 2015 paper and Nagao's programme (Chapter 6) recorded these verdicts:

- eq. (11)'s λ is right to 1% at $t = 2$, but $P = 1 - e^{-\lambda}$ fails by 18–22% in the low-degree window (`DEC-20260913-3b51c8`);
- the typed-image ratio is not $m!$ at any of seven cells (`DEC-20260913-2386b5`);
- trace-parity confinement explains the whole deficit at $t = 2$ (`DEC-20260913-92aa48`);
- Nagao's memory claim fails at $n = 571$, and 81 of the 120 flagged cells were an artifact of an incoherent memory pairing (`DEC-20260914-537000`);
- a published measurement claim, P3, fails under the corrected XOR product: the Semaev median is 2 in all eight cells, with the null at least one above it (`DEC-20260929-e37785`, reject_scoped).

Prime-field degree versus yield. A closed form for the first fall degree of a digit-map presentation, $d_{\text{ff}} = m2^{m-1} + \lfloor (s - 2^{m-1})/2 \rfloor + 1$, survived 26,000 instances with zero deviations. Its companion fall-dimension clause was refuted by a certificate at $p = 13$ (`DEC-20260904-1e27a2`). The generator degree is $m2^{m-1}$, not $2m$, and the "cells below rho" of a cost table had zero rows at D_0, so they were rejected (`DEC-20260904-d47cd2`).

Frobenius-stable factor bases. The stable subspaces are exactly the ideals of $\mathbb{F}_q[T]/(T^n - 1)$, and the orbit count is exact on 14 of 14 cells (`DEC-20260922-5796b8`, `EV-FROB-d336b0`). The orbit gain is n, a constant factor. At $n = 131$ there are only four stable subspaces, and the orbit trick *loses* 57.0–96.0 bits for $m = 2$–5; at $n = 163$ it loses 72.7–121.7 bits. Whether WDSat is blind to basis choice was never measured, because 19 stage-0 launches failed.

The first costing of descended systems against rho. The ICPERF lane priced the descended algebraic normal form against the whole rho budget on standard binary curves (`DEC-20260922-925da8`, `EV-ICPERF-784b25`). Forming the system alone exceeds rho by +6.3 bits at $n = 233$ and by +40.8 bits at $n = 131$ (ECC2K-130). Below $n \approx 300$ the obstacle is building the system, not solving it. The excess turns negative at $n = 409$ and 571 for $m \geq 4$, and $m = 2$ loses at every degree. This is the first September result priced against rho. It is a model with measured inputs, not a solve.

Lanes without a result. Several September lanes produced approvals and redesigns but no measured outcome:

- SAT on summation systems, trapdoor detection, invariant calibrators and endomorphism decomposition, mostly frozen stage-0 contracts approved on 13 September;
- the auxiliary-input census, whose run was ruled invalid as evidence because its controls were inadmissible and its escalation was generic to 256-bit primes (`DEC-20260923-ce06a6`);
- EcGFp5 symmetrisation, where no $m = 5$ ideal degree was measured by September (`DEC-20260923-d86ff2`), although the audit found that the curve's self-reported 323 bits recomputes to 322. A quotient solve on 1 November measured $D = 2^{16}$ at all three primes it ran (`research/ecgfp5_quotient_solve_20261101/`);
- characteristic-2 quasi-subfield work, which found zero violations of its bound in about 340,000 candidates and produced the derivation $\beta > 1/2$ for every completely splitting L (`KN-FIND-617d78`).

The certified binary refutations of Chapter 6 begin here: W_4 refutes 62/62 at $n = 17$ and 400/400 at $n = 19$ (`DEC-20260924-e61f3b`, `DEC-20260926-901ea1`).

35.4 What the summer established

Table 35.1 collects one outcome per theme. Four patterns run through the 1,402 decisions:

1. *No measured improvement over rho survived review.* Every July and August support that claimed a gain was weakened or revoked within days. The surviving gains are constant factors at toy scale.
2. *The theory of Chapter 5 was found by measurement first.* The yield ratios at or below 1, the census ruled blind, and the exhaustive monodromy checks all preceded the proofs that explain them.
3. *Most closures were instrument repairs.* A large share of the weaken and reject decisions say that a control, a metric or a criterion was wrong, not the hypothesis. This is the origin of the matched-null and replay rules of Chapter 4.
4. *The binary line reached a priced verdict only in late September.* The ICPERF costing and the W_4 certificates opened the fall-degree and Koblitz work that the October campaign measured.

These four patterns are the author's reading. The decisions cited above are the record.

month	theme	decision	verdict	recorded result
Jul	IC beats enumeration	`DEC-20260727-008`	weaken	relation cost $2287\times$ the group order at 36 bits
	structured factor base	`DEC-20260727-009`	weaken	$\alpha = 27, 52, 197, 386$ vs. random $27, 52, 199, 387$
	subresultant degree	`DEC-20260728-003`	weaken	$\beta_{\mathrm{deg}} = 0.5985$ vs. gate 0.3
	xedni lifting	`DEC-20260724-021`	reject_scoped	$\alpha = 3$
Aug	Semaev yield, $m = 2$	`DEC-20260804-9da614`	weaken	0.80, 0.98, 1.00 of heuristic at $p = 1009, 4001, 9001$
	BKK speedup	`DEC-20260806-26c0e8`	supersede claim	1.0–1.8×, not $(m + 1)/2$
	degree-split claw	`DEC-20260801-005`	weaken	$R = 0.0472$ (null) vs. 0.0311
	monodromy, $m = 4, 5$	`DEC-20260830-0f8def`	support	0 exceptions in 2,915,514 triples at $p = 103$
	Elkies augmentation	`DEC-20260804-d0c452`	reject_scoped	within 0.004 of null, 8 cells, 20–28 bits
	boolean d_{reg}, $n = 21$	`DEC-20260815-6b806c`	supersede	rank 265,950 of 268,674; not evidence
Sep	re-selected DP table	`DEC-20260906-1f5b71`	weaken	0.65–$0.69\,T$ at $a = 1/4$
	canonical lift D_1	`DEC-20260903-256c7c`	support	values exactly $\{0, 8\}$ on 1,035 pairs; no attack
	natural-statistic bias	`DEC-20260921-4516a0`	weaken	$\lambda = 0.0806 \pm 0.0020$ vs. ≤ 0.05
	Semaev degree loophole	`DEC-20260929-780751`	reject_scoped	$\beta_{\mathrm{work}} = 0.797$, 36 cells
	Frobenius-stable bases	`DEC-20260922-5796b8`	support	loss of 57.0–96.0 bits at $n = 131$
	ICPERF costing	`DEC-20260922-925da8`	support	+6.3 ($n = 233$) to +40.8 ($n = 131$) bits over rho

Table 35.1: The summer record at a glance: one outcome decision per theme, July–September 2026, with the number it recorded. Every row is discussed above. Verdicts are the ledger's, shortened: "weaken" includes `weaken_h_yield`, and "supersede claim" replaced the earlier BKK support. Index: `research/ecdlp_jul_sep_index_20261029/INDEX.json`.

36 What improved, and where the frontier stands

36.1 The measured frontier: one campaign, every cost priced

The frontier synthesis of 13 October assembled ratios from studies that used different curves, sizes and accounting. To replace it we ran one campaign (`research/ic_frontier_campaign_20261015/`, protocol frozen before any measured run). Every method family ran on the same fixed curve ladders: eight Koblitz curves chosen by subgroup size ($\log_2 r \approx 11$–39, $n \in \{13, 17, 19, 23, 37, 41, 43, 47\}$), six NIST-shaped $a = -3$ prime-order curves of 20–40 bits, and the five 2-adic curves of the Kummer-tower ladder. Each curve had 16 public hash-to-subgroup targets nobody knows the logarithm of, 2 measured rounds and a warm-up, and every answer was checked outside the solver. That is 4,848 executions, 3,232 of them measured: 3,178 verified and 54 exhausted, all of the latter in the shared-rank arm at $n = 41$ and $n = 47$. Every session is audited by re-executing every measured run and requiring identical counts. Index-calculus parameters (compact-orbit columns, negfold subspace dimension, prime base size) were chosen per size on a disjoint set of training targets, with grids widened until the optimum was interior. Every cost is reported across the pricing band of §4.3.

What the campaign shows.

- *No index-calculus arm beats generic rho on any curve, at any weight.* Signed-Frobenius rho is below every index-calculus arm on all eight Koblitz curves and across the whole band; rho with negation is below every prime-field arm on all eleven prime curves (Table 36.1). The one unresolved pair in the campaign is compact orbit against the *strong* rho reference at $n = 13$, whose scalar-multiplication set-up dominates on an 11-bit subgroup; the cheaper signed-Frobenius walk is below both.
- *Compact orbit is the closest index calculus.* It sits 3.8–15× the best rho with native work free and 4.9–17.8× at $w = 0.1$; the ratio does not shrink with size.
- *Prime-field index calculus falls further behind with size.* The two-summand pair-table arm goes from 10.5× to 101.8× rho (at $w = 0.1$) between 20 and 40 bits, the $r^{1/6}$ widening its $r^{2/3}$ cost against rho's $r^{1/2}$ predicts; the Kummer-tower arm is 28–159× rho.
- *Negfold root finding stops scaling.* It is 12–45× rho up to $n = 23$, about 1,900× at $n = 37$, and does not complete within $5 \cdot 10^7$ trials from $n = 43$ on (PROTOCOL A4b).
- *Inherited F4 is not competitive once its work is priced*: S rises from 16 (word XORs free) to about $2 \cdot 10^4$ at $w = 0.1$ at $n = 13$. The band weighs one 64-bit word XOR like a table lookup, which overstates F4 (its pinned calibration is about 0.002 GAE per XOR); even with XORs free it is above rho.
- *BSGS has the lowest time on several curves* because it spends $\sqrt{r}$ memory; it is reported on the time–memory Pareto set, not ranked against constant-memory methods.

Cost exponents. Table 36.2 fits total cost $\propto r^{\alpha}$ per arm over each size tier with `ecbench bound fit -weight` (two-stage bootstrap, 95% intervals; a tier needs at least four sizes for an exponent). For the prime pair table, index calculus grows faster than rho (the predicted 2/3 against 1/2), but under the band rule the difference is not established at every weight. On $n = 13$–23 both Koblitz intervals are wide because the automorphism count $2n$ changes with the size.
The rho arms are rho with negation (prime) and signed-Frobenius rho (Koblitz). No index-calculus arm has a fitted exponent below rho's at any weight, so no crossover with rho is supported by these data, extrapolated or otherwise. The fitted rho exponents (0.435–0.444) lie below the theoretical 1/2, which reads as a toy-size effect (fixed per-run overheads and the size-dependent automorphism count) rather than a property of rho; so the measured widening of the gap is a fit on small sizes, and the conclusion that it widens as $r^{1/6}$ rests on the generic-oracle bound, not on these fits alone. The sixty-three fits are in `research/ic_frontier_campaign_20261015/analysis/bounds/`.

Limits. Operation counts only (isolation L0); wall time and a second-host replay certificate need the planned RunPod phase. Parameters were selected on four training targets per size. The shared-rank arm

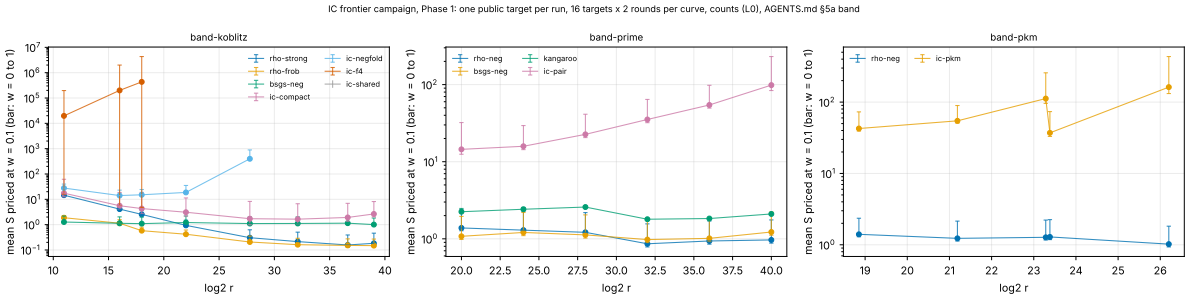


Figure 36.1: The campaign frontier: mean S per arm and curve at $w = 0.1$ GAE per native operation, the bar spanning $w = 0$ to $w = 1$; one public target per run, 16 targets × 2 rounds per curve, operation counts. Sources: `research/ic_frontier_campaign_20261015/analysis/band-*.json` (`ecbench band`).

Curve	$\log_2 r$	Arm	$w = 0$	$w = 0.1$	$w = 1$
Koblitz $n = 13$	11.0	compact orbit	6.9	9.3	27.1
		negfold	14.4	14.8	17.5
Koblitz $n = 17$	16.0	compact orbit	3.8	4.9	11.8
Koblitz $n = 19$	18.0	compact orbit	4.9	7.5	23.6
Koblitz $n = 23$	22.0	compact orbit	5.5	7.4	18.4
Koblitz $n = 37$	27.8	compact orbit	5.2	8.3	23.3
		negfold	1806	1927	2512
Koblitz $n = 43$	32.1	compact orbit	7.4	10.2	23.1
Koblitz $n = 47$	36.6	compact orbit	9.9	12.7	25.3
Koblitz $n = 41$	39.0	compact orbit	15.0	17.8	30.3
prime 20 bit	20	pair table	9.8	10.5	13.5
prime 24 bit	24	pair table	12.1	12.2	12.9
prime 28 bit	28	pair table	18.6	18.6	18.9
prime 32 bit	32	pair table	40.9	40.9	41.1
prime 36 bit	36	pair table	58.1	58.0	57.5
prime 40 bit	40	pair table	95.2	101.8	131.4
PKM 20 bit	18.9	Kummer tower	30.3	30.4	31.0
PKM 28 bit	26.2	Kummer tower	141.3	158.7	238.7

Table 36.1: Index calculus over the cheapest constant-memory generic arm on the same curve (signed-Frobenius rho on Koblitz curves, rho with negation on prime curves), mean S ratio at three band weights. Every run verified (32/32 per cell). Source: `research/ic_frontier_campaign_20261015/analysis/ratios_to_rho.tsv`.

leaves three categories of work uncounted and is reported as a lower bound only. Fitted cost exponents per arm and tier are in `research/ic_frontier_campaign_20261015/analysis/bounds/`.

36.2 The engineering record

Table 36.3 collects the measured improvements of the programme on the boundary each was measured on. Ratios are baseline/candidate; "stage" means one oracle call or one solver run; "complete" means a verified discrete logarithm with the stated phases.
Derived entries: $53.0 = 1/0.018861$; $17.5 = 1/0.057047$; $1.21 = 1/0.828$ with interval from $[0.783, 0.863]$; $18.3 = 1/0.0545$ with interval from $[0.047, 0.064]$; $1.77 = 300{,}021.9/169{,}069.8$; $1.20 = 1/0.830$ with interval from $[0.697, 0.962]$; $1.87 = 18.714/9.985$ (one session; the compact arm's S is a lower bound, so the ratio is an upper bound); 1.69 and 25.2 compare means from different sessions with different seeds and are descriptive; F5 and F6-IC ratios are reciprocals of the recorded candidate/baseline word-XOR ratios.

Arm	Tier	w	IC α	Rho α	Intervals
Prime pair table		0	0.620 [0.557, 0.663]	0.435 [0.331, 0.501]	separated
		0.1			separated
		1	0.594 [0.482, 0.665]	0.444 [0.334, 0.517]	overlap
Koblitz compact orbit	$n = 37$–47	0	0.589 [0.518, 0.743]	0.436 [0.358, 0.508]	
		0.1, 1			overlap
	$n = 13$–23	0	0.25	0.29	overlap
		0.1, 1			overlap
Kummer tower			0.717 [0.358, 0.940]	0.438 [0.332, 0.599]	overlap

Table 36.2: Fitted cost exponents α (total cost $\propto r^{\alpha}$) of each index-calculus arm and the cheapest constant-memory rho on the same curves, by size tier and band weight w; 95% intervals from a two-stage bootstrap. Intervals: whether the IC and rho intervals are separated or overlap at that weight. Rows without exponent values report only the interval comparison; no tier or weight is stated for the Kummer-tower row. Source: `research/ic_frontier_campaign_20261015/analysis/bounds/`.

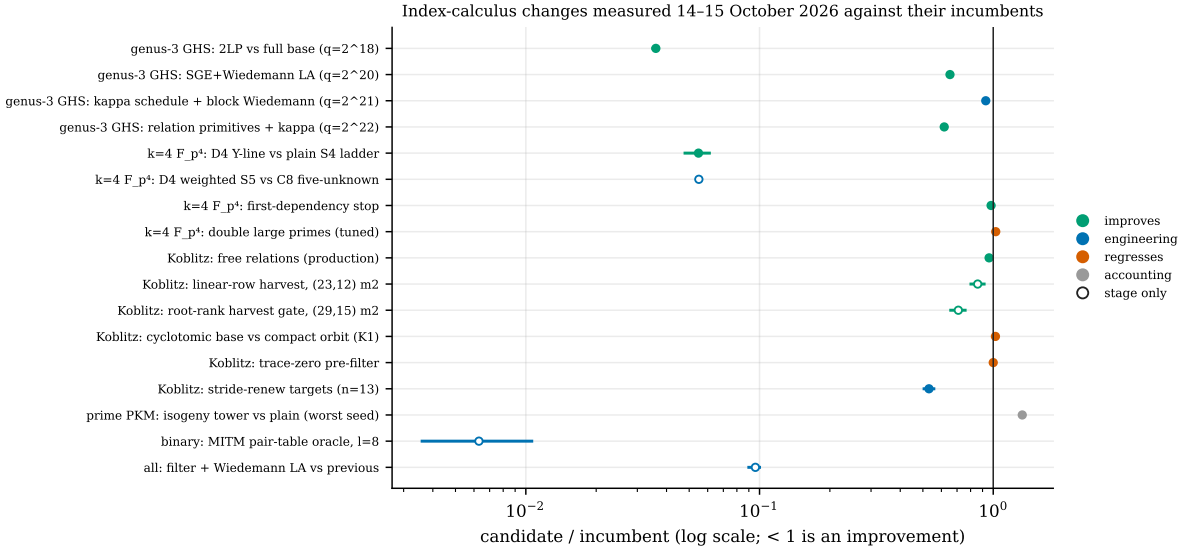


Figure 36.2: Candidate over incumbent for the changes of 14–15 October, with 95% intervals where the study reports one (the binary oracle's range is the spread over three runs). Filled: whole solve; open: one stage. Values: `papers/ecdlp-index-calculus-program/figures/ladder.tsv`; sources in Table 36.4.

36.3 The improvement ladder, 14–15 October

The repository's owner fixed the criterion for success on 15 October: a verified, measured improvement over the current index-calculus baseline is a success; each one becomes the new baseline; the gap to the matched rho is reported as the distance remaining, not as a pass/fail gate; and relabelling or accounting changes do not count. In the two days before that, two improvement rounds and a next-steps campaign landed about twenty measured changes, each paired against its own incumbent under a protocol frozen before data. Figure 36.2 and Table 36.4 list them, with the distance to rho that remains after each one. None is in the frontier campaign of §36.1, which predates them, and none was priced across the band; the boundaries are the ones the studies used, which differ from row to row.

Genus three: an exponent below rho's. The cubic GHS transfer [21] maps the DLP in $E_0(\mathbb{F}_{q^3})$ to the Jacobian of a genus-3 hyperelliptic curve over $\mathbb{F}_q$, where double-large-prime index calculus [22] costs $\tilde{O}(q^{2-2/g}) = \tilde{O}(q^{4/3})$ against rho's $q^{3/2}$. That is known; what is new is a priced, end-to-end, verified run of it on a family the repository can build ($4 \mid \#E_0(\mathbb{F}_{q^3})$, split arm, $q = 2^8$–2^{22}, every logarithm checked against the planted scalar). Adding double large primes changed the exponent from $q^{1.910}$ to $q^{1.385}$ $[1.379, 1.391]$, and keeping the large primes as columns, merging them with structured Gaussian elimination and finishing with Wiedemann moved it to $q^{1.380}$ and the constant to 0.653×. This was the

Change	Bnd.	Cl.	Ratio	Scope
Decomposition oracle: root finding				
S_3 roots, binary pair table vs matrix F4	St	E	383–1,535	$\mathbb{F}_{2^9}$, $\mathbb{F}_{2^{13}}$, $m = 2$
S_3 roots, prime vs `groebner_f4`	St	E	3.5–4.7·10^4	16–24-bit, $\lvert\mathcal{F}\rvert = 6$
S_3 roots vs Buchberger, NIST curves	St	E	3,021–5,802	P-192…P-521
`rf-mitm` vs PS-0 oracle	St	E	3,007–115,160	binary $\ell = 8..14$
Root-finding DLP vs msolve DLP	Cx	E	50–105	16–24-bit prime
Pair-table DLP vs F4 DLP	Cx	E	122	binary $n = 23$ (F4 wins $n \leq 13$)
Pair table vs subtraction	Cold	E	53.0	20-bit prime (bounded)
Frobenius pair table vs subtraction	Cold	E	17.5	Koblitz $n = 17$, priced
Root-finding K13 arm, first to best	Cold	E	1.69	38.82 → 22.97, across sessions
Windowed $m = 3$ scan	Cold	E	1.21	$[1.16, 1.28]$
Negfold root finding vs XOR-priced F4	Cold	E	18.3	$[15.6, 21.3]$, K13
Dense root index K1798 vs K1500 (probes)	On	E	1.77	$n = 83$, one point
Pair-table size 64 vs 32 abscissae	Cold	E	1.20	$[1.04, 1.43]$
Gröbner engines				
Inherited vs matrix F4 (matrix work)	St	E	198.6–222.1	K13, R13
F5 vs matrix F4 (word XORs)	St	E	1.55–1.56	K13, R13
F6-IC vs inherited F4 (word XORs)	St	E	1.22–1.30	$m = 13, 19$
`f4_fp` new vs previous	St	E	29.8–54.1	square systems, = msolve
Four-Russians routing	St	E	1.75–3.22	oracle matrices
Eager vs lazy inherited rewrite	St	E	1.40–1.62	dim 8–9
Quotient-curve $m = 2$ model	St	E	1.34–1.64	five cells
Factor bases and symmetry				
Compact orbit vs full orbit	Cold	E	≤ 1.87	Koblitz $n = 17$; compact is LB
FGHR D_4 vs Gaudry S_4, C_4	St	E	47.5–48.4	$\mathbb{F}_{269^4}$
`gaudry_quartic` solve	St	E	1.96–6.08	$d = 4..6$ (sweep recompute)
One-target online				
Compact-orbit IC vs strong rho	On	E	13.94	$n = 61$; rho without a table; retired (§36.3)
Accounting (costs that went up when priced)				
F4 word XORs priced	Cold	A	1/25.2	16.75 → 421.4, K13
F6-IC batched geometry priced	Cold	A	1/64.6–1/1,353	$m = 13, 19$; F4 unpriced (about 1/2.5 priced)

Table 36.3: Measured improvements, July–October 2026. Class: E = ENGINEERING, A = ACCOUNTING. Boundary: St = stage, Cx = complete solve outside ecbench, Cold = ecbench cold one-target S, On = online window after retained set-up. Sources: Appendix A.

Change	Cl.	Ratio [95%]	To rho
Genus 3 after the cubic GHS transfer, $E_0(\mathbb{F}_{q^3})$, cost in $\mathbb{F}_q$ multiplications			
Double large primes vs full base (04)	Ie	0.036 $[0.035, 0.036]$ at 2^{18}; $q^{1.910} \to q^{1.385}$	2.58× negation rho model at $q = 2^{20}$
Large primes as columns, SGE, Wiedemann (01)	Ie	0.653 $[0.647, 0.660]$ at 2^{20}; $q^{1.398} \to q^{1.380}$	—
$\kappa(q)$ schedule + block Wiedemann (06)	T	0.929 at 2^{21}	1.15× measured negation rho at 2^{20} (ref.)
Cheaper relation primitives + re-tuned κ (r3-01)	Ic	0.617 $[0.616, 0.619]$ at 2^{22}	no matched rho in this unit
$k = 4$ over $\mathbb{F}_{p^4}$, operation counts			
D_4 Y-line vs plain S_4, $p = 101$–1033	Ic	0.0548 $[0.048, 0.061]$	handover 2^{126}–2^{137} (d)
D_4-weighted S_5, four unknowns, vs five	E	1/18.5–1/17.8	handover $2^{135.6} \to 2^{118.9}$ (d)
First-dependency stop (`d4-ed`)	Ic	0.979 $[0.973, 0.986]$	no $k = 4$ run below rho
Double large primes, tuned	R	1.025 $[1.016, 1.037]$ (LA 0.827)	—
Koblitz, compact orbit and Gröbner oracle			
Free relations in production, $n = 41, 53, 61$	Ic	0.960 $[0.946, 0.976]$	3.8–15× rho-frob ($w = 0$, §36.1)
Linear-row harvest, $(23, 12)$, $m = 2$	Ic	0.858 $[0.803, 0.916]$	oracle stage; default on
Root-rank harvest gate, $(29, 15)$, $m = 2$	Ic	1/1.41 $[1/1.52, 1/1.32]$; end to end 1/1.20	oracle stage; default on
Gated fall-row augmentation	Ic	1/11.9 on barrel $n = 13$; 1 in production	production no-op
Equivariant Macaulay oracle	T	4.0 ($n = 5$) … 1/15.5 ($n = 13$)	not applicable to production
`stride-renew` targets, $n = 13$	E	0.531 $[0.506, 0.557]$	4.7× rho-strong at $n = 19$
Cyclotomic γ-weight base, K1	R	1.022 $[1.021, 1.023]$	yield-neutral
Trace-zero pre-filter	R	1.00003	vacuous on odd order
Prime fields and shared stages			
PKM isogeny tower, search charged	A	1.002–1.331	17–137× rho
Binary MITM pair-table oracle, $\ell = 8$	E	1/277–1/94 per target	no exponent claim
Filter + Wiedemann LA, $D = 2^{18}$	E	1/10.40; 1/1.93 vs CADO-NFS	LA is 0.1–1.9% of runs

Table 36.4: Changes measured 14–15 October 2026 against their own incumbents. Ratio: candidate/incumbent, < 1 better. Class, against the index-calculus incumbent (not the paper's ADVANCE, which is against rho): Ie = improves the exponent, Ic = improves the constant, E = ENGINEERING on one stage, R = regresses, T = trades, A = ACCOUNTING. "To rho": the distance left to the matched rho on the study's own boundary; "d" marks a derived or extrapolated value. Sources under `research/`: `ic_next10_20261014/`, `ic_improve_20261014/`, `ic_improve_20261015/`, `ic_improve3_20261015/` and the named study directories.

first index-calculus family in the programme whose fitted exponent lies below rho's (the $k = 4$ binary family of §36.4 is the second), and the first where measured IC cost fell below a rho arm: the item-7 rho (which walks at $1.75\sqrt{\ell}$ steps, about twice the ideal), at $q = 2^{15}$ (0.69 $[0.55, 0.83]$; no band pricing; not the strongest rho). Against the strongest rho see Chapter 26 (genus-3 round 4, the plane quartic and the validated Joux–Vitse sieve). Two further rounds kept going. A frozen schedule for the factor-base fraction $\kappa(q)$ plus block Wiedemann brought the whole solve to 0.605× the first double-large-prime arm at $q = 2^{21}$. Cheaper relation primitives (one-inversion Cantor addition, a Stickelberger pre-filter, reused exponentiations) then cut the relation phase to 0.47× without changing a single relation, and with κ re-tuned the whole solve to 0.617 $[0.616, 0.619]$ at $q = 2^{22}$, in a unit that also charges inversions. The distance to rho shrank from 2.58× the negation-map rho model at $q = 2^{20}$ (first double-large-prime arm) to 1.15× a measured negation-map rho there after the second round ($2.31 \cdot 10^{10}$ against $2.01 \cdot 10^{10}$ $\mathbb{F}_q$ multiplications; a reference on overlapping but not identical curves). The third round's 0.62× has no matched rho in its unit yet, so whether it crosses is not established. The local exponents sit at 1.38–1.41, still above 4/3, because the linear algebra grows faster than $q^{4/3}$ and the tuned κ falls with q. The family is narrow: the transfer needs the right 2-power torsion, and $\mathbb{F}_{q^3}$ curves are not deployed. *Next improvement*: a negation-map rho measured in the third round's unit on the same instances, then the run past $q = 2^{22}$ that would show a crossover or rule it out.

The $k = 4$ chain. The D_4 Macaulay system of §21.10 entered a complete ladder ($p = 101$–1033, 21 curves): 18.2× cheaper than plain S_4 at the same exponent, with the handover with negation rho extrapolated to 2^{126}–2^{137}. Writing the D_4-symmetrised S_5 in four weighted unknowns cut the per-solve cost another 17.8–18.5×, to $1.42 \cdot 10^9$ $\mathbb{F}_p$ multiplications, 854× below the plain form; in the 2-torsion frame that moves the derived handover from $2^{157.9}$ to $2^{118.9}$. Double large primes cut the linear algebra (0.827×) but regress the whole solve; stopping at the first dependency gains 2.1%. No $k = 4$ run below rho exists; every handover is an extrapolation of fitted exponents through a measured constant. *Next improvement*: put the four-unknown solve into the complete ladder, then keep double large primes' linear-algebra gain without its relation-phase cost.

Koblitz. Free relations—root-table self-collisions used as homogeneous rows—are now the production default (0.960× counted, 0.941× wall); the linear rows the Gröbner oracle used to discard are harvested (0.86–0.93× where the gate fires, and a root-rank gate then made it fire on more cells: 1/1.41 at $(29, 15)$, 1/1.20 end to end); the gated fall-row and equivariant passes pay only on overdetermined systems the production pipeline does not build. The cyclotomic base and the trace-zero filter were refuted. A census closes the free-relation lever at the rungs run: at $n = 61$, $K = 600$ the self-collisions span 49 independent relations of the 599 needed, and full rank would need $K \approx 2{,}050$. The distance to signed-Frobenius rho is still 3.8–15× cold at $w = 0$, and the fitted Koblitz ratio grows as about $r^{1/6}$ (slope 0.165 $[0.117, 0.213]$ per $\ln r$ on eight public curves, every native priced). *Next improvement*: relation yield per probe, which none of these changes touched.

The online result is retired. An equal-advice audit gave Bernstein–Lange preprocessing rho [4] a distinguished-point table no larger than the index-calculus pair table and ran both on the same 64 unseen targets at $n = 41$, 53 and 61. Rho with the table is 53–317× faster online (at $n = 41$, with exactly equal advice, 317 $[137, 739]$), and even at 1/1,024 of the bytes it stays 7.7–32× faster. The 13.9× online win of §15 and the later 7.3× and 12.3× hosted single-target points at $n = 71$ and $n = 83$ compared index calculus *with* a table against rho *without* one. They are measurements of the preprocessing model, not index-calculus improvements, and the online frontier belongs to preprocessing rho.

The reference moved too. Two rho studies shortened the distances above. On prime curves, choosing the better of $P + R$ and $P - R$ at each step takes 1.129× fewer steps under distinguished points (1.081–1.184); each step costs more, so the predicted net gain is 1.03–1.12× with 8–64 walks per inversion and a net loss (0.95) at 1,024. On ECC2K-130, signed σ steps cut GPU time per solve to 0.961–0.970 (derived; not rolled out). Neither is yet in a campaign arm. A comparison against a rho that has not adopted them would overstate index calculus's position by that much.

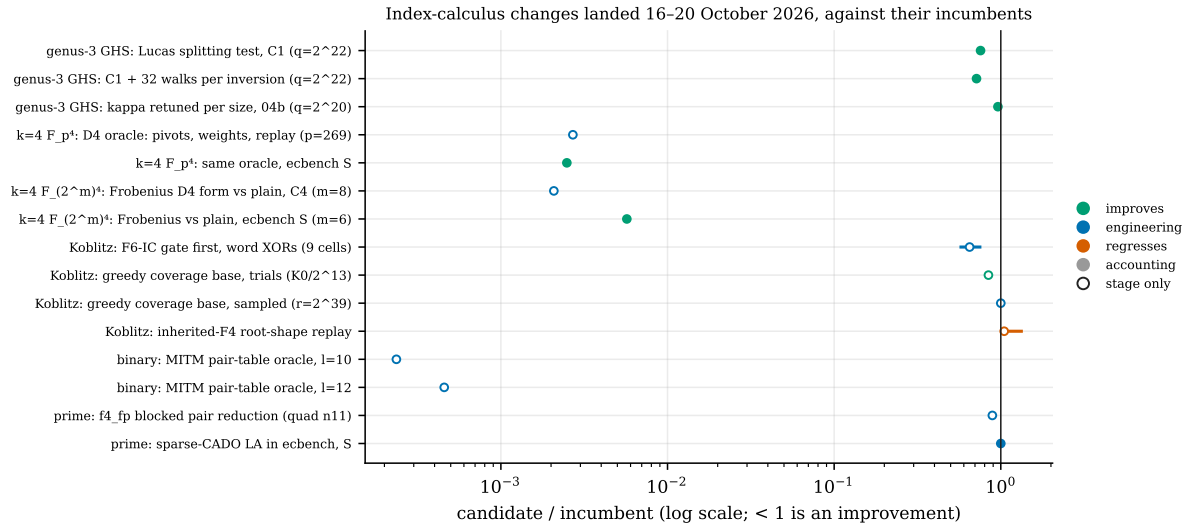


Figure 36.3: Candidate over incumbent for the changes that landed 16–20 October, as in Figure 36.2. Values: `papers/ecdlp-index-calculus-program/figures/ladder2.tsv`; sources in Table 36.5.

36.4 The ladder continued, 16–20 October

Work that reached `main` between 16 and 20 October, after the programme record was first assembled (some of it measured on 14–15 October), added a genus-3 round, two $k = 4$ oracles, a long binary-oracle ladder, Gröbner-engine work and a repository-wide validation sweep (Figure 36.3, Table 36.5). The reading rules of §36.3 apply: each row is against its own incumbent, on its study's boundary, unpriced across the band.

Genus three, round four. A field-level splitting test (a Lucas ladder on the Cardano resolvent's cubic character, with the same relations) removes most of the per-trial log factor (9.75 to 2.71 units per bit) and moves the whole solve to 0.756× at $q = 2^{22}$; with 32 walks sharing one inversion, to 0.715×, with local exponent 1.356 (fitted 1.368 over 2^{16}–2^{22}), 0.023 above 4/3. Chained with round three, the whole solve is about 0.44× the second round's at $q = 2^{22}$ (derived, in the inversion-charging unit). Rounds three and four ran no rho arm; a 23 October study then measured round 4 against the strongest rho at $q = 2^{14}$–2^{20}: under the registered pricing it is below rho at 2^{16} at every band weight and unresolved at the other sizes (geometric means 0.70–1.52), and above rho under the protocol's heavier sensitivity pricing; §26). Re-tuning κ on training again steepened the schedule and raised the exponent. A separate rebalance study measured IC at 0.16× (0.25× renormalised) a rho on $\mathrm{Jac}(C)$ at $q = 2^{14}$: that rho walks on the Jacobian (four walks per size, at 1.6–2.8 steps per $\sqrt{\ell}$ against an ideal 1.25), not on $E_0(\mathbb{F}_{q^3})$, so it is not the matched reference and the ratio is not a distance to rho. Memory now limits the ladder: $q = 2^{22}$ peaks at 3.9 GB of a 4.5 GB cap.

$k = 4$: two oracles and a closing gap. Over $\mathbb{F}_{p^4}$, sparsest pivots, the weighted grading $\sigma_4 = \pi^2$ and shape replay cut one D_4 decomposition from $2.55 \cdot 10^{10}$ to $6.89 \cdot 10^7$ $\mathbb{F}_p$ multiplications (370×; 648/648 solves exact) and ecbench S by 398–406×; index calculus is still $1.39 \cdot 10^6$× rho at the measured sizes and the model handover moves from 2^{135} to 2^{103}. In characteristic two, a Frobenius-reduced form of the same system (weights $(2, 2, 1, 1)$, Macaulay degree 11, a 384×448 matrix) is 481× below the plain form. End to end on $E(\mathbb{F}_{(2^m)^4})$ the ratio of IC to the negation-map rho model falls from 5,080 at $m = 6$ to 6.06 at $m = 14$ ($n \approx 2^{56}$; 4.3× a measured rho), with fitted slope -0.248 in $\log_2 n$—Gaudry's $n^{-1/4}$—and an extrapolated crossover near $n \approx 2^{68}$ (2^{63}–2^{69} under the model). A later red-team review found this crossover unsupported (Chapter 25; $m = 15$: 5.2× after corrections). After genus three, this is the nearest any index calculus in the programme has come to rho, and its extrapolated crossover lies where rho itself takes about 2^{34} steps, so the extrapolation can be tested directly. A Joux–Vitse $(n-1)$-decomposition at $n = 5$ on $E(\mathbb{F}_{p^5})$ completed verified DLPs at $4.5 \cdot 10^7$–$1.8 \cdot 10^8$× rho, with a crossover extrapolated to groups near 2^{291}.

Change	Cl.	Ratio [95%]	To rho
Genus 3 after the cubic GHS transfer			
Lucas splitting test (round 4, C1)	Ie	0.756 [0.755, 0.756] at 2^{22}; local 1.408 → 1.364	no rho arm since round 2
C1 + 32 walks per inversion	Ic	0.715 at 2^{22}; local 1.356	as above
Factor base retuned per size (04b)	Ic	0.961 at 2^{20}	0.16× rho on Jac(C) at 2^{14}, not on E_0
$k = 4$ over $\mathbb{F}_{p^4}$ and $\mathbb{F}_{(2^m)^4}$			
D_4 oracle: pivots, weighted form, replay, $p = 269$	E	1/370 [1/371.6, 1/367.9]	—
Same oracle, ecbench S	Ic	1/402 [1/406, 1/398]	$1.39 \cdot 10^6\times$ rho
Frobenius-reduced D_4 form, characteristic 2, $m = 8$	E	1/481 [1/483, 1/479]	—
Same, ecbench S ($m = 6$)	Ic	0.0057	6.1× rho model, 4.3× measured, at $n \approx 2^{56}$
Koblitz and binary oracles			
F6-IC gate-first order	E	1/1.73–1/1.34 word XORs	default on
Greedy coverage base, $K_0/2^{13}$	Ic	0.843 (expected trials)	gain 1.000 at $r = 2^{39}$
Inherited-F4 root-shape replay	R	1.015–1.328	switch stays off
Binary MITM oracle, $\ell = 9..12$	E	1/4,244–1/928 per target	relation collection $\Theta(2^n)$
Algebraic $m = 3, 4$ oracles vs MITM floor	R	slope 0.53–0.81 vs 0.21–0.36	$3 \cdot 10^2$–$3 \cdot 10^4\times$ floor
Prime fields and shared stages			
`f4_fp` blocked pair reduction	E	0.78–0.89 (wall, quad $n10, 11$)	beats msolve there
Sparse-CADO LA inside ecbench	E	S 1.000; LA wall 0.25–0.72	149× rho
S_3 solve on SECG small-x bases	—	table 2,600–313,000× cheaper	52–280 bits over rho
Column-touch conversion K (ECC2K-130 $m = 4$)	A	$+7.62/+10.32$ → $+9.78/+12.48$ bits ($w = 1$ floor; $+45.71$ at $w = 2$, unconverted)	accounting

Table 36.5: Changes that landed 16–20 October 2026, against their own incumbents; columns and classes as in Table 36.4. Sources under `research/`: `ic_improve4_20261015/`, `ic_improve_20261014/04b-genus3-2lp-rebalance/`, `gaudry_quartic_d4_cost_20261014/`, `gaudry_quartic_d4_char2_20261015/`, `joux_vitse_n1_20261015/`, `binary_mitm_oracle_l9_l10_20261015/`, `binary_mitm_oracle_l11_l12_20261016/`, `ic_m3_algebraic_vs_mitm_20261014/`, `secg_small_x_20261014/`, `greedy_factor_base_yield_20261014/`, `f6_gate_first_x86_20261016/`, `pdp_shape_replay_20261015/`, `f4_fp_blocked_pairs_20261016/`, `ic_linalg_pipeline_20261019/`, `k_conversion_20261018/`.

Binary oracles and their floor. The MITM pair-table oracle on a non-Koblitz curve reached $\ell = 12$ over $\mathbb{F}_{2^{37}}$ (2,474–4,244× over pairs-and-solve at $\ell = 9, 10$; 928–2,321× at $\ell = 11, 12$, the last as recomputed by the validation sweep; 0 disagreements on 256 targets). Against that floor, no algebraic oracle (inherited F4, F6-IC, SAT, split S_4 hybrids, a wide chained S_3 solver) grows more slowly in n: the cheapest is $3 \cdot 10^2\times$ the floor at $n = 13$ and $3.8 \cdot 10^4\times$ at $n = 31$ for $m = 3$, and the $m = 4$ chain grows at slope 0.805 against the floor's 0.333. A greedy coverage base needs 1.08–1.18× fewer trials on small Koblitz curves, but its gain falls to 1.0000 at $r = 2^{39}$. On the 33 SECG curves an S_3 solve over a small-x base, with no table, is 52–280 bits over rho.

The reference moved again. The strong signed-Frobenius rho of every Koblitz comparison executed 718 instructions per walk step after three rounds, down from 1,196; a fourth round of batched inversion took main from 707.2 to 679.7 Callgrind instructions per step at $n = 53$, with no established native-time gain (outputs identical; `research/rho_batchinv_20261022/`; steps per ideal 0.92–0.99 on 400 fixtures), and ecbench's standing `rho.signed_frobenius_strong` runs at 0.802× its former instruction count with identical GAE. On hyperelliptic Jacobians the programme's rho was slow at large N until full-width branch coefficients brought S_{walk} from 2.10 to 1.48; the negation fold saves only 0.97× at $N \geq 10^5$. Counted-unit comparisons (S) are unaffected by the instruction counts; wall-time comparisons against the older rho overstate index calculus by up to the same factor.

Corrections from the validation sweep. A repository-wide validation sweep (19–20 October) contradicted five recorded results, all now struck through in their sources with dated notes. Three bear on this paper. (i) The one-point $n = 83$ online ratio (18.5×, §9) and the hosted $n = 71/n = 83$ panels (7.3×, 12.3×) compared against rho with an empty table; with a Bernstein–Lange table matched to the index calculus's set-up time or memory, precomputed rho is faster in every matched $n = 83$ pair (3.7× time-matched and 40.6× memory-matched against $K = 3300$; 82–396× against $K = 600$). They join the $n = 61$ point as retired. (ii) The root-finder session's replay-all, recorded as 120/168 identical with the 48 mismatches confined to inherited F4, replays 72/168 on current `main`, with mismatches in all four IC arms, root finding included. The sweep later isolated the cause: an x86-only BMI2 rewrite in inherited F4 charges a different word-XOR count from the portable path, and with it disabled x86 and emulated aarch64 both replay 168/168 (`research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`). The 26 October addendum states the cause precisely and closes it (Chapter 41, item 4). (iii) The equal-bytes preprocessing control at $n = 73$ and $n = 83$ is built and frozen but not run. *Next improvement* (from the 18 October leads review): a large-memory host. The two exponent-level questions named then—the Weil-descent cell $(m, \ell) = (4, 6)$ and the PKM tower at $m = 3$, $N = 21$—have since run on rented hosts: the $(4, 6)$ cell refuted at degree 4 at $N = 54$, and the PKM cell finished degree 7 and stopped on memory inside degree 8, implying $D \geq 8$ but read as inconclusive (§§16.1,21.2). The $m = 4$, $N = 20$ cell remains; the $(3, 7)$ and $\ell = 8$ cells were since run with guessed variables (§36.5).

36.5 The ladder continued, 21–24 October

Work that reached `main` between 21 and 24 October falls into four groups: factor bases closed under cheap known-logarithm maps, a paired re-run of the Koblitz leaderboard recipe, a guess-and-determine attack on the Weil-descent degree question, and transfer measurements, among them a third family whose measured gap to rho is now single-digit (Table 36.6). The reading rules of §36.3 apply: each row is against its own incumbent and on its study's boundary. The companion ledger (`papers/ecdlp-ic-improvement-ledger/`) records the same period as a ratchet of incumbents; this section keeps to what moves the frontier or corrects it.

One mechanism, four factor bases. The week's measured gains on the index-calculus side are all the same idea: put a point and its images under maps of known logarithm—the Frobenius τ on a Koblitz curve, the CM endomorphism ϕ, small integer multiples, the cofactor $[h]$—into one column with a known coefficient, so the linear-algebra stage needs fewer relations. On Koblitz curves the $\mathbb{Z}[\tau]$ base needs $K+1$ relations instead of about one per orbit: child CPU of the complete process falls 1.9–4.4× at $n = 41$–61 and 6.2–10.5× at $n = 71$. These are measured against the repository's own tuned producer, on a host at load about 20 (child CPU is the less affected number), with three targets up to $n = 61$ and one at $n = 71$. The prime CM base ($j = 0$, $j = 1728$, $D = -7, -8$) cuts S 10–20×, and the integer-multiple

Change	Cl.	Ratio [95%]	To rho
Factor bases closed under known-log maps			
$\mathbb{Z}[\tau]$ small-norm multipliers, Koblitz $n = 41$–61	Ic	1/1.9–1/4.4 child CPU (3 targets, loaded host)	not measured
Same, wide path, $n = 71$	Ic	1/6.2 equal points; 1/10.5 retuned (1 target)	not measured
$\mathbb{Z}[\phi]$ multipliers, prime CM curves, 36–40 bit	Ic	1/10–1/20 in S	7–12×
Integer-multiple closure, random binary, degree 26–28	Ic	1/3.1–1/3.9 vs plain	6–12×
Cofactor-projected base (LEAD-19), $\mathbb{F}_{2^{15}}, \mathbb{F}_{2^{21}}, \mathbb{F}_{2^{27}}$	Ic	1/1.29 [1/1.52, 1/1.12], 1/1.58 [1/1.82, 1/1.35], 1/3.36 [1/4.02, 1/2.80]	6.1, 11.5, 14.4×
Koblitz leaderboard recipe (Table A, $n = 41$–61)			
Rule v4 recipe vs v3, paired, six sizes	Ic	0.953 [0.904, 0.998] at $n = 57$; five sizes not separated	9.9–43× at the canonical step
Weil descent, degree question			
Hybrid guess, $(m, \ell) = (3, 7)$, $k^* \leq 2$ guesses	E	1/116–1/260 of projected degree-5 closure	—
Implied ECC2K-130 $m = 4$ row (derived)	N	+8.43 (flat k) to +58.27 bits (linear k)	vs +10.32 ($w = 1$), +45.71 ($w = 2$)
Prime fields			
Kummer tower, 30 bit (new row)	N	—	158.9× [121.6, 221.4]
Coppersmith lattice on S_3, $\ell \leq 3$	N	0/10 recovered in all 96 cells with $\beta \geq 0.20$	ceiling $\beta = 1/4$ stands
Transfer and covers			
Joux–Vitse cover of $E(\mathbb{F}_{p^6})$, $p = 53$–1009	N	slope −2.05 $[-2.28, -1.83]$ in $\ln p$	1.59–1.98× at $p = 1009$ (LB; pooled rho constant, superseded by matched rho, §30.4)
Measured trial cost, genus-255 binary Jacobian	A	103 curve additions per trial ($2^{6.69}$)	$N = 130$ margin +4.71 → −1.83 bits
GHS covers of $E/\mathbb{F}_{q^4}$, odd q, census	N	genus floor 13 on odd-order curves	model exponent gap 0.154

Table 36.6: Changes that landed 21–24 October 2026, against their own incumbents; columns and classes as in Table 36.4 (N = null or diagnostic, no cost row). Sources under `research/`: `ic_improve_20261023/` (`README.md`, `PRIME_README.md`, `BINARY_README.md`), `lead19_projected_base/`, `ic_tool_program/rule/v4/`, `dreg_hybrid_guess_20261024/`, `jv_cover_p_scaling_20261024/`, `hcdlp_trial_cost_20261024/`, `ghs_cover_census_n4_20261024/`, `coppersmith_s3_theta2_20261023/`; and `docs/ic/CERTICOM131_IC_PROGRAM.md`.

base on random binary curves 3–4×; both at 36–40-bit or degree-26–28 sizes, where the best bases have 3–10 columns over 512–3,480 points and, as their sources note, tend to a baby-step giant-step table over multiples as the column count falls. The cofactor-projected base (LEAD-19) is the only one measured on leaderboard rows, with 20 repeats and the order stable across the band: 1.29–3.36× in cold S, leaving 6.1–14.4× to matched rho. Every source states that the exponent is unchanged; the gain is bounded by the size of the multiplier set and shrinks as linear algebra's share of the cost shrinks. The known-coefficient columns are those of GGMP's Frobenius-orbit bases [18] (§15); what is new here is extending them from automorphisms to small-norm and integer multipliers. Whether that extension is in the literature has not been checked. All four are ENGINEERING, and none is measured against a published index calculus; the $\mathbb{Z}[\tau]$ source notes that the summation-polynomial methods have no table-based counterpart at these parameters.

Koblitz: the recipe is near its optimum, the distance is larger. Rule v4 retuned columns, collection window and unit size on disjoint training seeds and re-ran Table A paired against rule v3 (32 pairs per size, 16 seeds, A/A and a positive control that detects a 10–15% slowdown). One size improves (0.953 $[0.904, 0.998]$ at $n = 57$) and five are not separated, so it is not a Table A improvement under its own rule. The free-relation lever costs 5–9% in this pipeline. More consequential is the reference: with rho priced in the same process at the canonical batched step, the cold distance of rule v4 is 9.2–42.1× (rule v3: 9.9–43.5×), 1.76–1.94× the as-built figure that rule v3 used. The ledger's rule v8 reads 4.67–18.07× on the same sizes in its own unit; the two are different domains in the sense of Definition 4.1 and are not combined here.

Weil descent: a hybrid closes one cell, the degree question stays open. The $(m, \ell) = (3, 7)$ cell had no refutation at degree 4. Fixing one or two variables restores degree 4: all branches refute with $k^* \leq 2$ on 10 of 10 unsatisfiable draws, at 1/116–1/260 of the projected degree-5 cost, in 3–6 GB (planted-solution and placebo controls passed). For the ECC2K-130 $m = 4$ row this changes less. Its often-quoted "+10.32 bits over rho if degree 5 holds" prices one column touch per Macaulay column ($w = 1$), a floor no elimination beats; at $w = 2$, the exponent the measured closure tracks, the same row is +45.71 bits, and the measured conversion $K = 4$–16 rho steps per touch raises both (§36.4). With the hybrid, the row is +8.43 bits if the number of guesses stays flat in ℓ and +58.27 bits if it grows linearly; one value of ℓ cannot separate them. The pre-registered $m = 3$ runs at $(n, \ell) = (22, 8)$ have since completed, with a series of guess sets (§19.1, Table 19.1): four guessed variables suffice on every draw measured, and three suffice on 7 of 12 draws when they include the summand-swap invariant. The count is 0 at $\ell = 6$, about 1 at $\ell = 7$ and at most 4 at $\ell = 8$, which still does not decide whether it stays bounded or grows. At the priced $m = 4$, $\ell = 33$ cell the hybrid beats degree 5 only if the count is at most 6 (`research/dreg_closure_guess_20261022/`, `research/dreg_closure_guess3_20261023/`). In no scenario does the row go below the degree-4 floor of +7.62 bits, so the route does not reach rho.

Prime fields: further behind. A new 30-bit Kummer-tower row in ecbench (160/160 verified, loaded host) is 158.9× $[121.6, 221.4]$ rho at $w = 0$, against 141× at 28 bits in the campaign; its source calls it "a measurement, not an improvement". Two falsifiers for the Certicom 131-bit programme (random-b binary degrees of regularity; structured bases) were null. A frozen-protocol test of a Coppersmith lattice on S_3 recovered no planted decomposition in any of 96 cells with $\beta \geq 0.20$ at 24–56 bits, with positive controls passing below, so the $\beta = 1/4$ ceiling stands at multiplicity ≤ 3.

A third family crosses rho: the Joux–Vitse cover. The genus-3 cover of $E(\mathbb{F}_{p^6})$ over $\mathbb{F}_{p^2}$ (full rational 2-torsion; a weak class, not prime-order curves) has two arms; the full account is §30.4. The plain arm, with Nagao decompositions, ran at $p = 53$–1009, three seeds per prime, every logarithm checked. Against a pooled rho constant ($S = 1.344$, measured at $p \leq 251$) its gap fell from a geometric mean of 620× at $p = 53$ to 1.73× at $p = 1009$ (1.59–1.98), slope -2.05 $[-2.28, -1.83]$ in $\ln p$, and the line met that constant near $p \approx 1330$ (an extrapolation). Rho was later run on the same instances. Against a matched negation rho the plain arm is above rho in group additions at $p = 1301$ (1.41 $[1.21, 1.63]$) and $p = 1451$ (1.16 $[1.02, 1.34]$), unresolved at $p = 1601$ (0.87 $[0.70, 1.18]$), and above it in $\mathbb{F}_p$ multiplications at all three (`research/jv_cover_negrho_rescore_20261101/README.md`), so the extrapolated crossing near 1330 does not hold. The arm still does not emit its unpriced F4 and Wiedemann work, so it stays a lower-bound comparison. A separate sieve arm on the same cover was measured against a matched negation rho at $p = 1511$ and validated: it is 0.0065× rho (155× $[125, 184]$ below it), and the order holds across the band.

Covers: censuses and a repriced counterfactual. Measuring the cost of one relation trial (a Cantor addition plus a smoothness test) on a genus-255 binary Jacobian gives 103 additions of a point over $\mathbb{F}_{2^{131}}$ ($2^{6.69}$), which turns the $N = 130$ GHS counterfactual's modelled margin over rho from +4.71 bits to −1.83 bits; only a trial below about 29 additions would preserve it, and the smoothness test alone costs 16.5. Two censuses fix what covers exist at toy sizes: for $E/\mathbb{F}_{q^4}$ in odd characteristic the genus floor is 13 on odd-order curves ($2 - 2/g$ against rho's 2 leaves an exponent gap of 0.154, constants excluded), and for binary curves over $\mathbb{F}_{2^{3l}}$ the genus-3 class is a 2^{-l} fraction (0.78% at $l = 7$).

The validation gate. On 24 October the leaderboard's gate ran for the first time: all 13 Table B rows (eight prime, five random binary) passed—every counted field replays identically, every scalar checks in an independent verifier, and IC stays above rho at every weight of the band (4.05–54.76×)—and all nine Table C rows failed; Table A was not in scope (`research/ic_validation_gate_20261024/`). The study staged its manifest rather than installing it, and the leaderboard now marks the Table B rows validated. The ledger paper, current to 23 October, records that no row had passed; for Table B that is now out of date. Validation says the numbers are what the archive says and the reference is fair, not that index calculus approaches rho.

What is still not measured. No arm in the programme has yet been compared with a published index calculus at the same (q, n, m): FGHR-style symmetrised summation polynomials over the standard base $\{P : x(P) \in \mathbb{F}_q\}$, solved by msolve, or WDSat for binary decompositions. (WDSat has been run as a solver on its authors' published instances, §16, but not inside an index calculus with relations, linear algebra and a rho comparison.) Every ratio in Tables 36.4–36.6 is against rho or against the programme's own previous arm. Defining and running that reference arm (open problem 10) is the precondition for any claim of improving on the state of the art.

36.6 Three frontiers

Cold, one target, set-up included. Unchanged: generic rho. Every measured index-calculus arm is above its matched rho (Figure 36.4); the closest complete count is root-finding negfold at 1.55× strong signed-Frobenius rho at $n = 13$; the closest lower-bound arms are 1.08× (compact orbit with a counted Frobenius table) and 6.97× (matrix F4/F5 against rho-negation at K13). Prime-field arms are 12–239× rho and diverge with size (the 30-bit Kummer-tower row of §36.5 is 159×). Off the prime and prime-degree binary fields, three families are within a single-digit factor of rho or go below it, all in weak classes: genus three after cubic GHS (below the strongest rho at $q = 2^{16}$ under the registered pricing, unresolved at the other sizes to 2^{20}, above it under a heavier sensitivity pricing), $k = 4$ over $\mathbb{F}_{(2^m)^4}$ (4.3× a measured rho at $n \approx 2^{56}$ on one target, 5.2× at $m = 15$ after corrections; crossover unsupported), and the Joux–Vitse cover of $E(\mathbb{F}_{p^6})$ (plain Nagao arm 1.16–1.41× a matched negation rho at $p = 1301$ and 1451, unresolved at 1601, a lower-bound comparison; its sieve arm is 0.0065× a matched negation rho at $p = 1511$, validated, §30.4).

Online, after retained set-up. Generic again: Bernstein–Lange preprocessing rho. The admitted compact-orbit point at Koblitz $n = 61$, 13.9× [9.8, 19.1] over strong rho without a table (cold median 83.4× rho), is 53–317× slower online than preprocessing rho given equal or less advice (§36.3), and the $n = 83$ points lose every pair run with rho given a table matched on set-up time or memory (§§36.4,15.5).

Stage. Moved substantially: the decomposition oracle is two to five orders of magnitude cheaper than at the start of the programme on the workloads where root finding applies, the $\mathbb{F}_p$ F4 is at msolve parity, the Boolean F4 family is one to two orders of magnitude cheaper in counted matrix work, and the $k = 4$ extension-field solve is 48× and then a further 370× cheaper over $\mathbb{F}_{p^4}$ (481× below the plain form in characteristic two). None of this, by itself, moves the cold frontier: the stages it leaves untouched—relation yield, factor-base size, linear algebra, table set-up—now dominate.

36.7 What would move the cold frontier

From the frontier synthesis and the problem records: (i) register the torsion-symmetrised bases and the $n = 83$ dense-root engine in ecbench under immutable method identities and emit cold and online

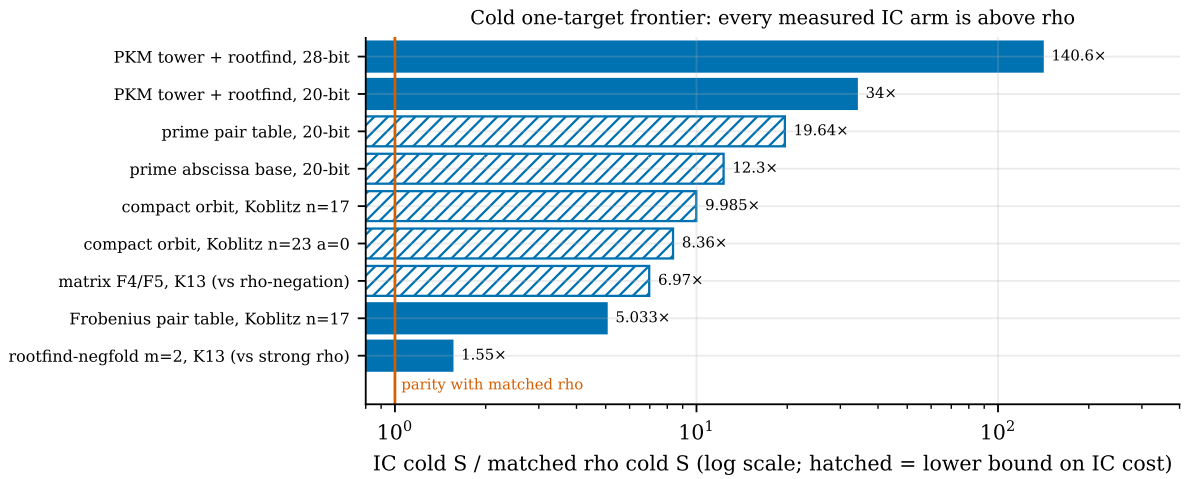


Figure 36.4: Cold one-target S of index-calculus candidates over their matched rho arm (each study's own estimator; curves differ, so bars are not ranked against each other). Hatched: lower bound on IC cost (unpriced native or solver work). Sources: `research/ecbench_frontier_20261013/`, `research/ecbench_rootfinding_assessment_20261012/`, `research/pkm_tower_ecbench_20261013/`, `research/ecbench_rootfind_faster_20261014/`.

records; (ii) resolve the inherited-F4 cross-host counter drift before using that counter in a bound; (iii) run a four-size native challenge against the standing rho bound—two sizes cannot establish an exponent; (iv) obtain an isolated cold session at $n = 83$ with every set-up phase and retained byte charged; (v) price F6-IC's batched geometry with a pinned conversion. Since then three items have moved to the front: (vi) run $k = 4$ over $\mathbb{F}_{(2^m)^4}$ at $m = 15$–17 against a measured negation-map rho, where the fitted trend was read as predicting parity near $n \approx 2^{68}$ (the red team found that unsupported, Chapter 25); (vii) measure the matched rho on $E_0(\mathbb{F}_{q^3})$ in the genus-3 rounds' unit (done on 23 October: below rho at 2^{16} under the registered pricing, unresolved elsewhere, §26); (viii) obtain a large-memory host for the $(4, 6)$ Weil-descent cell and the PKM tower. The 21–24 October work adds two more and re-weights one: (ix) measure rho on the Joux–Vitse cover family at $p \geq 503$ (or a size where it can be run) and emit the cover arm's unpriced counters, so that the p^{-2} trend can be ranked under the band and tested past $p \approx 1330$ (the matched rho now exists: for the sieve arm at $p = 1511$, and for the Nagao arm at $p = 1301$–1601, where the crossing near 1330 failed; the unpriced counters are still not emitted); (x) define and run a published-index-calculus reference arm (open problem 10); and (viii) is worth less than it looked for the binary question, since the $m = 4$ row at $w = 2$ is about +46 bits over rho even if degree 5 holds (§36.5). A result in the cold domain that beats rho at four sizes would be the first ADVANCE of the programme.

37 A map of unconventional routes and their lower bounds

The previous chapters measured index calculus as the literature defines it: a factor base, a decomposition system, a solver and linear algebra, priced against matched rho. The programme also kept a family of leads that change one of those ingredients more radically: an extra oracle, a tensor network in place of the decomposition system, a learned component in the solver, a statistic that might show that a factor base is not random, a lift to another curve model or another field. The lead atlas files them as the *exotic* family (`crypto-autoresearcher:knowledge/leads/ecdlp/exotic/`). This chapter maps the family, with one table of verdicts and standing, and then gives the lower bounds that most of the family runs into. The two lines with the most work behind them have their own chapters: the tensor-contraction route (Chapter 38) and the learning and pseudorandomness lines (Chapter 39).

Three conventions hold for all three chapters.

- *Standing.* Two items in this family carry a proof-carrying claim: `PCC-20261104-1df881` (the source audit of implicit tensor entries) and `PCC-20261104-458dda` (the structured-predicate integration report), both of standing *unverified* (§24.1). Every other figure is an unverified assertion of the producer named by its source. Several studies name their own validator or red-team report; that is recorded where it exists, but it is not a proof-carrying-claim standing and none of these studies has a `VALIDATION.json` gate row. No row of the index-calculus leaderboard comes from this family.
- *Costs.* No line in this family produced a counted cost of a complete discrete logarithm against matched rho. Where a study gives a ratio, it is a stage or structure statistic, and the pricing band of §4.3 is not applicable. Under the iterative success criterion none of the results is an improvement of the index-calculus baseline.
- *Scope.* Most measurements are at toy size: $p \leq 2^{14}$ for the structure instruments, $n \leq 17$ for the binary SAT studies, $p \leq 2039$ for the rank experiment. A toy-size null excludes only effects large enough to be seen at that size.

37.1 The exotic family in the atlas

On 6 November the exotic directory held 32 lead records. Their dispositions, read from the YAML files on crypto-autoresearcher main, are:

- 26 `open_next_step`;
- 2 `open_blocked` (advice leakage, the idea corpus);
- 2 `exhausted_at_scale` (the crypto-size candidate screen and H-PSEUDO);
- 1 `closed_obstructed` (label-valued augmented oracles);
- 1 `closed_dominated` (the holant decomposition-count network).

A disposition records what the programme does next, not whether the route works: most of the open leads have a named obstruction for the version that was tried and a narrower question left. Table 37.1 lists (with Table 37.2) every record with its best measured result as the record states it and the chapter that discusses it. The table is a map, not a ranking. Where the record's best result is "none", nothing was measured.

The family overlaps with work recorded elsewhere in this treatise. The August oracle closures, the canonical-lift statistic and xedni lifting are in Chapter 35; the transport, monodromy, oracle and lifting obstructions are in Chapter 8; the k-sum tables are in §22.2; the external tensor preprint is reviewed in §24.1; the H-PSEUDO correction is in Chapter 35 and Table B.1. Corrections specific to this family are collected in Table B.8.

37.2 Routes by which an encoding can be read

A sub-exponential algorithm for the ECDLP on a prime-order subgroup cannot be generic. Shoup's bound puts every algorithm that uses only the group law and equality at $\Omega(\sqrt{N})$, and with S bits of advice the preprocessing bound is $ST^2 \gtrsim N$ [8]. A candidate must therefore read the encoding: coordinates, the curve equation, the field. The sub-exponential desk evaluation of 1 November (`research/ecdlp_subexp_desk_20261101/README.md`) sorted the routes by which an encoding can be read and

Lead (LEAD-EXO-)	Disposition	Best result recorded; obstruction	Where
Oracles and query bounds			
augmented-oracle-ggm	closed, obstructed	label-valued and public-data oracles are Shoup-simulable; toy split saving $B/A = 0.0557$ (preliminary)	§37.4
coordinate-revealing-oracles	open	split off 4 Nov; "the GGM gives no bound" for x, y, full point, x-halving, field-valued jet and net, C_t, degree-D tests	§37.4
low-degree-query-lower-bound	open	H1 false as stated, H1′ proved; full rank in 39 of 39 cells, $p = 251$–2039; no toy-size violation	§37.5
low-degree-algebraic-query-bound	open	theorem programme; none	§37.5
ksum-conjecture-s5-oracle	open	a 4-SUM oracle on $\{P : x(P) < p^{1/5}\}$ needs $n^{\lceil k/2 \rceil}$ under the k-SUM conjecture; none measured	§37.3
four-list-ktree-join	open	$j = 2$ four-tree at exponent 0.4167 closed for character filters, toy $N \le 65539$	§37.3
invertible-hint-subset	open	none; Lange–Winterhof and Kiltz–Winterhof statements still to be read	§37.3
lange-winterhof-interpolation	open	none; theorem not yet recorded	§37.3
elliptic-code-list-decoding	open	none; counting bound $\binom{B}{M} p^{k-M-1} \ge 1$, $M = \sqrt{Bk}$	§37.3
hnp-advice-leakage	blocked	Cheon walk exponent 0.23 against rho's 0.50 only when $[\alpha^i]G$ is leaked	§37.3
Tensor networks			
hashed-factor-base-tensor-ecdl	open	none: 34 correctness tests; ECC2K-130 and ECCp-131 exceed the exact-leaf budget without a relation	Ch. 38
holant-decomposition-count-network	closed, dominated	digit-QTT rank in x-labelling at the ceiling (1024 at $p = 1021$), as random relabellings; NULL	§39.4
gag-holant-semaev-chain	open	none	§39.4
Learning			
ml-learned-models	open	learned WDSat order 16,097 conflicts against 3,092 (confounded, §39.3)	§39.1–39.3
Pseudorandomness and instruments			
dl-character-sum-pseudorandom	exhausted	H-PSEUDO: R consistent with 1 against a negation-closed null	Ch. 35
quadratic-phase-small-x-leakage	open	NULL at $N \approx 2^{12}$–2^{14} and one curve at 2^{16}	§39.4
coordinate-statistic-leakage	open	census lowered from strong to preliminary (CORR-20261026-c94668)	§39.4
weak-class-instrument-calibration	open	no instrument detects anomalous or small-k curves; ζ-stable base detected	§39.5
methodology-instruments	open	none; false-negative rate of the screens unmeasured	§39.5

Table 37.1: The exotic lead records on 6 November, part 1 of 2. Disposition and best result as each record states them; all figures producer-reported and unverified unless the text says otherwise. Source: `crypto-autoresearcher:knowledge/leads/ecdlp/exotic/`, one YAML file per row (`LEAD-EXO-`*row*`.yaml`).

Lead (LEAD-EXO-)	Disposition	Best result recorded; obstruction	Where
Lifts, transports, models			
global-lift-height	open	xedni exponent 3 for one family; short Mordell–Weil relations at $p \leq 31$, no decomposition step	Ch. 8
padic-jet-canonical-lift	open	digit identity exact 1360/1360 on hard lifts, whose cost is the lift	§39.6
elliptic-net-eds	open	charged exponent 1.97–2.80 against BSGS 1.17–1.32	§39.6
curve-model-representation	open	$d_{\mathrm{reg}} = 2$ for Weierstrass and Edwards at $m = 2, 3$ (toy)	§39.6
split-jacobian-projection	open	none; Hasse–Weil order inflation closure	§39.6
miller-pairing-cocycle	open	none; one unreviewed proposal	§39.6
torsor-deck-period	open	none; deck-orbit router no-go	§39.6
lattes-xline-dynamics	open	none; forward iterates are group operations	§39.6
lossy-object-propagation	open	none; orbit argument and rigidity theorem	§39.6
Screens, corpora and out of scope			
cryptoscale-candidate-screen	exhausted	thirteen batches, no admissible candidate	§37.2
ideas-corpus-algebraic-transforms	blocked	none; 15 active registry rows lack constructors	§37.2
minors-guess-and-determine	open	none; design reconstructed from abstracts	§37.2
quantum-ecdlp-resources	open	reversible secp256k1 addition, −0.124% on a sample island; not classical	§39.7

Table 37.2: The exotic lead records, part 2 of 2 (Table 37.1 continued). Same source and standing.

matched sixteen proposed mechanisms against the atlas with `lead_atlas.py -match`; every one of the sixteen landed on an existing lead. Its route table, as recorded at atlas snapshot `cfc1dae2`, gives one named obstruction per route:

- an algebraic factor base of size B over $\mathbb{F}_p$ has description degree at least $B/3$, and mean yield is $\binom{B+m-1}{m}/N$ for every base of size B (Chapter 5);
- Coppersmith lattices reach at most 0.3293 per S_3 block in three unknowns against a demand of 0.8889 at $(m, w) = (4, 2)$;
- the digit and subset-sum form uses only group operations and equality, so it is generic;
- enumerative, pair-table and meet-in-the-middle oracles have no sub-rho cell in the replicated cost model at any arity;
- p-adic lifts, jets, canonical and Teichmüller lifts carry no additive image of order ℓ when $\gcd(n, p) = 1$;
- homomorphic images in a multiplicative group factor through the embedding degree;
- index calculus in a genus-g Jacobian over $\mathbb{F}_p$ costs $p^{2-2/g} \geq p$ for $g \geq 2$;
- generic cost is an isogeny-class invariant, and automorphisms give $\sqrt{|\mathrm{Aut}|}$.

The README now carries a banner: it is a historical snapshot of 154 leads, a later reconciliation catalogued 160, and its "xedni is closed" statements are superseded by the conditional reading of Chapter 8. The route table is otherwise unchanged.

Three searches were broad screens rather than mechanisms. The crypto-size candidate screen (`GOAL-CRYPTO-001`) ran thirteen batches that ended `FINAL_NO_CANDIDATE` or `NO_ADMISSIBLE_NEXT_MECHANISM` over its checked source set (`EV-CRYPTO-010..013`); its record states the scale reached as "zero-compute admission sheets; no crypto-size run" and asks for a sensitivity audit showing that the screen could have admitted a known positive. The idea corpus has 15 active registry rows, each blocked on a constructor or theorem that no one supplied. The minors and guess-and-determine lead, which would extend the lit2026 W12 reproduction of §21, has a draft design reconstructed from abstract pages only. None of the three produced a candidate to measure.

37.3 Generic and conjecture-level obstructions

Five records in Table 37.1 are bounded by a statement about generic or structureless algorithms rather than by a measurement.

- *k-SUM.* The prime-field tier that needs a 4-SUM oracle on $S_B = \{P : x(P) < p^{1/5}\}$ in time well below $\sqrt{|S_B|}$ meets the k-SUM conjecture: structureless k-SUM needs $n^{\lceil k/2 \rceil}$. The one known escape, clustered 3SUM, needs additive energy that the coincidence and character-sum measurements exclude. The record's next step is to write the reduction as a conditional obstruction, with the caveat that k-SUM has near-linear linear-decision-tree complexity, so no Shoup-style proof is available. The measured k-sum tables are in §22.2.

- *Four-list joins.* A Wagner tree on $E(\mathbb{F}_p)$ needs a cheap filter h with $h(P+Q) = f(h(P), h(Q))$ often enough. For character filters under group-law or quasigroup combining, the $j = 2$ four-tree at exponent 0.4167 is closed at toy $N \le 65539$ (`crypto-autoresearcher:analysis/o2-sum-compatible-filters/`); the log-domain FFT scans covered 507 families and 13,528 combinations, with a maximum of 1.10× the null (`research/ic_prime_eth_leads_20261031/SYNTHESIS.md`, proposal 8). A planted log-mod-M character is the Pohlig–Hellman leak, not a coordinate-computable filter.
- *Invertible hint subsets and interpolation degree.* A dense subset of $E(\mathbb{F}_p)$ whose membership and logarithms are both computable from coordinates would beat the square-root bound by random self-reduction. The Lange–Winterhof and Kiltz–Winterhof degree bounds on polynomials interpolating the ECDL and ECDH maps are the expected closure, but both records say the theorem statements have not been read beyond their abstracts; whether they cover the statistics of Chapter 39 is open.
- *List decoding.* Using Guruswami–Sudan list decoding of elliptic codes as a decomposition oracle predicts hits only when $\binom{B}{M} p^{k-M-1} \ge 1$ with $M = \sqrt{Bk}$, which forces k of order B; nothing was run.
- *Advice.* The hidden-number conversion of a log-bit leak is deferred until some statistic has a measured nonzero advantage. The Cheon walk on leaked powers $[\alpha^i]G$ measured exponent 0.23 against rho's 0.50 from 24 bits up, but that is a different problem (the discrete logarithm with auxiliary inputs), not index calculus.

37.4 Augmented oracles after the 4 November split

Chapter 8 records the August result that jet, endomorphism, incidence, halving and net oracles are simulable in the generic group model. A red-team review of 4 November (`crypto-autoresearcher:docs/reports/leads-round2-20261104/augmented-oracle-review.md`) showed that the record used "oracle" for three kinds of object, and the owner split the lead into two the same day:

- *label oracles* (input a label, output a label or a bit fixed by group relations): endomorphism image, label-valued halving, incidence check over a label factor base, label-valued net. These are Shoup-simulable with overhead $\Theta(\log N)$ or $O(m)$, and stay closed in `LEAD-EXO-augmented-oracle-ggm`;
- *coordinate oracles* (input a label, output field data): x, y, full point, x-valued halving, field-valued jet and net, the threshold oracle C_t, degree-D coordinate tests. These moved to `LEAD-EXO-coordinate-revealing-oracles`, whose record says "the GGM gives no bound";
- *public-data "oracles"* with no instance input (factor-base membership of a field element): not augmentations at all, 0 queries.

The key step is Lemma 3 of the holant note (`crypto-autoresearcher:docs/reports/holant-h1-ggm-lemma-20261103/README.md`): on every curve the x-oracle cannot be simulated with $o(\sqrt{N'})$ group queries, N' the largest prime factor of N. Two x-queries give $x(G)$ and $x(Q)$, free local brute force gives k up to sign, and $O(\log N)$ group queries fix the sign. The review's sanity check: if a full-point oracle were simulable with $o(\sqrt{N})$ overhead, Shoup would bound concrete ECDLP by $\Omega(\sqrt{N})$ on every curve, which Smart's attack on anomalous curves contradicts. The same review found the endomorphism-image argument's premise "$\mathrm{End}_{\mathbb{F}_p}(E) = \mathbb{Z}$" false (Frobenius) and replaced it with an eigenvalue argument with the same exponent, and it read the earlier jet verdict "$C = 1$" as the structured-GGM verdict withdrawn by `CORR-20260727-003` (Table B.8).

The consequence for index calculus is the point of the split. Every index-calculus algorithm reads coordinates, so it is a coordinate-revealing algorithm, and the generic group model says nothing about it. What the open lead asks is whether coordinate access gives a *time-bounded* advantage on prime-order prime-field curves, which is the index-calculus question itself (`KN-OPEN-001`). The only measured number on either side is a toy split saving: on 7–8-bit curves at $m = 3$, an x-membership-filtered Semaev meet in the middle has the same yield as the unfiltered one in 40 of 40 cells and costs 0.0557 of its field operations, recorded as preliminary and as "not a bound".

37.5 The low-degree query lower bound

One way to keep a lower bound while admitting coordinate access is to allow only low-degree tests: the solver may ask whether $f(\text{coordinates of } X_1, \dots, X_t) = 0$ for a polynomial f of total degree at most D, with $X_i = [a_i]P + [b_i]Q$. The lead's conjecture H1 was success probability $O(T^2/N + TD/\sqrt{N})$. A desk note of 4 November (`crypto-autoresearcher:docs/reports/ecdlp-leads-desk-20261104/low-degree-desk.md`, claims tagged proved, cited or heuristic) settled the model.

- *Single-point tests do not leak* (proved): a degree-D test on one point is true for at most $2D$ ($3D$ with y) values of k.
- *H1 as stated is false* (proved, given a rank condition). The pulled-back function has degree up to $\Delta = \sum_i (2d_{x,i} + 3d_{y,i})b_i^2$ on E, and large b_i cost only $O(\log N)$ group operations. If the evaluation matrix of degree-$\leq D$ monomials on the N points $v(R)$ has full rank, one test answers "$k \in S$?" for any S, and binary search finds k with $\lceil \log_2 N \rceil$ tests; for $D = 3$ this needs $t \approx (6N)^{1/3}$ points, so $T = O(N^{1/3} \log N)$.
- *The restatement H1′ is proved*: success is at most $\left(\binom{T_g+2}{2} + \sum_j (\Delta_j + t_j) + 1\right)/N$, so sub-$\sqrt{N}$ success needs $\sum_j \Delta_j \gtrsim N$. The bound is meaningful for low-degree statistics of points with small multipliers.
- *The Kohel–Shparlinski bound does not suffice* for $t \geq 2$ with large multipliers: every bound in that paper scales with the degree of the pulled-back function, not with D. The note read the paper (ANTS-IV, 2000) in full and records what it proves.

The rank condition was then tested exactly under a protocol frozen before data (`crypto-autores earcher:docs/reports/lowdeg-rank-20261104/README.md`, `LOWDEG-RANK-20261104`). With random multipliers, full rank held in 39 of 39 decisive cells at $p = 251$, 509, 1021 and 2039, once the monomial count was about twice the number of candidates; the random-point null was full rank and the shift control stayed within its Riemann–Roch bound $3Dt + 1$. Small multipliers $b_i = 1..t$ failed in exactly the three cells with $\sum b_i^2 \approx 0.34$–$0.41N$, as H1′ predicts. The report's validator re-ranked 8 cells exactly, confirmed the verdict, and corrected the protocol's own reading: at the tested points the attack spends about 180–490 group operations, more than $\sqrt{N}$ (32–45), so *no toy-size violation of H1's query bound is shown*. By the validator's accounting the attack would beat $\sqrt{N}/D$ only for $D \geq 3$ and N from about 2^{28} ($D = 5$) to 2^{48} ($D = 3$), and never for $D = 2$, which rests on full rank persisting at $t \sim N^{1/D}$, untested.

The attack is not a cost. Building the test takes about N^{ω} local field operations, and a test that answers "$k \in S$?" for $|S| \approx N/2$ needs a polynomial with about $N/2$ monomials, so a plain logarithm table dominates it. The next step on the lead is a bound that charges the test's description length or local work, which, the desk note observes, would be a complexity conjecture rather than a generic-model theorem. For index calculus the lesson is the same as in §37.4: once local work on coordinates is free, query bounds stop constraining what a factor base can be.

37.6 The lit2026 W3 audit against generic bounds

The third bound is a check of the programme's own records against three recent generic lower bounds (`research/lit2026_w3_ggm_audit_20261011/README.md`, workstream W3 of the lit2026 programme, 11 October; the other workstreams are in §21). The three papers are:

- the structured generic-group model (SGGM) of Corrigan-Gibbs, Henzinger and Wu (ePrint 2026/384): an algorithm that exploits structure on a δ fraction of the group needs $\Omega(\min(\sqrt{q}, 1/\delta))$ group operations;
- the non-adaptive preprocessing bound of Dinur, Keller and Marmor (arXiv 2505.00894): success at most $3T^2/N + 4 \ln 2 \cdot ST/N$, so non-adaptive algorithms below $T = \sqrt{N}$ need $ST \gtrsim N$;
- the multi-instance bound of Akshima, Kiltz, Nishaburi, Nooripoor and Wiedijk (ePrint 2026/2253), read in abstract only.

The study maps an m-term decomposition oracle over a factor base F onto the SGGM's partial operation $\star$, so that the constrained labels are F and its j-sums, $j \leq m$, and $\delta \leq c \sum_{j=1}^{m} \binom{|F|+j-1}{j}/r$, with c the cofactor (loose) or 1. That mapping is the study's derivation, not a theorem of the paper, and the elliptic-curve section of 2026/384 could not be extracted from the copy that was read (a stated gap).

Table 37.3 gives the result. No configuration is below any bound, for $\min(\sqrt{r/A}, 1/\delta)$, for the explicit constant of the paper's Theorem 3.2 at $\varepsilon = 1/2$, or for the uniform-landing δ. The closest row, `icv1-fp14-tm43-158b6914/mitm_m2_negfold_walk_balanced`, is 38.9× above its bound, and the balanced $m = 2$ prime rows at 10–18 bits are 39–52× above. The informative number is the other one: *122 of 208 rows have* $\delta_{\text{upper}} = 1$. Their factor base's m-sums cover the subgroup, so the bound is 1 and certifies nothing; this covers every compact-orbit rung and every Koblitz $m = 3$ row with $c = h$. For those configurations the whole cost is in *finding* a decomposition, which the SGGM treats as a free evaluation of $\star$. The SGGM therefore splits the programme's configurations into sparse ones, which it bounds and which all sit 39× or more above the bound, and dense ones, which it cannot see. Thirteen rows show relations per trial above the counting ceiling, up to 1.2×; none is significant at the Bonferroni level (smallest $p = 0.0087$), so the study reads them as noise, not as an accounting bug.

Family	Rows	$\delta_{\text{upper}} < 1$	min measured / bound	below a bound
prime (Table B)	72	48	38.9	0
random binary (Table B)	35	23	56.8	0
Koblitz round 5 ($n \leq 41$)	72	13	355	0
Koblitz one target (Table A)	6	1	$2.84 \cdot 10^6$	0
Koblitz 32-target batch (Table C)	9	1	$7.99 \cdot 10^4$	0
Koblitz compact orbit $n37$–$n83$ (probes)	14	0	1077	0
All	208	86		0

Table 37.3: The lit2026 W3 audit: 208 index-calculus and precompute configurations against the SGGM bound. Producer-reported, from committed records, isolation L0; the compact-orbit rows are in probes, not group operations. Source: `research/lit2026_w3_ggm_audit_20261011/README.md`, Results §1.

The other two papers place the table arms. Every non-adaptive table arm in the repository sits on $ST \approx N/A$: the multiplier-closure arms at 0.739–1.432 (canonical entries times T over r/A, 27 arms) and the compact-orbit target extraction at 1.053–2.734. Dinur–Keller–Marmor say that is the floor for the class. Batch rho follows the $\sqrt{kq}$ multi-instance baseline: fitted slopes 0.521–0.585 with every bootstrap interval covering the Kuhn–Struik finite-k slope 0.533–0.539, on 744 plus 744 verified fresh logarithms. The audit also found three record problems (an $n = 61$ subgroup size recorded as $\log_2 r = 47.5407$ instead of 47.2109; excluded precompute defined differently by rung; two frozen targets at $n71$ and $n73$ about 0.0065–0.0075× as expensive as the rank-stage mean, each a $P \approx 0.006$–0.007 event); they are reported, not edited.

37.7 What the bounds leave

The three bounds agree. Generic and query models bound the label side of an algorithm and are silent on the side index calculus works on: free local work with coordinates. The SGGM is vacuous on the dense configurations that carry the cost, the augmented-oracle split moves every coordinate oracle outside Shoup's model, and the low-degree bound holds only for small multipliers, because free local interpolation breaks it otherwise. None of the three is evidence for or against index calculus below rho; each says where a lower bound would have to charge local work to say anything. The open leads in Table 37.1 are, in effect, attempts either to find such a charge or to find local work that a charge would not cover. The next chapter is the attempt with the strongest claim behind it.

38 The tensor-contraction route

§24.1 reviews the external claim in brief: the preprint of Akshima and coauthors, *A Subexponential Time Algorithm for Elliptic Curve Discrete Logarithm*, version 0.2 (Bochum/MPI, pinned at `1ee44e56`), claims prime-field ECDLP in $2^{O(b/(\log b)^{1/4})}$, $b = \lceil \log_2 p \rceil$, by hashed factor bases and exact Boolean tensor contraction. This chapter gives the full account of what the programme built and learned on that route: the dependence on the external theorem, the prototype and the native implementation, the challenge attempts, the twelve prime-field leads and their refutations, and the finite-size situation. The summary is one sentence. *Nothing on this route has a counted cost: no relation was found on ECC2K-130 or ECCp-131, no exponent moved, and the route's sub-exponential claim is conditional on an external result that is unverified.*

38.1 What the claim rests on

The Bochum preprint's Theorem 4 is conditional on its Theorem 2, which invokes Theorem D.1 of the OpenAI preprint *The Exponential Time Hypothesis is False* (17 October 2026, revised 25 October; pinned at `openai/math 7ef5c0cc`). D.1 contracts an s-edge bipartite Boolean tensor network, with degree at most $\lceil s^{9/10} \rceil$ and vertex tensors evaluated as black boxes, in $(s+L)^{O(1)} 2^{O(s/\sqrt{\log s})}$ time and polynomial space (`research/ic_prime_eth_leads_20261031/LEADS.md` §1).

The external result is unverified. The programme did not verify D.1, and as of the sources read here no one had. The lead note of 31 October records, from the public discussion it consulted, that no bug had been reported, that independent experts were examining the proof, that the Lean development in `openai/math` proves a weaker rate $(n(\log\log n)^2/\sqrt{\log n})$ and does not contain Theorem D.1, and that nobody had machine-checked that the formal statements match the paper; its status line is "claimed, under examination, no reported bug". The atlas's literature note on the preprint is tagged `unverified-proof` and states its proof status as "not checked here" (`crypto-autoresearcher: knowledge/literature/KN-LIT-caa51d.md`). The two in-session readings of D.1 and the D.4 recurrence described in §24.1 found no concrete gap; they are scoped readings, not a certification. Every statement in this chapter about the asymptotic behaviour of the route therefore inherits that condition, and a reader should treat the route's sub-exponential bound as conditional on an unverified theorem.

Where D.1 stops being inert. D.1's exponential term depends on the edge count only: vertices are black boxes, so a linear or low-rank vertex gets no discount, and edges are bits, so a p-ary digit network would need a p-adic analogue the paper does not supply (`research/ic_prime_eth_leads_20261031/LEADS.md` §1, facts 1–3). The recursion's admission condition $u_0 = \lambda/2^{17} \geq \lambda^{4/5}$ needs $\lambda \geq 2^{85}$, that is $s \geq 2^{2^{85}}$ edges at depth 0 (derived by a delegated audit; the Bochum paper quotes the weaker $2^{2^{17}}$). Below that threshold the algorithm is brute force over the network. The native implementation states the same fact in its own constants: the admission scale $u = \lfloor \lceil \log_2 s \rceil / 131072 \rfloor$ is zero for every practically addressable network, so the code always selects the manuscript's finite-size fallback (`docs/hashed-ecdl.md`).

38.2 The prototype

The first build (`research/btn_ecdl_20261029/README.md`, 29 October) was a Python prototype of the relation-finding step on the Koblitz curve $y^2 + xy = x^3 + 1$, with a plain bucket-elimination contractor standing in for the claimed algorithm. Its findings, all local and seeded:

- over $\mathbb{F}_{2^5}$ the network count of ordered pairs $A_1 + A_2 = R$ equals brute force (42 and 40 on two targets);
- over $\mathbb{F}_{2^7}$ the count is 112 against a brute-force 112, but plain elimination needs tables of 2^{34} entries (min-fill width 34, against 23 at $n = 5$), so the count conditions on the 7 bits of x_1 and takes 336 s; dropping the curve-membership predicate on P_2 as redundant gives 237 instead of 112, a recorded error;

Study	Recorded result
ECCp-131 input validation, 29 Oct	public parameters as runner input; 27 regression tests pass; the challenge command exits with `resource limit: AKS multiplicative order` before graph or relation search
Exact primality and functional witness, 30 Oct	AKS kept above `u64`, Pocklington proof nodes verified; ECCp-131 validation at 5,589 charged diagnostic steps; 34 tests; matched performance evaluation blocked
Checked hash prime, 30 Oct	a user-reported 54-minute set-up hang traced to the AKS branch above `u64`, not to Fourier-network construction; attribution corrected
Primality fast paths, 30 Oct	thirteen strong witnesses below 3317044064679887385961981, bounded Pocklington; original ECCp-131 input validates in 17,471 charged steps; 74 package tests; the challenge network constructs with 1,684,796 edges, 249,620 vertices, degree 1,032, 526 interface bits
CPU oracle panels, 30 Oct	two matched panels of 120 executions, byte-identical certificates; median 45.99 → 3.90 ms ($p = 65521$, $m = 3$, $\tau = 80$) and 80.54 → 4.12 ms ($m = 4$, $\tau = 64$); L0 on one EKS VM; the oracle is enumeration with cached MITM tables, not contraction
Counts-only protocol, 31 Oct	v2 supersedes v1, withdrawn before data; isolation forced from L2 to L0 because rho solves under 1 ms grade L1; no session has run
Entry-cost audit, 4 Nov	`PCC-20261104-1df881`, unverified: D.1 permits an exponential entry-work allowance; maps target-row budgets to entry traversal

Table 38.1: Engineering on the native hashed-ECDL implementation. All charged-step counts are uncalibrated admission diagnostics, not costs. Producer-reported. Sources: `docs/hashed-ecdl-eccp131-validation.md`; `research/hashed_ecdl_speed_20261029/`; `research/hashed_ecdl_hash_prime_20261030/`; `research/hashed_ecdl_primality_fast_paths_20261030/`; `research/hashed_ecdl_cpu_20261030/`; `research/hashed_ecdl_counts_20261031/`; `research/hashed_ecdl_entry_cost_20261104/`.

- a complete $m = 2$ index calculus over $\mathbb{F}_{2^5}$ ($r = 11$, $N = 16$) found 17 relations and recovered the planted logarithm; one contraction took about 340 s;
- on the ECC2K-130 field the addition network with two curve-membership predicates has 105,062 variables, 104,801 predicates and 524,008 edges; a witness from reference arithmetic satisfies all predicates and a one-bit flip violates one. Factor-base predicates are omitted at that size. This is an encoding check, not an attack.

The lead note reads the widths as 2^{5n}, five times the enumeration exponent. The README lists what the prototype does not show: the multiplier chain has $O(n^2)$ edges, the paper's $O(n)$ Fourier multiplier does not carry over to $\mathbb{F}_2[x]$ without a characteristic-2 variant, and the GPU contractor was never run on a GPU.

38.3 The native implementation and the challenge attempts

The Rust implementation of the complete algorithm, with the D.4 recursive backend, is in `crates/hashed-ecdl-tensor` and mirrored in the cryptanalysis repository (`docs/hashed-ecdl.md`). Its correctness record is in §24.1 (34 release tests, forced recursive and direct controls, a tiny complete relation pipeline). Table 38.1 lists the engineering studies that followed and what each did and did not establish.

No relation on either challenge. Two attempts at the published challenges stopped at resource limits. The ECCp-131 network compiles to 1,564,506 edges, 654,523 variables and degree 1,032 in the Python companion, which reaches its default $u_0 = 0$ exact-leaf budget without a witness; the native command first stopped in prime certification, and after the primality fixes constructs the network but has not produced a relation. The ECC2K-130 network has 2,694,680 edges and also exceeds the exact-leaf budget without a relation. The atlas records the state as "ECC2K-130 and ECCp-131 attempts exceed their exact-leaf budget without a relation; no runtime, S or S/rho figure" (`crypto-autoresearcher: knowledge/leads/ecdlp/exotic/LEAD-EXO-hashed-factor-base-tensor-ecdl.yaml`; `docs/hashed-ecdl.md`). The validation receipt states that the budget failures "do not prove nonexistence, a practical speedup, or a theorem obstruction". Both challenge scalars remain unresolved, and the independent correctness and complexity audit of the backend is the open problem `problems/hashed-ecdl-tensor-backend.md`.

The speed panels are not tensor measurements. The CPU panels of Table 38.1 are the one place on this route with matched timings. Their 11.8–19.6× median gain at $p = 65521$ is the cached meet-in-the-middle oracle against the original enumeration oracle, on three public known-answer fixtures; the README states that the panels do not establish a matched-rho advantage or "the paper's conditional tensor contraction hypothesis". The larger-field feasibility ladder expired at its scheduling deadline with no rows. The Rust port of the Go prototype and its ecbench arm `ic.tensor_contraction_v1` are described in §24.1; that arm has not been run.

38.4 Black-box balance

The lead note of 31 October (`research/ic_prime_eth_leads_20261031/LEADS.md` §§2, 4) puts the route in the cost model of the rest of this treatise. Let $s = b/m$ be the window, $|F| = 2^s$ and 2^{es} the per-attempt decomposition cost. Three statements follow, with Lean receipts in `lean/IcTensorRoute.lean` (standard axioms, no `sorry`) that cover the cost-model arithmetic only.

- *Ladder rule.* The relation phase $m!\,p^{(1+e)/m}$ is below $p^{1/2}$ iff $e < m/2 - 1$, the gate of §22.1.
- *Solver-saving floor.* An encoding has at least b interface bits; a Boolean solver costing $2^{n_{\text{eff}}(1-\delta)}$ meets the per-attempt budget only if $\delta \geq 1/2 + 1/m$. Tensor contraction under D.1 gives $\delta = 1/\sqrt{\log s}$, far below.
- *Black-box balance.* Letting m grow trades the $1/m$ of the budget against the m of the edge count; under the cost model $E(m) = b/m + cmb/g$ with $g = \sqrt{\log_2 b}$, the optimum is $m \approx (\log b)^{1/4}$, and a route below rho needs $g > 16c$, that is $\log_2 b > 256c^2$, with c the unspecified constant in D.1.

The note adds a concrete size: at $b = 256$ the balance gives $m = 2$, and the linear algebra alone costs $r^{3/2} \approx 2^{384}$ against rho's 2^{128}; beating rho on linear algebra needs $m > 6$, which the balance reaches only for b above about 2^{1296} (derived). The remaining levers it names are an improvement of the contractor's exponent on these particular networks, the open problem `problems/correlated-packet-common-kernel.md`, relation networks with $o(mb)$ edges, and structure that lets a solver skip a constant fraction of the interface.

38.5 Twelve prime-field leads and their refutations

The same note proposed twelve measurable leads, L01–L12, each with a decision rule. They were evaluated on 31 October and reviewed by two adversarial skeptics per lead; the skeptic digest is authoritative on standing, and the synthesis reconciles the two (`research/ic_prime_eth_leads_20261031/SYNTHESIS.md`; `research/ic_prime_eth_leads_20261031/leads/SKEPTIC_DIGEST.md`). The raw lead directories are on an external volume, and nothing in the synthesis was re-run. This section is the full account: it gives the standing per lead, what the leads measured and why six were refuted as stated; §23.6 summarises it for the prime-field chapter. The verdict of the synthesis is quoted: "Nothing moved the exponent question for prime-field ECDLP index calculus. Every result is a constant, a known negative, or an unfinished harness."

The errata of 2 November list the standing after review (`research/ic_prime_eth_leads_20261031/LEADS.md` §9): L03, L07 and L11 narrow, constant only; L04 and L09 contested and partial; *L01, L05, L06, L08, L10 and L12 refuted as stated*; L02 not run. For L05 the skeptic digest is more precise: what is refuted is the lead's claim to be *closed* ("refuted as closed", both skeptics); the harness and its numbers stand.

Table 38.2 gives the six refutations. The other five run leads carry numbers that are worth keeping as harness constants.

- *L03, sparse-parity bases.* Every hashed base (Bochum's affine hash, dense, sparse weight 3, 5, 7, biregular, window) matches a size-matched null to a paired ratio 1.000 ± 0.0003 at $p \approx 2^{14}$ to 2^{22}, $m = 2, 3$. Weight 3 loses about 10% at $m = 3$ from empty or heavy-tailed bases. This restates the conservation of Chapter 5: a size-matched null makes pairwise-independent hashes match by construction.
- *L04, backdoor census.* The offset $(m-1)s - B^*$ saturates at $2\log_2(\text{cap}) - 1$ (1, 3, 5 for caps 2, 4, 8) at $m = 3$, $s = 4..7$: the enumeration has moved inside the propagator. The supportable sentence is "no sub-$(m-1)s$ strong backdoor under cap-bounded window-branching propagation at $s \leq 7$".
- *L07, second moment.* Hashed bases are 1.000 ± 0.005 of the null; a shared negation-closed x-window gives 0.713/0.833/0.916 at $m = 4$, the symmetric-base mechanism already known.

- *L09, census tool.* Over 34 encodings the margin to D.1 admission is -16.11 to -16.33 bits of $\log_2 \lambda$; with D.1's constant set to 1, $s/\sqrt{\log_2 s}$ is 37× to 41,000× the enumeration exponent.
- *L11, multiplier encodings.* Propagation-complete against school multiplication: paired ratio 0.78 $[0.67, 0.92]$ at $b = 9$ only, 1.15× worse at $b = 6$, untested at $b \geq 12$.

The synthesis also records feasibility: $m = 3$ prime SAT is censored above $b \approx 12$ in every lead that tried it, so only the $b = 6$ and $b = 9$ diagonal cells are uncensored, and its closing recommendation is that the SAT and propagation lane for prime fields "should be closed for exponent purposes", with one item untested (a CaDiCaL external propagator with $\mathbb{F}_p$ Gaussian elimination).

38.6 Corrected sentences of the lead note

The lead note is left unchanged so that its original claims stay auditable; its §9 lists eight corrections and says that where the text conflicts, the errata win. Three of the corrected sentences concern the route itself.

- *"The Bochum route can never beat rho on any curve anyone will deploy, whatever D.1's constant is"* is wrong as stated. Lean proves $g > 16c$; the figure of more than 2^{256} bits needs $c \geq 1$, an outside assumption on an unspecified $O(\cdot)$; the general threshold is $\log_2 b > 256c^2$.
- *"By the paper's own gadget counts the network is at least* $\sim 10^5 b$ *edges per point addition"* is 9–30× high. Measured: 3,364 edges per slot-bit for the freed binary network, $1.17 \cdot 10^4$ for the unfreed prototype, and 11.9k edges per bit for the $m = 2$ prime network at $b = 131$. The exponent argument is unaffected.
- *The clustering loophole* (merging vertices to reduce s below the admission threshold) is closed: no row is D.1-admitted, hidden interface bits cost 2^h entry evaluations, $s_{\text{eff}} \geq (h/A)^{1.11}$, and the $b < 2^{36}$ closure is specific to $m = 3$ ($b < 2^{16}$ at $m = 4$); $m \geq 4$ is closed by `no_crossover`, not by the floor. The derivation, with heuristic steps labelled, is in `notes/clustering_closure.md`.

The other five corrections are the L10 Lean overclaims, path errors, the ladder feasibility limit, the L08 hand bound and the L08 margin; the last is in Table 38.2. The atlas carries the same corrections (Table B.8). The skeptics refuted all nine follow-up proposals as proposed, in one sentence each, and kept six cheap residues of them as harness constants; none touches an exponent.

38.7 Finite size

The finite-size review of 30 October (`research/hashed_ecdl_finite_size_20261030/README.md`) asked whether the hashed relation language can be useful below the recursion's onset. It argues that the onset is a guard in one algorithm, not a lower bound for solving these particular networks, and proposes five search strategies (listed in §24.1); nothing was implemented or measured under it. Its fourth route, replacing the global recursion schedule with checked local plans, is the closest to lowering the same recursion's onset and, in the review's words, "has the largest proof gap". The state assessment of the same day (`research/hashed_ecdl_leads_20261030/README.md`) put the priorities as an independent validation of the general contraction route and a complete-cost evaluation of the finite algorithms already delivered; it records the finite hashed track as implemented and development-tested (selectors, last-slot derivation, width and cut-set plans, partial relations, binary components) with "no admitted matched full-cost panel or evidence of an exponent improvement". The elimination campaign of 1 November and the rank-census design are in §24.1.

38.8 What the route has and has not shown

The route is the only claimed sub-exponential algorithm for prime-field ECDLP that the programme examined in detail, and it is external. What the programme established is:

- an independent native implementation of the complete algorithm, checked on small curves, with an exact contraction and certificate replay;
- that the literal constants select brute force on every network that can be written down, and that both published challenges exceed the exact budget without a relation;
- that under the lead note's cost model black-box use of D.1 needs $\log_2 b > 256c^2$ to cross rho, with Lean receipts for the arithmetic;

- twelve leads whose measured outcomes are constants or known negatives, six of them refuted as first stated.

What it has not established is any counted cost, any exponent, or the correctness of D.1. The route does not move a row of any table in this treatise. It remains open in the atlas with two next steps: the independent audit of correctness and complexity, and the frozen 84-case comparison panel, which is blocked on checkpoint transport and native execution access. If D.1 were withdrawn, the route's asymptotic claim would go with it and the implementation would remain an exact enumerator with a tensor interface.

Lead	As proposed or first read	After skeptic review
L01 char-2 gadget	a Schönhage-type gadget for $\mathbb{F}_{2^n}$; evaluator verdict "undecided" because $\ell = 7$ was censored	three finished $\ell = 7$ runs give slope about 2.58 bits of conflicts per window bit (reopen bar: CI below 2.5; ladder bar 0.5); the early ratio fall fits a constant padding floor of about 11k conflicts; CPU 3× worse at $\ell = 7$; only 4 of 11 products use the gadget
L05 policy pilot	closed after a pilot of an LLM policy-search harness	refuted as closed: renaming is a null experiment by construction in kissat; the ladder held $b = 8$ and varied ℓ, not e against s; 3 curves; the hooked CaDiCaL engine was never built; label "harness built, renaming null recorded, policy question untested"
L06 contraction width	closed: tree-decomposition width loses to enumeration	measured on a Python structural model, not a graph dump; dense gate-graph treewidth is the wrong proxy for D.4; the transform-word cut was effectively not run; surviving claim: plain tree-decomposition DP on the school CNF loses at $b \leq 16$ (lower bound 15–51 against 2–10.7)
L08 Koblitz transfer	char-2 gadget no worse than the integer one, margin 3.42×	that compared a freed binary gadget with an unfreed integer one; the skeptic's freed 133-bit integer multiplier has 14,618 edges instead of 37,030, so the margin is about 1.4×; hand bound $80d + 228\sqrt{d}$ fails at $d = 46$; supported bound about $175d$ for $d \leq 5000$
L10 Lean filings	"done; file as known results"	seven gaps between the lead text and the Lean statements, all confirmed: the cost model is an unproved hypothesis, `no_crossover` needs $c \geq 1$, the interface bound is an injectivity count; not fileable as drafted
L12 clause retention	incremental solving amortises learned clauses, $1/R = 40\times$ at $m = 2$, $b = 12$	the retained solver fills an $\lvert F\rvert^{m-1}$ table whose set-up is never charged against the explicit enumeration reference; fresh baseline censored at $m = 3$; target lists 48% duplicated; class retained set-up/advice, closed

Table 38.2: The six leads refuted as stated. Producer-reported, from the skeptic reconciliation; no lead has a validation record. Source: `research/ic_prime_eth_leads_20261031/SYNTHESIS.md` §3; `research/ic_prime_eth_leads_20261031/LEADS.md` §9.

39 Learning, pseudorandomness and curve models

This chapter collects three kinds of unconventional work that share one methodological lesson: an instrument that sees nothing excludes only what it could have seen. The first is learning inside the decomposition oracle, where a preregistered study measured the most a learned component could gain and found the gain unreachable. The second is a set of tests of whether the factor base looks random in discrete-logarithm order, and a calibration that showed which known-weak curves those tests can and cannot see. The third is the remainder of the lifts, transports and curve models not already in Chapter 8.

39.1 The ceiling for a learned decomposition oracle

Chapter 16 records that SAT, SMT and WDSat are enumerators on binary point decomposition. A study of 27 October (`research/pdp_ml_sat_ceiling_20261027/README.md`; protocol frozen at `8a85584fd` with one amendment before any result) asked the question a learned component would have to answer before anyone builds a graph network or an RL branching policy: how much could any learned component gain inside a SAT point-decomposition oracle, and does any cheap signal reach it? The setting is the symmetrised S_4 at $m = 3$, a native CDCL+XOR solver, the curves of the Trimoska corpus [39] and uniform targets, in two families: $n = 15$, $\ell = 5$ (3,000 targets) and $n = 17$, $\ell = 6$ (1,200 targets). Cost is in solver conflicts. Three levers were tested: triage (finish only the solves that score high), a fixed conflict cutoff, and branching order.

Table 39.1 gives the numbers. All 4,200 labels agree with exhaustive sorted-triple search. A refutation costs about one conflict per sorted factor-base triple, and every refutation lands within 4.1% of its family median, so the solver is an enumerator. The prize is real: a perfect UNSAT predictor, charged its telemetry conflicts, would raise relations per conflict by 5.2–9.0×. Nothing reaches it. Test AUC sits within the noise of 0.5 for every feature group (bits of x_r, their pairwise products, solver telemetry after 64 or 512 conflicts), on both families and both splits; the positive control at 0.72–0.79 shows that the learner works. The one interval excluding 0.5, telemetry at 512 conflicts on $n15\ell5$ at 0.556 $[0.518, 0.599]$, includes the "decided within 512 conflicts" flag and gains 0.99×. Train-fitted gains of up to 4.7× fall to about 1× held out, and telemetry trained at $n = 15$ scores 0.50 and 0.52 at $n = 17$. A cutoff gains nothing because a witness found early is no more likely than one found late. Default VSIDS is at the floor: an order learned from VSIDS activity on training refutations is at least 10× worse on every held-out target, because the variables that take part in conflicts are not the ones worth deciding on.

The study closed under its preregistered rule (no lever reached 1.5× with a lower bound above 1.2× on both families; the largest held-out lower bound was 0.92). Its prediction P4, that every branching arm would stay within 0.7–1.4× of the default, failed. The reading for index calculus is the product law the README states: even a perfect classifier is capped by relations × targets × oracle $\approx m \cdot 2^n$, so it removes a constant from the oracle (measured here at 5–9×) and cannot change the exponent. The run also found and fixed a soundness bug in the native solver: resuming after a budget stop answered wrongly in both directions; an audit found no published artefact that re-solved without a reset.

39.2 The ML dataset campaign

From 28 October to 1 November the programme built a native `crypto dataset` generator and custody pipeline for a corpus on which to train models of ECDLP and point-decomposition behaviour: Koblitz, ordinary binary and prime curves with paired twists, arities $m = 2$ to 15, bit sizes 23–31 and 47–61, multiple generators and keys, and every supported native method (`docs/ml-dataset/DESIGN.md`; more than forty `research/ml_dataset_*` directories). The design's planning arithmetic is 2,016 parameter cells and, at 16 curve seeds, 8 generators and 64 keys, 16,515,072 tasks per backend configuration; the stratified design keeps 165,150,720 method cells at full width. These are planned cells, not data.

Scope. The campaign inventory of 1 November (`research/ml_dataset_campaign_inventory_20261101/README.md`) records 124 selected terminal method cells in seven cohorts, under different protocols,

Quantity	$n15\ell5$	$n17\ell6$
UNSAT conflicts, median [IQR]	5781 [5738, 5819]	45649 [45539, 45781]
conflicts per sorted factor-base triple	0.963	0.998
perfect-predictor ceiling, relations per conflict	5.21–9.01×	5.75–6.69×
test AUC, all feature groups	0.485–0.556	0.461–0.513
permuted-label AUC	0.478–0.540	0.449–0.517
positive-control AUC (label flipped w.p. 0.25)	0.747, 0.785	0.717, 0.725
fixed-cutoff test gain	0.93, 1.05	0.91, 0.98
learned-activity branching order, conflicts / default	≥ 10.03	≥ 10.00

Table 39.1: The ML-aided SAT ceiling for point decomposition (symmetrised S_4, $m = 3$, binary toy families). Ranges are over the two splits and four feature groups. Producer-reported; toy stage diagnostic, no validation record. Source: `research/pdp_ml_sat_ceiling_20261027/README.md`.

seeds, methods and caps, and states that the count "is a coverage inventory, not a pooled estimate or a training set". Its labels are mostly method completion, bounded candidate counts and censored statuses. Examples: Koblitz $n = 23$, $m = 3$–15, 26 cells with 4 finite-domain negatives and 22 timeouts; Koblitz $n = 47$, $m = 4$, 16 verified method artifacts; $p = 2^{61} - 1$, $m = 2$, 2 verified enumerations and 2 root timeouts; $n = 26$–31, $m = 2$, 36 cells, 35 known-answer-replayed. A separate 252-cell compatibility matrix is development-only. The one model study (`research/ecdlp_dataset_models_20261102/REPORT.typ`) is a development pilot that by its own protocol may not select a winner or estimate generalisation: 960 cells, 819 training rows, and an arm that nearly fixes the label (`enumerate` completes on all 240 rows and recovers the scalar on none; `bsgs` solves 237 of 240). Only the double-large-prime arm shows signal, a Brier score of 0.050 against 0.192 for the majority baseline, which the permutation control removes (0.168). It ran in the session container, not through the experiment route.

Relevance to index calculus. The campaign has no model result, and it bears on index calculus only through the ceiling of §39.1: a learned predictor of decomposition success can at best remove the constant measured there. What it does contribute is a label discipline that the rest of this treatise also follows (`docs/ml-dataset/DESIGN.md`, "Scientific labels"):

- target creation by $Q = [k]G$ is not an independently solved ECDLP, and a verified decomposition is not an ECDLP solve;
- the first degree drop in one algorithm's trace is not the first fall degree, and the maximum degree visited is a solving-degree observation, not the last fall degree (Chapter 6);
- censored outcomes (timeouts, budget exhaustion, unsupported cells) are kept, not dropped;
- splits group all keys and generators on one curve, both twists, isomorphic and isogenous models and Frobenius-equivalent tasks into one component; random row splits are inadmissible.

39.3 Other learned components

Four smaller studies tested learned components directly; none improved on its incumbent.

- *Learned WDSat branching* (`cryptanalysis:experiments/ideas_learning_20261018/REPORT.md`, item 8). 88 solver calls with upstream WDSat `61c6ff3f`: on eight held-out targets the paper default with symmetry took 3,092 conflicts, and the block, interleaved, random and learned static orders without symmetry took 16,097, 16,785, 16,676 and 16,097. The learned policy selected the ordinary ascending order, so it *is* the block arm. The 16,097-against-3,092 gap is the symmetry switch, which the controls disable because the upstream symmetry code interprets fixed positions in the order; it is not a learned-policy effect (Table B.8). The atlas still quotes the comparison as "the only learned-branching negative"; read correctly, it shows that the learned order did not improve the ordinary order.
- *Learned pivot selection and program search* (`research/ideas_execution_20261018/EXECUTION.md`, items 9–10). A native learned F4 pivot selector returned the constant minimum-weight rule. A restricted search over group-law query kernels re-selected the incumbent pair-hash kernel; both took 820 counted units on the held-out curve.
- *Learned prefilter* (item 11). On 7,254 exact subgroup targets in 27 cells, the one-class prefilter rejected genuine decompositions and was disabled.

- *Degree surrogate* (item 12). Fitted to 1,718 archived degree-profile cells, the curve-held-out surrogate has degree MAE 0.10041, but with three calibration curves the honest 90% group-conformal interval is unbounded, so it cannot price a future run.

A reinforcement-learning pair selector for Buchberger's algorithm (`research/gbrl/PAPER.md`, undated, a paper-style self-report with no frozen protocol or ecbench record) reports -5.5% S-polynomial steps on cyclic-6 and, on held-out Semaev $m = 2$ systems, -3.9% steps with $+5.4\%$ arithmetic operations; at $m = 3$ it fails to generalise to a factor base of size 6 (496 steps against sugar's 459). It works in Buchberger only, not F4. The L05 pilot of Chapter 38 closed LLM policy search on variable renaming. The atlas's next step for the whole learning lead is a learnability control: predict a log bit from coordinates with a planted homomorphic leak as positive control and a random-relabelling null, at N up to 2^{24}.

39.4 Pseudorandomness tests

The question behind this line is whether a natural factor base, ordered by discrete logarithm, looks like a random set. Any detectable structure would be a lever: a relation yield above the counting heuristic, or a statistic that leaks log bits. The linear-phase test, H-PSEUDO, is told in Chapter 35, with its 4 November correction: the small-x base is negation-closed and the original null was not, and against a negation-closed null R is consistent with 1 (Table B.1). Three further instruments followed, each under a protocol frozen before data.

- *Quadratic phases* (QUADPHASE, `crypto-autoresearcher:docs/reports/quadphase-20261104/README.md`). The statistic is the maximum over (α, β) of $|\sum_{l \in S} e((\alpha l^2 + \beta l)/N)|/\sqrt{|S|}$ for the logs S of the first 512 points in (x, y) order, against 40 negation-closed random sets per curve. Verdict NULL: 9 of 9 decisive curves at $N \approx 2^{12}$–2^{14} inside the null's band (u from 0.125 to 0.85); planted control 3.8–5.2× the null maximum at the planted α_0; A/A difference at most 2.3% pooled. An independent validation recomputed all 385 rows with an FFT-free implementation to within $7 \cdot 10^{-15}$. The validator's probe says the test detects coherent single-phase leakage of about 0.3 of the set or more; weaker or multi-peak structure is untested. One descriptive curve at 2^{16} sat above all 5 of its null draws; a rerun against 40 draws (`crypto-autoresearcher:docs/reports/quadphase-2p16-20261104/README.md`) put it at the median, $u = 0.50$ (null mean 4.750, s.d. 0.229, maximum 5.581).
- *Digit-level tensor structure* (HOLANT-STEPB-QTT, `crypto-autoresearcher:docs/reports/holant-step-b-qtt-20261104/README.md`). The holant lead asked whether a basis change on the $\mathbb{F}_p$ bond alphabet makes the decomposition-count network cheap. Flattening and CP ranks are invariant under invertible bond-wise maps (checked at $p = 61$), so the test was reformulated as the quantised tensor-train rank of the transfer matrix M under binary digit splitting. NULL: from $L = 8$ the natural x-labelling and the random relabellings both sit at the rank ceiling 2^L (1024 at $p = 1021$), while the discrete-log labelling, the positive control, has rank 109–121. A low score on one anomalous curve at $p = 61$ was followed up on 63 anomalous and 63 ordinary curves at $37 \le p \le 127$ (`crypto-autoresearcher:docs/reports/anomalous-qtt-20261104/README.md`): NULL, permutation $p = 0.831$, the anomalous curves slightly above the ordinary ones.
- *Coordinate statistics.* The census that found every structured additive statistic PGL_2-unstable and every stable multiplicative coset statistic at the random level (`EV-ECDLP-81f4d4`) was lowered from strong to preliminary by `CORR-20261026-c94668` (Table B.7). The record's next step is the one natural family the census never measured, digit characters.

Together with H-PSEUDO, these say that at toy size the small-x factor base in log order looks like a random negation-closed set under linear and coherent quadratic phases and under digit-level tensor rank. They say it only for structure of each statistic's type, and the next section shows how narrow that is.

39.5 Calibration against known-weak curves

The weak-class calibration of 4 November (`crypto-autoresearcher:docs/reports/weakclass-20261104/README.md`, WEAKCLASS-20261104) asked whether these instruments see curves whose discrete logarithm is *known* to be easy: anomalous curves ($n = p$, Smart's attack), small embedding degree ($\mathrm{ord}_n(p) = 3$–6, MOV), and $j = 0$ curves with a factor base of whole ζ-orbits, each with a paired ordinary curve over the same prime, with p of 10, 12 and 14 bits and 3 weak curves per class per size. A cell detects only if all 3 class curves fall beyond the null's 5–95% band in the same direction at two sizes and the paired ordinary curves do not.

Instrument and run	anomalous	small k	$j = 0$, ζ-stable base
I1 character sum, WEAK-CLASS	invalid ($\bar{u} = 0.61$)	invalid (0.41)	invalid (0.12)
I1 rerun, calibrated A/A	blind (0.46)	blind (0.44)	blind (0.28)
I1, 36 curves per class	not detected (0.483, $p = 0.86$)	not detected (0.476, $p = 0.82$)	detected (0.233, $p < 10^{-5}$)
I2 $m = 2$ sumset yield	blind (0.56)	blind (0.44)	blind (0.64)
I3 QTT rank, $p \approx 2^{10}$	blind (0.36)	blind (0.44)	blind (0.60)

Table 39.2: Which instrument sees which known-weak class. Entries are the verdict and mean percentile $\bar{u}$ within the instrument's own null. All producer-reported; the third row's study names an independent validation that replayed all 324 rows. Sources: `crypto-autoresearcher:docs/reports/weakclass-20261104/README.md`; `crypto-autoresearcher:docs/reports/weakclass-i1-20261104/README.md`; `crypto-autoresearcher:docs/reports/paired-anomaly-20261104/README.md`.

Table 39.2 gives the outcome. I2 and I3 are blind to all three classes while their positive controls sit at $u = 0.00$ on every curve, so the instruments work and simply do not see Smart- or MOV-weak curves. I1 came out *invalid* in the first run: its preregistered A/A gate required even and odd null means to agree within 5% on every curve, while with 20+20 draws the sampling error alone is about 2%, and 5 of 54 curves exceeded it. The study attributes the fault to the gate, not the instrument (Table B.7). The rerun (`crypto-autoresearcher:docs/reports/weakclass-i1-20261104/README.md`) used 200 null draws per curve in two streams compared by a KS test (7 curves rejected against an allowance of 14) and found I1 valid and blind to all three classes. It also found a post-hoc anomaly: the paired ordinary curves sat low (mean $u = 0.276$, KS $p = 1.4 \cdot 10^{-4}$), which put the pairing baseline in doubt. A preregistered replication (`crypto-autoresearcher:docs/reports/paired-anomaly-20261104/README.md`, 108 paired and 108 independent curves) did not replicate it (Mann–Whitney $z = 0.45$, $p = 0.65$), and with 36 curves per class it gave the one detection in the table: the ζ-stable factor base of the $j = 0$ class sits low at every size (mean $u = 0.233$, $z = -4.79$). The study explains why: a log set closed under multiplication by the eigenvalue λ has fewer independent frequencies, so its character-sum maximum is lower. That reflects how the factor base was built, not how easy the discrete logarithm is.

The consequence is a scope note on every null above. With 36 curves per class at $p \le 2^{14}$, I1 shows no detected shift for anomalous or small-embedding-degree curves, with a minimum detectable shift of about 0.13 in u at 50% power; I2 and I3 are blind at 3 curves per class, I3 only at $p \approx 2^{10}$ where it is rank-saturated. So the instruments that found the small-x base random in log order would also have found it random on curves where the discrete logarithm is polynomial-time or sub-exponential. A NULL from them excludes structure of the statistic's own type and says nothing about whether a curve's discrete logarithm is easy. The methodology lead's next step, injecting planted signals into the real pipeline to measure each screen's false-negative rate, has not run.

39.6 Lifts, transports and curve models

Chapter 8 records the transport, lifting and twist obstructions, and its xedni section is now read as conditional; the atlas reopened the global-lift leads accordingly, with short Mordell–Weil relations found on function-field surfaces at $p \le 31$ but no decomposition step. The remaining lift and model records are short.

- *p-adic jets and canonical lifts.* For $\gcd(n, p) = 1$ every additive order-ℓ image of a jet or formal-group target is zero (an independently verified scoped gate). The digit identity $d(m\hat{S}) = m\, d(\hat{S})$ holds exactly, 1360 of 1360, but only on the hard global lift, whose cost is the lift itself; the computable Teichmüller contrast reads as chance, and a valuation profile collapses to the $\pm$ fibre indicator (`KN-FIND-85845f`, `KN-FIND-e8cfdc`, `KN-FIND-1ba0fe`). The September statistic is in Chapter 35.
- *Elliptic nets.* Somos identities hold for every k, so net relations encode the group law on one fibre and not k. On toy primes $p \in \{101, 431, 1601\}$ the Somos sieve's charged k-recovery exponent is 1.97–2.80

against BSGS's 1.17–1.32 at identical accounting, and the sieve-to-BSGS operation ratio grows from about 24× to 135× (`EV-NET-001`, preliminary).
- *Curve models.* Model-native solving degree is the same for Weierstrass Semaev and twisted Edwards systems, $d_{\mathrm{reg}} = 2$ at $m = 2$ and 3, with Edwards a constant factor larger (`EV-REP-001`, `EV-REP-002`, replicated, toy). The EcGFp5 representation audit and its correction `CORR-20261102-7e3a91` are in §24.2.
- *Split Jacobians.* Embedding E as a Poincaré factor of a genus-2 Jacobian cannot beat rho on the elliptic factor because the Hasse–Weil order inflation of the ambient Jacobian outweighs the genus-2 index-calculus saving (`KN-FIND-61347e`); the projected-smoothness counting gate finds no route with $\lambda, \mu \leq 0.45$.
- *Torsors and deck orbits.* For a generically Galois cover every rational deck invariant factors through the quotient, and a connected nontrivial cover has no rational section, so a target-compatible selector is a missing section (independently verified scoped gate).
- *Miller cocycles.* When $\gcd(N, p-1) = 1$ the N-th root of a Miller function is a canonical primitive computable from the point; whether it exposes k cheaply is an unreviewed proposal with no measurement.
- *Lattès dynamics.* Every forward iterate on the x-line is a group operation, so forward dynamics is generic, and orbit quotients give $\sqrt{|\mathrm{Aut}|}$ for units and nothing for non-units; two accumulator closures are recorded (`EV-ECDLP-0c0f51`, `EV-ECDLP-a8af77`).
- *Lossy tracked objects.* An orbit argument shows that no lossy object propagates deterministically under the full translation action, a rigidity theorem covers single-point objects, and the one measurement, at $\mathbb{F}_{17}$, is mixed (`EV-ECDLP-ce9b50`).

Each record is in its YAML file under `crypto-autoresearcher:knowledge/leads/ecdlp/exotic/`; none has a cost against rho.

39.7 Out of scope

Two lines of work in the same repositories are not classical ECDLP index calculus and are excluded from this treatise's accounting.

- *Quantum resources.* The reversible secp256k1 point-addition circuit for Shor's algorithm records a best score of 2,490,847,962 (average Toffoli count 1,736,993 times 1,434 qubits), −0.124% against its head, valid on a 9,024-shot sample island only (`crypto-autoresearcher:knowledge/leads/ecdlp/exotic/LEAD-EXO-quantum-ecdlp-resources.yaml`). It is a resource estimate for a quantum attack, not a classical cost.
- *Eigen-Séta* is an isogeny-based public-key encryption scheme (`research/eigen_seta_assessment_20261017/REPORT.md` and related directories), not an ECDLP method; its corrections are listed in Table B.8 for completeness. The neural distinguishers on Speck and AES (`research/neural_distinguishers_20261102/`) are symmetric cryptanalysis.

39.8 What the three chapters settle

None of the unconventional routes produced a counted cost of a discrete logarithm, so none changes the distance to rho of Chapter 40. What they settle is narrower and mostly methodological.

- Generic and query bounds do not constrain index calculus, because they leave local work with coordinates free: the SGGM is vacuous on 122 of 208 configurations, every coordinate oracle is outside Shoup's model, and the low-degree bound holds only for small multipliers (Chapter 37).
- The tensor route rests on an unverified external theorem, is brute force at every realisable size, and found no relation on ECC2K-130 or ECCp-131; six of its twelve prime-field leads were refuted as stated and none moved an exponent (Chapter 38).
- A perfect learned predictor would buy 5–9× in a SAT decomposition oracle at toy size and nothing in the exponent; no feature tested reaches it (§39.1).
- The pseudorandomness instruments find the small-x base random in log order at toy size, and they are blind to anomalous and small-embedding-degree curves; the only class they detect is a factor base built with a symmetry (§39.5).

The corrections these chapters state are: the learned-WDSat comparison is confounded with the symmetry switch; the I1 instrument was invalid in its first run because of a miscalibrated A/A gate; the paired-curve anomaly did not replicate; L01, L05, L06, L08, L10 and L12 were refuted as stated; three

sentences of the tensor lead note were corrected; the hashed-ECDL set-up hang was misattributed; and the low-degree protocol's own reading of a toy-size pass was narrowed by its validator. The open items are the independent audit of the tensor backend and D.1, a counted cost for any tensor arm, a lower bound that charges local work, the learnability control with a planted leak, and the false-negative rate of the structure screens.

40 The distance to rho, family by family

This chapter gives, for each curve family, the current distance between index calculus and a matched rho. Each figure carries its reference, its boundary and its validation status. Here a ratio is IC over rho, so greater than 1 means rho is cheaper. Chapter 36 gives the frontier campaign, the improvement ladders of 14–24 October, and the three frontiers (cold, online, stage). This chapter adds the work of 25 October to 3 November and states the position as it stands at the end of the programme. Under the proof-carrying-claims rule, every leaderboard row named here has a `pcc` measurement claim, and every such claim has standing *unverified* (`pcc.py standing`, 6 November); every figure without a claim id is a producer-reported value (an unverified assertion), whatever its validation-gate status.

40.1 Koblitz curves

The leaderboard's Table A measures a Koblitz descent: aimed $m = 3$ collection, then an $m = 2$ or $m = 3$ descent. It is measured cold on one unseen public point per process, with 64 points per size, at six sizes from $2^{36.6}$ to $2^{47.2}$. Figure 40.2 shows the rule at three baselines, and Table 40.1 gives its ratios. From v3 to v7 the rho stayed the same and index calculus became 1.52–1.89× faster; the board carries the v7 figures, validated (§41.2). In v8, rho's signed-Frobenius canonicalisation, batched 16 lanes at a time in the same binary, raised the gap by 1.30–1.58× at every size. All six v8 values lie inside ranges declared before the rows were taken, and the 408 rows of each run all pass the `vs_rho` check (`research/ic_tool_program/rule/v8/README.md`). The v3→v7 improvement lowered the constant, not the growth.

The priced campaign. The closest Koblitz arm, compact Frobenius orbits, is 3.8–15× signed-Frobenius rho at $w = 0$. Fully priced against fully priced rho, its best cell is 1.566 $[0.992, 2.347]$, and the ratio grows as $r^{0.165\,[0.117,0.213]}$. A later fully priced component on the same kind of cell, the lazy orbit-member index, is closer: 1.27 $[0.77, 1.80]$ over signed-Frobenius rho at $n = 29$, $r = 2^{15.4}$ (`research/ecbench_lazy_orbit_members_20261018/`; a component, not a board row, not validated). No point estimate is below 1; at $r = 2^{15.4}$ the intervals include 1.

$n = 83$. The cold pipeline uses the base K1107/root11, which confirmed a saving of 1.195 s $[0.957, 1.433]$ on a 115 s process (16 public points, GitHub Tier 1 host). The last cold ratio is the K1120 holdout's 55.4, from five points: cold IC over the same-point plain-rho online wall (rho has no set-up, so its online wall is its cold cost; Chapter 15 admits no cold ratio on a stricter reading). A matched precomputed-rho reproduction on 28 October gave the median online IC over rho of Table 40.2, for three ways of matching rho's precomputation. Cold, IC took 155.5 s against 36.96 s for plain rho over all 16 targets. In operation counts the order flips across the pricing band, so no ranking is made.

40.2 Prime curves

Generic track. The prime-field programme's owner decisions of 29 October split prime-field index calculus into two counted tracks (`research/prime_ic_program_20261029/DECISIONS.md`):

- a *generic* track, with table, pair-table and root-finding oracles, labelled "bounded by $\sqrt{r}$; cannot beat rho";
- an *algebraic* track, whose prime baseline is empty.

Table 40.3 gives the distances on the generic track. The campaign pair table's exponent against rho's is the $r^{2/3}$ against $r^{1/2}$ of Chapter 2. The k-sum oracle (`research/prime_ksum_table_20261028/`) is a negation-folded table of every signed 4-sum of a smaller base. It lowers mean S at $w = 0$ by 1.9× at 20 bits, rising to 7.2× at 40 bits (83.7 → 11.6), and its fitted exponent falls from 0.620 $[0.586, 0.650]$ to the tabulated value, against the theory value $k/(2k-1) = 4/7$. 2,016 runs were verified and replayed; the source notes that the method is generic, so its limit is baby-step giant-step. The new 20-bit baseline (`research/prime_base_oracle_tuning_20261029/`) raises the factor base from 32 to 45, which gives 0.818 $[0.773, 0.866]$ of the incumbent at $w = 0$; a second host (L2) replayed all 1,008 records identically and gave 0.836 $[0.806, 0.866]$. The same rule at 24 and 28 bits was a relabelling, and at 32 bits null.

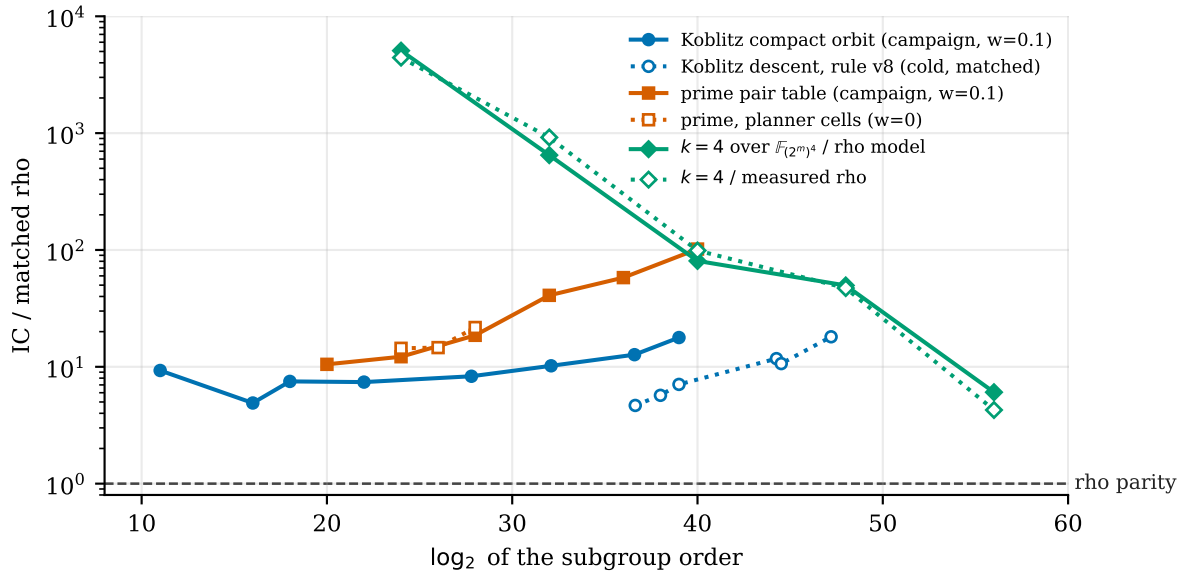


Figure 40.1: Distance to matched rho by family and size. Koblitz and prime curves: the frontier campaign's priced arms ($w = 0.1$, 16 targets × 2 rounds); open markers are the Koblitz rule at v8 (cold, 64 targets) and the prime planner cells ($w = 0$). $k = 4$ over $\mathbb{F}_{(2^m)^4}$: the characteristic-two ladder against the negation-map rho model (filled) and a measured rho (open); the $n \approx 2^{56}$ point is one target. Curves and units differ between families, so the lines are not ranked against each other. Sources: `figures/data-ledger.json`.

Algebraic track. The Kummer tower is 29–159× rho at $w = 0$ (29.4 at 24 bits, `research/pkm_tower_ecbench_20261013/README.md`; 158.9 [121.6, 221.4] at 30 bits). No algebraic prime baseline exists. Its solving degree is the subject of §6.5.

40.3 Binary curves of prime degree

Table B's random-binary rows are 10.6–38.2× matched rho at $\mathbb{F}_{2^{15}}$–$\mathbb{F}_{2^{27}}$, and they are validated. A cofactor-projected base cuts three of them by 1.29–3.36×, leaving 6.1–14.4×. That base is not yet a board row. For Certicom ECC2-131, the $m = 3$ floor with a free oracle is already above rho (Chapter 16).

40.4 Weak extension-field families

Only in these families (Figure 40.3, Table 40.4) does index calculus approach rho or go below it. The genus-3 hyperelliptic row and the Joux–Vitse sieve row are the programme's preregistered decisions of IC below the strongest rho, and the sieve row is the only one validated (Chapter 26, §30.4). It has no pcc claim, so its statement is an unverified assertion under the proof-carrying-claims rule. The plane-quartic row uses double large primes, structured elimination and Lanczos; its 501 IC units and 2,442 rho runs are all verified, and it is a measurement of a known result (Diem, Diem–Thomé).

40.5 Summary

On every prime-order curve of cryptographic shape that the programme measured (at the toy and medium sizes measured, $\leq 2^{47}$), rho is cheaper. Index calculus goes below rho only in weak extension-field families, which is where theory said it would. There the programme's contribution is matched, priced, verified measurement of known algorithms.

Rule	Rho	IC/rho	Slope	Status
v3	same as v7	5.23–20.97	0.176	
v7	same as v3	3.43–11.5		on the board; validated; claims `PCC-20261104-24b9af` and five more, unverified
v8	matched, same binary	4.67 [4.34, 5.07], 5.70, 7.07, 11.75, 10.65, 18.07 [15.98, 20.58]	0.168 [0.128, 0.208]	inside pre-declared ranges; `VALIDATION.json` present (replay and validator, `research/ic_tool_program/rule/v8/VALIDATION.json`); not on the board, which still carries v7; no pcc claim
v8	canonical step (model)	6.45–23.32		in v8's unit
v4	canonical step (model)	9.2–42.1		§36.5

Table 40.1: The Koblitz rule (leaderboard Table A) at each baseline. IC/rho: cold IC over cold rho on the same unseen public point, one point per process, 64 points per size; a range runs over the six sizes $2^{36.6}$ to $2^{47.2}$, and the v8 row lists all six values in the order the source gives them. Slope: of the gap in $\ln r$. The canonical-step rows are a model of rho, not a measured rho: rho priced at the canonical batched step. Source: `research/ic_tool_program/rule/v8/README.md`.

Rho precomputation	IC/rho	Interval
650 s table walks	40.9	[7.58, 141.9]
walks matched to IC set-up	4.10	[1.58, 24.8]
table matched to IC memory	33.1	[6.60, 74.4]

Table 40.2: Koblitz $n = 83$, base K1107/root11: median online IC over a precomputed rho, by how rho's precomputation is matched; matched reproduction of 28 October. L0 shared VM, one run, no replay certificate or validator report; public reproduction panel, not a holdout; producer-reported. Source: `research/n83_k1107_bl_reproduction_20261028/RESULT.md`.

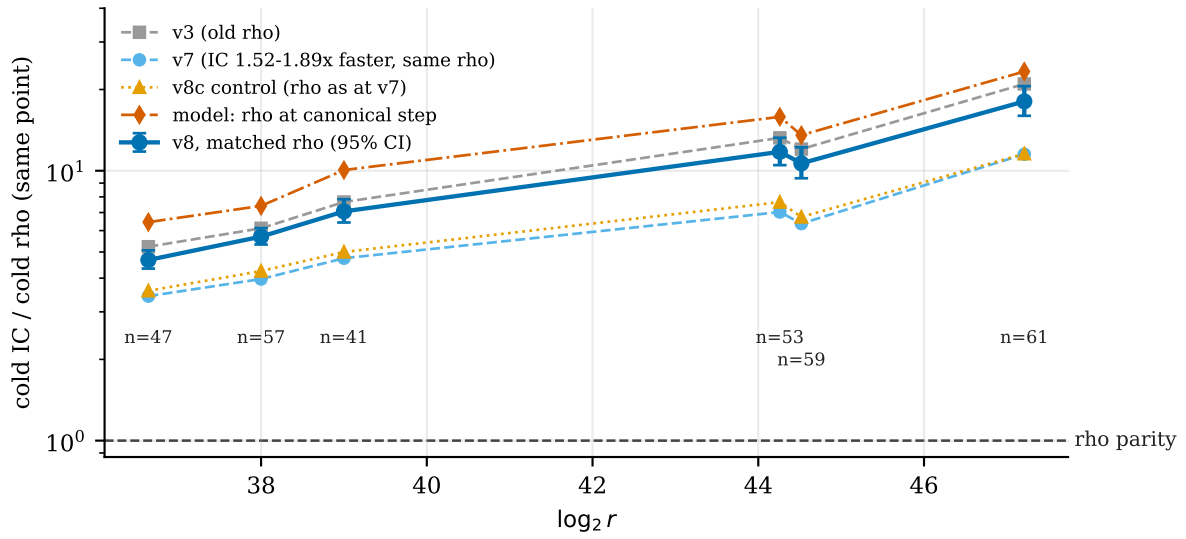


Figure 40.2: The Koblitz rule at three baselines: cold IC over cold rho on the same public point, 64 targets per size. From v3 to v7 the IC improved against the same rho. v8 runs the same IC against a matched, faster rho; its same-binary control, v8c, reproduces v7. The dash-dot line prices rho at the canonical step. Source: `research/ic_tool_program/rule/v8/README.md`.

Arm	Bits	IC/rho	Exponent	Status
Residual collector `res1eq`	36, 37	1.54, 1.64	rho's	`VALIDATION.json`; not an admitted row
Residual collector `res2`	36	2.59	rho's	improvement-log row; no validator
Table B prime rows	10–24	5.47–13.5		validated; claims unverified
Campaign pair table	20	9.8 ($w = 0.1$: 10.5)	0.620 $[0.557, 0.663]$	
	40	95.2 ($w = 0.1$: 101.8)		
k-sum table (28 October)	20	5.3	0.521 $[0.506, 0.538]$	L0; no validation
	40	13.1		
New 20-bit baseline	20	7.75 $[6.92, 8.70]$		validated component; incumbent 9.47 in the same session (board row 8.30)
Planner cells	23.5–29.6	12.6–25.8		supplementary cells

Table 40.3: Generic-track prime-field index calculus. IC/rho: cost over rho, as each source states it (Table B against matched negation rho; the new baseline against negation rho), at $w = 0$ unless marked; greater than 1 means rho is cheaper. Exponent: fitted cost exponent in r at $w = 0$; the campaign's rho with negation fits 0.435; "rho's" marks an arm at rho's exponent. The text's 0.620 $[0.586, 0.650]$ is the k-sum study's refit of the same pair-table arm. All rows except Table B carry no pcc claim and are producer-reported. The full prime-field list is Table 22.1. Sources: leaderboard Table B (claims `PCC-20261104-f4bc7f`, `-a93a81` and six more); `benchmarks/20261101-ic-residual-dp-eq/VALIDATION.json`; `research/ic_planner_exhaustive_20261026/README.md`; the campaign (§36.1); `research/prime_ksum_table_20261028/`; `research/prime_base_oracle_tuning_20261029/`; `research/ic_planner_validation_20261020/README.md`.

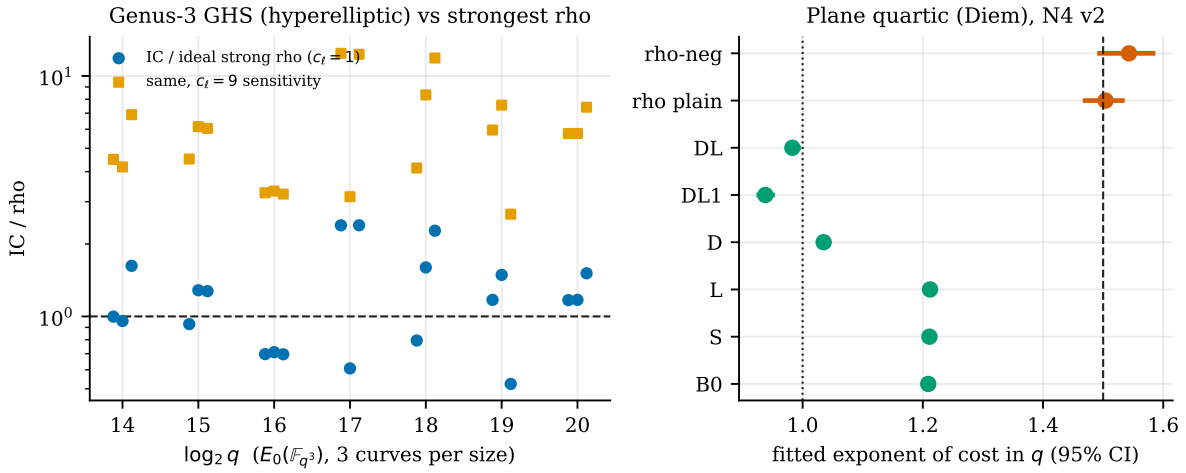


Figure 40.3: Weak families. *Left:* genus-3 GHS index calculus (hyperelliptic, double large primes, round 4) over the ideal strong rho on 21 instances of $E_0(\mathbb{F}_{q^3})$, in round 4's unit. Blue: one subgroup multiplication priced at one $\mathbb{F}_q$ multiplication, the registered unit. Orange: priced at nine, the protocol's sensitivity. Below the dashed line IC is cheaper. Source: `research/ghs3_vs_rho_20261023/TABLES.md`. *Right:* plane-quartic (Diem) index calculus, N4 v2. Fitted exponents of cold cost in q over k = 8–22 against plain and negation rho, at $w = 0$. Dotted line: $\tilde{O}(q)$. Dashed line: rho's $q^{3/2}$. Source: `research/quartic_n4v2_20261020/README.md`. Data: `figures/data-new.json`.

Family	Size	IC/rho	Rho	Exponents	Status
Genus 3, cubic GHS, hyperelliptic	$q = 2^{16}$	0.70, 0.71, 0.70	ideal rho	pooled slope 0.034 per bit, [−0.047, 0.105]	preregistered; at $c_\ell = 9$ every instance 2.7–12.4
Genus 3, cubic GHS, plane quartic (Diem)	$q =$ 2,097,091	0.005 $[0.004, 0.008]$	negation rho	0.983 $[0.975, 0.993]$ against 1.543	interpolated crossover $q^* = 555$ $[389, 752]$; L0; no validation file
Joux–Vitse cover, sieve route	$p = 1511$	0.00647 $[0.00545, 0.00799]$ (U1); 0.0156 (U2); 0.107–0.123 at $w = 0.1$, 1	matched negation rho (142 pooled runs)		validated (`VAL-20261103-c31c76`); weak class only; unemitted work upper-bounded; local run
Joux–Vitse cover, plain Nagao arm	$p = 1009$	1.59–1.98	pooled rho constant		lower bound; against matched negation rho above at $p = 1301$ (1.41) and 1451 (1.16), unresolved at 1601 (§30.4)
$k = 4$ over $\mathbb{F}_{(2^m)^4}$	$m = 15$	5.2			after corrections; crossover unsupported
GHS covers over $\mathbb{F}_{4096}$		549–2,039			exact genus-3 covers found by search; full transfer cost

Table 40.4: Weak extension-field families. IC/rho: cost over the stated rho reference; greater than 1 means rho is cheaper. Exponents: fitted cost exponent in q, IC against rho. Genus-3 hyperelliptic: at six other sizes the decision is unresolved; $c_\ell = 9$ prices one subgroup multiplication at nine $\mathbb{F}_q$ multiplications. Plane quartic: three caveats, the host was L0 under heavy load, there is no validation file, and the frozen baseline's records do not replay. Joux–Vitse sieve: U1 is group additions, U2 textbook $\mathbb{F}_p$ multiplications; the $w > 0$ cells price the cover's unemitted F4, descent and linear-algebra work as an upper bound; the order holds across the band in all four cells. Only the genus-3 and Joux–Vitse sieve rows ran under a frozen protocol, and no row has a pcc claim. Sources: Chapter 26; `research/quartic_n4v2_20261020/`; `research/jv_sieve_negrho_20261102/VALIDATION.json`; `research/jv_cover_negrho_rescore_20261101/README.md`; Chapter 25; `research/ghs_q16_genus3_subcovers_20261028/REPORT.md`.

Family	IC/rho	Qualifier
Joux–Vitse cover, sieve route, $p = 1511$	0.0065	matched negation rho; weak class; validated (`VAL-20261103-c31c76`); no pcc claim
Plane quartic via cubic GHS, $q \approx 2^{21}$	0.005	negation rho; weak family; L0; no validation file; producer-reported
Genus-3 hyperelliptic via GHS, $q = 2^{16}$	0.70	ideal strong rho; one size of seven; pricing-sensitive; producer-reported
Joux–Vitse cover, plain Nagao arm	1.16	matched negation rho at $p = 1451$; above rho at 1301 and 1451; producer-reported
Koblitz, lazy orbit-member index, $n = 29$	1.27 $[0.77, 1.80]$	fully priced; component, not a board row; not validated
Prime, residual collector `res1eq`, 36 bits	1.54	rho's exponent; `VALIDATION.json`, not an admitted row
Prime, residual collector `res2`, 36 bits	2.59	$w = 0$; improvement-log row; no validator
Koblitz rule v8, $2^{36.6}$	4.67	cold, matched, 64 targets; `VALIDATION.json`; not on the board
$k = 4$, $\mathbb{F}_{(2^m)^4}$, $m = 15$	5.2	two targets; crossover unsupported
Prime, k-sum table, 20 bits	5.3	generic track; not validated
Prime, Table B, 12 bits	5.47	validated; `PCC-20261104-f4bc7f` (unverified)
Random binary, Table B, 15 bits	10.6	validated; `PCC-20261104-876bdc` (unverified)
Koblitz $n = 83$	55.4	cold IC over same-point plain-rho wall, five points

Table 40.5: Smallest measured distances to rho, by family. A row without a pcc claim id is producer-reported (an unverified assertion).

41 Corrections and validation

A ratchet is worth only as much as its pawl. Several recorded results were corrected once the reference was matched, native work was priced, or the evidence was replayed. Each correction is recorded in the source with a dated note; frozen evidence is never rewritten. Figure 41.1 shows the before and after values. Most corrections in this chapter moved IC's position against rho *down*, and several moved it from apparently ahead of rho to behind it. Appendix B records two that moved it up: the byte-counter pricing rule and the matched-rho re-measurement of the Joux–Vitse sieve.

41.1 Claims that did not survive

1. Every online win is retired. The one-target online ladder compared IC *with* a retained table against rho *without* one (Table 41.1).

Given a distinguished-point table no larger than IC's pair table, Bernstein–Lange preprocessing rho [4] is 53–317× faster online at $n = 41, 53, 61$ on the same 64 targets (benchmarks row `20261014-ic-next10-09-equal-advice`). At $n = 83$, rho with a table matched to IC's set-up time or memory wins every pair: 3.74× time-matched and 40.61× memory-matched against K3300, and 81.5–395.5× against K600 (benchmarks row `20261016-n83-precomputed-rho-control`). Cold, the hosted $n = 83$ point is 133.7× rho. Those online rows measured what a table buys, and generic rho buys more per byte. The within-IC result survives: K1798's probe reduction over K1500 at $n = 83$ (from 300,021.9 to 169,069.8 probes per relation).

2. Pricing native work moved two arms. Table 41.2 gives both before and after.

The pricing change itself (#1774, 0.002304 GAE per word XOR) made every earlier F4 S a lower bound. A separate algorithmic change (#1862) raised counted work (`research/f4_pricing_jump_20261018/README.md`).

3. Matching the rho reference. Table 41.3 lists the changes to the rho side.

4. A replay regression, and its root cause. The root-finder ecbench session was recorded with a replay certificate of 120 of 168 identical records; the 48 mismatches were all in inherited F4's word-XOR counts. On `main` of 19 October it replayed only 72 of 168, with mismatches in all four index-calculus arms.

- *First diagnosis* (20 October). A cross-ISA replay found that with the x86-only BMI2 branch disabled, x86 reproduces all 168 of 168 records, and so does an aarch64 build under emulation (`research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`).
- *Cause, stated precisely* (26 October addendum). At the recorded commit, BMI2 availability decided whether F4's row specialisation was *eager* (every kept row rewritten at once and charged per row) or *lazy* (rewritten on first read). The two differ in the quantity charged, for example 301,961 against 300,531 word XORs. Since #2043 every host is eager and the two eager paths charge identical counts. Sessions now record the version of the F4 accounting they were made under (`research/validation_sweep_20261019/D-known-issues/ADDENDUM-20261026-item1-accounting.md`).

Archived sessions recorded with the lazy charge cannot replay on `main`. That is a change of definition, not a dependence on the host. Scalars, trial counts and charged group operations are unaffected.

5. Reported cells that compared unlike statistics. The u64-rootbatch rank and per-target cells compared a fastest run with a median, and were corrected from 2.95× and 6.09× to 2.47× and 5.00× (Chapter 15). The $n = 83$ stacked producer's "360 → 94.4" cold-gap history mixed a ratio from another run; paired within the run it is 305.8 → 94.4 (Chapter 15).

6. Results resting on faulty code or wrong references.

- *Plane-quartic IC.* Its S values of 650,896 to $3.79 \cdot 10^7$ rest on a degree-3 reduction in `research/ghs_cubic_quartic_ic_20261012/quartic_ic.rs` that merges distinct classes (reported in an open pull request). Recoveries are still checked against the planted scalar, but relation counts and S are unverified. Canonical Pic^0 keys later fixed the merge (`research/quartic_pic_canonical_2026101`

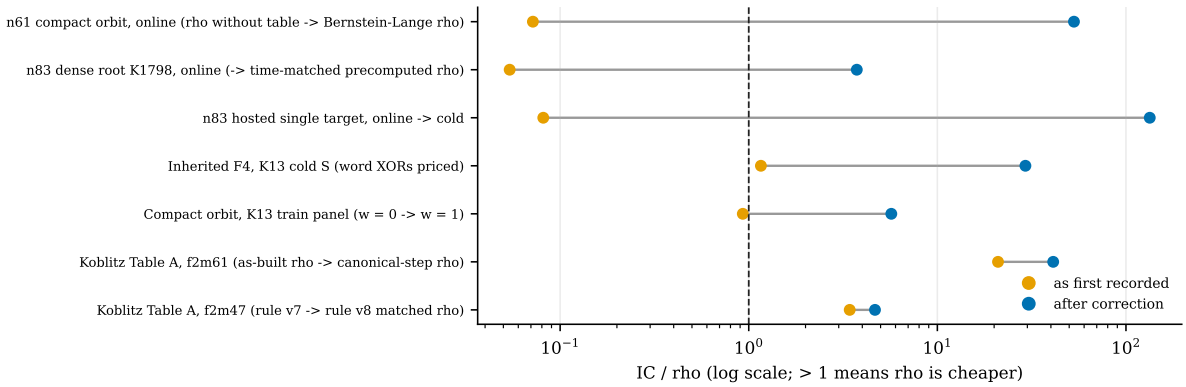


Figure 41.1: IC/rho as first recorded (orange) and after correction (blue), on the boundary the correction used. Values left of the dashed line had claimed IC ahead of rho. Sources: `figures/data-ledger.json`.

n	Reading	IC online over rho
61	strong rho, 64 targets	13.94× [9.81, 19.13]
83	dense root index	18.53×
71	hosted single target	7.29×
83	hosted single target	12.29×

Table 41.1: The retired online wins: IC's online speed-up over rho with no table, IC keeping its retained table.

`8/`). The N4 and N4 v2 measurements of Chapter 40 use the fixed code; the withdrawn values are the earlier toys only.

- *Volcano census crater.* A claim that IC is 22–25% cheaper on the volcano crater was withdrawn, because the crater's rho reference should have been the cubic-automorphism rho. Against it, IC is 84.9× rho at $w = 0$ (`research/volcano_ecdlp_census_20261017/REPORT.md`, erratum A3).

Reading. None of these corrections removes an IC improvement against its own incumbent. They correct the *distance to rho*, which is the quantity most exposed to an unmatched reference or an unpriced counter. This is why the gate of §41.2 asks for a matched, band-priced rho on every row.

41.2 Validation status

Since 19 October no row may enter the index-calculus leaderboard or `benchmarks/README.md` without three pieces of evidence, named by the evidence directory's `VALIDATION.json` (`docs/VALIDATION_GATE.md`):

- an independent replay certificate with verified scalars;
- an independent validator report that recomputes every headline number;
- a matched rho reference: the strongest available, in the same unit and the same binary, priced across the §5a band.

On 29 October, **19 of the 28 leaderboard rows are validated** (`docs/ic/leaderboard.json`, `validation_gate.status_counts`).

Table 41.4 gives the status by leaderboard table.

Validation says the numbers are what the archive says, and that the reference is fair. **No validated leaderboard row is below its reference.** The one validated measurement below a matched rho is the Joux–Vitse sieve route on the `jv_cover` weak class (0.0065×, $p = 1511$, `VAL-20261103-c31c76`; §30.4). It is not a leaderboard row, never has prime order, and has no pcc claim. The 19 validated board rows each have a pcc measurement claim, of standing unverified.

The theory of Part II has a different standard of evidence:

- written proofs, which still need expert review;
- independently checked certificates;
- Lean for the finite and algebraic identities only.

Each theorem there names its tier.

Arm	Reading	Rho	IC
Inherited F4, $n = 13$	Boolean word XORs unpriced	$S = 14.4$	$S = 16.7$
	priced		$S = 421.4$
Compact orbit, $n = 13$	earlier, lower bound		1.08× rho
	strong rho, $w = 0$		0.93
	strong rho, $w = 0.1$		1.53
	strong rho, $w = 1$		5.7×

Table 41.2: The two arms moved by pricing native work. Inherited F4, whose unpriced S was close to rho's: `research/ecbench_rootfind_faster_20261014/`. Compact orbit at the three band weights w; the pair is unresolved inside the band (`research/ic_frontier_campaign_20261015/select/SELECTION.md`).

Change	Quantity	Scope	Factor
Rule v8, matched rho	Koblitz gap raised	every size	1.30–1.58×
Standing rho improved	instruction count	`rho.signed_frobenius_strong`	0.802×
	instructions per step	strong Koblitz rho	718 instead of 1,196
Better of $P \pm R$	steps cut	prime side	1.129×
Signed σ steps	GPU time cut to	ECC2K-130	0.961–0.970

Table 41.3: Changes to the rho reference and their factors. Rule v8: Chapter 40. The improved ecbench standing rho keeps identical GAE, so counted-unit comparisons are unaffected; wall-time comparisons against the older rho overstate IC by up to that factor. The last two are rho studies, neither yet in a campaign arm.

Table	Scope	Rows	Status	Rho reference	IC/rho
B	prime and random binary, one target	13	validated 24 October	matched negation-map	5.47–38.21×
				across the band	4.05–54.76×
A	Koblitz descent, one target	6	validated at baseline v7	as-built, same process	3.43–11.5×
				matched, rule v8	4.67–18.07×
C	32-target batch	9	pre-gate	none matched, no band	
—	components outside A–C	(2)	validated		

Table 41.4: Validation status of the leaderboard on 29 October. Table B validated on 24 October (`research/ic_validation_gate_20261024/`): 1,257 runs replay identically in a fresh process; 1,578 of 1,578 negative controls are rejected by an independent verifier; 52 headline numbers recompute to relative 10^{-9}; index calculus stays above rho at every band weight. Table A (64 targets per size) validated at baseline v7 (`research/ic_tool_program/rule/v7/`); the sweep flags the reference as only partly adequate, and the rule v8 row is from Chapter 40 (rule v8 also carries a `VALIDATION.json`, `research/ic_tool_program/rule/v8/VALIDATION.json`; the board has not been regenerated on it). Table C's replays and recomputations pass, but no matched rho with a band exists for them. The component row is not counted in the 28 rows; it names two of several validated components (`docs/ic/leaderboard.json` has more, among them the generator-orbit and aimed-gather components): the prime-field base-size tuning and the post-Gröbner linear stage (`research/prime_base_oracle_tuning_20261029/VALIDATION.json`, `research/postgb_linear_stage_20261026/validation/VALIDATION.json`).

Part VI

Outlook

42 Open problems and proposed experiments

1. The solving degree of Weil-descended summation systems. Every subexponential claim for prime-degree binary curves rests on a bound for this degree (§3). No growing lower bound is proved; Chapter 6 certifies $d_F \geq 5$ at $n = 33$–45 and the census of Chapter 7 gives ≥ 4 at $n = 17$–24. The proof half of the question is problem 12; this item is its measurement half. *Experiment:* measure first-fall, last-fall and solving degree of S_{m+1} restricted to ℓ-dimensional subspaces over a grid of (n, ℓ, m) with $m \geq 3$, at least four values of n per (ℓ, m), against the semi-regular prediction, with multi-graded regularity (one block per summand) as a second model. *Settles:* whether the degree grows with n at fixed (ℓ, m); a growth consistent with zero would revive the Petit–Quisquater and Semaev heuristics, a positive slope would close them for this model.

2. Cold index calculus against rho with every operation priced. No measured arm on a prime-order or prime-degree binary curve is below matched rho on the cold one-target boundary once native work is priced on both sides (§36). *Experiment:* the frontier campaign of §36 extended to the largest sizes each arm can reach, with exponents fitted per arm and the crossover reported only where the fitted exponents separate across the whole pricing band. *Settles:* whether any implemented arm has an exponent below $1/2$ in r on these families.

3. Relation yield is now the dominant stage. Root finding made the decomposition oracle cheap; what remains is the number of trials per relation, governed by the factor-base density. *Experiment:* for each factor-base family, the measured trials-per-relation law against r and base size, compared with the counting prediction $m!\, r/N^{m-1}$; the residual (e.g. the unexplained factor 5 in the ECC2K-130 extrapolation, §15) is the target.

4. Root finding beyond two summands on prime curves. The two-summand oracle is prior art in mechanism (McGuire–Mueller) and costs $\Theta(N)$ per trial; three summands cost a pair table of N^2 entries. A generic table of signed 4-sums already moved the generic exponent to 0.521 (Chapter 40), and any membership structure built from group operations stays inside the $\sqrt{r}$ floor (§32.1); a single-target generic arm with unknown base logarithms costs at least 1.5× rho (Proposition 32.1). *Question:* is there a sub-N^2 *non-generic* (coordinate-algebraic) membership test for S_4 over a prime-field base—for example from the D_m-invariant coordinates of 2-torsion curves—whose cost escapes that floor? A negative answer with a counting argument would be as useful as a positive one.

5. The preprocessing regime. Index calculus with a retained table beats rho without one, and loses by 53–317× to Bernstein–Lange rho with an equal or smaller table (§36.3). *Experiment:* measure the advice–time curve (S, T) of the compact-orbit arm across sizes against preprocessing rho's $S\,T^2 \approx N$; an index-calculus online result counts only if it lies below that curve at equal S.

6. Symmetry at $k = 4$, end to end. The D_4 reduction and its four-unknown weighted form moved a *modelled* handover over $\mathbb{F}_{p^4}$ down from about 2^{150} (§§36.3,36.4). Those single-point models are superseded: the E32 ladder, validated, measured $1.5 \cdot 10^3\times$ its strongest matched rho at $n = 2^{42.4}$, falling as $n^{-0.24}$, with a fitted (not measured) crossover at $2^{85.5}$ $[2^{79.2}, 2^{94.3}]$ (Chapter 31). In characteristic two a Frobenius-reduced form brought the gap to 4.3× a measured rho at $n \approx 2^{56}$ on one target (6.1× against the rho model, §36.4). A red-team review (Chapter 25) found the extrapolated crossover near $n \approx 2^{68}$ unsupported, not refuted: the only clean point, $m = 15$, is 5.2× rho after corrections, and the crossover extrapolates to $m^* \approx 17.6$ (16.3–19.2), at or above the edge of the data. *Experiment:* the frozen replication plan (eight targets per arm, $m = 14$–17, about 30 h), with a measured negation-map rho at every rung, cold and priced across the band; a measured crossover would be an ADVANCE on a weak family, and its absence would locate the term the model misses.

7. Pricing and calibration infrastructure. Rankings now require every arm's native work to be counted (AGENTS.md §5a). *Work:* count the three uncounted categories of the shared-rank arm;

calibrate every ladder size on one fixed host; register the $n = 83$ dense root index in the harness. (The cross-host F4 counter drift is closed: since #2043 every host takes the eager path, and sessions record the version of the F4 accounting; Chapter 41, item 4.)

8. Genus three after cubic GHS, to the crossover. The one family whose fitted exponent (about $q^{1.4}$) lies below rho's $q^{3/2}$ was 1.15× a measured negation-map rho at $q = 2^{20}$ before rounds three and four brought the whole solve to about 0.44× the second round's at $q = 2^{22}$ (derived, in the inversion-charging unit; §§36.3,36.4), and is now below the strongest rho at $q = 2^{16}$ under the registered pricing, unresolved at the other sizes to 2^{20}, and above it under a heavier sensitivity pricing, with no measurable pooled trend (§26). What would settle it is the same comparison at $q = 2^{21}$–2^{24} on instances of one cofactor, with the subgroup multiplication's price measured rather than chosen; its local exponent is down to 1.356, still above 4/3 because the linear algebra grows faster than $q^{4/3}$, and $q = 2^{23}$ needs more than the 4.5 GB cap. *Experiment:* measure negation-map rho on the same instances in the latest unit (inversions and mod-ℓ multiplications priced), and run both past $q = 2^{22}$, in the harness and across the pricing band. A measured size crossover there would be an ADVANCE against rho on a narrow and undeployed family; the programme's one validated below-rho measurement (the Joux–Vitse sieve, problem 11) is accounting at a single size.

9. Plane quartics: Diem's $\tilde{O}(q)$. Genus-3 GHS covers whose Jacobian is that of a smooth plane quartic admit Diem's index calculus [9, 11] at $\tilde{O}(q)$, against $\tilde{O}(q^{4/3})$ for the hyperelliptic route and $q^{3/2}$ for rho on $E(\mathbb{F}_{q^3})$. The repository builds such quartics directly ($m = 2$ GHS, toy index calculus at $q \leq 11$) and through Smith's $(2, 2, 2)$-isogeny from the hyperelliptic case (18.6% of random curves); a four-workstream programme was scoped on 18 October (`research/plane_quartic_program_20261018/PROGRAM.md`). *Experiment:* the hyperelliptic ladder's instances, carried to plane quartics where Smith's conditions pass, with Diem's relation search and the same linear algebra, against the same matched rho. The 23 October literature round ranks this branch as its one exponent candidate (LEAD-15, `research/pdp_literature_leads_20261022/`). *Since then:* the arithmetic has been rebuilt with canonical class keys, which resolve the degree-3 merges (§4.5). On the direct quartic route, index calculus is below a matched plain rho from $q = 4051$, with fitted exponent 1.083 against 1.485, and N4 v2 extends this against a negation-map rho over $q = 2^8$–2^{22}, at exponent 0.983, and 0.005× rho at $q \approx 2^{21}$ (L0, no validation file; §26). *Still open:* a validation of N4 v2 on an isolated host with a replayable baseline; index calculus on the quartics reached through Smith's isogeny; and a record-scale run (for example $q \approx 2^{40}$, $\ell \approx 2^{120}$) where rho is out of reach and the published Diem–Thomé timings are the comparator.

10. A published index calculus as the reference arm. Every ratio in this treatise is against rho or the programme's own previous arm; none is against the state of the art in index calculus at the same (q, n, m) (§36.5). *Work:* fix, per curve and size, the standard base $\{P : x(P) \in \mathbb{F}_q\}$, the FGHR-symmetrised summation polynomial (2-torsion and, on Koblitz curves, Frobenius invariants), msolve as the solver and WDSat for binary decompositions, and run it in ecbench on the existing ladders (binary $n = 13$–37, Koblitz $n = 41$–61, $k = 4$ extension fields). *Settles:* whether any engineering result of §§9–26 improves on published index calculus, as opposed to on this programme's own baseline. Stage ratios quoted against the in-house F4 say nothing about published Gröbner engines until this is done.

11. The Joux–Vitse cover, to the crossover. On 24 October the plain Nagao arm of the cover of $E(\mathbb{F}_{p^6})$ was 1.6–2.0× a pooled rho constant at $p = 1009$ (a lower bound), closing as about p^{-2}, with the line meeting rho near $p \approx 1330$ by extrapolation (§36.5). The November studies ran rho on the same instances (§30.4), and the extrapolation does not hold. Against a matched negation rho the plain arm is above rho in group additions at $p = 1301$ (1.405) and 1451 (1.160) and unresolved at 1601 (0.870 $[0.697, 1.183]$), and above it in $\mathbb{F}_p$ multiplications at all three (`research/jv_cover_negrho_rescore_20261101/`). A factor-base line search makes it 1.71× cheaper, below rho in additions (0.791–0.548) but above in multiplications (1.900–1.318), so it is unranked across the band; its modelled crossovers are $p^* \approx 1185$ in additions and 1799 in multiplications, with no validation file (`research/jv_cover_fbline_20261102/`). The *sieve* arm is 0.0065× a matched negation rho at $p = 1511$ in group additions (155× $[125, 184]$ below it; 64× in $\mathbb{F}_p$ multiplications; at most 0.12× across the band), on the weak class only, validated as accounting (`VAL-20261103-c31c76`); it has no `pcc` claim record, so its statement is a validation-gate result and otherwise an unverified assertion. Four questions remain:

- whether the sieve arm's order survives once its F4, descent and linear-algebra additions are emitted and priced, rather than bounded;

- whether the Nagao line-search row validates, in both units;
- why the two arms differ by about $30\times$ at $p = 1009$;
- whether both results hold on the isolated lab rather than a local run.

Experiment: emit those counters on both arms, and run both arms and rho at $p \approx 1300$–2000 on the isolated lab, across the band. On a weak, undeployed class this would turn a validated accounting result into a priced result.

12. A growing lower bound on the last fall degree. Chapter 6 proves the first fall degree constant and certifies $d_F \geq 5$ on named binary instances, but no presentation has a proved lower bound that grows with n or k. *Proof route:* construct degree-c pseudo-solutions, i.e. linear functionals on $B_{\leq c}$ that kill $W_c(F)$ and not 1, for a family of instances, generalising the closure certificates of Proposition 6.4. The regular closure pattern observed (exactly $4n$ new cubic dimensions per round, then a stall) and the iterated-system lower-bound technique of [38] are the starting points. *Settles:* the open problem of HKY [23] and Kousidis–Wiemers [29]; a bound $\omega(\sqrt{n \ln n})$ would invalidate the subexponential conclusion of Semaev's analysis for these systems.

13. Certificates at scale. *Experiment:* certify $1 \notin W_4$ (and, where it holds, $1 \notin W_5$) on about ten random and ten polynomial subspaces per n for $n = 33$–51, publish the certificates, and extend the guarded chain to $B = 4$–8 by computation. *Settles:* whether the step $4 \to 5$ is a population property, and where $5 \to 6$ occurs; the exact d_F of the guarded chain beyond $B = 3$.

14. The prime-field Kummer tower, re-measured. The session values $d_F = 3, 4, 4, 4, 4, 5, 5, 5, 5, 6$ at $t = 1..10$ were never committed. *Experiment:* regenerate stage A at $t = 1..14$ with closure certificates and score the registered prediction $d_F = 4 + \lfloor (t-2)/4 \rfloor$; extrapolated F4 times are about 1.7 h, 17 h and 7 days at $t = 12$–14 (`research/prime_ic_program_20261029/`). *Settles:* whether the proved constant first fall degree 4 sits beside a last fall degree that keeps growing in the setting PKM introduced.

43 Conclusion

Eighteen weeks of work on elliptic-curve index calculus, from July to early November 2026, produced four kinds of result.

Theory.

- *What is fixed for every curve.* The summation cover's monodromy is $(\mathbb{Z}/2)^{m-2}$ everywhere. Its fibres split completely where relation search operates. The mean yield at a given base size is a double count. So no choice of curve or base geometry can raise the *mean* decomposition yield at matched base size. Coverage, the cost of the membership oracle and the difficulty of the solve remain free, and the engineering of Part III works on those.
- *What is open.* For three presentations of the decomposition system the first fall degree is a constant of the presentation, and the last fall degree is not bounded by it:
  - proved for the guarded chain;
  - certified on named instances of binary Weil descent;
  - session-reported (raw data not committed) for prime-field Kummer towers.

  The subexponential claims for binary curves assume exactly the step these results show is not automatic. Whether the last fall degree grows without bound is unproved.
- *What hardness rests on* (Chapter 7). The literal Petit–Quisquater Assumption 2 fails on all 119 census instances, PKM's "$\Delta = 10$ suffices" is contradicted at toy sizes, and a kernel-checked left-kernel theorem removes the rank condition for a single target. The census figures are unverified assertions.
- *The generic floor* (Chapter 32). A single-target index calculus whose base logarithms are unknown costs at least 1.5× rho, and 1.875× with two base elements, the constant at which the measured generic incumbent sits (a derivation in the generic group model, checked by Monte Carlo).

Engineering. The decomposition oracle became about 400–$5 \cdot 10^4\times$ cheaper per decomposition against F4 where root finding applies. The Gröbner engines improved by one to two orders in counted work. Factor bases closed under known-logarithm maps, a k-sum table, and set-up and linear algebra engineering each cut a constant. None of this moved an exponent below rho's on a prime or prime-degree binary field. The generic k-sum oracle moved its own exponent from 0.62 to 0.52, but its limit is baby-step giant-step, and the collision floor keeps any generic single-target arm with unknown base logarithms at least 1.5× rho.

Measurement. At a matched reference, priced across a band and replayed, rho is cheaper on every prime-order curve the programme measured:

- on Koblitz curves of 37–47 bits, 3.4–11.5× on the validated board (rule v7) and 4.7–18× against rule v8's same-binary rho;
- on prime curves, 5.5–13.5× on the validated rows of 10–24 bits, 5.3–13.1× for the k-sum table and 10.5–102× for the campaign pair table at 20–40 bits, and 1.3–1.7× for the generic residual collectors at 36–42 bits;
- 10.6–38× on random binary curves.

Nineteen leaderboard rows passed a validation gate that demands independent replay, recomputation and a matched rho. None of them is below its reference. Off the board, one validated measurement is below a matched rho: the Joux–Vitse sieve route on the weak class at $p = 1511$, at 0.0065× a matched negation rho in group additions and 0.016× in $\mathbb{F}_p$ multiplications, with the order holding across the band (`VAL-20261103-c31c76`; no `pcc` claim). The validator classes it as accounting: a local run, a sampled replay, and the F4, descent and linear-algebra additions bounded rather than emitted. Every online win disappeared under an equal-advice audit, and the corrections forced by matching, pricing and replaying lengthened the measured distance more often than any single improvement shortened it.

Where index calculus wins. In weak extension-field families ($E_0/\mathbb{F}_{q^3}$ through genus-3 covers, Joux–Vitse covers of $E(\mathbb{F}_{p^6})$, and perhaps $k = 4$ in characteristic two) the measured cost crosses or approaches rho, as theory predicts. The plane-quartic route measured exponent ≈ 1 against rho's 1.5, at 0.005× negation rho near $q = 2^{21}$ (L0, no validation file). On the Joux–Vitse cover the sieve route is the

validated row above, while the plain Nagao arm is above a matched negation rho at $p = 1301$ and 1451 and unresolved at 1601, so its extrapolated crossing near $p \approx 1330$ does not hold. These are known algorithms; the programme's contribution there is fair measurement.

What would change the picture. Three items, in order of what they would settle:

1. *A proved lower bound on the last fall degree that grows with n*, for any presentation of the binary decomposition system. It would answer the question left open by Huang, Kosters and Yeo and by Kousidis and Wiemers, and it is the only result that bears on the subexponential claims.
2. *More certified instances* at $n = 33$–51, with public certificates, to show whether the last fall degree reaches 6.
3. *The frozen $k = 4$ replication* at $m = 14$–17, the cheapest experiment that could put a measured size crossover (an ADVANCE) on record. It would be in a weak family. The one validated below-rho measurement, the Joux–Vitse sieve at $p = 1511$, is accounting and below rho at the one size matched; emitting and pricing its F4, descent and linear-algebra work (open problem 11) is the cheapest way to turn it into a priced result.

Nothing here concerns a deployed curve. The open problems of Chapter 42 state each remaining question as the experiment or proof that would settle it.

A Artefacts

Every number in this treatise is copied from a committed artefact, except the Kummer-tower last fall degrees of §6.5, which are session-reported (raw data not committed). The treatise ran one re-verification (§6.3) and no new measurement. Artefact paths are relative to the `crypto` repository unless prefixed `crypto-autoresearcher:` or `cryptanalysis:`. Each chapter names its own sources; the lists below give the main directories per part.

Companion documents.

- `papers/ecdlp-index-calculus-program/` [3] is the chronological record, with study catalogues for both repositories (about 1,400 directories) and the list of open pull requests. Where this treatise reuses its chapters (background, literature, measurement, root finding, Gröbner engines, Koblitz, binary and prime curves, transfers, frontier, open problems), the text is that record's as of 26 October, updated where later evidence changed a number.
- `papers/ecdlp-ic-improvement-ledger/` [2] is the ratchet of incumbents to 23 October.
- `research/paper_opportunities_20261029/` is the assessment that identified the fall-degree results of Chapter 6 as the programme's strongest theory, and that holds the certificate re-verification.

Theory (Part II).

- `crypto-autoresearcher:papers/semaev-conservation-specialization/` (Chapter 5).
- `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/`: `theorem-dossier.md`, `results/`, and `verify/` (`cert_check.py`, `b_sweep.py`).
- `research/guarded_fall_degree_20261028/`: `proofs/`, `extensions/`, `native/`, `formal/`.
- `research/prime_fall_degree_20261027/DERIVED.md`.
- `crypto-autoresearcher:papers/nagao-lemma4-note/` and `crypto-autoresearcher:docs/reports/2026-10-22-semaev-first-fall-replay/`.
- `crypto-autoresearcher:knowledge/leads/ecdlp/` (Chapter 8), `research/twist_obstruction_20261026/` and `research/lift_dense_factor_base_20261027/`.
- Hardness (Chapter 7): `research/ic_weak_assumptions_20261102/`, `research/ic_weakpoints_20261102/`, `research/ecdlp_weakness_census_20261104/`, `research/ecdlp_weakness_lean_20261104/` and `docs/ecdlp-hardness-corpus/`.
- Obstructions (Chapter 8): `research/jksst_xedni_audit_20261101/`.

Algorithms and measurements (Part III).

- Engines and instruments (Chapters 11–14): `research/f4_harvest_remeasure_20261024/`.
- Binary curves (Chapters 17–20): `research/n83_crossfamily_current_20261025/`, `research/icb_large_prime_n53_n61_20261029/` and `research/ic_m2_torsion_n41_20261101/`.
- Prime fields (Chapters 21 and 22–24): `research/ic_residual_scale_20261101/`, `research/ic_byte_counter_pricing_20261101/`, `research/prime_ksum_pooled_20261029/`, `research/ic_prime_eth_leads_20261031/`, `research/semaev_alternatives_audit_20261104/`, `research/curve25519_autolab_20261104/` and `research/ecgfp5_quotient_solve_20261101/`.
- Transfers (Chapters 27–31): `research/jv_cover_negrho_rescore_20261101/`, `research/jv_cover_fbline_20261102/`, `research/jv_sieve_negrho_20261102/` (the validated sieve row), `research/jv_sieve_line_vs_rho_20261103/`, `research/ic_acct1_cover_bitop_20261021/`, `research/conductor_gap_cm_pairs_20261010/` and `benchmarks/20261031-e32-handover-ladder/`.

The rho side (Part IV).

- `research/curve25519_rho_parity_bound_20261104/` (the collision floor), `research/ecbench_calibration_20261002/`, `research/rho_assurance_20261016/`, `research/rho_negation_audit_20261020/`, `research/cheon_ecbench_20261022/` and `docs/bounds/FRONTIER.md`.

The summer record (Chapter 35).

- `research/ecdlp_jul_sep_index_20261029/`: `INDEX.json`, `INDEX.tsv`, `SUMMARY.json`, `build_index.py`. Each row cites its `crypto-autoresearcher:ledger/decisions/` file and the `experiments/EXP-*/runs/` directories of its targets.

The frontier (Part V) and late measurements.

- `research/gaudry_k4_char2_redteam_20261026/REPORT.md`.
- `research/quartic_n4v2_20261020/` and `research/ghs3_vs_rho_20261023/`.
- `research/prime_ksum_table_20261028/` and `research/prime_base_oracle_tuning_20261029/`.
- `research/prime_ic_program_20261029/`.
- `research/n83_k1107_bl_reproduction_20261028/`.
- `research/ghs_q16_genus3_subcovers_20261028/`.
- `research/postgb_linear_stage_20261026/`.
- `research/ic_validation_gate_20261024/`, `docs/ic/LEADERBOARD.md` and `docs/ic/leaderboard.json`.
- Unconventional routes (Chapters 37–39): `crypto-autoresearcher:knowledge/leads/ecdlp/exotic/`.
- Corrections (Appendix B): the directories named in each row of its tables.

Figures. The figures of the programme record and the ledger are copied with their data (`figures/data-program.json`, `figures/data-ledger.json`); their plotting scripts live in those papers' directories. The new figures (fall degrees, weak families) are rendered by `figures/make_fig_fall_degrees.py` and `figures/make_fig_weak.py` from `figures/data-new.json`. That file holds recorded values only, each with its source.

B Corrections, retractions and withdrawn readings

This appendix collects the corrections, retractions, errata, withdrawn claims and reversed verdicts recorded in the program's evidence directories and ledgers. Where a chapter tells a correction at length, the entry here is a one-line pointer. Chapter 41 and §4.5 hold the corrections that moved index calculus against rho on the leaderboard (the retired online wins, the priced word XORs, rule v8, the cross-host F4 counter drift, the volcano-crater and plane-quartic withdrawals); the rho-side corrections are told in Chapters 33 and 34, and the unconventional-route corrections in Chapters 38 and 39. They are not repeated here.

Each entry gives what was first read, the record that read it, what replaced the reading and when, and the source. Three conventions apply throughout.

- *Standing.* Unless a row says *validated*, the correction is itself a producer's record, dated and additive (the frozen evidence it corrects is left in place), and carries no proof-carrying-claim verdict. Under the rule that every producer may be sabotaging the work, such a row is an unverified assertion of its producer, as is the reading it replaced.
- *Direction.* A correction of the rho reference or of the pricing is an accounting change. None of the rows below is an improvement of index calculus, and none moves a validated leaderboard row below its reference.
- *Cross-references.* Seven corrections are carried in the main text (Table B.1) and listed here with a pointer only.

B.1 Binary and Koblitz fields

Table B.2 lists the binary-field items. Two need a sentence beyond the table.

Torsion anchors. The merged report of `research/ic_m2_cycle_leads_20261030` (PR #3009) stated that torsion anchors "cost 1.5× ($n = 23$) to 16× ($n = 31$) on cofactor-4 curves". A frozen-protocol test at $n = 41$ found the on/off trial ratio 0.975 $[0.641, 1.420]$, and a post-hoc replicate of 32 paired targets at $n = 31$ gave 13.3× $[10.3, 17.5]$: the penalty exists at $n = 31$, is about 1.5× at $n = 23$ and is absent at $n = 41$, so the range is not to be quoted as a property of torsion anchors. A companion study on $a = 0$, $n = 37$ found a penalty of 6.3–14.5× at every cofactor-to-base ratio from 0.30 to 10.84. Its README also records that the earlier bootstrap intervals (PRs #3126 and #3138) were drawn with `mawk`'s non-reproducible `srand`; recomputed with a seeded generator they move by at most 0.02. The earlier $n = 37$ figure 0.134 $[0.082, 0.205]$ was off/on; it is the same measurement as the on/off 7.46 $[4.90, 11.92]$ and is not a reversal (`research/ic_m2_torsion_n41_20261101/REPORT.md`, `research/ic_m2_torsion_ratio_n37_20261101/REPORT.md`).

The $k = 2$ collision excess. On $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{19}}$ ($r = 130{,}873$, birthday expectation 5.004) the archived census read informative-to-birthday ratios of 3.397 and 7.593 at $k = 2$ and called them a k-specific collision excess. The census counted a colliding pair as informative unless its difference lay in the span of one kernel vector per generator, but the factor base is the union of $\langle \tau, -1 \rangle$ orbits, so negation identities and τ-shifted kernel relations were counted as informative. Replaying the archived code classifies 16 of 17 and 33 of 38 pairs as structural; the genuine-to-expected ratios are then 0.20 and 1.00 at $k = 2$ and 0.40 and 0.80 at $k = 3$, none above the band (`crypto-autoresearcher: analysis/ecdlp2m-iteration-20261009/certbin-k2-collision-audit/README.md`).

B.2 Prime fields

Table B.3 lists the prime-field items; the first needs a paragraph.

Byte counters priced as operations. The §5a band priced `residual_table_bytes_uncharged`, a byte count of the residual table, at w GAE per byte. On the residual arms it was 83–99% of S at $w = 1$,

Item	Original reading	Correction	Main text
H-PSEUDO character-sum constant	Fourier-flatness offset $R \approx 1.28$–1.34, $C(p) \sim p^{0.055}$–$p^{0.079}$, open	4 Nov: the (x, y)-keyed small-x base is negation-closed and the null was not; against a negation-closed null R is consistent with 1 (pooled 56 curves 1.005 ± 0.008); verdict *explained* under a non-rejection rule, so a residual up to about +9% at 2^{20} is not excluded	Chapter 35
Xedni lifting	"closed" in exact form	1 Nov: the Jacobson–Koblitz–Silverman–Stein–Teske obstruction is conditional (Lean audit); overbroad readings refuted; algorithm-specific bridges still open	Chapters 8, 35
$\ell = 8$ hybrid-guess cells	the $\ell = 8$ guess run had not completed	22 Oct: four guessed variables bring every $\ell = 8$ draw back to degree 4; 23 Oct: three do not, for that guess set	§16.1, §36
Planner weight rules	`ic-planner/2`, `/3`: $1.41N_0$ (prime, $w = 1$), $0.71c_0$ (Koblitz, $w \geq 0.3$)	26 Oct: both failed their preregistered out-of-sample tests; `ic-planner/4` withdraws them	§21
EcGFp5 symmetrisation	no $m = 5$ ideal degree measured	1 Nov: $D = 2^{16} = 65{,}536$ at all three primes	Chapter 35
$n = 83$ Weil-descent refutations	F4, F5, F6-IC, crossbred, SAT-descent and MQ-FES "proved UNSAT" on planted relations	24 Oct: `FieldStructure` truncated the field table to one 64-bit word; builders now refuse such fields; every earlier $n = 83$ engine refutation is void	§12.8 (summary in §15.5)
Joux–Vitse sieve arm	19 Oct comparisons withheld by the 20 Oct review; replication failed its rho A/A rule	2–3 Nov: matched negation-map rho at $p = 1511$; sieve below rho in every cell of the band; *validated* (`VAL-20261103-c31c76`); no pcc claim, so its statement is an unverified assertion	§26

Table B.1: Corrections carried into the main text, listed here by pointer. Sources: `crypto-autoresearcher: docs/reports/hpseudo-symnull-20261104/README.md`; `research/jksst_xedni_audit_20261101/`; `research/dreg_closure_guess_20261022/RESULTS.md`, `research/dreg_closure_guess3_20261023/RESULTS.md`; `research/ic_planner_weight_rule_20261021/README.md`, `research/ic_planner_weight_rule_binkob_20261026/README.md`; `research/ecgfp5_quotient_solve_20261101/README.md`; `research/n83_msweep_20261024/README.md`; `research/jv_sieve_negrho_20261102/`.

Item	Original reading	Correction	Source
Torsion anchors, cofactor 4	PR #3009: 1.5–16× more trials	1 Nov: 0.975 [0.641, 1.420] at $n = 41$; not a general property (see text)	`research/ic_m2_torsion_n41_20261101/`
Semaev 2015, $t = 2$	first fall degree "always 4" (§4.5); corpus inference that these rows refute $d_{F4} \leq d_{ff}$	2 Nov: $d_{ff} = 2$ on 200 of 200 instances by a trace certificate; the solver needs degree 2 on 154 and 3 on 46, never 4, so the sharp assumption fails on 46/200 only	`research/semaev_t2_trace_20261102/REPORT.md`; cf. Corollary 6.12
Hamming ball B_3, $n = 19$	$R = 1.10$ against random subspaces	1.10 came from one noisy subspace denominator; against 8 matched subspaces $R = 0.27$ [0.13, 0.46] and 0.32 [0.14, 0.55]; *not decided*	`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-nonlinear-frob-stable-bases.yaml`
SAT answers in `EV-ICPERF-390707`	five unverified SAT answers	they are decompositions on the quadratic twist $y^2 + xy = x^3 + 1$; instrument not invalidated, SAT-over-Gröbner gap direction reproduced	`crypto-autoresearcher:ledger/evidence/EV-ICPERF-390707.yaml`
Certified $k = 2$ collisions	17 and 38 pairs against 5.004: a k-specific excess	instrument artefact; 16/17 and 33/38 structural (see text)	certbin audit (see text)
Learned factor-base selection	producer: "about 9 bits, identical across the band" behind rho	5–6 Nov: a pinned-calibration point; ridge pick 1.8/14.8/17.7 bits ($n = 17$) and 3.3/15.1/17.9 ($n = 19$) behind strong rho at $w = 0/0.1/1$; the lead records independent validation	`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-learned-cost-model-factor-base-selection.yaml`
Root linear-row harvest (#1862)	raised counted word XORs about 7% (one workload, K_0, $n = 13$)	24 Oct: the sign depends on size: S up 9.6% and 7.0% at $n = 11, 13$, down 0.3% and 4.1% at $n = 17, 19$; every priced F4 arm stays 7–1,250× strong rho	`research/f4_harvest_remeasure_20261024/README.md`

Table B.2: Binary and Koblitz corrections not in the main text. All are producer-reported except the learned-selection row, whose atlas lead records a `VALIDATION.json`; the treatise did not open that repository. The already-recorded binary corrections (September reversals, the retired $n = 61$ 13.94× and $n = 83$ 18.53× online readings, Table A's understated gap, the contradicted degree-4 floor) are in Chapter 41 and §4.5.

Item	Original reading	Correction	Source
Residual collectors at $w > 0$	res4/rho 41–58× at $w = 1$; C1 met in 6 of 8 cells	1 Nov, option B: res4/rho 3.6–4.8×; C1 met in 2 of 8	`research/ic_byte_counter_pricing_20261101/`
Small-root lattice reach	$\varepsilon(S_3) = 0.8408$, best margin -0.0481 at $m = 4$	24 Oct (`CORR-20261024-b2a799`): wrong exponent in `smallroot_reach.py`, $\varepsilon_{HG} = \varepsilon_{TOOL}/n$; $\varepsilon(S_3) = 0.2803$, margin -0.6086 at $(4, 2)$, best -0.5161 at $m = 32$	`crypto-autoresearcher:analysis/smallroot-reach/CORRECTION.md`
Oracle gate of the Semaev-alternatives memo	"$e < 1$ at $m = 4$"	4 Nov: not implied by its own cost model; a generic 2+2 MITM passes it ($s = 1.93$) at cost $p^{2/3}$; correct gate $s < m - 3$, i.e. $e < (m - 3)/4$	`research/semaev_alternatives_audit_20261104/README.md`
Cofactor-cleared prime base	$G_0 = 6.180$ [5.915, 6.456], pass line met at every weight	3 Nov: verdict `invalid_controls` by the frozen rule (positive control's lower limit $1.149 < 1.2$); not to be cited as a gain	`research/ic_cofactor_clear_ab_20261103/README.md`
Multi-target Koblitz figures	closest batch figure 1.17× rho	accounting: 6.51× batch rho at $k = 32$ (Kuhn–Struik reference), 6.38× with step and build priced; one target 43.7×	`research/ic_rho_koblitz_20260923/README.md`

Table B.3: Prime-field corrections, with one Koblitz accounting row that belongs with them because it changed a rho reference rule. All are producer-reported; the small-root correction is a single-agent computation that has had no independent review.

and it alone made their $w > 0$ distance to rho 17–58× rather than 2.6–5×. On 1 November the owner chose option B: the band prices operations only, and `band.rs` lists the byte counters without pricing them. Re-running every committed weighted comparison under the rule turned all 24 residual-versus-planner comparisons of `research/ic_prime_residual_lp_20261026/` from "unresolved inside the band" to "residual cheaper at every weight". It also resolved 36 of 136 comparisons in the planner-exhaustive study and 2 of 6 in the Koblitz weight-rule study, and reversed no direction. Its effect cut both ways: the pre-registered criterion C1 of `research/ic_residual_scale_20261101/` is met in 2 of 8 cells, not 6 of 8, because the `eq` candidate's apparent gain at $w = 1$ rested on byte counts. The $w = 0$ results and the 2.59× leaderboard row are unchanged (`research/ic_byte_counter_pricing_20261101/README.md`).

B.3 Engines and measurement

Several corrections changed the instrument, not a result (Table B.4). Five are told at length in Chapter 33 and only named here: calibration round 1 (§33.1), the independent-validation host class and the statistics v2 intervals with censored rho (§33.8), the bounds frontier v1→v2 (§33.3), and the $n = 29$ generator-orbit panel, which is a Frobenius-folded BSGS (§32.3). They matter because intervals and references quoted earlier in this treatise were computed under the older rules.

- *Accounting v2* (23 October) sized the strong Koblitz reference to r. Its mean S falls to 0.182 [0.171, 0.196], 0.223 [0.201, 0.248] and 0.678 [0.499, 0.891] of the frozen id at $n = 13$, 19 and 37, and its fixed set-up share from 83.3%, 71.3% and 30.2% to 39.9%, 29.3% and 4.3%. Any small-n comparison against the frozen strong id read rho too expensive by up to that factor. The report states that this is a correction of the reference instrument, not a faster algorithm (`research/ecbench_accounting_v2_20261023/REPORT.typ`).
- *In-house F4 is not a published engine.* A desk review found that the repository's F4 needs 10^3–10^7 more operations than a brute grid on the same PKM systems, so root-finding-over-F4 ratios

Item	Original reading	Correction	Source
Strong Koblitz rho id	frozen id, set-up 30–83% of S	23 Oct: v2 sized to r, S ratio 0.182–0.678 (see text)	`research/ecbench_accounting_v2_20261023/`
Runner PR #2455	IC online window "about 15% lower"	23 Oct: counts identical in 10 of 10 pairs; online ratio 1.067, not distinguished; the 18.4–38.8× shorter small-arm walls are an allocator effect (`malloc_trim`, `/proc` reads)	`research/ecbench_allocator_confound_20261023/`
E42 Macaulay tails	preregistered control range $[900, 960]$ re-allocations	3 Nov: the range was mis-derived (measured 245.6); E42's kill fired with W unranked; E42b re-registered before any ranking data	`research/e42_macaulay_tails_20261103/REPORT.typ`

Table B.4: Instrument and measurement corrections not told elsewhere. All producer-reported; none changes an IC count except where the row says so. Calibration round 1, independent validation, statistics v2 and the bounds frontier v1 are in Chapter 33; the $n = 29$ panel is in §32.3.

such as those of §9 (for example 383×) are not evidence about published Gröbner solvers; degree measurements made with msolve are unaffected (`research/ic_sweeper_20261024/README.md`).

B.4 Rho references

Table 41.3 gives the rho-side changes listed with the leaderboard corrections. Table B.5 adds those that moved study readings and are not told elsewhere; the negation-map audit (§33.5), the withdrawn Cheon/rho readings on the P-224 shape (§32.4), the Curve25519 `wwalk` re-baseline (§32.2) and the hyperelliptic rho reference (§34.5) are in Chapters 32–34. One binary defect underlies the first two rows: the build that wrote the crater-automorphism and prime-automorphism sessions left the `rho.cubic`/`rho.quartic` class set-up (16–58 GAE per solve) out of each record's total (`research/class_setup_impact_20261024/`).

B.5 Transfers and covers

The transfer audit's v2 certificates (amendment 3, 26 October) changed the baseline itself. The model's $\sqrt{\pi r/4}$ is already rho with negation; dividing it again by $\sqrt{c}$ counted negation twice and put every baseline 0.5 bit too low. The amendment also gave rho on a binary cover's Jacobian image the Frobenius class of the baseline curve (Koblitz cover ratios had been overstated by $\frac{1}{2}\log_2(n/d)$ bits), required a margin of 8 bits before a price counts as below the baseline, and kept the verdict `refuted` only for a carried route priced as a route; formula sweeps, identity-path prices and unconstructed maps are now `unknown`. The regeneration first overwrote the v1 certificates; they were restored byte for byte in `v1/` on 27 October (`research/transfer_audit_20261011/PROTOCOL.md`). The other transfer items are in Table B.6.

B.6 Structure: conductor gaps, curve traits and instruments

Conductor gaps. The conductor-gap series has the longest correction trail. The CM-pairs study first stated that "the Lean-provable theorem" gave the null for generic methods. On 16 October a sheriff review (`SHF-20261016-3caa6f`, finding L5) noted that `lean/GenericInvariance.lean` covers only algorithms whose control flow depends on equality answers and a random tape, that is BSGS-type algorithms, while ecbench's rho and kangaroo hash the point representation. On 17 October a second review (`SHF-20261017-e4d7f6`, L-N2) withdrew the conclusion "no implemented method sees the gap", and on 28 October (`SHF-20261028-0f3f64`, PPN-1) the banner that still carried it was corrected. The open rho reading at $\ell = 8209$, 1.22 $[1.005, 1.46]$ from 4 targets, was then excluded by 12,288

Item	Original reading	Correction	Source
Automorphism rho, prime	cubic/negation 0.611, quartic/negation 0.791	24 Oct: 0.647 $[0.543, 0.770]$ and 0.819 $[0.722, 0.929]$; the quartic interval no longer contains $1/\sqrt{2}$, so finding 1 holds for cubic only	`research/prime_rho_automorphism_20261015/README.md`
Volcano crater controls	$j = 0$ 0.697, $j = 1728$ 0.707	24 Oct: 0.716 $[0.598, 0.857]$ and 0.723 $[0.621, 0.843]$; both still exclude 1	`research/isogeny-volcano-cost_20261017/README.md`
GLV-invariant matched walk (E27)	distances of §9.24	30 Oct: every row 1.448–1.456× further from rho at $w = 0$; *validated*; told in §34.1	`research/e27_walk_20261030/REPORT.typ`
Order-24 walk (E29)	crossing with order 6 at $w = 1$ (§9.33)	with its 40.6 unpriced mod-n operations per step priced, order 24 loses at every weight; the crossing was a one-op-per-step artefact	GLV note, §9.39 (E34)
Verschiebung split	round 1: $\lambda = t - 1$ via V^n, ratio 0.49, "not established"	round 2: $V^n = [2^n] \circ \pi^{-1}$ is n doublings plus square roots; ratio ≥ 2.5 at $c_{\mathrm{dbl}} = 8$, a provable loss	`research/isogeny_loop_glv_20261011/README.md`

Table B.5: Corrections to rho references used by studies. GLV note: `research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`, whose §9.19 also records seven textual errata found by the validation gate with no result, verdict or ranking changed. All rows except E27 are producer-reported.

paired solves under a fresh seed: any floor/crater difference is below about 1.2% there. In the easy-gap tier ($\ell \approx 2^{25}$) the frozen interval 0.917 $[0.851, 0.988]$ excluded 1, but a preregistered twin round put the decisive floor-twin/crater-twin ratio at 1.040 $[0.966, 1.119]$: the shift attaches to the coordinate models, not to the endomorphism ring (`research/conductor_gap_cm_pairs_20261010/README.md`, `research/conductor_gap_rho_large_sample_20261018/README.md`). The index-calculus side is in §26; the remaining structural items are in Table B.7.

B.7 Unconventional routes

None of the rows in Table B.8 opens or closes a route by itself. Two changed a status in the other direction from the usual one: the jet correction removed a closure that had no derivation behind it, and the JKSST audit (Table B.1) turned an exact-form closure into a conditional one. A withdrawn closure is a correction, not evidence for the route.

B.8 Papers, notes and provenance

Table B.9 lists corrections to papers and notes, and provenance flags that bar a record from being cited as evidence.

The LLL degeneracy note. The note reported two curve-specific failure modes: secp256k1 failing LLL at 256 bits where P-256 and brainpoolP256r1 succeed, and a uniform failure at 384 bits and above. Both were overflow bugs in the `f64` Gram–Schmidt step (n^4 exceeds the largest `f64`), and were withdrawn on 16 October (`SHF-20261016-3caa6f`). On the fixed code every distinct seed pair at 256 and 384 bits recovers the key, secp256k1 included (3 of 3); P-521 at $k_{\mathrm{bits}} = 384$ recovers 0 of 3 on scaled `f64` and 3 of 3 with 2048-bit fixed-point Gram–Schmidt. On 17 October the mechanism sentence for P-521 ("catastrophic cancellation") was withdrawn as well (`SHF-20261017-e4d7f6`), because with the scaling P-521's diagonal is exactly 2^{500} and the described subtraction is exact (`papers/lll-degeneracy-note/README.md`). The note has no index-calculus content; it is listed because it is the clearest case here of a curve "property" that was a numerical bug.

Item	Original reading	Correction	Source
Transfer-audit baseline	§4 baseline; `refuted` for any route not below it	26 Oct: every baseline +0.5 bit; 8-bit margin; `refuted` narrowed (see text)	`research/transfer_audit_20261011/`
Cover descent at the genus floor	846 `cover_descent` rows `formula_advantage`; earlier reading: all rest on a genus no GHS cover has	priced at the genus the construction has: 599 remain (576 at $k = 6$, 23 at $k = 14$), 41 marginal, 206 boundary; rows stay `proposed`	`research/cover_descent_genus_floor_20261015/README.md`
Hilbert stop on the weak-class cover (B0)	counted S falls 2.53–2.64×	23 Oct: relabelling; unpriced F4-driver operations rose 8.3–8.5×; the repaired form B is engineering (1.85–1.88×)	`research/ic_cover_weakclass_20261023/REPORT.md`
GHS cover census, $n = 4$	preregistered: floor 21 for $E(K)[2] = \mathbb{Z}/2$; odd-class minima in $\{13, 25\}$	both failed; derived floor 25; odd class 13 iff $j \notin \mathbb{F}_{q^2}$, else 37, never 25; one check bug fixed	`research/ghs_cover_census_n4_20261024/PROTOCOL.md`
Isogeny seed, curve25519	seed 1093 cheaper than seed 0	3 Nov: V2 failed (mean S ratio $0.641 > 0.6$); interval $[0.379, 1.084]$ includes 1; gain not established	`research/curve25519_isogeny_seed_20261103/README.md`
CM canonical lift (lane L9)	closure argument as first written	`CORR-20260810-930c42`: closes only ordinary prime-field Serre–Tate/global CM models of fixed arity 2–9; point-lifting routes give $\Pr \le p^{-1+o(1)}$, exponent 1 against rho's 1/2	`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-cm-canonical-lift-heights.yaml`
GHS magic number	legacy `ec_trapdoor::magic_number` (plain orbit)	13 Oct: MMT definition; consequences for ECC2K-130 are in §15 and §36.5	`research/weak_isogeny_classes_20261013/REPORT.md`

Table B.6: Corrections in the transfer and cover studies. All producer-reported.

Item	Original reading	Correction	Source
Conductor gap, CM pairs	"no implemented method sees the gap"; null is a theorem for generic methods	16, 17, 28 Oct: theorem covers BSGS-type arms only; banner withdrawn and qualified (see text)	`research/conductor_gap_cm_pairs_20261010/`
Rho at $\ell = 8209$; easy-gap tier	1.22 $[1.005, 1.46]$; post hoc 1.17 $[1.00, 1.38]$	18 Oct: first excluded; second attributed to coordinate models (1.040 $[0.966, 1.119]$)	`research/conductor_gap_rho_large_sample_20261018/`
Round-1 crater/floor set-up gap	prime floor/crater S ratio 3.10	round 2, A0: a closure counting defect (the floor paid 3.56× the needed set-up); ratio 1.00	`research/isogeny_conductor_gap_round2_20261011/README.md`
Drastic prime gap, $j = 0$	floor/crater rho quotient 0.5286	round 2, 32 fresh targets: 0.9468 $[0.6622, 1.3474]$ (99%); target-sampling noise	`research/drastic_prime_conductor_gap_round2_20261012/README.md`
NIST skip criterion	coefficient test separates the skipped curves	25 Oct: the smaller P-224 root rejects the published model, and the two models are isomorphic; representation-dependent, not an intrinsic weakness	`research/nist_skip_criterion_20261025/REPORT.md`
Coordinate census `EV-ECDLP-81f4d4`	strength *strong*	26 Oct (`CORR-20261026-c94668`): lowered to *preliminary*, with other records of the same sheriff case	`crypto-autoresearcher:ledger/corrections/CORR-20261026-c94668.yaml`
Weak-class instrument I1	*invalid* on all three classes	4 Nov: the A/A gate (5% on 20+20 draws) was miscalibrated; with a calibrated gate I1 is valid and blind; the paired-curve anomaly did not replicate (108+108 curves)	`crypto-autoresearcher:docs/reports/weakclass-i1-20261104/`, `crypto-autoresearcher:docs/reports/paired-anomaly-20261104/`
Polynomial-ring form (FHJRV)	"unavailable" unless b is a square	2 Nov (`CORR-20261102-7e3a91`): holds for the x (Kummer) class only; for $u = x/y$ on a double-odd curve the quotient is a polynomial ring	`crypto-autoresearcher:ledger/corrections/CORR-20261102-7e3a91.yaml`

Table B.7: Structural corrections. All producer-reported, the sheriff findings included.

Item	Original reading	Correction	Source
Jet oracle	"simulable with $C = 1$, closed at exponent 1/2", a derivation "valid at all scales"; close jet candidates by theorem	27 Jul (`CORR-20260727-003`): withdrawn, no derivation exists; closure countermanded; simulability open; `EV-GGM-002` supersedes `EV-GGM-001`	`crypto-autoresearcher:ledger/corrections/CORR-20260727-003.yaml`
Planted generic-group null	Stage 1 null on planted $\mathbb{Z}/n$, n of 16–20 bits	`EV-ECDLP-8a5106`: O-ARTIFACT, the kangaroo check M4 missed its 0.10 band; decision *refine*	`crypto-autoresearcher:ledger/evidence/EV-ECDLP-8a5106.yaml`
ETH-route leads	twelve leads as written in `LEADS.md`	2 Nov errata; told in §38.6	`research/ic_prime_eth_leads_20261031/LEADS.md` §9
Learned WDSat branching	learned order 16,097 held-out conflicts against 3,092	confounded with the symmetry switch; told in §39.3	`cryptanalysis:experiments/ideas_learning_20261018/REPORT.md`
Hashed-ECDL set-up hang	user report: Fourier-network construction impractical	30 Oct: the AKS branch above `u64`; told in Table 38.1	`research/hashed_ecdl_hash_prime_20261030/README.md`
Eigen-Séta (not ECDLP)	`walk` variant; NIST categories 1/3/5	11 Oct: `walk` insecure (5/5 ciphertexts decrypted from public data); 17 Oct: categories hold only in gate count, unqualified claim withdrawn	`research/eigen_seta_prototype_20261011/README.md`, `research/eigen_seta_assessment_20261017/REPORT.md`

Table B.8: Corrections on unconventional routes, besides H-PSEUDO and xedni (Table B.1). All producer-reported.

Item	Original reading	Correction	Source
LLL degeneracy note	secp256k1 and $\geq$ 384-bit curves fail LLL	withdrawn 16–17 Oct (see text)	`papers/lll-degeneracy-note/`
Structural-completeness paper	"the theorem is proven by case analysis"	17 Oct (`SHF-20261017-e4d7f6`, P-N4–N6): "argued"; §8 holds proof sketches; check counts recounted	`papers/structural-completeness/README.md`
Isogeny-class rho paper	no test fired; inferential null	19 Oct review: false (one B1 generator, two P1 cells, one P0 cell flagged; Kruskal–Wallis $p = 0.039$); KS defect, missing tie corrections; null provisional, p-values not recomputed	`papers/isogeny-class-rho/REVIEW-20261019.md`
Chung–Costello–Smith Recover	displayed Eqs. (11), (12)	22 Oct: in this implementation's convention the literal equations give 1,257 wrong signed divisors and 3 $D = 0$ cases of 1,260; corrected equations recover all 1,260	`research/kummer_recover_20261022/REPORT.md`
`secpkk1` ladder (lit2026 W11(e))	published 8/12/16-bit challenge cards	the cards use the opposite prime-search direction from the paper's own Algorithm 3	`research/lit2026_w11_generic_engineering_20261011/README.md`
Teske trapdoor evidence	`EV-ECTD-8c098d` rho-substitution prose	fabrication defect disclosed: the stated empirical justification has no supporting data	`crypto-autoresearcher:ledger/evidence/EV-ECTD-8c098d.yaml`
Kernel-of-reduction design	`DEC-20261023-26a6de` cites `KN-LIT-7593`	the citation resolves to an unrelated AES paper	`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-kernel-of-reduction-lemma.yaml`
Synthetic volcano study	volcano and ECDLP figures	invented data by its own banner; not evidence	`research/synthetic_volcano_ecdlp_20261018/README.md`
Frobenius-quotient follow-up	reported SAT runs	supplied as a note; models and logs not committed; not reviewed evidence	`research/frobenius_quotient_followup_20260926/README.md`
Scalar-3 Frobenius gate	new screen for $q = 16$	the mechanism is a rediscovery of the scalar-3/Jordan screen of `research/cubic_extension_transfer_20261025/`; only its extension to $q = 16$ is new	`research/ghs_scalar3_frobenius_gate_20261027/REPORT.md`

Table B.9: Corrections to papers and notes, and provenance flags. All producer-reported or sheriff-reported.

What the list shows. Three patterns recur. First, a reference or a pricing rule was corrected more often than an index-calculus count: the class set-up, the strong Koblitz id, the negation map, the byte counter and the transfer baseline each changed the distance to rho with no change to the attacking side. Second, a preregistered control or gate that failed, as in the Cheon, cofactor-clearing and E42 rows, voided the verdict or the ranking whatever the measured quantity said; the rule was applied as frozen. Third, several "properties" of curves or classes (the LLL failures, the conductor-gap rho shifts, the NIST coefficient test) turned out to be properties of a number format, a coordinate model or a representation. As in Chapter 41, no correction listed here moved a generic or leaderboard index-calculus row below a matched rho. The one row in this appendix that ends below rho is the Joux–Vitse sieve route, which is restricted to the `jv_cover` weak class (full rational 2-torsion, never prime order) and is an accounting result against a newly matched reference.

Bibliography

[1] Alessandro Amadori, Federico Pintore, and Massimiliano Sala. On the discrete logarithm problem for prime-field elliptic curves. *Finite Fields and Their Applications*, 51:168–182, 2018.

[2] Anonymous. Ratcheting elliptic-curve index calculus toward Pollard rho. Research draft, `papers/ecdlp-ic-improvement-ledger/` in the `crypto` repository, 2026.

[3] Anonymous. Elliptic-curve index calculus against Pollard rho: A survey of results, a measured frontier with every cost priced, and open problems. Research draft, `papers/ecdlp-index-calculus-program/` in the `crypto` repository, 2026.

[4] Daniel J. Bernstein and Tanja Lange. Computing small discrete logarithms faster. In *Progress in Cryptology – INDOCRYPT 2012*, volume 7668 of *LNCS*, pages 317–338, 2012.

[5] Alessio Caminata and Elisa Gorla. Solving degree, last fall degree, and related invariants. *Journal of Symbolic Computation*, 2023. arXiv:2112.05579.

[6] Matthew Clegg, Jeffery Edmonds, and Russell Impagliazzo. Using the Groebner basis algorithm to find proofs of unsatisfiability. In *Proceedings of the 28th ACM Symposium on Theory of Computing (STOC)*, pages 174–183, 1996.

[7] Don Coppersmith. Solving homogeneous linear equations over GF(2) via block Wiedemann algorithm. *Mathematics of Computation*, 62(205):333–350, 1994.

[8] Henry Corrigan-Gibbs and Dmitry Kogan. The discrete-logarithm problem with preprocessing. In *Advances in Cryptology – EUROCRYPT 2018*, LNCS. Springer, 2018. Cryptology ePrint Archive, Paper 2017/1113.

[9] Claus Diem. An index calculus algorithm for plane curves of small degree. In *Algorithmic Number Theory – ANTS VII*, volume 4076 of *LNCS*, pages 543–557, 2006.

[10] Claus Diem. On the discrete logarithm problem in elliptic curves. *Compositio Mathematica*, 147(1): 75–104, 2011.

[11] Claus Diem and Emmanuel Thomé. Index calculus in class groups of non-hyperelliptic curves of genus three. *Journal of Cryptology*, 21(4):593–611, 2008.

[12] Jean-Charles Faugère. A new efficient algorithm for computing Gröbner bases (F4). *Journal of Pure and Applied Algebra*, 139(1–3):61–88, 1999.

[13] Jean-Charles Faugère. A new efficient algorithm for computing Gröbner bases without reduction to zero (F5). In *Proceedings of ISSAC 2002*, pages 75–83. ACM, 2002.

[14] Jean-Charles Faugère, Pierrick Gaudry, Louise Huot, and Guénaël Renault. Using symmetries in the index calculus for elliptic curves discrete logarithm. *Journal of Cryptology*, 27(4):595–635, 2014.

[15] Michael D. Fried and Moshe Jarden. *Field Arithmetic*. Springer, 3rd edition, 2008.

[16] Steven D. Galbraith and Pierrick Gaudry. Recent progress on the elliptic curve discrete logarithm problem. *Designs, Codes and Cryptography*, 78(1):51–72, 2016.

[17] Steven D. Galbraith and Shishay W. Gebregiyorgis. Summation polynomial algorithms for elliptic curves in characteristic two. In *Progress in Cryptology – INDOCRYPT 2014*, LNCS. Springer, 2014. Cryptology ePrint Archive, Paper 2014/806.

[18] Steven D. Galbraith, Robert Granger, Simon-Philipp Merz, and Christophe Petit. On index calculus algorithms for subfield curves. In *Selected Areas in Cryptography – SAC 2020*, volume 12804 of *LNCS*. Springer, 2021. Cryptology ePrint Archive, Paper 2020/1315.

[19] Robert P. Gallant, Robert J. Lambert, and Scott A. Vanstone. Improving the parallelized Pollard lambda search on anomalous binary curves. *Mathematics of Computation*, 69(232):1699–1705, 2000.

[20] Pierrick Gaudry. Index calculus for abelian varieties of small dimension and the elliptic curve discrete logarithm problem. *Journal of Symbolic Computation*, 44(12):1690–1702, 2009.

[21] Pierrick Gaudry, Florian Hess, and Nigel P. Smart. Constructive and destructive facets of Weil descent on elliptic curves. *Journal of Cryptology*, 15(1):19–46, 2002.

[22] Pierrick Gaudry, Emmanuel Thomé, Nicolas Thériault, and Claus Diem. A double large prime variation for small genus hyperelliptic index calculus. *Mathematics of Computation*, 76(257):475–492, 2007.

[23] Ming-Deh A. Huang, Michiel Kosters, and Sze Ling Yeo. Last fall degree, HFE, and Weil descent attacks on ECDLP. In *Advances in Cryptology – CRYPTO 2015*, LNCS. Springer, 2015. Cryptology ePrint Archive, Paper 2015/573.

[24] Ming-Deh A. Huang, Michiel Kosters, Yun Yang, and Sze Ling Yeo. On the last fall degree of zero-dimensional Weil descent systems. *Journal of Symbolic Computation*, 87, 2018. arXiv:1505.02532.

[25] Michael J. Jacobson, Jr., Alfred Menezes, and Andreas Stein. Solving elliptic curve discrete logarithm problems using Weil descent. *Journal of the Ramanujan Mathematical Society*, 16(3):231–260, 2001. Also University of Waterloo CORR 2001-31.

[26] David Jao, Stephen D. Miller, and Ramarathnam Venkatesan. Do all elliptic curves of the same order have the same difficulty of discrete log? In *Advances in Cryptology – ASIACRYPT 2005*, volume 3788 of *LNCS*, pages 21–40. Springer, 2005.

[27] Antoine Joux and Vanessa Vitse. Cover and decomposition index calculus on elliptic curves made practical. In *Advances in Cryptology – EUROCRYPT 2012*, volume 7237 of *LNCS*, pages 9–26. Springer, 2012.

[28] Michiel Kosters and Sze Ling Yeo. Notes on summation polynomials. arXiv:1503.08001, 2015.

[29] Stavros Kousidis and Andreas Wiemers. On the first fall degree of summation polynomials. *Journal of Mathematical Cryptology*, 2019. doi:10.1515/jmc-2017-0022; arXiv:1906.05594.

[30] Gary McGuire and Daniela Mueller. A new index calculus algorithm for the elliptic curve discrete logarithm problem and summation polynomial evaluation. Cryptology ePrint Archive, Paper 2017/1262, 2017.

[31] Koh-ichi Nagao. Complexity of ECDLP under the first fall degree assumption. Cryptology ePrint Archive, Paper 2015/984, 2015.

[32] Christophe Petit and Jean-Jacques Quisquater. On polynomial systems arising from a Weil descent. In *Advances in Cryptology – ASIACRYPT 2012*, volume 7658 of *LNCS*, pages 451–466. Springer, 2012.

[33] Christophe Petit, Michiel Kosters, and Ange Messeng. Algebraic approaches for the elliptic curve discrete logarithm problem over prime fields. In *Public-Key Cryptography – PKC 2016*, volume 9615 of *LNCS*, pages 3–18. Springer, 2016.

[34] John M. Pollard. Monte Carlo methods for index computation (mod p). *Mathematics of Computation*, 32(143):918–924, 1978.

[35] Igor Semaev. Summation polynomials and the discrete logarithm problem on elliptic curves. Cryptology ePrint Archive, Paper 2004/031, 2004.

[36] Igor Semaev. New algorithm for the discrete logarithm problem on elliptic curves. Cryptology ePrint Archive, Paper 2015/310, 2015.

[37] Michael Shantz and Edlyn Teske. Solving the elliptic curve discrete logarithm problem using Semaev polynomials, Weil descent and Gröbner basis methods – an experimental study. In *Number Theory and Cryptography*, volume 8260 of *LNCS*, pages 94–107. Springer, 2013.

[38] Matthias Johann Steiner. Solving degree bounds for iterated polynomial systems. arXiv:2310.03637, 2023.

[39] Monika Trimoska, Sorina Ionica, and Gilles Dequen. Parity (xor) reasoning for the index calculus attack. In *Principles and Practice of Constraint Programming – CP 2020*, volume 12333 of *LNCS*, pages 774–790. Springer, 2020.

[40] Paul C. van Oorschot and Michael J. Wiener. Parallel collision search with cryptanalytic applications. *Journal of Cryptology*, 12(1):1–28, 1999.

[41] M. D. Velichka, M. J. Jacobson, Jr., and A. Stein. Computing discrete logarithms in the Jacobian of high-genus hyperelliptic curves over even characteristic finite fields. *Mathematics of Computation*, 83(286):935–963, 2014.

[42] Mark Velichka. Improvements to index calculus algorithms for solving the hyperelliptic curve discrete logarithm problem over characteristic two finite fields. Master's thesis, University of Calgary, 2008.

[43] Douglas H. Wiedemann. Solving sparse linear equations over finite fields. *IEEE Transactions on Information Theory*, 32(1):54–62, 1986.

[44] Michael J. Wiener and Robert J. Zuccherato. Faster attacks on elliptic curve cryptosystems. In *Selected Areas in Cryptography – SAC 1998*, volume 1556 of *LNCS*, pages 190–200. Springer, 1999.